

Engineering Activity Detail Reference

A 23-stage semiconductor program template — every engineering activity's purpose, work flow, inputs, relationship to the stage's key deliverables, risks, effort and criteria.

STAGES	ACTIVITIES	KEY DELIVERABLES	PROGRAM SPAN	TOTAL EFFORT
23	257	167	132 _w	3,653 _{M/M}

Reference basis: new-architecture 4nm-class AI accelerator, 2.5D packaging with 2xHBM3.
Generated from docs/activity-details.json · joins on activity id against docs/stage-template-v2.json.

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DEFINE

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DEF

DEF-01

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DEF-02

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DEF-03

PPA target definition — frequency, power envelope, die area budget

DEF-04

Die size, yield and product cost model (wafer, mask, package, test, ASP)

DEF-05

Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)

DEF-06

Competitive benchmarking and product positioning

DEF-07

Feasibility assessment against candidate technology nodes

DEF-08

Program schedule, resource and staffing plan drafting

DEF-09

Business case preparation and funding approval package

02

Architecture

ARCH

ARCH-01

System-level performance modelling and workload simulation

ARCH-02

Compute / memory / interconnect partitioning; monolithic vs chiplet decision

ARCH-03

Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)

ARCH-04

Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E)

ARCH-05

Power domain, clock domain and DVFS architecture; UPF intent draft

ARCH-06

PPA budget allocation per block

ARCH-07

Security and safety architecture — secure boot, root of trust, fusing

ARCH-08

Chip-level floorplan intent, pin and bump budget

ARCH-09

Architecture specification authoring and review

ARCH-10

Block microarchitecture specification kickoff

TECHNOLOGY & IP ENABLEMENT

03 **Technology & Foundry Selection** TECH w0-w14

TECH-01	Foundry and node long-list to short-list evaluation	TECH-06	Foundry legal engagement — NDA, design agreement, IP licensing frame
TECH-02	Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)	TECH-07	Foundry roadmap and risk-production timing alignment
TECH-03	Density / performance / leakage benchmarking against product targets (DTCO study)	TECH-08	Second-source and node-migration strategy assessment
TECH-04	Wafer price, mask and NRE quotation; MPW and volume pricing	TECH-09	OSAT and backend supply chain preliminary alignment
TECH-05	Fab capacity and tapeout slot reservation; hot-lot policy agreement		

04 PDK & Design Enablement **PDK**

w8–w44

PDK-01	PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	PDK-08	EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR
PDK-02	Design rule manual review; restricted and recommended rule disposition	PDK-09	Foundry reference flow bring-up and internal methodology deck
PDK-03	Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	PDK-10	Rule deck, QRC tech file and DRC/LVS deck version control
PDK-04	Memory compiler evaluation and instance generation (SRAM, register file)	PDK-11	Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)
PDK-05	Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage	PDK-12	Compute farm, EDA license and storage capacity planning
PDK-06	Custom / pushed-rule memory instance decision, scope and schedule impact assessment	PDK-13	Golden design environment release and flow regression
PDK-07	IO, ESD and latch-up library qualification		

05 IP Strategy & Readiness **IPR**

w4–w26

IPR-01	Product requirement → IP requirement decomposition (IP bill of materials)	IPR-06	IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)
IPR-02	Reuse inventory audit — internal IP already available on the target node	IPR-07	Licence negotiation, PO issue and delivery-date commitment
IPR-03	Make / buy / reuse decision per IP block	IPR-08	Porting / hardening scope and schedule for IP not proven on the node
IPR-04	Vendor RFI / RFQ and technical evaluation	IPR-09	IP delivery schedule alignment to RTL and PD integration windows
IPR-05	Silicon-proven status check per IP on the selected node and process option	IPR-10	IP maturity risk rating and second-source contingency

06 Custom, AMS & Memory IP Development **AMS**

w18–w60

AMS-01	AMS IP specification and budget allocation (jitter, BER, PSRR, area)	AMS-09	Redundancy and repair scheme integration with the BISR architecture
AMS-02	PLL / clock generator design and closure	AMS-10	Custom memory view generation and compiler-equivalent characterisation across corners
AMS-03	SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)	AMS-11	Schematic design and pre-layout simulation
AMS-04	LDO, bandgap and power-management cell design	AMS-12	Custom layout with DRC / LVS / antenna and fill closure
AMS-05	Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing	AMS-13	Post-layout extracted simulation across PVT and Monte Carlo
AMS-06	Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse	AMS-14	Reliability checks — EM/IR, aging, ESD, latch-up
AMS-07	Custom array layout and pushed-rule DRC closure with foundry approval	AMS-15	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
AMS-08	Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation	AMS-16	AMS-digital co-simulation and integration support

07 Test Chip / MPW Shuttle **TC**

w22–w62

TC-01	Test chip objective definition and risk-item selection	TC-05	Shuttle tapeout and fabrication
TC-02	Test chip design and integration (IP under test, ring oscillators, PCM structures)	TC-06	Test chip board and lab setup preparation
TC-03	MPW shuttle slot booking and data submission scheduling	TC-07	Silicon characterisation and correlation to simulation
TC-04	Test chip physical implementation and signoff	TC-08	Feedback into the production design — model correction and margin decisions

DESIGN & VERIFY

08 **RTL Design & Integration** **RTL**

w20–w52

RTL-01	Block microarchitecture specification
RTL-02	Block-level RTL implementation
RTL-03	Third-party and internal IP integration — wrappers, glue, configuration
RTL-04	Chip-level integration and top assembly
RTL-05	Clock, reset and power intent (UPF) implementation

RTL-06	Lint, CDC and RDC closure
RTL-07	Register map / RDL definition and header generation
RTL-08	Trial synthesis feedback loop on RTL PPA
RTL-09	CI build, nightly regression and release management
RTL-10	Specification change control and ECO board

09 **Verification** **DV**

w26–w66

DV-01	Verification plan and coverage model definition
DV-02	UVM environment and VIP bring-up
DV-03	Block-level constrained-random and directed testing
DV-04	Chip-level and system-level scenario testing
DV-05	Formal property verification — control logic, connectivity, security
DV-06	Low-power (UPF) verification
DV-07	AMS / mixed-signal co-simulation

DV-08	Emulation and FPGA prototype bring-up — capacity, partitioning, speed
DV-09	Performance and bandwidth validation against the architecture model
DV-10	Coverage closure and regression stability management
DV-11	Gate-level simulation, functional and timing-annotated
DV-12	Bug triage and disposition board

10 **DFT Architecture** **DFT**

w18–w58

DFT-01	DFT architecture definition — scan style, compression ratio, hierarchical DFT
DFT-02	Test coverage and test-time target negotiation with product and test engineering
DFT-03	MBIST / BIRA / BISR architecture for embedded memories
DFT-04	Boundary scan and JTAG (1149.1 / 1687), TAP and debug access architecture
DFT-05	On-chip clock controller design for at-speed testing
DFT-06	Scan insertion and DFT DRC during synthesis

DFT-07	ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
DFT-08	Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
DFT-09	Scan compression and chain routing feasibility with physical design
DFT-10	eFuse, chip ID and memory repair infrastructure
DFT-11	Silicon debug hooks — trace, observability, design-for-debug

IMPLEMENT

11 **Logic Synthesis & Netlist Drops** **SYN**

w42–w66

SYN-01	SDC constraint development and validation
SYN-02	Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
SYN-03	N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow
SYN-04	Physical-aware synthesis with congestion feedback from the PD flow setup
SYN-05	N1 netlist drop — first quality turn against PD placement and congestion feedback
SYN-06	Power optimisation — clock gating, operand isolation, Vt mix

SYN-07	Low-power synthesis — isolation and level-shifter insertion against UPF
SYN-08	N2 netlist drop — timing and congestion converged, late RTL changes absorbed
SYN-09	FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point
SYN-10	Formal equivalence checking per drop, RTL versus netlist
SYN-11	Timing, area and power reporting against budget, per drop
SYN-12	Netlist handoff and QoR delta review with physical design, per drop

12 **Physical Design** PD

w46–w76

PD-01	Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity	PD-08	Detailed routing and DRC convergence
PD-02	Floorplan, macro placement and partition definition	PD-09	Multi-corner multi-mode timing closure
PD-03	Power delivery network design and early IR analysis	PD-10	Signal and power integrity iteration — crosstalk, EM
PD-04	Bump and RDL planning with chip-package co-design feedback	PD-11	Chip power model (CPM/CPS) extraction and handoff to package co-verification
PD-05	Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	PD-12	Scan chain reordering and DFT-aware routing
PD-06	Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified	PD-13	Final turn on the FFN — full closure, no functional change admitted
PD-07	Clock tree synthesis with skew and jitter budgeting, refined across turns	PD-14	Functional and timing ECO implementation
		PD-15	Chip finishing — seal ring, dummy and metal fill, alignment marks
		PD-16	Hierarchical block closure and top-level assembly

13 **Signoff** SO

w62–w78

SO-01	Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed	SO-07	Reliability verification — ESD, latch-up, soft error / FIT
SO-02	Multi-corner multi-mode static timing analysis and closure on the final turn	SO-08	DFM and lithography checks, CMP and hotspot analysis
SO-03	Full-chip DRC, LVS, antenna and density verification	SO-09	Final formal equivalence and LVS netlist consistency
SO-04	EM and IR-drop signoff, static and dynamic	SO-10	Waiver review board and foundry waiver alignment
SO-05	Power and signal integrity signoff — crosstalk, noise	SO-11	Gate-level simulation with final SDF
SO-06	Chip-package-system co-analysis signoff review against SIPI results	SO-12	Signoff corner correlation against foundry decks

MANUFACTURE

14 **Tapeout & Mask Release (MTO)** T0

w78–w86

T0-01	Final GDSII / OASIS assembly on the FFN closure database, with layer map verification	T0-07	FEOL mask order confirmation and mask shop scheduling
T0-02	Final full-chip verification re-run on the released database	T0-08	BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
T0-03	Tapeout checklist completion and owner sign-off	T0-09	BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack
T0-04	Open-issue risk assessment and waiver acceptance	T0-10	BEOL layer data preparation and MTO release
T0-05	Go / No-Go decision meeting	T0-11	BEOL mask order confirmation and full mask set completion tracking
T0-06	FEOL layer data preparation and MTO release to the mask shop		

15 **Fabrication** FAB

w79–w98

FAB-01	FEOL mask set fabrication, inspection and qualification	FAB-05	Back-end-of-line processing
FAB-02	Wafer start on FEOL mask availability	FAB-06	Inline metrology and defect inspection monitoring
FAB-03	BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing	FAB-07	E-test / PCM data review
FAB-04	Front-end wafer processing	FAB-08	Wafer acceptance test and lot disposition
		FAB-09	Hot-lot management and WIP tracking
		FAB-10	Wafer shipment and logistics to sort and assembly

INTEGRATE

16 Package & Substrate Design

PKGD

w24–w76

- PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count
- PKGD-02 Bump map, pitch and power-ground planning with physical design
- PKGD-03 Interposer / RDL routing design
- PKGD-04 Substrate stack-up, escape routing and package DRC
- PKGD-05 Package routing for signal integrity — length matching, reference plane and via design

- PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)
- PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
- PKGD-08 Test vehicle requirement definition and handoff to PTV
- PKGD-09 Substrate and interposer supplier selection and lead-time booking
- PKGD-10 OSAT selection and assembly process definition
- PKGD-11 Package design freeze, DRC and tooling release

17 Package Test Vehicle (CPI / TTV)

PTV

w36–w88

- PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress
- PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon
- PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement
- PTV-04 Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures
- PTV-05 TV die fabrication and interposer / substrate vehicle build
- PTV-06 OSAT assembly of vehicles with process window DOE

- PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)
- PTV-08 Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity
- PTV-09 Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model
- PTV-10 Board-level reliability on vehicles — temperature cycle, drop, bend
- PTV-11 Daisy-chain continuity test and assembly yield learning
- PTV-12 Feedback into package design and assembly process window freeze

18 Chip-Package-System Co-Verification

SIPI

w50–w76

- SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode
- SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
- SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
- SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
- SIPI-05 Decoupling capacitor budget and placement optimisation across die, package and board
- SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring

- SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
- SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)
- SIPI-09 Power-aware timing correlation — voltage drop back-annotated into STA
- SIPI-10 Electro-thermal co-analysis coupled with the package thermal model
- SIPI-11 Co-verification signoff review and criteria disposition with design and package teams

19 Assembly & Package Build

ASSY

w98–w106

- ASSY-01 Known-good-die sort and selection
- ASSY-02 HBM stack procurement and incoming inspection
- ASSY-03 Die attach, micro-bump and thermo-compression bonding
- ASSY-04 Interposer and substrate assembly, underfill, molding
- ASSY-05 Lid / TIM attach and ball attach

- ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology
- ASSY-07 Assembly yield analysis and process tuning
- ASSY-08 Package-level open / short and continuity test
- ASSY-09 Unit build for bring-up, qualification and customer samples

VALIDATE & RAMP

20 Validation Hardware / EVB EV

w52–w90

EVB-01	Validation platform requirements and topology definition
EVB-02	EVB schematic design — power tree, clocking, host interface, connectors
EVB-03	PCB layout with high-speed channel SI/PI simulation
EVB-04	PCB fabrication and assembly
EVB-05	Power delivery, VRM and telemetry design and bring-up

EVB-06	Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers
EVB-07	Board bring-up with socketed or dummy parts
EVB-08	Thermal solution and cooling for the lab platform
EVB-09	Lab instrumentation reservation and test rack build
EVB-10	Minimum host-side enablement for power-on (BSP handoff boundary)

21 Test Development TEST

w54–w96

TEST-01	Test plan and coverage strategy — wafer sort, final test, system-level test
TEST-02	ATE platform selection and tester time booking
TEST-03	Probe card design, fabrication and qualification
TEST-04	Load board / DUT board design, fabrication and bring-up
TEST-05	Wafer sort test program development
TEST-06	Final / package test program development

TEST-07	Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps
TEST-08	DFT pattern porting to ATE format and pattern debug
TEST-09	Test time and cost optimisation — parallelism, site count
TEST-10	Correlation between ATE, bench and system results
TEST-11	Test data infrastructure — STDF collection and yield database

22 Silicon Bring-up BU

w102–w120

BU-01	Sample receipt, incoming inspection and board mounting
BU-02	Power-on, power sequencing and basic health check
BU-03	Reset, clocking and PLL lock validation
BU-04	Boot, firmware load and functional smoke test
BU-05	Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
BU-06	Memory subsystem and bandwidth validation

BU-07	Shmoo across voltage, frequency and temperature
BU-08	Performance validation against the architecture model
BU-09	Silicon debug of anomalies and failure analysis requests
BU-10	Errata capture, workaround definition and documentation
BU-11	Respin versus metal-fix decision analysis — ECO scope, cost, schedule
BU-12	Customer sample readiness and release package

23 Qualification & Production MP

w106–w132

MP-01	Qualification plan definition against JEDEC / AEC standards
MP-02	Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
MP-03	ESD (HBM, CDM) and latch-up qualification
MP-04	Package qualification — MSL, drop, bend, board-level reliability
MP-05	Yield learning, failure pareto and defect analysis
MP-06	Production test program release and guard-band validation

MP-07	Test time reduction and multi-site conversion
MP-08	Process corner and split-lot validation
MP-09	Production readiness review and change control (PCN) setup
MP-10	Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
MP-11	Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety
MP-12	Documentation release — datasheet, user guide, errata, application notes

DEFINE

01 Product Definition

DEF

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w0 – w8	8 _w	9	28 _{M/M}	5 _{of 9}		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
DEF-01	Market and customer requirements consolidation		4w	4 M/M	DEF-D1	Product requirements document (PRD)
DEF-02	Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)		5w	6 M/M	DEF-D2	Target specification — PPA and KPI table
DEF-03	PPA target definition — frequency, power envelope, die area budget		4w	3 M/M	DEF-D3	Product cost and margin model
DEF-04	Die size, yield and product cost model (wafer, mask, package, test, ASP)		4w	3 M/M	DEF-D4	Feasibility report
DEF-05	Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)		3w	2 M/M	DEF-D5	Program charter, staffing and budget plan
DEF-06	Competitive benchmarking and product positioning		4w	2 M/M	DEF-D6	Kickoff Go / No-Go decision record
DEF-07	Feasibility assessment against candidate technology nodes		4w	3 M/M		
DEF-08	Program schedule, resource and staffing plan drafting		4w	3 M/M		
DEF-09	Business case preparation and funding approval package		3w	2 M/M		

01 DEF · PRODUCT DEFINITION

DEF-01

Market and customer requirements consolidation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	4.0 M/M	~4.3 FTE	7	yes	Requirements lead
w0-w4 in stage	planned	while it runs	2 in parallel	no float	Stage lead approves

PURPOSE

Turn what customers and the market are asking for into **one ranked, signed requirement set** that the whole program is measured against.

Everything downstream is an answer to this list — the PPA targets, the cost model, the feasibility call, the architecture itself. Until it is baselined and signed, every one of those answers is provisional. A requirement that arrives in week 30 costs roughly a hundred times what it costs here, and the difference is paid in silicon.

WORK FLOW

7 STEPS · 2 LANES

w0	w1	w2	w3	w4
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Stakeholder map and interview plan	0.5w	main
2	Anchor-customer interviews — two accounts, six sessions	1.5w	main
3	Market and analyst input, segment sizing	1w	parallel
4	Requirement extraction, de-duplication, register build	1w	main
5	Conflict resolution and MoSCoW prioritisation workshop	0.5w	main
6	Traceability tagging to candidate KPIs and PPA targets	0.5w	parallel
7	Stakeholder review and baseline sign-off	0.5w	main

Seven steps add up to 5.5 weeks of work inside a 4-week window — steps 3 and 6 run alongside the main sequence. That is why the activity TAT and the sum of its steps do not match, and why pulling either parallel step onto the critical path costs a further 1.5 weeks.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Customer RFI / RFQ responses
- Field feedback and errata from the previous generation
- Analyst forecasts and segment sizing
- Competitor datasheets and public benchmarks
- Sales pipeline and committed-volume view

PRODUCES

- Interview notes, per account
- Requirement register — ID, owner, priority, acceptance measure
- Conflict and deferral log
- MoSCoW prioritisation record
- Baseline sign-off sheet

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-D1	PRODUCES	Product requirements document (PRD). The register is the PRD's body — this activity is where the deliverable's content comes from, not merely a contributor to it.
DEF-D2	FEEDS	Target specification — PPA and KPI table. Each ranked requirement becomes a KPI line that DEF-03 turns into a number. An untraced target is a target nobody asked for.
DEF-D3	INFORMS	Product cost and margin model. Supplies the volume, segment and ASP assumptions the model is built on.
DEF-D4	FEEDS	Feasibility report. DEF-07 tests <i>this</i> list against the candidate nodes — a feasibility study run against a different list proves nothing.
DEF-D6	GATES	Kickoff Go / No-Go decision record. The gate reads the sign-off sheet. An unsigned register is a No-Go condition, not a footnote.

RISKS AND WATCH-OUTS

- ◆ **Requirements not signed off by all stakeholders.** Sales agreeing verbally is not a signature, and the Go / No-Go gate reads signatures.
- ◆ **Late customer additions after baseline.** With no change-control channel agreed at sign-off, the register keeps moving and nothing downstream can freeze.
- ◆ **Requirements written as solutions.** "Use HBM3E" is an answer; "819 GB/s sustained" is a requirement. Solutions here remove the architect's options in ARCH-02.
- ◆ **No acceptance measure.** A requirement that cannot be tested resurfaces at qualification, when the only remaining fix is a respin or a waiver.
- ◆ **One anchor account dominating the list.** Requirements that fit a single customer and no segment turn a product into a bespoke part carrying a product's cost structure.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Interviews and market input	1.5	● Program charter drafted and funded to study	DEF-02 Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)
Register build and traceability	1.5	● Anchor customers identified, NDAs executed	DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)
Workshops and review	0.5	● Previous-generation field feedback available	DEF-06 Competitive benchmarking and product positioning
Documentation and sign-off	0.5		DEF-07 Feasibility assessment against candidate technology nodes
Total	4.0		ARCH-01 System-level performance modelling and workload simulation
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA	
Requirements lead Owns the register and the baseline		● Every requirement has an owner, a priority and an acceptance measure	IPR-01 Product requirement → IP requirement decomposition (IP bill of materials)
Customer engagement One per anchor account, through the interview window		● Conflicts resolved, or deferred with a named decision owner	
Product cost analyst Volume, segment and ASP assumptions		● Register baselined and signed by product, engineering and sales	
Feasibility liaison Flags requirements the node cannot carry		DEPENDS ON <i>Nothing upstream — this is the program's first activity. Its preconditions are the entry criteria above.</i>	MEASURED BY
Stage lead Approves the baseline			● Requirements baselined against the expected count ● Share carrying an acceptance measure ● Change requests raised after baseline

DEF-02

Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	6.0 M/M	~5.2 FTE	7	yes	Workload architect
w1–w6 in stage	planned	while it runs	2 in parallel	no float	Architecture liaison approves

PURPOSE

The requirement register says what customers want. This says **what the silicon has to do** — the workloads it will actually run, and the KPI numbers derived from them.

A KPI with no workload behind it is a marketing number. ARCH-01 spends fourteen weeks simulating against whatever this activity hands over; if the traces are not representative of the deployed mix, the architecture is optimised for a workload nobody runs, and nothing in verification will catch it.

WORK FLOW

7 STEPS · 2 LANES

w1	w2	w3	w4	w5	w6
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MAIN SEQUENCE

1	2	4	6	7
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Use-case shortlist drawn from the requirement register	0.5w	main
2	Workload trace capture — LLM inference, vision, recommendation	1.5w	main
3	Reference runs on the previous generation for a baseline	1w	parallel
4	KPI derivation — TOPS, TOPS/W, tokens/s, time-to-first-token, batch scaling	1.5w	main
5	Precision and quantisation policy — INT8, FP8, mixed	1w	parallel
6	Sensitivity study — which KPI moves with which architectural knob	1w	main
7	Workload suite packaging and handoff to architecture	0.5w	main

The main sequence fills the 5-week window exactly. Steps 3 and 5 ride alongside it — the reference runs need lab time rather than analyst time, and the precision policy can be settled while traces are still being captured.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Requirement register from DEF-01
- Customer-supplied models and traces, where the NDA allows
- Previous-generation silicon and its lab platform
- Public model zoo and framework benchmarks
- Deployment mix estimates from the segment sizing

PRODUCES

- Workload suite — traces, models, and the harness to run them
- KPI derivation worksheet, one row per requirement
- Precision and quantisation policy note
- Sensitivity table — KPI against architectural knob
- Baseline measurements on the previous generation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-D2	PRODUCES	Target specification — PPA and KPI table. The KPI half of the deliverable is derived here; DEF-03 supplies the PPA half.
DEF-D1	FEEDS	Product requirements document. The use-case list and the deployment mix are written back into the PRD as the context the requirements sit in.
DEF-D4	FEEDS	Feasibility report. Feasibility is tested against these KPI numbers, not against the requirement prose.

RISKS AND WATCH-OUTS

- ◆ **Traces that flatter the design.** A workload suite assembled from what runs well on the previous generation will not find the case that breaks the new one.
- ◆ **Customer models unavailable under NDA.** Substituting a public proxy is legitimate; not recording that a proxy was used is not.
- ◆ **Deployment mix guessed rather than sized.** The architecture is tuned to the mix; a wrong mix misallocates silicon area that cannot be recovered later.
- ◆ **Precision policy left open.** INT8 versus FP8 changes the datapath, the memory footprint and the KPI at once, and the decision gets more expensive every week it waits.
- ◆ **KPIs quoted without conditions.** TOPS with no batch size, no sequence length and no precision is not a number anyone can design against.

EFFORT	SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Trace capture and reference runs		2.0	- Requirement register at v0.9 or better - Lab access to the previous-generation platform - NDA coverage for any customer-supplied model	DEF-03 PPA target definition — frequency, power envelope, die area budget
KPI derivation		1.5		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)
Sensitivity study		1.0		DEF-07 Feasibility assessment against candidate technology nodes
Precision and quantisation policy		1.0		ARCH-01 System-level performance modelling and workload simulation
Suite packaging and handoff		0.5		DV-09 Performance and bandwidth validation against the architecture model
Total		6.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES		~5.2 FTE	- Every KPI traces to a named workload and stated conditions - Workload suite runs end to end on the reference platform - Architecture has accepted the handoff as modellable	- KPIs traceable to a workload - Workload suite runtime on the reference platform - Spread between the sensitivity study and the eventual ARCH-01 model
Workload architect Owns the suite and the KPI derivation				
ML performance engineer Trace capture, reference runs, precision study				
Systems analyst Deployment mix and sensitivity analysis				
Customer engagement Obtains models and confirms representativeness				
Architecture liaison Takes the handoff and confirms it is modellable			DEPENDS ON	
			DEF-01 Market and customer requirements consolidation	

DEF-03

PPA target definition — frequency, power envelope, die area budget

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	7	yes	PPA lead
w3–w7 in stage	planned	while it runs	2 in parallel	no float	Stage lead approves

PURPOSE

Convert the KPI set into the three numbers the design is actually held to — **frequency, power envelope, die area** — and allocate them so every block owner inherits a share rather than a slogan.

An unallocated target is not a target. "75 W board power" binds nobody until it is decomposed into a die budget, a per-domain budget and a per-block budget with an owner against each. That decomposition is what makes an overrun visible in week 20 instead of week 60.

WORK FLOW

7 STEPS · 2 LANES

w3	w4	w5	w6	w7
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Frequency target from the KPI set and expected IPC	0.8w	main
2	Power envelope decomposition — board, package, die, per-domain	1w	main
3	Die area budget from the block list and the reuse inventory	1w	main
4	Multi-Vt and operating voltage strategy, first pass	0.5w	parallel
5	PPA sensitivity against the candidate nodes	0.8w	parallel
6	Per-block allocation with a named owner against each budget	0.8w	main
7	Target review with architecture and physical design	0.5w	main

Allocation is the step that matters and it sits late, because it cannot start until the block list is stable. Steps 4 and 5 run in parallel with the budget work — both are inputs to the review rather than to the budgets themselves.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- KPI table from DEF-02
- Block list and reuse inventory from the architecture sketch
- Node PPA data from TECH-03
- Previous-generation power and area actuals
- Thermal and board constraints from the requirement register

PRODUCES

- Frequency target with the IPC assumption behind it
- Power budget tree — board to per-domain
- Die area budget by block
- Voltage and multi-Vt strategy note
- Per-block allocation sheet with owners

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-D2	PRODUCES	Target specification — PPA and KPI table. The PPA half of the deliverable. DEF-02 supplies the KPI half; the two are published as one table.
DEF-D4	FEEDS	Feasibility report. These are the numbers DEF-07 asks each candidate node to carry.
DEF-D3	INFORMS	Product cost and margin model. The die area budget is the model's largest single input.

RISKS AND WATCH-OUTS

- ◆ **Targets set from ambition rather than from the node.** A frequency target the library cannot reach turns into an eighteen-month argument that ends in a respin or a reduced bin.
- ◆ **Power budget with no margin.** A budget allocated to 100% at definition has nothing left for the ECO storm at physical design, when power always grows.
- ◆ **Area budget without the reuse inventory.** Budgets set before knowing which blocks are reused assign area to work that will not happen and starve work that will.
- ◆ **Allocation without owners.** An unowned budget is a budget nobody defends, and it is spent by whoever moves first.
- ◆ **Targets frozen before DEF-07 reports.** Feasibility can only return a verdict against targets; setting them beyond reach makes the verdict meaningless.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Frequency and power budgets	1.0	KPI table available from DEF-02	DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)
Die area budget	0.8	Block list stable enough to allocate against	DEF-07 Feasibility assessment against candidate technology nodes
Node sensitivity	0.5	At least two candidate nodes characterised by TECH-03	ARCH-06 PPA budget allocation per block
Per-block allocation	0.5		SYN-01 SDC constraint development and validation
Review and arbitration	0.3		PD-09 Multi-corner multi-mode timing closure
Total	3.0	EXIT CRITERIA - Frequency, power and area targets each carry a stated assumption - Every block has an allocated budget and a named owner - Architecture and physical design have signed the review	MEASURED BY - Share of the die budget allocated to a named owner - Margin held back against each budget - Delta between these targets and the ARCH-06 allocation
WHO IT TAKES	~3.3 FTE	DEPENDS ON DEF-02 Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s) DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D) TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	
PPA lead Owns the budget tree and the allocation			
Architect Frequency and IPC assumptions, block list			
Power analyst Envelope decomposition and voltage strategy			
Physical design liaison Reality-checks the frequency and area against the node			
Stage lead Arbitrates when blocks contest the same budget			

01 DEF · PRODUCT DEFINITION

DEF-04

Die size, yield and product cost model
(wafer, mask, package, test, ASP)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	7	no	Product cost analyst
w3–w7 in stage	planned	while it runs	2 in parallel	has float	Finance partner approves

PURPOSE

Price the product **before anyone commits to building it**. Wafer, mask, package, test and yield fold into one model that says what the part costs at volume and whether the margin survives.

The cost model is also the veto. A design that meets every KPI and misses the cost ceiling is not a product, and the only cheap moment to discover that is here — while the die size is still an estimate rather than a floorplan.

WORK FLOW

7 STEPS · 2 LANES

w3	w4	w5	w6	w7
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Die area estimate from the block list and the reuse inventory	0.8w	main
2	Wafer cost and gross die per wafer, per node candidate	0.5w	main
3	Defect density and yield model per node candidate	1w	main
4	Package and substrate cost — interposer, substrate layers, HBM stacks	0.8w	parallel
5	Test cost model — sort and final, with a test-time assumption	0.5w	parallel
6	Volume, ASP and margin roll-up	1w	main
7	Sensitivity — the three assumptions the model is most exposed to	0.8w	main

The sensitivity step is not optional decoration. A 2.5D part's cost is dominated by yield, interposer supply and HBM price, and each of those is an assumption rather than a quote at this point in the program.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Die area budget from DEF-03
- Wafer, mask and NRE quotations from TECH-04
- Package architecture assumption from the requirement register
- HBM and substrate price indications
- Volume forecast and ASP from the segment sizing

PRODUCES

- Cost model workbook — wafer, package, test, yield
- Gross die and yield curves per node candidate
- Margin roll-up at the forecast volume
- Sensitivity analysis on the three dominant assumptions
- Cost ceiling statement for the design

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-D3	PRODUCES	Product cost and margin model. This activity is the deliverable — the workbook and its assumptions are handed over intact, not summarised.
DEF-D4	FEEDS	Feasibility report. A node that meets the PPA targets and breaks the cost ceiling is recorded as infeasible, and this model is what says so.
DEF-D6	GATES	Kickoff Go / No-Go decision record. The funding decision is a margin decision. A model that does not close is a No-Go condition.

RISKS AND WATCH-OUTS

- ◆ **Yield assumed from a mature node.** Defect density at a node in early ramp is not the number in the foundry's marketing chart, and the gap lands entirely in margin.
- ◆ **Package cost treated as a rounding error.** On a 2.5D part with two HBM stacks the package can approach the die cost; modelling it as an adder hides the largest controllable line.
- ◆ **Test cost with no test time.** Test cost is test time multiplied by tester rate; without a time assumption from DFT the line is decorative.
- ◆ **Single-point ASP.** An ASP with no downside scenario turns the margin into a number nobody can be held to.
- ◆ **Die size estimate carried without a tolerance.** An area estimate that later grows 15% moves gross die per wafer non-linearly, and the model reports it as a small change.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Yield model per node	1.0	● Die area budget available from DEF-03	DEF-07 Feasibility assessment against candidate technology nodes
Cost roll-up and margin	0.8	● Wafer and mask quotations received from TECH-04	DEF-09 Business case preparation and funding approval package
Package and test cost lines	0.8	● Volume forecast agreed with sales	TECH-04 Wafer price, mask and NRE quotation; MPW and volume pricing
Sensitivity analysis	0.5		MP-05 Yield learning, failure pareto and defect analysis
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.3 FTE	● Cost model closes at the forecast volume with stated margin	● Margin at forecast volume against the target
Product cost analyst Owns the model and its assumptions		● Every assumption is named, sourced and dated	● Share of model inputs backed by a quotation rather than an estimate
Yield engineer Defect density and yield curves per node		● Sensitivity identifies the three exposures and their range	● Cost delta carried by the sensitivity range
Package cost liaison Interposer, substrate and HBM pricing		DEPENDS ON	
Test engineering liaison Test time and tester rate assumptions		DEF-01 Market and customer requirements consolidation	
Finance partner Volume, ASP and margin treatment		DEF-03 PPA target definition — frequency, power envelope, die area budget	
		TECH-04 Wafer price, mask and NRE quotation; MPW and volume pricing	

01 DEF · PRODUCT DEFINITION

DEF-05

Memory and interface bandwidth requirements
(HBM stacks, PCIe/CXL lanes, D2D)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w w2–w5 in stage	2.0 M/M planned	~2.9 FTE while it runs	6 2 in parallel	no has float	Memory systems architect Architecture liaison approves

PURPOSE

Fix the bandwidth the workloads need at each level of the hierarchy, and turn that into **the interface count the package and the floorplan will have to carry**.

Bandwidth is where a specification becomes physical. Two HBM stacks rather than one is not a line in a document; it is a larger interposer, a different substrate, a bigger power envelope and a different cost model. Deciding it here is cheap and deciding it at architecture is not.

WORK FLOW6 STEPS · 2 LANES

w2	w3	w4	w5
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MAIN SEQUENCE

1	2	5	6
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Bandwidth demand per workload, from the DEF-02 traces	0.8w	main
2	Memory hierarchy sizing — on-die SRAM, HBM capacity and channels	1w	main
3	Host interface sizing — PCIe / CXL generation and lane count	0.5w	parallel
4	Die-to-die interface requirement, if a chiplet split is a candidate	0.5w	parallel
5	Translation into a pin and bump budget	0.8w	main
6	Requirement review with architecture and package	0.5w	main

Step 5 is the one that binds. Bandwidth stated in GB/s commits nobody; the same number stated as a bump count commits the floorplan, the package and the cost model at once.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Workload traces and KPI table from DEF-02
- Requirement register — capacity and latency requirements
- HBM generation availability and roadmap
- Interface standard revisions in flight (PCIe, CXL, UCIe)
- Previous-generation bandwidth utilisation data

PRODUCES

- Bandwidth demand table, per workload and per hierarchy level
- Memory configuration recommendation — capacity, stacks, channels
- Host and die-to-die interface requirement
- Pin and bump budget estimate
- Review record with architecture and package

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-01

FEEDS

Product requirements document. Bandwidth and capacity enter the PRD as requirements with numbers rather than as adjectives.

DEF-02

FEEDS

Target specification. The bandwidth figures become KPI lines, and the bump budget becomes an area constraint on the PPA half.

RISKS AND WATCH-OUTS

- ◆ **Bandwidth sized on peak rather than sustained.** Peak numbers size an interface nobody reaches and a cost nobody wanted.
- ◆ **Capacity and bandwidth conflated.** HBM stack count is often set by capacity while the argument is conducted in GB/s; the two constraints do not move together.
- ◆ **Interface standard still in revision.** Committing to a generation whose specification is not final transfers a standards-body schedule into the program's schedule.
- ◆ **Bump budget never derived.** A bandwidth requirement that is not translated into bumps is discovered as a floorplan problem at PD-04, when the package is already ordered.
- ◆ **Chiplet option left open too long.** A die-to-die interface is a large architectural commitment; deferring it past architecture freeze forces the conservative choice by default.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Hierarchy sizing and bandwidth analysis	1.0	- Workload traces available from DEF-02 - Requirement register carries capacity and latency requirements	DEF-03 PPA target definition — frequency, power envelope, die area budget
Interface sizing	0.5	HBM and interface roadmaps available under NDA	ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)
Bump budget translation and review	0.5		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E)
Total	2.0		PKGD-02 Bump map, pitch and power-ground planning with physical design
WHO IT TAKES	~2.9 FTE	EXIT CRITERIA	MEASURED BY
Memory systems architect Owns the hierarchy sizing and the bandwidth table		- Bandwidth stated as sustained, per workload, with conditions - Memory configuration and interface count recommended with a reason - Bump budget accepted as buildable by package	- Bandwidth headroom against sustained demand - Bump budget against the package's stated capability - Interface revisions still open at architecture freeze
Interface architect Host and die-to-die interface selection			
Workload analyst Sustained demand extraction from the traces		DEPENDS ON	
Package liaison Confirms the bump budget is buildable		DEF-01 Market and customer requirements consolidation	
Architecture liaison Takes the requirement into ARCH-03		DEF-02 Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)	

01 DEF · PRODUCT DEFINITION

DEF-06

Competitive benchmarking and product positioning

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	2.0 M/M	~2.2 FTE	6	no	Product marketing lead
w0–w4 in stage	planned	while it runs	1 in parallel	has float	Stage lead approves

PURPOSE

Establish where the part lands **against what will be shipping when it ships** — not against what ships today.

A benchmark against the current generation is a comfortable exercise and a useless one. The program's silicon arrives thirty months out; the comparison that matters is against the parts that will be launching in that window, projected from public roadmaps and node cadence.

WORK FLOW

6 STEPS · 2 LANES

w0	w1	w2	w3	w4
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MAIN SEQUENCE

1	2	4	5	6
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Competitor set definition and launch-window mapping	0.5w	main
2	Public benchmark, datasheet and disclosure collection	1w	main
3	Normalisation to comparable metrics and conditions	1w	parallel
4	Projection to our own launch window	1w	main
5	Positioning statement and capability gap list	1w	main
6	Review with product and sales	0.5w	main

Normalisation runs in parallel because it is slow, thankless and independent of collection — vendors quote TOPS at different precisions, batch sizes and power points, and none of the published numbers compare until they are restated.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Competitor datasheets, disclosures and conference material
- Public benchmark submissions (MLPerf and equivalents)
- Foundry node cadence and process roadmaps
- Analyst launch forecasts
- Field intelligence from sales

PRODUCES

- Competitor set with launch-window estimates
- Normalised comparison table with stated conditions
- Projection of the competitive field at our launch
- Positioning statement
- Capability gap list, ranked

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-01	FEEDS	Product requirements document. The gap list becomes requirements — a gap nobody writes down is a gap nobody closes.
DEF-03	INFORMS	Product cost and margin model. Competitive positioning is what an ASP assumption is defended with.

RISKS AND WATCH-OUTS

- ◆ **Benchmarking against today's field.** The comparison that decides the product is against parts launching thirty months out, and those numbers have to be projected rather than read.
- ◆ **Vendor numbers taken at face value.** TOPS quoted at a precision, batch size and power point chosen to flatter is not a comparable number until it is restated.
- ◆ **Gap list with no owner.** A capability gap that does not become a requirement is an observation, and observations do not change silicon.
- ◆ **Positioning built on a single axis.** A part positioned only on TOPS/W loses to one positioned on total cost of ownership, and the argument is not had until launch.
- ◆ **Competitor launch dates treated as fixed.** A slip in the competitive field is as consequential as a slip in ours, and neither is knowable to a quarter.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Collection and competitor set	0.8	● Product concept defined well enough to name a competitor set	DEF-01Market and customer requirements consolidation
Normalisation and projection	0.8	● Access to analyst material and benchmark submissions	DEF-04Die size, yield and product cost model (wafer, mask, package, test, ASP)
Positioning and gap list	0.5	● Launch-window assumption from the program schedule	DEF-09Business case preparation and funding approval package
Total	2.0		
WHO IT TAKES	~2.2 FTE	EXIT CRITERIA	MEASURED BY
Product marketing lead Owns the competitor set and the positioning		● Every comparison is normalised to stated conditions	● Competitors covered against the expected launch field
Performance analyst Normalisation and projection		● Field projected to our launch window, not read from today	● Share of comparisons normalised rather than quoted
Technology analyst Node cadence and roadmap reading		● Gap list handed to DEF-01 as candidate requirements	● Gaps converted into tracked requirements
Sales liaison Field intelligence and account-level reality			
Stage lead Turns the gap list into requirements		DEF-01Market and customer requirements consolidation	

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DEF-07

Feasibility assessment against candidate technology nodes

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	7	yes	Feasibility lead
w4–w8 in stage	planned	while it runs	2 in parallel	no float	Stage lead approves

PURPOSE

Test the requirement set, the PPA targets and the cost model against real nodes and return a verdict: **buildable as specified, buildable with named concessions, or not buildable**.

Feasibility is the only activity in the stage whose output can be "no". Its value is entirely in being allowed to say so — a feasibility study run after the node is chosen and the budget is committed is a formality, and formalities do not stop programs that should be stopped.

WORK FLOW

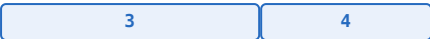
7 STEPS · 2 LANES

w4	w5	w6	w7	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Candidate node shortlist agreed with TECH	0.5w	main
2	PPA achievability per node — density, frequency, leakage	1.3w	main
3	IP availability check per node, ahead of IPR	0.8w	parallel
4	Cost model re-run per node	0.5w	parallel
5	Package and thermal feasibility at the target power	0.8w	main
6	Concession list — what must be given up, per node	0.8w	main
7	Verdict and recommendation	0.8w	main

The concession list is the deliverable that gets used. A binary verdict tells the decision room nothing it can act on; a list saying "this node, at 200 MHz less, with the PHY bought rather than built" is a decision it can take.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Requirement register from DEF-01
- PPA targets from DEF-03
- Cost model from DEF-04
- Node characterisation and DTCO study from TECH-03
- IP availability indications from vendors and internal inventory

PRODUCES

- Node-by-node achievability matrix
- IP availability finding per node
- Thermal and package feasibility note
- Concession list per node
- Feasibility verdict with a named recommendation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-D4	PRODUCES	Feasibility report. This activity is the deliverable. The verdict, the matrix and the concession list are published together — the verdict alone is not usable.
DEF-D6	GATES	Kickoff Go / No-Go decision record. A No-Go recommendation here is the only mechanism in the stage that stops the program, and it must reach the room unedited.

RISKS AND WATCH-OUTS

- ◆ **Feasibility run after the node is already chosen.** The study then documents a decision rather than informing one, and its independence is gone.
- ◆ **Optimism at the boundary.** A target reachable "with aggressive closure" is a target reachable in a slide and not in a timing report.
- ◆ **IP availability taken from the vendor's roadmap.** A block promised on the node and not yet ported is a schedule risk being recorded as an asset.
- ◆ **Thermal checked at typical rather than at the envelope.** Feasibility at average power says nothing about the part at its rated envelope, which is the condition it is sold at.
- ◆ **No concession list.** A bare yes or no gives the decision room no third option, and the third option is usually the right one.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
PPA achievability per node	1.0	● PPA targets published by DEF-03	DEF-09 Business case preparation and funding approval package
IP and cost checks	0.8	● Cost model closing in DEF-04	TECH-01 Foundry and node long-list to short-list evaluation
Package and thermal feasibility	0.5	● At least two nodes characterised by TECH-03	ARCH-01 System-level performance modelling and workload simulation
Concession list and verdict	0.8		IPR-03 Make / buy / reuse decision per IP block
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.3 FTE	● Every target has an achievability finding per candidate node	● Targets assessed against every candidate node
Feasibility lead		● Concessions named and costed, not described	● Concessions accepted at the gate against those raised
Owns the verdict and the concession list		● Verdict recorded with a recommendation and a dissent record	● Findings later contradicted by ARCH-01 modelling
PPA analyst		DEPENDS ON	
Achievability per node against the DEF-03 targets		DEF-01 Market and customer requirements consolidation	
IP strategist		DEF-03 PPA target definition — frequency, power envelope, die area budget	
Availability and porting status per node		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)	
Thermal and package liaison		TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	
Feasibility at the rated power envelope			
Stage lead			
Carries the recommendation into the gate			

01 DEF · PRODUCT DEFINITION

DEF-08

Program schedule, resource and staffing plan drafting

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	7	no	Program manager
w4–w8 in stage	planned	while it runs	2 in parallel	has float	Finance partner approves

PURPOSE

Turn the milestone profile into a dated plan for **this** program, and say what it takes in people — then set that against the people who actually exist.

A schedule that assumes a headcount curve the organisation cannot staff is not a plan; it is a forecast of a slip. The value of this activity is the comparison, not the Gantt chart: effort rolled up by stage and discipline, against the capacity on hand.

WORK FLOW

7 STEPS · 2 LANES

w4	w5	w6	w7	w8
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MAIN SEQUENCE

1	2	3	4	6
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IN PARALLEL

5	7
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#	STEP	TAT	LANE
1	Profile selection and kickoff date	0.5w	main
2	Stage baseline adjustment for this program's scope	0.8w	main
3	Effort roll-up by stage and by discipline	1w	main
4	Headcount curve set against the organisation's real capacity	1w	main
5	External lead-time overlay — substrate, probe card, IP, shuttle	0.8w	parallel
6	Risk buffer placement and critical path identification	0.8w	main
7	Schedule review and commitment	0.5w	parallel

The lead-time overlay runs in parallel and changes more dates than the internal plan does. A twenty-week probe card and a twenty-week substrate are ordered against dates this activity sets, and both sit outside the organisation's control once placed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Milestone profile and stage baselines
- Effort figures per activity from the template
- Organisation capacity and current program commitments
- Supplier lead times — substrate, interposer, probe card, IP, shuttle
- Feasibility concessions from DEF-07

PRODUCES

- Program schedule with dated stage boundaries and milestones
- Effort roll-up by stage and discipline
- Headcount curve against available capacity
- Lead-time overlay and order-by dates
- Critical path and buffer placement record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-05	PRODUCES	Program charter, staffing and budget plan. The schedule, the headcount curve and the effort roll-up are the charter's operative content.
DEF-06	GATES	Kickoff Go / No-Go decision record. The gate approves a schedule and a headcount. An unresourced plan is a No-Go condition.

RISKS AND WATCH-OUTS

- ◆ **Headcount curve nobody can staff.** A peak of 200 engineers in a 120-engineer organisation is a slip already committed to, recorded as a plan.
- ◆ **Template baselines used unadjusted.** The profile describes a typical program; a derivative or a first-of-node program is not typical, and the deltas belong in the plan.
- ◆ **Lead times discovered rather than planned.** Probe card and substrate lead times run to twenty weeks; found late they move wafer-out, not the purchasing date.
- ◆ **Buffer at the end.** A buffer placed after the last stage protects nothing; buffers belong in front of the gates that cannot move.
- ◆ **Critical path asserted rather than derived.** On a 23-stage profile with parallel enablement workstreams the critical path is rarely the design spine, and assuming it is misdirects the attention that matters.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Effort roll-up by stage and discipline	1.0	● Milestone profile selected ● Scope stable enough to adjust baselines against ● Organisation capacity data available	DEF-09 Business case preparation and funding approval package
Headcount curve against capacity	0.8		TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement
Lead-time overlay	0.5		IPR-09 IP delivery schedule alignment to RTL and PD integration windows
Buffer and critical path	0.5		TEST-02 ATE platform selection and tester time booking
Review and commitment	0.3		PKGD-09 Substrate and interposer supplier selection and lead-time booking
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.3 FTE	● Schedule dated from a committed kickoff ● Headcount curve reconciled against real capacity, gaps named ● Critical path derived and buffers placed against the fixed gates	● Peak headcount against available capacity ● Buffer weeks held in front of fixed gates ● Long-lead items with an order-by date on the plan
Program manager Owns the schedule, the buffers and the critical path			
Resource manager Capacity, headcount curve and hiring plan			
Discipline leads Confirm the effort roll-up for their stages		DEF-01 Market and customer requirements consolidation	
Procurement liaison Supplier lead times and order-by dates		DEF-07 Feasibility assessment against candidate technology nodes	
Finance partner Converts effort into budget			

01 DEF · PRODUCT DEFINITION

DEF-09

Business case preparation and funding approval package

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	2.0 M/M	~2.9 FTE	6	yes	Program manager
w5–w8 in stage	planned	while it runs	1 in parallel	no float	Executive sponsor approves

PURPOSE

Assemble what the stage produced into **the case the funding decision is made on**, and put a named recommendation in front of the people who can say no.

This activity writes nothing new. Its work is selection and honesty: the feasibility verdict, the cost model and the resourced schedule carried into the room intact, with the downside scenario shown rather than described. A business case that only closes in the base case has not been reviewed.

WORK FLOW

6 STEPS · 2 LANES



#	STEP	TAT	LANE
1	Financial model assembly — NRE, opex, revenue, breakeven	1w	main
2	Scenario set — base, downside, upside	0.8w	parallel
3	Risk register summary written for the decision room	0.5w	main
4	Recommendation and the ask, stated as a number	0.5w	main
5	Executive review pack	0.8w	main
6	Decision meeting and record	0.3w	main

The decision meeting is a quarter of a week and the preparation is three weeks, which is the correct ratio. A gate that is argued in the room rather than before it is a gate whose outcome depends on who attends.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Cost and margin model from DEF-04
- Feasibility verdict and concession list from DEF-07
- Resourced schedule and headcount curve from DEF-08
- Positioning and gap list from DEF-06
- Program risk register

PRODUCES

- Financial model — NRE, opex, revenue, breakeven
- Scenario set with the downside made explicit
- Risk summary for the decision room
- Recommendation with a stated ask
- Decision record with owners, conditions and dissent

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DEF-D6	PRODUCES	Kickoff Go / No-Go decision record. This activity is the deliverable — the record, its conditions and its dissent, not the slides that preceded it.
DEF-D5	FEEDS	Program charter, staffing and budget plan. The charter is only funded once this decision lands; the approved figure is written back into it.

RISKS AND WATCH-OUTS

- ◆ **Base case only.** A case that closes at forecast volume and is never shown at downside volume transfers the risk to whoever inherits the program.
- ◆ **Conditions accepted verbally.** A Go granted "subject to" something nobody wrote down is a Go, and the condition is never revisited.
- ◆ **Dissent unrecorded.** The objection nobody minuted is the objection that resurfaces at first silicon with nothing to point at.
- ◆ **Feasibility concessions edited on the way to the room.** A concession list softened in the pack is the one failure mode that makes the whole stage worthless.
- ◆ **Decision owners unclear.** A gate where nobody is identified as able to say no is a gate that cannot say no.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Financial model assembly	0.8	● Cost model closed by DEF-04	ARCH-01System-level performance modelling and workload simulation
Scenario set	0.5	● Feasibility verdict issued by DEF-07	
Review pack	0.5	● Resourced schedule available from DEF-08	TECH-06Foundry legal engagement — NDA, design agreement, IP licensing frame
Decision meeting and record	0.3		IPR-07Licence negotiation, PO issue and delivery-date commitment
Total	2.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~2.9 FTE	● Case shown at base and downside, not base alone	● Scenarios presented against those prepared
Program manager Owns the pack, the ask and the record		● Decision recorded with named owners, conditions and dissent	● Conditions attached to the decision and their owners
Finance partner Financial model and scenario set		● Approved budget written back into the program charter	● Days from pack circulation to decision
Product lead Revenue, positioning and the market case		DEPENDS ON	
Stage lead Carries the feasibility verdict unedited		DEF-04Die size, yield and product cost model (wafer, mask, package, test, ASP)	
Executive sponsor Chairs the decision and owns the outcome		DEF-06Competitive benchmarking and product positioning	
		DEF-07Feasibility assessment against candidate technology nodes	
		DEF-08Program schedule, resource and staffing plan drafting	

DEFINE

02 Architecture

ARCH

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w6 – w24	18 _w	10	111 _{M/M}	6 of 10

ENGINEERING ACTIVITIES

ARCH-01	System-level performance modelling and workload simulation	14w	28 _{M/M}
ARCH-02	Compute / memory / interconnect partitioning; monolithic vs chiplet decision	8w	14 _{M/M}
ARCH-03	Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	8w	12 _{M/M}
ARCH-04	Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	6w	6 _{M/M}
ARCH-05	Power domain, clock domain and DVFS architecture; UPF intent draft	7w	9 _{M/M}
ARCH-06	PPA budget allocation per block	4w	4 _{M/M}
ARCH-07	Security and safety architecture — secure boot, root of trust, fusing	6w	7 _{M/M}
ARCH-08	Chip-level floorplan intent, pin and bump budget	5w	5 _{M/M}
ARCH-09	Architecture specification authoring and review	9w	14 _{M/M}
ARCH-10	Block microarchitecture specification kickoff	6w	12 _{M/M}

KEY DELIVERABLES

ARCH-D1	Architecture specification
ARCH-D2	Performance model and workload analysis report
ARCH-D3	Block partitioning and PPA budget table
ARCH-D4	Interface and protocol definition document
ARCH-D5	Power / clock / reset architecture and UPF intent
ARCH-D6	Chip-level block diagram with pin and bump budget
ARCH-D7	Architecture Freeze review package

ARCH-01

System-level performance modelling and workload simulation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w	28.0 M/M	~8.7 FTE	10	yes	Performance architect
w0–w14 in stage	planned	while it runs	3 in parallel	no float	Silicon correlation engineer approves

PURPOSE

Build the model the architecture is **argued with**. Every partitioning, sizing and interface decision in this stage is settled by simulation against the DEF-02 workload suite rather than by opinion.

Fourteen weeks and 28 M/M make this the largest activity in the stage by a wide margin, and that is the right shape. Being wrong in a model costs a re-run; being wrong in RTL costs a quarter; being wrong in silicon costs a respin. The model's credibility is the stage's credibility — the moment nobody trusts it, every decision reverts to seniority.

WORK FLOW

10 STEPS · 2 LANES

w0	w3	w6	w9	w12	w14
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MAIN SEQUENCE

1	2	4	5	7	9	10
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IN PARALLEL

3	6	8
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#	STEP	TAT	LANE
1	Model scope and fidelity decision — analytical, trace-driven or cycle-approximate	1w	main
2	Baseline model construction — compute, memory, interconnect	2.5w	main
3	Workload suite integration from DEF-02	1w	parallel
4	Correlation against previous-generation silicon	1.5w	main
5	Design-space exploration sweeps — cluster count, SRAM capacity, HBM channels	3w	main
6	Interconnect and NoC topology studies	2w	parallel
7	Power model overlay and TOPS/W projection	2w	main
8	Bottleneck analysis and roofline characterisation	1.5w	parallel
9	Sensitivity analysis and a stated confidence interval	3w	main
10	Model handoff and regression harness for later stages	1w	main

Correlation sits early on purpose. A model that has never been checked against silicon produces numbers, not answers, and every sweep run before correlation has to be re-run afterwards.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Workload suite and KPI table from DEF-02

PRODUCES

- Performance model and its source

- PPA targets from DEF-03
 - Bandwidth requirements from DEF-05
 - Previous-generation silicon and its measured performance
 - Node library characterisation from PDK-03
- Correlation report against previous-generation silicon
 - Design-space exploration results, per sweep
 - Power and TOPS/W projection
 - Sensitivity analysis with a confidence statement
 - Regression harness for downstream re-runs

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D2	PRODUCES	Performance model and workload analysis report. The model, its correlation and its confidence statement are the deliverable — the report without the model is an assertion.
ARCH-D1	FEEDS	Architecture specification. Every quantitative claim in the specification traces back to a sweep in this model, or it does not belong in the specification.
ARCH-D3	FEEDS	Block partitioning and PPA budget table. The sweeps are what decide cluster count and SRAM capacity before ARCH-02 draws the boundaries.

RISKS AND WATCH-OUTS

- ◆ **Model never correlated against silicon.** An uncorrelated model has an unknown error bar, and every decision taken on it inherits that error silently.
- ◆ **Fidelity chosen for comfort.** An analytical model that runs in seconds will answer the wrong question quickly; a cycle-accurate model that takes a week per sweep will answer nothing in time to matter.
- ◆ **Sweeps run on an unrepresentative workload.** The model inherits whatever bias DEF-02 built into the suite, and amplifies it across every architectural decision.
- ◆ **Power model bolted on at the end.** TOPS/W is a KPI; a performance model with no power overlay cannot answer the question the product is sold on.
- ◆ **No confidence statement.** A projection quoted as a single number invites the organisation to plan against a precision the model does not have.

EFFORT SPLIT	28 M/M	ENTRY CRITERIA	FEEDS INTO
Model construction	7.0	Workload suite delivered and runnable from DEF-02	ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision
Design-space exploration	6.0	PPA targets published by DEF-03	ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)
Power model overlay	4.0	Previous-generation platform available for correlation	ARCH-06 PPA budget allocation per block
NoC and interconnect studies	3.5		ARCH-09 Architecture specification authoring and review
Bottleneck and sensitivity analysis	3.5		DV-09 Performance and bandwidth validation against the architecture model
Silicon correlation	3.0		
Handoff and regression harness	1.0		
Total	28.0		
WHO IT TAKES	~8.7 FTE		
Performance architect Owns the model, its fidelity and its confidence statement		EXIT CRITERIA - Model correlated against silicon with a stated error band - Every architectural decision in the stage traced to a sweep - Regression harness handed over so later stages can re-run it	MEASURED BY - Correlation error against previous-generation silicon - Architectural decisions supported by a sweep - Model re-run turnaround once a parameter changes
Modelling engineers Model construction, sweeps and the regression harness		DEPENDS ON	
Workload analyst Suite integration and result interpretation		DEF-02 Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)	
Power modelling engineer Power overlay and TOPS/W projection		DEF-03 PPA target definition — frequency, power envelope, die area budget	
Silicon correlation engineer Previous-generation measurement and correlation		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)	

ARCH-02

Compute / memory / interconnect partitioning; monolithic vs chiplet decision

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	14.0 M/M	~7.6 FTE	8	yes	Chief architect
w2–w10 in stage	planned	while it runs	3 in parallel	no float	DFT liaison approves

PURPOSE

Decide what sits on which die and where the block boundaries fall — including **whether this is one die or several**.

The monolithic-versus-chiplet call is the most expensive decision in the stage. It sets the package, the die-to-die interface, the test strategy, the yield model and the cost model at once, and after Architecture Freeze it is effectively irreversible: unwinding it means redoing the floorplan, the package design and the test hardware together.

WORK FLOW

8 STEPS · 2 LANES

w2	w4	w6	w8	w10
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MAIN SEQUENCE

1	2	3	6	8
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IN PARALLEL

4	5	7
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#	STEP	TAT	LANE
1	Block decomposition from the functional requirements	1w	main
2	Reticle limit and die size feasibility check	0.5w	main
3	Monolithic and chiplet options built as costed candidates	2w	main
4	Yield and cost comparison per candidate, with the DEF-04 model	1.5w	parallel
5	Die-to-die interface implication study	1w	parallel
6	Partition boundary definition — what crosses, at what bandwidth	3w	main
7	Known-good-die and test implication review	1w	parallel
8	Partitioning decision and record	1.5w	main

Boundary definition takes three of the eight weeks because it is the part that gets skipped. Deciding "two dies" is quick; deciding exactly which signals cross at what bandwidth and latency is the work that makes the decision real.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Design-space exploration results from ARCH-01
- Cost and yield model from DEF-04
- Reticle limit and node constraints from TECH-02
- Package architecture options from the requirement register
- Test strategy assumptions from DFT-01

PRODUCES

- Block decomposition and hierarchy
- Costed partitioning candidates
- Partition boundary specification — signals, bandwidth, latency
- Yield and cost comparison per candidate
- Partitioning decision record with dissent

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D3	PRODUCES	Block partitioning and PPA budget table. The partitioning half of the deliverable; ARCH-06 supplies the budget half against these blocks.
ARCH-D1	FEEDS	Architecture specification. The partition and its boundaries are the specification's structural spine — every other section is written against them.
ARCH-D6	FEEDS	Chip-level block diagram with pin and bump budget. ARCH-08 draws what this activity decides.

RISKS AND WATCH-OUTS

- ◆ **Chiplet chosen for the roadmap rather than for this product.** A die-to-die interface costs area, power and a test strategy; taking it on to be ready for the next program charges this one.
- ◆ **Boundaries drawn without bandwidth.** A partition that names blocks but not the traffic between them is redrawn at PD-02 when the floorplan cannot carry it.
- ◆ **Reticle limit checked against the estimate, not the growth.** Die area grows through implementation; a partition that fits at architecture and not at signoff is a partition that fails late.
- ◆ **Test implications deferred.** Known-good-die coverage on a chiplet part is a DFT and test-hardware programme of its own, and it is priced after the decision is taken.
- ◆ **Decision recorded without the dissent.** The objection nobody minuted resurfaces at first silicon with nothing to point at.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Candidate construction	4.0	● First design-space results available from ARCH-01	ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)
Boundary definition	3.5	● Cost model usable for candidate comparison	ARCH-06 PPA budget allocation per block
Yield and cost comparison	2.5	● Reticle and node constraints confirmed by TECH	ARCH-08 Chip-level floorplan intent, pin and bump budget
Die-to-die interface study	2.0		ARCH-10 Block microarchitecture specification kickoff
Test implication review	1.0		PKG0-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count
Decision record	1.0		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT
Total	14.0	EXIT CRITERIA ● Partitioning decided and recorded with the alternatives it beat ● Every boundary carries a bandwidth and latency budget ● Test and yield implications priced, not deferred	MEASURED BY ● Candidates evaluated against the one chosen ● Boundaries carrying an agreed bandwidth budget ● Partition changes raised after Architecture Freeze
WHO IT TAKES	~7.6 FTE	DEPENDS ON ARCH-01 System-level performance modelling and workload simulation DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP) TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)	
Chief architect Owns the partitioning decision and its record			
Physical architect Reticle, floorplan and boundary feasibility			
Cost and yield analyst Candidate comparison against the DEF-04 model			
Interconnect architect Boundary traffic, bandwidth and latency			
DFT liaison Known-good-die and test strategy implications			

ARCH-03

Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w4–w12 in stage	12.0 M/M planned	~6.5 FTE while it runs	8 3 in parallel	yes no float	Memory systems architect Workload analyst approves

PURPOSE

Define how data moves and where it lives — **dataflow, on-die SRAM capacity and organisation, HBM channel count** — for each workload class the product serves.

An inference accelerator is a memory system with arithmetic attached. A hierarchy that does not match the reuse pattern of the workload shows up as a utilisation number nobody can lift afterwards: not by clock speed, not by more compute, and not by any amount of software.

WORK FLOW

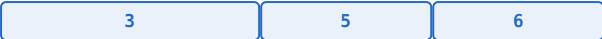
8 STEPS · 2 LANES

w4	w6	w8	w10	w12
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Dataflow selection per workload class — weight-stationary, output-stationary, hybrid	1.5w	main
2	On-die SRAM capacity sizing against measured reuse distance	2w	main
3	SRAM banking, port and bandwidth organisation	1.5w	parallel
4	HBM channel count and address interleaving scheme	1.5w	main
5	Cache and scratchpad policy	1w	parallel
6	Prefetch and data-movement engine requirement	1w	parallel
7	Hierarchy simulation against the ARCH-01 model	2w	main
8	Hierarchy specification and handoff	1w	main

Sizing precedes simulation and simulation revises sizing — the two weeks in step 7 include at least one full turn back through step 2. A hierarchy accepted on its first simulation has almost certainly not been swept.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Workload traces and reuse analysis from DEF-02
- Bandwidth requirements from DEF-05
- Design-space results from ARCH-01
- Partition boundaries from ARCH-02
- Memory compiler capability from PDK-04

PRODUCES

- Dataflow specification per workload class
- SRAM capacity, banking and port organisation
- HBM channel count and interleaving scheme
- Cache and scratchpad policy note
- Hierarchy simulation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D1	PRODUCES	Architecture specification. The memory hierarchy chapter is authored here and lands in the specification whole; ARCH-09 assembles rather than rewrites it.
ARCH-D3	FEEDS	Block partitioning and PPA budget table. SRAM capacity is the largest single area line in the budget, and it is decided here.
ARCH-D2	FEEDS	Performance model and workload analysis report. The hierarchy sweeps are run in the ARCH-01 model and reported with it.

RISKS AND WATCH-OUTS

- ◆ **SRAM sized on average reuse.** Reuse distance has a tail; a capacity chosen at the median misses the workloads that define the product's worst case.
- ◆ **Banking decided after capacity.** Capacity and bandwidth are separate constraints, and a bank organisation that cannot feed the compute wastes the area it took.
- ◆ **Memory compiler capability assumed.** The instance the hierarchy needs may not exist on the node — PDK-05 is where that is found, and it is found after this decision unless someone asks early.
- ◆ **Dataflow chosen once for all workloads.** A single dataflow optimised for one class starves another, and the deployment mix decides which mistake is cheaper.
- ◆ **Interleaving left to implementation.** Address interleaving decides whether HBM channels are actually parallel; deferred, it becomes a bandwidth shortfall nobody predicted.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
SRAM capacity and organisation	3.0	● Reuse analysis available from the DEF-02 traces	ARCH-06 PPA budget allocation per block
Dataflow selection	2.5	● Partition boundaries stable from ARCH-02	ARCH-08 Chip-level floorplan intent, pin and bump budget
Hierarchy simulation	2.5	● Memory compiler capability indicated by PDK-04	ARCH-10 Block microarchitecture specification kickoff
HBM channels and interleaving	1.5		
Cache, scratchpad and prefetch policy	1.5		PDK-05 Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage
Specification and handoff	1.0		
Total	12.0	EXIT CRITERIA ● Capacity and bandwidth justified by simulation, not by precedent ● Dataflow chosen per workload class with the trade recorded ● Memory instance requirements handed to PDK and AMS	AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing
WHO IT TAKES	~6.5 FTE		RTL-01 Block microarchitecture specification
Memory systems architect Owns the hierarchy and its specification			
Dataflow architect Dataflow selection and compute-memory matching		DEPENDS ON	MEASURED BY
Performance modelling engineer Hierarchy sweeps in the ARCH-01 model		DEF-02 Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)	● Utilisation achieved in simulation against the roofline
Memory IP liaison Compiler capability and instance feasibility		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)	● SRAM area against the ARCH-06 allocation
Workload analyst Reuse distance and access-pattern analysis		ARCH-01 System-level performance modelling and workload simulation	● Hierarchy changes raised after RTL freeze
		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	

ARCH-04

Interface and protocol selection
(PCIe gen, CXL, UCIe, HBM3/3E)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w3–w9 in stage	6.0 M/M planned	~4.3 FTE while it runs	8 3 in parallel	no has float	Interface architect Physical architect approves

PURPOSE

Choose the external interfaces and **pin their revisions** — PCIe generation, CXL profile, HBM generation, die-to-die if the part is a chiplet — and turn each into a controller and PHY requirement IPR can go and buy.

Interfaces are where the program touches standards bodies and IP vendors, neither of which runs on the program's calendar. A generation selected before its specification is ratified imports someone else's schedule into this one, and the import is silent until the PHY is late.

WORK FLOW

8 STEPS · 2 LANES

w3	w5	w7	w9
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Interface requirement extraction from DEF-05	0.5w	main
2	Standard revision status and ratification timing review	1w	main
3	Host interface selection — PCIe generation and CXL profile	1.5w	main
4	HBM generation selection and controller requirement	1w	parallel
5	Die-to-die interface selection, if a chiplet split survives ARCH-02	1w	parallel
6	Compliance and interoperability requirement definition	1w	parallel
7	Controller and PHY requirement handoff to IP strategy	1w	main
8	Interface definition document and review	2w	main

The handoff in step 7 is what makes this activity urgent rather than merely important. `IPR-04` cannot evaluate a PHY vendor against a generation nobody has chosen, and PHY licence lead times run to a quarter.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Bandwidth and interface sizing from DEF-05
- Partitioning decision from ARCH-02
- Standards body revision status and ratification schedules
- Vendor PHY roadmaps under NDA
- Node PHY availability from TECH-03

PRODUCES

- Interface selection record with the revision pinned
- Controller and PHY requirement specification
- Compliance and interoperability requirement list
- Die-to-die interface requirement, where applicable
- Interface definition document

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D4	PRODUCES	Interface and protocol definition document. This activity is the deliverable — the selections, their revisions and the requirements derived from them.
ARCH-D1	FEEDS	Architecture specification. The interface chapter is imported from the definition document rather than restated, so the two cannot drift.
ARCH-D6	FEEDS	Chip-level block diagram with pin and bump budget. Lane counts and PHY footprints are what ARCH-08 allocates bumps against.

RISKS AND WATCH-OUTS

- ◆ **Specification not ratified at selection.** Designing to a draft revision transfers a standards-body schedule into the program, and the transfer is invisible until compliance testing.
- ◆ **PHY assumed available on the node.** A PHY proven on the previous node is not a PHY on this one; IPR-05 is where that is confirmed, and it is confirmed after this choice.
- ◆ **Compliance requirements written late.** Interoperability testing needs hardware, test suites and lab time booked against dates set here.
- ◆ **Lane count set by bandwidth alone.** Lanes cost bumps, area and power; a count chosen without ARCH-08 in the room is a count the floorplan will renegotiate.
- ◆ **Die-to-die selection deferred past ARCH-02.** The interface is a large architectural commitment, and deferring it forces the conservative default.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Interface selection studies	2.0	● Interface sizing available from DEF-05	ARCH-08 Chip-level floorplan intent, pin and bump budget
Standards and ratification review	1.0	● Partitioning decided by ARCH-02	IPR-01 Product requirement → IP requirement decomposition (IP bill of materials)
Compliance requirement definition	1.0	● Vendor PHY roadmaps accessible under NDA	IPR-04 Vendor RFI / RFQ and technical evaluation
IP requirement handoff	1.0		AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY)
Definition document and review	1.0		SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
Total	6.0	EXIT CRITERIA ● Every interface pinned to a named, ratified revision ● Controller and PHY requirements handed to IPR ● Compliance obligations listed with the hardware they need	TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test
WHO IT TAKES	~4.3 FTE	DEPENDS ON	MEASURED BY ● Interfaces pinned to a ratified revision ● Days from selection to IPR vendor engagement ● Interface changes raised after Architecture Freeze
Interface architect Owns the selections and the definition document		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)	
Standards liaison Revision status, ratification timing and compliance		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
IP strategist Turns requirements into vendor evaluation criteria		TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	
SerDes and PHY specialist PHY feasibility on the selected node			
Physical architect Bump and area cost of the lane count			

ARCH-05

Power domain, clock domain and DVFS architecture; UPF intent draft

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w7–w14 in stage	9.0 M/M planned	~5.6 FTE while it runs	8 3 in parallel	no has float	Power architect Verification liaison approves

PURPOSE

Define the power and clock structure the whole implementation inherits — **domains, rails, DVFS operating points, clock crossings, reset topology** — and write the first UPF intent.

Power intent is one of the few architectural artifacts consumed literally by tools. A domain boundary drawn here becomes isolation and level-shifter cells at SYN-06 and a verification obligation at DV-06. Drawn late, or drawn differently by two teams, it forces both to be redone.

WORK FLOW

8 STEPS · 2 LANES

w7	w9	w11	w13	w14
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MAIN SEQUENCE

1	2	4	7	8
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	Power domain partitioning against the block list	1.5w	main
2	Voltage rail and regulator topology	1w	main
3	DVFS operating point definition and switching policy	1.5w	parallel
4	Clock domain architecture and crossing inventory	1.5w	main
5	Reset topology and sequencing	1w	parallel
6	Power management controller requirement	1w	parallel
7	UPF intent draft	2w	main
8	Review with physical design and verification	1w	main

The crossing inventory in step 4 is the artifact `RTL-06` closes CDC against. An inventory that is incomplete at Architecture Freeze becomes a CDC finding in week 40, when the fix is an RTL change under change control.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Power envelope and per-domain budget from DEF-03
- Block list and partitioning from ARCH-02
- DVFS requirements from the workload analysis
- Library and IO capability from PDK-03
- Package power delivery constraints from PKGD-06

PRODUCES

- Power domain definition and boundary map
- Voltage rail and regulator topology
- DVFS operating point table and policy
- Clock domain architecture with a crossing inventory
- Reset topology and sequence specification
- UPF intent file, first draft

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D5	PRODUCES	Power / clock / reset architecture and UPF intent. This activity is the deliverable, and the UPF file is the part of it tools consume directly.
ARCH-D1	FEEDS	Architecture specification. The power and clock chapter, and the constraint every block owner designs against.
ARCH-D3	FEEDS	Block partitioning and PPA budget table. Power domains are the unit the power budget is allocated in.

RISKS AND WATCH-OUTS

- ◆ **Domains drawn for elegance rather than for control.** Every domain boundary costs isolation cells, level shifters and verification states; a domain that is never independently gated is pure overhead.
- ◆ **Crossing inventory incomplete.** A clock crossing missing from the inventory is a crossing nobody constrains and nobody verifies, and it fails intermittently in silicon.
- ◆ **DVFS policy without measured operating points.** Operating points chosen from the library rather than from silicon leave either performance or margin on the table.
- ◆ **UPF drafted after RTL starts.** Power intent that arrives mid-RTL forces retrofitting into blocks already written, and the retrofit is never as clean as the design.
- ◆ **Reset sequencing left to bring-up.** A reset order discovered in the lab is discovered on silicon that cannot be changed.

EFFORT SPLIT	9 M/M	ENTRY CRITERIA	FEEDS INTO
Power domain partitioning	2.0	● Power budget per domain available from DEF-03	ARCH-06 PPA budget allocation per block
UPF intent draft	2.0	● Block list stable from ARCH-02	ARCH-09 Architecture specification authoring and review
Clock domain architecture	1.5	● Library and IO low-power capability confirmed by PDK-03	RTL-05 Clock, reset and power intent (UPF) implementation
DVFS definition	1.5		
Reset and power management controller	1.5		SYN-06 Power optimisation — clock gating, operand isolation, Vt mix
Review	0.5		DV-06 Low-power (UPF) verification
Total	9.0		PD-03 Power delivery network design and early IR analysis
WHO IT TAKES	~5.6 FTE	EXIT CRITERIA ● Every clock crossing on the inventory with a named synchroniser strategy ● UPF intent parses in the target tools, not only on paper ● Physical design and verification have accepted the architecture	MEASURED BY ● Clock crossings inventoried against those later found by CDC ● Power domains that are actually independently gated ● UPF revisions after RTL freeze
Power architect Owns the domains, the rails and the UPF intent		DEPENDS ON	
Clocking architect Clock domain architecture and crossing inventory		DEF-03 PPA target definition — frequency, power envelope, die area budget	
Low-power methodology engineer UPF authoring and tool consumption		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
Physical design liaison Rail feasibility and PDN implication			
Verification liaison Confirms the intent is verifiable as written			

ARCH-06

PPA budget allocation per block

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	4.0 M/M	~4.3 FTE	7	yes	PPA lead
w12–w16 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Divide the stage's PPA targets across the real block list, so **every block owner starts implementation with a number rather than an aspiration**.

This is DEF-03's allocation done again, and done properly. DEF-03 allocated against an estimated block list; this allocates against the partitioning ARCH-02 settled and the hierarchy ARCH-03 sized. It is the version RTL, synthesis and physical design are actually held to, and the version an overrun is measured against.

WORK FLOW

7 STEPS · 2 LANES

w12	w13	w14	w15	w16
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
---	---

#	STEP	TAT	LANE
1	Reconcile the DEF-03 targets against the ARCH-02 block list	0.8w	main
2	Area allocation per block, from the floorplan intent	1w	main
3	Power allocation per block and per domain	1w	main
4	Frequency and timing budget per clock domain	0.8w	parallel
5	Margin reservation for ECO and late growth	0.5w	parallel
6	Owner assignment and explicit acceptance per block	0.8w	main
7	Budget publication and change-control rule	0.5w	main

Step 6 is acceptance, not notification. A budget a block owner has not agreed to is a budget that will be exceeded and disputed, and the dispute happens at `PD-09` when there is no time for it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- PPA targets from DEF-03
- Block list and partitioning from ARCH-02
- Memory hierarchy sizing from ARCH-03
- Power domain definition from ARCH-05
- Floorplan intent from ARCH-08

PRODUCES

- Per-block area, power and timing budget
- Margin reservation record
- Owner acceptance sheet per block
- Change-control rule for budget movement
- Reconciliation note against the DEF-03 targets

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D3	PRODUCES	Block partitioning and PPA budget table. The budget half of the deliverable, published against the blocks ARCH-02 defined.
ARCH-D1	FEEDS	Architecture specification. The budget table is referenced by the specification rather than copied, so a budget change does not need a specification revision.
ARCH-D7	GATES	Architecture Freeze review package. A block without an accepted budget is an open item at the freeze gate.

RISKS AND WATCH-OUTS

- ◆ **Budgets allocated to 100%.** A budget with no margin has nothing left for the ECO storm at physical design, where power and area always grow.
- ◆ **Allocation without acceptance.** An unaccepted budget is defended by nobody and consumed by whoever moves first.
- ◆ **No change-control rule.** Budgets that can be moved quietly stop being budgets; movement has to cost a conversation.
- ◆ **Reconciliation against DEF-03 skipped.** If the architecture's total quietly exceeds the definition target, the cost model is wrong and nobody has been told.
- ◆ **Timing budget by clock domain only.** Domains contain blocks with very different closure difficulty, and a flat budget hides which one is impossible.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Area and power allocation	2.0	Block list final from ARCH-02	ARCH-09 Architecture specification authoring and review
Reconciliation against DEF-03	0.8	Memory hierarchy sized by ARCH-03	ARCH-10 Block microarchitecture specification kickoff
Timing budget	0.5	Floorplan intent available from ARCH-08	RTL-01 Block microarchitecture specification
Owner acceptance	0.5		SYN-07 Low-power synthesis — isolation and level-shifter insertion against UPF
Publication and change control	0.3	EXIT CRITERIA - Every block carries an accepted area, power and timing budget - Margin reserved and stated, not assumed - Change-control rule published and agreed	PD-02 Floorplan, macro placement and partition definition
Total	4.0		MEASURED BY - Blocks with an accepted budget against the block count - Margin held back as a share of each budget - Architecture total against the DEF-03 target
WHO IT TAKES	~4.3 FTE	DEPENDS ON DEF-03 PPA target definition — frequency, power envelope, die area budget ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels) ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft ARCH-08 Chip-level floorplan intent, pin and bump budget	
PPA lead Owns the allocation, the margin and the change-control rule			
Chief architect Arbitrates when blocks contest the same budget			
Block owners Accept or contest their allocation			
Physical architect Confirms the area allocation against the floorplan intent			
Program manager Records acceptance and carries gaps into the gate			

ARCH-07

Security and safety architecture — secure boot, root of trust, fusing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	7.0 M/M	~5.1 FTE	7	no	Security architect
w6–w12 in stage	planned	while it runs	3 in parallel	has float	Compliance liaison approves

PURPOSE

Define how the part **boots trustworthily, protects its assets and identifies itself** — secure boot chain, root of trust, key storage, fuse map and lifecycle states — and the debug lockdown that goes with them.

Security is architectural or it is absent. A root of trust cannot be retrofitted onto a design with no immutable boot path and no fuse plan. The tension worth naming early is that the debug access DFT and bring-up want is the same access an attacker wants, and only an architecture can resolve it.

WORK FLOW

7 STEPS · 2 LANES

w6	w8	w10	w12
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	Threat model and asset inventory	1w	main
2	Secure boot chain and immutable ROM definition	1.5w	main
3	Root of trust and key storage selection	1w	parallel
4	Fuse map, chip identity and lifecycle state definition	1w	main
5	Debug and test access lockdown policy, with DFT	1w	parallel
6	Cryptographic accelerator requirement	0.8w	parallel
7	Security architecture note and review	2.5w	main

Step 5 is a negotiation, not a decision taken alone. `DFT-04` needs JTAG and trace access for silicon debug, and every path it opens is a path this activity has to close in production lifecycle states.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Security requirements from the requirement register
- Customer and regulatory obligations
- Block list and boot architecture from ARCH-02
- eFuse and OTP capability from PDK-02
- DFT access architecture intent from DFT-04

PRODUCES

- Threat model and asset inventory
- Secure boot chain specification
- Root of trust and key storage architecture
- Fuse map and lifecycle state definition
- Debug lockdown policy

- Cryptographic accelerator requirement

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D1	PRODUCES	Architecture specification. The security chapter is authored here in full — boot chain, lifecycle states and lockdown policy — and lands in the specification whole.
ARCH-D5	FEEDS	Power / clock / reset architecture. Secure boot constrains the reset sequence: the root of trust must run before anything else is released from reset.

RISKS AND WATCH-OUTS

- ◆ **Security started after the block list is fixed.** A root of trust needs its own reset domain, its own ROM and its own fuse path; adding them later disturbs three architectures at once.
- ◆ **Debug access left open in production.** The lifecycle state that locks JTAG is the one nobody tests, because testing it makes the part undebuggable.
- ◆ **Fuse map without a lifecycle model.** Fuses are one-time; a map with no defined state transitions produces parts that cannot be returned to a testable state.
- ◆ **Key storage decided by IP availability.** Choosing the root of trust because a block was available is choosing a threat model by accident.
- ◆ **No named security owner past architecture.** Security decisions get eroded one waiver at a time through implementation unless someone owns them all the way to qualification.

EFFORT SPLIT	7 M/M	ENTRY CRITERIA	FEEDS INTO
Boot chain and root of trust	2.5	Security requirements present in the requirement register	ARCH-09 Architecture specification authoring and review
Threat model and asset inventory	1.5	Block list available from ARCH-02	ARCH-10 Block microarchitecture specification kickoff
Fuse map and lifecycle states	1.0	eFuse and OTP capability confirmed by PDK-02	DFT-04 Boundary scan and JTAG (1149.1 / 1687), TAP and debug access architecture
Debug lockdown negotiation	1.0		DFT-10 eFuse, chip ID and memory repair infrastructure
Cryptographic accelerator requirement	0.5		RTL-04 Chip-level integration and top assembly
Review	0.5		MP-11 Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety
Total	7.0		
WHO IT TAKES	~5.1 FTE		
Security architect Owns the threat model, the boot chain and the lockdown policy		EXIT CRITERIA - Threat model reviewed and every asset mapped to a control - Lifecycle states defined with their transitions and their tests - Debug lockdown agreed with DFT rather than imposed on it	MEASURED BY - Assets in the inventory mapped to an implemented control - Lifecycle state transitions with a defined test - Security waivers accepted after Architecture Freeze
Boot and firmware architect Boot sequence and immutable ROM content		DEPENDS ON DEF-01 Market and customer requirements consolidation ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
DFT liaison Negotiates debug access against lockdown			
Fuse and lifecycle engineer Fuse map, chip identity, state transitions			
Compliance liaison Customer and regulatory obligations			

ARCH-08

Chip-level floorplan intent, pin and bump budget

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	5.0 M/M	~4.3 FTE	7	no	Physical architect
w10–w15 in stage	planned	while it runs	3 in parallel	has float	Chief architect approves

PURPOSE

Draw the chip as a **physical object** for the first time — block placement intent, IO ring, bump field — and publish the pin and bump budget the package will be designed against.

The bump map is the first artifact the die and the package share, and the first hard constraint on bandwidth. An interface needing more bumps than the field can supply is an interface that has to shrink, and having that argument here costs a meeting. Having it at PD-04, with the substrate already ordered, costs the substrate.

WORK FLOW

7 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE

1	2	3	7
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Block placement intent from the partitioning and the hierarchy	1w	main
2	IO ring and PHY placement plan	1w	main
3	Bump field definition — pitch, count, power-to-signal ratio	1.5w	main
4	Signal bump allocation per interface	0.8w	parallel
5	Power delivery bump budget against the power envelope	1w	parallel
6	Package feasibility check with package design	0.8w	parallel
7	Block diagram and bump budget publication	1.5w	main

The power-to-signal ratio in step 3 is the number that decides the argument. On a part at this power envelope most of the bump field is power delivery, and the signal budget is what is left over rather than what was asked for.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Partitioning and boundaries from ARCH-02
- Memory hierarchy and HBM channel count from ARCH-03
- Interface lane counts from ARCH-04
- Power envelope from DEF-03
- Package architecture options from PKGD-01

PRODUCES

- Chip-level block diagram with placement intent
- IO ring and PHY placement plan
- Bump field definition and map
- Signal and power bump allocation
- Package feasibility finding

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D6	PRODUCES	Chip-level block diagram with pin and bump budget. This activity is the deliverable, and the bump map is the half of it the package team consumes directly.
ARCH-D1	FEEDS	Architecture specification. The physical view chapter — the first place the specification stops being a block diagram and becomes a die.
ARCH-D3	FEEDS	Block partitioning and PPA budget table. Placement intent is what ARCH-06 allocates die area against.

RISKS AND WATCH-OUTS

- ◆ **Bump budget derived after interfaces are fixed.** Lane counts set without a bump budget produce a floorplan that cannot be built and an interface that has to be cut late.
- ◆ **Power delivery bumps underestimated.** Too few power bumps is an IR-drop problem discovered at SIPI-04, when the bump map is already in the package design.
- ◆ **Placement intent treated as decoration.** Physical design inherits this drawing; an intent that ignores routing feasibility is redrawn at PD-01 and the schedule pays for both.
- ◆ **Package feasibility checked against a generic assumption.** Bump pitch and field size depend on the actual interposer and substrate technology, which PKGD-01 may not have settled yet.
- ◆ **HBM PHY placement left open.** HBM PHYs must sit against the interposer edge facing their stacks; deciding this late constrains everything else on the die.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Bump field definition and allocation	2.0	Partitioning decided by ARCH-02	ARCH-06 PPA budget allocation per block
Placement intent	1.3	Interface lane counts available from ARCH-04	ARCH-09 Architecture specification authoring and review
Power bump budget	0.8	Package architecture candidates known from PKGD-01	PKGD-02 Bump map, pitch and power-ground planning with physical design
Package feasibility check	0.5		
Publication	0.5		
Total	5.0		
WHO IT TAKES	~4.3 FTE		
Physical architect Owns the placement intent and the bump field		EXIT CRITERIA - Bump field defined with an explicit power-to-signal ratio - Every interface has an allocated bump count package has accepted - Placement intent reviewed by physical design as buildable	PD-01 Flow setup and MMMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity
IO and PHY architect Ring organisation and PHY placement			PD-04 Bump and RDL planning with chip-package co-design feedback
Power delivery engineer Power bump budget against the envelope		DEPENDS ON	SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
Package design liaison Feasibility against the interposer and substrate		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	MEASURED BY - Signal bumps allocated against those requested - Power-to-signal ratio against the envelope requirement - Bump map revisions after Architecture Freeze
Chief architect Arbitrates bandwidth against bump budget		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	
		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	
		DEF-03 PPA target definition — frequency, power envelope, die area budget	

ARCH-09

Architecture specification authoring and review

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w w9–w18 in stage	14.0 M/M planned	~6.8 FTE while it runs	8 3 in parallel	yes no float	Chief architect Program manager approves

PURPOSE

Write the document the rest of the program builds against, and run it through the review that ends in **Architecture Freeze**.

The specification is not a record of decisions already taken; writing it is what exposes the decisions nobody took. Every downstream stage reads this document, and every ambiguity left in it becomes an argument in RTL, a bug in verification, or a waiver at signoff. The consistency pass matters as much as the authoring.

WORK FLOW

8 STEPS · 2 LANES

w9	w11	w13	w15	w17	w18
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MAIN SEQUENCE

1	2	5	6	8
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IN PARALLEL

3	4	7
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#	STEP	TAT	LANE
1	Specification structure and section ownership assignment	0.5w	main
2	Section authoring across the architecture team	3.5w	main
3	Consistency pass — every number traced to its source activity	1.5w	parallel
4	Open-issue register and disposition	1w	parallel
5	Internal technical review, two rounds	2w	main
6	Downstream readiness review with RTL, DV, DFT and physical design	1.5w	main
7	Freeze package assembly	1w	parallel
8	Architecture Freeze gate	1.5w	main

The downstream readiness review in step 6 is the one that predicts the program. Architecture reviewing its own specification confirms it is coherent; RTL, DV and DFT reviewing it confirms it is implementable, and those are different questions.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Every other ARCH activity's output
- Performance model results from ARCH-01
- PPA budget table from ARCH-06
- Feasibility concessions from DEF-07
- Open issues carried from the architecture risk register

PRODUCES

- Architecture specification
- Open-issue register with disposition per item
- Consistency and traceability record
- Downstream readiness review findings
- Architecture Freeze package and decision record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D1	PRODUCES	Architecture specification. This activity assembles and closes it. Chapters authored in ARCH-03, ARCH-05 and ARCH-07 arrive whole; the work here is coherence, not rewriting.
ARCH-D7	PRODUCES	Architecture Freeze review package. The specification, the open-issue disposition and the downstream readiness findings, assembled for the gate.
ARCH-D2	FEEDS	Performance model and workload analysis report. Published alongside the specification, because the specification's numbers are only defensible with it.

RISKS AND WATCH-OUTS

- ◆ **Freeze declared with open issues undispositioned.** An open issue carried past freeze is a decision deferred to whoever hits it first in RTL.
- ◆ **Consistency pass skipped under schedule pressure.** The same number appearing differently in two chapters is the defect that costs the most and is found the latest.
- ◆ **Downstream teams reviewing after freeze.** A readiness review that happens after the gate cannot change anything, and becomes a complaint rather than a review.
- ◆ **Specification written as a narrative.** Implementable specifications are tables, interfaces and constraints; prose is where ambiguity survives review.
- ◆ **Freeze treated as a date rather than a condition.** A gate that passes because it was scheduled to teaches the organisation that gates do not mean anything.

EFFORT SPLIT	14 M/M	EXIT CRITERIA	FEEDS INTO
Section authoring	6.0	● Every open issue dispositioned — closed, deferred with an owner, or accepted as risk ● Downstream readiness review completed before the gate, not after ● Freeze decision recorded with conditions and dissent	ARCH-10 Block microarchitecture specification kickoff
Technical and readiness reviews	3.5		RTL-01 Block microarchitecture specification
Consistency and traceability pass	2.0		DV-01 Verification plan and coverage model definition
Open-issue disposition	1.5		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT
Freeze package and gate	1.0		SYN-01 SDC constraint development and validation
Total	14.0	DEPENDS ON	PD-01 Flow setup and MMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity
WHO IT TAKES	~6.8 FTE	ARCH-01 System-level performance modelling and workload simulation	MEASURED BY
Chief architect Owns the specification and the freeze recommendation		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
Section authors Architecture leads writing their own chapters		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	
Technical editor Consistency, traceability and structure		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E)	
Downstream leads RTL, DV, DFT and PD readiness review		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
Program manager Open-issue disposition and the gate record		ARCH-06 PPA budget allocation per block	
ENTRY CRITERIA		ARCH-07 Security and safety architecture — secure boot, root of trust, fusing	● Open issues dispositioned at the gate against those raised ● Specification numbers traceable to a source activity ● Specification revisions issued after freeze
● All architectural decisions taken, or explicitly listed as open ● PPA budget accepted by block owners in ARCH-06 ● Downstream leads available to review before the gate		ARCH-08 Chip-level floorplan intent, pin and bump budget	

ARCH-10

Block microarchitecture specification kickoff

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	12.0 M/M	~8.7 FTE	8	yes	Block architects
w12–w18 in stage	planned	while it runs	3 in parallel	no float	Chief architect approves

PURPOSE

Turn each architectural block into a **microarchitecture specification its RTL owner can implement from** — pipeline structure, interfaces, control, register map and the verification hooks.

The architecture specification says what each block does; this says how. Starting it inside the architecture stage rather than at RTL kickoff is what lets RTL-02 begin with something to write against instead of a paragraph — and it is the cheapest place to find out that two blocks disagree about the interface between them.

WORK FLOW

8 STEPS · 2 LANES

w12	w14	w16	w18
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Block specification template and review cadence	0.5w	main
2	Block owner assignment and kickoff	0.5w	main
3	Pipeline and datapath definition per block	2w	main
4	Block interface and protocol definition against the NoC	1.5w	parallel
5	Control, configuration and register map, first pass	1.5w	parallel
6	Verification hook and observability requirement, with DV	1w	parallel
7	Microarchitecture review, first round	1.5w	main
8	Handoff to RTL implementation	1.5w	main

Step 4 runs in parallel and is where the interesting failures live. Two block owners each writing their side of a shared interface will disagree, and finding that here costs a meeting rather than an integration debug at `RTL-04`.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Architecture specification chapters from ARCH-09
- Partitioning and boundaries from ARCH-02
- Memory hierarchy specification from ARCH-03
- PPA budget per block from ARCH-06
- Verification strategy intent from DV-01

PRODUCES

- Microarchitecture specification per block
- Block interface and protocol definitions
- Register map first pass, per block
- Verification hook and observability requirements
- RTL implementation handoff package

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ARCH-D1	FEEDS	Architecture specification. Block specifications are children of it — they refine rather than restate, and any conflict is resolved in favour of the architecture specification.
ARCH-D3	FEEDS	Block partitioning and PPA budget table. Microarchitecture is where a block owner first tests whether the allocated budget is achievable.
ARCH-D7	GATES	Architecture Freeze review package. A block with no microarchitecture owner is an open item at the gate — the specification cannot be implemented by nobody.

RISKS AND WATCH-OUTS

- ◆ **Block owners not assigned.** A block whose specification has no author is a block whose RTL starts by inventing one, usually in week three of implementation.
- ◆ **Interfaces written from one side.** Each owner specifying their own side of a shared interface produces two consistent documents that disagree with each other.
- ◆ **Register map deferred to RTL.** The register map is the software contract; defining it late means firmware development starts late and the two never converge.
- ◆ **Verification hooks added after the design.** Observability designed in is cheap; observability retrofitted costs area, timing and an RTL change under control.
- ◆ **Microarchitecture that ignores the budget.** A pipeline designed without reference to ARCH-06's allocation is a budget overrun scheduled for synthesis.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Pipeline and datapath definition	4.0	Architecture specification chapters stable enough to refine	RTL-01 Block microarchitecture specification
Interface definition	2.5	Block owners assigned and available	RTL-02 Block-level RTL implementation
Register map first pass	2.0	PPA budget accepted per block in ARCH-06	RTL-07 Register map / RDL definition and header generation
Microarchitecture reviews	2.0		
Verification hooks	1.0		
Handoff	0.5		DV-01 Verification plan and coverage model definition
Total	12.0	EXIT CRITERIA - Every block has a specification and a named owner - Shared interfaces agreed by both sides, not by each - RTL leads have accepted the handoff as implementable	DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT
WHO IT TAKES	~8.7 FTE		MEASURED BY
Block architects Author their block's microarchitecture specification			Blocks with a specification and an owner at freeze
Interconnect architect Arbitrates shared interface definitions		DEPENDS ON	Shared interfaces signed by both sides
RTL leads Receive the handoff and confirm it is implementable		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	Interface defects found at RTL integration
Verification liaison Observability and verification hook requirements		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	
Chief architect Resolves conflicts against the architecture specification		ARCH-06 PPA budget allocation per block	
		ARCH-09 Architecture specification authoring and review	

03 Technology & Foundry Selection

TECH

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w0 – w14	14 _w	9	26 _{M/M}	4 _{of 9}		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
TECH-01	Foundry and node long-list to short-list evaluation		5w	4 M/M	TECH-D1	Technology selection report and decision record
TECH-02	Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)		5w	4 M/M	TECH-D2	Process option / flavour sheet agreed with foundry
TECH-03	Density / performance / leakage benchmarking against product targets (DTCO study)		7w	6 M/M	TECH-D3	Wafer, mask and NRE cost sheet
TECH-04	Wafer price, mask and NRE quotation; MPW and volume pricing		6w	3 M/M	TECH-D4	Executed foundry design agreement (DA) and NDA
TECH-05	Fab capacity and tapeout slot reservation; hot-lot policy agreement		5w	2 M/M	TECH-D5	Capacity and tapeout slot commitment
TECH-06	Foundry legal engagement — NDA, design agreement, IP licensing frame		8w	2 M/M	TECH-D6	Node risk assessment — maturity, defect density, yield learning curve
TECH-07	Foundry roadmap and risk-production timing alignment		4w	1.5 M/M		
TECH-08	Second-source and node-migration strategy assessment		4w	1.5 M/M		
TECH-09	OSAT and backend supply chain preliminary alignment		5w	2 M/M		

TECH-01

Foundry and node long-list to short-list evaluation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	4.0 M/M	~3.5 FTE	7	yes	Technology strategist
w0–w5 in stage	planned	while it runs	2 in parallel	no float	Chief architect approves

PURPOSE

Narrow the field of foundries and nodes to a short-list the program can actually evaluate, and **record why the others were dropped**.

Every other decision in this stage is conditional on this one, and so is most of the enablement band: IP availability, PDK maturity, library qualification and capacity all become node-specific the moment the short-list closes. The screen has to be quick and it has to be written down — a selection nobody can reconstruct is one that gets reopened at every setback.

WORK FLOW

7 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
---	---

#	STEP	TAT	LANE
1	Long-list construction — foundries, nodes and their availability windows	0.8w	main
2	Screening criteria weighted against the product requirements	0.5w	main
3	Foundry engagement and formal information requests	1.5w	main
4	Previous-program experience and relationship review	0.5w	parallel
5	Geopolitical, export-control and supply-continuity screen	0.8w	parallel
6	Short-list down to two or three candidates	1w	main
7	Selection recommendation and decision record	1.3w	main

Step 3 is the pacing item and almost none of it is our work — information requests sit with the foundry's technical marketing team, and the response time is theirs. Starting the engagement in week one is the only lever the program has over it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Product requirements and PPA targets from DEF-01 and DEF-03
- Volume forecast and cost ceiling from the business case
- Foundry public roadmaps and node cadence
- Previous-program experience with each foundry
- Export control and supply-continuity guidance

PRODUCES

- Foundry and node long-list with availability windows
- Weighted screening criteria
- Short-list of two or three candidates
- Screening record — what was dropped and why
- Selection recommendation and decision record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D1

PRODUCES

Technology selection report and decision record. This activity owns the deliverable; TECH-03 supplies the technical evidence it rests on.

TECH-D6

FEEDS

Node risk assessment. The supply-continuity and geopolitical screen is one of the risk assessment's inputs, and it is gathered here.

RISKS AND WATCH-OUTS

- ◆ **Short-list of one.** A screen that arrives at a single candidate has no negotiating position on price, capacity or terms, and the foundry knows it.
- ◆ **Selection driven by relationship.** The foundry the last program used is a legitimate input and a poor criterion; making it the criterion means the screen was theatre.
- ◆ **Availability window not checked.** A node that is right for the product but reaches risk production after the tapeout date is not a candidate, however good the numbers are.
- ◆ **Export control screened late.** A candidate that fails compliance after technical evaluation has consumed weeks of the stage for nothing.
- ◆ **Criteria written after the data arrives.** Weighting the screen once the responses are in produces the answer somebody already wanted.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Foundry engagement	1.3	● Product requirements and PPA targets available at draft quality	TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)
Screening criteria and long-list	0.8	● Volume forecast good enough to discuss with a foundry	TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)
Short-listing	0.8	● Legal cover in place for preliminary discussion	TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame
Continuity and compliance screen	0.5		TECH-07 Foundry roadmap and risk-production timing alignment
Decision record	0.8		DEF-07 Feasibility assessment against candidate technology nodes
Total	4.0	EXIT CRITERIA	IPR-05 Silicon-proven status check per IP on the selected node and process option
WHO IT TAKES	~3.5 FTE	● Short-list of at least two candidates, with the dropped options recorded	PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis
Technology strategist Owns the screen, the short-list and the recommendation		● Criteria weighted before the responses were read	
Foundry relationship manager Engagement and information requests		● Recommendation carries availability windows, not only technical fit	
Program manager Availability windows against the program schedule			
Compliance liaison Export control and supply-continuity screen		DEPENDS ON	
Chief architect Confirms the criteria reflect what the product needs		DEF-01 Market and customer requirements consolidation	
			MEASURED BY
			● Candidates carried to short-list against those screened
			● Weeks from first engagement to short-list
			● Selection reopened after the decision record

TECH-02

Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	4.0 M/M	~3.5 FTE	7	yes	Technology strategist
w4–w9 in stage	planned	while it runs	3 in parallel	no float	Foundry technical account manager approves

PURPOSE

Choose the **flavour of the node**, not just the node — performance or density library, the multi-Vt menu, the metal stack, backside power delivery, and any analog or high-voltage adders the product needs.

A node name is not a specification. Two programs on the same node with different flavour sheets get different density, different leakage and different routing resource, and the difference is larger than most architectural decisions. This sheet is also what the foundry prices against, so it has to close before TECH-04 can finish.

WORK FLOW

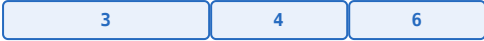
7 STEPS · 2 LANES

w4w5w6w7w8w9

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Process flavour options against the PPA targets — HPC, HD, or mixed	1w	main
2	Multi-Vt menu selection and its leakage implication	1w	main
3	Backside power delivery evaluation, where the node offers it	0.8w	parallel
4	RF, high-voltage and analog adder requirements	0.5w	parallel
5	Metal stack selection and routing resource implication	1w	main
6	Option cost adders priced with TECH-04	0.5w	parallel
7	Flavour sheet agreement with the foundry	2w	main

Two of the five weeks sit in step 7, which is signature time rather than engineering time. The sheet is a contractual artifact: once agreed it governs what the PDK contains and what the wafer costs, and changing it later reopens both.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Short-list from TECH-01
- PPA targets from DEF-03
- DTCO findings as they emerge from TECH-03
- Reticle and die size direction from ARCH-02

PRODUCES

- Flavour option comparison against the PPA targets
- Multi-Vt menu selection with leakage projection
- Metal stack selection and routing resource note
- Analog, RF and high-voltage adder list

- Analog and PHY requirements from ARCH-04
- Process option / flavour sheet agreed with the foundry

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D2	PRODUCES	Process option / flavour sheet agreed with foundry. This activity is the deliverable, and it is contractual rather than advisory.
TECH-D3	FEEDS	Wafer, mask and NRE cost sheet. Every option on the sheet carries a price adder, so the flavour decides a meaningful part of the wafer cost.
TECH-D1	FEEDS	Technology selection report. The selection is only complete when the flavour is named — a node without a flavour sheet is not a selection anyone can build on.

RISKS AND WATCH-OUTS

- ◆ **Flavour chosen for density alone.** A high-density library that cannot reach the frequency target trades an area saving for a performance failure, and the trade is discovered at trial synthesis.
- ◆ **Metal stack chosen without a congestion study.** Too few routing layers is the defect that surfaces at PD-06, when the only remaining fix is a floorplan change.
- ◆ **Multi-Vt menu wider than the flow supports.** Every extra Vt is a library to characterise, qualify and maintain; menus are chosen for leakage and paid for in methodology.
- ◆ **Backside power taken for the roadmap.** A first-generation backside PDN buys power integrity and costs schedule risk, and the trade belongs to this product rather than the next one.
- ◆ **Adders discovered after the sheet is agreed.** An analog or high-voltage requirement that arrives late is a sheet amendment, a price change and a PDK delta at once.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Flavour and Vt studies	1.5	Short-list closed by TECH-01	TECH-04 Wafer price, mask and NRE quotation; MPW and volume pricing
Metal stack selection	0.8	PPA targets published by DEF-03	PDK-02 Design rule manual review; restricted and recommended rule disposition
Backside PDN and adder evaluation	0.8	Early DTCO results available from TECH-03	PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)
Sheet agreement	1.0		PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)
Total	4.0	EXIT CRITERIA - Flavour sheet agreed and signed, not merely discussed - Metal stack backed by a congestion study rather than by precedent - Every analog and adder requirement captured before agreement	SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
WHO IT TAKES	~3.5 FTE	DEPENDS ON TECH-01 Foundry and node long-list to short-list evaluation DEF-03 PPA target definition — frequency, power envelope, die area budget	PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
Technology strategist Owns the flavour sheet and its agreement			MEASURED BY - Options on the sheet against those the design actually uses - Sheet amendments raised after agreement - Routing layers against the congestion study's recommendation
Library and CAD lead Multi-Vt menu and library implications			
Physical design liaison Metal stack and routing resource			
Analog lead RF, high-voltage and analog adder requirements			
Foundry technical account manager Availability of each option on the node			

TECH-03

Density / performance / leakage benchmarking against product targets (DTCO study)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	6.0 M/M	~3.7 FTE	8	yes	DTCO lead
w3–w10 in stage	planned	while it runs	3 in parallel	no float	Foundry technical contact approves

PURPOSE

Measure the candidate nodes against **this product's targets** rather than against the foundry's brochure — density, achievable frequency, leakage, SRAM Vmin and routing resource, on circuits that resemble the design.

This is the technical evidence the selection stands on. Published node comparisons are made on somebody else's circuits at somebody else's operating point; a DTCO study run on representative structures is the only way to know whether the PPA targets survive the node, and it is what DEF-07 tests feasibility against.

WORK FLOW

8 STEPS · 2 LANES

w3	w5	w7	w9	w10
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Benchmark circuit set definition, representative of the design	0.8w	main
2	Standard cell density and area scaling study	1.5w	main
3	Ring oscillator and critical-path frequency study	1.5w	main
4	Leakage and dynamic power characterisation across the Vt menu	1.3w	parallel
5	SRAM bitcell density and Vmin study	1.3w	parallel
6	Routing congestion proxy study on the candidate metal stacks	1w	parallel
7	Node comparison against the DEF-03 targets	1.5w	main
8	DTCO findings and recommendation	1.8w	main

Three of the eight steps run in parallel because they use different data and different people — power characterisation, SRAM and congestion are separable studies. What cannot be parallelised is step 7: comparison needs all three to have landed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Short-list from TECH-01
- PPA targets from DEF-03
- Foundry-supplied preliminary libraries and models
- Previous-generation design as a benchmark reference

PRODUCES

- Benchmark circuit set and its rationale
- Density and area scaling results per node
- Frequency and leakage characterisation per Vt
- SRAM density and Vmin findings

- Memory compiler capability indications
- Routing congestion proxy results
 - DTCO findings and node recommendation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D1	FEEDS	Technology selection report and decision record. The evidence half of the report — TECH-01 records the decision, this activity is why the decision is defensible.
TECH-D6	FEEDS	Node risk assessment. Where a target is only just met, this study is what says so, and that margin is a node risk.
TECH-D2	FEEDS	Process option / flavour sheet. The Vt menu and metal stack on the sheet are chosen from these results.

RISKS AND WATCH-OUTS

- ◆ **Benchmark circuits unrepresentative.** A study run on ring oscillators alone measures the node and not the design; the answer is precise and irrelevant.
- ◆ **Preliminary libraries treated as final.** Early PDK models move, sometimes by more than the difference between the candidates, and a comparison made on 0.5 data can invert.
- ◆ **SRAM left out.** On an accelerator the array is a large fraction of the die; a node comparison without a bitcell and Vmin study has skipped the dominant area and the dominant Vmin constraint.
- ◆ **Congestion never modelled.** Density that cannot be routed is not density, and the routing resource question is decided by the metal stack chosen in TECH-02.
- ◆ **Results reported without margin.** A target met exactly in a DTCO study will not be met in implementation, and reporting the number without the margin invites the wrong conclusion.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Density and frequency studies	2.0	● Short-list available from TECH-01	TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)
Leakage and SRAM characterisation	1.5	● Preliminary libraries released under NDA	
Node comparison	1.0	● PPA targets published by DEF-03	TECH-07 Foundry roadmap and risk-production timing alignment
Congestion proxy study	0.8		
Findings and recommendation	0.8		
Total	6.0		
WHO IT TAKES	~3.7 FTE		
DTCO lead Owns the study, its circuits and its findings		EXIT CRITERIA ● Every target has a measured result per candidate node ● Findings state margin, not only pass or fail ● Benchmark set documented well enough to re-run on a later PDK	DEF-03 PPA target definition — frequency, power envelope, die area budget
Library characterisation engineer Density, frequency and leakage measurement		DEPENDS ON TECH-01 Foundry and node long-list to short-list evaluation DEF-03 PPA target definition — frequency, power envelope, die area budget	DEF-07 Feasibility assessment against candidate technology nodes
Memory design liaison SRAM bitcell density and Vmin study			PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)
Physical design liaison Congestion proxy and routing resource			ARCH-01 System-level performance modelling and workload simulation
Foundry technical contact Preliminary models, decks and interpretation			MEASURED BY ● Targets measured against targets asserted ● Margin reported on each met target ● Findings later contradicted by trial synthesis

TECH-04

Wafer price, mask and NRE quotation; MPW and volume pricing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	3.0 M/M	~2.2 FTE	7	no	Procurement lead
w5–w11 in stage	planned	while it runs	3 in parallel	has float	Program manager approves

PURPOSE

Put a **price on the technology** — wafers at the forecast volume, the mask set, the NRE, the MPW shuttle, and every adder the flavour sheet carries — and hand it to the cost model as quotations rather than estimates.

The cost model in DEF-04 is only as good as this activity. A model built on published estimates produces a margin nobody can defend at the funding gate; a model built on quotations produces one that survives the first conversation with finance.

WORK FLOW

7 STEPS · 2 LANES

w5

w7

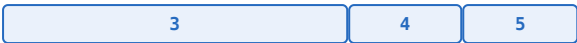
w9

w11

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Volume and wafer-start forecast from the business case	0.5w	main
2	Wafer price request and negotiation	2w	main
3	Mask set and NRE quotation per node and flavour	1.5w	parallel
4	MPW shuttle pricing and slot cost	0.5w	parallel
5	Option adder pricing against the TECH-02 sheet	0.5w	parallel
6	Volume tier and price-break structure	2.5w	main
7	Cost sheet publication into the DEF-04 model	1w	main

The tier structure in step 6 is worth more than the headline price. A wafer price that only applies above a volume the product will not reach in its first year is a price the cost model should not be using.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Short-list and flavour sheet from TECH-01 and TECH-02
- Volume forecast and ramp profile from the business case
- Die size estimate from DEF-04
- Mask layer count implied by the metal stack
- MPW shuttle calendar from the foundry

PRODUCES

- Wafer price quotation with tier structure
- Mask set and NRE quotation per candidate
- MPW shuttle pricing and slot cost
- Option adder pricing
- Cost sheet published into the DEF-04 model

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D3

PRODUCES

Wafer, mask and NRE cost sheet. This activity is the deliverable, and its value is that the numbers are quoted rather than estimated.

TECH-D1

FEEDS

Technology selection report. Cost is a selection criterion; a node that wins on PPA and loses on price is a decision, not a formality.

RISKS AND WATCH-OUTS

- ◆ **Price quoted at a volume the product will not reach.** Tier structures are where the real cost lives, and a headline price at ten times the forecast volume is a number the model should reject.
- ◆ **Mask cost estimated from layer count alone.** EUV layers dominate mask cost, and the count of them depends on the flavour sheet rather than on the total layer count.
- ◆ **Adders priced after the sheet is agreed.** Once TECH-02 closes, the adders are committed; pricing them afterwards removes the option of dropping one.
- ◆ **Currency and term exposure ignored.** A quotation with no validity period or currency term is not a quotation the cost model can carry to a funding gate.
- ◆ **Second-source pricing never requested.** Without a comparison the negotiation has no floor, and the first price is the last price.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Price negotiation	1.3	● Short-list closed and flavour options known ● Volume forecast agreed with sales and finance ● Die size estimate available from DEF-04	DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)
Mask, NRE and shuttle quotations	0.8		DEF-07 Feasibility assessment against candidate technology nodes
Tier structure	0.5		DEF-09 Business case preparation and funding approval package
Publication into the cost model	0.5		TC-03 MPW shuttle slot booking and data submission scheduling
Total	3.0	EXIT CRITERIA	MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
WHO IT TAKES	~2.2 FTE	● Every cost line quoted, with a validity period and currency term ● Tier structure reflects the actual ramp, not a notional volume ● Cost sheet loaded into the DEF-04 model, not summarised into it	MEASURED BY
Procurement lead			● Cost lines backed by a quotation rather than an estimate ● Quoted price against the tier the program will actually reach ● Cost sheet revisions after the funding gate
Owns the negotiation and the quotations			
Product cost analyst			
Feeds the sheet into the DEF-04 model		DEPENDS ON	
Technology strategist		TECH-01	Foundry and node long-list to short-list evaluation
Confirms what each adder buys technically		TECH-02	Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)
Finance partner		DEF-04	Die size, yield and product cost model (wafer, mask, package, test, ASP)
Currency, validity and term treatment			
Program manager			
Volume and ramp profile assumptions			

TECH-05

Fab capacity and tapeout slot reservation; hot-lot policy agreement

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	2.0 M/M	~1.7 FTE	6	no	Foundry relationship manager
w8–w13 in stage	planned	while it runs	2 in parallel	has float	Product engineering liaison approves

PURPOSE

Convert the program's schedule into **reserved fab capacity** — a wafer-start profile, a tapeout slot, hot-lot terms and an engineering-lot allocation — and get it committed in writing.

Capacity is not bought at tapeout; it is reserved a year before, against a date that will move. The commitment matters most in its escalation path: when the schedule slips or the market tightens, what the program has is whatever was written down, and a verbal understanding is worth exactly nothing at that moment.

WORK FLOW

6 STEPS · 2 LANES

w8	w9	w10	w11	w12	w13
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Wafer-start profile derived from the program schedule	0.5w	main
2	Capacity request and allocation discussion	1.5w	main
3	Tapeout slot reservation against the DEF-08 date	1w	main
4	Hot-lot policy and priority terms	1w	parallel
5	Engineering-lot and risk-wafer allocation	0.8w	parallel
6	Capacity commitment letter and escalation path	2w	main

Steps 4 and 5 look like details and behave like insurance. Hot-lot priority is what turns an 18-week fab cycle into an 11-week one at `FAB-04`, and it is only available to programs that negotiated it before they needed it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Program schedule and tapeout date from DEF-08
- Wafer volume forecast and ramp profile
- Foundry capacity availability at the node
- Hot-lot policy terms offered by the foundry
- Engineering and risk-lot requirements from the test plan

PRODUCES

- Wafer-start profile against the program schedule
- Capacity allocation agreement
- Tapeout slot reservation
- Hot-lot policy and priority terms
- Engineering and risk-lot allocation
- Capacity commitment letter with an escalation path

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

- TECH-D5

PRODUCES

Capacity and tapeout slot commitment. This activity is the deliverable, and it is worth only as much as the escalation path written into it.
- TECH-D3

INFORMS

Wafer, mask and NRE cost sheet. Committed volume is what the price tier is granted against, so the two documents have to agree.

RISKS AND WATCH-OUTS

- ◆ **Slot reserved against a date that has not been committed.** A tapeout slot booked against an optimistic schedule is a slot that will be missed and re-queued behind whoever is ready.
- ◆ **Hot-lot terms left informal.** Priority granted by relationship evaporates when a larger customer needs the same capacity in the same week.
- ◆ **No escalation path.** A commitment with no named escalation contact is a commitment enforced by goodwill, which is not a mechanism.
- ◆ **Engineering lots forgotten.** Test-program development and characterisation need silicon before the production ramp, and those wafers have to be allocated separately.
- ◆ **Capacity sized to the base case only.** An upside scenario the fab cannot supply is a revenue forecast the program cannot deliver on.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Capacity allocation discussion	0.8	● Program schedule dated by DEF-08	T0-07 FEOL mask order confirmation and mask shop scheduling
Tapeout slot reservation	0.5	● Volume and ramp forecast agreed	FAB-02 Wafer start on FEOL mask availability
Hot-lot and engineering-lot terms	0.5	● Design agreement executed or in final redline	FAB-09 Hot-lot management and WIP tracking
Commitment letter	0.3		TC-03 MPW shuttle slot booking and data submission scheduling
Total	2.0	EXIT CRITERIA	MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
WHO IT TAKES	~1.7 FTE	● Capacity committed in writing, with an escalation contact named ● Hot-lot terms explicit rather than understood ● Engineering and risk lots allocated separately from production	MEASURED BY
Foundry relationship manager Owns the capacity negotiation and the commitment		DEPENDS ON	● Capacity committed against capacity requested
Program manager Wafer-start profile and tapeout date		TECH-01 Foundry and node long-list to short-list evaluation	● Weeks of hot-lot priority secured
Operations planner Ramp profile and engineering-lot needs		TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame	● Slot date against the DEF-08 tapeout date
Procurement lead Contractual terms and escalation path		DEF-08 Program schedule, resource and staffing plan drafting	
Product engineering liaison Risk-wafer and characterisation requirements			

TECH-06

Foundry legal engagement — NDA, design agreement, IP licensing frame

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w2–w10 in stage	2.0 M/M planned	~1.1 FTE while it runs	8 3 in parallel	yes no float	Legal counsel Compliance officer approves

PURPOSE

Put the **legal frame** in place that everything technical depends on — NDA, design agreement, PDK and library licensing, the foundry IP frame, and liability terms — and execute it.

This is the longest activity in the stage at eight weeks and one of the smallest at 2 M/M, which is exactly the shape it should have: the elapsed time belongs to two legal departments and almost none of the work belongs to the program. It is on the critical path anyway, because production PDK access, memory compilers and foundry IP all sit behind the design agreement.

WORK FLOW

8 STEPS · 2 LANES

w2	w4	w6	w8	w10
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MAIN SEQUENCE

1	2	4	7	8
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	NDA execution, to open technical discussion	1w	main
2	Design agreement scope definition	1w	main
3	PDK and library licensing terms	1.5w	parallel
4	Foundry IP licensing frame — memory compilers, IO, PHY	2.5w	main
5	Liability, indemnity and yield-responsibility terms	1.5w	parallel
6	Export control and compliance review	1w	parallel
7	Legal negotiation and redline cycles	2.5w	main
8	Execution and distribution to the teams that need it	1w	main

Step 1 is deliberately first and deliberately short. Nothing technical can be discussed before the NDA executes, so every week it takes is a week subtracted from `TECH-03` — which is why the NDA is chased separately from the agreement it precedes.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Short-list from TECH-01
- Corporate legal templates and negotiation positions
- Foundry standard agreement terms
- Export control and compliance guidance
- IP requirements from ARCH-04 and IPR-01

PRODUCES

- Executed NDA
- Design agreement, scoped and executed
- PDK and library licence terms
- Foundry IP licensing frame
- Compliance and export-control record

- Distribution record — who is covered to access what

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D4

PRODUCES

Executed foundry design agreement (DA) and NDA. This activity is the deliverable, and until it executes the enablement band cannot legally start.

TECH-D5

FEEDS

Capacity and tapeout slot commitment. Capacity is committed under the design agreement; the commitment has no standing without it.

RISKS AND WATCH-OUTS

- ◆ **NDA treated as a formality and queued behind the agreement.** Technical evaluation is blocked until it executes, and every week lost here is taken out of the DTCO study.
- ◆ **Foundry IP frame left out of scope.** Memory compilers and IO libraries are licensed separately from the PDK; discovering that at PDK-04 stops library qualification.
- ◆ **Distribution not tracked.** An executed agreement nobody circulated leaves engineers unable to download the PDK they are waiting for, and the delay looks technical.
- ◆ **Yield responsibility left ambiguous.** Who owns a process excursion is a term, not a conversation to have after it happens.
- ◆ **Redline cycles unbudgeted.** Two legal departments will take longer than either forecasts, and the program's plan should assume the longer number.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Negotiation support	0.8	● Short-list narrow enough to engage legally	TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement
IP and licensing frame	0.5	● Corporate legal position and templates available	PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis
Terms review	0.5	● Compliance guidance for the candidate foundries	PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)
Execution and distribution	0.3		IPR-07 Licence negotiation, PO issue and delivery-date commitment
Total	2.0		TC-03 MPW shuttle slot booking and data submission scheduling
WHO IT TAKES	~1.1 FTE	EXIT CRITERIA	MEASURED BY
Legal counsel Owns the agreements and the negotiation		● NDA and design agreement executed, not merely agreed	● Weeks from engagement to executed NDA
Foundry relationship manager Drives the redline cycles to closure		● PDK, library and foundry IP access all covered by scope	● Access requests blocked on missing scope
Procurement lead Commercial terms and licensing scope		● Access distributed to the teams that need it, and tracked	● Redline cycles against those forecast
CAD and methodology lead States what PDK and library access is needed		DEPENDS ON	
Compliance officer Export control and regulatory review		TECH-01 Foundry and node long-list to short-list evaluation	

03 TECH · TECHNOLOGY & FOUNDRY SELECTION

TECH-07

Foundry roadmap and risk-production timing alignment

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	1.5 M/M	~1.6 FTE	5	no	Technology strategist
w1–w5 in stage	planned	while it runs	2 in parallel	has float	Product cost analyst approves

PURPOSE

Establish where each candidate node **actually is on its maturity curve** — risk production timing, defect density trend, yield learning rate, capacity ramp — and turn that into a risk assessment the program plans against.

A node chosen at the right time is a node in volume production when the program taped out. A node chosen too early gives the program the foundry's learning curve as a yield problem, and the schedule to absorb it was never in the plan.

WORK FLOW

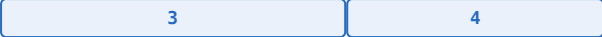
5 STEPS · 2 LANES

w1	w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Node maturity and risk-production timing review	1w	main
2	Defect density trend and yield learning curve request	1w	main
3	Foundry roadmap alignment against the program schedule	1w	parallel
4	Capacity ramp profile at the node	0.8w	parallel
5	Node risk assessment and mitigation options	2w	main

Step 2 asks for data the foundry treats as sensitive, and how much of it arrives is itself a signal. A foundry that will not share a defect density trend under NDA is telling the program something about the trend.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Short-list from TECH-01
- Foundry roadmap and node cadence under NDA
- Defect density and yield learning data where available
- Program schedule and tapeout date from DEF-08
- DTCO findings as they emerge from TECH-03

PRODUCES

- Node maturity assessment per candidate
- Defect density trend and yield learning curve
- Roadmap alignment note against the program schedule
- Capacity ramp profile
- Node risk assessment with mitigation options

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D6

PRODUCES

Node risk assessment — maturity, defect density, yield learning curve. This activity is the deliverable, and it is what the yield assumption in the cost model should be built on.

TECH-D1

FEEDS

Technology selection report. Maturity is a selection criterion in its own right, separate from the PPA the node can deliver.

RISKS AND WATCH-OUTS

- ◆ **Marketing defect density used as a planning number.** The published curve is a target; the yield the program gets is the curve at its own tapeout date, which is a different point.
- ◆ **Risk production date taken as the availability date.** Risk production is when the node exists, not when it is ready for a large die at volume, and the gap can be two quarters.
- ◆ **Roadmap alignment checked once.** Foundry roadmaps move; an alignment done at selection and never revisited is a stale assumption by tapeout.
- ◆ **No mitigation options developed.** Identifying a maturity risk without a mitigation — a later slot, a shuttle, a conservative design rule set — leaves the program with a finding and no action.
- ◆ **Large-die yield inferred from small-die data.** Defect density scales against area brutally, and a curve measured on small test die flatters a 600 mm² part.

EFFORT SPLIT	1.5 M/M	ENTRY CRITERIA	FEEDS INTO
Maturity and roadmap review	0.5	- Short-list available from TECH-01 - NDA executed far enough to receive roadmap data - Program tapeout date estimated by DEF-08	TECH-08 Second-source and node-migration strategy assessment
Defect density and yield data	0.5		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)
Risk assessment and mitigations	0.5		DEF-07 Feasibility assessment against candidate technology nodes
Total	1.5		TC-01 Test chip objective definition and risk-item selection
WHO IT TAKES	~1.6 FTE	EXIT CRITERIA	
Technology strategist Owns the maturity and risk assessment		- Yield curve stated at the program's own tapeout date, not in general - Every maturity risk carries at least one mitigation option - Assessment handed to DEF-04 as the model's yield input	FAB-06 Inline metrology and defect inspection monitoring
Yield engineer Defect density interpretation and area scaling			MP-05 Yield learning, failure pareto and defect analysis
Foundry technical contact Roadmap, ramp and learning-curve data			
Program manager Alignment against the program schedule			
Product cost analyst Turns the yield curve into a cost assumption		DEPENDS ON	MEASURED BY
		TECH-01 Foundry and node long-list to short-list evaluation	- Risks identified with a mitigation against those without - Yield assumption against the cost model's requirement - Roadmap revisions after selection

TECH-08

Second-source and node-migration strategy assessment

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	1.5 M/M	~1.6 FTE	5	no	Technology strategist
w9–w13 in stage	planned	while it runs	2 in parallel	has float	Program manager approves

PURPOSE

Decide, deliberately, **whether the program accepts single-source risk** — and if it does not, what dual-sourcing or migration readiness would cost in area, schedule and IP.

Most leading-edge programs end up single-source, and that is a defensible answer. What is not defensible is arriving there by default. This activity prices the alternative so that the acceptance is a decision with a number attached rather than a shrug.

WORK FLOW

5 STEPS · 2 LANES

w9	w10	w11	w12	w13
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Second-source viability screen at the selected node class	1w	main
2	Migration cost and schedule estimate	1.3w	main
3	IP portability assessment across foundries	0.8w	parallel
4	Design rule and library delta analysis	0.8w	parallel
5	Strategy recommendation — dual-source, migration-ready, or single-source accepted	1.8w	main

Migration readiness is a design-time choice, not a later option: it constrains which IP can be used and how aggressively design rules can be pushed. Deciding after RTL starts means the decision has already been made by everyone who chose an IP.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Selected node and flavour from TECH-01 and TECH-02
- IP bill of materials direction from IPR-01
- Design rule sets for alternative foundries
- Migration experience from previous programs
- Supply-continuity findings from TECH-01

PRODUCES

- Second-source viability finding
- Migration cost and schedule estimate
- IP portability assessment
- Design rule and library delta analysis
- Sourcing strategy recommendation and decision record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D1

FEEDS

Technology selection report and decision record. Single-source acceptance belongs in the selection record, priced, rather than being implied by its absence.

TECH-D6

FEEDS

Node risk assessment. Supply concentration is a node risk, and its mitigation is either a second source or an accepted exposure.

RISKS AND WATCH-OUTS

- ◆ **Single-source accepted by default.** A decision nobody made is a decision nobody owns, and it surfaces during a supply disruption as a surprise.
- ◆ **Migration cost estimated without IP portability.** Foundry-specific memory compilers, IO and PHY are the expensive part of a migration, and they are usually the part left out of the estimate.
- ◆ **Migration readiness assumed to be free.** Designing to the intersection of two rule sets costs density and frequency on the node actually used.
- ◆ **Decision taken after IP selection.** Once IPR-03 has committed to foundry-specific IP, migration readiness is no longer available at the price this study quotes.
- ◆ **Second source screened on capability alone.** Capability without allocated capacity is not a second source; it is a phone number.

EFFORT SPLIT

1.5 M/M

ENTRY CRITERIA

DEPENDS ON

Migration cost and schedule estimate

0.5

- Node and flavour selected
- IP bill of materials direction available from IPR-01
- Design rule sets accessible for at least one alternative

TECH-01 Foundry and node long-list to short-list evaluation

Second-source viability screen

0.5

TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)

IP portability and rule delta

0.3

TECH-07 Foundry roadmap and risk-production timing alignment

Strategy recommendation

0.3

Total

1.5

EXIT CRITERIA

WHO IT TAKES

~1.6 FTE

Technology strategist

Owns the sourcing recommendation

IP strategist

Portability of the IP bill of materials

Physical design liaison

Design rule and library delta impact

Procurement lead

Capacity and commercial viability of an alternative

Program manager

Carries the accepted exposure into the risk register

- Migration cost estimated with IP portability included
- Recommendation recorded, including the option of accepting single source
- Accepted exposure entered on the program risk register with an owner

FEEDS INTO

IPR-03 Make / buy / reuse decision per IP block

IPR-10 IP maturity risk rating and second-source contingency

DEF-09 Business case preparation and funding approval package

MEASURED BY

- Migration cost estimated against the exposure accepted
- IP blocks assessed for portability
- Whether the sourcing decision is recorded or implied

TECH-09

OSAT and backend supply chain preliminary alignment

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w w9–w14 in stage	2.0 M/M planned	~1.7 FTE while it runs	7 3 in parallel	no has float	Backend supply chain lead Operations planner approves

PURPOSE

Open the **backend supply chain** early — OSAT capability and capacity, substrate and interposer suppliers, the foundry-to-OSAT logistics model, and who owns known-good die — so package design starts against real constraints.

On a 2.5D part the backend is not a downstream detail; it is a second supply chain with its own lead times, its own capacity constraints and its own qualification. Advanced packaging capacity in particular is allocated far ahead, and a program that starts asking at PKGD-10 is asking late.

WORK FLOW

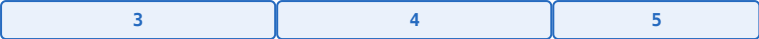
7 STEPS · 2 LANES

w9	w10	w11	w12	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	OSAT long-list against the package architecture assumption	0.8w	main
2	Advanced packaging capability and capacity screen	1.5w	main
3	Foundry-to-OSAT logistics and interface model	1w	parallel
4	Substrate and interposer supplier landscape	1w	parallel
5	Known-good-die handoff responsibility model	0.8w	parallel
6	Preliminary capacity indication and engagement	1.5w	main
7	Backend supply chain finding and handoff to package design	1.3w	main

Step 5 decides an argument before it happens. When a packaged unit fails, whether the die or the assembly is responsible is a contractual question, and settling it while nobody has lost money is far easier than settling it afterwards.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture assumption from the requirement register
- Selected foundry and node from TECH-01
- Volume and ramp profile from the business case
- OSAT and substrate supplier capability data
- Interposer technology options from the foundry

PRODUCES

- OSAT long-list with capability assessment
- Advanced packaging capacity indication
- Foundry-to-OSAT logistics model
- Substrate and interposer supplier landscape
- Known-good-die responsibility model
- Backend supply chain findings for package design

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TECH-D5	FEEDS	Capacity and tapeout slot commitment. Front-end capacity without matching backend capacity delivers wafers that cannot be assembled.
TECH-D3	INFORMS	Wafer, mask and NRE cost sheet. Interposer and assembly pricing is the backend half of the technology cost, and the cost model needs both.
TECH-D1	FEEDS	Technology selection report. On a 2.5D part the choice of foundry constrains the interposer options, so backend capability belongs in the selection record.

RISKS AND WATCH-OUTS

- ◆ **Advanced packaging capacity assumed available.** CoWoS-class capacity is allocated quarters ahead and to a small number of customers; assuming access is the single most common backend planning error.
- ◆ **OSAT chosen after package design starts.** Assembly process capability constrains the package design; choosing the OSAT afterwards means redesigning to fit.
- ◆ **Known-good-die responsibility left open.** The first failed assembly lot is the wrong moment to discover that nobody agreed who owns the die.
- ◆ **Substrate lead time discovered rather than planned.** Twenty-week substrate lead times set the order date, and the order date is derived from a wafer-out date nobody has committed yet.
- ◆ **Logistics model ignored.** Wafers crossing borders between fab, sort and assembly carry customs, export control and transit time that the schedule rarely shows.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Capability and capacity screen	0.8	● Package architecture assumption stated in the requirement register	PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count
OSAT and supplier engagement	0.5	● Foundry selected, so interposer options are known	PKGD-09 Substrate and interposer supplier selection and lead-time booking
Logistics and supplier landscape	0.5	● Volume and ramp profile available	
Handoff to package design	0.3		
Total	2.0		
WHO IT TAKES	~1.7 FTE	EXIT CRITERIA	
Backend supply chain lead Owns the OSAT and substrate engagement		● Advanced packaging capacity indicated, not assumed ● Known-good-die responsibility model agreed in principle ● Findings handed to PKGD before package architecture is chosen	PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress
Package architect States the capability the package will need			ASSY-01 Known-good-die sort and selection
Procurement lead Capacity indication and commercial engagement			
Quality liaison Known-good-die responsibility and failure ownership			
Operations planner Logistics, transit and customs model			
		DEPENDS ON	MEASURED BY
		TECH-01 Foundry and node long-list to short-list evaluation	● Advanced packaging capacity indicated against the ramp requirement
		DEF-01 Market and customer requirements consolidation	● Weeks between backend engagement and package architecture selection ● Substrate lead time against the wafer-out date

04

PDK & Design Enablement

PDK

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w8 – w44	36 _w	13	97 M/M	7 of 13		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
PDK-01	PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis		34w cont.	14 M/M	PDK-D1	PDK readiness dashboard — version, release date, open gap list
PDK-02	Design rule manual review; restricted and recommended rule disposition		6w	5 M/M	PDK-D2	Qualified library list with .lib / LEF / GDS views
PDK-03	Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)		8w	9 M/M	PDK-D3	Memory PPA gap analysis and custom-instance decision record
PDK-04	Memory compiler evaluation and instance generation (SRAM, register file)		9w	10 M/M	PDK-D4	EDA tool and version matrix (qualified and frozen)
PDK-05	Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage		6w	7 M/M	PDK-D5	Internal reference flow and methodology guide
PDK-06	Custom / pushed-rule memory instance decision, scope and schedule impact assessment		4w	3 M/M	PDK-D6	Signoff corner definition agreed with foundry
PDK-07	IO, ESD and latch-up library qualification		6w	5 M/M	PDK-D7	Compute and license capacity plan
PDK-08	EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR		10w	12 M/M	PDK-D8	Golden environment release notes
PDK-09	Foundry reference flow bring-up and internal methodology deck		10w	12 M/M		
PDK-10	Rule deck, QRC tech file and DRC/LVS deck version control		6w cont.	4 M/M		
PDK-11	Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)		6w	5 M/M		
PDK-12	Compute farm, EDA license and storage capacity planning		6w	3 M/M		
PDK-13	Golden design environment release and flow regression		8w	8 M/M		

PDK-01

PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
34w cont. w0-w34 in stage, runs throughout	14.0 M/M planned	~1.8 FTE while it runs	8 3 in parallel	yes no float	PDK owner Physical design liaison approves

PURPOSE

Track what the foundry actually ships, version by version, and turn each release into **a delta the program can act on** — what changed, what it costs, and whether anything already done has to be redone.

A PDK is not a thing you receive; it is a sequence. Design starts on 0.5 because waiting for production would cost two quarters, and the 0.5-to-1.0 delta then arrives in the middle of physical design. Whether that delta is a two-week re-characterisation or a reopened timing signoff depends entirely on whether anyone was watching.

WORK FLOW

8 STEPS · 2 LANES

w0	w7	w14	w21	w28	w34
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Release calendar and version tracking framework	1w	main
2	0.1 to 0.5 delta impact analysis	4w	main
3	Gap list maintained against what the design actually needs	3w	parallel
4	0.5 to 1.0 delta impact analysis	5w	main
5	Library and deck re-qualification per version	6w	parallel
6	Production PDK adoption and version freeze decision	4w	main
7	Version freeze policy and exception handling	3w	parallel
8	Readiness dashboard maintenance across the stage	20w	main

Step 8 runs for twenty of the thirty-four weeks and is the activity most likely to be dropped when the team is busy. It is also the only artifact that tells the program which of its assumptions have quietly expired.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Foundry PDK release calendar and notes
- Selected node and flavour sheet from TECH-01 and TECH-02
- Design agreement access from TECH-06
- Library and deck versions in use across the teams

PRODUCES

- PDK readiness dashboard, maintained
- Delta impact analysis per version transition
- Open gap list against the design's requirements
- Re-qualification record per library and deck

- Design status — what has been characterised against which version
- Version freeze decision and exception log

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D1	PRODUCES	PDK readiness dashboard. This activity is the deliverable, and it is a living document rather than a report — its value is entirely in being current.
PDK-D2	FEEDS	Qualified library list. Every version transition potentially invalidates a qualification, and this activity is what says which ones.
PDK-D8	GATES	Golden environment release notes. The environment cannot be released against a version that has not been assessed and frozen.

RISKS AND WATCH-OUTS

- ◆ **Design starts on a version the foundry calls preliminary and nobody re-checks.** The gap between 0.5 and production has reopened timing signoff on more than one program.
- ◆ **Delta analysis done on the release notes alone.** Foundry notes describe what changed; only re-running the design's own characterisation says what it costs here.
- ◆ **Teams on different versions.** Synthesis on one library revision and physical design on another produces a mismatch that looks like a tool bug for weeks.
- ◆ **Freeze declared without an exception path.** A hard freeze with no route for a critical fix gets broken quietly, which is worse than breaking it openly.
- ◆ **Dashboard maintained until the team gets busy.** A stale readiness dashboard is more dangerous than none, because it is still trusted.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Dashboard maintenance	4.0	Design agreement executed so PDK access is legal	PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)
Delta impact analyses	5.0	Node and flavour selected	PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control
Gap list against design needs	2.0	Release calendar obtained from the foundry	PDK-13 Golden design environment release and flow regression
Re-qualification coordination	2.0		SYN-01 SDC constraint development and validation
Freeze policy and exceptions	1.0		PD-01 Flow setup and MMMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity
Total	14.0	EXIT CRITERIA - Every version transition assessed before adoption, not after - All teams on one recorded version, with exceptions logged - Production PDK adopted and frozen with an exception path defined	S0-10 Waiver review board and foundry waiver alignment
WHO IT TAKES	~1.8 FTE	DEPENDS ON TECH-01 Foundry and node long-list to short-list evaluation TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders) TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame	MEASURED BY - Days between a foundry release and its delta assessment - Teams on the recorded version against the total - Signoff reopened by a PDK change
PDK owner Owns the dashboard, the deltas and the freeze decision			
CAD methodology engineer Re-qualification and deck version alignment			
Library engineer Library re-characterisation per version			
Foundry technical contact Release calendar, notes and interpretation			
Physical design liaison States what a delta costs in the flow			

PDK-02

Design rule manual review; restricted and recommended rule disposition

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	5.0 M/M	~3.6 FTE	6	no	CAD methodology lead
w1-w7 in stage	planned	while it runs	2 in parallel	has float	Foundry technical contact approves

PURPOSE

Read the design rule manual as a **constraint on this design** rather than as a document — identify which restricted rules bite, dispose of every recommended rule, and get the exceptions requested early.

Restricted and recommended rules are where a node's real density lives. A design that adopts every recommended rule is safe and large; one that ignores them is dense and fails DFM. The disposition is a series of deliberate trades, and making them here is far cheaper than discovering them as violations at physical verification.

WORK FLOW

6 STEPS · 2 LANES

w1	w3	w5	w7
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	DRM ingestion and change log against the previous node	1w	main
2	Restricted rule identification and impact assessment	1.5w	main
3	Recommended rule disposition — adopt, waive or defer	1.5w	main
4	Density and spacing rule impact on the floorplan	1w	parallel
5	Rule exception request preparation for the foundry	1w	parallel
6	DRM review record and distribution to the design teams	2w	main

Exception requests in step 5 run in parallel because the foundry's response time is measured in weeks. Submitting them at the end of the review rather than as each one is identified adds a month to the answer.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Design rule manual for the selected node and flavour
- Previous-node DRM for comparison
- Floorplan intent from ARCH-08
- Custom and analog design intent from AMS-01
- Foundry exception request process

PRODUCES

- Rule change log against the previous node
- Restricted rule impact assessment
- Recommended rule disposition record
- Exception requests submitted to the foundry
- DRM review record for the design teams

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D1	FEEDS	PDK readiness dashboard. An outstanding exception request is a readiness gap, and it belongs on the dashboard with a date.
PDK-D5	FEEDS	Internal reference flow and methodology guide. Rule dispositions become flow settings — what the tools are told to enforce and what they are told to allow.

RISKS AND WATCH-OUTS

- ◆ **Recommended rules adopted wholesale.** Safe, and it costs density the product was sized against; the trade should be made per rule, not once.
- ◆ **Restricted rules discovered at physical verification.** A restricted rule that bites is a layout change, and finding it at S0-02 means finding it after routing closed.
- ◆ **Exception requests batched to the end.** Foundry turnaround runs to weeks per request; batching them converts a parallel wait into a serial one.
- ◆ **Analog rules read only by the digital team.** Pushed-rule and analog-specific sections matter to AMS-07, and they are the sections a digital-led review skips.
- ◆ **Disposition never distributed.** A decision the layout engineers never saw is a decision that does not exist.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Rule impact assessment	2.0	● DRM released for the selected node and flavour	PDK-09 Foundry reference flow bring-up and internal methodology deck
Disposition decisions	1.5	● Floorplan intent available from ARCH-08	PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control
Exception preparation	0.8	● Exception request process agreed with the foundry	AMS-07 Custom array layout and pushed-rule DRC closure with foundry approval
Review record and distribution	0.8		PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
Total	5.0		S0-06 Chip-package-system co-analysis signoff review against SIPI results
WHO IT TAKES	~3.6 FTE	EXIT CRITERIA	MEASURED BY
CAD methodology lead Owns the disposition and the review record		● Every recommended rule dispositioned with a reason, not defaulted	● Recommended rules dispositioned against those in the DRM
Physical design representative Floorplan and routing impact of each rule		● Exception requests submitted, with expected response dates	● Exception requests answered before physical design starts
Custom layout representative Analog and pushed-rule sections		● Disposition distributed to digital and custom teams alike	● Rule violations at signoff traceable to a missed disposition
DFM engineer Density, spacing and manufacturability trades			
Foundry technical contact Exception requests and interpretation		TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)	
		PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	

PDK-03

Standard cell library selection and qualification
(track height, multi-Vt, multi-bit flops)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	9.0 M/M	~4.9 FTE	7	yes	Library lead
w4-w12 in stage	planned	while it runs	3 in parallel	no float	Foundry library contact approves

PURPOSE

Choose the standard cell libraries the design will be built from and **qualify them rather than assume them** — track height, Vt menu, multi-bit flops, and the timing models every downstream tool will trust.

Library selection is a PPA decision disguised as a procurement one. Track height sets density and drive strength; the Vt menu sets leakage; multi-bit flops set clock power. And a timing model that is wrong at one corner produces a design that closes in the tool and fails in silicon, which is the failure mode nobody catches until bring-up.

WORK FLOW

7 STEPS · 2 LANES

w4	w6	w8	w10	w12
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	Track height and library family selection	1w	main
2	Multi-Vt menu qualification against the flavour sheet	1.5w	main
3	Multi-bit flop and special cell evaluation	1.5w	parallel
4	Timing model quality check across corners	2w	main
5	Physical view and abstract consistency check	1.5w	parallel
6	Library re-characterisation where the foundry data is thin	2w	parallel
7	Qualification record and library freeze	3.5w	main

Step 4 is the step that earns the activity. Foundry timing models are qualified by the foundry against the foundry's test cases; checking them at this design's corners and slew ranges is what catches the model error before it becomes a silicon failure.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Flavour sheet from TECH-02
- DTCO findings from TECH-03
- PDK release under tracking from PDK-01
- PPA targets and clock power budget from ARCH-06
- Foundry library release notes and characterisation data

PRODUCES

- Library family and track height selection record
- Qualified Vt menu with leakage projections
- Multi-bit and special cell evaluation
- Timing model quality report across corners
- Library qualification record and freeze

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D2	PRODUCES	Qualified library list with .lib / LEF / GDS views. The standard cell half of the deliverable; PDK-04 adds memory and PDK-07 adds IO.
PDK-D1	FEEDS	PDK readiness dashboard. An unqualified library is a readiness gap that blocks synthesis, and it shows on the dashboard as one.
PDK-D6	FEEDS	Signoff corner definition. The corners a library is characterised at bound the corners signoff can be run at.

RISKS AND WATCH-OUTS

- ◆ **Timing models trusted without a quality check.** A model error at one corner produces a design that closes in the tool and fails on silicon — the most expensive class of library defect.
- ◆ **Track height chosen from the previous program.** Density and drive strength trade differently at each node; inheriting the choice inherits an answer to a question nobody asked here.
- ◆ **Vt menu wider than the flow can maintain.** Every Vt is a characterisation, a qualification and a maintenance burden across every version transition.
- ◆ **Multi-bit flops evaluated after synthesis starts.** Clock power is largely decided by flop banking, and retrofitting it means re-running synthesis and re-closing timing.
- ◆ **LEF and GDS inconsistency.** An abstract that disagrees with the layout produces DRC violations at the very end, after routing has been built on the abstract.

EFFORT SPLIT	9 M/M	ENTRY CRITERIA	FEEDS INTO
Library re-characterisation	3.0	● Flavour sheet agreed by TECH-02	PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)
Timing model quality checks	2.0	● Library release available under the design agreement	PDK-13 Golden design environment release and flow regression
Vt menu qualification	1.5	● PPA and clock power budget available from ARCH-06	ARCH-01 System-level performance modelling and workload simulation
Special cell evaluation	1.0		
Qualification record and freeze	1.5		
Total	9.0	EXIT CRITERIA ● Timing models checked at this design's corners, not only the foundry's ● Vt menu frozen with a maintenance owner per corner ● Abstract and layout views confirmed consistent	ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft
WHO IT TAKES	~4.9 FTE	DEPENDS ON	SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
Library lead Owns the selection, the qualification and the freeze		TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)	PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Characterisation engineer Timing model checks and re-characterisation		TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	
Synthesis liaison Vt menu and multi-bit flop implications		PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	MEASURED BY ● Corners with a validated timing model ● Library defects found downstream against those found here ● Vt menu size against the flow's maintenance capacity
Physical design liaison Track height, abstracts and routing resource			
Foundry library contact Release data, known issues and fixes			

PDK-04

Memory compiler evaluation and instance generation (SRAM, register file)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	10.0 M/M	~4.8 FTE	7	yes	Memory library engineer
w6-w15 in stage	planned	while it runs	2 in parallel	no float	Foundry memory contact approves

PURPOSE

Get the memory compiler licensed, evaluate what it can actually produce, and **generate the instances the design asked for** — with every view a downstream tool needs.

On an inference accelerator the memory array is a large fraction of the die, and the compiler decides most of it. What the compiler can generate is a hard boundary on the architecture: a configuration outside its range is not a compiler setting, it is a custom design programme in AMS-05 costing four months.

WORK FLOW

7 STEPS · 2 LANES

w6w8w10w12w14w15

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Memory requirement collection from the hierarchy definition	0.5w	main
2	Compiler licence and access setup	1w	main
3	Compiler capability screen against the requirement list	1.5w	main
4	Instance generation across the required configurations	2.5w	main
5	Redundancy and repair option evaluation	1.5w	parallel
6	Compiler view completeness check — .lib, LEF, GDS, CDL, models	1.5w	parallel
7	Instance library assembly and release	3.5w	main

Step 3 is the one with a schedule consequence. A capability gap found here opens the <code>PDK-06</code> decision with fifteen weeks of runway; found at instance generation, it opens the same decision with three.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Memory hierarchy and capacity requirements from ARCH-03
- Compiler licence terms from TECH-06
- PDK version under tracking from PDK-01
- BISR and repair architecture intent from DFT-03
- Block memory budgets from ARCH-06

PRODUCES

- Memory requirement list per block
- Compiler capability screen and gap list
- Generated instances across required configurations
- Redundancy and repair option evaluation
- Instance library with complete views

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D2	PRODUCES	Qualified library list. The memory half of the deliverable — instances with every view a downstream tool consumes.
PDK-D3	FEEDS	Memory PPA gap analysis and custom-instance decision record. The capability screen is where the gap first appears; PDK-05 quantifies it.
PDK-D1	FEEDS	PDK readiness dashboard. Compiler access and instance availability are tracked readiness items in their own right.

RISKS AND WATCH-OUTS

- ◆ **Compiler licence assumed to be covered by the PDK agreement.** Memory compilers are licensed separately, and discovering that here stops the activity for weeks.
- ◆ **Capability screened after generation starts.** The gap is the same size either way; what differs is how much runway the custom path has.
- ◆ **Views generated but never checked.** A missing or inconsistent view surfaces at PD-02 when the macro cannot be placed, and the compiler run has to be repeated.
- ◆ **Redundancy chosen without DFT.** Repair schemes have to match the BISR architecture in DFT-03, and a mismatch is found at pattern generation.
- ◆ **Instance count explosion.** Generating every configuration a block owner asks for produces a library nobody can characterise or maintain; consolidation is cheaper here than later.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Instance generation	3.5	Memory requirements available from ARCH-03	PDK-05 Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage
Capability screen	2.0	Compiler licence executed under TECH-06	PDK-13 Golden design environment release and flow regression
View completeness checks	1.5	PDK version stable enough to generate against	AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing
Redundancy option evaluation	1.5		DFT-03 MBIST / BIRA / BISR architecture for embedded memories
Library assembly and release	1.5		RTL-03 Third-party and internal IP integration — wrappers, glue, configuration
Total	10.0	EXIT CRITERIA - Every required configuration either generated or recorded as a gap - All views present and checked, not merely produced - Redundancy scheme aligned with the DFT architecture	PD-02 Floorplan, macro placement and partition definition
WHO IT TAKES	~4.8 FTE	DEPENDS ON ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels) TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	MEASURED BY - Configurations generated against those requested - Views complete against the tool requirement list - Instance count against what the flow can maintain
Memory library engineer Owns compiler access, generation and the instance library			
Memory architect liaison Requirement list and configuration consolidation			
DFT liaison Redundancy and repair option alignment			
CAD methodology engineer View completeness and tool consumption			
Foundry memory contact Compiler capability, known issues and updates			

PDK-05

Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w14–w20 in stage	7.0 M/M planned	~5.1 FTE while it runs	6 2 in parallel	yes no float	Memory characterisation engineer Block owners approves

PURPOSE

Measure what the generated instances actually deliver — **density, Vmin, access time, leakage** — and set it against the budget each block was allocated.

This is the activity that decides whether the custom memory path opens. The compiler produces an instance that meets the specification; the question here is whether it meets the *budget*, and the answer is frequently no by a margin large enough to matter but small enough to be argued about.

WORK FLOW

6 STEPS · 2 LANES

w14

w16

w18

w20

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Block memory budget extraction from ARCH-06	0.5w	main
2	Density measurement per generated instance	1.5w	main
3	Vmin and access time characterisation	1.5w	main
4	Leakage and dynamic power measurement	1w	parallel
5	Area efficiency against the block budget	1w	parallel
6	Gap analysis and shortfall quantification	2.5w	main

The gap analysis is quantified rather than described on purpose. "The compiler misses the budget" starts an argument; "the L2 array misses its density budget by 14%, which is 3.1 mm² of die" starts a decision.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Generated instances from PDK-04
- Block memory budgets from ARCH-06
- Vmin and operating voltage requirements from ARCH-05
- Access time requirements from the hierarchy specification
- DTCO SRAM findings from TECH-03

PRODUCES

- Density measurement per instance
- Vmin and access time characterisation results
- Leakage and dynamic power measurements
- Area efficiency comparison against budget
- Quantified gap analysis per block

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-03

PRODUCES

Memory PPA gap analysis and custom-instance decision record. The gap analysis half — PDK-06 adds the decision that follows from it.

PDK-02

FEEDS

Qualified library list. An instance that fails characterisation does not belong on the qualified list, whatever the compiler says.

RISKS AND WATCH-OUTS

- ◆ **Density measured on the datasheet rather than the instance.** Compiler datasheets quote best-case configurations; the instance the design needs is rarely one of them.
- ◆ **Vmin characterised at nominal only.** Vmin is a distribution, and the number that matters is at the corner and the sigma the product is sold at.
- ◆ **Gap described rather than quantified.** A shortfall without a number in square millimetres cannot be traded against schedule in PDK-06.
- ◆ **Measurement late enough to remove the option.** Characterisation completing after the AMS window has closed makes the custom path unavailable regardless of the gap.
- ◆ **Access time measured without the wire load.** An instance that meets access time standalone and misses it in place has not been characterised usefully.

EFFORT SPLIT	7 M/M	ENTRY CRITERIA	FEEDS INTO
Vmin and access time characterisation	3.0	Instances generated and released by PDK-04	PDK-06 Custom / pushed-rule memory instance decision, scope and schedule impact assessment
Density measurement	1.5	Block budgets published by ARCH-06	AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing
Gap analysis	1.5	Characterisation environment available	
Power measurement	1.0		
Total	7.0		
WHO IT TAKES	~5.1 FTE	EXIT CRITERIA	
Memory characterisation engineer Owns the measurements and the gap analysis		- Every instance measured against its block's budget - Gaps quantified in area, power and Vmin, not described - Results delivered with enough runway for PDK-06 to act	AMS-08 Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation
Memory library engineer Instance generation and configuration control			ARCH-06 PPA budget allocation per block
PPA lead Budget comparison and shortfall quantification			PD-02 Floorplan, macro placement and partition definition
Circuit design liaison Vmin interpretation and assist implications		DEPENDS ON	MEASURED BY
Block owners Confirm the budget and accept or contest the gap		PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)	Instances characterised against those released
		ARCH-06 PPA budget allocation per block	- Gap quantified in die area - Weeks of runway left for the custom decision

PDK-06

Custom / pushed-rule memory instance decision, scope and schedule impact assessment

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	6	yes	PDK owner
w19–w23 in stage	planned	while it runs	2 in parallel	no float	Chief architect approves

PURPOSE

Decide what to do about the memory shortfall — **push rules, build a custom array, or change the architecture** — and hand the chosen scope to AMS with a schedule attached.

This is a gate, not a study. The compiler instance missed the budget; the options each cost something different and one of them has to be taken. Deferring the decision does not preserve the options — the custom array path needs its full window in AMS-05 through AMS-10, and every week of deferral removes one from the end.

WORK FLOW

6 STEPS · 2 LANES

w19	w20	w21	w22	w23
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Shortfall review against the block budgets	0.5w	main
2	Option definition — pushed rules, custom array, or architecture change	1w	main
3	Schedule impact assessment against the AMS window	1w	main
4	Foundry appetite for pushed rules and the approval path	0.8w	parallel
5	Cost and risk comparison of the options	0.8w	parallel
6	Decision and scope handoff to AMS	1.5w	main

Step 4 runs in parallel and can veto an option outright. A pushed-rule array the foundry will not approve is not an option, and finding that after the decision is taken costs the whole custom window.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Quantified gap analysis from PDK-05
- AMS capacity and window from the program schedule
- Foundry pushed-rule policy and approval process
- Architecture flexibility from ARCH-03
- Cost impact of the die area at stake from DEF-04

PRODUCES

- Option definition with cost, schedule and risk per path
- Foundry pushed-rule appetite finding
- Schedule impact assessment against the AMS window
- Decision record with the alternatives rejected
- Scope handoff package to AMS

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D3

PRODUCES

Memory PPA gap analysis and custom-instance decision record. The decision half of the deliverable, and the record downstream stages will point at.

PDK-D1

GATES

PDK readiness dashboard. An undecided memory gap is the most consequential open item the dashboard can carry — it holds both library closure and the AMS plan.

RISKS AND WATCH-OUTS

- ◆ **Decision deferred to preserve options.** Deferral removes the custom option rather than preserving it, because the window is fixed at the far end by physical design.
- ◆ **Pushed rules chosen without foundry commitment.** Approval is a process with its own duration, and an unapproved pushed-rule array is a layout nobody can tape out.
- ◆ **Architecture change dismissed too quickly.** Reducing the array is sometimes cheaper than building it; the option deserves the same costing as the other two.
- ◆ **Scope handed to AMS without a schedule.** A custom instance request with no date is a request that competes with the SerDes for the same engineers and loses.
- ◆ **Decision taken without the block owner.** The owner who accepted the budget in ARCH-06 is the one who has to accept the concession here.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Option definition	1.0	● Gap quantified by PDK-05	AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing
Schedule impact assessment	0.8	● AMS window and capacity known from the program plan	
Foundry approval path	0.5	● Foundry pushed-rule policy understood	AMS-06 Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse
Decision record and handoff	0.8		AMS-07 Custom array layout and pushed-rule DRC closure with foundry approval
Total	3.0	EXIT CRITERIA	ARCH-06 PPA budget allocation per block
WHO IT TAKES	~3.3 FTE	● Decision taken and recorded with the rejected options priced	PD-02 Floorplan, macro placement and partition definition
PDK owner Owns the decision and its record		● Foundry approval path confirmed if pushed rules are chosen	
Memory circuit lead Custom array scope and feasibility		● Scope handed to AMS with a date, not only a requirement	
Program manager Schedule impact and AMS capacity		DEPENDS ON	MEASURED BY
Foundry technical contact Pushed-rule appetite and approval path		PDK-05 Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage	● Weeks from gap quantification to decision
Chief architect Arbitrates when the option is an architecture change		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	● Options costed against options considered
		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)	● AMS window remaining at handoff

PDK-07

IO, ESD and latch-up library qualification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	5.0 M/M	~3.6 FTE	6	no	IO library engineer
w8–w14 in stage	planned	while it runs	2 in parallel	has float	SI liaison approves

PURPOSE

Qualify the **IO, ESD and latch-up** collateral — the library that sits at the die edge and decides whether the part survives handling, and whether its interfaces meet their electrical specifications.

IO qualification is unglamorous and it is where parts die. An ESD strategy chosen late constrains the pad ring after the bump map is fixed; a latch-up rule missed produces a part that fails qualification for reasons no functional test would ever find.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	IO requirement collection from the interface and bump plans	0.5w	main
2	IO library selection against the interface list	1.5w	main
3	ESD strategy and clamp selection	1.5w	main
4	Latch-up rule review and guard ring strategy	1w	parallel
5	IO timing and signal integrity model quality check	1w	parallel
6	Qualification record and release	2.5w	main

The ESD strategy in step 3 has to precede pad ring layout, which in turn precedes the bump map freeze in `ARCH-08`. Late ESD is a pad ring change, and a pad ring change is a bump map change.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Interface list from ARCH-04
- Bump and pad ring plan from ARCH-08
- Foundry IO library and ESD rule set
- Latch-up and reliability requirements from the qualification plan
- Package and board ESD requirements

PRODUCES

- IO library selection per interface
- ESD strategy and clamp selection record
- Latch-up and guard ring strategy
- IO timing and SI model quality report
- IO qualification record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D2	PRODUCES	Qualified library list. The IO and ESD half of the deliverable — the cells at the die edge, qualified rather than assumed.
PDK-D6	FEEDS	Signoff corner definition. ESD and latch-up verification conditions are part of what signoff has to run, and they are defined here.

RISKS AND WATCH-OUTS

- ◆ **ESD strategy chosen after the pad ring.** Clamp placement and rail topology constrain the ring; retrofitting means redoing the ring and the bump map with it.
- ◆ **IO models unchecked.** IO timing models feed both STA and the channel simulation in SIPI-07; an error propagates into two signoffs at once.
- ◆ **Latch-up rules read as a formality.** Guard ring strategy affects area at the die edge, and a late change competes with the interfaces for the same space.
- ◆ **Package and board ESD requirements ignored.** The part's ESD rating is a system property; qualifying only the die produces a rating the system cannot use.
- ◆ **Library selected per interface with no consolidation.** Several IO families in one ring multiplies the qualification and the pad ring complexity for little gain.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
IO library selection	1.5	● Interface list available from ARCH-04	PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)
ESD strategy	1.5	● Pad ring and bump plan drafted in ARCH-08	PD-01 Flow setup and MMMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity
Model quality checks	1.0	● Foundry IO library and ESD rules released	SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring
Qualification record	1.0		SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
Total	5.0		S0-07 Reliability verification — ESD, latch-up, soft error / FIT
WHO IT TAKES	~3.6 FTE	EXIT CRITERIA	MP-03 ESD (HBM, CDM) and latch-up qualification
IO library engineer Owns the selection and the qualification record		● ESD strategy fixed before the pad ring is committed	
ESD engineer Strategy, clamps and rail topology		● IO models validated for both timing and channel use	
Reliability engineer Latch-up and guard ring requirements		● Latch-up strategy reviewed against the qualification plan	
Physical design liaison Pad ring and area implications			
SI liaison IO model quality for channel and timing use			
		DEPENDS ON	MEASURED BY
		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	● IO families in the ring against those qualified
		ARCH-08 Chip-level floorplan intent, pin and bump budget	● Model errors found downstream
		PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	● ESD rating achieved against the system requirement

PDK-08

EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	8	no	CAD tools lead
w10–w20 in stage	planned	while it runs	3 in parallel	has float	IT infrastructure approves

PURPOSE

Decide which tool at which version each flow step runs on, **qualify that combination on the node**, and freeze it — so that a result from one team means the same thing as a result from another.

Tool versions are a correctness question, not an IT one. A node this advanced needs certified versions for extraction, timing and physical verification, and mixing versions across teams produces disagreements that consume weeks before anyone suspects the tools rather than the design.

WORK FLOW

8 STEPS · 2 LANES

w10	w12	w14	w16	w18	w20
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Tool and version matrix definition per flow step	1w	main
2	Vendor node certification status review	1w	main
3	Synthesis and place-and-route version qualification	2.5w	main
4	STA and signoff tool qualification	2w	parallel
5	Physical verification and EM/IR tool qualification	2w	parallel
6	Interoperability and database version check across tools	1.5w	parallel
7	Version freeze and upgrade exception policy	4w	main
8	Tool matrix publication to every team	1.5w	main

Step 6 catches what per-tool qualification cannot. Each tool can be certified on the node and still disagree with the next one about a database format or a parasitic convention, and only a cross-tool check finds it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Selected node and flavour from TECH
- Vendor certification matrices for the node
- Foundry reference flow tool requirements
- Existing internal tool estate and licences

PRODUCES

- Tool and version matrix per flow step
- Vendor certification status record
- Qualification results per tool
- Interoperability check findings

- Flow step definitions from PDK-09
- Version freeze and upgrade exception policy

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D4	PRODUCES	EDA tool and version matrix (qualified and frozen). This activity is the deliverable, and the freeze is the part that matters.
PDK-D5	FEEDS	Internal reference flow and methodology guide. The flow is written against specific tool versions; the guide and the matrix are read together.
PDK-D7	FEEDS	Compute and license capacity plan. Which tools and how many concurrent seats is what the licence plan is built from.

RISKS AND WATCH-OUTS

- ◆ **Teams on different versions.** Two teams with different STA versions will disagree about timing, and the disagreement is diagnosed as a design problem for weeks first.
- ◆ **Certification status assumed.** Vendor node certification lags node release; an uncertified version produces results the foundry will not accept at signoff.
- ◆ **Interoperability never checked.** Per-tool qualification passes and the handoff between them fails, usually on parasitic or database conventions.
- ◆ **Freeze with no upgrade path.** A critical tool fix will arrive mid-program, and a policy that has no route to accept it will be broken informally.
- ◆ **Matrix published once.** A matrix nobody re-reads after month three is a matrix that describes an estate nobody is running.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Synthesis and P&R qualification	3.0	● Node and flavour selected	PDK-09 Foundry reference flow bring-up and internal methodology deck
STA and signoff qualification	2.5	● Vendor certification matrices available	PDK-12 Compute farm, EDA license and storage capacity planning
Physical verification and EM/IR qualification	2.5	● Foundry reference flow requirements known	PDK-13 Golden design environment release and flow regression
Interoperability checks	2.0	EXIT CRITERIA ● Every flow step has a qualified, named tool version ● Cross-tool interoperability checked, not assumed ● Freeze published with an exception route for critical fixes	SYN-01 SDC constraint development and validation
Matrix and freeze policy	2.0		PD-01 Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
Total	12.0	DEPENDS ON TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders) PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
WHO IT TAKES	~5.2 FTE		MEASURED BY ● Flow steps with a qualified version against the total ● Cross-tool defects found here against those found downstream ● Version exceptions granted against the policy
CAD tools lead Owns the matrix, the qualification and the freeze			
Flow engineers Per-tool qualification on the node			
Signoff methodology engineer STA and physical verification qualification			
EDA vendor account contacts Certification status and fixes			
IT infrastructure Deployment and version enforcement			

PDK-09

Foundry reference flow bring-up and internal methodology deck

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	8	yes	Methodology lead
w12–w22 in stage	planned	while it runs	3 in parallel	no float	Foundry flow contact approves

PURPOSE

Take the foundry's reference flow, **adapt it into a methodology this design can actually run**, and prove it on a representative block before the whole team depends on it.

A reference flow is a starting point written for a generic design. This one has hierarchical partitions, hard macros, multiple power domains and a memory subsystem the reference never contemplated. The pilot in the last step is what turns an adapted flow into a trusted one.

WORK FLOW

8 STEPS · 2 LANES

w12	w14	w16	w18	w20	w22
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Foundry reference flow ingestion	1w	main
2	Gap analysis against internal methodology and design needs	1.5w	main
3	Synthesis and place-and-route flow adaptation	2.5w	main
4	Signoff flow adaptation and deck integration	2w	parallel
5	Low-power flow integration against the UPF intent	1.5w	parallel
6	Hierarchical and block-level flow definition	2w	parallel
7	Methodology deck authoring	2w	main
8	Flow pilot on a representative block	3w	main

Three of the ten weeks are the pilot, and it is the step under most pressure to be skipped. A flow that has never been run end to end on a real block is a flow whose first user is doing the debugging on the critical path.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Foundry reference flow and scripts
- Tool matrix from PDK-08
- Rule and extraction decks from PDK-10
- UPF intent from ARCH-05
- Hierarchical partitioning from ARCH-02

PRODUCES

- Adapted synthesis, P&R and signoff flows
- Low-power flow integration
- Hierarchical and block-level flow definition
- Internal methodology deck
- Pilot results on a representative block

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D5	PRODUCES	Internal reference flow and methodology guide. This activity is the deliverable, and the pilot is what makes it credible.
PDK-D8	FEEDS	Golden environment release notes. The environment in PDK-13 packages this flow; the two are released together.
PDK-D1	FEEDS	PDK readiness dashboard. Flow readiness is tracked alongside library and deck readiness — a design cannot start on any one of them alone.

RISKS AND WATCH-OUTS

- ◆ **Pilot skipped under schedule pressure.** The flow's first real user then debugs it while on the critical path, which is the most expensive place to do it.
- ◆ **Reference flow adopted rather than adapted.** A generic flow on a hierarchical design with hard macros produces results nobody can close.
- ◆ **Low-power flow bolted on late.** UPF handling touches synthesis, placement and verification; adding it after the flow is built means rebuilding it.
- ◆ **Methodology deck written after the flow changes.** A guide that lags the scripts teaches the wrong flow to every engineer who joins.
- ◆ **Pilot block unrepresentative.** A small clean block pilots nothing; the pilot has to include a macro, a domain crossing and real congestion.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Flow adaptation	4.0	● Tool matrix qualified by PDK-08	PDK-13 Golden design environment release and flow regression
Pilot on a representative block	3.0	● Decks under version control from PDK-10	SYN-01 SDC constraint development and validation
Signoff flow and deck integration	2.0	● UPF intent available from ARCH-05	PD-01 Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
Methodology deck	1.5		
Low-power flow integration	1.5		
Total	12.0		
WHO IT TAKES	~5.2 FTE		
Methodology lead Owns the flow, the deck and the pilot		EXIT CRITERIA ● Flow run end to end on a representative block, not a toy one ● Methodology deck matches the scripts as released ● Hierarchical and block flows both defined, not only the top	S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
Flow engineers Synthesis, P&R and signoff adaptation			DFT-06 Scan insertion and DFT DRC during synthesis
Low-power methodology engineer UPF flow integration		DEPENDS ON	MEASURED BY ● Pilot blocks run end to end ● Flow defects found in the pilot against those found in production use ● Deck and script version agreement
Physical design pilot engineer Runs the pilot on a real block		PDK-02 Design rule manual review; restricted and recommended rule disposition	
Foundry flow contact Reference flow support and known issues		PDK-08 EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
		PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control	
		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	

PDK-10

Rule deck, QRC tech file and DRC/LVS deck version control

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w cont. w6–w12 in stage, runs throughout	4.0 M/M planned	~2.9 FTE while it runs	6 2 in parallel	no has float	CAD methodology engineer Foundry deck contact approves

PURPOSE

Put every rule deck, extraction file and verification runset **under version control**, validate them on known-good structures, and make deck changes a notified event rather than a surprise.

Decks are the quiet dependency. A DRC runset revision that arrives without notice changes what passes verification, and if two teams are running different revisions the disagreement looks like a layout problem. Version control here is what makes a signoff result reproducible six months later.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Deck inventory and version baseline	1w	main
2	Version control and release process definition	1w	main
3	QRC tech file and extraction deck validation	1.5w	parallel
4	DRC and LVS deck validation on known-good structures	1.5w	main
5	Deck change notification and impact process	1w	parallel
6	Continuous deck version tracking across the stage	2.5w	main

Validation on known-good structures in step 4 is what makes a deck trustworthy. A runset that has never been tested against a layout with a known answer will report both false positives and, far worse, false negatives.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Foundry rule decks, runsets and tech files
- PDK version tracking from PDK-01
- Tool matrix from PDK-08
- Known-good test structures from previous programs
- Rule dispositions from PDK-02

PRODUCES

- Deck inventory with version baseline
- Version control and release process
- Extraction deck validation results
- DRC and LVS validation results
- Change notification process and log

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D5

FEEDS

Internal reference flow and methodology guide. A flow is only reproducible if the decks it calls are versioned; the guide names the versions.

PDK-D8

FEEDS

Golden environment release notes. The environment pins deck versions, and this activity is what makes pinning possible.

PDK-D1

FEEDS

PDK readiness dashboard. Deck currency and validation status are readiness items on the dashboard.

RISKS AND WATCH-OUTS

- ◆ **Decks used from a shared directory.** An unversioned deck makes every result unreproducible, and the problem is invisible until two results disagree.
- ◆ **Extraction deck unvalidated.** Parasitic extraction errors move timing by amounts that look like design problems, and they are found after weeks of closure work.
- ◆ **Deck updates applied silently.** A runset revision mid-closure changes what passes, and nobody can explain why yesterday's clean run is dirty today.
- ◆ **No known-good structures.** Without a layout whose answer is known, deck validation is an assertion.
- ◆ **Tracking stops when the stage ends.** Decks keep moving through signoff; a process that stops at PDK 1.0 leaves the most critical period unmanaged.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Deck validation	1.5	● Foundry decks released under the design agreement	PDK-09 Foundry reference flow bring-up and internal methodology deck
Version control process	1.0	● PDK version tracking established by PDK-01	PDK-13 Golden design environment release and flow regression
Continuous tracking	1.0	● Known-good test structures available	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Change notification process	0.5		S0-10 Waiver review board and foundry waiver alignment
Total	4.0		PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
WHO IT TAKES	~2.9 FTE	EXIT CRITERIA	MEASURED BY
CAD methodology engineer Owns deck version control and the release process		● Every deck versioned and pinned, none used from a shared path	● Decks under version control against the inventory
Physical verification engineer DRC and LVS deck validation		● Extraction and verification decks validated against known answers	● Deck-induced result changes explained on the first day
Extraction engineer QRC tech file and parasitic deck validation		● Change notification process running, with a log	● Reproducibility of a signoff run six months later
PDK owner Change notification against the version tracking		DEPENDS ON	
Foundry deck contact Deck releases, fixes and known issues		PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	
		PDK-02 Design rule manual review; restricted and recommended rule disposition	

PDK-11

Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	5.0 M/M	~3.6 FTE	6	no	Signoff methodology lead
w20–w26 in stage	planned	while it runs	2 in parallel	has float	CAD infrastructure approves

PURPOSE

Agree with the foundry **exactly which corners and derates signoff will be run at** — PVT combinations, modes, and the OCV, AOCV or POCV methodology — and write it down before anyone closes timing against it.

A corner set is a contract. Sign off at corners the foundry does not recognise and the foundry will not accept the result; sign off at more corners than necessary and every closure iteration costs runtime and licence capacity the program has not budgeted for.

WORK FLOW

6 STEPS · 2 LANES

w20	w22	w24	w26
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3	6
---	---

#	STEP	TAT	LANE
1	Corner and mode enumeration from the operating conditions	1w	main
2	PVT corner set proposal	1.5w	main
3	OCV, AOCV or POCV derate methodology selection	1.5w	parallel
4	Foundry alignment on the corner set and derates	2w	main
5	Signoff condition documentation	1.5w	main
6	Corner count against runtime and licence cost	1w	parallel

Step 6 exists because corner count multiplies. Nine corners across three modes is twenty-seven signoff runs per iteration, and the compute plan in `PDK-12` is sized against whatever number this activity settles on.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Operating conditions and DVFS points from ARCH-05
- Library characterisation corners from PDK-03
- IO and ESD conditions from PDK-07
- Foundry signoff requirements and recommended derates
- Runtime and licence capacity constraints

PRODUCES

- Corner and mode enumeration
- PVT corner set, agreed with the foundry
- Derate methodology selection and rationale
- Signoff condition document
- Corner count impact on runtime and licences

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-06	PRODUCES	Signoff corner definition agreed with foundry. This activity is the deliverable, and "agreed" is the operative word — an internal corner set has no standing.
PDK-07	FEEDS	Compute and license capacity plan. Corner count multiplied by modes is the single largest driver of signoff compute demand.

RISKS AND WATCH-OUTS

- ◆ **Corner set agreed after closure starts.** Adding a corner mid-closure invalidates the work done, and the addition usually comes from the foundry.
- ◆ **Derate methodology chosen for margin.** OCV is conservative and expensive in area and power; POCV is tighter and demands better data. Choosing without the data is choosing margin blindly.
- ◆ **Corner count unbounded.** Every corner is a full signoff run per iteration; a set nobody trimmed becomes a schedule problem at 50-02.
- ◆ **Modes forgotten.** Corners multiply by modes, and a mode discovered late — a test mode, a low-power mode — multiplies the whole set.
- ◆ **Library corners narrower than the signoff set.** Signing off at a corner the library was never characterised at is signing off against extrapolation.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Corner and mode definition	1.5	● Operating conditions and DVFS points defined by ARCH-05	PDK-12 Compute farm, EDA license and storage capacity planning
Foundry alignment	1.5	● Library characterisation corners known from PDK-03	SYN-01 SDC constraint development and validation
Derate methodology	1.0	● Foundry signoff requirements available	PD-09 Multi-corner multi-mode timing closure
Documentation	1.0		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Total	5.0		S0-12 Signoff corner correlation against foundry decks
WHO IT TAKES	~3.6 FTE	EXIT CRITERIA	MEASURED BY
Signoff methodology lead Owns the corner set and the derate methodology		● Corner set agreed with the foundry in writing	● Corners agreed against corners later added
STA lead Corner and mode enumeration, runtime implications		● Every signoff corner covered by library characterisation	● Signoff runs per closure iteration
Library engineer Confirms characterisation covers the proposed corners		● Runtime and licence impact quantified and accepted	● Corners covered by library characterisation
Foundry technical contact Agreement on corners and recommended derates		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
CAD infrastructure Runtime and licence capacity impact		PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	
		PDK-07 IO, ESD and latch-up library qualification	

PDK-12

Compute farm, EDA license and storage capacity planning

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	3.0 M/M	~2.2 FTE	6	no	CAD infrastructure lead
w16–w22 in stage	planned	while it runs	2 in parallel	has float	IT operations approves

PURPOSE

Model what the program will actually need in **compute, EDA licences and storage** across its peak, and procure it against a lead time rather than discovering it during timing closure.

Compute is a schedule input disguised as an IT line item. Timing closure at PD-09 runs multi-corner across a 600 mm² die; if the farm or the licence pool caps concurrency, every iteration takes longer and the closure burn-down flattens for reasons nobody attributes to infrastructure.

WORK FLOW

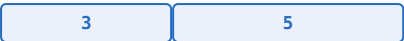
6 STEPS · 2 LANES

w16	w18	w20	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Compute demand model per stage and per flow step	1.5w	main
2	EDA licence demand model against peak concurrency	1.5w	main
3	Storage and data retention requirement	0.8w	parallel
4	Farm capacity plan against procurement lead time	1.5w	main
5	Licence negotiation support and pooling strategy	1w	parallel
6	Capacity plan publication and review	1.5w	main

The demand model is built per flow step rather than per team because that is how contention actually happens: three teams running the same tool at the same point in the schedule contend for one licence pool, whatever the org chart says.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Flow step definitions and runtimes from PDK-09
- Corner and mode count from PDK-11
- Tool matrix from PDK-08
- Program schedule and stage overlap from DEF-08
- Existing farm and licence estate

PRODUCES

- Compute demand model per stage and flow step
- EDA licence demand model at peak concurrency
- Storage and retention requirement
- Farm capacity plan with procurement dates
- Licence pooling strategy

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D7

PRODUCES

Compute and license capacity plan. This activity is the deliverable, and its value is that it is dated against lead times.

PDK-D4

FEEDS

EDA tool and version matrix. Which tools are pooled and at what seat count is decided against the matrix.

RISKS AND WATCH-OUTS

- ◆ **Demand modelled per team rather than per flow step.** Contention happens when several teams hit the same tool at the same time, which an org-chart model cannot see.
- ◆ **Peak sized to the average.** Signoff and closure peaks are several times the steady state, and a farm sized to the average flattens the burn-down exactly when it matters.
- ◆ **Licences procured without lead time.** Additional seats are a commercial negotiation with a duration; needing them next week is needing them next quarter.
- ◆ **Storage forgotten.** Multi-corner signoff databases are large, and a farm that runs out of storage stops as hard as one that runs out of cores.
- ◆ **Plan never reviewed against actual usage.** A capacity plan is a forecast; without a comparison to actuals nobody learns that it was wrong until closure.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Demand modelling	1.5	● Flow steps and runtimes known from PDK-09	SYN-11 Timing, area and power reporting against budget, per drop
Capacity plan	0.8	● Corner and mode count settled by PDK-11	PD-09 Multi-corner multi-mode timing closure
Licence strategy	0.5	● Program schedule available from DEF-08	S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
Publication and review	0.3		DV-10 Coverage closure and regression stability management
Total	3.0		
WHO IT TAKES	~2.2 FTE	EXIT CRITERIA	MEASURED BY
CAD infrastructure lead Owns the demand model and the capacity plan		● Peak demand modelled per flow step, not per team	● Peak concurrency modelled against peak observed
Flow engineers Runtime and concurrency data per flow step		● Procurement dated against lead time, not against need date	● Licence denials during closure
Procurement lead Licence negotiation and lead times		● Plan published with a review cadence against actual usage	● Procurement lead time against the need date
Program manager Stage overlap and peak timing			
IT operations Farm, storage and deployment			
		PDK-08 EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
		PDK-09 Foundry reference flow bring-up and internal methodology deck	
		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
		DEF-08 Program schedule, resource and staffing plan drafting	

PDK-13

Golden design environment release and flow regression

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	8.0 M/M	~4.3 FTE	7	yes	Methodology lead
w24–w32 in stage	planned	while it runs	3 in parallel	no float	IT operations approves

PURPOSE

Package everything this stage produced into **one environment every team runs from** — tools, decks, libraries, scripts — prove it with a regression, and release it with known issues stated.

Enablement only becomes real when it is a single thing an engineer can load. Until then the program has a qualified library here, a validated deck there and a flow in someone's home directory, and every new engineer assembles a slightly different version of the design environment.

WORK FLOW

7 STEPS · 2 LANES

w24	w26	w28	w30	w32
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
---	---	---

#	STEP	TAT	LANE
1	Environment scope definition — tools, decks, libraries, scripts	1w	main
2	Golden environment assembly	2w	main
3	Flow regression suite definition	1.5w	parallel
4	Regression run on the reference block	2w	main
5	Result comparison against the pilot baseline	1.5w	parallel
6	Release notes and known-issue list	1.5w	parallel
7	Environment release and team onboarding	3w	main

Step 5 is what makes the release trustworthy. If the golden environment produces different results from the `PDK-09` pilot on the same block, something in the packaging changed the answer, and finding that after release means everyone's results are suspect.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Qualified libraries from PDK-03, PDK-04 and PDK-07
- Validated decks from PDK-10
- Tool matrix from PDK-08
- Adapted flow and methodology deck from PDK-09
- Frozen PDK version from PDK-01

PRODUCES

- Golden design environment, versioned
- Flow regression suite
- Regression results against the pilot baseline
- Release notes and known-issue list
- Onboarding material and team access

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PDK-D8	PRODUCES	Golden environment release notes. This activity is the deliverable — the environment itself, with what is in it and what is known to be wrong with it.
PDK-D1	GATES	PDK readiness dashboard. The dashboard closes when the environment releases; until then every readiness item is still individually tracked.
PDK-D5	FEEDS	Internal reference flow and methodology guide. The guide describes the flow; this environment is the flow, packaged and versioned.

RISKS AND WATCH-OUTS

- ◆ **Environment released without a regression.** An untested package is a package whose first regression is run by whoever is closest to a deadline.
- ◆ **Results differ from the pilot and the difference is not chased.** A packaging change that moves results silently invalidates the pilot's credibility and everything built on it.
- ◆ **Known issues omitted.** An environment released as clean produces bug reports for problems the team already knew about, and the reports go to the wrong place.
- ◆ **No onboarding.** A golden environment nobody was shown how to use is a directory people copy from and modify, which is where divergence starts.
- ◆ **Released and then not maintained.** Decks and PDK versions keep moving; an environment frozen at release drifts away from what signoff will require.

EFFORT SPLIT	8 M/M	EXIT CRITERIA	FEEDS INTO
Environment assembly	2.5	- Regression reproduces the pilot baseline, or the difference is explained - Known issues published with the release, not after it - Teams onboarded onto the environment rather than copying it	SYN-01 SDC constraint development and validation
Regression suite and runs	2.5		PD-01 Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
Onboarding	1.5		SO-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
Release notes	1.0		DFT-06 Scan insertion and DFT DRC during synthesis
Baseline comparison	0.5		AMS-12 Custom layout with DRC / LVS / antenna and fill closure
Total	8.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~4.3 FTE	PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	- Regression deviation from the pilot baseline - Teams on the golden environment against those running private copies - Known issues published against those later reported as new
Methodology lead Owns the environment, its release and its maintenance		PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	
CAD engineers Assembly, versioning and deployment		PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)	
Regression owner Suite definition and baseline comparison		PDK-07 IO, ESD and latch-up library qualification	
Physical design representative Confirms the environment is usable for real work		PDK-08 EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
IT operations Deployment and access		PDK-09 Foundry reference flow bring-up and internal methodology deck	
ENTRY CRITERIA		PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control	
- Libraries qualified and frozen - Decks validated and under version control - Flow piloted successfully by PDK-09			

05 IP Strategy & Readiness

IPR

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w4 – w26	22 _w	10	47 M/M	6 of 10
ENGINEERING ACTIVITIES				KEY DELIVERABLES
IPR-01	Product requirement → IP requirement decomposition (IP bill of materials)		5w 5 M/M	IPR-D1 IP bill of materials with make / buy / reuse decision per block
IPR-02	Reuse inventory audit — internal IP already available on the target node		4w 3 M/M	IPR-D2 Vendor evaluation matrix and selection record
IPR-03	Make / buy / reuse decision per IP block		5w 4 M/M	IPR-D3 IP readiness report — silicon-proven status and maturity level per IP
IPR-04	Vendor RFI / RFQ and technical evaluation		9w 10 M/M	IPR-D4 IP deliverable acceptance checklist
IPR-05	Silicon-proven status check per IP on the selected node and process option		7w 6 M/M	IPR-D5 Executed licences and POs with committed delivery dates
IPR-06	IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)		7w 6 M/M	IPR-D6 IP delivery schedule folded into the program plan
IPR-07	Licence negotiation, PO issue and delivery-date commitment		12w 5 M/M	IPR-D7 IP risk register and contingency plan
IPR-08	Porting / hardening scope and schedule for IP not proven on the node		5w 4 M/M	
IPR-09	IP delivery schedule alignment to RTL and PD integration windows		4w 2 M/M	
IPR-10	IP maturity risk rating and second-source contingency		4w 2 M/M	

IPR-01

Product requirement → IP requirement
decomposition (IP bill of materials)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	5.0 M/M	~4.3 FTE	6	yes	IP strategist
w0–w5 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Decompose the product into **the IP it actually needs** — every block, with its specification, its interfaces and the requirement it traces back to — and publish it as a bill of materials the program can be held to.

An IP-BOM derived from the last program's list is the most common way to arrive at the wrong one. What this product needs follows from its requirements and its architecture, and the blocks that differ from last time are exactly the ones with the longest lead times.

WORK FLOW

6 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE

1	2	5	6
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Requirement-to-IP decomposition against the requirement register	1w	main
2	Block-level IP identification from the architecture sketch	1.5w	main
3	Interface and protocol IP identification	0.8w	parallel
4	Foundry IP identification — memory compilers, IO, standard cells	0.8w	parallel
5	Per-IP specification stub — function, performance, interfaces	1.5w	main
6	Bill of materials assembly and traceability check	1w	main

Step 6's traceability check is what separates a list from a bill of materials. An IP that traces to no requirement is an IP nobody needs; a requirement that traces to no IP is a gap nobody has staffed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Requirement register from DEF-01
- Architecture sketch and block list direction from ARCH-02
- Interface selections from ARCH-04
- Previous-program IP list as a starting reference
- Foundry IP catalogue for the selected node

PRODUCES

- Requirement-to-IP traceability matrix
- IP bill of materials, block by block
- Per-IP specification stubs
- Interface and foundry IP identification
- Traceability gap list

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-01	PRODUCES	IP bill of materials with make / buy / reuse decision per block. The bill half of the deliverable; IPR-03 adds the decision against each line.
IPR-03	FEEDS	IP readiness report. Readiness is assessed per line of this bill, so the bill's completeness bounds the report's.

RISKS AND WATCH-OUTS

- ◆ **Bill inherited from the previous program.** The blocks that differ are the ones with the longest lead times, and inheriting the list hides exactly those.
- ◆ **Foundry IP omitted.** Memory compilers, IO and standard cells are IP with licences and lead times, and a bill that lists only third-party blocks understates the procurement.
- ◆ **Specification stubs too thin to evaluate against.** A vendor cannot be assessed against "a PCIe PHY"; the stub has to carry performance, interface and process detail.
- ◆ **Traceability skipped.** Without it, blocks appear because someone assumed them and requirements go unserved because nobody noticed.
- ◆ **Bill frozen before architecture settles.** A bill written before ARCH-02 partitions will miss the die-to-die interface entirely if the part turns out to be a chiplet.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Requirement decomposition	1.5	● Requirement register available at v0.9 from DEF-01	IPR-02 Reuse inventory audit — internal IP already available on the target node
Block and interface identification	1.5	● Architecture direction known well enough to name blocks	IPR-03 Make / buy / reuse decision per IP block
Specification stubs	1.0	● Foundry IP catalogue accessible	IPR-04 Vendor RFI / RFQ and technical evaluation
Bill assembly and traceability	1.0		TECH-08 Second-source and node-migration strategy assessment
Total	5.0	EXIT CRITERIA	PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)
WHO IT TAKES	~4.3 FTE	● Every IP traces to a requirement, and every requirement to an IP or a gap	MEASURED BY
IP strategist		● Foundry IP included alongside third-party and internal blocks	● IP lines traceable to a requirement
Owns the bill of materials and its traceability		● Specification stubs detailed enough to evaluate a vendor against	● Requirements with no IP against them
Chief architect			● Bill changes after the architecture freeze
Confirms the block list and its interfaces		DEPENDS ON	
Interface architect		DEF-01 Market and customer requirements consolidation	
Protocol and PHY IP identification		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
CAD and library lead		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	
Foundry IP — compilers, IO, cells			
Program manager			
Carries the bill into procurement planning			

IPR-02

Reuse inventory audit — internal IP already available on the target node

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	5	no	IP strategist
w1–w5 in stage	planned	while it runs	2 in parallel	has float	Silicon validation liaison approves

PURPOSE

Audit what the organisation **already owns and can actually reuse on this node** — not what it owns in general, which is a different and much longer list.

Reuse is where the schedule is won, and it is also where optimism concentrates. A block proven on the previous node is not available on this one until it has been ported, re-characterised and re-verified, and that work is a project with a duration. The audit's job is to separate genuine reuse from hopeful reuse.

WORK FLOW

5 STEPS · 2 LANES

w1	w2	w3	w4	w5
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MAIN SEQUENCE

1	2	5
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Internal IP inventory against the bill of materials	0.8w	main
2	Node availability check per internal block	1.3w	main
3	Collateral completeness audit — views, testbenches, documentation	1w	parallel
4	Maintenance and ownership status per block	0.8w	parallel
5	Reuse classification — direct, port required, or rebuild	2w	main

Step 4 asks who still maintains each block, and the answer is often nobody. An unmaintained block is technically reusable and practically a rebuild, because the first bug found has no owner.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- IP bill of materials from IPR-01
- Internal IP catalogue and repositories
- Previous-program silicon results per block
- Node and flavour selection from TECH
- Team ownership records

PRODUCES

- Internal IP inventory mapped to the bill
- Node availability finding per block
- Collateral completeness audit
- Ownership and maintenance status
- Reuse classification per block

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D1	FEEDS	IP bill of materials. Reuse classification is one of the three answers the make / buy / reuse column can take, and it is decided here.
IPR-D3	FEEDS	IP readiness report. An internal block on the wrong node is exactly as unready as an external one, and belongs in the same report.

RISKS AND WATCH-OUTS

- ◆ **Reuse claimed on the previous node.** Availability is node-specific; a block not yet ported is a porting project, not a reuse.
- ◆ **Collateral assumed complete.** Missing multi-corner .lib files or an absent testbench turns reuse into re-characterisation and re-verification.
- ◆ **Unmaintained blocks counted as reuse.** A block with no owner has no one to fix its first bug, and the fix lands on the integration team.
- ◆ **Audit run on the catalogue rather than the repository.** Catalogues describe intent; repositories contain what actually exists.
- ◆ **Silicon history ignored.** A block that shipped with an erratum is reusable with that erratum, and the erratum belongs in the audit.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Node availability assessment	1.0	● Bill of materials available from IPR-01	IPR-03 Make / buy / reuse decision per IP block
Inventory against the bill	0.8	● Access to internal IP repositories	IPR-08 Porting / hardening scope and schedule for IP not proven on the node
Collateral audit	0.8	● Node selected so availability can be assessed	ARCH-06 PPA budget allocation per block
Reuse classification	0.5		
Total	3.0	EXIT CRITERIA	
WHO IT TAKES	~3.3 FTE	● Every internal block classified as direct reuse, port, or rebuild	MEASURED BY
IP strategist Owns the audit and the reuse classification		● Collateral gaps listed per block, not assumed away	● Blocks classified as direct reuse against those later ported
IP librarian Inventory, repositories and collateral		● Blocks with no maintainer flagged as such	● Collateral gaps found here against those found at integration
Design leads Ownership and maintenance status per block		DEPENDS ON	● Blocks with a named maintainer
Verification liaison Testbench and DV collateral completeness		IPR-01 Product requirement → IP requirement decomposition (IP bill of materials)	
Silicon validation liaison Errata and known issues from previous silicon		TECH-01 Foundry and node long-list to short-list evaluation	

IPR-03

Make / buy / reuse decision per IP block

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	4.0 M/M	~3.5 FTE	6	yes	IP strategist
w5–w10 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Take each line of the bill and decide, with a reason, whether it is **made, bought or reused** — and record what the rejected options would have cost.

Make/buy is a schedule decision more than a cost one. Building a PHY internally is cheaper on paper and costs nine months in AMS-03; buying it costs a licence and a vendor dependency. Neither is wrong, and the program only gets to argue about it once.

WORK FLOW

6 STEPS · 2 LANES

w5	w6	w7	w8	w9	w10
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Decision criteria definition — cost, schedule, differentiation, risk	0.8w	main
2	Differentiation screen — which blocks are the product's own value	1w	main
3	Internal capacity check against the make candidates	1w	parallel
4	Cost and schedule comparison per line	1.5w	main
5	Foundry-specific versus portable IP consideration	0.8w	parallel
6	Decision per line and record of rejected options	1.8w	main

Step 3 is the check that changes answers. A make decision assumes engineers who are also assigned to the custom SRAM and the PLL; capacity, not capability, is what usually decides.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- IP bill of materials from IPR-01
- Reuse classification from IPR-02
- AMS capacity and schedule from the program plan
- Sourcing strategy direction from TECH-08
- Cost model assumptions from DEF-04

PRODUCES

- Decision criteria and their weighting
- Differentiation screen result
- Make / buy / reuse decision per line
- Rejected option record with cost and schedule
- Internal capacity commitment for make decisions

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D1	PRODUCES	IP bill of materials with make / buy / reuse decision per block. The decision half of the deliverable, with the rejected options priced rather than forgotten.
IPR-D7	FEEDS	IP risk register. Every buy decision creates a vendor dependency and every make decision creates a capacity dependency; both belong on the register.

RISKS AND WATCH-OUTS

- ◆ **Make decided on capability rather than capacity.** The team can build it; whether it can build it alongside everything else it is already committed to is the real question.
- ◆ **Differentiating IP bought.** Buying the block that is the product's actual advantage buys a product the competitor can buy too.
- ◆ **Commodity IP built.** Building a standard interface consumes analog engineers who are the program's scarcest resource, for no differentiation.
- ◆ **Foundry-specific IP chosen without reference to sourcing strategy.** It forecloses the migration option TECH-08 was pricing, and usually after that study concludes.
- ◆ **Decisions recorded without the alternatives.** Every setback reopens a decision whose reasoning nobody can reconstruct.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Cost and schedule comparison	1.5	● Bill of materials complete from IPR-01 ● Reuse classification available from IPR-02 ● AMS capacity known from the program plan	IPR-04 Vendor RFI / RFQ and technical evaluation
Differentiation screen	1.0		IPR-07 Licence negotiation, PO issue and delivery-date commitment
Capacity check	0.8		IPR-08 Porting / hardening scope and schedule for IP not proven on the node
Decision record	0.8		AMS-01 AMS IP specification and budget allocation (jitter, BER, PSRR, area)
Total	4.0	EXIT CRITERIA	AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)
WHO IT TAKES	~3.5 FTE	● Every line decided with a criterion attached, not by default ● Make decisions backed by a named capacity commitment ● Rejected options priced and recorded	MEASURED BY
IP strategist		DEPENDS ON	● Lines decided against lines in the bill ● Make decisions with a capacity commitment behind them ● Decisions reopened after IP Plan Freeze
Owns the decisions and their record		IPR-01 Product requirement → IP requirement decomposition (IP bill of materials)	
Chief architect		IPR-02 Reuse inventory audit — internal IP already available on the target node	
Differentiation screen — what is the product's own value		TECH-08 Second-source and node-migration strategy assessment	
AMS lead			
Internal capacity for make candidates			
Procurement lead			
Buy-side cost and vendor landscape			
Program manager			
Schedule impact and capacity commitment			

05 IPR · IP STRATEGY & READINESS

IPR-04

Vendor RFI / RFQ and technical evaluation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	10.0 M/M	~4.8 FTE	7	yes	IP strategist
w6–w15 in stage	planned	while it runs	3 in parallel	no float	Program manager approves

PURPOSE

Run a real evaluation of the vendors behind every buy decision — **technical capability, deliverable quality, support model and track record** — rather than choosing from a datasheet.

A PHY vendor is a nine-month dependency. What matters is less the block's peak specification than whether the deliverables arrive complete, whether the support engineer answers, and whether the last three customers taped out on schedule. None of that is on the datasheet.

WORK FLOW

7 STEPS · 2 LANES

w6	w8	w10	w12	w14	w15
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MAIN SEQUENCE

1	2	3	7
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Vendor long-list per IP line	0.8w	main
2	RFI issue and response collection	2w	main
3	Technical evaluation against the specification stubs	2.5w	main
4	Reference customer checks and tapeout track record	1.5w	parallel
5	Support model and escalation path assessment	1w	parallel
6	RFQ issue and commercial comparison	1.8w	parallel
7	Evaluation matrix and vendor selection	3.8w	main

Step 4 is the step most often skipped and the one that predicts the outcome best. A vendor's last three customers know whether the deliverables arrived complete and on time, and they will usually say.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Buy decisions from IPR-03
- Specification stubs from IPR-01
- Node and process option from TECH-02
- Interface requirements from ARCH-04
- Vendor landscape and previous-program experience

PRODUCES

- Vendor long-list per IP line
- RFI and RFQ responses
- Technical evaluation against the stubs
- Reference customer findings
- Vendor evaluation matrix and selection record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-02	PRODUCES	Vendor evaluation matrix and selection record. This activity is the deliverable, and the matrix is what a later dispute is settled against.
IPR-05	FEEDS	Executed licences and POs. Selection precedes negotiation; IPR-07 contracts with whoever this activity chose.
IPR-03	FEEDS	IP readiness report. Vendor track record is part of an IP's maturity rating, not separate from it.

RISKS AND WATCH-OUTS

- ◆ **Evaluation from the datasheet.** Peak specifications are marketing; deliverable completeness and support responsiveness are what the integration actually depends on.
- ◆ **Reference checks skipped.** The vendor's previous customers know what the program is about to find out, and asking costs an hour.
- ◆ **Single vendor evaluated.** An evaluation with one candidate is a justification, and the commercial position that follows is weak.
- ◆ **Support model unexamined.** A block with no named support engineer becomes the integration team's problem the first time it misbehaves.
- ◆ **Node-specific capability assumed from a different node.** A PHY proven at the previous node is not the same product; IPR-05 is where that is confirmed.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Technical evaluation	3.5	● Buy decisions made in IPR-03	IPR-05 Silicon-proven status check per IP on the selected node and process option
RFI and RFQ handling	2.0	● Specification stubs detailed enough to evaluate against	IPR-06 IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)
Reference checks	1.5	● NDAs in place with candidate vendors	IPR-07 Licence negotiation, PO issue and delivery-date commitment
Evaluation matrix and selection	2.0		AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY)
Support model assessment	1.0	EXIT CRITERIA	
Total	10.0	● At least two candidates evaluated per significant line ● Reference customers contacted for the critical blocks ● Selection recorded against the matrix, with support model assessed	MEASURED BY ● Candidates evaluated per line ● Reference checks completed on critical blocks ● Vendor changes after selection
WHO IT TAKES	~4.8 FTE	DEPENDS ON	
IP strategist		IPR-01	Product requirement → IP requirement decomposition (IP bill of materials)
Owns the evaluation and the selection		IPR-03	Make / buy / reuse decision per IP block
Technical evaluators		ARCH-04	Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)
Assessment against the specification stubs			
Procurement lead			
RFQ, commercial comparison and terms			
Integration lead			
Deliverable completeness and support expectations			
Program manager			
Reference checks on schedule performance			

05 IPR · IP STRATEGY & READINESS

IPR-05

Silicon-proven status check per IP on the selected node and process option

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	6.0 M/M	~3.7 FTE	7	yes	IP strategist
w8–w15 in stage	planned	while it runs	2 in parallel	no float	Procurement lead approves

PURPOSE

Establish, for each bought and reused block, **whether it has actually run in silicon on this node and this process option** — and if not, what it would take to get it there.

"Silicon-proven" is the most abused phrase in IP procurement. It can mean proven on a different node, on a different process option, at a different speed grade, or in a test chip that was never characterised. This activity replaces the phrase with a fact, and the fact is what the integration schedule should be built on.

WORK FLOW7 STEPS · 2 LANES

w8w10w12w14w15

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Proven-status definition — what the program will accept as evidence	0.8w	main
2	Evidence request per block — test chip, production part, characterisation data	1.5w	main
3	Node and process option match verification	1.5w	main
4	Speed grade and operating condition match check	1w	parallel
5	Errata and known-issue review per block	1w	parallel
6	Maturity level assignment per block	1.3w	main
7	Readiness report and gap list	2w	main

Step 1 is not bureaucracy. Without an agreed evidence standard, every vendor's claim is accepted at its own definition, and the readiness report becomes a collection of adjectives.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vendor selections from IPR-04
- Reuse classification from IPR-02
- Node and process option from TECH-02
- Vendor silicon evidence and characterisation data
- Errata from previous programs and vendor disclosures

PRODUCES

- Evidence standard the program accepts
- Silicon evidence per block, with node and option match
- Speed grade and condition match findings
- Errata and known-issue summary
- Maturity level per block and a gap list

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D3	PRODUCES	IP readiness report — silicon-proven status and maturity level per IP. This activity is the deliverable, and it is the one that replaces a claim with a fact.
IPR-D7	FEEDS	IP risk register. Every block below the maturity bar is a risk with a name, an owner and a mitigation.
IPR-D6	FEEDS	IP delivery schedule. An unproven block needs a hardening window, and the window has to be in the schedule.

RISKS AND WATCH-OUTS

- ◆ **Vendor claim accepted at the vendor's definition.** Without an evidence standard, "silicon-proven" means whatever is most convenient.
- ◆ **Node matched, process option not.** The same node with a different flavour sheet is a different library, different rules and often a different PHY.
- ◆ **Errata not requested.** Every silicon-proven block has errata; a block presented with none has either not been characterised or is not being fully disclosed.
- ◆ **Assessment done after the licence is signed.** The finding has no commercial consequence once the PO is placed, and the leverage is gone.
- ◆ **Speed grade ignored.** A PHY proven at a lower rate on the same node is an engineering project, not a proven block.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Evidence assessment	2.5	● Vendor selections available from IPR-04	IPR-06 IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)
Node and option verification	1.5	● Node and process option fixed by TECH-02	IPR-07 Licence negotiation, PO issue and delivery-date commitment
Errata review	1.0	● Vendor NDAs allow access to characterisation data	IPR-08 Porting / hardening scope and schedule for IP not proven on the node
Maturity assignment and report	1.0		IPR-10 IP maturity risk rating and second-source contingency
Total	6.0		TC-02 Test chip design and integration (IP under test, ring oscillators, PCM structures)
WHO IT TAKES	~3.7 FTE	EXIT CRITERIA	
IP strategist Owns the evidence standard and the readiness report		● Every block carries a maturity level backed by evidence, not by claim	RTL-03 Third-party and internal IP integration — wrappers, glue, configuration
Technical evaluators Evidence assessment per block		● Node and process option match verified per block	
AMS lead PHY and analog block evidence interpretation		● Findings available before licences are signed in IPR-07	
Silicon validation liaison Errata review and characterisation data reading			
Procurement lead Uses findings as negotiating position before signature			
		DEPENDS ON	MEASURED BY
		IPR-02 Reuse inventory audit — internal IP already available on the target node	● Blocks with evidence against blocks with a claim
		IPR-04 Vendor RFI / RFQ and technical evaluation	● Node and option mismatches found
		TECH-02 Process option and flavour selection (HPC/HD, multi-Vt menu, backside PDN, RF/HV adders)	● Errata disclosed against errata found later

IPR-06

IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	6.0 M/M	~3.7 FTE	7	no	IP integration lead
w11–w18 in stage	planned	while it runs	3 in parallel	has float	Vendor technical contacts approves

PURPOSE

Define **exactly what a complete IP delivery contains** — every view, model, testbench and document — and get that list agreed before it becomes a contractual obligation.

Most IP delivery disputes are not about whether the block works; they are about whether a delivery counted as a delivery. A block that arrives without multi-corner timing views or without its verification collateral has arrived in a form the program cannot integrate, and the argument that follows is unwinnable if the checklist was never agreed.

WORK FLOW

7 STEPS · 2 LANES

w11w13w15w17w18

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Deliverable checklist definition per IP class	1w	main
2	View and model requirements against the tool matrix	1.5w	main
3	Verification collateral requirements with DV	1w	parallel
4	Documentation, errata and integration guide requirements	1w	parallel
5	Acceptance criteria and incoming inspection procedure	1.5w	main
6	Checklist agreement with each vendor	2w	parallel
7	Checklist publication and handoff to procurement	3w	main

Step 5 turns the checklist into something enforceable. A list of expected files is a wish; an acceptance procedure that says what is checked, by whom, and what happens when it fails is a term.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vendor selections from IPR-04
- Tool matrix and view requirements from PDK-08
- Verification methodology needs from DV-01
- Integration requirements from RTL-03
- Previous-program IP delivery disputes

PRODUCES

- Deliverable checklist per IP class
- View and model requirement list
- Verification collateral requirements
- Acceptance criteria and incoming inspection procedure
- Vendor-agreed checklists

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D4	PRODUCES	IP deliverable acceptance checklist. This activity is the deliverable, and its value is entirely in being agreed before the PO rather than after the delivery.
IPR-D5	FEEDS	Executed licences and POs. The checklist becomes a contractual schedule, which is the only form in which it is enforceable.

RISKS AND WATCH-OUTS

- ◆ **Checklist agreed after the PO.** Once the commercial terms are closed, a missing view is a favour rather than an obligation.
- ◆ **Views specified without the tool matrix.** A .lib set that does not cover the signoff corners in PDK-11 is incomplete in a way nobody notices until closure.
- ◆ **Verification collateral omitted.** A block delivered without its testbench has to be verified from scratch, and that effort was never in the DV plan.
- ◆ **No incoming inspection.** Deliverables accepted without a check are found incomplete weeks later, when the vendor's engineer has moved on.
- ◆ **One checklist for all IP classes.** A memory compiler, a PHY and a controller have genuinely different deliverable sets, and a single list serves none of them.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Checklist and view requirements	2.0	● Vendor selections made in IPR-04	IPR-07Licence negotiation, PO issue and delivery-date commitment
Acceptance criteria and inspection	1.5	● Tool matrix available from PDK-08	IPR-09IP delivery schedule alignment to RTL and PD integration windows
Verification collateral requirements	1.0	● Signoff corner intent known from PDK-11	RTL-03Third-party and internal IP integration — wrappers, glue, configuration
Vendor agreement and publication	1.5		PD-02Floorplan, macro placement and partition definition
Total	6.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.7 FTE	● Checklist agreed with each vendor before the PO is placed	● Vendors with an agreed checklist before PO
IP integration lead		● Acceptance procedure names who checks what, and when	● Deliveries rejected at incoming inspection
Owns the checklist and the acceptance procedure		● Views cover the signoff corners the program will actually run	● Missing views discovered after acceptance
CAD methodology engineer		DEPENDS ON	
View and model requirements against the tools		IPR-04Vendor RFI / RFQ and technical evaluation	
Verification lead		PDK-08EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
Collateral requirements and what DV will need		DV-01Verification plan and coverage model definition	
Procurement lead			
Makes the checklist a contractual schedule			
Vendor technical contacts			
Agree the list is deliverable as written			

05 IPR · IP STRATEGY & READINESS

IPR-07

Licence negotiation, PO issue and delivery-date commitment

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	5.0 M/M	~1.8 FTE	8	yes	Procurement lead
w10–w22 in stage	planned	while it runs	3 in parallel	no float	Finance partner approves

PURPOSE

Turn selections into **executed licences and purchase orders with committed delivery dates**, including the deliverable checklist as a contractual schedule.

Twelve weeks for 5 M/M — this is the stage's longest activity and among its smallest in effort, because the elapsed time belongs to two legal departments and a procurement process. It is on the critical path regardless: nothing arrives until it executes, and RTL-03 has integration windows that assume it did.

WORK FLOW

8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Commercial terms definition per IP line	1.5w	main
2	Licence model negotiation — per-use, per-project, royalty	2.5w	main
3	Delivery date commitment and milestone schedule	2w	main
4	Deliverable checklist attached as a contractual schedule	1w	parallel
5	Support, maintenance and escalation terms	1.5w	parallel
6	Liability, indemnity and IP warranty terms	1.5w	parallel
7	Legal redline cycles	3.5w	main
8	Execution, PO issue and distribution	2.5w	main

Step 3 is the one the program cares about and the one commercial negotiation treats as secondary. A licence with excellent terms and no committed delivery date has bought a block that arrives when the vendor is ready.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vendor selections from IPR-04
- Readiness findings from IPR-05 as negotiating position
- Deliverable checklists from IPR-06
- Integration window requirements from the program schedule
- Corporate legal templates and positions

PRODUCES

- Commercial terms per IP line
- Executed licences with delivery commitments
- Purchase orders issued
- Support and escalation terms
- Distribution record of what has been licensed to whom

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D5	PRODUCES	Executed licences and POs with committed delivery dates. This activity is the deliverable, and the committed date is the part the schedule depends on.
IPR-D6	FEEDS	IP delivery schedule. Committed dates are what IPR-09 aligns against the integration windows.
IPR-D4	GATES	IP deliverable acceptance checklist. The checklist only has force once it is attached to an executed agreement.

RISKS AND WATCH-OUTS

- ◆ **Licence executed without a committed delivery date.** The block then arrives when the vendor is ready, and the integration window absorbs the difference.
- ◆ **Readiness findings not used in negotiation.** A block that is not proven on the node is a commercial position; signing before IPR-05 reports discards it.
- ◆ **Checklist not attached.** An agreed checklist outside the contract is an understanding, and understandings do not survive a disputed delivery.
- ◆ **Redline duration underestimated.** Two legal departments take longer than either forecasts, and the program's plan should assume the longer figure.
- ◆ **Support terms omitted.** A block with no support commitment becomes unsupported precisely when integration needs help.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Negotiation support	2.0	● Vendor selections final from IPR-04	IPR-09 IP delivery schedule alignment to RTL and PD integration windows
Terms and licence model	1.5	● Readiness findings available from IPR-05	RTL-03 Third-party and internal IP integration — wrappers, glue, configuration
Delivery commitment	0.8	● Deliverable checklists agreed in IPR-06	AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)
Execution and distribution	0.8		PD-02 Floorplan, macro placement and partition definition
Total	5.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~1.8 FTE	● Every licence carries a committed delivery date, not an intention ● Deliverable checklist attached as a contractual schedule ● Access distributed and tracked to the teams that need it	● Licences with a committed delivery date ● Weeks from selection to execution ● Deliveries received against the committed date
Procurement lead Owns the negotiation, the POs and the terms		DEPENDS ON	
Legal counsel Licence structure, liability and redlines		IPR-04 Vendor RFI / RFQ and technical evaluation	
IP strategist Technical terms and the deliverable schedule		IPR-05 Silicon-proven status check per IP on the selected node and process option	
Program manager Delivery dates against integration windows		IPR-06 IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)	
Finance partner Licence model, royalty exposure and budget		TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame	

05 IPR · IP STRATEGY & READINESS

IPR-08

Porting / hardening scope and schedule for IP not proven on the node

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w w14–w19 in stage	4.0 M/M planned	~3.5 FTE while it runs	6 2 in parallel	no has float	IP strategist Program manager approves

PURPOSE

For every block that is **not proven on this node**, scope the work that gets it there — porting, hardening, re-characterisation, re-verification — and put it in the schedule as a project with an owner.

Unproven IP does not become proven by being licensed. Someone ports it, re-characterises it at the program's corners and re-verifies it, and that someone is either the vendor on a schedule the program does not control, or the program's own engineers who are already committed elsewhere.

WORK FLOW

6 STEPS · 2 LANES

w14	w15	w16	w17	w18	w19
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Unproven block list from the readiness report	0.5w	main
2	Porting scope per block — layout, characterisation, verification	1.5w	main
3	Vendor versus internal execution decision per block	1w	parallel
4	Effort and duration estimate per block	1.3w	main
5	Dependency mapping against the integration windows	0.8w	parallel
6	Hardening plan with owners and dates	1.8w	main

Step 3 decides who absorbs the risk. Vendor-executed hardening is on a schedule the program cannot influence; internally executed hardening competes with `AMS-03` for the same analog engineers. Both are real answers and neither is free.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Readiness report and gap list from IPR-05
- Reuse classification from IPR-02
- Make / buy decisions from IPR-03
- AMS capacity from the program plan
- Integration window requirements from RTL and PD

PRODUCES

- Unproven block list with porting scope
- Execution decision per block — vendor or internal
- Effort and duration estimates
- Dependency map against integration windows
- Hardening plan with owners and dates

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D3	FEEDS	IP readiness report. A block's maturity level is only actionable alongside the plan that raises it, and this activity supplies the plan.
IPR-D6	FEEDS	IP delivery schedule. A hardening window is part of the delivery date, and omitting it produces a schedule that cannot hold.

RISKS AND WATCH-OUTS

- ◆ **Hardening treated as vendor overhead.** It is a project with a duration; leaving it off the plan means the block arrives on a date nobody derived.
- ◆ **Internal execution assumed available.** The engineers who would port a PHY are the engineers building the PLL and the custom array, and they cannot do all three.
- ◆ **Re-verification omitted from scope.** Porting changes the block; the verification collateral has to be re-run, and that effort belongs in the DV plan.
- ◆ **Characterisation corners not matched to the program's.** A block re-characterised at the vendor's corners still cannot be signed off at the program's.
- ◆ **Plan without dates.** A hardening plan with owners and no dates does not integrate with the program schedule and is therefore not tracked.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Porting scope definition	1.5	● Readiness report available from IPR-05	IPR-09 IP delivery schedule alignment to RTL and PD integration windows
Effort and duration estimation	1.0	● AMS capacity known from the program plan	IPR-10 IP maturity risk rating and second-source contingency
Execution decision	0.8	● Integration windows known from RTL and PD planning	AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY)
Hardening plan	0.8		DV-02 UVM environment and VIP bring-up
Total	4.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.5 FTE	● Every unproven block has a scoped plan with an owner and a date	● Unproven blocks with a dated plan
IP strategist		● Internal execution decisions backed by a capacity commitment	● Internal hardening committed against AMS capacity
Owns the hardening plan		● Re-verification scope included, not assumed	● Hardening completed against plan
AMS lead		DEPENDS ON	
Internal capacity and analog porting scope		IPR-02 Reuse inventory audit — internal IP already available on the target node	
Verification lead		IPR-03 Make / buy / reuse decision per IP block	
Re-verification scope and collateral		IPR-05 Silicon-proven status check per IP on the selected node and process option	
Vendor technical contacts			
Vendor-executed hardening scope and schedule			
Program manager			
Dates, owners and integration window alignment			

05 IPR · IP STRATEGY & READINESS

IPR-09

IP delivery schedule alignment to RTL and PD integration windows

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	2.0 M/M	~2.2 FTE	6	yes	Program manager
w17–w21 in stage	planned	while it runs	2 in parallel	no float	Procurement lead approves

PURPOSE

Line up **every IP delivery date against the window that actually needs it** — RTL integration, synthesis, floorplan — and surface the mismatches while they can still be fixed.

An IP schedule is only meaningful against the design schedule. A PHY delivered in week 40 is early for RTL and late for the floorplan, because PD-02 needs the abstract long before RTL-03 needs the model. Alignment means matching each deliverable to the window that consumes it, not the block to a date.

WORK FLOW

6 STEPS · 2 LANES

w17	w18	w19	w20	w21
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MAIN SEQUENCE

1	2	3	5
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IN PARALLEL

4	6
---	---

#	STEP	TAT	LANE
1	Integration window extraction — what each stage needs and when	0.8w	main
2	Committed delivery date collection per block and per deliverable	1w	main
3	Deliverable-level alignment — abstracts before models before collateral	1.3w	main
4	Hardening plan overlay from IPR-08	0.8w	parallel
5	Mismatch identification and escalation list	1w	main
6	Schedule integration into the program plan	0.8w	parallel

Step 3 is where the useful detail lives. Vendors quote one date for a block; the design needs the LEF abstract for floorplan months before it needs the multi-corner .lib for signoff, and asking for the abstract early is often free.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Committed delivery dates from IPR-07
- Hardening plan from IPR-08
- RTL integration windows from the program schedule
- Floorplan and macro placement needs from PD planning
- Deliverable checklist from IPR-06

PRODUCES

- Integration window map per consuming stage
- Deliverable-level delivery schedule
- Hardening overlay
- Mismatch and escalation list
- IP schedule folded into the program plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D6

PRODUCES

IP delivery schedule folded into the program plan. This activity is the deliverable, and "folded in" is the operative part — a separate IP schedule is one nobody tracks.

IPR-D7

FEEDS

IP risk register. Every mismatch that cannot be closed becomes a tracked risk with a mitigation.

RISKS AND WATCH-OUTS

- ◆ **Block-level dates instead of deliverable-level.** One date per block hides that the abstract is needed months before the model, and the earlier need is the binding one.
- ◆ **IP schedule maintained separately.** A schedule outside the program plan is not reviewed at program reviews and drifts unnoticed.
- ◆ **Mismatches recorded without escalation.** A known gap with no owner and no escalation is a gap that is discovered again at integration.
- ◆ **Hardening overlay omitted.** Delivery dates for unproven blocks are meaningless without the porting window that follows them.
- ◆ **Floorplan needs underestimated.** Physical design needs abstracts and blockages very early; planning against RTL's needs alone produces a late floorplan.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Deliverable-level alignment	0.8	● Committed dates available from IPR-07	RTL-03 Third-party and internal IP integration — wrappers, glue, configuration
Window extraction and date collection	0.5	● Hardening plan available from IPR-08	RTL-09 CI build, nightly regression and release management
Mismatch identification	0.5	● Integration windows known from the program schedule	PD-02 Floorplan, macro placement and partition definition
Schedule integration	0.3		PD-04 Bump and RDL planning with chip-package co-design feedback
Total	2.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~2.2 FTE	● Alignment done per deliverable, not per block ● Every mismatch has an owner and an escalation path ● IP schedule visible inside the program plan, not beside it	● Deliverables aligned to a consuming window ● Mismatches closed against those raised ● IP deliveries late against the window that needed them
Program manager Owns the alignment and the escalation list		DEPENDS ON	
IP strategist Deliverable-level requirements per block		IPR-06 IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)	
RTL integration lead Integration window requirements		IPR-07 Licence negotiation, PO issue and delivery-date commitment	
Physical design lead Abstract and macro availability needs		IPR-08 Porting / hardening scope and schedule for IP not proven on the node	
Procurement lead Escalation with vendors on mismatches		DEF-08 Program schedule, resource and staffing plan drafting	

05 IPR · IP STRATEGY & READINESS

IPR-10

IP maturity risk rating and second-source contingency

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	2.0 M/M	~2.2 FTE	5	no	IP strategist
w18–w22 in stage	planned	while it runs	2 in parallel	has float	Chief architect approves

PURPOSE

Rate every IP line for **maturity and supply risk**, and decide where a contingency is worth having — a second source, an internal fallback, or an accepted exposure with a trigger.

Some risk here is unavoidable and pretending otherwise is what makes it dangerous. The point is not to eliminate single-vendor dependencies but to know which ones would stop the program, and to have decided in advance what would be done about each.

WORK FLOW

5 STEPS · 2 LANES

w18	w19	w20	w21	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Risk rating per IP line — maturity, vendor, schedule, supply	1w	main
2	Critical dependency identification — which failures stop the program	1w	main
3	Second-source viability per critical block	0.8w	parallel
4	Internal fallback option assessment	0.8w	parallel
5	Contingency decision and trigger definition per critical block	2w	main

Step 5 defines triggers, not intentions. "We would build it internally if the vendor slipped" is worthless without the date at which that decision has to be taken to still be executable.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Readiness report and maturity levels from IPR-05
- Hardening plan from IPR-08
- Delivery schedule and mismatches from IPR-09
- Vendor evaluation matrix from IPR-04
- Sourcing strategy from TECH-08

PRODUCES

- Risk rating per IP line
- Critical dependency list
- Second-source viability findings
- Internal fallback assessment
- Contingency plan with triggers and decision dates

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

IPR-D7

PRODUCES

IP risk register and contingency plan. This activity is the deliverable, and the triggers are what make it a plan rather than a list.

IPR-D3

FEEDS

IP readiness report. Maturity and risk are read together — a mature block from a fragile vendor is not a low-risk line.

RISKS AND WATCH-OUTS

- ◆ **Register without triggers.** A contingency with no decision date cannot be executed, because by the time the need is obvious the option has expired.
- ◆ **Every line rated as critical.** A register where everything is critical directs attention nowhere; the point is to identify the few that stop the program.
- ◆ **Second source assessed on capability alone.** An alternative vendor without a licence, an evaluation and a delivery date is not a contingency.
- ◆ **Internal fallback assumed.** A fallback that depends on engineers already committed to the critical path is not a fallback.
- ◆ **Register written and not reviewed.** IP risk changes as deliveries land; a register reviewed once at IP Plan Freeze is stale by RTL integration.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Risk rating	0.8	● Readiness levels assigned by IPR-05	RTL-10 Specification change control and ECO board
Critical dependency analysis	0.5	● Delivery schedule aligned by IPR-09	DEF-09 Business case preparation and funding approval package
Contingency options	0.5	● Sourcing strategy known from TECH-08	AMS-01 AMS IP specification and budget allocation (jitter, BER, PSRR, area)
Trigger definition	0.3		
Total	2.0	EXIT CRITERIA	
WHO IT TAKES	~2.2 FTE	● Critical dependencies identified and separated from the rest ● Every critical block has a contingency with a trigger date ● Register has a review cadence, not just an author	MEASURED BY ● Critical dependencies with a triggered contingency ● Register reviews held against those planned ● Contingencies executed against those defined
IP strategist Owns the register and the contingency plan			
Program manager Critical dependency identification and triggers			
Procurement lead Second-source viability and commercial options		IPR-05 Silicon-proven status check per IP on the selected node and process option	
AMS lead Internal fallback feasibility and capacity		IPR-08 Porting / hardening scope and schedule for IP not proven on the node	
Chief architect Confirms which failures actually stop the program		IPR-09 IP delivery schedule alignment to RTL and PD integration windows	
		TECH-08 Second-source and node-migration strategy assessment	

06

Custom, AMS & Memory IP Development

AMS

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w18 – w60	42 _w	16	296 _{M/M}	8 _{of 16}

ENGINEERING ACTIVITIES				KEY DELIVERABLES	
AMS-01	AMS IP specification and budget allocation (jitter, BER, PSRR, area)		6w	8 _{M/M}	AMS-D1 AMS IP specifications and design review packages
AMS-02	PLL / clock generator design and closure		26w	32 _{M/M}	AMS-D2 Characterisation reports across PVT and Monte Carlo
AMS-03	SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)		34w	60 _{M/M}	AMS-D3 Custom SRAM instance specification with Vmin and sigma-yield report
AMS-04	LDO, bandgap and power-management cell design		20w	22 _{M/M}	AMS-D4 Custom memory views characterised to compiler equivalence
AMS-05	Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing		10w	14 _{M/M}	AMS-D5 Hard macro GDS with abstract views (LEF, .lib, CDL, UPF, wreal/Verilog model)
AMS-06	Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse		10w	12 _{M/M}	AMS-D6 Per-macro DRC / LVS clean signoff, pushed rules approved by foundry
AMS-07	Custom array layout and pushed-rule DRC closure with foundry approval		16w	22 _{M/M}	AMS-D7 Reliability report — EM/IR, ESD, latch-up, aging
AMS-08	Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation		10w	12 _{M/M}	AMS-D8 Integration guide with known limitations and errata
AMS-09	Redundancy and repair scheme integration with the BISR architecture		6w	6 _{M/M}	
AMS-10	Custom memory view generation and compiler-equivalent characterisation across corners		10w	12 _{M/M}	
AMS-11	Schematic design and pre-layout simulation		16w	20 _{M/M}	
AMS-12	Custom layout with DRC / LVS / antenna and fill closure		18w	26 _{M/M}	
AMS-13	Post-layout extracted simulation across PVT and Monte Carlo		16w	20 _{M/M}	
AMS-14	Reliability checks — EM/IR, aging, ESD, latch-up		10w	10 _{M/M}	

AMS-15	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)	8w	8 M/M
AMS-16	AMS-digital co-simulation and integration support	16w cont.	12 M/M

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-01

AMS IP specification and budget allocation (jitter, BER, PSRR, area)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	8.0 M/M	~5.8 FTE	6	no	AMS architect
w0-w6 in stage	planned	while it runs	2 in parallel	has float	AMS lead approves

PURPOSE

Turn each custom block from a name on the make list into **a specification with budgets** — jitter, BER, PSRR, area, power — that a circuit designer can start against and a reviewer can hold them to.

Analog specifications are budgets before they are targets. Total jitter is divided between the PLL, the distribution network and the PHY; each owner gets a share, and a share nobody allocated becomes an argument at integration when the eye closes and no one can say whose fault it is.

WORK FLOW

6 STEPS · 2 LANES

w0	w2	w4	w6
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Custom block list confirmation from the make decisions	0.5w	main
2	System-level budget decomposition — jitter, noise, BER	1.5w	main
3	Per-block specification authoring	2w	main
4	Area and power allocation per block	1w	parallel
5	PSRR and supply sensitivity requirements	1w	parallel
6	Specification review and owner acceptance	2w	main

Step 2 is the one that prevents the integration argument. A jitter budget divided explicitly between PLL, distribution and PHY gives each owner a number; an unallocated total gives each owner a reason to believe the problem is somewhere else.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Make decisions from IPR-03
- Custom memory scope from PDK-06
- Interface specifications from ARCH-04
- Power and clock architecture from ARCH-05
- Block PPA budgets from ARCH-06

PRODUCES

- Custom block list with owners
- System-level budget decomposition
- Per-block specifications with jitter, BER, PSRR and area budgets
- Area and power allocation per block
- Specification review and acceptance record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D1	PRODUCES	AMS IP specifications and design review packages. This activity is the deliverable — the specifications every other AMS activity is executed and reviewed against.
AMS-D8	FEEDS	Integration guide. The specification is what the integration guide's limitations are stated relative to.

RISKS AND WATCH-OUTS

- ◆ **Budgets stated as totals rather than allocations.** An unallocated jitter budget produces three designers each assuming the others left them room.
- ◆ **Specifications written without the PHY vendor.** Where a PHY is bought, its jitter tolerance bounds what the PLL must deliver, and the two have to be written together.
- ◆ **Area allocated before the architecture is understood.** Analog area is set by matching, isolation and routing, and an allocation made from a digital estimate is not usable.
- ◆ **PSRR requirements omitted.** Supply sensitivity is what turns a working block into a failing one once the digital neighbours switch, and it is rarely specified up front.
- ◆ **No owner acceptance.** A specification the designer has not accepted is one they will design around rather than to.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Per-block specification authoring	3.0	● Make decisions available from IPR-03	AMS-02 PLL / clock generator design and closure
Budget decomposition	2.0	● Custom memory scope handed over by PDK-06	AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)
Area and power allocation	1.5	● Interface and clock architecture defined by ARCH-04 and ARCH-05	AMS-04 LDO, bandgap and power-management cell design
Review and acceptance	1.5		AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing
Total	8.0	EXIT CRITERIA	AMS-11 Schematic design and pre-layout simulation
WHO IT TAKES	~5.8 FTE	● Every system budget decomposed into per-block allocations	MEASURED BY
AMS architect		● Each specification accepted by its named owner	● Blocks with an accepted specification
Owns the budgets and the specification set		● Area and power allocations reconciled against ARCH-06	● System budgets fully allocated
Block circuit owners			● Specification changes after design start
Accept or contest their specification		DEPENDS ON	
Interface architect		IPR-03 Make / buy / reuse decision per IP block	
Interface-side requirements and PHY interaction		PDK-06 Custom / pushed-rule memory instance decision, scope and schedule impact assessment	
Power architect		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E)	
Supply, PSRR and rail assumptions		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
AMS lead		ARCH-06 PPA budget allocation per block	
Review chair and arbitration			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-02

PLL / clock generator design and closure

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
26w	32.0 M/M	~5.3 FTE	10	yes	PLL circuit designer
w4–w30 in stage	planned	while it runs	3 in parallel	no float	AMS lead approves

PURPOSE

Design the **clock generation** the whole chip runs on — PLLs, their loop dynamics, their supply immunity and their jitter — and close them across process, voltage and temperature.

A PLL is small, slow to build and impossible to work around. Twenty-six weeks for a block of a few hundred microns square looks disproportionate until the first silicon that will not lock over temperature, at which point the entire program is waiting on a circuit nobody can patch.

WORK FLOW

10 STEPS · 2 LANES

w4	w10	w16	w22	w28	w30
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MAIN SEQUENCE

1	2	3	4	7	8	10
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IN PARALLEL

5	6	9
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#	STEP	TAT	LANE
1	Architecture selection — integer, fractional, ring or LC	2w	main
2	Loop dynamics and phase noise budget	2w	main
3	VCO design and tuning range definition	4w	main
4	Charge pump, loop filter and divider design	4w	main
5	Supply regulation and PSRR strategy	3w	parallel
6	Lock detection, calibration and startup sequencing	3w	parallel
7	Layout with matching, shielding and isolation	6w	main
8	Post-layout jitter closure across PVT	5w	main
9	Clock distribution interface definition with physical design	3w	parallel
10	Characterisation and macro release	3w	main

Step 8 is where the schedule risk lives. Post-layout jitter is routinely worse than pre-layout by more than the margin, and the fix is a layout change that takes the six weeks of step 7 again.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Jitter and PSRR budgets from AMS-01
- Clock architecture and DVFS points from ARCH-05
- Reference clock specification from the system requirements
- Node device models and PDK from PDK-01
- Supply and rail topology from ARCH-05

PRODUCES

- PLL architecture selection and loop design
- VCO and loop component designs
- Supply regulation and calibration scheme
- Post-layout jitter results across PVT
- Characterised PLL macro with views

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D1	FEEDS	AMS IP specifications and design review packages. The PLL review package is the largest single item in the set.
AMS-D2	FEEDS	Characterisation reports across PVT and Monte Carlo. Jitter across PVT is the headline number in the report.
AMS-D5	FEEDS	Hard macro GDS with abstract views. The PLL is one of the macros AMS-15 abstracts and hands to physical design.

RISKS AND WATCH-OUTS

- ◆ **Post-layout jitter worse than budget.** The commonest analog schedule failure, and the fix is a layout iteration measured in weeks, not days.
- ◆ **PSRR ignored until integration.** A PLL that meets jitter on a clean supply and fails beside switching digital logic has met a specification nobody can use.
- ◆ **Tuning range too narrow across corners.** A VCO that cannot cover the DVFS range at the slow corner is a DVFS policy that has to be reduced in silicon.
- ◆ **Lock time and startup unspecified.** Boot sequencing depends on lock time, and discovering it at bring-up delays every subsequent test.
- ◆ **One PLL designed for several very different uses.** Sharing one design across HBM, SerDes and core clocking usually means it is optimal for none.

EFFORT SPLIT	32 M/M	ENTRY CRITERIA	FEEDS INTO
VCO and loop design	9.0	Jitter and PSRR budgets accepted in AMS-01	AMS-11 Schematic design and pre-layout simulation
Layout	7.0	Clock architecture and DVFS points fixed by ARCH-05	AMS-12 Custom layout with DRC / LVS / antenna and fill closure
Post-layout jitter closure	6.0	Device models available from the tracked PDK	AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
Supply regulation and calibration	5.0		
Characterisation and release	3.0		
Distribution interface	2.0		
Total	32.0		
WHO IT TAKES	~5.3 FTE	EXIT CRITERIA	
PLL circuit designer Owns the architecture, the loop and the closure		- Jitter closed post-layout across PVT, not pre-layout - Lock time and startup sequence characterised and documented - Macro released with views and a behavioural model	AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
Analog layout engineer Matching, shielding and isolation			PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Verification engineer, AMS Behavioural model and co-simulation			BU-03 Reset, clocking and PLL lock validation
Clocking architect Distribution interface and jitter budget		DEPENDS ON	MEASURED BY
AMS lead Review gates and schedule		AMS-01 AMS IP specification and budget allocation (jitter, BER, PSRR, area)	- Post-layout jitter against budget - Tuning range coverage at the slow corner - Layout iterations after first post-layout simulation
		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
		PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-03

SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
34w w4–w38 in stage	60.0 M/M planned	~7.7 FTE while it runs	10 3 in parallel	yes no float	SerDes and PHY lead Vendor technical contact approves

PURPOSE

Deliver the **high-speed interfaces** — PCIe or CXL, die-to-die, HBM PHY — either designed from scratch or hardened from a vendor's deliverable onto this node and this flavour sheet.

At thirty-four weeks and sixty man-months this is the largest single activity in the entire template, and it is the one most likely to decide the tapeout date. Even a bought PHY is not free: hardening, integration, and closing its analog margins on this process option is a project, and the vendor's schedule for it is not the program's.

WORK FLOW10 STEPS · 2 LANES

w4	w11	w18	w25	w32	w38
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MAIN SEQUENCE

1	2	4	6	7	8	10
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IN PARALLEL

3	5	9
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#	STEP	TAT	LANE
1	PHY strategy per interface — build, buy hardened, or buy soft	2w	main
2	Vendor deliverable ingestion and gap assessment	3w	main
3	Equalisation and link training architecture	4w	parallel
4	TX and RX front-end design or adaptation	7w	main
5	Clock recovery and per-lane timing	5w	parallel
6	HBM PHY integration and training sequence	6w	main
7	Layout, matching and lane pitch closure	8w	main
8	Post-layout BER and eye closure across PVT	6w	main
9	Compliance simulation against the interface specification	4w	parallel
10	Macro release and integration support	2w	main

Step 7 is eight weeks of layout because lane pitch is a floorplan constraint as much as a circuit one — the PHY has to fit the bump pitch `ARCH-08` defined, and a pitch mismatch found here forces a change at the die edge.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Interface selections and revisions from ARCH-04
- Vendor deliverables and licences from IPR-07

PRODUCES

- PHY strategy per interface
- Vendor deliverable gap assessment

- Bump pitch and PHY placement from ARCH-08
 - Channel budgets from SIPI-08
 - Jitter budget from AMS-01
- TX/RX front-end and clock recovery designs
 - Post-layout BER and eye results across PVT
 - Compliance simulation results
 - Released PHY macros with integration support

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D2	PRODUCES	Characterisation reports across PVT and Monte Carlo. The PHY's eye and BER characterisation is the report's most consequential section.
AMS-D5	FEEDS	Hard macro GDS with abstract views. The PHY macros are the largest and most constrained of the set.
AMS-D8	FEEDS	Integration guide with known limitations and errata. PHY limitations — training corner cases, unsupported modes — are what integration most needs stated.

RISKS AND WATCH-OUTS

- ◆

Bought PHY assumed to need no work. Hardening, integration and margin closure on this flavour sheet is a multi-quarter project regardless of who designed the circuit.
- ◆

Lane pitch incompatible with the bump map. Discovered at layout, it forces a change at the die edge where there is least room to make one.
- ◆

Compliance simulated late. A PHY that closes its own eye and fails the interface specification's compliance mask has to be reworked with no schedule left.
- ◆

Channel budget not shared with SI. The PHY closes against an assumed channel; if SIPI-07 extracts a worse one, the margin was never real.
- ◆

Vendor support model untested until integration. The first hard question is when a support relationship is discovered to be a ticketing system.

EFFORT SPLIT	60 M/M	ENTRY CRITERIA	FEEDS INTO
TX and RX front-end design	14.0	- Interface selections pinned by ARCH-04 - Vendor licences executed where PHYs are bought - Bump pitch and PHY placement known from ARCH-08	AMS-12 Custom layout with DRC / LVS / antenna and fill closure
Layout and pitch closure	13.0		AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
Post-layout BER and eye closure	11.0		AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
HBM PHY integration	8.0	EXIT CRITERIA - BER and eye closed post-layout across PVT at the specified channel - Compliance simulated against the interface specification, not only internally - Macros released with a stated limitation list	SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
Clock recovery and per-lane timing	7.0		SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)
Compliance simulation	5.0		PD-04 Bump and RDL planning with chip-package co-design feedback
Macro release and support	2.0	DEPENDS ON AMS-01 AMS IP specification and budget allocation (jitter, BER, PSRR, area) ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E) ARCH-08 Chip-level floorplan intent, pin and bump budget IPR-07 Licence negotiation, PO issue and delivery-date commitment	BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Total	60.0		MEASURED BY
WHO IT TAKES	~7.7 FTE		- Eye margin against the compliance mask - Lane pitch against the bump map constraint - Weeks of vendor-side schedule the program cannot influence
SerDes and PHY lead			
Owns the strategy, the design and the closure			
Analog circuit designers			
TX/RX front end and clock recovery			
Analog layout engineers			
Lane pitch, matching and shielding			
SI liaison			
Channel budget and compliance conditions			
Vendor technical contact			
Hardening support and deliverable gaps			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-04

LDO, bandgap and power-management cell design

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
20w	22.0 M/M	~4.8 FTE	8	no	Power management circuit designer
w6–w26 in stage	planned	while it runs	2 in parallel	has float	Reliability engineer approves

PURPOSE

Build the **power management cells** everything else depends on — LDOs, bandgap references, bias generation — quietly and correctly, because every other block's specification assumes them.

These blocks are small, unglamorous and load-bearing. A bandgap that drifts moves every analog threshold on the die; an LDO with insufficient PSRR at the switching frequency turns the PLL's careful jitter budget into noise. They are specified against by everyone and reviewed by almost nobody.

WORK FLOW

8 STEPS · 2 LANES

w6	w10	w14	w18	w22	w26
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MAIN SEQUENCE

1	2	3	6	7	8
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Supply topology and regulation requirement per domain	1.5w	main
2	Bandgap reference design and trim strategy	3.5w	main
3	LDO design — load range, stability, PSRR	5w	main
4	Bias generation and distribution	3w	parallel
5	Startup, sequencing and brown-out behaviour	2.5w	parallel
6	Layout with matching and supply isolation	4w	main
7	Post-layout stability and PSRR verification across PVT	4w	main
8	Trim and calibration procedure definition for test	2w	main

Step 8 is easy to leave until later and expensive to. A trim procedure that test engineering only sees at `TEST-05` becomes a test-time cost, and a bandgap with no defined trim is a part with a distribution nobody can tighten.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Supply and rail topology from ARCH-05
- PSRR requirements from AMS-01
- Load profiles from the digital power estimate
- Node device models and precision element availability
- Test and trim capability from TEST-01

PRODUCES

- Supply topology and regulation plan
- Bandgap reference design with trim strategy
- LDO designs with stability and PSRR results
- Bias generation and distribution scheme
- Trim and calibration procedure for production test

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D2	FEEDS	Characterisation reports across PVT and Monte Carlo. Bandgap drift and LDO PSRR across corners belong in the same report as the PLL and PHY results.
AMS-D5	FEEDS	Hard macro GDS with abstract views. Power management cells are delivered as macros like any other analog block.
AMS-D7	FEEDS	Reliability report. LDO pass devices and bandgaps carry EM and aging exposure that belongs in the reliability assessment.

RISKS AND WATCH-OUTS

- ◆ **PSRR specified at DC.** The frequency that matters is where the digital load switches, and a regulator flat at DC can be transparent at that frequency.
- ◆ **Stability verified only at nominal load.** LDO compensation has to hold across the full load range including the transitions, which is where it usually does not.
- ◆ **Bandgap trim undefined.** An untrimmed reference has a distribution wide enough to move every analog threshold, and there is no way to tighten it in production.
- ◆ **Brown-out and startup behaviour unspecified.** The part's behaviour at supply ramp is discovered at bring-up, on silicon that cannot be changed.
- ◆ **Reviewed by nobody.** Blocks everyone assumes and nobody reviews are the ones that fail in a way that looks like someone else's problem.

EFFORT SPLIT	22 M/M	ENTRY CRITERIA	FEEDS INTO
LDO design	6.0	Supply topology defined by ARCH-05	AMS-12 Custom layout with DRC / LVS / antenna and fill closure
Bandgap and trim strategy	4.5	PSRR and load requirements available from AMS-01	AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
Layout	4.0	Node precision element characterisation available	AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up
Post-layout verification	3.5		AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
Bias generation	2.5		SIPI-05 Decoupling capacitor budget and placement optimisation across die, package and board
Trim procedure for test	1.5		TEST-05 Wafer sort test program development
Total	22.0	EXIT CRITERIA - PSRR verified at the digital switching frequency, not only at DC - Stability held across the full load range post-layout - Trim procedure agreed with test engineering	MEASURED BY - PSRR at the switching frequency against requirement - Load range over which stability is proven - Reference distribution before and after trim
WHO IT TAKES	~4.8 FTE	DEPENDS ON AMS-01 AMS IP specification and budget allocation (jitter, BER, PSRR, area) ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
Power management circuit designer Owns LDOs, bandgap and bias			
Analog layout engineer Matching, isolation and supply routing			
Power architect Topology, domains and load profiles			
Test engineering liaison Trim and calibration procedure			
Reliability engineer EM and aging on pass devices			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-05

Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	14.0 M/M	~6.1 FTE	8	yes	Memory circuit architect
w2–w12 in stage	planned	while it runs	3 in parallel	no float	PDK owner approves

PURPOSE

Architect the **custom SRAM instance** the compiler could not deliver — bitcell choice, array organisation, sense amplifier topology and replica timing — against the density and Vmin the block budget demands.

This activity exists only because PDK-06 opened it. That decision came with a quantified gap and a fixed window, and the architecture chosen here determines whether the rest of the custom memory chain can close inside it. A conservative architecture will not recover the density; an aggressive one may not close Vmin.

WORK FLOW8 STEPS · 2 LANES

w2	w4	w6	w8	w10	w12
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MAIN SEQUENCE

1	2	3	4	7
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IN PARALLEL

5	6	8
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#	STEP	TAT	LANE
1	Scope confirmation against the PDK-06 decision and gap	0.8w	main
2	Bitcell selection — foundry cell, pushed rules, or custom	1.8w	main
3	Array organisation — banking, segmentation, aspect ratio	2w	main
4	Sense amplifier topology and offset budget	2w	main
5	Replica timing and self-timed path design	1.8w	parallel
6	Peripheral circuit architecture — decoders, drivers, IO	1.5w	parallel
7	Density projection against the block budget	3.5w	main
8	Architecture review and freeze	1.5w	parallel

Step 7 closes the loop with `PDK-05`. If the projected density still misses the budget, the answer is to reopen the architecture now rather than to discover it after sixteen weeks of layout in `AMS-07`.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Custom instance decision and gap from PDK-06
- Density and Vmin measurements from PDK-05
- Memory hierarchy requirements from ARCH-03
- Block memory budget from ARCH-06

PRODUCES

- Bitcell selection with rule implications
- Array organisation and banking scheme
- Sense amplifier topology and offset budget
- Replica timing scheme

- Foundry bitcell options and pushed-rule policy
- Density projection against budget
- Architecture freeze record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D3	PRODUCES	Custom SRAM instance specification with Vmin and sigma-yield report. The specification half of the deliverable; AMS-08 supplies the yield half.
AMS-D1	FEEDS	AMS IP specifications and design review packages. The array's architecture review is one of the stage's formal gates.

RISKS AND WATCH-OUTS

- ◆ **Architecture that does not recover the gap.** A conservative organisation produces a custom array with the compiler's density and none of the compiler's maintenance advantages.
- ◆ **Bitcell pushed beyond foundry appetite.** A cell the foundry will not approve is a layout that cannot tape out, and the approval path runs through PDK-06.
- ◆ **Sense amplifier offset budget optimistic.** Offset drives Vmin more than any other single choice, and an optimistic budget is discovered in AMS-08 when it is far too late to change topology.
- ◆ **Replica timing not proven across corners.** Self-timed paths that track at nominal and diverge at the corner produce read failures that look random.
- ◆ **Architecture frozen without the density projection.** Freezing before step 7 means committing sixteen weeks of layout to an organisation that may not close the gap.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Array organisation	4.0	● Custom instance decision taken by PDK-06 with a quantified gap	AMS-06 Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse
Sense amplifier and offset budget	3.5	● Compiler measurements available from PDK-05	AMS-07 Custom array layout and pushed-rule DRC closure with foundry approval
Bitcell selection	2.5	● Foundry pushed-rule appetite known	AMS-08 Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation
Replica timing	2.0		AMS-09 Redundancy and repair scheme integration with the BISR architecture
Density projection and review	2.0		AMS-10 Custom memory view generation and compiler-equivalent characterisation across corners
Total	14.0	EXIT CRITERIA ● Projected density closes the gap the decision was taken on ● Bitcell choice within the foundry's stated approval appetite ● Architecture frozen with an offset and Vmin budget attached	
WHO IT TAKES	~6.1 FTE	DEPENDS ON PDK-05 Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage PDK-06 Custom / pushed-rule memory instance decision, scope and schedule impact assessment ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels) ARCH-06 PPA budget allocation per block	MEASURED BY ● Projected density against the PDK-05 gap ● Offset budget against what the topology can deliver ● Architecture reopened after freeze
Memory circuit architect Owns the array architecture and its freeze			
Bitcell and device engineer Cell selection and rule implications			
Sense amplifier designer Topology and offset budget			
Memory layout lead Density projection and floorplan feasibility			
PDK owner Foundry pushed-rule approval path			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-06

Read / write assist circuit design for Vmin
— WL boost, negative bitline, VDD collapse

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	7	no	Memory circuit designer
w10–w20 in stage	planned	while it runs	2 in parallel	has float	Memory architect approves

PURPOSE

Design the **assist circuits that buy Vmin** — wordline boost, negative bitline, VDD collapse — and decide how much of each the array actually needs.

Assist is what makes an aggressive array workable, and it is not free: each technique costs area, power and complexity, and some of them interact badly. The design question is not which assist is best but which combination reaches the Vmin target at the least cost, verified statistically rather than at nominal.

WORK FLOW

7 STEPS · 2 LANES

w10	w12	w14	w16	w18	w20
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MAIN SEQUENCE

1	2	5	6	7
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Vmin gap analysis against the AMS-05 architecture	1w	main
2	Assist technique selection and interaction study	2w	main
3	Wordline boost circuit design	2w	parallel
4	Negative bitline circuit design	2w	parallel
5	VDD collapse and read-assist timing	2w	main
6	Assist control, timing and calibration	2w	main
7	Vmin verification with assist enabled, across corners	3w	main

Steps 3 and 4 run in parallel and step 2 decides whether both are needed. Write assist and read assist pull in opposite directions on the cell, and enabling both without studying the interaction can leave Vmin worse than either alone.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Array architecture and Vmin budget from AMS-05
- Vmin requirements from ARCH-05 operating points
- Bitcell characterisation data
- Node assist circuit precedents from the foundry
- Power budget for the array from ARCH-06

PRODUCES

- Vmin gap analysis with assist requirement
- Assist technique selection and interaction findings
- Wordline boost, negative bitline and collapse circuits
- Assist control and calibration scheme
- Vmin verification results with assist enabled

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

- AMS-D3

FEEDS

Custom SRAM instance specification with Vmin and sigma-yield report. The Vmin the instance is specified at is the Vmin with assist enabled, and this activity produces it.
- AMS-D2

FEEDS

Characterisation reports. Assist-enabled Vmin across corners is one of the report's core tables.

RISKS AND WATCH-OUTS

- ◆ **Assist techniques combined without an interaction study.** Read and write assist push the cell in opposite directions; enabling both blindly can make Vmin worse.
- ◆ **Assist verified at nominal.** Vmin is a statistical quantity; verification at nominal proves nothing about the tail that decides yield.
- ◆ **Boost levels beyond reliability limits.** Wordline boost stresses the cell, and a level chosen for Vmin alone shortens life in a way AMS-14 will find late.
- ◆ **Assist timing not calibrated.** Fixed assist timing that works at one corner and misfires at another produces intermittent failures indistinguishable from a marginal cell.
- ◆ **Area cost of assist unbudgeted.** Assist circuits sit in the periphery and can consume the density the custom array was built to recover.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Assist circuit design	4.5	● Array architecture frozen by AMS-05	AMS-07 Custom array layout and pushed-rule DRC closure with foundry approval
Vmin verification	3.0	● Vmin target set from the operating points in ARCH-05	AMS-08 Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation
Assist control and calibration	2.5	● Bitcell characterisation data available	AMS-10 Custom memory view generation and compiler-equivalent characterisation across corners
Interaction study	2.0		AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up
Total	12.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.2 FTE	● Vmin target met statistically, not at nominal	● Vmin with assist against the target
Memory circuit designer		● Assist interactions studied rather than assumed	● Area cost of assist against the recovered density
Owns the assist circuits and their verification		● Boost levels within the reliability limits AMS-14 will check	● Assist-related failures found in silicon
Bitcell and device engineer			
Cell response to assist and reliability limits		DEPENDS ON	
Statistical analysis engineer		AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing	
Vmin verification across the distribution		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
Reliability engineer			
Boost stress and aging implications			
Memory architect			
Area and power cost against the recovered density			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-07

Custom array layout and pushed-rule DRC closure with foundry approval

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w w18–w34 in stage	22.0 M/M planned	~6.0 FTE while it runs	8 2 in parallel	yes no float	Memory layout lead Foundry technical contact approves

PURPOSE

Lay out the custom array at **pushed rules, and get the foundry to approve it** — because a layout the foundry will not sign is a layout that cannot tape out.

Sixteen weeks of layout is what the density recovery actually costs. Pushed rules are the mechanism, and they carry an approval process with its own duration and its own possibility of refusal. Running layout and approval in sequence would not fit the window, so they run together, which means designing against rules that are not yet certain.

WORK FLOW

8 STEPS · 2 LANES

w18	w22	w26	w30	w34
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MAIN SEQUENCE

1	2	3	6	7	8
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Pushed-rule set proposal and foundry pre-alignment	2w	main
2	Bitcell array layout at pushed rules	4w	main
3	Peripheral and assist circuit layout	3.5w	main
4	Dummy, fill and density compliance	2w	parallel
5	Foundry approval submission and iteration	5w	parallel
6	DRC closure including pushed-rule waivers	3w	main
7	LVS and antenna closure	2w	main
8	Layout freeze and extraction handoff	1.5w	main

Step 5 runs in parallel out of necessity rather than choice. Waiting for approval before laying out would not fit the window, so the array is laid out against rules that may still be refused — which is why step 1's pre-alignment matters more than its two weeks suggest.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Array architecture from AMS-05
- Assist circuits from AMS-06
- Pushed-rule policy and approval path from PDK-06
- Design rule manual and exceptions from PDK-02

PRODUCES

- Pushed-rule set with foundry pre-alignment
- Bitcell array and peripheral layout
- Density and fill compliance record
- Foundry approval record and waivers

- Density target from the block budget
- DRC, LVS and antenna clean layout

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D6	PRODUCES	Per-macro DRC / LVS clean signoff, pushed rules approved by foundry. This activity is the deliverable, and the approval is the half that cannot be produced internally.
AMS-D3	FEEDS	Custom SRAM instance specification. The achieved density is a layout result, and it is what the specification finally states.

RISKS AND WATCH-OUTS

- ◆ **Foundry refuses a pushed rule after layout is built on it.** The mitigation is pre-alignment in step 1, and it is the single most valuable two weeks in the activity.
- ◆ **Density target missed despite the pushed rules.** The array then costs sixteen weeks and delivers the compiler's density, which is the worst available outcome.
- ◆ **Approval iterations underestimated.** Each foundry round trip runs to weeks, and the five-week allowance assumes no more than two.
- ◆ **Fill and density compliance left to the end.** Fill interacts with the array's matching and can undo careful layout if applied late.
- ◆ **Layout frozen before extraction is proven.** An array that extracts badly has to be re-laid out, and the extraction handoff is where that is found.

EFFORT SPLIT	22 M/M	ENTRY CRITERIA	FEEDS INTO
Array layout	7.0	● Array architecture frozen by AMS-05	AMS-08 Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation
Peripheral and assist layout	5.5	● Assist circuits available from AMS-06	
Foundry approval iteration	3.5	● Foundry pushed-rule approval path agreed in PDK-06	AMS-10 Custom memory view generation and compiler-equivalent characterisation across corners
DRC and LVS closure	4.0		AMS-12 Custom layout with DRC / LVS / antenna and fill closure
Fill and density compliance	2.0	EXIT CRITERIA ● Every pushed rule approved by the foundry in writing ● Achieved density meets the target the decision was taken on ● Layout DRC, LVS and antenna clean, and extractable	AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
Total	22.0		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
WHO IT TAKES	~6.0 FTE	DEPENDS ON AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing AMS-06 Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse PDK-02 Design rule manual review; restricted and recommended rule disposition PDK-06 Custom / pushed-rule memory instance decision, scope and schedule impact assessment	MEASURED BY ● Pushed rules approved against those proposed ● Achieved density against the target ● Foundry approval iterations against the allowance
Memory layout lead Owns the layout and the density result			
Layout engineers Array, periphery and assist layout			
PDK owner Foundry approval submission and iteration			
Physical verification engineer DRC, LVS and antenna closure			
Foundry technical contact Pushed-rule review and approval			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-08

Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	7	no	Statistical analysis engineer
w20–w30 in stage	planned	while it runs	2 in parallel	has float	AMS lead approves

PURPOSE

Prove the array works **at the tail of the distribution rather than at nominal** — sigma-Vmin, bitcell variation, sense amplifier offset — using importance sampling because brute-force Monte Carlo cannot reach six sigma.

A memory array of this size fails at the tail by definition: with hundreds of millions of cells, the six-sigma cell is not a hypothetical, it is present in every part. Nominal simulation says nothing about it, and the yield the cost model assumed depends entirely on where that tail sits.

WORK FLOW

7 STEPS · 2 LANES

w20	w22	w24	w26	w28	w30
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Statistical methodology selection — importance sampling, scaled sigma	1.5w	main
2	Bitcell variation model validation against silicon data	2w	main
3	Sigma-Vmin analysis for read, write and retention	3w	main
4	Sense amplifier offset distribution analysis	2w	parallel
5	Peripheral and timing variation contribution	1.5w	parallel
6	Array-level yield projection with redundancy applied	2w	main
7	Statistical margin report and sign-off	1.5w	main

Step 2 is the step that makes the rest meaningful. A variation model that has not been validated against silicon produces a six-sigma number with an unknown error, and the number is then used to justify a yield assumption in the cost model.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Array architecture from AMS-05
- Assist circuits and Vmin results from AMS-06
- Layout parasitics from AMS-07
- Foundry variation models and silicon data
- Redundancy scheme from AMS-09

PRODUCES

- Statistical methodology and its validation
- Bitcell variation model validation record
- Sigma-Vmin results for read, write and retention
- Offset and peripheral variation contributions
- Array yield projection with redundancy
- Statistical margin report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D3	PRODUCES	Custom SRAM instance specification with Vmin and sigma-yield report. The yield half of the deliverable, and the half the cost model depends on.
AMS-D2	FEEDS	Characterisation reports across PVT and Monte Carlo. The statistical results sit alongside the corner results, and the two say different things.

RISKS AND WATCH-OUTS

- ◆ **Brute-force Monte Carlo used at the tail.** Reaching six sigma by direct sampling is computationally impossible; a run that terminates has not reached the tail it claims.
- ◆ **Variation model unvalidated.** The six-sigma answer inherits the model's error, and nobody knows its size.
- ◆ **Redundancy assumed to cover the tail.** Repair fixes a bounded number of failing bits; a distribution problem produces more than the repair scheme can absorb.
- ◆ **Retention overlooked.** Read and write Vmin get the attention; retention failures at low voltage and high temperature are found in qualification.
- ◆ **Result reported without the methodology.** A sigma number quoted without stating how it was obtained cannot be compared or challenged.

EFFORT SPLIT	12 M/M	EXIT CRITERIA	FEEDS INTO
Sigma-Vmin analysis	4.0	● Sigma targets met for read, write and retention	AMS-10 Custom memory view generation and compiler-equivalent characterisation across corners
Variation model validation	2.5	● Variation model validated against silicon, not assumed	AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up
Offset and peripheral analysis	2.5	● Yield projection handed to the cost model with its methodology	DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)
Yield projection	2.0		MP-05 Yield learning, failure pareto and defect analysis
Report and sign-off	1.0		
Total	12.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~5.2 FTE	AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing	● Sigma achieved against target for each failure mode
Statistical analysis engineer Owns the methodology and the sigma results		AMS-06 Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse	● Variation model error against silicon
Memory circuit designer Interprets results back into circuit changes		AMS-07 Custom array layout and pushed-rule DRC closure with foundry approval	● Array yield projection against the cost model assumption
Device and bitcell engineer Variation model validation against silicon		AMS-09 Redundancy and repair scheme integration with the BISR architecture	
Yield engineer Array yield projection and cost model input		PDK-05 Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage	
AMS lead Signs the statistical margin report			
ENTRY CRITERIA			
● Array architecture and assist circuits complete			
● Layout parasitics available from AMS-07			
● Foundry variation models and silicon data accessible			

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AMS-09

Redundancy and repair scheme integration with the BISR architecture

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w16–w22 in stage	6.0 M/M planned	~4.3 FTE while it runs	6 2 in parallel	no has float	Memory circuit designer Memory architect approves

PURPOSE

Build **redundancy and repair into the array** and make it match the BISR architecture DFT is designing — spare rows and columns, fuse allocation, repair algorithm, and the interface between them.

Redundancy is worth several points of yield and only if the two halves agree. The array provides spares and a repair interface; DFT-03 provides the BIST engine and the repair analysis. A mismatch between them is found at pattern generation, when both are effectively frozen.

WORK FLOW6 STEPS · 2 LANES

w16w18w20w22

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Redundancy scheme selection — row, column, or both	1w	main
2	Spare allocation against the projected defect distribution	1.5w	main
3	Repair interface definition with the BISR architecture	1.3w	main
4	Fuse and repair storage allocation	1w	parallel
5	Repair algorithm agreement with DFT	1w	parallel
6	Redundancy verification and yield contribution analysis	2.3w	main

Steps 3 and 5 are joint work with `DFT-03` rather than handoffs. Both sides can specify a repair scheme that is internally consistent and mutually incompatible, and only writing them together prevents it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Array architecture from AMS-05
- BISR architecture intent from DFT-03
- Defect density projection from TECH-07
- Fuse map and lifecycle model from ARCH-07
- Yield target from the cost model

PRODUCES

- Redundancy scheme with spare allocation
- Repair interface specification, agreed with DFT
- Fuse and repair storage allocation
- Repair algorithm agreement
- Redundancy yield contribution analysis

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D3	FEEDS	Custom SRAM instance specification. Repairable and unrepairable yield are different numbers, and the specification has to state which it quotes.
AMS-D8	FEEDS	Integration guide. The repair interface is one of the integration points most likely to be misunderstood, and it belongs in the guide.

RISKS AND WATCH-OUTS

- ◆ **Repair interface specified independently by both sides.** Two consistent specifications that disagree with each other, discovered at ATPG when both are frozen.
- ◆ **Spare allocation not matched to the defect distribution.** Row spares against a column-clustered defect mode repair nothing.
- ◆ **Fuse capacity insufficient.** Repair data has to be stored; an allocation made without counting the worst-case repair set produces parts that cannot be repaired.
- ◆ **Redundancy credited twice.** Yield projections that apply repair in both the array analysis and the product model double-count the benefit.
- ◆ **Repair algorithm assumed rather than agreed.** The BIST engine implements one algorithm; the array assumes another, and the mismatch is silent until silicon.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Redundancy scheme and spare allocation	2.0	● Array architecture available from AMS-05	AMS-08 Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation
Repair interface with DFT	1.5	● BISR architecture intent available from DFT-03	AMS-10 Custom memory view generation and compiler-equivalent characterisation across corners
Yield contribution analysis	1.5	● Defect density projection from TECH-07	DFT-03 MBIST / BIRA / BISR architecture for embedded memories
Fuse allocation and algorithm agreement	1.0		DFT-10 eFuse, chip ID and memory repair infrastructure
Total	6.0	EXIT CRITERIA	TEST-05 Wafer sort test program development
WHO IT TAKES	~4.3 FTE	● Repair interface written jointly with DFT, not in parallel ● Spare allocation matched to the projected defect modes ● Fuse capacity sized against the worst-case repair set	MEASURED BY
Memory circuit designer Owns the redundancy scheme and the array side		DEPENDS ON	● Repair interface mismatches found at pattern generation
DFT engineer BISR architecture and the repair algorithm		AMS-05 Custom SRAM instance architecture — bitcell selection, array organisation, sense amp and replica timing	● Yield gain from repair against projection
Yield engineer Defect distribution and repair effectiveness		DFT-03 MBIST / BIRA / BISR architecture for embedded memories	● Fuse capacity against worst-case repair need
Fuse and lifecycle engineer Repair storage allocation		TECH-07 Foundry roadmap and risk-production timing alignment	
Memory architect Area cost of spares against yield gain			

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AMS-10

Custom memory view generation and compiler-equivalent characterisation across corners

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	7	yes	Memory characterisation engineer
w28–w38 in stage	planned	while it runs	3 in parallel	no float	Signoff methodology engineer approves

PURPOSE

Generate the **views the custom array needs to behave like a compiler instance** — timing, power, physical, functional — characterised across the same corners, so nothing downstream has to treat it as a special case.

A custom array that cannot be dropped into the flow like a compiler macro has not finished. Synthesis, place-and-route, timing signoff and verification all expect a standard view set; anything missing becomes a manual exception that has to be remembered at every iteration by everyone.

WORK FLOW

7 STEPS · 2 LANES

w28w30w32w34w36w38

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	View requirement definition against the tool matrix	1w	main
2	Timing characterisation across the signoff corners	3w	main
3	Power and leakage characterisation	2w	parallel
4	Physical abstract, blockage and pin view generation	1.5w	parallel
5	Functional and behavioural model generation	1.5w	parallel
6	Compiler-equivalence check against a reference instance	2w	main
7	View release and integration verification	4w	main

Step 6 is the test that matters. Dropping the custom instance into the flow in place of a compiler instance and confirming the tools behave identically is what proves the view set is complete — reading the file list does not.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Layout and extraction from AMS-07
- Vmin and assist results from AMS-06
- Statistical results from AMS-08
- Signoff corner definition from PDK-11

PRODUCES

- Timing views across the signoff corners
- Power and leakage views
- Physical abstract and blockage views
- Functional and behavioural models

- Tool view requirements from PDK-08
- Compiler-equivalence check results
- Released view set

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D4	PRODUCES	Custom memory views characterised to compiler equivalence. This activity is the deliverable, and equivalence is the standard it is measured against.
AMS-D5	FEEDS	Hard macro GDS with abstract views. The custom array is delivered into the macro set on the same terms as the analog blocks.

RISKS AND WATCH-OUTS

- ◆ **Characterised at fewer corners than signoff runs.** A missing corner turns into an extrapolation at S0-02, which the foundry will not accept.
- ◆ **Equivalence assumed from the file list.** A complete-looking view set can still behave differently in the tools, and only a substitution test finds it.
- ◆ **Behavioural model diverging from the circuit.** Verification passes against a model that does not match the array, and the mismatch reaches silicon.
- ◆ **Abstract blockages wrong.** Over-blocking wastes routing resource; under-blocking produces shorts found at S0-02.
- ◆ **Views released before the layout freezes.** Characterising against a layout that then changes means doing the whole activity twice.

EFFORT SPLIT	12 M/M	EXIT CRITERIA	FEEDS INTO
Timing characterisation	4.0	● Characterised at every corner signoff will run ● Equivalence proven by substitution, not by inspection ● Views released only after the layout freeze	AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
Power and leakage characterisation	2.5		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
Equivalence checking	2.0		PD-02 Floorplan, macro placement and partition definition
Physical and functional views	2.0		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Release and integration verification	1.5		DV-11 Gate-level simulation, functional and timing-annotated
Total	12.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~5.2 FTE	AMS-06 Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse	● Corners characterised against signoff corners ● Equivalence test results against a reference instance ● View defects found downstream
Memory characterisation engineer Owns the view set and its characterisation		AMS-07 Custom array layout and pushed-rule DRC closure with foundry approval	
CAD methodology engineer View requirements and equivalence testing		AMS-08 Statistical margin and yield analysis — importance sampling, sigma-Vmin, bitcell variation	
Memory circuit designer Behavioural model fidelity		AMS-09 Redundancy and repair scheme integration with the BISR architecture	
Physical design liaison Abstract and blockage correctness		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
Signoff methodology engineer Corner coverage against the signoff set			
ENTRY CRITERIA			
		● Layout frozen and extractable from AMS-07 ● Signoff corners defined by PDK-11 ● Tool view requirements known from PDK-08	

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AMS-11

Schematic design and pre-layout simulation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	20.0 M/M	~5.4 FTE	7	no	Analog design leads
w6–w22 in stage	planned	while it runs	2 in parallel	has float	PDK owner approves

PURPOSE

Run the **schematic-level design and pre-layout simulation** that every analog block passes through — the common engineering discipline underneath the PLL, the PHY, the regulators and the array periphery.

This is the shared substrate rather than a block: topology exploration, device sizing, pre-layout corner simulation and the design reviews that gate a block into layout. Separating it makes the schedule honest — a block that has not passed pre-layout review is not ready for the sixteen weeks of layout that follow.

WORK FLOW

7 STEPS · 2 LANES

w6	w10	w14	w18	w22
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MAIN SEQUENCE

1	2	3	4	7
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IN PARALLEL

5	6
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#	STEP	TAT	LANE
1	Design environment and testbench framework setup	1.5w	main
2	Topology exploration per block	3w	main
3	Device sizing and operating point definition	3.5w	main
4	Pre-layout corner simulation	3w	main
5	Pre-layout Monte Carlo and mismatch analysis	2.5w	parallel
6	Testbench and regression suite per block	2.5w	parallel
7	Pre-layout design review and layout entry gate	5w	main

Step 7 is a gate rather than a meeting. Layout is the longest and least reversible part of an analog block, and letting a schematic that has not closed at the corners enter layout guarantees that the corner failure is found after the layout is built.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Block specifications from AMS-01
- Node device models from the tracked PDK
- Design environment and simulator versions from PDK-08
- Reference designs from previous programs
- Reliability rules and limits from the DRM

PRODUCES

- Design environment and testbench framework
- Topology selection per block
- Sized schematics with operating points
- Pre-layout corner and Monte Carlo results
- Per-block regression suites
- Layout entry gate records

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D1	FEEDS	AMS IP specifications and design review packages. The pre-layout review package is the formal record that a block was fit to enter layout.
AMS-D2	FEEDS	Characterisation reports. Pre-layout corner results are the baseline post-layout results are compared against.

RISKS AND WATCH-OUTS

- ◆ **Blocks entering layout without passing the gate.** Sixteen weeks of layout on a schematic that fails at the corner is the most expensive way to discover a topology problem.
- ◆ **Corner simulation run at typical mismatch.** Mismatch dominates analog corner behaviour, and a corner run without it is a nominal run in disguise.
- ◆ **Testbenches written per block by different people.** Inconsistent testbenches make results incomparable across blocks and unusable as a regression.
- ◆ **Device models used from an unfrozen PDK.** Analog is far more model-sensitive than digital, and a model update mid-design can invalidate the sizing.
- ◆ **No regression suite.** Every subsequent change is then verified by hand, and the coverage of that verification degrades under schedule pressure.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Device sizing and operating points	5.5	Block specifications accepted in AMS-01	AMS-12 Custom layout with DRC / LVS / antenna and fill closure
Topology exploration	4.5	Device models available from a tracked PDK version	AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
Pre-layout corner simulation	4.0	Simulator versions qualified by PDK-08	AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up
Testbench and regression suites	3.0		AMS-16 AMS–digital co-simulation and integration support
Monte Carlo and mismatch	2.0		
Design reviews	1.0		
Total	20.0		
WHO IT TAKES	~5.4 FTE	EXIT CRITERIA - Every block passes a pre-layout gate before entering layout - Corner simulation includes mismatch, not only process corners - Regression suite exists per block and runs automatically	MEASURED BY - Blocks passing the layout entry gate first time - Corner failures found pre-layout against post-layout - Regression coverage per block
Analog design leads Topology, sizing and pre-layout closure per block		DEPENDS ON	
AMS methodology engineer Environment, testbench framework and regressions		AMS-01 AMS IP specification and budget allocation (jitter, BER, PSRR, area)	
Design reviewers Layout entry gate		PDK-01 PDK version roadmap tracking (0.1 → 0.5 → 1.0 → production) and delta impact analysis	
AMS architect Confirms designs meet their allocated budgets		PDK-08 EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
PDK owner Model version control for analog design			

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AMS-12

Custom layout with DRC / LVS / antenna and fill closure

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
18w w20–w38 in stage	26.0 M/M planned	~6.3 FTE while it runs	8 2 in parallel	yes no float	Analog layout lead Physical design liaison approves

PURPOSE

Lay out every analog and custom block to **DRC, LVS and antenna clean**, with the matching, shielding and fill discipline analog requires — and get the layouts extractable.

Eighteen weeks is the largest single block of time in the stage after the PHY, and it is largely irreducible. Analog layout is where matching, isolation and parasitic control are actually implemented, and none of it can be automated or parallelised the way digital place-and-route can.

WORK FLOW

8 STEPS · 2 LANES

w20	w24	w28	w32	w36	w38
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MAIN SEQUENCE

1	2	3	6	7	8
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Floorplan and pin placement per macro	2w	main
2	Matching-critical device layout	4w	main
3	Shielding, guarding and isolation structures	3w	main
4	Supply and bias routing with EM compliance	3w	parallel
5	Fill and density compliance without disturbing matching	2.5w	parallel
6	DRC closure per macro	4w	main
7	LVS and antenna closure	3w	main
8	Extraction setup and layout freeze	2w	main

Step 5 is where careful layout is most often undone. Fill applied without regard to matching structures changes the parasitics the whole design was tuned against, and the effect is only seen in post-layout simulation.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Sized schematics from AMS-11
- PLL, PHY and regulator designs from AMS-02 to AMS-04
- Design rules and exceptions from PDK-02
- Macro placement intent from ARCH-08 and PD-01
- Golden environment and decks from PDK-13

PRODUCES

- Macro floorplans and pin placement
- Matching-critical device layouts
- Shielding and isolation structures
- EM-compliant supply and bias routing
- DRC, LVS and antenna clean layouts
- Extraction-ready frozen layouts

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D6	PRODUCES	Per-macro DRC / LVS clean signoff. Every analog macro's physical signoff, alongside the pushed-rule approvals from AMS-07.
AMS-D5	FEEDS	Hard macro GDS with abstract views. The layout is the GDS half of every macro delivery.

RISKS AND WATCH-OUTS

- ◆ **Fill applied late and without care.** It disturbs the parasitics the design was tuned against, and the damage appears only in post-layout results.
- ◆ **Matching implemented by intent rather than by construction.** Common-centroid and dummy structures are a layout discipline; describing them in a review is not implementing them.
- ◆ **EM checked only at signoff.** Analog supply routing carries high current density in small geometries, and an EM failure found at S0-03 is a layout rebuild.
- ◆ **Pin placement not agreed with physical design.** A macro whose pins face the wrong way costs routing resource at the top level for the life of the design.
- ◆ **Layout frozen without an extraction check.** A macro that extracts slowly or badly stalls every post-layout simulation that follows.

EFFORT SPLIT	26 M/M	ENTRY CRITERIA	FEEDS INTO
Matching-critical layout	7.0	Schematics passed the AMS-11 layout entry gate	AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
DRC closure	5.0	Design rules and exceptions available from PDK-02	AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up
Shielding and isolation	4.5	Macro placement intent known from ARCH-08	AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
LVS and antenna closure	4.0		PD-02 Floorplan, macro placement and partition definition
Supply and bias routing	3.5		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Fill and extraction setup	2.0	EXIT CRITERIA - Every macro DRC, LVS and antenna clean - Fill applied without disturbing matching structures - Pin placement agreed with physical design before freeze	MEASURED BY - Macros clean at first physical verification run - Post-layout degradation against pre-layout results - EM violations found at signoff
Total	26.0	DEPENDS ON AMS-02 PLL / clock generator design and closure AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY) AMS-04 LDO, bandgap and power-management cell design AMS-11 Schematic design and pre-layout simulation PDK-02 Design rule manual review; restricted and recommended rule disposition PDK-13 Golden design environment release and flow regression	
WHO IT TAKES	~6.3 FTE		
Analog layout lead Owns layout quality and the freeze			
Layout engineers Per-macro implementation			
Analog designers Matching intent and parasitic sensitivity			
Physical verification engineer DRC, LVS and antenna closure			
Physical design liaison Pin placement and top-level integration			

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AMS-13

Post-layout extracted simulation across PVT and Monte Carlo

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	20.0 M/M	~5.4 FTE	7	yes	Analog design leads
w22–w38 in stage	planned	while it runs	2 in parallel	no float	AMS lead approves

PURPOSE

Simulate every block **with its layout parasitics extracted**, across process, voltage and temperature and across the Monte Carlo distribution — and close whatever the layout took away.

Post-layout is where analog designs are actually judged. Parasitics reduce bandwidth, add mismatch, degrade PSRR and move every margin in the wrong direction. The gap between pre- and post-layout is the real measure of layout quality, and closing it is what the sixteen weeks are for.

WORK FLOW

7 STEPS · 2 LANES

w22	w26	w30	w34	w38
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Extraction flow setup and parasitic model selection	1.5w	main
2	Post-layout functional simulation per block	3w	main
3	PVT corner closure post-layout	4w	main
4	Monte Carlo and mismatch analysis post-layout	3w	parallel
5	Pre- to post-layout degradation analysis	2w	parallel
6	Layout iteration requests and re-simulation	4.5w	main
7	Characterisation data assembly	3w	main

Step 6 is planned iteration rather than contingency. Analog blocks essentially always need at least one layout turn after post-layout simulation, and a schedule with no allowance for it is a schedule that will slip on the first block.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Frozen layouts and extraction from AMS-12
- Pre-layout baselines from AMS-11
- Signoff corner definition from PDK-11
- Extraction decks from PDK-10
- Block specifications and budgets from AMS-01

PRODUCES

- Extraction flow and parasitic model selection
- Post-layout functional and corner results per block
- Monte Carlo and mismatch results
- Degradation analysis pre- to post-layout
- Layout iteration requests
- Assembled characterisation data

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D2

PRODUCES

Characterisation reports across PVT and Monte Carlo. This activity is the deliverable — the numbers every downstream consumer of these macros relies on.

AMS-D1

FEEDS

AMS IP specifications and design review packages. Post-layout results are what each block's final review is held against.

RISKS AND WATCH-OUTS

- ◆ **No layout iteration in the plan.** At least one turn per block is normal; a schedule with none slips on the first block and never recovers.
- ◆ **Corner closure on a subset of blocks.** The block skipped for time is the one that fails in silicon, and it is usually skipped because it looked simple.
- ◆ **Parasitic model too coarse.** A model that misses coupling produces post-layout results that are optimistic in exactly the cases that matter.
- ◆ **Degradation not analysed.** Knowing a block passed post-layout is less useful than knowing how much margin the layout consumed, which is what predicts the next block.
- ◆ **Monte Carlo run at reduced sample count.** Cutting samples to save runtime narrows the distribution and flatters the tail.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
PVT corner closure	5.5	● Layouts frozen and extractable from AMS-12	AMS-14Reliability checks — EM/IR, aging, ESD, latch-up
Layout iteration and re-simulation	4.5	● Pre-layout baselines available from AMS-11	AMS-15Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
Post-layout functional simulation	3.5	● Extraction decks validated by PDK-10	S0-02Multi-corner multi-mode static timing analysis and closure on the final turn
Monte Carlo and mismatch	3.0		BU-07Shmoo across voltage, frequency and temperature
Characterisation assembly	2.5		
Extraction setup	1.0		
Total	20.0		
WHO IT TAKES	~5.4 FTE	EXIT CRITERIA	MEASURED BY
Analog design leads		● Every block closed post-layout at every signoff corner	● Blocks closed post-layout at all corners
Own post-layout closure per block		● Degradation from pre-layout quantified per block	● Margin lost from pre- to post-layout
Extraction engineer		● Characterisation data complete enough to release the macros	● Layout iterations per block against the allowance
Flow setup and parasitic model selection			
Characterisation engineer			
Corner and Monte Carlo data assembly		AMS-11Schematic design and pre-layout simulation	
Analog layout lead		AMS-12Custom layout with DRC / LVS / antenna and fill closure	
Executes iteration requests		PDK-10Rule deck, QRC tech file and DRC/LVS deck version control	
AMS lead		PDK-11Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
Schedules iterations and arbitrates margin trades			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-14

Reliability checks — EM/IR, aging, ESD, latch-up

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	10.0 M/M	~4.3 FTE	7	no	Reliability engineer
w28–w38 in stage	planned	while it runs	3 in parallel	has float	Quality liaison approves

PURPOSE

Check every analog macro for the failure modes that **appear over time rather than at test** — electromigration, IR, aging, ESD and latch-up — because none of them show up in a functional simulation.

Analog blocks carry high current densities in small geometries and run continuously for the product's life. A design that passes every corner and fails EM has passed everything the design flow checks and none of what qualification will.

WORK FLOW

7 STEPS · 2 LANES

w28w30w32w34w36w38

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Reliability requirement definition per macro	1w	main
2	Electromigration analysis on supply and signal routing	2.5w	main
3	IR drop analysis within macros	1.5w	parallel
4	Aging and lifetime simulation at the rated conditions	2.5w	main
5	ESD path verification through the macro	2w	parallel
6	Latch-up and guard structure review	1.5w	parallel
7	Reliability report and waiver disposition	4w	main

Step 4 is the one that needs the product's actual mission profile. Aging simulated at nominal conditions for a nominal lifetime says nothing about a part that runs at the top of its DVFS range continuously, which is exactly how an accelerator is used.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Post-layout results and parasitics from AMS-13
- Reliability rules from PDK-02 and the DRM
- ESD strategy from PDK-07
- Mission profile and lifetime requirements from the qualification plan
- Assist boost levels from AMS-06

PRODUCES

- Reliability requirements per macro
- EM analysis results on supply and signal routing
- In-macro IR drop results
- Aging and lifetime simulation results
- ESD path verification and latch-up review
- Reliability report with waiver dispositions

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D7

PRODUCES

Reliability report — EM/IR, ESD, latch-up, aging. This activity is the deliverable, and it is what qualification will be checked against.

AMS-D8

FEEDS

Integration guide. Reliability limits — maximum current, boost duty cycle, operating conditions — are integration constraints and belong in the guide.

RISKS AND WATCH-OUTS

- ◆ **Aging simulated at nominal conditions.** An accelerator runs near its limit continuously; a nominal mission profile understates the stress by a wide margin.
- ◆ **EM checked on supplies only.** High-speed signal routing in a PHY carries current densities that matter, and signal EM is routinely omitted.
- ◆ **ESD path assumed from the strategy.** The strategy defines the intent; only tracing the path through the actual layout confirms it exists.
- ◆ **Assist boost stress not fed back.** The boost levels AMS-06 chose for Vmin have an aging cost, and the two activities have to reconcile.
- ◆ **Waivers granted without a lifetime calculation.** A reliability waiver without a quantified life impact is a decision nobody can defend at qualification.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Aging and lifetime simulation	2.5	● Post-layout results available from AMS-13	AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
Electromigration analysis	2.5	● Mission profile and lifetime requirements defined	S0-07 Reliability verification — ESD, latch-up, soft error / FIT
ESD path verification	2.0	● ESD strategy fixed by PDK-07	MP-03 ESD (HBM, CDM) and latch-up qualification
IR and latch-up review	1.5		MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Report and waiver disposition	1.5		
Total	10.0		
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA	MEASURED BY
Reliability engineer Owns the analysis and the report		● Aging simulated against the product's real mission profile	● Macros passing EM and aging at rated conditions
Analog designers Interpret findings back into design changes		● EM checked on signal routing as well as supplies	● Waivers with a quantified lifetime impact
ESD engineer Path verification and clamp effectiveness		● Every waiver carries a quantified lifetime impact	● Reliability findings at qualification traceable to a missed check
Layout engineers Implement routing and guard structure fixes			
Quality liaison Waiver disposition against qualification requirements			
		DEPENDS ON	
		AMS-06 Read / write assist circuit design for Vmin — WL boost, negative bitline, VDD collapse	
		AMS-12 Custom layout with DRC / LVS / antenna and fill closure	
		AMS-13 Post-layout extracted simulation across PVT and Monte Carlo	
		PDK-07 IO, ESD and latch-up library qualification	

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-15

Hard macro abstraction and view generation
(LEF, .lib, CDL, GDS, behavioural model)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w32–w40 in stage	8.0 M/M planned	~4.3 FTE while it runs	8 3 in parallel	yes no float	AMS integration lead Verification liaison approves

PURPOSE

Wrap every finished block into a **hard macro physical design can place and verification can simulate** — GDS, abstract, timing views, netlist, power intent and a behavioural model.

This is the handoff the whole stage exists to make. The macro set has to arrive before PD-02 places it, and the abstracts specifically have to arrive well before the models do, because floorplan needs the shape long before signoff needs the timing.

WORK FLOW8 STEPS · 2 LANES

w32	w34	w36	w38	w40
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MAIN SEQUENCE

1	2	3	7	8
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	View requirement confirmation with physical design and verification	1w	main
2	GDS finalisation and macro boundary definition	1.5w	main
3	Abstract generation — blockages, pins, routing layers	1.5w	main
4	Timing view generation from characterisation data	2w	parallel
5	CDL netlist and UPF power intent generation	1.5w	parallel
6	Behavioural and wreal model release	1.5w	parallel
7	Integration verification on a trial placement	2w	main
8	Macro release and version control	2w	main

Step 7 is worth the two weeks. Placing every macro in a trial floorplan and running the flow over it finds abstract errors, pin conflicts and missing views before the whole physical design team is depending on them.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Frozen layouts from AMS-12
- Characterisation data from AMS-13
- Custom memory views from AMS-10
- Reliability constraints from AMS-14
- View and tool requirements from PDK-08 and PD-01

PRODUCES

- Finalised GDS per macro
- Physical abstracts with blockages and pins
- Timing and power views
- CDL netlists and UPF power intent
- Behavioural and wreal models

- Released, version-controlled macro set

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D5	PRODUCES	Hard macro GDS with abstract views. This activity is the deliverable, and it is the stage's milestone — AMS Macro Handoff.
AMS-D4	FEEDS	Custom memory views characterised to compiler equivalence. The custom array joins the macro set here on the same terms as the analog blocks.
AMS-D2	FEEDS	Characterisation reports. The views are generated from the characterisation data and have to stay consistent with it.

RISKS AND WATCH-OUTS

- ◆ **Abstracts delivered with the models.** Floorplan needs the shape months before signoff needs the timing; a single delivery date serves the later need and misses the earlier one.
- ◆ **Trial placement skipped.** Abstract and pin errors are then found by the physical design team, at the point where the whole floorplan depends on them.
- ◆ **Views inconsistent with the characterisation.** A timing view generated from stale data produces a design closed against numbers the silicon does not have.
- ◆ **Macro released without version control.** Two teams on different macro revisions produce a mismatch that is diagnosed as a tool problem.
- ◆ **Power intent omitted.** A macro with no UPF view breaks low-power verification and forces a manual exception at every run.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Timing and power view generation	2.5	Layouts frozen by AMS-12	PD-01 Flow setup and MMMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity
Abstract generation	1.5	Characterisation complete from AMS-13	PD-02 Floorplan, macro placement and partition definition
Trial placement verification	1.5	View requirements confirmed with PD and DV	SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
GDS finalisation	1.0		DV-07 AMS / mixed-signal co-simulation
Netlist, UPF and models	1.0		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Release and version control	0.5		
Total	8.0	EXIT CRITERIA - Abstracts released ahead of models, on the schedule PD needs - Trial placement run and clean before general release - Every macro version-controlled and its views mutually consistent	MEASURED BY - Macros released against the PD need date - Abstract and view defects found at trial placement - Macro revisions issued after release
WHO IT TAKES	~4.3 FTE	DEPENDS ON	
AMS integration lead Owns the macro set and its release		AMS-10 Custom memory view generation and compiler-equivalent characterisation across corners	
Characterisation engineer Timing and power view generation		AMS-12 Custom layout with DRC / LVS / antenna and fill closure	
Layout engineer GDS finalisation and abstract generation		AMS-13 Post-layout extracted simulation across PVT and Monte Carlo	
Physical design liaison Trial placement and abstract validation		AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up	
Verification liaison Behavioural model fidelity and release			

06 AMS · CUSTOM, AMS & MEMORY IP DEVELOPMENT

AMS-16

AMS—digital co-simulation and integration support

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w cont. w26–w42 in stage, runs throughout	12.0 M/M planned	~3.3 FTE while it runs	7 2 in parallel	no has float	AMS verification engineer Validation liaison approves

PURPOSE

Keep the analog and digital halves talking — **co-simulation, integration support and the errata that come out of it** — through the whole period when the macros are being integrated.

Analog blocks fail at their boundaries: a reset that arrives before a rail is stable, a calibration sequence nobody scheduled, a control interface each side interpreted differently. Co-simulation is where those are found, and the integration guide is where they stop being rediscovered.

WORK FLOW

7 STEPS · 2 LANES

w26	w30	w34	w38	w42
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
---	---

#	STEP	TAT	LANE
1	AMS co-simulation environment setup	2w	main
2	Behavioural model correlation against circuit simulation	2.5w	main
3	Boundary and control interface co-verification	3w	main
4	Startup, calibration and sequencing co-simulation	2.5w	parallel
5	Integration support to RTL and physical design	5w	parallel
6	Errata capture and integration guide authoring	3w	main
7	Bring-up support material and debug hooks	5.5w	main

Step 5 runs across the whole window and is the reason this activity is marked continuous. Integration questions do not arrive in a block; they arrive one at a time for four months, and the engineer who can answer them has to still be assigned.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Behavioural models from AMS-15
- Circuit simulation results from AMS-13
- RTL integration environment from RTL-03
- Verification methodology from DV-07
- Startup and sequencing requirements from ARCH-05

PRODUCES

- AMS co-simulation environment
- Model correlation results
- Boundary and control interface verification results
- Startup and calibration co-simulation results
- Integration guide with known limitations
- Errata list and bring-up debug material

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

AMS-D8

PRODUCES

Integration guide with known limitations and errata. This activity is the deliverable, and it is what stops the same integration question being asked four times.

AMS-D1

FEEDS

AMS IP specifications and design review packages. Where co-simulation contradicts a specification, the specification is what gets corrected.

RISKS AND WATCH-OUTS

- ◆ **Behavioural models never correlated.** Digital verification then passes against a model that does not match the circuit, and the mismatch reaches silicon undetected.
- ◆ **Boundary conditions unverified.** Reset ordering, rail sequencing and enable timing are where analog blocks actually fail, and none of them are internal to the block.
- ◆ **Support engineer unassigned.** Integration questions arrive for months; without a named engineer they are answered by guesswork or not at all.
- ◆ **Errata captured in email.** A limitation known to one integrator and nobody else is rediscovered by every other integrator in turn.
- ◆ **Calibration sequence not co-simulated.** A calibration that works in the circuit and is never exercised from the digital side is first tried at bring-up.

EFFORT SPLIT

12 M/M

ENTRY CRITERIA

FEEDS INTO

Integration support	4.0
Boundary and interface co-verification	2.5
Model correlation	2.0
Integration guide and errata	1.5
Bring-up support material	1.0
Environment setup	1.0
Total	12.0

WHO IT TAKES

~3.3 FTE

AMS verification engineer

Owns co-simulation and model correlation

AMS integration engineer

Integration support and errata capture

Analog designers

Answer boundary and behaviour questions

RTL integration lead

Digital-side boundary definition

Validation liaison

Bring-up debug hooks and support material

- Behavioural models released by AMS-15
- RTL integration environment available from RTL-03
- Co-simulation tools qualified by PDK-08

EXIT CRITERIA

- Every behavioural model correlated against circuit simulation
- Boundary and sequencing verified, not assumed
- Integration guide published and maintained, not held in email

DEPENDS ON

- AMS-11** Schematic design and pre-layout simulation
- AMS-13** Post-layout extracted simulation across PVT and Monte Carlo
- AMS-15** Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)
- RTL-03** Third-party and internal IP integration — wrappers, glue, configuration

- RTL-03** Third-party and internal IP integration — wrappers, glue, configuration
- DV-07** AMS / mixed-signal co-simulation
- PD-02** Floorplan, macro placement and partition definition
- BU-03** Reset, clocking and PLL lock validation
- BU-05** Interface bring-up — PCIe/CXL link training, HBM training, die-to-die

MEASURED BY

- Models correlated against those released
- Integration questions answered within a working day
- Errata found at bring-up that co-simulation could have caught

07

Test Chip / MPW Shuttle

TC

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w22 – w62	40 _w	8	83 _{M/M}	6 _{of 8}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
TC-01	Test chip objective definition and risk-item selection		5w 4 M/M	TC-D1 Test chip specification and risk coverage matrix
TC-02	Test chip design and integration (IP under test, ring oscillators, PCM structures)		12w 18 M/M	TC-D2 Test chip GDS and shuttle submission record
TC-03	MPW shuttle slot booking and data submission scheduling		4w 1 M/M	TC-D3 Test chip silicon and characterisation report
TC-04	Test chip physical implementation and signoff		12w 20 M/M	TC-D4 Silicon-to-model correlation report
TC-05	Shuttle tapeout and fabrication		16w 4 M/M	TC-D5 Design guidance and margin decisions for the production chip
TC-06	Test chip board and lab setup preparation		9w 8 M/M	
TC-07	Silicon characterisation and correlation to simulation		12w 20 M/M	
TC-08	Feedback into the production design — model correction and margin decisions		6w 8 M/M	

07 TC · TEST CHIP / MPW SHUTTLE

TC-01

Test chip objective definition and risk-item selection

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	4.0 M/M	~3.5 FTE	6	yes	Test chip lead
w0–w5 in stage	planned	while it runs	2 in parallel	no float	Device and technology engineer approves

PURPOSE

Decide what the test chip is **for** — which specific risks it retires — and select the structures that will actually answer those questions rather than the ones that are easy to include.

A test chip with no objective becomes a collection of everything anyone wanted to try, and it answers nothing in time to matter. The discipline is subtractive: name the two or three uncertainties that would change a production decision, and carry only what measures them.

WORK FLOW

6 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Objective definition — which production decisions this chip informs	1w	main
2	Risk item selection and prioritisation	1.5w	main
3	Schedule feasibility — does the answer arrive before the decision	1w	parallel
4	Structure selection per risk item	1w	main
5	Cost and area budget for the shuttle die	0.8w	parallel
6	Specification and risk coverage matrix	1.5w	main

Step 3 can cancel the activity, and should be allowed to. A test chip whose results land after the production design has committed is an expense with no decision attached, and finding that out in week three is far better than in week forty.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Node risk assessment from TECH-07
- IP readiness gaps from IPR-05
- Custom memory decision from PDK-06
- AMS block risk list from AMS-01
- Production design schedule from DEF-08

PRODUCES

- Test chip objective statement
- Prioritised risk item list
- Schedule feasibility finding
- Structure selection per risk
- Test chip specification and risk coverage matrix

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-D1

PRODUCES

Test chip specification and risk coverage matrix. This activity is the deliverable, and the matrix is what every later result is reported against.

TC-D5

FEEDS

Design guidance and margin decisions. The guidance can only address risks this activity chose to carry — everything else stays unmeasured.

RISKS AND WATCH-OUTS

- ◆ **Scope grows until the shuttle is missed.** Every team has something worth measuring, and a chip that carries all of them misses the slot that made it useful.
- ◆ **Results arrive after the decision.** A test chip is only worth its cost if the answer lands while the production design can still act on it.
- ◆ **Structures chosen for measurability rather than for risk.** Ring oscillators are easy and tell you about the node; the risk is usually in a specific circuit nobody has built before.
- ◆ **No decision rule attached to a result.** Measuring something without agreeing in advance what each outcome would change produces data and no decisions.
- ◆ **Area budget set after structure selection.** Shuttle die area is expensive and fixed; selecting first and budgeting second guarantees a painful subtraction later.

EFFORT SPLIT

4 M/M

ENTRY CRITERIA

DEPENDS ON

Risk item selection

1.3

- Node risk assessment available from TECH-07

TECH-07

Foundry roadmap and risk-production timing alignment

Structure selection

1.0

- IP and memory gaps known from IPR-05 and PDK-06

IPR-05

Silicon-proven status check per IP on the selected node and process option

Specification and coverage matrix

1.0

- Production design schedule dated by DEF-08

PDK-06

Custom / pushed-rule memory instance decision, scope and schedule impact assessment

Objective definition

0.8

Total**4.0**

EXIT CRITERIA

AMS-01

AMS IP specification and budget allocation (jitter, BER, PSRR, area)

WHO IT TAKES

~3.5 FTE

Test chip lead

Owns the objective, the scope and the coverage matrix

Chief architect

Confirms which risks would change a production decision

AMS lead

Analog risk items and the structures that measure them

Program manager

Schedule feasibility against the production decision dates

Device and technology engineer

PCM and process monitor structure selection

- Every carried structure traces to a named risk and a decision rule
- Schedule feasibility confirmed against the decision dates
- Area and cost budget fixed before structures are finalised

FEEDS INTO

TC-02

Test chip design and integration (IP under test, ring oscillators, PCM structures)

TC-03

MPW shuttle slot booking and data submission scheduling

TC-04

Test chip physical implementation and signoff

TC-07

Silicon characterisation and correlation to simulation

MEASURED BY

- Structures traceable to a risk item
- Risks covered against risks identified
- Shuttle die area against budget

07 TC · TEST CHIP / MPW SHUTTLE

TC-02

Test chip design and integration (IP under test, ring oscillators, PCM structures)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	18.0 M/M	~6.5 FTE	8	yes	Test chip design lead
w2-w14 in stage	planned	while it runs	3 in parallel	no float	Layout lead approves

PURPOSE

Design and integrate the vehicle itself — **the IP under test, its access wrappers, the process monitors and the measurement structures** — as a small chip that has to work first time.

A test chip gets one shuttle. There is no bring-up iteration and no respin, so the access paths, the monitors and the pad ring have to be right on submission. That makes it a small design with an unusually high verification burden relative to its size.

WORK FLOW

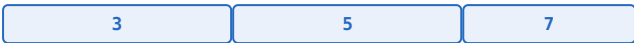
8 STEPS · 2 LANES

w2w5w8w11w14

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	IP-under-test integration plan	1w	main
2	Device-under-test wrapper and access path design	3w	main
3	Ring oscillator and process monitor array	2w	parallel
4	Measurement and PCM structure design	3w	main
5	Test structure library assembly	2w	parallel
6	Top-level integration and pad assignment	2.5w	main
7	Pad ring and ESD for the shuttle die	1.5w	parallel
8	Verification of access paths and measurement modes	2.5w	main

Step 8 gets two and a half weeks for a chip that is mostly structures, because an access path that does not work makes the structure behind it unmeasurable. Every mode has to be exercised in simulation, since there is no second chance in silicon.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test chip specification and coverage matrix from TC-01
- IP-under-test deliverables from AMS and IPR
- Process monitor structure definitions from the foundry
- Shuttle die area and pad count limits from TC-03
- PDK and libraries from PDK-13

PRODUCES

- IP-under-test integration with access wrappers
- Ring oscillator and process monitor arrays
- Measurement and PCM structures
- Top-level netlist with pad assignment
- Access path and measurement mode verification results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-D2	FEEDS	Test chip GDS and shuttle submission record. The design half — TC-04 turns it into a submittable database.
TC-D1	FEEDS	Test chip specification and risk coverage matrix. Where a structure cannot be built as specified, the matrix is what gets updated.

RISKS AND WATCH-OUTS

- ◆ **Access paths unverified.** A structure that cannot be reached is a structure that was not measured, and the failure is only discovered in the lab.
- ◆ **Pad count exceeded.** Shuttle die have few pads; a measurement scheme needing more forces multiplexing that has to be designed and verified.
- ◆ **IP under test modified for the vehicle.** A block wrapped or altered to fit the test chip is no longer the block the production design will use.
- ◆ **Monitors placed without regard to position.** Process variation across the die is part of what is being measured, and monitors clustered in one corner cannot see it.
- ◆ **No redundancy in critical structures.** A single instance of the structure that matters most means one defect loses the answer entirely.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
DUT wrappers and access paths	4.5	● Specification and coverage matrix from TC-01	TC-04 Test chip physical implementation and signoff
Measurement and PCM structures	4.0	● IP-under-test available in a form that can be instantiated	TC-06 Test chip board and lab setup preparation
Top-level integration	3.5	● Shuttle die area and pad limits known	TC-07 Silicon characterisation and correlation to simulation
Verification	3.0		
Ring oscillator arrays	2.0		
Pad ring and ESD	1.0		
Total	18.0		
WHO IT TAKES	~6.5 FTE	EXIT CRITERIA	MEASURED BY
Test chip design lead Owns the integration and the access architecture		● Every structure reachable through a verified access path ● Monitors distributed to see across-die variation ● Pad budget met without unverified multiplexing	● Structures with a verified access path ● Measurement modes exercised in simulation ● Pad count against the shuttle limit
Analog designers IP-under-test wrappers and measurement paths		DEPENDS ON	
Device engineer Process monitor and PCM structure design		TC-01 Test chip objective definition and risk-item selection	
Verification engineer Access path and measurement mode verification		TC-03 MPW shuttle slot booking and data submission scheduling	
Layout lead Pad ring, area and structure placement		PDK-13 Golden design environment release and flow regression	

07 TC · TEST CHIP / MPW SHUTTLE

TC-03

MPW shuttle slot booking and data submission scheduling

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	1.0 M/M	~1.1 FTE	5	no	Program manager
w5–w9 in stage	planned	while it runs	2 in parallel	has float	Physical design lead approves

PURPOSE

Book the shuttle slot and **work backwards from its submission date** — because the shuttle calendar is fixed, and missing a slot costs a quarter rather than a week.

One man-month across four weeks: almost all of this is scheduling and paperwork. Its importance is out of proportion to its effort, because the submission date it fixes is what every other test chip activity is planned against.

WORK FLOW

5 STEPS · 2 LANES

w5	w6	w7	w8	w9
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Shuttle calendar review and slot options	1w	main
2	Foundry or broker engagement and pricing	0.8w	parallel
3	Slot booking against the production decision dates	1w	main
4	Data format and submission requirement collection	0.8w	parallel
5	Submission schedule and checklist working backwards from the slot	2w	main

Step 5 is the useful output. A shuttle slot is a date on which a compliant database must exist; scheduling backwards from it is what tells `TC-04` when physical verification actually has to finish.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test chip objective and schedule feasibility from TC-01
- Shuttle calendars from the foundry or broker
- Capacity and pricing terms from TECH-05
- Production decision dates from DEF-08
- Submission format requirements

PRODUCES

- Shuttle slot options and pricing
- Booked slot with confirmation
- Data format and submission requirements
- Backward-planned submission schedule
- Submission checklist

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-D2

FEEDS

Test chip GDS and shuttle submission record. The booking and its requirements are half the submission record; the database is the other half.

RISKS AND WATCH-OUTS

- ◆ **Slot booked without checking the production decision date.** An available slot that returns silicon too late buys nothing.
- ◆ **Submission requirements collected late.** Shuttle submissions have their own format and layer rules, and discovering them near the date forces rushed rework.
- ◆ **Backward schedule not enforced.** A submission date that is not translated into a physical verification deadline gets missed by exactly the amount by which signoff runs long.
- ◆ **No contingency slot identified.** A missed shuttle costs a quarter, and knowing the next available date in advance changes how the schedule is managed.
- ◆ **Cost approved after booking.** Shuttle die area is charged by area and layer count; booking before the budget is agreed produces an invoice nobody planned.

EFFORT SPLIT	1 M/M	ENTRY CRITERIA	FEEDS INTO
Slot booking	0.4	● Objective and schedule feasibility confirmed by TC-01 ● Shuttle calendars available ● Budget authority for the shuttle cost	TC-02 Test chip design and integration (IP under test, ring oscillators, PCM structures)
Calendar review	0.3		TC-04 Test chip physical implementation and signoff
Submission schedule	0.3		TC-05 Shuttle tapeout and fabrication
Total	1.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~1.1 FTE	● Slot confirmed in writing with a submission date ● Submission requirements collected and understood ● Backward schedule agreed with physical design	● Weeks between slot booking and submission ● Submission date against the production decision date ● Contingency slot identified
Program manager Owns the slot booking and the backward schedule		DEPENDS ON	
Foundry relationship manager Shuttle engagement and confirmation		TC-01 Test chip objective definition and risk-item selection	
Test chip lead Confirms the slot serves the objective		TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement	
Procurement Pricing and purchase authorisation		TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame	
Physical design lead Confirms the backward schedule is achievable			

07 TC · TEST CHIP / MPW SHUTTLE

TC-04

Test chip physical implementation and signoff

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	20.0 M/M	~7.2 FTE	8	yes	Test chip physical lead
w12–w24 in stage	planned	while it runs	3 in parallel	no float	Program manager approves

PURPOSE

Implement the test chip physically and **sign it off to shuttle submission standard** — the same DRC, LVS and antenna closure a production die needs, on a fraction of the schedule.

The vehicle is small and the standard is not reduced. A shuttle rejects a non-compliant database, and the rejection costs the slot. What compresses the schedule is that timing closure is easy here; what does not compress is physical verification, which has to be as clean as production.

WORK FLOW

8 STEPS · 2 LANES

w12	w15	w18	w21	w24
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MAIN SEQUENCE

1	2	4	6	8
---	---	---	---	---

IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Floorplan and structure placement	1.5w	main
2	Place and route	2.5w	main
3	Power delivery network for measurement accuracy	2w	parallel
4	Timing closure on access and control paths	2.5w	main
5	Fill, seal ring and shuttle frame compliance	1.5w	parallel
6	Physical verification — DRC, antenna, density	3w	main
7	LVS closure	1.5w	parallel
8	GDS assembly and shuttle submission	2.5w	main

Step 3 exists because a test chip measures small signals. A power delivery network adequate for function but noisy in the measurement bands turns a characterisation result into an argument about the vehicle rather than the device.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test chip netlist and structures from TC-02
- Submission requirements and date from TC-03
- PDK, libraries and decks from PDK-13
- Shuttle frame and seal ring rules from the foundry
- Measurement accuracy requirements from TC-01

PRODUCES

- Floorplan and routed database
- Measurement-grade power delivery network
- Physical verification clean reports
- Shuttle frame compliant GDS
- Submission package and record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-D2

PRODUCES

Test chip GDS and shuttle submission record. This activity is the deliverable, and the submission record is what proves the slot was met.

TC-D3

FEEDS

Test chip silicon and characterisation report. Measurement quality is determined here — a noisy vehicle produces uncertain results.

RISKS AND WATCH-OUTS

- ◆ **Submission rejected on frame or format compliance.** Shuttle frames have their own rules, and a rejection costs the slot rather than a re-run.
- ◆ **PDN adequate for function and inadequate for measurement.** The results then carry vehicle noise that cannot be separated from device behaviour.
- ◆ **Physical verification compressed because the chip is small.** Size does not reduce the standard, and a DRC violation is as fatal here as on a production die.
- ◆ **Structures placed without regard to the measurement plan.** Placement affects what across-die variation can be seen, and it is decided here rather than in the lab.
- ◆ **No margin against the submission date.** Everything before this activity can slip and this one cannot, so its schedule needs the buffer.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	DEPENDS ON
Place and route	5.0	● Netlist and structures complete from TC-02	TC-02 Test chip design and integration (IP under test, ring oscillators, PCM structures)
Physical verification	4.5	● Submission date and format requirements from TC-03	TC-03 MPW shuttle slot booking and data submission scheduling
Timing closure	4.0	● Golden environment available from PDK-13	PDK-13 Golden design environment release and flow regression
GDS assembly and submission	2.5		
Power delivery network	2.5		
Fill and seal ring	1.5		
Total	20.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~7.2 FTE	● Database clean against production physical verification standards	TC-05 Shuttle tapeout and fabrication
		● Shuttle frame and format compliance confirmed before submission	TC-07 Silicon characterisation and correlation to simulation
		● Submission made on the booked date	MEASURED BY
Test chip physical lead Owns implementation and submission			● Submission accepted at first attempt
Physical design engineers Place, route and closure			● Days of margin against the submission date
Physical verification engineer DRC, LVS, antenna and frame compliance			● Measurement-band noise on the PDN
Analog layout engineer Measurement-grade PDN and shielding			
Program manager Submission date enforcement			

07 TC · TEST CHIP / MPW SHUTTLE

TC-05

Shuttle tapeout and fabrication

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w w22–w38 in stage	4.0 M/M planned	~1.1 FTE while it runs	6 2 in parallel	yes no float	Program manager Test chip lead approves

PURPOSE

Wait, visibly — **mask, fabricate, dice and deliver** — while keeping the program informed about a schedule it has almost no ability to influence.

Sixteen weeks and four man-months. The work here is tracking and logistics rather than engineering, and the value is that the rest of the program knows what date to plan against. A shuttle lot moves for reasons that have nothing to do with this program, and being told late is the avoidable part.

WORK FLOW

6 STEPS · 2 LANES

w22	w26	w30	w34	w38
-----	-----	-----	-----	-----

MAIN SEQUENCE

1	2	4	6
---	---	---	---

IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Data submission and acceptance confirmation	1w	main
2	Shuttle mask fabrication	4w	main
3	WIP tracking and schedule communication	2w	parallel
4	Wafer processing	8w	main
5	E-test and PCM data review as it becomes available	1.5w	parallel
6	Dicing, packaging and delivery	3w	main

Step 5 matters more than its 1.5 weeks suggest. PCM and e-test data arrives before packaged parts do, and it answers some of the process questions the chip was built for weeks earlier than the lab can.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Submitted GDS from TC-04
- Shuttle slot confirmation from TC-03
- Foundry WIP reporting
- Packaging requirements from TC-06
- Delivery logistics arrangements

PRODUCES

- Submission acceptance confirmation
- WIP status reporting against the program schedule
- E-test and PCM data
- Packaged test chip parts
- Delivery record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-03

FEEDS

Test chip silicon and characterisation report. The silicon itself, plus the PCM data that arrives with it.

RISKS AND WATCH-OUTS

- ◆ **Shuttle lot slips and the program hears late.** The slip cannot be prevented; the surprise can, and the surprise is what damages downstream plans.
- ◆ **Packaging not arranged in advance.** Diced die with no package and no socket cannot be measured, and arranging it after delivery adds weeks.
- ◆ **PCM data ignored until parts arrive.** Some process questions are answered by e-test alone, weeks earlier and at no additional cost.
- ◆ **Insufficient parts requested.** Shuttle quantities are small; a characterisation plan needing more parts than were ordered cannot be run.
- ◆ **Logistics and customs not planned.** Cross-border shipment of engineering samples carries paperwork that stops parts at a border for weeks.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
WIP tracking and communication	1.5	● GDS submitted and accepted by the shuttle ● Packaging and socket requirements defined by TC-06 ● Delivery logistics arranged	TC-07 Silicon characterisation and correlation to simulation
PCM and e-test review	1.0		
Submission and acceptance	0.8		MEASURED BY ● Delivery date against the forecast at submission ● Parts delivered against parts planned ● Days between a lot status change and the program hearing about it
Logistics and delivery	0.8		
Total	4.0	EXIT CRITERIA ● Parts delivered in the quantity the characterisation plan needs ● PCM and e-test data reviewed before parts arrive ● Schedule communicated at every status change, not only at delivery	
WHO IT TAKES	~1.1 FTE	DEPENDS ON	
Program manager WIP tracking and schedule communication		TC-03 MPW shuttle slot booking and data submission scheduling	
Foundry relationship manager Lot status and escalation		TC-04 Test chip physical implementation and signoff	
Operations planner Packaging, logistics and delivery			
Device engineer E-test and PCM data review			
Test chip lead Confirms part quantities against the characterisation plan			

07 TC · TEST CHIP / MPW SHUTTLE

TC-06

Test chip board and lab setup preparation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	8.0 M/M	~3.9 FTE	6	no	Characterisation engineer
w24–w33 in stage	planned	while it runs	2 in parallel	has float	Analog designer approves

PURPOSE

Have the **board, sockets, instruments and automation ready before the parts arrive**, so that characterisation starts on the day of delivery rather than six weeks later.

This is the small-scale version of the same mistake the production program makes with EVB. Parts arriving into a lab with no board is a delay that consumes the schedule advantage the test chip was supposed to buy, and it is entirely avoidable by starting during fabrication.

WORK FLOW

6 STEPS · 2 LANES

w24	w26	w28	w30	w32	w33
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Measurement requirement definition from the coverage matrix	1w	main
2	Board design — supplies, bias, measurement paths	2.5w	main
3	Socket or probe card selection and lead time	1.5w	parallel
4	Board fabrication and assembly	2.5w	main
5	Measurement automation and data capture scripting	1.5w	parallel
6	Board bring-up and instrument setup	3w	main

Step 3 has a lead time that can exceed the whole activity. Sockets for an unusual package are made to order, and a socket ordered when the parts ship arrives after they do.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Coverage matrix and measurement plan from TC-01
- Pad assignment and access modes from TC-02
- Package and die dimensions from TC-05
- Lab instrument availability
- Measurement accuracy requirements

PRODUCES

- Measurement requirement specification
- Test chip board design and assembled boards
- Socket or probe card
- Measurement automation and capture scripts
- Bring-up record and instrument setup

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-03

FEEDS

Test chip silicon and characterisation report. The measurement quality the report can claim is bounded by the board and the automation built here.

RISKS AND WATCH-OUTS

- ◆ **Board started when parts ship.** The lab then waits, and the schedule advantage the test chip existed to buy is spent waiting for a PCB.
- ◆ **Socket lead time discovered late.** Custom sockets are made to order and can take longer than the board itself.
- ◆ **Measurement automation deferred.** Manual measurement of a full characterisation matrix is slow enough to consume the window and inconsistent enough to weaken the data.
- ◆ **Board noise floor above the signal of interest.** A board adequate for function cannot measure a low-jitter clock or a small offset, and the limitation is found with the parts in hand.
- ◆ **Instruments not reserved.** A shared high-end instrument booked by another program stops characterisation as effectively as a missing board.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Board design	2.5	- Coverage matrix and measurement plan available from TC-01 - Pad assignment fixed by TC-02 - Package format known from TC-05	TC-07 Silicon characterisation and correlation to simulation
Bring-up and instrument setup	2.0		EVB-01 Validation platform requirements and topology definition
Measurement automation	1.5		
Board fabrication	1.0		
Socket procurement	1.0	EXIT CRITERIA	MEASURED BY
Total	8.0	- Board bring-up complete before parts are delivered - Sockets in hand, not on order - Automation able to run the matrix without manual steps	- Days between part delivery and first measurement - Board noise floor against the smallest signal of interest - Matrix points automated against total
WHO IT TAKES	~3.9 FTE		
Characterisation engineer Owns the measurement setup and automation			
Board designer Test chip board design and bring-up			
Lab manager Instrument reservation and socket procurement		TC-01 Test chip objective definition and risk-item selection	
Test chip lead Confirms the setup covers the matrix		TC-02 Test chip design and integration (IP under test, ring oscillators, PCM structures)	
Analog designer Measurement path accuracy requirements			

07 TC · TEST CHIP / MPW SHUTTLE

TC-07

Silicon characterisation and correlation to simulation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	20.0 M/M	~7.2 FTE	8	yes	Characterisation engineer
w28–w40 in stage	planned	while it runs	3 in parallel	no float	Failure analysis engineer approves

PURPOSE

Measure the silicon and **correlate it against the models the production design is being built on** — which is the only reason the chip exists.

Characterisation without correlation produces a data set. Correlation produces a statement of the form "the model is optimistic by 8% on this parameter at this corner", and that statement is what TC-08 converts into margin decisions the production design can act on.

WORK FLOW

8 STEPS · 2 LANES

w28	w31	w34	w37	w40
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Incoming inspection and basic functionality check	1.5w	main
2	IP-under-test characterisation across the matrix	3w	main
3	Ring oscillator frequency and leakage measurement	2.5w	parallel
4	PCM and device parameter extraction	3w	main
5	Corner correlation setup against simulation	2w	parallel
6	Across-wafer and lot-to-lot variability analysis	2w	main
7	Failure analysis on outliers	1.5w	parallel
8	Correlation results assembly	2.5w	main

This window overlaps the tail of fabrication because the stage as templated does not allow a clean sequence — PCM and e-test data are worked first, and device characterisation follows once packaged parts arrive. It is worth noting that the stage's forty weeks are tight for its own chain.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Packaged parts and PCM data from TC-05
- Measurement setup from TC-06
- Coverage matrix from TC-01
- Simulation results the models predicted
- Foundry PCM reference data

PRODUCES

- Functional and characterisation results per structure
- Ring oscillator and leakage measurements
- Extracted device parameters
- Variability analysis across wafer and lot
- Failure analysis findings
- Silicon-to-model correlation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-D3	PRODUCES	Test chip silicon and characterisation report. The measurement half of the deliverable.
TC-D4	PRODUCES	Silicon-to-model correlation report. The half that makes the measurements actionable — a delta against what was predicted, per parameter and corner.

RISKS AND WATCH-OUTS

- ◆ **Characterisation without correlation.** A data set with no comparison to prediction cannot change a design decision, and changing decisions was the point.
- ◆ **Sample size too small for variability.** Across-wafer and lot-to-lot variation is often the question, and a handful of parts cannot answer it.
- ◆ **Measurement uncertainty larger than the effect.** A correlation claim that sits inside the measurement error is not a finding.
- ◆ **Outliers discarded.** The parts that behave unexpectedly are usually the informative ones, and discarding them removes the tail the yield model depends on.
- ◆ **Results delivered too late for TC-08.** Correlation completed after the production design has committed is a report nobody can act on.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
IP-under-test characterisation	6.0	● Parts delivered from TC-05	TC-08 Feedback into the production design — model correction and margin decisions
PCM and device extraction	4.0	● Measurement setup ready from TC-06	AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
Ring oscillator measurement	3.0	● Simulation predictions available for comparison	PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)
Variability analysis	3.0		S0-10 Waiver review board and foundry waiver alignment
Correlation assembly	2.0	EXIT CRITERIA	
Failure analysis	2.0	● Every measured parameter compared against its prediction	
Total	20.0	● Measurement uncertainty stated alongside each correlation claim	
WHO IT TAKES	~7.2 FTE	● Results delivered with time for TC-08 to act	MEASURED BY
Characterisation engineer		DEPENDS ON	● Parameters correlated against parameters measured
Owens the measurements and the data quality		TC-01 Test chip objective definition and risk-item selection	● Measurement uncertainty against effect size
Device engineer		TC-02 Test chip design and integration (IP under test, ring oscillators, PCM structures)	● Parts characterised against the sample size the plan required
PCM extraction and process parameter analysis		TC-04 Test chip physical implementation and signoff	
Analog designers		TC-05 Shuttle tapeout and fabrication	
IP-under-test interpretation against simulation		TC-06 Test chip board and lab setup preparation	
Modelling engineer			
Correlation setup and delta analysis			
Failure analysis engineer			
Outlier investigation			

07 TC · TEST CHIP / MPW SHUTTLE

TC-08

Feedback into the production design — model correction and margin decisions

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w34–w40 in stage	8.0 M/M planned	~5.8 FTE while it runs	6 2 in parallel	yes no float	Test chip lead Program manager approves

PURPOSE

Convert correlation findings into **changes the production design actually makes** — corrected models, revised margins, methodology updates — and track that they were adopted.

This is where a test chip earns or wastes its cost. A correlation report that circulates and changes nothing has bought an expensive confirmation of what everyone assumed. The output has to be specific: this model corrected by this factor, this margin increased by this amount, owned by this person, adopted by this date.

WORK FLOW

6 STEPS · 2 LANES

w34w36w38w40

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Correlation gap analysis — where model and silicon disagree	1w	main
2	Model correction and re-characterisation requests	1.5w	main
3	Design rule and methodology update proposals	1w	parallel
4	Margin decisions for the production design	1.5w	main
5	Production design change requests	1w	parallel
6	Guidance publication and adoption tracking	2w	main

Step 6 is the one that distinguishes a useful test chip from an expensive one. Publishing guidance is easy; tracking that physical design actually changed its margin and that the model was actually re-characterised is what turns the finding into a change.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Correlation results from TC-07
- Production design status and open margins
- Model and library data from PDK-03
- AMS post-layout results from AMS-13
- Production design change control process from RTL-10

PRODUCES

- Correlation gap analysis
- Model correction requests and re-characterisation scope
- Methodology and rule update proposals
- Margin decisions for the production design
- Adoption tracking record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TC-D5

PRODUCES

Design guidance and margin decisions for the production chip. This activity is the deliverable, and adoption tracking is what makes it more than a report.

TC-D4

FEEDS

Silicon-to-model correlation report. The gap analysis is published with the correlation, so the finding and its consequence stay together.

RISKS AND WATCH-OUTS

- ◆ **Guidance published and not adopted.** The most common way a test chip wastes its cost: everyone reads the report and nobody changes a margin.
- ◆ **Findings arrive after the design has committed.** A margin decision for a block that has already closed is a decision that cannot be taken.
- ◆ **Model corrections requested but not scheduled.** Re-characterisation is work; a correction with no owner and no date does not happen.
- ◆ **Only bad news acted on.** Where silicon is better than the model, the margin released is real and usually left on the table.
- ◆ **Guidance stated as observation.** "Leakage is higher than modelled" is not actionable; "increase the leakage budget for the compute cluster by 12%" is.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Model correction	2.5	● Correlation results available from TC-07 ● Production design status known well enough to act on ● Change control process available from RTL-10	PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)
Margin decisions	2.0		AMS-13 Post-layout extracted simulation across PVT and Monte Carlo
Gap analysis	1.5		PD-09 Multi-corner multi-mode timing closure
Guidance and adoption tracking	1.0		S0-10 Waiver review board and foundry waiver alignment
Change requests	1.0		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)
Total	8.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.8 FTE	● Every finding converted into a specific, owned action ● Margin released where silicon beat the model, not only tightened where it did not ● Adoption tracked to completion, not just published	● Findings converted into owned actions ● Actions adopted against actions raised ● Margin released where silicon outperformed the model
Test chip lead Owns the guidance and its adoption tracking			
Modelling and library engineers Model corrections and re-characterisation			
PPA lead Margin decisions against the production budgets			
Physical design lead Adopts margin changes into closure			
Program manager Change requests through the ECO board		TC-07 Silicon characterisation and correlation to simulation	

08

RTL Design & Integration

RTL

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w20 – w52	32 _w	10	410 _{M/M}	6 _{of 10}

ENGINEERING ACTIVITIES

RTL-01	Block microarchitecture specification	10w	40 _{M/M}
RTL-02	Block-level RTL implementation	24w	180 _{M/M}
RTL-03	Third-party and internal IP integration — wrappers, glue, configuration	16w	45 _{M/M}
RTL-04	Chip-level integration and top assembly	12w	35 _{M/M}
RTL-05	Clock, reset and power intent (UPF) implementation	10w	18 _{M/M}
RTL-06	Lint, CDC and RDC closure	16w	30 _{M/M}
RTL-07	Register map / RDL definition and header generation	8w	10 _{M/M}
RTL-08	Trial synthesis feedback loop on RTL PPA	12w	20 _{M/M}
RTL-09	CI build, nightly regression and release management	30w cont.	20 _{M/M}
RTL-10	Specification change control and ECO board	30w cont.	12 _{M/M}

KEY DELIVERABLES

RTL-D1	Block and top-level RTL release, tagged
RTL-D2	Register map / RDL and generated headers
RTL-D3	Integration testbench and build system
RTL-D4	Lint / CDC / RDC clean reports with waiver list
RTL-D5	UPF power intent file
RTL-D6	IP integration report and version manifest
RTL-D7	RTL Freeze package

08 RTL · RTL DESIGN & INTEGRATION

RTL-01

Block microarchitecture specification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	40.0 M/M	~17.4 FTE	7	yes	Block owners
w0–w10 in stage	planned	while it runs	3 in parallel	no float	Chief architect approves

PURPOSE

Complete the **microarchitecture specification for every block** — pipeline, interfaces, control, register map — to the level of detail an implementer can write code against without inventing anything.

ARCH-10 started these inside architecture; this finishes them. The distinction matters: a block whose specification is still being written while its RTL is being written produces code that documents itself, and the specification then follows the code instead of leading it.

WORK FLOW

7 STEPS · 2 LANES

w0	w2	w4	w6	w8	w10
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
---	---	---

#	STEP	TAT	LANE
1	Specification completion plan and block-owner assignment	1w	main
2	Pipeline and datapath detail per block	3w	main
3	Control and state machine specification	2.5w	parallel
4	Block interface and protocol detail against the NoC	2.5w	main
5	Exception, error and corner-case behaviour	2w	parallel
6	Verification hook and observability requirements with DV	1.5w	parallel
7	Specification review and implementation entry gate	3.5w	main

Step 5 is the section most often left thin and the one verification spends the most time on. Corner-case behaviour that the specification does not define is behaviour the implementer chooses and the verification engineer has no basis to check.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Architecture specification and block chapters from ARCH-09
- Microarchitecture drafts from ARCH-10
- PPA budgets per block from ARCH-06
- Verification strategy from DV-01
- Interface definitions from ARCH-04

PRODUCES

- Block microarchitecture specifications, complete
- Control and state machine definitions
- Interface and protocol detail per block
- Exception and corner-case behaviour definitions
- Implementation entry gate records

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D1	FEEDS	Block and top-level RTL release. The specification is what the release is checked against; without it, review has nothing to compare code to.
RTL-D7	GATES	RTL Freeze package. A block whose specification was never completed cannot be declared frozen against anything.

RISKS AND WATCH-OUTS

- ◆ **Specification written alongside the code.** The document then records what was built rather than defining it, and verification loses its independent reference.
- ◆ **Corner cases left undefined.** The implementer decides, verification cannot check, and the disagreement surfaces at chip-level integration.
- ◆ **Interfaces specified from one side.** Two blocks each documenting their own view of a shared protocol produces two consistent documents that disagree.
- ◆ **Entry gate skipped for schedule.** Coding against an incomplete specification is faster for two weeks and slower for two months.
- ◆ **Observability requirements added later.** Debug and coverage hooks designed in cost little; retrofitted they cost timing and an RTL change under control.

EFFORT SPLIT	40 M/M	ENTRY CRITERIA	FEEDS INTO
Pipeline and datapath detail	14.0	● Architecture specification frozen by ARCH-09	RTL-02Block-level RTL implementation
Control and state machines	9.0	● Block owners assigned and available	RTL-04Chip-level integration and top assembly
Interface detail	7.0	● PPA budgets accepted per block	RTL-07Register map / RDL definition and header generation
Exception and corner-case behaviour	5.0		DV-01Verification plan and coverage model definition
Reviews and entry gate	5.0	EXIT CRITERIA	DV-03Block-level constrained-random and directed testing
Total	40.0	● Every block passes an entry gate before implementation starts	
WHO IT TAKES	~17.4 FTE	● Corner-case behaviour defined, not left to the implementer	MEASURED BY
Block owners	Author their block's specification	● Shared interfaces agreed by both sides	● Blocks passing the entry gate before coding starts
RTL lead	Owns the entry gate and specification quality	DEPENDS ON	● Specification changes raised during implementation
Interconnect architect	Arbitrates shared interface definitions	ARCH-06PPA budget allocation per block	● Corner cases defined against those found in verification
Verification leads	Observability and checkability requirements	ARCH-09Architecture specification authoring and review	
Chief architect	Resolves conflicts against the architecture specification	ARCH-10Block microarchitecture specification kickoff	

08 RTL · RTL DESIGN & INTEGRATION

RTL-02

Block-level RTL implementation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
24w	180.0 M/M	~32.6 FTE	8	yes	Block designers
w4–w28 in stage	planned	while it runs	3 in parallel	no float	Configuration manager approves

PURPOSE

Write the **synthesisable logic for every block**, to a coding standard that survives lint, CDC, synthesis and gate-level simulation without rework.

At 180 man-months this is the largest single activity in the entire template — nearly 5% of the program's effort in one line. It is also the one most exposed to specification churn: every change upstream lands here, multiplied by the number of blocks that inherit it.

WORK FLOW

8 STEPS · 2 LANES

w4	w9	w14	w19	w24	w28
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MAIN SEQUENCE

1	2	4	6	8
---	---	---	---	---

IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Coding standard, directory structure and naming conventions	1.5w	main
2	Block RTL implementation, first pass	9w	main
3	Block-level self-checking and assertion insertion	4w	parallel
4	Block RTL review against specification	3.5w	main
5	Parameterisation and configurability implementation	3w	parallel
6	Bug fix and refinement against block verification	7w	main
7	Block-level PPA refinement against trial synthesis	4w	parallel
8	Block freeze and release tagging	3w	main

Step 6 is seven weeks of fixing rather than writing, and it is the honest half of RTL. First-pass code is written in step 2; what makes it correct is the loop against block-level verification, and shortening that loop is where schedule pressure does the most damage.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Block specifications from RTL-01
- Coding and lint standards from the methodology
- IP integration requirements from RTL-03
- Verification feedback from DV-03
- Trial synthesis results from RTL-08

PRODUCES

- Block RTL source, tagged
- Embedded assertions and self-checking
- Parameterised and configurable blocks
- Block review records
- Block release tags with known issues

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D1	PRODUCES	Block and top-level RTL release, tagged. The block half of the deliverable; RTL-04 assembles the top.
RTL-D4	FEEDS	Lint / CDC / RDC clean reports. Clean reports are a property of the code, and the coding standard set here largely determines them.

RISKS AND WATCH-OUTS

- ◆ **Specification churn arriving mid-implementation.** Every upstream change lands here multiplied by the blocks affected, which is why RTL-10 exists.
- ◆ **Coding standard imposed after code exists.** Retrofitting a standard across 180 man-months of code is a project nobody schedules and everyone half-does.
- ◆ **Assertions treated as verification's job.** Assertions written by the implementer capture intent; written by someone else they capture observed behaviour.
- ◆ **Blocks declared complete without verification feedback.** First-pass code is not correct code, and a block frozen before its verification loop closes freezes its bugs.
- ◆ **Configurability added everywhere.** Parameterisation nobody uses multiplies the verification space at no benefit, and the cost lands on DV-03.

EFFORT SPLIT	180 M/M	ENTRY CRITERIA	FEEDS INTO
Block implementation	70.0	● Block specifications passed the RTL-01 entry gate	RTL-04 Chip-level integration and top assembly
Bug fix and refinement	45.0	● Coding standard and build environment available	RTL-06 Lint, CDC and RDC closure
Assertions and self-checking	25.0	● Block-level verification environment starting up in DV-02	RTL-08 Trial synthesis feedback loop on RTL PPA
Block review	20.0		DV-03 Block-level constrained-random and directed testing
PPA refinement	12.0		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
Parameterisation	8.0		
Total	180.0	EXIT CRITERIA	
WHO IT TAKES	~32.6 FTE	● Every block reviewed against its specification, not only linted	MEASURED BY
Block designers		● Blocks frozen only after their verification loop has closed	● Blocks frozen against blocks planned
Implement, review and refine their blocks		● Releases tagged with known issues listed	● Bugs found per block after freeze
RTL lead			● Specification changes absorbed during implementation
Coding standard, review discipline and freeze		DEPENDS ON	
Verification engineers		RTL-01 Block microarchitecture specification	
Feedback loop through block-level testing		RTL-09 CI build, nightly regression and release management	
Synthesis liaison			
Trial synthesis feedback on PPA			
Configuration manager			
Release tagging and traceability			

RTL-03

Third-party and internal IP integration
— wrappers, glue, configuration

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	45.0 M/M	~12.2 FTE	8	yes	IP integration lead
w8–w24 in stage	planned	while it runs	3 in parallel	no float	Procurement liaison approves

PURPOSE

Bring every third-party, foundry and internal IP into the design — **wrappers, glue, configuration and version control** — and prove each one behaves as its documentation claims.

IP integration fails at the boundary rather than in the block. A PHY that meets its specification and a controller that meets its specification can still disagree about reset ordering, and the disagreement is found here or it is found in the lab.

WORK FLOW

8 STEPS · 2 LANES

w8	w12	w16	w20	w24
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MAIN SEQUENCE

1	2	3	5	8
---	---	---	---	---

IN PARALLEL

4	6	7
---	---	---

#	STEP	TAT	LANE
1	IP receipt, incoming inspection against the acceptance checklist	1.5w	main
2	Integration environment and wrapper framework	2w	main
3	Per-IP wrapper and glue implementation	5w	main
4	Configuration, tie-off and parameter setting per IP	3w	parallel
5	Boundary and reset sequencing verification per IP	4w	main
6	AMS macro integration with behavioural models	3.5w	parallel
7	IP version manifest and update handling	2.5w	parallel
8	Integration report and issue escalation to vendors	3.5w	main

Step 1 is the acceptance check `IPR-06` defined, and it is the last point at which an incomplete delivery is the vendor's problem. Accepting a delivery without inspecting it transfers the missing views to the integration team.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- IP deliveries against the schedule from IPR-09
- Acceptance checklists from IPR-06
- AMS macros and models from AMS-15 and AMS-16
- Block RTL from RTL-02
- Integration requirements from the architecture specification

PRODUCES

- Incoming inspection records per delivery
- IP wrappers and glue logic
- Per-IP configuration and tie-off
- Boundary and reset sequencing verification results
- IP version manifest

- Integration report and vendor issue log

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D6	PRODUCES	IP integration report and version manifest. This activity is the deliverable, and the manifest is what makes an IP-related bug traceable to a version.
RTL-D1	FEEDS	Block and top-level RTL release. Integrated IP is part of the release, and its versions are part of the tag.

RISKS AND WATCH-OUTS

- ◆ **Deliveries accepted without inspection.** Missing views and collateral then become the integration team's problem rather than the vendor's.
- ◆ **Boundary behaviour assumed from documentation.** Reset ordering, enable timing and calibration sequences are where IP integration actually fails.
- ◆ **Version manifest not maintained.** An IP bug that cannot be traced to a version cannot be reported usefully or fixed reproducibly.
- ◆ **IP updates absorbed silently.** A vendor drop applied without impact assessment can change behaviour verification has already signed off.
- ◆ **Late IP compressing integration.** The integration window is fixed at its far end by RTL-04; a late delivery does not move it, it shortens it.

EFFORT SPLIT	45 M/M	ENTRY CRITERIA	FEEDS INTO
Wrapper and glue implementation	14.0	● IP delivery schedule aligned by IPR-09	RTL-04 Chip-level integration and top assembly
Boundary and sequencing verification	11.0	● Acceptance checklists agreed in IPR-06	RTL-06 Lint, CDC and RDC closure
AMS macro integration	7.0	● Integration environment available from RTL-09	DV-03 Block-level constrained-random and directed testing
Configuration and tie-off	5.0		
Incoming inspection	4.0		DV-07 AMS / mixed-signal co-simulation
Version manifest and reporting	4.0		PD-02 Floorplan, macro placement and partition definition
Total	45.0		
WHO IT TAKES	~12.2 FTE	EXIT CRITERIA	MEASURED BY
IP integration lead Owns integration, inspection and the manifest		● Every delivery inspected against its checklist on receipt ● Boundary and reset behaviour verified, not assumed from documentation ● Version manifest complete and maintained through updates	● Deliveries inspected on receipt ● Boundary defects found here against those found at chip level ● IP issues traceable to a manifest version
Integration engineers Wrappers, glue and configuration			
Verification engineers Boundary and sequencing verification			
AMS integration engineer Analog macro integration and co-simulation		IPR-06 IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)	
		IPR-07 Licence negotiation, PO issue and delivery-date commitment	
		IPR-09 IP delivery schedule alignment to RTL and PD integration windows	
		AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)	
Procurement liaison Vendor escalation on delivery and quality issues		AMS-16 AMS-digital co-simulation and integration support	

08 RTL · RTL DESIGN & INTEGRATION

RTL-04

Chip-level integration and top assembly

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	35.0 M/M	~12.7 FTE	8	yes	Chip integration lead
w16–w28 in stage	planned	while it runs	3 in parallel	no float	Software liaison approves

PURPOSE

Assemble the blocks and the IP into **one top-level design that elaborates, connects and boots** — the first point at which the chip exists as a single object.

Chip-level integration is where every optimistic assumption made in isolation meets every other one. Connectivity, clock and reset distribution, address maps and configuration all become concrete here, and the defects found are the ones no block-level test could have seen.

WORK FLOW8 STEPS · 2 LANES

w16	w19	w22	w25	w28
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MAIN SEQUENCE

1	2	5	7	8
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IN PARALLEL

3	4	6
---	---	---

#	STEP	TAT	LANE
1	Top-level hierarchy and connectivity definition	1.5w	main
2	NoC and interconnect integration	3w	main
3	Clock and reset distribution implementation	2.5w	parallel
4	Address map and configuration space assembly	2w	parallel
5	Top-level elaboration and connectivity checking	2.5w	main
6	Boot and bring-up path integration	2.5w	parallel
7	Chip-level smoke tests and first boot	2.5w	main
8	Integration issue closure and top release	2.5w	main

Step 7 — the first chip-level boot in simulation — is the moment the design becomes real. It is also slow, because the first boot exercises paths that no block test touched, and each failure takes a full-chip simulation to reproduce.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Block RTL releases from RTL-02
- Integrated IP from RTL-03
- Power and clock intent from RTL-05
- Register map from RTL-07
- Top-level architecture from ARCH-09

PRODUCES

- Top-level hierarchy and connectivity
- Integrated NoC and interconnect
- Clock and reset distribution
- Assembled address map and configuration space
- Chip-level smoke test results
- Top-level RTL release

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D1	PRODUCES	Block and top-level RTL release, tagged. The top half of the deliverable — the object every downstream stage actually consumes.
RTL-D7	FEEDS	RTL Freeze package. Freeze is declared on the top-level release, so its state is what the package describes.

RISKS AND WATCH-OUTS

- ◆ **Connectivity checked by inspection.** A chip this size has too many connections to review; automated connectivity checking is the only reliable method.
- ◆ **Integration started before blocks are stable.** Integrating moving blocks produces defects that are re-found every time a block updates.
- ◆ **Address map assembled from block documentation.** Generated maps agree with the code; hand-assembled maps agree with the documentation, which is a different thing.
- ◆ **First boot deferred.** The boot path exercises reset, clocking, configuration and the fabric together, and delaying it delays every defect that only appears in combination.
- ◆ **Clock and reset distribution treated as a wiring task.** It carries the crossing inventory from ARCH-05 and the CDC obligations that follow from it.

EFFORT SPLIT	35 M/M	ENTRY CRITERIA	FEEDS INTO
NoC and interconnect integration	9.0	● Blocks stable enough to integrate from RTL-02	RTL-06 Lint, CDC and RDC closure
Chip-level smoke tests and boot	7.0	● IP integrated and boundary-verified in RTL-03	RTL-08 Trial synthesis feedback loop on RTL PPA
Connectivity checking	6.0	● Register map generated by RTL-07	DV-04 Chip-level and system-level scenario testing
Clock and reset distribution	5.0		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
Address map assembly	4.0		DFT-06 Scan insertion and DFT DRC during synthesis
Issue closure and release	4.0		
Total	35.0		
WHO IT TAKES	~12.7 FTE		MEASURED BY
Chip integration lead Owns the top level and its release		● Connectivity verified automatically, not by inspection	● Connectivity defects found by checking against by simulation
Integration engineers Hierarchy, connectivity and fabric integration		● Chip boots in simulation before freeze is contemplated	● Weeks from first integration to first successful boot
Clock and reset owner Distribution implementation		● Top-level release tagged with its full IP manifest	● Blocks integrated against blocks released
Verification engineers Chip-level smoke tests and first boot			
Software liaison Boot path and configuration space expectations			
		DEPENDS ON	
		RTL-02 Block-level RTL implementation	
		RTL-03 Third-party and internal IP integration — wrappers, glue, configuration	
		RTL-05 Clock, reset and power intent (UPF) implementation	
		RTL-07 Register map / RDL definition and header generation	

RTL-05

Clock, reset and power intent (UPF) implementation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	18.0 M/M	~7.8 FTE	7	no	Low-power implementation lead
w6–w16 in stage	planned	while it runs	3 in parallel	has float	Physical design liaison approves

PURPOSE

Implement the **clock, reset and power intent** the architecture defined — domain boundaries, isolation, retention, crossing synchronisers — and produce a UPF file the tools accept.

Power intent is where architecture becomes an obligation on every downstream tool. Synthesis inserts isolation cells from it, verification checks against it, and physical design implements it. An intent file that is wrong or incomplete propagates the same error into all three.

WORK FLOW

7 STEPS · 2 LANES

w6	w8	w10	w12	w14	w16
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Domain boundary implementation against the ARCH-05 definition	2w	main
2	Isolation, level shifter and retention strategy implementation	2.5w	main
3	Clock domain crossing synchroniser implementation	2w	parallel
4	Reset sequencing and synchronisation implementation	1.5w	parallel
5	UPF authoring and tool acceptance check	2.5w	main
6	Power management controller RTL	2w	parallel
7	Intent review with synthesis, verification and physical design	3w	main

Step 5 includes a tool acceptance check because UPF that reads correctly to a human and fails to elaborate in synthesis is common. The file has to be exercised in the actual flow, not reviewed as a document.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Power and clock architecture from ARCH-05
- Crossing inventory from ARCH-05
- Block list and hierarchy from RTL-01
- UPF methodology from PDK-09
- Library low-power cell availability from PDK-03

PRODUCES

- Implemented domain boundaries
- Isolation, level shifter and retention implementation
- CDC synchroniser implementation
- Reset sequencing logic
- UPF power intent file, tool-accepted
- Power management controller RTL

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D5	PRODUCES	UPF power intent file. This activity is the deliverable, and "tool-accepted" is the standard rather than "reviewed".
RTL-D4	FEEDS	Lint / CDC / RDC clean reports. The synchroniser implementation is what CDC closure in RTL-06 is checking.

RISKS AND WATCH-OUTS

- ◆ **UPF written as a document.** Intent that has never been elaborated in synthesis is intent whose errors are found by the first person who runs the flow.
- ◆ **Crossings implemented without reference to the inventory.** A synchroniser strategy chosen per crossing by each designer produces inconsistency CDC will flag hundreds of times.
- ◆ **Retention strategy decided late.** Retention cells cost area and change the physical implementation; adding them after floorplan is expensive.
- ◆ **Reset sequencing implemented from the block view.** Reset order is a chip-level property, and blocks each releasing in their own order produces a boot that works in simulation and not in silicon.
- ◆ **Intent reviewed only by its author.** Synthesis, verification and physical design each consume it differently, and each has to confirm it works for them.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
Isolation and retention implementation	5.0	● Power and clock architecture fixed by ARCH-05	RTL-04 Chip-level integration and top assembly
Domain boundary implementation	4.0	● Crossing inventory available	RTL-06 Lint, CDC and RDC closure
UPF authoring and acceptance	3.5	● Low-power cells qualified in PDK-03	DV-06 Low-power (UPF) verification
CDC synchroniser implementation	2.5	EXIT CRITERIA	SYN-06 Power optimisation — clock gating, operand isolation, Vt mix
Power management controller	2.0		PD-03 Power delivery network design and early IR analysis
Intent review	1.0	● UPF elaborates in synthesis, not only reviews well ● Every crossing on the inventory has an implemented synchroniser ● Intent accepted by synthesis, verification and physical design	
Total	18.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~7.8 FTE	ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	● UPF elaboration errors at first synthesis run ● Crossings implemented against the inventory ● Intent revisions after RTL freeze
Low-power implementation lead Owns domains, intent and the UPF file		RTL-01 Block microarchitecture specification	
Clock and reset owner Synchroniser and reset sequencing implementation		PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	
Synthesis liaison Confirms the intent elaborates and infers correctly			
Verification liaison Confirms the intent is checkable			
Physical design liaison Confirms the intent is implementable			

08 RTL · RTL DESIGN & INTEGRATION

RTL-06

Lint, CDC and RDC closure

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	30.0 M/M	~8.1 FTE	7	yes	RTL quality lead
w12–w28 in stage	planned	while it runs	2 in parallel	no float	CAD engineer approves

PURPOSE

Drive **lint, clock domain crossing and reset domain crossing to closure** — every violation either fixed or waived with a reason someone signed.

These checks find the class of defect that simulation cannot: a crossing without a synchroniser works in every simulation and fails intermittently in silicon. The discipline is in the waivers — an unexamined waiver list is where the surviving bugs live.

WORK FLOW

7 STEPS · 2 LANES

w12	w16	w20	w24	w28
-----	-----	-----	-----	-----

MAIN SEQUENCE

1	2	4	6	7
---	---	---	---	---

IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Lint, CDC and RDC methodology and rule set definition	1.5w	main
2	Block-level lint closure	3.5w	main
3	Block-level CDC closure	3.5w	parallel
4	Chip-level CDC closure against the crossing inventory	4w	main
5	RDC closure and reset domain analysis	3w	parallel
6	Waiver review and disposition	3w	main
7	Closure reporting and regression setup	4w	main

Step 4 is chip-level rather than block-level because most real crossings are between blocks. Block-level CDC finds the easy ones; the crossing that fails in silicon is usually one neither block owner considered theirs.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Block and top RTL from RTL-02 and RTL-04
- Crossing inventory from ARCH-05
- Synchroniser implementation from RTL-05
- Lint and CDC tool setup from PDK-08
- Waiver policy from the methodology

PRODUCES

- Lint, CDC and RDC rule sets
- Block and chip-level closure results
- Waiver list with dispositions and signatures
- Crossing inventory reconciliation
- Closure regression in the CI flow

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D4

PRODUCES

Lint / CDC / RDC clean reports with waiver list. This activity is the deliverable, and the waiver list is the part that needs review rather than filing.

RTL-D7

GATES

RTL Freeze package. Freeze with open CDC violations is freezing a class of defect that will not appear until silicon.

RISKS AND WATCH-OUTS

- ◆ **Waivers accumulated rather than reviewed.** A waiver list nobody re-reads is where the surviving crossings hide, and they are the hardest silicon bugs to diagnose.
- ◆ **CDC run only at block level.** The crossings that matter are usually between blocks, and neither owner runs the check that would find them.
- ◆ **Closure not regressed.** Clean at one release and dirty at the next, with nobody noticing until the freeze package is assembled.
- ◆ **Rule set weakened to reduce noise.** Turning off the rule that produces the most violations is the fastest way to a clean report and the slowest way to a working chip.
- ◆ **Crossing inventory not reconciled.** Crossings found by the tool but absent from ARCH-05's inventory mean the architecture and the implementation disagree.

EFFORT SPLIT

30 M/M

ENTRY CRITERIA

FEEDS INTO

Chip-level CDC closure	8.0
Block-level lint closure	6.0
Block-level CDC closure	6.0
RDC closure	4.0
Waiver review	3.0
Reporting and regression	3.0
Total	30.0

WHO IT TAKES

~8.1 FTE

RTL quality lead

Owns lint, CDC and RDC closure

CDC specialist

Chip-level analysis and waiver review

Block owners

Block-level closure of their own violations

Clock and reset owner

Inventory reconciliation

CAD engineer

Tool setup and regression integration

- Blocks stable enough to lint meaningfully
- Synchronisers implemented in RTL-05
- Tools set up and qualified by PDK-08

EXIT CRITERIA

- Every violation fixed or waived with a signed reason
- Chip-level CDC clean, not only block-level
- Closure regressed in CI so it stays clean

DEPENDS ON

- RTL-02** Block-level RTL implementation
- RTL-04** Chip-level integration and top assembly
- RTL-05** Clock, reset and power intent (UPF) implementation
- PDK-08** EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR

- RTL-10** Specification change control and ECO board
- SYN-01** SDC constraint development and validation
- DV-11** Gate-level simulation, functional and timing-annotated
- S0-09** Final formal equivalence and LVS netlist consistency

MEASURED BY

- Waivers reviewed against waivers granted
- Crossings found at chip level that block level missed
- Closure regression failures after first clean

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D2	PRODUCES	Register map / RDL and generated headers. This activity is the deliverable, generated from one source rather than maintained in several.
RTL-D1	FEEDS	Block and top-level RTL release. Generated register RTL is part of every block, so the map's release cadence is tied to the RTL's.

RISKS AND WATCH-OUTS

- ◆ **Map maintained in more than one place.** RTL, headers and documentation drift apart, and the driver eventually writes to an offset that moved.
- ◆ **Software reviews the map after freeze.** An awkward interface then gets worked around in the driver, and the workaround becomes the contract nobody documented.
- ◆ **Side effects undocumented.** A read that clears a status bit is a behaviour firmware has to know about, and discovering it in the lab costs days per instance.
- ◆ **Security access rules omitted.** Registers that should be locked in production lifecycle states are locked late or not at all.
- ◆ **Reset values unspecified.** Registers whose reset value is whatever the implementer chose produce a boot sequence that depends on undocumented behaviour.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Per-block register definition	3.5	● Block specifications available from RTL-01	RTL-02Block-level RTL implementation
Generation flow	2.0	● Address map intent from the architecture specification	RTL-04Chip-level integration and top assembly
Address map allocation	1.5	● RDL tooling selected and available	DV-03Block-level constrained-random and directed testing
Access policy and side effects	1.5		BU-04Boot, firmware load and functional smoke test
Release and change control	1.5		
Total	10.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.4 FTE	● RTL, headers and documentation generated from one source	● Artifacts generated against artifacts hand-maintained
Register map owner Owns the description source and the generation flow		● Map reviewed by firmware before it is frozen	● Register changes after software review
Block owners Define their block's registers and side effects		● Side effects and reset values specified for every register	● Side effects documented against those found in bring-up
Firmware lead Reviews the map from the driver's perspective		DEPENDS ON	
Security architect Access rules per lifecycle state		RTL-01Block microarchitecture specification	
Verification liaison UVM register model requirements		ARCH-07Security and safety architecture — secure boot, root of trust, fusing	
		ARCH-09Architecture specification authoring and review	

08 RTL · RTL DESIGN & INTEGRATION

RTL-08

Trial synthesis feedback loop on RTL PPA

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	20.0 M/M	~7.2 FTE	7	no	Synthesis engineer
w10–w22 in stage	planned	while it runs	3 in parallel	has float	RTL lead approves

PURPOSE

Run **trial synthesis continuously against the evolving RTL** and feed timing, area and power back to block owners while the code can still change cheaply.

RTL that has never been synthesised is RTL whose PPA is unknown. A block that misses its area budget by 30% is a redesign if found at SYN-02 and an afternoon if found here, and the difference is only whether anyone was running synthesis during implementation.

WORK FLOW

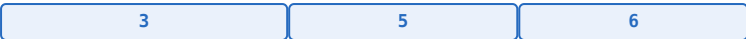
7 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Trial synthesis flow and constraint bootstrap	1.5w	main
2	Per-block trial synthesis and QoR baseline	3w	main
3	Timing feedback to block owners	2.5w	parallel
4	Area and budget tracking against ARCH-06	2w	main
5	Power and clock gating effectiveness feedback	2w	parallel
6	Congestion and routability early indication	2w	parallel
7	Recurring trial synthesis cadence and reporting	5.5w	main

Step 7 is the activity's real shape — a weekly cadence rather than a one-off study. A single trial synthesis is a snapshot; a weekly one is a feedback loop, and only the loop changes how blocks are written.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Block RTL from RTL-02
- Constraints bootstrap from SYN-01
- Library and PDK from PDK-13
- PPA budgets from ARCH-06
- Floorplan intent from ARCH-08

PRODUCES

- Trial synthesis flow and constraints
- Per-block QoR baselines
- Timing, area and power feedback reports
- Budget tracking against ARCH-06
- Congestion and routability indications

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D1

FEEDS

Block and top-level RTL release. A block released without a trial synthesis result has an unknown PPA, and the release quality reflects that.

RTL-D7

FEEDS

RTL Freeze package. The package states where each block sits against its budget, and this activity is where that number comes from.

RISKS AND WATCH-OUTS

- ◆ **Trial synthesis run once.** A snapshot tells a block owner where they were; only a cadence tells them whether they are improving.
- ◆ **Feedback delivered without constraints.** Timing results against bootstrap constraints that do not resemble the real ones send block owners chasing paths that do not matter.
- ◆ **Area overrun tolerated because it is early.** Blocks rarely shrink after implementation, and an overrun accepted at trial synthesis is an overrun accepted permanently.
- ◆ **Congestion ignored.** Routability problems are cheaper to fix in RTL structure than in placement, and the early indication is the only chance to do so.
- ◆ **Results not tied to the budget.** QoR reported in absolute terms rather than against ARCH-06 tells nobody whether the block is acceptable.

EFFORT SPLIT

20 M/M

ENTRY CRITERIA

FEEDS INTO

Recurring trial synthesis and reporting

7.0

- Blocks synthesisable, even if incomplete

RTL-02

Block-level RTL implementation

Per-block synthesis and baselines

4.5

- Constraint bootstrap available from SYN-01

SYN-02

Technology mapping and optimisation — multi-Vt, multi-bit, useful skew

Timing feedback

3.0

- Library and environment available from PDK-13

SYN-07

Low-power synthesis — isolation and level-shifter insertion against UPF

Area and budget tracking

2.5

Power feedback

1.5

EXIT CRITERIA

Congestion indication

1.5

PD-01

Flow setup and MMMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity

Total

20.0

- Every block has a current QoR result against its budget
- Feedback delivered on a cadence, not on request
- Overruns escalated rather than noted

MEASURED BY

WHO IT TAKES

~7.2 FTE

Synthesis engineer

Owens the trial flow and the reporting cadence

Block owners

Act on the feedback in their RTL

PPA lead

Budget tracking and escalation on overruns

Physical design liaison

Congestion and routability interpretation

RTL lead

Enforces that feedback is acted on

DEPENDS ON

RTL-02

Block-level RTL implementation

ARCH-06

PPA budget allocation per block

PDK-13

Golden design environment release and flow regression

- Blocks with a current QoR result
- Area against budget at RTL freeze
- Weeks between a block change and its next QoR number

08 RTL · RTL DESIGN & INTEGRATION

RTL-09

CI build, nightly regression and release management

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
30w cont. w2–w32 in stage, runs throughout	20.0 M/M planned	~2.9 FTE while it runs	7 3 in parallel	no has float	Configuration and release manager RTL lead approves

PURPOSE

Run the **build, regression and release machinery** the whole stage depends on — continuous integration, nightly regressions, tagged releases and a reproducible environment.

This is infrastructure, and its value shows only in its absence. Without a nightly build, a break introduced on Tuesday is found on Friday by whoever needed the design; without release tagging, a bug cannot be reproduced against the version that had it.

WORK FLOW7 STEPS · 2 LANES

w2	w8	w14	w20	w26	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Build system and dependency management setup	2.5w	main
2	Continuous integration and per-commit checking	3w	main
3	Nightly regression definition and scheduling	2.5w	parallel
4	Release tagging and version manifest process	2w	main
5	Compute and licence capacity management for regressions	2w	parallel
6	Failure triage and notification workflow	2.5w	parallel
7	Continuous operation and maintenance across the stage	22.5w	main

Twenty-two of the thirty weeks are step 7, which is maintenance rather than construction. A CI system that is not maintained degrades into one everyone ignores, and an ignored red build is worse than no build at all.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- RTL source from RTL-02, RTL-03 and RTL-04
- Verification regressions from DV-10
- Compute and licence capacity from PDK-12
- Tool versions from PDK-08
- Configuration management policy

PRODUCES

- Build system and dependency management
- Continuous integration with per-commit checks
- Nightly regression suite and schedule
- Release tagging and version manifests
- Failure triage and notification workflow

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

RTL-D3

PRODUCES

Integration testbench and build system. This activity is the deliverable, and its measure is that every release is reproducible from a tag.

RTL-D1

FEEDS

Block and top-level RTL release. Releases are produced and tagged by this machinery; without it a release is a directory copy.

RISKS AND WATCH-OUTS

- ◆ **Red builds tolerated.** Once a broken build is normal, the signal is gone and every subsequent break is invisible.
- ◆ **Regression runtime exceeding the night.** A nightly that takes thirty hours is not nightly, and the feedback loop silently doubles.
- ◆ **Releases produced by hand.** An untagged release cannot be reproduced, and an IP bug found against it cannot be reported precisely.
- ◆ **Capacity not managed.** Regressions competing with verification for the same licences slow both, usually at the point in the schedule where both matter most.
- ◆ **Triage unowned.** Failures nobody is assigned to investigate accumulate until the regression result is meaningless.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous operation and maintenance	8.0	Repository and branching model agreed	RTL-02 Block-level RTL implementation
CI and per-commit checking	4.0	Tool versions available from PDK-08	RTL-04 Chip-level integration and top assembly
Build system setup	3.0	Compute capacity allocated by PDK-12	DV-10 Coverage closure and regression stability management
Nightly regression setup	2.0	EXIT CRITERIA	SYN-01 SDC constraint development and validation
Triage workflow	1.5	Every release reproducible from a tag	MEASURED BY
Release tagging process	1.5	Nightly regression completing within the night	Days the build stays red
Total	20.0	Failures triaged within a working day, with an owner	Nightly regression runtime
WHO IT TAKES	~2.9 FTE	DEPENDS ON	Releases reproducible from their tag
Configuration and release manager Owns builds, tags and manifests		PDK-08 EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
CI engineer Continuous integration and regression infrastructure		PDK-12 Compute farm, EDA license and storage capacity planning	
Verification liaison Regression content and runtime			
CAD infrastructure Compute and licence capacity			
RTL lead Enforces build discipline			

08 RTL · RTL DESIGN & INTEGRATION

RTL-10

Specification change control and ECO board

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
30w cont. w2–w32 in stage, runs throughout	12.0 M/M planned	~1.7 FTE while it runs	7 2 in parallel	yes no float	Program manager Configuration manager approves

PURPOSE

Control what changes and when — **specification change requests, ECO board decisions and the freeze itself** — so the design converges rather than oscillates.

Every stage downstream assumes RTL stops moving at some point. Change control is how that point is created: not by refusing changes, but by making each one visible, costed and decided by people who can see what it affects. The freeze is this activity's output, and its credibility is the whole program's.

WORK FLOW

7 STEPS · 2 LANES

w2	w8	w14	w20	w26	w32
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Change control process and ECO board definition	1.5w	main
2	Change request intake and impact assessment workflow	2w	main
3	ECO board operation and decision records	20w	main
4	Change tracking against blocks, verification and synthesis	3w	parallel
5	Freeze criteria definition and readiness tracking	2.5w	parallel
6	RTL Freeze package assembly	3w	main
7	Freeze declaration and post-freeze exception policy	3.5w	main

Step 3 runs for twenty weeks and is a standing meeting rather than a task. Its discipline is impact assessment: a change that looks like an afternoon in one block can invalidate a month of verification, and the board exists so that cost is visible before the change is approved.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Change requests from every source — architecture, verification, synthesis, IP
- Verification status from DV-12
- Trial synthesis and PPA status from RTL-08
- Lint and CDC closure status from RTL-06
- Freeze criteria from the program plan

PRODUCES

- Change control process and ECO board charter
- Change request log with impact assessments
- ECO board decision records
- Freeze criteria and readiness tracking
- RTL Freeze package and declaration

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

- RTL-D7

PRODUCES

RTL Freeze package. This activity is the deliverable — the freeze, its criteria, its exceptions and the change history behind it.
- RTL-D1

GATES

Block and top-level RTL release. After freeze, a release is only produced through an approved exception.

RISKS AND WATCH-OUTS

- ◆ **Freeze declared by date rather than by criteria.** A gate that passes because it was scheduled teaches everyone that gates mean nothing.
- ◆ **Impact assessment skipped for small changes.** The cost of a change is rarely in the block that changes; it is in the verification that has to be re-run.
- ◆ **Post-freeze exceptions ungoverned.** Changes will be needed after freeze; without a policy they happen informally and nobody knows what the design contains.
- ◆ **Board without downstream representation.** Verification and synthesis are the ones who pay for most changes, and a board without them approves costs it cannot see.
- ◆ **Change log not reconciled with the release.** A freeze package that does not match what was actually built is a package nobody can rely on.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
ECO board operation	5.0	● Process agreed and board membership named ● Change intake path published to every team ● Freeze criteria defined in the program plan	SYN-09 FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point
Impact assessment workflow	2.5		DV-12 Bug triage and disposition board
Freeze package assembly	1.5		PD-14 Functional and timing ECO implementation
Change tracking	1.5		T0-03 Tapeout checklist completion and owner sign-off
Process definition	1.0		
Freeze declaration and policy	0.5		
Total	12.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~1.7 FTE	● Freeze declared against criteria, not against a date ● Every change carries an impact assessment before approval ● Post-freeze exception policy published and used	● Changes with an impact assessment before approval ● Freeze criteria met at declaration ● Post-freeze exceptions against the policy
Program manager Chairs the ECO board and owns the freeze			
RTL lead Impact assessment on the design side			
Verification lead Impact assessment on verification cost			
Synthesis and PD representatives Downstream cost of each change			
Configuration manager Change log reconciliation with releases			
		DEPENDS ON	
		RTL-06 Lint, CDC and RDC closure	
		RTL-08 Trial synthesis feedback loop on RTL PPA	
		DEF-08 Program schedule, resource and staffing plan drafting	

DESIGN & VERIFY

09 Verification

DV

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w26 – w66	40 _w	12	650 _{M/M}	7 _{of 12}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
DV-01	Verification plan and coverage model definition		8w 25 _{M/M}	DV-D1 Verification plan (vPlan) and coverage model
DV-02	UVM environment and VIP bring-up		14w 60 _{M/M}	DV-D2 UVM testbenches and integrated VIP
DV-03	Block-level constrained-random and directed testing		30w 200 _{M/M}	DV-D3 Regression and coverage dashboards
DV-04	Chip-level and system-level scenario testing		22w 110 _{M/M}	DV-D4 Formal proof reports with assumption list
DV-05	Formal property verification — control logic, connectivity, security		16w 40 _{M/M}	DV-D5 Low-power verification report
DV-06	Low-power (UPF) verification		10w 20 _{M/M}	DV-D6 Emulation platform and system test suite
DV-07	AMS / mixed-signal co-simulation		12w 20 _{M/M}	DV-D7 Gate-level simulation report
DV-08	Emulation and FPGA prototype bring-up — capacity, partitioning, speed		20w 55 _{M/M}	DV-D8 DV closure signoff package
DV-09	Performance and bandwidth validation against the architecture model		12w 30 _{M/M}	
DV-10	Coverage closure and regression stability management		26w cont. 45 _{M/M}	
DV-11	Gate-level simulation, functional and timing-annotated		10w 20 _{M/M}	
DV-12	Bug triage and disposition board		34w cont. 25 _{M/M}	

09 DV · VERIFICATION

DV-01

Verification plan and coverage model definition

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	25.0 M/M	~13.6 FTE	6	yes	Verification lead
w0–w8 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Write the plan that says **what will be verified, how, and how anyone will know it is done** — feature by feature, with a coverage model behind each claim.

Verification is the largest line of effort in the program, and without a plan it is 650 man-months of activity with no definition of completion. The coverage model is what turns "we tested it" into a measurable statement, and the features nobody writes into the plan are the features nobody verifies.

WORK FLOW

6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Feature extraction from the architecture and block specifications	1.5w	main
2	Verification strategy per feature — simulation, formal, emulation	1.5w	main
3	Coverage model definition — functional, code, assertion	2w	main
4	Closure criteria and sign-off definition per block	1w	parallel
5	Resource, compute and schedule modelling	1w	parallel
6	Verification plan review with design and architecture	3w	main

Step 6 is a review by the people who wrote the specification, and its purpose is to find features the plan missed. Verification reviewing its own plan confirms it is coherent; architecture reviewing it confirms it is complete, and incompleteness is the failure mode that matters.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Architecture specification from ARCH-09
- Block specifications from RTL-01
- Register map from RTL-07
- IP verification collateral requirements from IPR-06
- Compute capacity model from PDK-12

PRODUCES

- Feature list extracted from the specifications
- Verification strategy per feature
- Coverage model — functional, code and assertion
- Closure criteria per block and at chip level
- Resource and schedule model
- Reviewed verification plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D1

PRODUCES

Verification plan (vPlan) and coverage model. This activity is the deliverable, and the coverage model is what every later closure claim is measured against.

DV-D8

GATES

DV closure signoff package. Closure is defined here; a signoff against criteria written later is a signoff against a moving target.

RISKS AND WATCH-OUTS

- ◆ **Plan written from the RTL rather than the specification.** It then verifies what was built rather than what was specified, and the two differ exactly where the bugs are.
- ◆ **Coverage model that measures activity rather than intent.** Code coverage says lines were executed; functional coverage says features were exercised, and only the second answers the question.
- ◆ **Closure criteria left vague.** "High coverage" is not a criterion anyone can sign, and it becomes whatever the schedule allows.
- ◆ **Architecture not in the review.** The features the plan missed are exactly the ones only the specification's author would notice.
- ◆ **Resource model omitted.** A plan that needs more compute or more engineers than exist is a plan whose closure date is fiction.

EFFORT SPLIT	25 M/M	ENTRY CRITERIA	FEEDS INTO
Coverage model definition	8.0	● Architecture specification frozen by ARCH-09	DV-02 UVM environment and VIP bring-up
Feature extraction	5.0	● Block specifications available from RTL-01	DV-03 Block-level constrained-random and directed testing
Strategy per feature	5.0	● Compute capacity model available from PDK-12	DV-05 Formal property verification — control logic, connectivity, security
Plan review	4.0		DV-10 Coverage closure and regression stability management
Closure criteria	2.0		IPR-06 IP deliverable checklist review (GDS, LEF, multi-corner .lib, CDL, UPF, DV collateral, errata)
Resource modelling	1.0		
Total	25.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~13.6 FTE	● Every specified feature has a verification strategy and a coverage point ● Closure criteria written and agreed before verification starts ● Plan reviewed by architecture and design, not only by verification	● Features with a coverage point against features specified ● Closure criteria signed before verification starts ● Features added to the plan after it was reviewed
Verification lead Owns the plan, the coverage model and the criteria		DEPENDS ON	
Block verification owners Per-block strategy and closure criteria		ARCH-09 Architecture specification authoring and review	
Chief architect Confirms feature completeness against the specification		RTL-01 Block microarchitecture specification	
RTL lead Confirms the plan is verifiable as the design is built		RTL-07 Register map / RDL definition and header generation	
Program manager Resource and schedule feasibility			

09 DV · VERIFICATION

DV-02

UVM environment and VIP bring-up

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w	60.0 M/M	~18.6 FTE	8	yes	Verification architect
w4–w18 in stage	planned	while it runs	3 in parallel	no float	Register map owner approves

PURPOSE

Build the **verification environment** — UVM agents, scoreboards, sequences, VIP integration — that every test in the program will run inside.

The environment is infrastructure with a long lead time and no visible output until it works. Sixty man-months buys agents for every interface, scoreboards that actually check, and reusable sequences. Testbench bring-up running late is the single most common reason verification starts late, and it starts late invisibly.

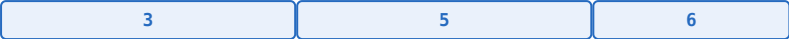
WORK FLOW8 STEPS · 2 LANES

w4w7w10w13w16w18

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Environment architecture and reuse strategy	1.5w	main
2	Agent development per interface	4w	main
3	Third-party VIP integration and configuration	3w	parallel
4	Scoreboard and reference model development	4w	main
5	Sequence library and stimulus infrastructure	3w	parallel
6	Register model integration from the RDL source	2w	parallel
7	Environment bring-up on the first block	2.5w	main
8	Environment release and documentation	2w	main

Step 4 is where verification quality is decided. A scoreboard that compares against a reference model finds bugs; one that only checks protocol legality confirms the design does something rather than the right thing.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Verification plan from DV-01
- Block and interface specifications from RTL-01
- Register description source from RTL-07
- VIP licences from IPR-07
- Methodology standards and reuse libraries

PRODUCES

- Environment architecture and reuse strategy
- UVM agents per interface
- Integrated third-party VIP
- Scoreboards and reference models
- Sequence library
- Register model integration

- Released environment with documentation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D2

PRODUCES

UVM testbenches and integrated VIP. This activity is the deliverable — the environment every subsequent verification activity runs inside.

DV-D1

FEEDS

Verification plan and coverage model. Coverage is collected by the environment, so what it can measure bounds what the plan can claim.

RISKS AND WATCH-OUTS

- ◆ **Bring-up slower than planned and invisible.** Environment work has no test-pass metric, so slippage is not visible until testing should have started.
- ◆ **Scoreboards that check protocol rather than function.** A legal transaction carrying wrong data passes, and the bug survives to chip level.
- ◆ **No reuse strategy.** Every block building its own environment multiplies effort and makes results incomparable across blocks.
- ◆ **Register model hand-written.** A model that diverges from the RDL source produces tests that pass against the wrong offsets.
- ◆ **VIP licences short.** Verification concurrency is capped by VIP seats, and the cap is discovered when regressions start queuing.

EFFORT SPLIT	60 M/M	ENTRY CRITERIA	FEEDS INTO
Agent development	16.0	● Verification plan available from DV-01	DV-03 Block-level constrained-random and directed testing
Scoreboard and reference models	16.0	● Interface specifications stable enough to build agents against	DV-04 Chip-level and system-level scenario testing
VIP integration	10.0	● VIP licences executed in IPR-07	DV-06 Low-power (UPF) verification
Sequence library	8.0		DV-07 AMS / mixed-signal co-simulation
Environment bring-up	5.0	EXIT CRITERIA	DV-10 Coverage closure and regression stability management
Register model and documentation	5.0	● Environment running on a real block, not on a stub ● Scoreboards checking against reference models ● Environment documented and reused rather than rebuilt per block	
Total	60.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~18.6 FTE	DV-01 Verification plan and coverage model definition	● Blocks running in the shared environment against total ● Bugs found by scoreboards against by manual inspection ● Environment bring-up against plan
Verification architect Owns the environment architecture and reuse		RTL-01 Block microarchitecture specification	
Agent and VIP engineers Interface agents and VIP integration		RTL-07 Register map / RDL definition and header generation	
Reference model engineers Scoreboards and functional models		IPR-07 Licence negotiation, PO issue and delivery-date commitment	
Methodology engineer Standards, reuse libraries and documentation			
Register map owner RDL-generated register model			

09 DV · VERIFICATION

DV-03

Block-level constrained-random and directed testing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
30w	200.0 M/M	~29.0 FTE	8	yes	Block verification owners
w8–w38 in stage	planned	while it runs	2 in parallel	no float	Methodology engineer approves

PURPOSE

Verify every block against its specification with **constrained-random and directed testing**, and drive each one to its coverage closure criteria.

Two hundred man-months — the largest activity in the template — spread across thirty weeks and every block in the design. This is where most bugs are found and most verification time is spent, and its efficiency is set almost entirely by decisions taken in DV-01 and DV-02.

WORK FLOW

8 STEPS · 2 LANES

w8	w14	w20	w26	w32	w38
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MAIN SEQUENCE

1	2	3	5	6	8
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IN PARALLEL

4	7
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#	STEP	TAT	LANE
1	Per-block testbench instantiation from the shared environment	2.5w	main
2	Directed test development for specified behaviour	5w	main
3	Constrained-random stimulus and constraint development	6w	main
4	Assertion development and binding	4w	parallel
5	Corner-case and error-injection testing	5w	main
6	Coverage analysis and hole closure per block	6w	main
7	Debug and bug reporting loop with block owners	8w	parallel
8	Block-level closure and sign-off	5.5w	main

Step 7 runs in parallel across the whole window because debugging is continuous rather than a phase. It is also where the largest single share of the effort goes — finding a bug is usually quicker than proving what caused it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Verification environment from DV-02
- Block RTL releases from RTL-02
- Coverage model and closure criteria from DV-01
- Block specifications from RTL-01
- Regression capacity from DV-10

PRODUCES

- Per-block testbenches and test suites
- Directed and constrained-random tests
- Assertions bound to block interfaces
- Corner-case and error-injection results
- Coverage analysis and hole closure records
- Block-level sign-off records

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D2	FEEDS	UVM testbenches and integrated VIP. Per-block testbenches extend the shared environment and are delivered with it.
DV-D3	FEEDS	Regression and coverage dashboards. Block coverage is the largest contributor to the dashboards DV-10 maintains.
DV-D8	FEEDS	DV closure signoff package. Block sign-off is the substance of chip-level closure; the package aggregates it.

RISKS AND WATCH-OUTS

- ◆ **Coverage closed by adding coverage points rather than tests.** Redefining the model to match what was tested produces a closure number that measures nothing.
- ◆ **Constrained-random left unconstrained.** Random stimulus without well-designed constraints spends compute exploring illegal or uninteresting space.
- ◆ **Corner cases skipped because the specification is silent.** The gap belongs back in RTL-01, not quietly into the untested set.
- ◆ **Debug loop bottlenecked on block owners.** Verification finds faster than design fixes, and the queue is where the schedule quietly goes.
- ◆ **Blocks signed off against different standards.** Without enforced criteria from DV-01, each block closes to whatever its owner considered enough.

EFFORT SPLIT	200 M/M	ENTRY CRITERIA	FEEDS INTO
Debug and bug reporting	55.0	● Environment released by DV-02	DV-04 Chip-level and system-level scenario testing
Constrained-random development	40.0	● Block RTL available from RTL-02	DV-10 Coverage closure and regression stability management
Coverage analysis and closure	35.0	● Coverage model and criteria set by DV-01	DV-12 Bug triage and disposition board
Directed test development	30.0		RTL-02 Block-level RTL implementation
Corner-case and error injection	22.0	EXIT CRITERIA	
Assertions and sign-off	18.0	● Every block closed against the DV-01 criteria, not its own	
Total	200.0	● Coverage holes closed by tests, not by model changes	MEASURED BY
WHO IT TAKES	~29.0 FTE	● Sign-off records complete per block	● Blocks closed against plan
Block verification owners		DEPENDS ON	● Coverage achieved against the model
Own their block's testing and closure		DV-01 Verification plan and coverage model definition	● Bug find and fix rates over time
Verification engineers		DV-02 UVM environment and VIP bring-up	
Test development, coverage and debug		RTL-02 Block-level RTL implementation	
Block designers			
Bug fixing and specification clarification			
Verification lead			
Closure standard and sign-off consistency			
Methodology engineer			
Constraint and assertion quality			

09 DV · VERIFICATION

DV-04

Chip-level and system-level scenario testing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
22w	110.0 M/M	~21.7 FTE	8	yes	Chip verification lead
w14–w36 in stage	planned	while it runs	3 in parallel	no float	Software liaison approves

PURPOSE

Verify the chip as **one system running realistic scenarios** — boot, configuration, traffic, power transitions, error recovery — rather than as a collection of verified blocks.

Blocks that each pass in isolation routinely fail together. Chip-level verification is where arbitration, ordering, deadlock, reset sequencing and end-to-end data integrity are actually tested, and none of those are visible from inside a block.

WORK FLOW

8 STEPS · 2 LANES

w14	w19	w24	w29	w34	w36
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MAIN SEQUENCE

1	2	3	6	8
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IN PARALLEL

4	5	7
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#	STEP	TAT	LANE
1	Chip-level environment assembly	2.5w	main
2	Boot and configuration scenario development	3w	main
3	End-to-end data path and integrity testing	4w	main
4	Interconnect stress, arbitration and deadlock testing	3.5w	parallel
5	Power state transition and DVFS scenario testing	3w	parallel
6	Error injection and recovery at system level	3.5w	main
7	Realistic workload scenarios from the DEF-02 suite	3.5w	parallel
8	Chip-level coverage closure and sign-off	9w	main

Step 7 uses the same workloads `DEF-02` assembled and `ARCH-01` modelled. Running them in simulation is slow and worth it: it is the only place the design, the model and the product's actual intent are compared against each other before silicon.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Top-level RTL from RTL-04
- Block sign-offs from DV-03
- Workload suite from DEF-02
- Power state definitions from ARCH-05
- Chip-level coverage model from DV-01

PRODUCES

- Chip-level verification environment
- Boot and configuration scenarios
- End-to-end data integrity results
- Interconnect stress and deadlock findings
- Power transition scenario results
- Chip-level coverage closure and sign-off

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-08	FEEDS	DV closure signoff package. Chip-level closure is the part of the package that speaks to whether the system works, rather than the parts.
DV-03	FEEDS	Regression and coverage dashboards. Chip-level scenarios are the slowest and most informative regressions on the dashboard.

RISKS AND WATCH-OUTS

- ◆ **Started too late because blocks are not ready.** Chip-level defects take longest to find and fix, and compressing the window pushes them into silicon.
- ◆ **Scenarios written by verification alone.** Realistic use comes from the workload suite and from software; invented scenarios test what verification imagined.
- ◆ **Deadlock and arbitration untested.** These fail only under contention, which random block-level stimulus does not create.
- ◆ **Power transitions verified functionally only.** Retention, isolation and sequencing across a real power state change are where low-power designs actually fail.
- ◆ **Simulation too slow to run meaningful scenarios.** If a boot takes a day, the scenario set shrinks to what fits, and emulation in DV-08 is the answer.

EFFORT SPLIT	110 M/M	ENTRY CRITERIA	FEEDS INTO
End-to-end and integrity testing	26.0	● Top-level RTL boots in RTL-04	DV-08 Emulation and FPGA prototype bring-up — capacity, partitioning, speed
Chip-level coverage closure	20.0	● Blocks largely closed in DV-03	DV-09 Performance and bandwidth validation against the architecture model
Interconnect stress and deadlock	16.0	● Workload suite available from DEF-02	DV-10 Coverage closure and regression stability management
Boot and configuration scenarios	16.0		DV-12 Bug triage and disposition board
Error injection and recovery	16.0		BU-04 Boot, firmware load and functional smoke test
Workload scenarios and environment	16.0		
Total	110.0		
WHO IT TAKES	~21.7 FTE		
Chip verification lead			
Owns chip-level scenarios and closure			
System verification engineers			
Scenario development and debug			
Interconnect verification specialist			
Arbitration, ordering and deadlock			
Low-power verification engineer			
Power state transition scenarios			
Software liaison			
Boot and configuration realism			
		EXIT CRITERIA	MEASURED BY
		● Realistic workload scenarios run, not only synthetic ones	● Scenarios run against scenarios planned
		● Deadlock and arbitration tested under contention	● Chip-level bugs found against block-level
		● Chip-level coverage closed against the DV-01 model	● Simulation throughput on the boot scenario
		DEPENDS ON	
		DV-01 Verification plan and coverage model definition	
		DV-02 UVM environment and VIP bring-up	
		DV-03 Block-level constrained-random and directed testing	
		RTL-04 Chip-level integration and top assembly	

09 DV · VERIFICATION

DV-05

Formal property verification — control logic, connectivity, security

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	40.0 M/M	~10.9 FTE	7	no	Formal verification lead
w10–w26 in stage	planned	while it runs	2 in parallel	has float	Verification lead approves

PURPOSE

Use **formal proof where simulation is weak** — connectivity, control logic, arbiters, security properties, register access — and prove exhaustively what testing can only sample.

Formal answers a different question. Simulation shows that the cases tried worked; a proof shows that no case exists that does not. For arbiters, protocol state machines and security isolation, the space is too large to sample and the failure modes are exactly the ones random stimulus misses.

WORK FLOW

7 STEPS · 2 LANES

w10	w14	w18	w22	w26
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MAIN SEQUENCE

1	2	4	6	7
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Target selection — where formal beats simulation	1.5w	main
2	Property specification per target	3w	main
3	Connectivity and structural proof at chip level	2.5w	parallel
4	Control logic and arbiter proofs	3.5w	main
5	Security and isolation property proofs	2.5w	parallel
6	Assumption review and over-constraint checking	2.5w	main
7	Proof closure, bounded results and reporting	5.5w	main

Step 6 is what separates a proof from a false claim. A property that passes because assumptions excluded the failing case has proved nothing, and reviewing assumptions is the only way to know which of the two happened.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Verification plan target selection from DV-01
- Block RTL from RTL-02
- Security architecture properties from ARCH-07
- Interface protocol definitions from ARCH-04
- Formal tool setup from PDK-08

PRODUCES

- Formal target list with rationale
- Property specifications per target
- Connectivity and structural proof results
- Control and arbiter proof results
- Security property proof results
- Assumption list and proof closure report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D4

PRODUCES

Formal proof reports with assumption list. This activity is the deliverable, and the assumption list is what makes the proofs auditable.

DV-D8

FEEDS

DV closure signoff package. Formally proved properties are a different kind of evidence from coverage, and the package should distinguish them.

RISKS AND WATCH-OUTS

- ◆ **Over-constrained proofs.** Assumptions that exclude the failing case produce a green result and no information, and only assumption review catches it.
- ◆ **Formal applied where simulation is better.** Datapath and workload behaviour are poor formal targets; applying it there consumes specialist time for little return.
- ◆ **Bounded results reported as proofs.** A property proved to depth twenty is not proved; the bound has to be stated with the result.
- ◆ **Properties written from the RTL.** A property derived from the implementation proves the implementation matches itself.
- ◆ **Security properties omitted.** Isolation and access control are exactly the properties simulation samples poorly and formal proves well.

EFFORT SPLIT	40 M/M	ENTRY CRITERIA	FEEDS INTO
Property specification	10.0	● Formal targets selected in DV-01	DV-10Coverage closure and regression stability management
Control and arbiter proofs	9.0	● Block RTL stable enough to prove against	DV-12Bug triage and disposition board
Proof closure and reporting	8.0	● Formal tools qualified by PDK-08	S0-09Final formal equivalence and LVS netlist consistency
Assumption review	6.0		
Security property proofs	4.0		
Connectivity proofs	3.0		
Total	40.0		
WHO IT TAKES	~10.9 FTE	EXIT CRITERIA	MEASURED BY
Formal verification lead		● Every proof reported with its assumptions and its bound	● Properties fully proved against bounded
Owns target selection and proof quality		● Assumptions reviewed for over-constraint, not just listed	● Assumptions reviewed against assumptions made
Formal engineers		● Security properties proved rather than sampled	● Bugs found by formal that simulation had missed
Property specification and proof closure			
Security architect		DEPENDS ON	
Security and isolation properties		DV-01Verification plan and coverage model definition	
Block designers		RTL-02Block-level RTL implementation	
Assumption review and counterexample interpretation		ARCH-07Security and safety architecture — secure boot, root of trust, fusing	
Verification lead		PDK-08EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
Integrates formal results into closure evidence			

09 DV · VERIFICATION

DV-06

Low-power (UPF) verification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	20.0 M/M	~8.7 FTE	7	no	Low-power verification engineer
w18–w28 in stage	planned	while it runs	3 in parallel	has float	Block designers approves

PURPOSE

Verify the design **against its power intent** — isolation, retention, level shifting, power sequencing — because none of it is visible in a normal functional simulation.

A design that passes every functional test can still corrupt state on a power-down, drive an isolated domain, or fail to retain what it needed. Low-power verification simulates the domains actually powering down, and it is the only place these failures appear before silicon.

WORK FLOW

7 STEPS · 2 LANES

w18	w20	w22	w24	w26	w28
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MAIN SEQUENCE

1	2	3	7
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	UPF-aware simulation environment setup	1.5w	main
2	Static UPF structural checking	1.5w	main
3	Isolation and level shifter verification	2w	main
4	Retention and state restoration verification	2w	parallel
5	Power sequencing and controller verification	2w	parallel
6	Power-aware regression integration	1.5w	parallel
7	Low-power coverage closure and reporting	5w	main

Step 4 is the one that finds real bugs. Retention verification requires actually powering a domain down in simulation and checking what survives; a design that has never been simulated through a power cycle has never been tested for the failure mode.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- UPF power intent from RTL-05
- Power state definitions from ARCH-05
- Verification environment from DV-02
- Power management controller RTL from RTL-05
- Low-power tool support from PDK-08

PRODUCES

- UPF-aware simulation environment
- Static UPF structural check results
- Isolation and level shifter verification results
- Retention verification results
- Power sequencing verification results
- Low-power coverage report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D5

PRODUCES

Low-power verification report. This activity is the deliverable, and it is the only evidence that the power architecture works as intended.

DV-D3

FEEDS

Regression and coverage dashboards. Power-aware regressions belong in the same dashboard as functional ones, not in a side report.

RISKS AND WATCH-OUTS

- ◆ **Only static checking performed.** Structural UPF checks find missing cells; only simulation finds a design that corrupts state when the domain actually goes down.
- ◆ **Retention verified for the intended state only.** What is not retained matters as much as what is, and a design that retains too little fails on resume.
- ◆ **Power sequences tested in isolation.** The failure is usually a power transition during traffic, not a clean idle transition.
- ◆ **Power-aware simulation excluded from regression.** It is slow, so it gets run once; a later RTL change then breaks it silently.
- ◆ **UPF used for verification differing from synthesis.** Two versions of the intent means verifying one design and building another.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Isolation and retention verification	7.0	● UPF available and tool-accepted from RTL-05 ● Power states defined by ARCH-05 ● Environment available from DV-02	DV-10Coverage closure and regression stability management
Power sequencing verification	4.0		DV-12Bug triage and disposition board
Coverage closure and reporting	3.5		S0-09Final formal equivalence and LVS netlist consistency
Environment setup	2.5		BU-02Power-on, power sequencing and basic health check
Static checking	1.5		
Regression integration	1.5	EXIT CRITERIA	MEASURED BY
Total	20.0	● Domains simulated powering down, not only statically checked ● Retention verified for both what is kept and what is not ● Power-aware tests running in the regular regression	● Power states simulated against states defined ● Retention defects found pre-silicon ● Power-aware tests in the nightly regression
WHO IT TAKES	~8.7 FTE	DEPENDS ON	
Low-power verification engineer Owns power-aware verification and closure		DV-02UVM environment and VIP bring-up	
Low-power implementation lead UPF correctness and intent alignment		RTL-05Clock, reset and power intent (UPF) implementation	
Verification methodology engineer Power-aware regression integration		ARCH-05Power domain, clock domain and DVFS architecture; UPF intent draft	
Power architect Confirms the verified states match the architecture			
Block designers Retention and isolation defect fixes			

09 DV · VERIFICATION

DV-07

AMS / mixed-signal co-simulation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	20.0 M/M	~7.2 FTE	7	no	AMS verification engineer
w16–w28 in stage	planned	while it runs	3 in parallel	has float	Verification lead approves

PURPOSE

Verify the **boundary between the analog macros and the digital design** — control interfaces, calibration sequences, startup ordering — with the analog represented by models correlated to its circuits.

Analog blocks fail where they meet digital. A PLL that locks perfectly and a controller that sequences perfectly can still disagree about when lock is valid. Co-simulation is where those disagreements appear, and the alternative is finding them at bring-up with an oscilloscope.

WORK FLOW7 STEPS · 2 LANES

w16	w19	w22	w25	w28
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MAIN SEQUENCE

1	2	3	7
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Co-simulation environment and model integration	2w	main
2	Model fidelity review against AMS characterisation	2w	main
3	Control and configuration interface verification	2.5w	main
4	Calibration and training sequence verification	2.5w	parallel
5	Startup, lock and ready-signalling verification	2w	parallel
6	Analog fault and degraded-mode behaviour	1.5w	parallel
7	Co-simulation closure and findings report	5.5w	main

Step 2 decides whether any of this is meaningful. Verification against a behavioural model that does not match the circuit proves the digital side works with a fiction, and the correlation in `AMS-16` is what makes the model a proxy rather than a guess.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Behavioural models from AMS-15
- Correlation results from AMS-16
- Integrated AMS macros from RTL-03
- Startup and sequencing requirements from ARCH-05
- Verification environment from DV-02

PRODUCES

- AMS co-simulation environment
- Model fidelity review findings
- Control and configuration interface results
- Calibration and training sequence results
- Startup and ready-signalling results
- Co-simulation findings report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D2	FEEDS	UVM testbenches and integrated VIP. The co-simulation environment extends the shared testbench rather than standing beside it.
DV-D8	FEEDS	DV closure signoff package. Mixed-signal boundary closure is a distinct claim in the package, because it rests on models rather than on RTL alone.

RISKS AND WATCH-OUTS

- ◆ **Models never correlated.** Digital passes against a fiction, and the mismatch reaches silicon as a bring-up failure nobody predicted.
- ◆ **Calibration sequences untested from the digital side.** A calibration that works when the analog engineer drives it manually is first exercised by firmware at bring-up.
- ◆ **Ready and lock signalling assumed.** When a lock signal becomes valid is a contract; each side assuming a different answer produces intermittent boot failures.
- ◆ **Degraded modes unverified.** What the digital side does when a PLL fails to lock is behaviour that only matters in the field, where it matters a great deal.
- ◆ **Co-simulation too slow to regress.** Run once and never again, it cannot catch a later change on either side.

EFFORT SPLIT		20 M/M	ENTRY CRITERIA		FEEDS INTO		
Control and calibration verification		7.0	● Behavioural models released by AMS-15 ● AMS macros integrated in RTL-03 ● Co-simulation tools available and qualified		DV-10		Coverage closure and regression stability management
Model fidelity review		4.0			DV-12		Bug triage and disposition board
Startup and signalling verification		3.5			BU-03		Reset, clocking and PLL lock validation
Environment and model integration		3.0			BU-05		Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Findings report		1.5					
Degraded-mode testing		1.0					
Total		20.0	EXIT CRITERIA				
WHO IT TAKES ~7.2 FTE		● Every model correlated against its circuit before being relied on ● Calibration sequences exercised from the digital side ● Degraded and fault behaviour verified, not assumed		MEASURED BY			
				● Models correlated against models used ● Boundary defects found here against at bring-up ● Calibration sequences verified against those implemented			
		AMS verification engineer Owns co-simulation and model fidelity					
		Analog designers Model accuracy and expected behaviour					
		Digital verification engineers Control and calibration sequence testing					
Firmware liaison Calibration sequences as software will drive them		DEPENDS ON					
		DV-02	UVM environment and VIP bring-up				
		RTL-03	Third-party and internal IP integration — wrappers, glue, configuration				
		AMS-15	Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)				
		AMS-16	AMS-digital co-simulation and integration support				

09 DV · VERIFICATION

DV-08

Emulation and FPGA prototype bring-up — capacity, partitioning, speed

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
20w w6–w26 in stage	55.0 M/M planned	~11.9 FTE while it runs	8 3 in parallel	no has float	Emulation lead Program manager approves

PURPOSE

Stand up **emulation and FPGA prototyping** — partitioning, capacity, speed, debug — so that scenarios too long for simulation can actually be run.

A boot that takes a day in simulation takes minutes in emulation, and workload scenarios that are impossible in one are routine in the other. Emulation is also the platform firmware development runs on, which makes its availability date a dependency for a team outside this stage entirely.

WORK FLOW8 STEPS · 2 LANES

w6w10w14w18w22w26

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Platform selection and capacity sizing	2w	main
2	Design partitioning across the platform	3.5w	main
3	Memory and interface modelling on the platform	3w	parallel
4	Bring-up and first boot on emulation	4w	main
5	Debug and visibility infrastructure	3w	parallel
6	Performance and throughput tuning	2.5w	parallel
7	Long-scenario and workload execution	8w	main
8	Platform handover to firmware and validation	2.5w	main

Step 8 makes this activity a dependency for people outside verification. Firmware cannot start on real workloads until the platform runs, and the handover date belongs in the program schedule rather than in the verification plan alone.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Top-level RTL from RTL-04
- Emulator or FPGA capacity and licences
- Memory and interface models for the platform
- Workload suite from DEF-02
- Debug requirements from validation and firmware

PRODUCES

- Platform selection and capacity plan
- Design partitioning for the platform
- Memory and interface models
- Working emulation platform with boot
- Debug and visibility infrastructure

- Long-scenario and workload results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D6	PRODUCES	Emulation platform and system test suite. This activity is the deliverable, and it serves firmware and validation as much as verification.
DV-D3	FEEDS	Regression and coverage dashboards. Emulation regressions cover scenarios simulation cannot reach, and belong in the same view.

RISKS AND WATCH-OUTS

- ◆ **Partitioning consuming more time than expected.** Splitting a large design across emulator capacity is difficult and iterative, and it is routinely underestimated.
- ◆ **Capacity contended with another program.** Emulator capacity is expensive and shared; a program without a booked allocation runs when the other one is not.
- ◆ **Debug visibility inadequate.** A platform that runs fast and cannot be probed produces failures that cannot be diagnosed.
- ◆ **Handover date not in the program schedule.** Firmware plans against it, and treating it as an internal verification milestone means nobody outside knows when it slips.
- ◆ **Models diverging from the design.** Memory and interface models built for the platform can drift from the RTL, and results then reflect the model.

EFFORT SPLIT	55 M/M	ENTRY CRITERIA	FEEDS INTO
Design partitioning	14.0	● Top-level RTL elaborating from RTL-04	DV-04 Chip-level and system-level scenario testing
Bring-up and first boot	12.0	● Emulator or FPGA capacity booked	DV-09 Performance and bandwidth validation against the architecture model
Memory and interface modelling	10.0	● Workload suite available from DEF-02	DV-10 Coverage closure and regression stability management
Debug infrastructure	8.0	EXIT CRITERIA	BU-04 Boot, firmware load and functional smoke test
Long-scenario execution	7.0	● Design boots on the platform at usable speed	
Handover and tuning	4.0	● Debug visibility sufficient to diagnose a failure	
Total	55.0	● Platform handed over to firmware on a scheduled date	MEASURED BY
WHO IT TAKES	~11.9 FTE	DEPENDS ON	● Emulation speed against target
Emulation lead		DV-02 UVM environment and VIP bring-up	● Weeks from RTL availability to first platform boot
Emulation engineers		RTL-04 Chip-level integration and top assembly	● Scenarios run on emulation that simulation could not
Partitioning, modelling and tuning		DEF-02 Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)	
Debug infrastructure engineer			
Visibility and trace on the platform			
Firmware liaison			
Handover requirements and platform use			
Program manager			
Capacity booking and handover date			

09 DV · VERIFICATION

DV-09

Performance and bandwidth validation
against the architecture model

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w w22–w34 in stage	30.0 M/M planned	~10.9 FTE while it runs	7 2 in parallel	no has float	Performance verification lead Chief architect approves

PURPOSE

Check that the design **actually delivers the performance the architecture promised** — bandwidth, latency, utilisation — against the same workloads ARCH-01 modelled.

Functional correctness and performance are separate questions, and passing every test says nothing about the second. A design that computes correctly at 60% of modelled throughput is a product that misses its headline KPI, and finding that in silicon leaves no options.

WORK FLOW

7 STEPS · 2 LANES

w22	w25	w28	w31	w34
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MAIN SEQUENCE

1	2	3	6	7
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Performance measurement infrastructure in the testbench	1.5w	main
2	Workload execution on emulation and simulation	3w	main
3	Bandwidth and utilisation measurement against the model	2.5w	main
4	Latency and queueing analysis	2w	parallel
5	Bottleneck identification and root cause	2.5w	parallel
6	Correlation against the ARCH-01 performance model	2.5w	main
7	Findings, recommendations and model reconciliation	2.5w	main

Step 6 closes a loop opened at architecture. Where the RTL and the model disagree, one of them is wrong — and finding out which is what tells the program whether it has a design problem or a modelling problem.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Emulation platform from DV-08
- Workload suite from DEF-02
- Performance model and predictions from ARCH-01
- Chip-level environment from DV-04
- KPI targets from DEF-02

PRODUCES

- Performance measurement infrastructure
- Workload execution results
- Bandwidth, utilisation and latency measurements
- Bottleneck analysis with root causes
- Model correlation results
- Findings and recommendations

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D8	FEEDS	DV closure signoff package. Performance evidence is part of tapeout readiness — functional closure alone does not say the product meets its KPIs.
DV-D6	FEEDS	Emulation platform and system test suite. The performance workloads become part of the standing suite the platform runs.

RISKS AND WATCH-OUTS

- ◆ **Performance treated as a bring-up activity.** Discovered in silicon, a throughput shortfall has no remaining fix except software and marketing.
- ◆ **Measured only on synthetic traffic.** Synthetic patterns exercise the interconnect; real workloads exercise the design, and only the second predicts the product.
- ◆ **Model and RTL disagreement left unresolved.** Whichever is wrong, not knowing which means every downstream projection is unreliable.
- ◆ **Bottlenecks found without root cause.** Knowing throughput is low is not actionable; knowing which queue saturates is.
- ◆ **No time left to act.** Findings that arrive after RTL freeze can only be documented, not fixed.

EFFORT SPLIT	30 M/M	ENTRY CRITERIA	FEEDS INTO
Bandwidth and utilisation measurement	8.0	● Emulation platform running from DV-08	DV-10Coverage closure and regression stability management
Workload execution	6.0	● Chip-level environment available from DV-04	ARCH-01System-level performance modelling and workload simulation
Bottleneck analysis	5.0	● Performance model predictions available from ARCH-01	BU-08Performance validation against the architecture model
Model correlation	5.0		
Latency analysis	3.0	EXIT CRITERIA	
Findings and reconciliation	3.0	● Real workloads measured, not only synthetic traffic	MEASURED BY
Total	30.0	● Every model disagreement resolved as design or model error	● Measured throughput against modelled
WHO IT TAKES	~10.9 FTE	● Findings delivered with time to act on them	● Bottlenecks with an identified root cause
Performance verification lead		DEPENDS ON	● Weeks between findings and RTL freeze
Owns measurement and correlation		DV-04Chip-level and system-level scenario testing	
Performance architect		DV-08Emulation and FPGA prototype bring-up — capacity, partitioning, speed	
Model comparison and reconciliation		ARCH-01System-level performance modelling and workload simulation	
Emulation engineers		DEF-02Workload / use-case profiling and target KPI derivation (TOPS, TOPS/W, tokens/s)	
Workload execution at usable speed			
Interconnect designers			
Bottleneck root cause			
Chief architect			
Decides between design change and model correction			

09 DV · VERIFICATION

DV-10

Coverage closure and regression stability management

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
26w cont. w12–w38 in stage, runs throughout	45.0 M/M planned	~7.5 FTE while it runs	8 3 in parallel	yes no float	Regression owner CAD infrastructure approves

PURPOSE

Keep the regressions running and the **coverage moving toward closure** — triage failures, manage capacity, publish where every block stands, and close the holes that remain.

Coverage closure is not a phase at the end; it is a curve that either bends or does not. This activity owns the curve: which holes are real, which tests would close them, and whether the regression is stable enough for the numbers to mean anything.

WORK FLOW8 STEPS · 2 LANES

w12	w18	w24	w30	w36	w38
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MAIN SEQUENCE

1	2	4	5	8
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IN PARALLEL

3	6	7
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#	STEP	TAT	LANE
1	Regression suite definition and tiering	2w	main
2	Regression infrastructure and capacity management	2.5w	main
3	Failure triage process and ownership	2.5w	parallel
4	Coverage collection, merging and reporting	3w	main
5	Coverage hole analysis and test targeting	5w	main
6	Regression stability and flake elimination	4w	parallel
7	Dashboard publication and closure tracking	4w	parallel
8	Coverage closure and sign-off assembly	13.5w	main

Step 6 is unglamorous and load-bearing. A regression with intermittent failures trains everyone to ignore red results, and once that habit forms the regression has stopped being a signal.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Coverage model and criteria from DV-01
- Test suites from DV-03 through DV-09
- Regression capacity from PDK-12 and RTL-09
- Build and CI infrastructure from RTL-09
- Closure criteria per block

PRODUCES

- Tiered regression suites
- Regression infrastructure and capacity plan
- Failure triage process with ownership
- Merged coverage reporting
- Hole analysis and targeted test list
- Coverage dashboards and closure sign-off

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

- DV-D3

PRODUCES

Regression and coverage dashboards. This activity is the deliverable, and its value is that the numbers are current and trusted.
- DV-D8

PRODUCES

DV closure signoff package. Closure is assembled here from the evidence every other DV activity produced.

RISKS AND WATCH-OUTS

- ◆

Flaky regressions tolerated. Once red results are routinely ignored, the regression has stopped providing information and nobody notices when it matters.
- ◆

Coverage merged without stability. Numbers from an unstable regression are noise presented as a trend.
- ◆

Holes closed by waiving rather than testing. A hole waived without analysis is untested behaviour recorded as covered.
- ◆

Capacity insufficient at peak. Regression queues at the point of closure directly extend the closure date.
- ◆

Closure signed with the curve still climbing. Coverage still rising at sign-off means the plan's completion criteria were not what stopped the work.

EFFORT SPLIT	45 M/M	ENTRY CRITERIA	FEEDS INTO
Coverage hole analysis and targeting	12.0	- Coverage model defined by DV-01 - Test suites producing coverage from DV-03 - Regression capacity allocated	DV-12 Bug triage and disposition board
Closure and sign-off assembly	10.0		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback
Regression stability	8.0		T0-03 Tapeout checklist completion and owner sign-off
Coverage collection and reporting	6.0		S0-09 Final formal equivalence and LVS netlist consistency
Triage process	5.0		
Infrastructure and dashboards	4.0		
Total	45.0		
WHO IT TAKES	~7.5 FTE	EXIT CRITERIA	MEASURED BY
Regression owner Owns suites, stability and capacity		- Regression stable enough that a red result is believed - Every hole analysed before it is closed or waived - Coverage curve flat at sign-off, not still climbing	- Regression flake rate - Coverage against the model at sign-off - Holes analysed against holes waived
Coverage lead Hole analysis, targeting and closure		DV-01 Verification plan and coverage model definition	
Verification lead Closure sign-off and package assembly		DV-03 Block-level constrained-random and directed testing	
Block verification owners Triage and close their own failures		RTL-09 CI build, nightly regression and release management	
CAD infrastructure Compute and licence capacity		PDK-12 Compute farm, EDA license and storage capacity planning	

09 DV · VERIFICATION

DV-11

Gate-level simulation, functional and timing-annotated

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	20.0 M/M	~8.7 FTE	7	yes	Gate-level simulation lead
w28–w38 in stage	planned	while it runs	3 in parallel	no float	Physical design liaison approves

PURPOSE

Simulate the **gate-level netlist**, first functionally and then with real timing, to catch what RTL simulation structurally cannot — X-propagation, reset behaviour, timing-dependent failures and synthesis or DFT-induced differences.

RTL simulation is optimistic about X and blind to timing. Gate-level simulation is slow, painful and the last check before the design becomes masks — it is also where scan insertion, clock gating and synthesis transformations are proved not to have changed behaviour.

WORK FLOW

7 STEPS · 2 LANES

w28	w30	w32	w34	w36	w38
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MAIN SEQUENCE

1	2	3	7
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IN PARALLEL

4	5	6
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#	STEP	TAT	LANE
1	Gate-level environment and netlist bring-up	1.5w	main
2	Zero-delay functional gate-level simulation	2w	main
3	X-propagation and reset initialisation analysis	2w	main
4	SDF back-annotated timing simulation	2.5w	parallel
5	DFT and scan mode simulation	2w	parallel
6	Power-aware gate-level simulation	1.5w	parallel
7	Failure debug and closure	4.5w	main

Step 3 finds the class of bug RTL simulation is structurally incapable of seeing. RTL X-optimism hides uninitialised state; at gate level the X propagates, and a design that boots in RTL can fail to come out of reset in gates.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Gate-level netlist from SYN-01
- SDF timing from PD-09
- Scan-inserted netlist from DFT-06
- Verification environment from DV-02
- Power intent from RTL-05

PRODUCES

- Gate-level simulation environment
- Functional gate-level results
- X-propagation and reset initialisation findings
- SDF timing-annotated simulation results
- Scan and DFT mode simulation results
- Gate-level simulation report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D7	PRODUCES	Gate-level simulation report. This activity is the deliverable, and it is a tapeout checklist item rather than an optional confirmation.
DV-D8	FEEDS	DV closure signoff package. Gate-level results close the gap between what was verified and what will be manufactured.

RISKS AND WATCH-OUTS

- ◆ **Skipped because it is slow.** The failures it finds — X propagation, reset, timing-dependent behaviour — have no other pre-silicon detector.
- ◆ **Run only in functional mode.** Scan and test modes are how the part is tested at ATE, and a scan mode that does not work makes the part untestable.
- ◆ **Started after netlist churn has settled.** Waiting for a stable netlist means starting too late; running early on an imperfect netlist finds environment problems while there is time.
- ◆ **X failures suppressed rather than fixed.** Forcing initial values to make the simulation run hides exactly the uninitialised state the check exists to find.
- ◆ **Timing simulation on a subset of corners.** Timing-dependent failures appear at specific corners, and the corner skipped is the one that fails.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Failure debug and closure	7.0	● Gate-level netlist available from SYN-01	DV-12 Bug triage and disposition board
Functional gate-level simulation	4.0	● Scan insertion complete in DFT-06	DFT-08 Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
X-propagation analysis	3.5	● SDF available from PD-09 for timing runs	T0-02 Final full-chip verification re-run on the released database
SDF timing simulation	3.0		S0-09 Final formal equivalence and LVS netlist consistency
Scan and DFT mode simulation	1.5	EXIT CRITERIA	
Power-aware simulation	1.0	● X-propagation issues fixed rather than suppressed	
Total	20.0	● Scan and test modes simulated, not only functional	MEASURED BY
WHO IT TAKES	~8.7 FTE	● Timing simulation run at the corners that matter	● Modes simulated against modes the part will use
Gate-level simulation lead			● X failures fixed against suppressed
Owns the runs and the closure		DEPENDS ON	● Gate-level defects found that RTL simulation could not
Verification engineers		DV-02 UVM environment and VIP bring-up	
Debug and failure analysis		SYN-01 SDC constraint development and validation	
Synthesis liaison		DFT-06 Scan insertion and DFT DRC during synthesis	
Netlist and transformation questions		RTL-05 Clock, reset and power intent (UPF) implementation	
DFT engineer			
Scan and test mode simulation			
Physical design liaison			
SDF generation and timing questions			

09 DV · VERIFICATION

DV-12

Bug triage and disposition board

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
34w cont. w6–w40 in stage, runs throughout	25.0 M/M planned	~3.2 FTE while it runs	8 3 in parallel	yes no float	Verification lead Validation liaison approves

PURPOSE

Run the **bug disposition board** — triage every defect, decide what gets fixed, and make the risk of what does not explicit and owned.

Every program reaches tapeout with open bugs. The question is whether they were assessed and accepted or simply ran out of time, and the difference is entirely this activity. Its output at the end is the errata list bring-up will work against and the risk statement the Go / No-Go decision reads.

WORK FLOW

8 STEPS · 2 LANES

w6	w13	w20	w27	w34	w40
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Bug tracking, severity and triage process definition	1.5w	main
2	Disposition board charter and membership	1w	main
3	Continuous triage and disposition	22w	main
4	Bug rate and closure trend analysis	4w	parallel
5	Deferred bug risk assessment	4w	parallel
6	Escape analysis on late-found defects	3w	parallel
7	Open bug list and risk statement for tapeout	5w	main
8	DV closure recommendation	4.5w	main

Step 6 is the loop that improves the process rather than the design. Asking why a bug found in week 34 was not found in week 12 usually points at a missing coverage point or an untested mode, and fixing that finds the next one earlier.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Defects from every DV activity
- Coverage and closure status from DV-10
- Change control decisions from RTL-10
- Silicon debug capability from DFT-11
- Tapeout criteria from the program plan

PRODUCES

- Bug tracking and triage process
- Disposition board decisions
- Bug rate and closure trend analysis
- Deferred bug risk assessments
- Escape analysis findings
- Open bug list and risk statement

- DV closure recommendation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DV-D8	FEEDS	DV closure signoff package. The open bug list and its risk statement are the part of the package the Go / No-Go decision actually reads.
DV-D3	FEEDS	Regression and coverage dashboards. Bug find and closure rates sit alongside coverage as the two curves that say whether verification is converging.

RISKS AND WATCH-OUTS

- ◆ **Bugs deferred without risk assessment.** An unassessed open bug at tapeout is a risk nobody quantified being accepted by nobody in particular.
- ◆ **Severity assigned by the finder.** Consistent severity requires a board; without one, the same defect is critical in one block and minor in another.
- ◆ **Bug rate still rising at closure.** A find rate that has not turned over means the design has more defects than have been found, whatever the coverage says.
- ◆ **No escape analysis.** The process that let a bug survive to week 34 will let the next one survive too.
- ◆ **Errata list assembled after tapeout.** Bring-up needs the known-issue list on day one, and assembling it later means debugging problems that were already known.

EFFORT SPLIT	25 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous triage and disposition	10.0	Bug tracking system and severity scheme agreed	T0-04 Open-issue risk assessment and waiver acceptance
Open bug list and risk statement	4.0	Board membership named and meeting	T0-05 Go / No-Go decision meeting
Deferred bug risk assessment	4.0	Tapeout criteria defined in the program plan	BU-10 Errata capture, workaround definition and documentation
Trend analysis	3.0		
Escape analysis	2.0		
Closure recommendation	2.0		S0-09 Final formal equivalence and LVS netlist consistency
Total	25.0		
WHO IT TAKES	~3.2 FTE	EXIT CRITERIA	MEASURED BY
Verification lead Chairs the disposition board		- Every deferred bug carries an assessed, owned risk - Bug find rate turned over before closure is recommended - Errata list ready for bring-up on day one	- Deferred bugs with a risk assessment - Bug find rate trend at closure - Escapes analysed against escapes found
RTL lead Fix feasibility and design-side impact			
Chief architect Severity and product impact of deferred defects		DEPENDS ON	
Program manager Risk acceptance and tapeout criteria		DV-01 Verification plan and coverage model definition DV-10 Coverage closure and regression stability management	
Validation liaison Turns deferred bugs into bring-up watch items		RTL-10 Specification change control and ECO board	

DESIGN & VERIFY

10 DFT Architecture

DFT

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w18 – w58	40 _w	11	130 _{M/M}	5 _{of 11}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
DFT-01	DFT architecture definition — scan style, compression ratio, hierarchical DFT		8w 12 _{M/M}	DFT-D1 DFT architecture specification and coverage plan
DFT-02	Test coverage and test-time target negotiation with product and test engineering		5w 4 _{M/M}	DFT-D2 MBIST / BISR insertion and test collateral
DFT-03	MBIST / BIRA / BISR architecture for embedded memories		10w 14 _{M/M}	DFT-D3 JTAG / IJTAG description files (BSDL, ICL, PDL)
DFT-04	Boundary scan and IJTAG (1149.1 / 1687), TAP and debug access architecture		8w 10 _{M/M}	DFT-D4 ATPG pattern sets with coverage report
DFT-05	On-chip clock controller design for at-speed testing		7w 8 _{M/M}	DFT-D5 DFT DRC clean report
DFT-06	Scan insertion and DFT DRC during synthesis		10w 14 _{M/M}	DFT-D6 Pattern validation (GLS) report and ATE-ready pattern files
DFT-07	ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure		16w 28 _{M/M}	DFT-D7 DFT signoff entry for the tapeout checklist
DFT-08	Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)		10w 14 _{M/M}	
DFT-09	Scan compression and chain routing feasibility with physical design		10w 10 _{M/M}	
DFT-10	eFuse, chip ID and memory repair infrastructure		6w 6 _{M/M}	
DFT-11	Silicon debug hooks — trace, observability, design-for-debug		8w 10 _{M/M}	

10 DFT · DFT ARCHITECTURE

DFT-01

DFT architecture definition — scan style, compression ratio, hierarchical DFT

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w0–w8 in stage	12.0 M/M planned	~6.5 FTE while it runs	6 2 in parallel	yes no float	DFT architect Physical design liaison approves

PURPOSE

Decide **how the chip will be tested** before it is built to be testable — scan style, compression ratio, hierarchical partitioning, and the test modes the design will support.

Every one of these choices constrains the design rather than following it. Compression ratio decides pattern volume and therefore test cost; hierarchical DFT decides whether a block can be tested independently; and both have to be settled before SYN-01 writes constraints against them.

WORK FLOW

6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Test strategy definition — what is tested where and how	1.5w	main
2	Scan architecture and chain organisation	2w	main
3	Compression architecture and ratio selection	1.5w	main
4	Hierarchical DFT partitioning against the design hierarchy	1.5w	parallel
5	Test mode definition and mode control	1.5w	parallel
6	Architecture specification and review	3w	main

Step 3 is a cost decision disguised as an architecture one. Compression ratio sets pattern volume, pattern volume sets tester memory and test time, and test time is a per-unit cost for the life of the product.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Design hierarchy and partitioning from ARCH-02
- Coverage and test-time targets from DFT-02
- Memory list and BISR intent from ARCH-03
- Test platform constraints from TEST-02
- Debug access requirements from ARCH-07

PRODUCES

- Test strategy statement
- Scan architecture and chain organisation
- Compression architecture and ratio
- Hierarchical DFT partitioning
- Test mode definition
- DFT architecture specification

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-01

PRODUCES

DFT architecture specification and coverage plan. This activity is the deliverable, and everything else in the stage is executed against it.

DFT-07

GATES

DFT signoff entry for the tapeout checklist. An architecture that was never fixed cannot be signed off against.

RISKS AND WATCH-OUTS

- ◆ **Architecture decided after synthesis starts.** Scan insertion and constraints depend on it; deciding late means re-running both.
- ◆ **Compression ratio chosen without the test cost model.** Ratio trades pattern volume against silicon area, and only TEST-09's cost model says which side to favour.
- ◆ **Hierarchical DFT ignored on a large die.** Flat scan on a design this size produces pattern volumes and runtimes that become their own schedule problem.
- ◆ **Test modes defined without the security architecture.** Every test mode is an access path, and ARCH-07 has to lock the ones that should not exist in production.
- ◆ **Architecture written and not reviewed by test engineering.** A scheme that is elegant in design and unrunnable on the chosen ATE is discovered at TEST-08.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Scan and compression architecture	5.0	● Design hierarchy known from ARCH-02	DFT-03 MBIST / BIRA / BISR architecture for embedded memories
Architecture specification and review	3.0	● Coverage and test-time targets agreed in DFT-02	DFT-05 On-chip clock controller design for at-speed testing
Test strategy definition	2.0	● Test platform direction available from TEST-02	DFT-06 Scan insertion and DFT DRC during synthesis
Hierarchical partitioning	1.0		DFT-09 Scan compression and chain routing feasibility with physical design
Test mode definition	1.0		SYN-01 SDC constraint development and validation
Total	12.0	EXIT CRITERIA	TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test
WHO IT TAKES	~6.5 FTE	● Architecture fixed before synthesis writes constraints against it ● Compression ratio chosen against a test cost model ● Specification reviewed by test and product engineering	MEASURED BY
DFT architect Owns the architecture and its specification		DEPENDS ON	● Architecture decisions taken before synthesis starts ● Compression ratio against pattern volume budget ● Architecture changes after freeze
Test engineering liaison ATE feasibility and test-time implications		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
Product engineering liaison Coverage and cost targets		ARCH-09 Architecture specification authoring and review	
Security architect Test mode access and lockdown		DFT-02 Test coverage and test-time target negotiation with product and test engineering	
Physical design liaison Chain routing and compression feasibility			

10 DFT · DFT ARCHITECTURE

DFT-02

Test coverage and test-time target negotiation with product and test engineering

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	4.0 M/M	~3.5 FTE	5	no	DFT architect
w2-w7 in stage	planned	while it runs	2 in parallel	has float	Program manager approves

PURPOSE

Agree the **coverage and test-time targets** with the people who pay for them — product engineering, who own quality, and test engineering, who own the tester seconds.

Coverage and test time pull against each other, and neither belongs to DFT alone. A defect-per-million target sets required coverage; a cost-per-unit target sets allowable test time; and the architecture in DFT-01 is chosen to sit between them. Negotiating this first is what makes the architecture decidable.

WORK FLOW5 STEPS · 2 LANES

w2w3w4w5w6w7

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Quality target translation — DPPM to required coverage	1w	main
2	Test time budget from the product cost model	1w	main
3	Fault model selection — stuck-at, transition, cell-aware	1w	parallel
4	Coverage target per fault model and per block class	1w	parallel
5	Target agreement and record	3w	main

Step 3 has a large downstream cost. Cell-aware patterns catch defects the other models miss and multiply pattern count doing it, so the decision belongs here with the cost model in the room rather than at `DFT-07` when the patterns are being generated.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Quality and DPPM targets from the product requirements
- Test cost budget from DEF-04
- Defect density projection from TECH-07
- Test platform capability from TEST-02
- Previous-generation coverage and escape data

PRODUCES

- Coverage target derived from the DPPM requirement
- Test time budget
- Fault model selection
- Per-block-class coverage targets
- Agreed target record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-01	FEEDS	DFT architecture specification and coverage plan. The coverage plan half — the numbers DFT-01's architecture is chosen to reach.
DFT-04	FEEDS	ATPG pattern sets with coverage report. The report is measured against the targets set here, not against what the tool happened to reach.

RISKS AND WATCH-OUTS

- ◆ **Coverage target set by convention.** "99% stuck-at" carried from the last program may be far from what this product's DPPM requirement needs.
- ◆ **Test time budget not derived from cost.** A test time nobody costed becomes whatever the patterns need, and the per-unit cost is discovered at production.
- ◆ **Fault model chosen at pattern generation.** Adding cell-aware late multiplies pattern volume against a tester memory budget already fixed.
- ◆ **Targets agreed without product engineering.** The organisation that owns field quality has to own the coverage number, or it will be renegotiated when patterns run long.
- ◆ **No per-block differentiation.** Memory-dominated blocks and control logic need different approaches, and a single flat target over-tests one and under-tests the other.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Quality target translation	1.5	● DPPM and quality requirements available	DFT-01DFT architecture definition — scan style, compression ratio, hierarchical DFT
Test time budget	1.0	● Test cost budget from the DEF-04 model	DFT-07ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
Fault model selection	1.0	● Defect density projection from TECH-07	TEST-01Test plan and coverage strategy — wafer sort, final test, system-level test
Target agreement	0.5		TEST-09Test time and cost optimisation — parallelism, site count
Total	4.0		
WHO IT TAKES	~3.5 FTE	EXIT CRITERIA	MEASURED BY
DFT architect Owns the target negotiation and record		● Coverage target derived from DPPM, not inherited	● Coverage target against DPPM requirement
Product engineering DPPM and field quality requirements		● Test time budget traced to a per-unit cost	● Test time budget against the cost model
Test engineering Test time, tester memory and cost		● Targets signed by product and test engineering	● Targets renegotiated during pattern generation
Yield engineer Defect density and fault model relevance			
Program manager Records the agreement and its cost implication		DEF-04Die size, yield and product cost model (wafer, mask, package, test, ASP)	
		TECH-07Foundry roadmap and risk-production timing alignment	

10 DFT · DFT ARCHITECTURE

DFT-03

MBIST / BIRA / BISR architecture for embedded memories

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	14.0 M/M	~6.1 FTE	8	yes	DFT memory lead
w6–w16 in stage	planned	while it runs	3 in parallel	no float	Verification engineer approves

PURPOSE

Architect **memory test and repair** — MBIST engines, BIRA analysis, BISR repair and the fuse path — for a design where memory is most of the die.

On an accelerator the arrays dominate both area and defect exposure, and they cannot be tested by scan. MBIST is how they are tested and repair is how the yield is recovered; both have to be designed jointly with AMS-09, because the array provides the spares and this activity provides the engine that uses them.

WORK FLOW

8 STEPS · 2 LANES

w6	w8	w10	w12	w14	w16
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MAIN SEQUENCE

1	2	4	7	8
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	Memory inventory and test requirement per instance	1.5w	main
2	MBIST architecture and engine allocation	2.5w	main
3	Algorithm selection per memory type	1.5w	parallel
4	BIRA and repair analysis architecture	2w	main
5	BISR and fuse path integration with ARCH-07	2w	parallel
6	Repair interface agreement with the memory designers	1.5w	parallel
7	MBIST insertion and integration	2.5w	main
8	Architecture verification and specification	1.5w	main

Step 6 is the joint decision with <code>AMS-09</code>. Both sides can specify an internally consistent repair scheme that the other cannot implement, and writing the interface together is the only reliable way to avoid it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Memory instances from PDK-04 and AMS-10
- Redundancy scheme from AMS-09
- Fuse map and lifecycle from ARCH-07
- Coverage targets from DFT-02
- Repair yield expectations from the yield model

PRODUCES

- Memory inventory with test requirements
- MBIST architecture and engine allocation
- Algorithm selection per memory type
- BIRA and BISR architecture
- Fuse path integration
- MBIST insertion and verification results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D2	PRODUCES	MBIST / BISR insertion and test collateral. This activity is the deliverable, including the repair collateral test engineering will run.
DFT-D1	FEEDS	DFT architecture specification. Memory test is a large part of the overall architecture, not a separate subject.

RISKS AND WATCH-OUTS

- ◆ **Repair interface specified separately from the array.** Two consistent specifications that disagree, found at pattern generation when both are frozen.
- ◆ **Algorithms chosen without the failure modes.** March algorithms differ in what they catch; a default choice misses the defect modes this array actually exhibits.
- ◆ **Fuse capacity insufficient.** Repair data has to be stored, and an allocation made without counting the worst-case repair set produces unrepairable parts.
- ◆ **MBIST runtime unbudgeted.** Memory test on an array-dominated die can dominate test time, and it is often costed after the architecture is fixed.
- ◆ **Repair not verified end to end.** A repair path proven in pieces and never exercised from failure through analysis to fuse blow fails at production test.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
MBIST architecture and insertion	5.0	Memory inventory available from PDK-04	DFT-06 Scan insertion and DFT DRC during synthesis
BIRA and BISR architecture	3.5	Redundancy scheme from AMS-09 in progress	DFT-10 eFuse, chip ID and memory repair infrastructure
Repair interface and fuse path	2.5	Fuse map and lifecycle model from ARCH-07	AMS-09 Redundancy and repair scheme integration with the BISR architecture
Algorithm selection	1.5		
Architecture verification	1.5		
Total	14.0	EXIT CRITERIA	TEST-05 Wafer sort test program development
WHO IT TAKES	~6.1 FTE	- Repair interface written jointly with the memory designers - Repair path verified end to end, not in pieces - MBIST runtime inside the test time budget	MP-05 Yield learning, failure pareto and defect analysis
DFT memory lead Owns MBIST, BIRA and BISR architecture			MEASURED BY
Memory circuit designers Redundancy and repair interface			MBIST runtime against the test time budget
Fuse and lifecycle engineer Fuse path and capacity		DEPENDS ON	- Repair interface mismatches at pattern generation - Repair yield recovered against projection
Test engineering liaison MBIST runtime and repair flow at ATE		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT	
Verification engineer Repair path end-to-end verification		PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)	
		AMS-09 Redundancy and repair scheme integration with the BISR architecture	
		ARCH-07 Security and safety architecture — secure boot, root of trust, fusing	

10 DFT · DFT ARCHITECTURE

DFT-04

Boundary scan and IJTAG (1149.1 / 1687),
TAP and debug access architecture

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w4–w12 in stage	10.0 M/M planned	~5.4 FTE while it runs	6 2 in parallel	no has float	DFT access lead Physical design liaison approves

PURPOSE

Design the **access network** — boundary scan, IJTAG, TAP controller and the debug paths — that every test, every repair and every silicon debug session runs through.

The TAP is the door into the chip. It is used at wafer sort, at package test, during bring-up and in the field, and it is also the door ARCH-07 needs to lock in production. Designing it is a negotiation between access and security rather than a purely technical exercise.

WORK FLOW

6 STEPS · 2 LANES

w4	w6	w8	w10	w12
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MAIN SEQUENCE

1	2	5	6
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Access architecture — TAP, IJTAG network topology	1.5w	main
2	Boundary scan implementation per IEEE 1149.1	1.5w	main
3	IJTAG instrument network and description files	2w	parallel
4	Debug and trace access path definition	1.5w	parallel
5	Security lockdown integration per lifecycle state	1.5w	main
6	Access network verification and description file generation	3.5w	main

Step 5 is where the negotiation with ARCH-07 lands. Every path this activity opens for debug is a path security has to close in production lifecycle states, and the lock has to be testable without making the part permanently undebuggable.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- DFT architecture from DFT-01
- Security lockdown policy from ARCH-07
- Debug requirements from validation and BU-09
- Package pin availability from ARCH-08
- IJTAG tooling from PDK-08

PRODUCES

- TAP and IJTAG network architecture
- Boundary scan implementation
- Instrument network with ICL and PDL descriptions
- Debug and trace access paths
- Lifecycle-state lockdown integration
- BSDL and description files

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-03	PRODUCES	JTAG / IJTAG description files (BSDL, ICL, PDL). This activity is the deliverable, and these files are what board test and ATE consume.
DFT-01	FEEDS	DFT architecture specification. The access network is how every other DFT feature is reached.

RISKS AND WATCH-OUTS

- ◆ **Debug access designed out for security.** A part that cannot be debugged in the field is a part whose first field failure has no diagnostic path.
- ◆ **Lockdown untested.** The lifecycle transition that locks the TAP is the one nobody exercises, because exercising it makes the part undebuggable.
- ◆ **Description files generated late.** BSDL and ICL are consumed by board test and ATE; generating them near tapeout leaves no time to validate them.
- ◆ **Access network verified only in functional mode.** The network exists to work when the design does not, and that is the mode that needs proving.
- ◆ **Pin budget for debug not reserved.** Trace ports need pins, and a bump map fixed without them means debug through a narrow channel.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Access network verification and files	3.0	● DFT architecture available from DFT-01	DFT-08 Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
TAP and IJTAG architecture	2.5	● Security lockdown policy defined by ARCH-07	DFT-10 eFuse, chip ID and memory repair infrastructure
Boundary scan implementation	1.5	● Pin and bump budget known from ARCH-08	DFT-11 Silicon debug hooks — trace, observability, design-for-debug
Debug and trace paths	1.5		TEST-08 DFT pattern porting to ATE format and pattern debug
Security lockdown integration	1.5		BU-03 Reset, clocking and PLL lock validation
Total	10.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.4 FTE	● Lockdown transitions tested, not only implemented	● Lifecycle lockdown transitions tested
DFT access lead		● Description files generated and validated before tapeout	● Description files validated before tapeout
Security architect		● Debug access preserved for the lifecycle states that need it	● Debug access available in the states validation needs
Validation liaison		DEPENDS ON	
Debug and trace requirements for bring-up		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT	
Verification engineer		ARCH-07 Security and safety architecture — secure boot, root of trust, fusing	
Physical design liaison		ARCH-08 Chip-level floorplan intent, pin and bump budget	
Pin and routing budget for the network			

10 DFT · DFT ARCHITECTURE

DFT-05

On-chip clock controller design for at-speed testing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w8–w15 in stage	8.0 M/M planned	~5.0 FTE while it runs	6 2 in parallel	no has float	DFT engineer, at-speed Physical design liaison approves

PURPOSE

Design the **on-chip clock controllers** that generate at-speed test clocks — because transition-fault testing needs clock pulses the tester cannot supply at frequency.

Stuck-at testing runs slowly and catches static defects. The defects that matter for a high-frequency part are timing-related, and catching them needs launch-capture pulses at functional speed. The tester cannot produce them, so the chip has to produce them for itself.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	At-speed test requirement per clock domain	1w	main
2	OCC architecture and PLL interface	1.5w	main
3	Launch-capture sequencing and pulse control	1.5w	main
4	Multi-domain and cross-domain at-speed strategy	1.5w	parallel
5	OCC insertion and integration	1.5w	parallel
6	At-speed clocking verification	3w	main

Step 4 is the difficult part. Single-domain at-speed test is well understood; testing paths that cross domains at speed requires the domains to have a defined phase relationship in test mode, which is a constraint on the clocking architecture rather than on DFT.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Clock architecture and domains from ARCH-05
- DFT architecture from DFT-01
- PLL interface from AMS-02
- Test platform clocking capability from TEST-02
- Transition fault coverage targets from DFT-02

PRODUCES

- At-speed test requirements per domain
- OCC architecture and PLL interface
- Launch-capture sequencing control
- Cross-domain at-speed strategy
- Inserted and verified OCC logic

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D1

FEEDS

DFT architecture specification. At-speed clocking is what makes transition-fault coverage achievable, so it belongs in the architecture.

DFT-D4

FEEDS

ATPG pattern sets with coverage report. Transition patterns cannot be generated for domains the OCC cannot clock at speed.

RISKS AND WATCH-OUTS

- ◆ **At-speed test not supported on every domain.** Coverage then has a hole exactly where the fastest logic is, which is where transition defects concentrate.
- ◆ **Cross-domain paths untested at speed.** These are among the most timing-marginal paths in the design and the hardest to cover.
- ◆ **OCC interaction with the PLL unverified.** A controller that cannot get a clean clock from the PLL in test mode produces patterns that fail on good silicon.
- ◆ **Power during at-speed test ignored.** Scan shift followed by at-speed capture creates switching activity far above functional, and the IR drop can cause failures on good parts.
- ◆ **OCC inserted late.** It sits in the clock path, so adding it after clock tree synthesis means redoing the tree.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
At-speed clocking verification	2.5	● Clock architecture and domains fixed by ARCH-05	DFT-06 Scan insertion and DFT DRC during synthesis
OCC architecture and interface	2.0	● DFT architecture available from DFT-01	DFT-07 ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
Launch-capture sequencing	1.5	● PLL test-mode behaviour known from AMS-02	SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
Cross-domain strategy	1.0		
Insertion and integration	1.0		
Total	8.0		
WHO IT TAKES	~5.0 FTE	EXIT CRITERIA	
DFT engineer, at-speed Owns OCC architecture and insertion		● At-speed clocking available on every domain that needs coverage	PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Clocking architect Domain phase relationships in test mode		● Cross-domain at-speed strategy defined and verified	MEASURED BY
PLL designer PLL behaviour and interface in test mode		● OCC inserted before clock tree synthesis, not after	● Domains with at-speed capability against domains needing it
Test engineering liaison Tester clocking capability and limits		DEPENDS ON	● Cross-domain paths covered at speed
Physical design liaison Clock tree impact of OCC insertion		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT	● Test-mode IR drop against the functional case
		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
		AMS-02 PLL / clock generator design and closure	

10 DFT · DFT ARCHITECTURE

DFT-06

Scan insertion and DFT DRC during synthesis

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	14.0 M/M	~6.1 FTE	7	yes	DFT implementation engineer
w14–w24 in stage	planned	while it runs	3 in parallel	no float	DFT architect approves

PURPOSE

Insert the **scan chains, compression and test logic** into the design during synthesis, and drive DFT design-rule checking to clean.

This is where the architecture becomes logic. DFT DRC violations — uncontrollable clocks, unresettable flops, blocked scan paths — reduce coverage silently, and each one is a set of faults ATPG will never be able to target.

WORK FLOW

7 STEPS · 2 LANES

w14w16w18w20w22w24

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Scan insertion setup and configuration	1.5w	main
2	Scan chain insertion and stitching	2w	main
3	Compression logic insertion	1.5w	parallel
4	Test point insertion for coverage improvement	1.5w	parallel
5	DFT DRC analysis and violation closure	3w	main
6	Scan chain integrity verification	1.5w	parallel
7	Insertion sign-off and netlist handoff	3.5w	main

Step 5 is the substance. Every DFT DRC violation is a region of logic ATPG cannot control or observe, and the coverage loss is silent — the tool simply reports a lower number without saying which violation caused it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- DFT architecture from DFT-01
- MBIST insertion from DFT-03
- OCC logic from DFT-05
- Synthesis flow from SYN-01
- Netlist from SYN-02

PRODUCES

- Scan insertion configuration
- Stitched scan chains
- Inserted compression logic
- Test points for coverage improvement
- DFT DRC clean results
- Scan-inserted netlist

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D5

PRODUCES

DFT DRC clean report. This activity is the deliverable, and clean means every violation fixed or explicitly accepted with its coverage cost.

DFT-D4

FEEDS

ATPG pattern sets with coverage report. Coverage achievable at DFT-07 is bounded by how clean this activity got.

RISKS AND WATCH-OUTS

- ◆ **DRC violations accepted without measuring their coverage cost.** Each one silently removes faults from the targetable set, and the total is invisible until ATPG reports a low number.
- ◆ **Test points inserted without area budget.** They improve coverage and cost area and timing, and adding them late competes with closure.
- ◆ **Chain balance ignored.** Unbalanced chains set shift time by the longest one, and shift time is most of the test time.
- ◆ **Insertion done once and not re-run.** Netlist changes after insertion leave chains stale, and the mismatch appears at pattern validation.
- ◆ **Chain integrity unverified.** A broken chain makes everything behind it untestable, and finding that at ATE rather than in simulation costs a silicon turn.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
DFT DRC closure	5.0	DFT architecture fixed by DFT-01	DFT-07 ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
Scan insertion and stitching	3.5	MBIST and OCC logic available	
Sign-off and handoff	2.5	Synthesis flow running from SYN-01	DFT-09 Scan compression and chain routing feasibility with physical design
Compression insertion	1.5		
Test point insertion	1.0		
Chain integrity verification	0.5		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback
Total	14.0		
WHO IT TAKES	~6.1 FTE		
DFT implementation engineer Owns insertion and DRC closure		EXIT CRITERIA - Every DRC violation fixed or accepted with a measured coverage cost - Chains balanced against the shift time budget - Chain integrity verified in simulation before ATPG	DV-11 Gate-level simulation, functional and timing-annotated PD-12 Scan chain reordering and DFT-aware routing
Synthesis engineer Flow integration and netlist handoff		DEPENDS ON DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT DFT-03 MBIST / BIRA / BISR architecture for embedded memories DFT-05 On-chip clock controller design for at-speed testing SYN-01 SDC constraint development and validation PKD-09 Foundry reference flow bring-up and internal methodology deck	MEASURED BY - DRC violations closed against accepted - Coverage cost of accepted violations - Chain length imbalance against budget
Block designers Fix DRC violations in their logic			
Physical design liaison Chain ordering and routing feasibility			
DFT architect Accepts violations with their coverage cost			

DFT-07

ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	28.0 M/M	~7.6 FTE	8	yes	ATPG engineer
w22–w38 in stage	planned	while it runs	3 in parallel	no float	Product engineering approves

PURPOSE

Generate the **ATPG patterns** — stuck-at, transition and cell-aware — and drive coverage to the targets DFT-02 agreed, inside the pattern volume the tester can hold.

Two constraints pull against each other here: coverage has to reach the quality target and pattern volume has to fit tester memory and test time. Every technique that raises one raises the other, and closing both is the work.

WORK FLOW

8 STEPS · 2 LANES

w22	w26	w30	w34	w38
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MAIN SEQUENCE

1	2	3	6	8
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IN PARALLEL

4	5	7
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#	STEP	TAT	LANE
1	ATPG setup, constraints and test mode configuration	2w	main
2	Stuck-at pattern generation and coverage analysis	3w	main
3	Transition and at-speed pattern generation	3.5w	main
4	Cell-aware pattern generation	2.5w	parallel
5	Untestable fault analysis and coverage justification	2.5w	parallel
6	Pattern volume reduction and compaction	3w	main
7	Coverage closure against the DFT-02 targets	2w	parallel
8	Pattern set assembly and release	4.5w	main

Step 5 is what makes a coverage number honest. Faults reported as untestable are either genuinely untestable — tied off, redundant — or untestable because of a DRC violation, and only analysis distinguishes a real limit from a fixable one.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Scan-inserted netlist from DFT-06
- Coverage targets and fault models from DFT-02
- Test time and memory budget from TEST-09
- Library fault models from PDK-03
- OCC and at-speed setup from DFT-05

PRODUCES

- ATPG setup and constraints
- Stuck-at, transition and cell-aware pattern sets
- Untestable fault analysis
- Compacted pattern volumes
- Coverage report against targets
- Released pattern sets

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D4

PRODUCES

ATPG pattern sets with coverage report. This activity is the deliverable, and the coverage report is what the quality target is measured against.

DFT-D7

FEEDS

DFT signoff entry for the tapeout checklist. Coverage against target is one of the checklist's quantitative entries.

RISKS AND WATCH-OUTS

- ◆ **Coverage target missed and accepted late.** The remedies — test points, DRC fixes, architecture changes — all need an RTL or netlist change, and none are available near tapeout.
- ◆ **Pattern volume exceeding tester memory.** Patterns that do not fit have to be split across loads, which multiplies test time and cost.
- ◆ **Untestable faults accepted without analysis.** A fixable DRC violation reported as an untestable fault is coverage given away.
- ◆ **Cell-aware added late.** It multiplies pattern count against a budget already set, forcing a choice between coverage and test time with no room to negotiate.
- ◆ **Patterns generated on a netlist that then changes.** Any netlist change after generation invalidates the set, and regeneration takes weeks.

EFFORT SPLIT	28 M/M	ENTRY CRITERIA	FEEDS INTO
Pattern generation	10.0	● Scan-inserted netlist available and DRC clean from DFT-06 ● Coverage targets set by DFT-02 ● Tester memory and test time budget known from TEST-09	DFT-08 Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
Pattern volume reduction	6.0		TEST-08 DFT pattern porting to ATE format and pattern debug
Untestable fault analysis	4.5		T0-03 Tapeout checklist completion and owner sign-off
Pattern set assembly and release	4.0		MP-06 Production test program release and guard-band validation
Coverage closure	3.5		
Total	28.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~7.6 FTE	● Coverage meets the DFT-02 targets per fault model ● Pattern volume inside tester memory without splitting ● Untestable faults analysed rather than accepted	● Coverage achieved against target per fault model ● Pattern volume against tester memory ● Untestable faults analysed against accepted
ATPG engineer Owns pattern generation and coverage closure			
DFT architect Untestable fault analysis and coverage decisions			
Test engineering liaison Pattern volume, memory and test time			
Block designers Fix DRC violations blocking coverage			
Product engineering Accepts final coverage against the quality target		DEPENDS ON	
		DFT-02 Test coverage and test-time target negotiation with product and test engineering	
		DFT-05 On-chip clock controller design for at-speed testing	
		DFT-06 Scan insertion and DFT DRC during synthesis	
		TEST-09 Test time and cost optimisation — parallelism, site count	

DFT-08

Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	14.0 M/M	~6.1 FTE	7	yes	DFT validation engineer
w28–w38 in stage	planned	while it runs	3 in parallel	no float	DFT architect approves

PURPOSE

Prove the patterns **actually run** — simulate them against the netlist with timing, convert them to tester format, and hand ATE a set that works.

A pattern set that passes in ATPG and fails in gate-level simulation is a pattern set that would have failed on silicon. This activity is the last place a pattern problem is cheap, and it is also where the format conversion for the tester is proved rather than assumed.

WORK FLOW

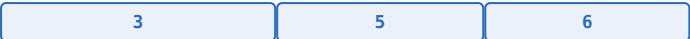
7 STEPS · 2 LANES

w28	w30	w32	w34	w36	w38
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Pattern simulation environment setup	1.5w	main
2	Zero-delay pattern validation by gate-level simulation	2w	main
3	Timing-annotated pattern simulation	2w	parallel
4	Pattern failure debug and regeneration	2.5w	main
5	STIL and WGL format conversion	1.5w	parallel
6	Tester format and timing set validation	1.5w	parallel
7	Pattern release and DFT tapeout sign-off	4w	main

Step 3 catches what zero-delay cannot: patterns that violate timing at the capture edge. On an at-speed pattern set this is not a formality — a launch-capture sequence that closes in ATPG and fails in timing simulation would have produced yield loss on good die.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Pattern sets from DFT-07
- Gate-level netlist and SDF from PD-09
- Simulation environment from DV-11
- Tester format requirements from TEST-08
- Timing sets and tester capability from TEST-02

PRODUCES

- Pattern simulation environment
- Zero-delay and timing-annotated validation results
- Regenerated patterns for failures
- STIL and WGL format files
- Tester format validation results
- Released ATE-ready pattern set

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D6	PRODUCES	Pattern validation (GLS) report and ATE-ready pattern files. This activity is the deliverable — patterns proved to run rather than assumed to.
DFT-D7	PRODUCES	DFT signoff entry for the tapeout checklist. DFT signs the checklist on validated patterns, not on generated ones.

RISKS AND WATCH-OUTS

- ◆ **Validation skipped for schedule.** A pattern failure then appears at ATE bring-up, where the diagnosis competes with everything else happening at first silicon.
- ◆ **Only zero-delay simulation run.** Timing-related pattern failures are exactly what at-speed patterns are meant to detect, and they are invisible without SDF.
- ◆ **Format conversion assumed.** STIL and WGL translation has tool-specific quirks, and a set that converts cleanly can still time incorrectly on the tester.
- ◆ **Regeneration cost underestimated.** A pattern failure requiring regeneration puts the whole set back through DFT-07, which takes weeks not days.
- ◆ **Timing sets not validated against the actual tester.** A pattern that assumes an edge placement the ATE cannot produce fails on hardware for a reason no simulation shows.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Pattern release and sign-off	4.0	● Pattern sets released by DFT-07	T0-03 Tapeout checklist completion and owner sign-off
Failure debug and regeneration	3.5	● Netlist and SDF available from PD-09	TEST-08 DFT pattern porting to ATE format and pattern debug
Gate-level validation	2.5	● Simulation environment available from DV-11	MP-06 Production test program release and guard-band validation
Timing-annotated simulation	2.0		
Format conversion and validation	2.0		
Total	14.0	EXIT CRITERIA ● Patterns validated with timing, not only zero-delay ● Format conversion validated against the actual tester capability ● DFT tapeout checklist entry signed on validated patterns	BU-02 Power-on, power sequencing and basic health check
WHO IT TAKES	~6.1 FTE	DEPENDS ON	MEASURED BY
DFT validation engineer Owns pattern validation and release		DFT-04 Boundary scan and IJTAG (1149.1 / 1687), TAP and debug access architecture	● Patterns validated against patterns released
Gate-level simulation engineer Simulation environment and debug		DFT-07 ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure	● Failures found in validation against at ATE bring-up
ATPG engineer Pattern regeneration where validation fails		DV-11 Gate-level simulation, functional and timing-annotated	● Regeneration cycles required
Test engineering Format conversion and tester timing sets		PD-09 Multi-corner multi-mode timing closure	
DFT architect Signs the tapeout checklist entry			

10 DFT · DFT ARCHITECTURE

DFT-09

Scan compression and chain routing feasibility with physical design

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	10.0 M/M	~4.3 FTE	6	no	DFT physical lead
w16–w26 in stage	planned	while it runs	3 in parallel	has float	Clock architect approves

PURPOSE

Work with physical design on whether the **scan chains and compression network can actually be routed** — and fix the plan before routing rather than after.

Compression networks connect every chain to a shared decompressor and compactor, which produces long routes across the die competing with functional signals for the same resource. A compression ratio that is optimal on paper and unroutable in practice is discovered at PD-06, when the remedies are expensive.

WORK FLOW6 STEPS · 2 LANES

w16	w18	w20	w22	w24	w26
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Chain routing feasibility analysis with the floorplan	1.5w	main
2	Compression network topology against physical partitioning	2w	main
3	Chain ordering and reordering strategy	1.5w	parallel
4	Congestion impact assessment	1.5w	parallel
5	Test clock distribution feasibility	1.5w	parallel
6	Physical DFT plan agreement with physical design	6.5w	main

Step 3 is where the two disciplines usually collide. DFT wants chains ordered for balance and diagnosis; physical design wants them ordered for routing. Reordering after placement is the standard compromise, and agreeing it here avoids arguing about it during closure.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- DFT architecture and compression from DFT-01
- Scan insertion from DFT-06
- Floorplan and partitioning from PD-01 and PD-02
- Routing resource from the metal stack in TECH-02
- Congestion analysis from PD-06

PRODUCES

- Chain routing feasibility findings
- Compression topology aligned to physical partitions
- Chain ordering and reordering strategy
- Congestion impact assessment
- Test clock distribution plan
- Agreed physical DFT plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D1

FEEDS

DFT architecture specification. An architecture that cannot be routed is not an architecture, and this is where that is established.

DFT-D5

FEEDS

DFT DRC clean report. Physical feasibility findings often send work back into insertion, so the two are closed together.

RISKS AND WATCH-OUTS

- ◆ **Compression ratio unroutable.** A ratio chosen for pattern volume alone produces a network that consumes routing resource the functional design needs.
- ◆ **Chain reordering breaking diagnosis.** Physical reordering improves routing and can destroy the chain-to-block mapping diagnosis depends on.
- ◆ **Test clock distribution treated as functional.** Scan shift clocks reach every flop and have their own skew and power requirements.
- ◆ **Feasibility assessed after routing starts.** The findings then arrive when the only remaining remedy is to reduce coverage.
- ◆ **Hierarchical partition boundaries ignored.** Chains crossing physical partitions carry a routing and timing cost that a logical view does not show.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Compression topology and feasibility	3.5	DFT architecture and compression defined by DFT-01	DFT-07 ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
Physical DFT plan agreement	2.5	Floorplan available from PD-01	PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
Chain ordering strategy	1.5	Scan insertion in progress in DFT-06	
Congestion assessment	1.5		
Test clock distribution	1.0		PD-12 Scan chain reordering and DFT-aware routing
Total	10.0	EXIT CRITERIA - Compression network confirmed routable at the chosen ratio - Chain reordering strategy preserves diagnosis capability - Physical DFT plan agreed before detailed routing starts	MEASURED BY - Routing resource used by the compression network - Chain reordering impact on diagnosis resolution - Feasibility findings raised before routing starts
WHO IT TAKES	~4.3 FTE	DEPENDS ON DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT DFT-06 Scan insertion and DFT DRC during synthesis PD-01 Flow setup and MMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity	
DFT physical lead Owns the physical DFT plan			
Physical design engineers Routing feasibility and congestion			
DFT architect Trades compression ratio against routability			
Diagnosis engineer Chain-to-block mapping for failure analysis			
Clock architect Test clock distribution			

10 DFT · DFT ARCHITECTURE

DFT-10

eFuse, chip ID and memory repair infrastructure

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w12–w18 in stage	6.0 M/M planned	~4.3 FTE while it runs	7 3 in parallel	no has float	Fuse and infrastructure engineer Verification engineer approves

PURPOSE

Build the **fuse and identity infrastructure** — eFuse arrays, chip ID, repair storage, lifecycle state — that repair, traceability and security all depend on.

Fuses are written once and read forever. They store memory repair, trim values, chip identity and lifecycle state, and every one of those consumers has its own requirements. Getting the capacity or the read path wrong produces parts that cannot be repaired, identified or secured.

WORK FLOW

7 STEPS · 2 LANES

w12	w14	w16	w18
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Fuse consumer inventory — repair, trim, ID, lifecycle, security	1w	main
2	Capacity allocation against worst-case need	1w	main
3	eFuse array integration and programming path	1.5w	parallel
4	Fuse read, shadow register and distribution design	1.5w	main
5	Chip identity and traceability scheme	1w	parallel
6	Lifecycle state encoding with ARCH-07	1w	parallel
7	Fuse infrastructure verification	2.5w	main

Step 2 is the step that fails silently. Every consumer asks for a modest allocation, the total is set once, and the worst-case repair set — which is a distribution rather than an average — exceeds it on the parts that most needed repairing.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Repair storage requirement from DFT-03 and AMS-09
- Trim requirements from AMS-04
- Lifecycle and security states from ARCH-07
- Traceability requirements from MP-09
- eFuse capability from PDK-02

PRODUCES

- Fuse consumer inventory
- Capacity allocation with worst-case margin
- eFuse array integration and programming path
- Fuse read and distribution logic
- Chip identity scheme
- Verified fuse infrastructure

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D2	FEEDS	MBIST / BISR insertion and test collateral. Repair is only useful if the repair data can be stored and read back, which this activity provides.
DFT-D3	FEEDS	JTAG / IJTAG description files. Fuse programming and readback run through the access network, and the instrument descriptions cover them.

RISKS AND WATCH-OUTS

- ◆ **Capacity sized to the average repair need.** Repair need is a distribution, and the parts needing most repair are exactly the ones the allocation fails.
- ◆ **Programming path untested.** Fuses are one-time; a programming path that does not work correctly produces parts that are permanently wrong.
- ◆ **Fuse read timing in the boot path unconsidered.** Repair and trim values have to be distributed before the logic that uses them is released from reset.
- ◆ **Traceability scheme designed late.** Chip identity is needed for yield analysis and field returns, and retrofitting it means parts that cannot be traced.
- ◆ **Lifecycle encoding without a test path.** A lifecycle transition that cannot be exercised in test is a transition whose correctness is unknown until the field.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Fuse infrastructure verification	2.0	● Repair storage requirements known from DFT-03	TEST-05 Wafer sort test program development
Read and distribution design	1.5	● Lifecycle states defined by ARCH-07	MP-09 Production readiness review and change control (PCN) setup
Array integration and programming	1.0	● eFuse capability confirmed by PDK-02	
Capacity allocation	0.8		
Identity and lifecycle encoding	0.8		BU-02 Power-on, power sequencing and basic health check
Total	6.0	EXIT CRITERIA ● Capacity sized against worst-case, not average, repair need ● Programming and read paths verified before tapeout ● Lifecycle transitions exercisable in test	MP-06 Production test program release and guard-band validation
WHO IT TAKES	~4.3 FTE	DEPENDS ON	MEASURED BY ● Fuse capacity against worst-case repair need ● Programming path failures found pre-silicon ● Lifecycle transitions with a test path
Fuse and infrastructure engineer Owns the array, capacity and read path			
DFT memory lead Repair storage requirements			
Security architect Lifecycle state encoding and protection		DFT-03 MBIST / BIRA / BISR architecture for embedded memories	
Product engineering Traceability and identity requirements		DFT-04 Boundary scan and IJTAG (1149.1 / 1687), TAP and debug access architecture	
Verification engineer Fuse infrastructure verification		ARCH-07 Security and safety architecture — secure boot, root of trust, fusing	
		PDK-02 Design rule manual review; restricted and recommended rule disposition	

10 DFT · DFT ARCHITECTURE

DFT-11

Silicon debug hooks — trace, observability, design-for-debug

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	10.0 M/M	~5.4 FTE	6	no	DFT debug lead
w6–w14 in stage	planned	while it runs	2 in parallel	has float	Verification engineer approves

PURPOSE

Design the **observability the lab will need** — trace, triggers, internal visibility, design-for-debug — because at bring-up the only view into the chip is the one designed in months earlier.

Silicon debug is constrained entirely by what was built. A failure that cannot be observed cannot be diagnosed, and the difference between a two-day debug and a two-month one is usually whether someone put a trigger and a trace port where it was needed.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE

1	2	5	6
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Debug requirement collection from validation and design	1w	main
2	Observability architecture — trace, triggers, snapshot	2w	main
3	Trace port and bandwidth budget	1w	parallel
4	Trigger and cross-trigger network	1.5w	parallel
5	Internal state snapshot and scan dump capability	1.5w	main
6	Debug infrastructure verification and documentation	3.5w	main

Step 5 is the cheapest debug capability available. Scan chains already reach every flop; adding the ability to halt and dump their contents turns the test infrastructure into a full internal state view for almost no extra area.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Debug requirements from validation and BU-09
- Access network from DFT-04
- Scan infrastructure from DFT-06
- Pin and bump budget from ARCH-08
- Security lockdown policy from ARCH-07

PRODUCES

- Debug requirement list
- Observability architecture
- Trace port and bandwidth budget
- Trigger and cross-trigger network
- Scan dump and snapshot capability
- Debug infrastructure documentation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

DFT-D3	FEEDS	JTAG / IJTAG description files. Debug instruments are part of the access network and are described alongside it.
DFT-D1	FEEDS	DFT architecture specification. Design-for-debug is part of the DFT architecture rather than a separate concern.

RISKS AND WATCH-OUTS

- ◆ **Debug traded away for area.** The saving is small and the cost is paid at bring-up, in weeks, by a different team.
- ◆ **Requirements collected from design rather than validation.** The people who will debug the silicon know what they need to see; the people who built it usually assume it works.
- ◆ **Trace bandwidth inadequate.** A trace port too narrow to capture the event of interest produces a debug capability that cannot answer the question.
- ◆ **Debug infrastructure unverified.** It is used first when the chip is misbehaving, which is the worst moment to discover the debug path is broken.
- ◆ **No documentation.** A capability the lab does not know exists is a capability that was not built.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Debug infrastructure verification and documentation	3.5	● Debug requirements collected from validation	BU-09 Silicon debug of anomalies and failure analysis requests
Observability architecture	2.5	● Access network architecture from DFT-04	BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Trigger network	1.5	● Pin budget known from ARCH-08	TEST-10 Correlation between ATE, bench and system results
Snapshot and scan dump	1.5		MP-05 Yield learning, failure pareto and defect analysis
Trace port budget	1.0		
Total	10.0	EXIT CRITERIA ● Debug paths verified before tapeout, not first used at bring-up ● Trace bandwidth sufficient for the events validation named ● Capabilities documented for the people who will use them	MEASURED BY ● Debug requirements met against those collected ● Debug paths verified pre-silicon ● Bring-up issues diagnosed with built-in observability
WHO IT TAKES	~5.4 FTE	DEPENDS ON	
DFT debug lead Owns observability architecture and documentation		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT	
Validation liaison States what the lab will need to see		DFT-04 Boundary scan and IJTAG (1149.1 / 1687), TAP and debug access architecture	
Design leads Identify internal state worth observing		ARCH-07 Security and safety architecture — secure boot, root of trust, fusing	
Security architect Debug access under lifecycle lockdown		ARCH-08 Chip-level floorplan intent, pin and bump budget	
Verification engineer Debug infrastructure verification			

IMPLEMENT

11 Logic Synthesis & Netlist Drops

SYN

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w42 – w66	24 _w	12	136 _{M/M}	6 _{of 12}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
SYN-01	SDC constraint development and validation		8w18 _{M/M}	SYN-D1N0 flow-flush netlist for PD flow setup
SYN-02	Technology mapping and optimisation — multi-Vt, multi-bit, useful skew		8w16 _{M/M}	SYN-D2N1 and N2 netlist drops with QoR delta reports
SYN-03	N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow		4w6 _{M/M}	SYN-D3FFN — final full netlist, release-tagged
SYN-04	Physical-aware synthesis with congestion feedback from the PD flow setup		7w14 _{M/M}	SYN-D4Validated SDC constraint set per mode and corner
SYN-05	N1 netlist drop — first quality turn against PD placement and congestion feedback		5w12 _{M/M}	SYN-D5Synthesis QoR report per drop against PPA targets
SYN-06	Power optimisation — clock gating, operand isolation, Vt mix		7w12 _{M/M}	SYN-D6Formal equivalence clean report per drop
SYN-07	Low-power synthesis — isolation and level-shifter insertion against UPF		5w8 _{M/M}	SYN-D7Power intent implementation report
SYN-08	N2 netlist drop — timing and congestion converged, late RTL changes absorbed		5w12 _{M/M}	SYN-D8Physical design handoff package per drop
SYN-09	FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point		4w10 _{M/M}	
SYN-10	Formal equivalence checking per drop, RTL versus netlist		20w cont.12 _{M/M}	
SYN-11	Timing, area and power reporting against budget, per drop		22w cont.10 _{M/M}	
SYN-12	Netlist handoff and QoR delta review with physical design, per drop		22w cont.6 _{M/M}	

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-01

SDC constraint development and validation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	18.0 M/M	~9.8 FTE	6	yes	Constraints lead
w0–w8 in stage	planned	while it runs	2 in parallel	no float	Signoff methodology engineer approves

PURPOSE

Write the **constraints the whole implementation is optimised and signed off against** — clocks, generated clocks, IO timing, exceptions, test modes — and validate them rather than assume them.

Constraints are the specification synthesis and physical design actually read. A missing clock definition produces a design optimised for a path nobody cares about; a wrong false path produces one that closes in the tool and fails in silicon. Every hour spent validating them here saves days at PD-09.

WORK FLOW6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Mode and corner enumeration against the operating conditions	1.5w	main
2	Clock and generated clock definition across the design	2w	main
3	Exception, false path and multicycle specification	1.5w	parallel
4	IO and interface timing constraints	2w	main
5	Test mode and scan constraints	1.5w	parallel
6	Constraint validation and quality checking	2.5w	main

Step 6 is not review, it is tooling. Constraint quality checkers find unconstrained paths, conflicting exceptions and clocks that reach nothing, and they find them in minutes — which is why every drop should be re-checked rather than checked once.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Clock architecture and DVFS points from ARCH-05
- Interface timing requirements from ARCH-04
- Corner and mode definition from PDK-11
- DFT test modes from DFT-01
- RTL hierarchy from RTL-04

PRODUCES

- Mode and corner enumeration
- Clock and generated clock definitions
- Exception and multicycle specification
- IO and interface constraints
- Test mode constraints
- Constraint validation reports

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-04	PRODUCES	Validated SDC constraint set per mode and corner. This activity is the deliverable, and "validated" is what distinguishes it from a file of assertions.
SYN-05	FEEDS	Synthesis QoR report per drop. QoR is only meaningful against constraints; an unconstrained path reports as passing.

RISKS AND WATCH-OUTS

- ◆ **Unconstrained paths reported as clean.** A path with no constraint is a path the tool ignores, and it appears in the report as no violation at all.
- ◆ **Exceptions written to close timing.** A false path added because the path would not close is a silicon failure being converted into a green report.
- ◆ **Constraints diverging between synthesis and signoff.** Two constraint sets means optimising against one and being judged against the other.
- ◆ **Test mode constraints written late.** Scan shift and at-speed modes have their own timing, and a design closed only in functional mode fails at ATE.
- ◆ **IO constraints assumed from the interface specification.** Actual budgets come from the package and channel analysis in SIPI-07, not from the protocol document.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
Clock and generated clock definitions	5.0	● Clock architecture fixed by ARCH-05 ● Corner and mode definition available from PDK-11	SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew
Constraint validation	4.5		SYN-03 NO flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow
IO and interface constraints	3.5	● RTL hierarchy stable enough to constrain	
Exception specification	3.0		SYN-11 Timing, area and power reporting against budget, per drop
Test mode constraints	2.0		PD-01 Flow setup and MMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity
Total	18.0	EXIT CRITERIA ● No unconstrained paths in the design ● Every exception justified by a design reason, not by a violation ● One constraint set used by both synthesis and signoff	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
WHO IT TAKES	~9.8 FTE	DEPENDS ON	MEASURED BY
Constraints lead Owns the SDC set and its validation		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	● Unconstrained paths at each drop ● Exceptions with a documented design reason ● Constraint differences between synthesis and signoff
Timing engineers Clock definitions and exception specification		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
DFT engineer Test mode constraints		DFT-01 DFT architecture definition — scan style, compression ratio, hierarchical DFT	
Physical design liaison Confirms constraints are implementable		RTL-04 Chip-level integration and top assembly	
Signoff methodology engineer Constraint consistency between synthesis and signoff			

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-02

Technology mapping and optimisation
— multi-Vt, multi-bit, useful skew

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w2–w10 in stage	16.0 M/M planned	~8.7 FTE while it runs	6 2 in parallel	yes no float	Synthesis lead PPA lead approves

PURPOSE

Map the RTL onto the qualified library and **optimise it for the PPA budget** — Vt mix, multi-bit banking, useful skew, datapath structure.

Technology mapping is where architecture becomes gates and where most of the achievable PPA is decided. The same RTL mapped two ways can differ by 20% in area and a hundred millivolts in achievable frequency, and the difference is entirely in how this activity is configured.

WORK FLOW

6 STEPS · 2 LANES

w2	w4	w6	w8	w10
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Library and mapping strategy selection	1.5w	main
2	Baseline mapping and optimisation	2.5w	main
3	Datapath and arithmetic optimisation	1.5w	parallel
4	Multi-bit banking and useful skew configuration	2w	main
5	Area recovery on non-critical paths	1.5w	parallel
6	QoR analysis and optimisation iteration	2w	main

Step 5 is where area is won without cost. Most of a design is not on the critical path, and recovering area and leakage there is free performance-wise — but it only happens if someone configures the flow to do it rather than optimising everything for speed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Qualified libraries from PDK-03
- Constraints from SYN-01
- RTL from RTL-02 and RTL-04
- PPA budgets from ARCH-06
- Flow and methodology from PDK-09

PRODUCES

- Mapping strategy and library configuration
- Baseline mapped netlist
- Datapath optimisation results
- Multi-bit and useful skew configuration
- Area recovery results
- QoR analysis per iteration

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-05	FEEDS	Synthesis QoR report per drop. This activity produces the QoR the report describes, drop after drop.
SYN-02	FEEDS	N1 and N2 netlist drops. Each drop is this activity's output at a point in time, with whatever optimisation was enabled.

RISKS AND WATCH-OUTS

- ◆ **Everything optimised for speed.** Area and leakage on non-critical paths are given away for performance nobody asked for, and the die grows.
- ◆ **Multi-bit banking configured late.** Clock power depends heavily on it, and adding it after closure means re-timing the design.
- ◆ **Useful skew used to hide a timing problem.** It buys real margin and also makes the clock tree harder to build; using it to rescue a path that should be restructured moves the problem to PD-07.
- ◆ **Library subset used without knowing it.** Synthesis using only part of the Vt menu produces avoidable leakage, and nobody notices because the design closes.
- ◆ **QoR compared drop to drop without normalising.** A netlist with more RTL in it is not directly comparable to the last one, and the trend becomes noise.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	FEEDS INTO
Baseline mapping and optimisation	5.0	- Libraries qualified and frozen by PDK-03 - Constraints available from SYN-01 - Flow available from PDK-09	SYN-04 Physical-aware synthesis with congestion feedback from the PD flow setup
QoR analysis and iteration	4.0		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback
Multi-bit and useful skew	3.0		SYN-06 Power optimisation — clock gating, operand isolation, Vt mix
Datapath optimisation	2.0	EXIT CRITERIA - Area recovered on non-critical paths, not only speed optimised - Vt menu used across its range, not defaulted - QoR reported normalised so drops are comparable	SYN-11 Timing, area and power reporting against budget, per drop
Area recovery	2.0		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Total	16.0		MEASURED BY - Area and leakage against the ARCH-06 budget - Clock power against the multi-bit projection - QoR trend across drops
WHO IT TAKES	~8.7 FTE	DEPENDS ON	
Synthesis lead		SYN-01 SDC constraint development and validation	
Owns mapping strategy and QoR		PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	
Synthesis engineers		PDK-09 Foundry reference flow bring-up and internal methodology deck	
Flow configuration and optimisation		RTL-02 Block-level RTL implementation	
Library engineer			
Vt menu and special cell usage			
Block owners			
RTL restructuring where synthesis cannot close			
PPA lead			
QoR against the ARCH-06 budgets			

SYN-03

N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	6.0 M/M	~6.5 FTE	5	yes	Synthesis lead
w3–w7 in stage	planned	while it runs	1 in parallel	no float	Configuration manager approves

PURPOSE

Release the **No flow-flush netlist** — functionally incomplete, structurally representative — so physical design can build and debug its flow against a real database months before the design is ready.

No exists to be wrong in the right way. It has the right hierarchy, the right macros and roughly the right size, and it does not need to work. Its entire purpose is to let PD-01 find its flow problems on a throwaway netlist rather than on the one the schedule depends on.

WORK FLOW

5 STEPS · 2 LANES

w3	w4	w5	w6	w7
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	N0 scope definition — what must be representative, what may be stubbed	1w	main
2	Black-box and stub strategy for incomplete blocks	1w	main
3	Constraint bootstrap for N0	1w	parallel
4	Elaboration and structural sanity checking	1w	main
5	Release and physical design handover	1w	main

Step 1 is the whole activity. N0 has to be representative in the dimensions PD's flow cares about — hierarchy, macro count, instance count, pin count — and can be arbitrarily wrong in everything else. Getting that distinction right is what makes it cheap.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- RTL in whatever state exists from RTL-02
- Macro list and abstracts from AMS-15
- Constraints from SYN-01
- Flow setup requirements from PD-01
- Hierarchy from ARCH-02

PRODUCES

- N0 scope definition and stub strategy
- Black-boxed and stubbed netlist
- Bootstrap constraint set
- Structural sanity check results
- Released N0 netlist with a stated limitation list

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-01	PRODUCES	N0 flow-flush netlist for PD flow setup. This activity is the deliverable, and its value is entirely in arriving early rather than in being correct.
SYN-08	FEEDS	Physical design handoff package per drop. N0 is the first handoff, and it sets the format every later one follows.

RISKS AND WATCH-OUTS

- ◆ **N0 delayed until it is good.** The point is to be early; a correct N0 delivered late has become N1 and left PD's flow undebugged.
- ◆ **Not representative in the dimensions that matter.** A netlist with a tenth of the instances exercises none of the runtime and capacity problems PD needs to find.
- ◆ **Limitations not stated.** Physical design chasing timing on a stubbed netlist wastes a week before someone explains it was never going to close.
- ◆ **Stub strategy that changes the hierarchy.** A hierarchy that differs from the real one means the flow is set up for a structure that will not arrive.
- ◆ **Treated as a milestone rather than a tool.** N0 is scaffolding; celebrating it as progress distracts from the fact that no real design has been synthesised.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Scope definition and stubs	2.0	Enough RTL exists to elaborate a representative hierarchy	PD-01 Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
Elaboration and structural checks	1.5	Macro abstracts available from AMS-15	PD-02 Floorplan, macro placement and partition definition
Release and handover	1.5	PD flow setup ready to receive a netlist	SYN-12 Netlist handoff and QoR delta review with physical design, per drop
Bootstrap constraints	1.0		
Total	6.0	EXIT CRITERIA	
WHO IT TAKES	~6.5 FTE	- Representative in hierarchy, instance count and macro content - Limitations stated explicitly with the release - Delivered on the date PD's flow setup needs it, not when it is good	MEASURED BY
Synthesis lead Owns N0 scope and release			Days from PD flow setup start to N0 availability
Physical design lead States what must be representative			Representativeness against the dimensions PD named
Integration engineer Stub and black-box implementation		DEPENDS ON	Flow problems found on N0 against on N1
Constraints engineer Bootstrap constraint set		SYN-01 SDC constraint development and validation	
Configuration manager Release tagging and limitation list		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew	
		AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)	
		RTL-02 Block-level RTL implementation	

SYN-04

Physical-aware synthesis with congestion feedback from the PD flow setup

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	14.0 M/M	~8.7 FTE	6	no	Synthesis engineers
w6–w13 in stage	planned	while it runs	2 in parallel	has float	CAD engineer approves

PURPOSE

Synthesise **with the floorplan in the loop** — physical-aware mapping, congestion feedback, placement-driven restructuring — so the netlist that reaches physical design is already routable.

Logic synthesis without physical information optimises for a wire model that does not exist. On a large die the difference between estimated and actual wire delay is large enough to invalidate the optimisation, and the correction happens either here or as weeks of congestion work at PD-06.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12	w13
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Physical-aware flow setup with the floorplan	1.5w	main
2	Floorplan-aware mapping and optimisation	2w	main
3	Macro and blockage awareness in placement	1.5w	parallel
4	Congestion feedback loop with physical design	2w	main
5	Placement-driven logic restructuring	1.5w	parallel
6	QoR comparison against non-physical synthesis	1.5w	main

Step 6 answers whether the added runtime is worth it. Physical-aware synthesis is slower; comparing its QoR against the wire-load-model result is what justifies the cost, and occasionally shows the floorplan is the problem rather than the logic.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Floorplan from PD-02
- Mapped netlist from SYN-02
- Congestion analysis from PD-06
- Macro placement and blockages from PD-02
- Constraints from SYN-01

PRODUCES

- Physical-aware synthesis flow
- Floorplan-aware mapped netlist
- Congestion feedback findings
- Restructured logic in congested regions
- QoR comparison against non-physical synthesis

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-05	FEEDS	Synthesis QoR report per drop. Physical-aware results are the ones that predict what physical design will actually achieve.
SYN-08	FEEDS	Physical design handoff package per drop. A netlist synthesised against the floorplan is a materially better handoff than one synthesised against a wire model.

RISKS AND WATCH-OUTS

- ◆ **Floorplan too immature to synthesise against.** Physical-aware synthesis against a floorplan that then changes produces optimisation for a layout that never exists.
- ◆ **Congestion feedback one-directional.** Synthesis restructures and physical design re-replaces; without both, each optimises against the other's stale result.
- ◆ **Runtime cost not justified.** Physical-aware flows are slow, and without the QoR comparison nobody knows whether the extra hours bought anything.
- ◆ **Restructuring breaking hierarchy.** Aggressive cross-boundary optimisation improves QoR and makes the netlist harder to close hierarchically and harder to equivalence-check.
- ◆ **Blockages not modelled.** Macros and their keep-out regions dominate placement on this die, and synthesis blind to them produces logic that cannot be placed where it assumed.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Congestion feedback loop	4.0	● Floorplan available and stable from PD-02	SYN-05N1 netlist drop — first quality turn against PD placement and congestion feedback
Floorplan-aware mapping	3.5	● Mapped netlist available from SYN-02	SYN-08N2 netlist drop — timing and congestion converged, late RTL changes absorbed
Flow setup	2.5	● Congestion analysis capability in PD-06	PD-05Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Logic restructuring	2.0		PD-06Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
QoR comparison	2.0	EXIT CRITERIA	
Total	14.0	● Netlist synthesised against a floorplan PD is actually using	
WHO IT TAKES	~8.7 FTE	● Congestion feedback flowing in both directions	
Synthesis engineers		● Runtime cost justified by a measured QoR improvement	
Physical-aware flow and restructuring			
Physical design engineers		DEPENDS ON	MEASURED BY
Floorplan, congestion analysis and feedback		SYN-02Technology mapping and optimisation — multi-Vt, multi-bit, useful skew	● QoR improvement against non-physical synthesis
Synthesis lead		PD-02Floorplan, macro placement and partition definition	● Congestion at PD-05 against prediction
QoR comparison and flow decisions			● Runtime cost per drop
Floorplan owner			
Macro placement and blockage data			
CAD engineer			
Runtime and capacity for physical-aware runs			

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-05

N1 netlist drop — first quality turn against PD placement and congestion feedback

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	12.0 M/M	~10.4 FTE	6	yes	Synthesis lead
w8–w13 in stage	planned	while it runs	2 in parallel	no float	PPA lead approves

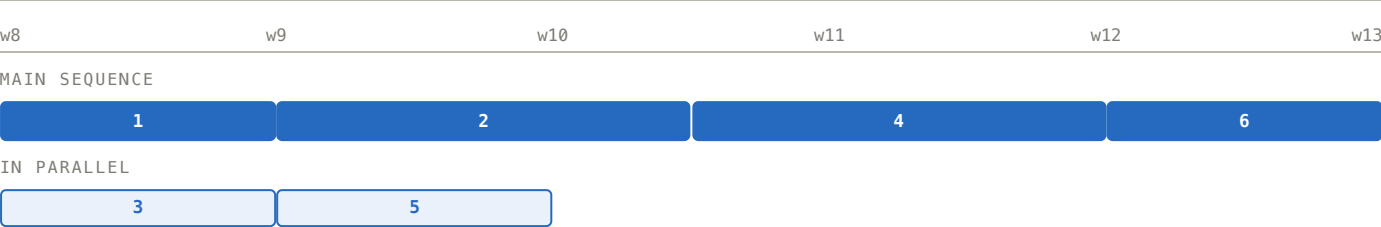
PURPOSE

Release the **N1 drop — the first netlist meant to be closed** — against PD's feedback from its flow setup, and establish the QoR baseline every later drop is measured against.

No proved the flow ran. N1 is where the design's actual QoR first becomes visible: real timing, real area, real congestion. The number it establishes is the one PD-05 will work from and the one everyone will argue about for the next three months.

WORK FLOW

6 STEPS · 2 LANES



#	STEP	TAT	LANE
1	RTL drop intake and delta review against N0	1w	main
2	Synthesis run and block-level closure	1.5w	main
3	Physical design feedback incorporation	1w	parallel
4	QoR delta analysis against the N0 baseline	1.5w	main
5	Issue list for the next drop	1w	parallel
6	Release and handoff to physical design	1w	main

Step 5 is what makes the drop cadence a process rather than a sequence of releases. Each drop should hand forward a named list of what it could not fix and why, so the next one starts from a position rather than from scratch.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- RTL release from RTL-02
- Physical-aware results from SYN-04
- PD flow setup feedback from PD-01
- Constraints from SYN-01
- PPA budgets from ARCH-06

PRODUCES

- N1 netlist
- QoR delta analysis against N0
- Physical design feedback incorporation record
- Issue list carried to N2
- Handoff package for PD turn 1

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-02

PRODUCES

N1 and N2 netlist drops with QoR delta reports. The first of the two quality drops, with the baseline it establishes.

SYN-05

FEEDS

Synthesis QoR report per drop. N1 sets the baseline the QoR trend is measured from.

RISKS AND WATCH-OUTS

- ◆ **N1 released before RTL is worth synthesising.** A drop taken from unstable RTL produces a baseline that means nothing and a PD turn that chases moving targets.
- ◆ **QoR delta not analysed.** Knowing the numbers without knowing what changed them leaves the next drop guessing about what to fix.
- ◆ **Issue list not carried forward.** N2 then rediscovers the same problems, and the drop cadence loses its cumulative benefit.
- ◆ **PD feedback arriving after the drop.** Feedback from flow setup has to reach synthesis before N1 is run, not after it is released.
- ◆ **Baseline set on a partial design.** A QoR baseline missing significant blocks flatters the trend and hides the growth that follows.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Synthesis run and closure	4.0	● RTL stable enough that a drop is meaningful	SYN-08 N2 netlist drop — timing and congestion converged, late RTL changes absorbed
QoR delta analysis	3.0	● PD flow setup complete and feedback available from PD-01	SYN-11 Timing, area and power reporting against budget, per drop
Feedback incorporation	2.5	● Constraints validated in SYN-01	SYN-12 Netlist handoff and QoR delta review with physical design, per drop
Intake and release	2.5		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Total	12.0		
WHO IT TAKES	~10.4 FTE	EXIT CRITERIA	MEASURED BY
Synthesis lead Owns the drop, its QoR and its issue list		● QoR baseline established against a substantially complete design	● QoR at N1 against the ARCH-06 budget
Synthesis engineers Run, closure and delta analysis		● Delta against N0 explained, not only reported	● Design completeness at the baseline
Physical design lead Feedback into the drop and acceptance out of it		● Issue list handed forward to N2	● Issues carried forward against issues found
RTL lead Confirms RTL readiness for the drop			
PPA lead Baseline against the ARCH-06 budgets			
		DEPENDS ON	
		SYN-01 SDC constraint development and validation	
		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew	
		SYN-04 Physical-aware synthesis with congestion feedback from the PD flow setup	
		PD-01 Flow setup and MMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity	
		RTL-02 Block-level RTL implementation	

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-06

Power optimisation — clock gating, operand isolation, Vt mix

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	12.0 M/M	~7.4 FTE	6	no	Power optimisation engineer
w9–w16 in stage	planned	while it runs	2 in parallel	has float	Library engineer approves

PURPOSE

Reduce power where it is actually spent — **clock gating, operand isolation, Vt mix** — driven by real switching activity rather than by defaults.

Power optimisation without activity data is guesswork. Clock gating inserted everywhere costs area and saves little; inserted where the data says the logic is idle it saves a meaningful fraction of dynamic power. The activity data comes from verification, which makes this a cross-stage dependency more than a synthesis setting.

WORK FLOW

6 STEPS · 2 LANES

w9	w11	w13	w15	w16
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Power analysis baseline and hot-spot identification	1.5w	main
2	Clock gating insertion and efficiency analysis	2w	main
3	Vt mix optimisation for leakage	1.5w	parallel
4	Operand isolation and datapath gating	2w	main
5	Activity-driven optimisation with real switching data	1.5w	parallel
6	Power QoR and comparison against the budget	1.5w	main

Step 5 is what separates power optimisation from power decoration. Switching activity from the `DV-04` workload simulations tells the tool which logic is actually idle; without it, gating is inserted by structural heuristic and saves a fraction of what it could.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Switching activity from DV-04 workload simulations
- Power budgets from ARCH-06
- Mapped netlist from SYN-02
- Library Vt menu from PDK-03
- Clock architecture from ARCH-05

PRODUCES

- Power analysis baseline and hot spots
- Clock gating insertion and efficiency report
- Vt mix optimisation results
- Operand isolation implementation
- Power QoR against budget

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-05

FEEDS

Synthesis QoR report per drop. Power is one of the three QoR axes, and the only one that needs external activity data to measure honestly.

SYN-07

FEEDS

Power intent implementation report. Dynamic power optimisation sits alongside the UPF implementation from SYN-07 in the same report.

RISKS AND WATCH-OUTS

- ◆ **Optimisation driven by default activity.** Uniform switching assumptions produce gating where it saves nothing and none where it would.
- ◆ **Clock gating efficiency unmeasured.** Gates inserted without measuring how often they actually gate cost area and leakage for no benefit.
- ◆ **Vt mix skewed to low-Vt for timing.** Leakage grows non-linearly, and a design closed on low-Vt everywhere blows its static power budget.
- ◆ **Power measured at the wrong operating point.** The number that matters is at the DVFS point the product runs at, not at nominal.
- ◆ **Optimisation deferred to the final turn.** Power changes timing; a design closed on timing and then power-optimised has to be re-closed.

EFFORT SPLIT

12 M/M

ENTRY CRITERIA

FEEDS INTO

Clock gating

3.5

- Switching activity available from DV-04

SYN-08

N2 netlist drop — timing and congestion converged, late RTL changes absorbed

Vt mix optimisation

2.5

- Power budgets published by ARCH-06

SYN-11

Timing, area and power reporting against budget, per drop

Operand isolation

2.5

- Mapped netlist available from SYN-02

Power analysis

2.0

EXIT CRITERIA

PD-09

Multi-corner multi-mode timing closure

QoR comparison

1.5

- Optimisation driven by real activity data, not defaults
- Clock gating efficiency measured, not assumed
- Power reported at the operating point the product uses

S0-04

EM and IR-drop signoff, static and dynamic

Total**12.0**

MEASURED BY

WHO IT TAKES

~7.4 FTE

Power optimisation engineer

Owns dynamic and leakage optimisation

Synthesis engineers

Flow configuration and insertion

Verification liaison

Switching activity from real workloads

PPA lead

Power against the ARCH-06 budget

Library engineer

Vt menu usage and leakage characterisation

DEPENDS ON

SYN-02

Technology mapping and optimisation — multi-Vt, multi-bit, useful skew

ARCH-06

PPA budget allocation per block

DV-04

Chip-level and system-level scenario testing

PDK-03

Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)

- Dynamic power against budget
- Clock gating efficiency
- Leakage against the Vt mix projection

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-07

Low-power synthesis — isolation and level-shifter insertion against UPF

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w w10–w15 in stage	8.0 M/M planned	~7.0 FTE while it runs	6 2 in parallel	no has float	Low-power synthesis engineer Library engineer approves

PURPOSE

Implement the **power intent in the netlist** — isolation cells, level shifters, retention flops, always-on paths — from the UPF, and prove the netlist and the intent still agree.

UPF describes what should happen; synthesis is where it becomes cells. An isolation cell missing at a domain boundary is a design that corrupts data on power-down, and it is invisible in any functional check that does not model power. The consistency check between netlist and intent is the only reliable detector.

WORK FLOW

6 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	UPF ingestion and elaboration in the synthesis flow	1w	main
2	Isolation and level shifter inference and insertion	1.5w	main
3	Always-on and feed-through path handling	1w	parallel
4	Retention cell insertion and control	1.5w	main
5	UPF and netlist consistency checking	1w	parallel
6	Power intent implementation report	1w	main

Step 5 runs in parallel and is the one that must not be skipped. The netlist and the UPF are two descriptions of the same intent, and any divergence between them propagates into physical design and verification as a defect nobody can attribute.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- UPF from RTL-05
- Low-power cells from PDK-03
- Power domain definition from ARCH-05
- Mapped netlist from SYN-02
- Low-power flow from PDK-09

PRODUCES

- Elaborated UPF in the synthesis flow
- Inserted isolation and level shifter cells
- Always-on path implementation
- Retention cell insertion and control logic
- UPF-netlist consistency results
- Power intent implementation report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-D7

PRODUCES

Power intent implementation report. This activity is the deliverable — what was inferred, what was inserted, and where intent and netlist agree.

SYN-D6

FEEDS

Formal equivalence clean report per drop. Low-power cells complicate equivalence checking, and the two activities have to agree on how they are handled.

RISKS AND WATCH-OUTS

- ◆ **Isolation inferred but not verified.** A missing cell at one boundary corrupts state on power-down and appears in no functional test.
- ◆ **Retention cells inserted without physical planning.** They have a different footprint and power connectivity, and adding them late disturbs the floorplan.
- ◆ **Always-on paths not identified.** Logic that must survive a domain shutting down and is not marked always-on is powered down with it.
- ◆ **UPF and netlist diverging.** Two descriptions of the same intent, and downstream tools reading different ones.
- ◆ **Level shifters omitted between voltage domains.** The design functions in simulation and fails at the voltage extremes in silicon.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Isolation and level shifter insertion	2.5	● UPF available and tool-accepted from RTL-05 ● Low-power cells qualified in PDK-03 ● Low-power flow available from PDK-09	SYN-10 Formal equivalence checking per drop, RTL versus netlist
Retention cell insertion	2.0		DV-06 Low-power (UPF) verification
UPF elaboration	1.5		PD-03 Power delivery network design and early IR analysis
Consistency checking	1.0		S0-09 Final formal equivalence and LVS netlist consistency
Implementation report	1.0		
Total	8.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~7.0 FTE	● Every domain boundary has verified isolation ● UPF and netlist consistent, checked rather than assumed ● Retention and always-on requirements handed to physical design	● Domain boundaries with verified isolation ● UPF-netlist consistency violations ● Low-power cells inserted against intent
Low-power synthesis engineer Owns intent implementation in the netlist			
Low-power implementation lead UPF correctness and intent alignment			
Physical design liaison Retention and always-on physical implications			
Verification liaison Confirms the implementation is verifiable			
Library engineer Low-power cell availability and characterisation		DEPENDS ON	
		RTL-05 Clock, reset and power intent (UPF) implementation	
		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew	
		PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)	
		PDK-09 Foundry reference flow bring-up and internal methodology deck	

SYN-08

N2 netlist drop — timing and congestion converged, late RTL changes absorbed

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	12.0 M/M	~10.4 FTE	6	yes	Synthesis lead
w14–w19 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Release the **N2 drop with everything enabled** — full optimisation, power, low-power intent, physical awareness — and state honestly how much closure risk remains.

N2 is the last drop before the FFN and the last chance to change anything structurally. Its QoR is the best prediction available of what the final turn will achieve, and the closure risk statement it carries is what tells the program whether the tapeout date is credible.

WORK FLOW

6 STEPS · 2 LANES

w14	w15	w16	w17	w18	w19
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	RTL and ECO intake since N1	1w	main
2	Full synthesis with all optimisations enabled	1.5w	main
3	Late RTL change impact assessment	1w	parallel
4	Congestion and timing convergence check	1.5w	main
5	Physical design turn-1 feedback incorporation	1w	parallel
6	Release with a closure risk statement	1w	main

Step 6 is the honest part. A drop released with "timing is converging" tells nobody anything; one released with "forty paths in two blocks remain, all in the compute cluster, requiring a restructure or a frequency concession" gives the program a decision.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- RTL updates and ECOs since N1
- Turn 1 results and feedback from PD-05
- Power optimisation from SYN-06
- Low-power implementation from SYN-07
- Physical-aware results from SYN-04

PRODUCES

- N2 netlist with full optimisation
- Late change impact assessment
- Congestion and timing convergence analysis
- Turn-1 feedback incorporation record
- Closure risk statement

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-D2	PRODUCES	N1 and N2 netlist drops with QoR delta reports. The second quality drop, and the last structurally changeable one.
SYN-D5	FEEDS	Synthesis QoR report per drop. The N2 QoR is the best available prediction of what the final turn will reach.

RISKS AND WATCH-OUTS

- ◆ **Closure risk understated.** An optimistic risk statement at N2 removes the program's last opportunity to take a frequency or feature concession while it is still cheap.
- ◆ **Late RTL changes absorbed without assessment.** A functional change arriving between N1 and N2 can invalidate the turn-1 closure work entirely.
- ◆ **All optimisations enabled for the first time at N2.** Enabling power, low-power and physical awareness together for the first time produces a QoR nobody can attribute.
- ◆ **Turn-1 feedback not incorporated.** N2 then repeats N1's physical problems, and turn 2 spends its time on the same issues.
- ◆ **Treated as the final netlist.** N2 is not the FFN, and freezing behaviour around it removes the RTL freeze discipline that SYN-09 depends on.

EFFORT SPLIT	12 M/M	EXIT CRITERIA	FEEDS INTO
Full synthesis run	4.0	Closure risk stated specifically, with paths, blocks and options	SYN-09 FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point
Convergence check	3.0	Turn-1 feedback incorporated rather than deferred	SYN-11 Timing, area and power reporting against budget, per drop
Change impact assessment	2.5	Late changes assessed before inclusion	PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
Release and risk statement	2.5		S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
Total	12.0		
WHO IT TAKES	~10.4 FTE	DEPENDS ON	MEASURED BY
Synthesis lead Owns the drop and the closure risk statement		SYN-04 Physical-aware synthesis with congestion feedback from the PD flow setup	Closure risk at N2 against outcome at the final turn
Synthesis engineers Full-optimisation run and convergence analysis		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback	Late changes assessed against absorbed
Physical design lead Turn-1 feedback and turn-2 acceptance		SYN-06 Power optimisation — clock gating, operand isolation, Vt mix	QoR delta from N1
RTL lead Late change control and impact		SYN-07 Low-power synthesis — isolation and level-shifter insertion against UPF	
Program manager Consumes the risk statement for schedule decisions		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	
ENTRY CRITERIA			
- Turn 1 complete with feedback from PD-05 - Power and low-power optimisation available from SYN-06 and SYN-07 - RTL changes since N1 controlled through RTL-10			

SYN-09

FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	10.0 M/M	~10.9 FTE	6	yes	Synthesis lead
w19–w23 in stage	planned	while it runs	2 in parallel	no float	Physical design lead approves

PURPOSE

Release the **FFN — the final full netlist** — against frozen RTL, with constraints frozen and equivalence proved, and declare that nothing functional changes after it.

The FFN is a commitment more than a deliverable. Everything after it is physical: PD-13 closes on this netlist and S0-02 signs off on what PD produces. A functional change after the FFN restarts both, which is why the freeze around it has to be real.

WORK FLOW

6 STEPS · 2 LANES

w19	w20	w21	w22	w23
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	RTL freeze confirmation and final intake	0.8w	main
2	Final synthesis run	1.3w	main
3	Final constraint freeze	1w	parallel
4	Final equivalence and UPF consistency verification	1w	main
5	Handoff package assembly	0.8w	parallel
6	FFN release and functional freeze declaration	1w	main

Step 1 is a confirmation rather than a formality. If `RTL-10` has not actually declared freeze, the FFN is a snapshot of a moving design, and the final turn in `PD-13` will be closing against something that changes underneath it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Frozen RTL from RTL-10
- Constraints from SYN-01
- Optimisation configuration from SYN-06 and SYN-07
- N2 results and closure risk from SYN-08
- Equivalence methodology from SYN-10

PRODUCES

- Final synthesis run and netlist
- Frozen constraint set
- Final equivalence and UPF consistency results
- Handoff package for the final turn
- FFN release with functional freeze declaration

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-D3 **PRODUCES** **FFN — final full netlist, release-tagged.** This activity is the deliverable, and the milestone the stage ends on.

SYN-D4 **GATES** **Validated SDC constraint set.** Constraints freeze with the FFN — closing against moving constraints is closing against nothing.

SYN-D8 **FEEDS** **Physical design handoff package per drop.** The final handoff, and the one PD-13 closes against.

RISKS AND WATCH-OUTS

- ◆ **FFN released against RTL that is not actually frozen.** The final turn then closes against a moving netlist, and every change restarts it.
- ◆ **Constraints still changing after the FFN.** A design closed against one constraint set and signed off against another has not been signed off.
- ◆ **Equivalence not proved on the final netlist.** The last synthesis run is the one most likely to contain a transformation error, and the least likely to be checked under schedule pressure.
- ◆ **Functional freeze declared and not enforced.** Post-FFN functional ECOs are sometimes unavoidable and must be exceptional, visible and costed.
- ◆ **Handoff package incomplete.** The final turn needs constraints, UPF, DFT collateral and abstracts together; a missing item stops closure at its start.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Final synthesis run	3.5	● RTL freeze declared by RTL-10	PD-13 Final turn on the FFN — full closure, no functional change admitted
Equivalence and consistency	2.5	● N2 closure risk understood from SYN-08	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Release and freeze declaration	2.0	● Equivalence methodology proven across earlier drops	S0-09 Final formal equivalence and LVS netlist consistency
Constraint freeze	1.0		T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
Handoff package	1.0		
Total	10.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~10.9 FTE	● Equivalence proved on the actual FFN, not on a proxy ● Constraints frozen with the netlist ● Functional freeze declared and its exception process published	● Functional changes after the FFN ● Equivalence result on the final netlist ● Handoff completeness at the final turn's start
Synthesis lead Owns the FFN and the freeze declaration		DEPENDS ON	
Constraints lead Constraint freeze		SYN-08 N2 netlist drop — timing and congestion converged, late RTL changes absorbed	
Equivalence engineer Final LEC and consistency		SYN-10 Formal equivalence checking per drop, RTL versus netlist	
RTL lead Confirms RTL freeze is real		RTL-10 Specification change control and ECO board	
Physical design lead Accepts the handoff for the final turn			

SYN-10

Formal equivalence checking per drop, RTL versus netlist

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
20w cont. w4–w24 in stage, runs throughout	12.0 M/M planned	~2.6 FTE while it runs	8 3 in parallel	no has float	Equivalence engineer Synthesis lead approves

PURPOSE

Prove that **every netlist still means what the RTL meant** — formal equivalence per drop, including through DFT insertion, clock gating and low-power transformations.

Synthesis rewrites the design, and the tools that do it are not infallible. Equivalence checking is the only evidence that the netlist implements the RTL that verification actually verified; without it, a transformation error is discovered in silicon as behaviour nobody can trace to a bug.

WORK FLOW

8 STEPS · 2 LANES

w4	w8	w12	w16	w20	w24
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Equivalence methodology and tool setup	1.5w	main
2	Block-level equivalence per drop	2w	main
3	Black-box and constraint handling for incomplete drops	2w	parallel
4	Chip-level equivalence per drop	3w	main
5	DFT and clock gating equivalence handling	2w	parallel
6	Non-equivalence debug and resolution	2.5w	main
7	Per-drop equivalence reporting	2w	parallel
8	Continuous checking across the drop sequence	11w	main

Step 5 is where most of the difficulty lives. Scan insertion and clock gating change the netlist in ways that are correct and not structurally equivalent, and setting up the checker to accept those and reject everything else is the skilled part of the activity.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- RTL from RTL-02 and RTL-04
- Netlists from every synthesis drop
- Scan-inserted netlist from DFT-06
- UPF and low-power implementation from SYN-07
- Equivalence tools from PDK-08

PRODUCES

- Equivalence methodology and setup
- Block and chip-level equivalence results per drop
- DFT and clock gating handling configuration
- Non-equivalence debug records
- Per-drop equivalence reports

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-06

PRODUCES

Formal equivalence clean report per drop. This activity is the deliverable, and "per drop" is what makes it a control rather than a final check.

SYN-03

GATES

FFN — final full netlist. An FFN that has not been proved equivalent is a netlist nobody can claim was verified.

RISKS AND WATCH-OUTS

- ◆ **Equivalence run only on the final netlist.** A transformation error introduced at N1 is then found at the FFN, when the debug competes with tapeout.
- ◆ **Non-equivalences waived to make progress.** Each waiver is a place where the netlist may not implement the verified RTL, and they accumulate quietly.
- ◆ **DFT transformations not modelled.** The checker reports thousands of differences and everyone stops reading the report.
- ◆ **Black-box handling too permissive.** Boxing out a block to get a clean result excludes exactly the logic that was not checked.
- ◆ **Debug capacity underestimated.** A single real non-equivalence on a design this size can take a week to localise, and there is rarely a week available.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous checking	4.0	● RTL and first netlists available	SYN-09FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point
Non-equivalence debug	3.0	● Equivalence tools qualified by PDK-08	
Chip-level equivalence	2.0	● DFT insertion methodology known from DFT-06	S0-09Final formal equivalence and LVS netlist consistency
Methodology and setup	1.5		T0-02Final full-chip verification re-run on the released database
DFT and low-power handling	1.5		
Total	12.0		
WHO IT TAKES	~2.6 FTE	EXIT CRITERIA	MEASURED BY
Equivalence engineer Owns methodology, runs and debug		● Every drop checked, not only the final one	● Drops with a clean equivalence result
Synthesis engineers Transformation explanations and fixes		● Non-equivalences resolved rather than waived	● Non-equivalences waived against resolved
DFT engineer Scan transformation handling		● Chip-level equivalence clean on the FFN	● Debug time per non-equivalence
Low-power engineer UPF-related transformation handling			
Synthesis lead Waiver decisions and reporting		DEPENDS ON	
		SYN-02Technology mapping and optimisation — multi-Vt, multi-bit, useful skew	
		SYN-07Low-power synthesis — isolation and level-shifter insertion against UPF	
		DFT-06Scan insertion and DFT DRC during synthesis	
		PDK-08EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	

11 SYN · LOGIC SYNTHESIS & NETLIST DROPS

SYN-11

Timing, area and power reporting against budget, per drop

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
22w cont. w2–w24 in stage, runs throughout	10.0 M/M planned	~2.0 FTE while it runs	6 2 in parallel	no has float	PPA reporting engineer Program manager approves

PURPOSE

Report **timing, area and power against the budget after every drop**, and escalate when a block is diverging rather than when it has already blown its allocation.

QoR reported in absolute terms tells nobody anything. Reported against the ARCH-06 allocation and trended across drops, it says which blocks are converging, which are not, and how much of the gap is still recoverable — which is the only version a program manager can act on.

WORK FLOW

6 STEPS · 2 LANES

w2w7w12w17w22w24

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Reporting framework and budget ingestion	1.5w	main
2	Per-drop timing report and analysis	2w	main
3	Budget comparison and escalation rules	2w	parallel
4	Per-drop area and power report	2w	main
5	Trend analysis across the drop sequence	2w	parallel
6	Continuous reporting across the stage	16.5w	main

Step 5 is the part that predicts. A block 10% over budget and improving is a different problem from one 5% over and flat, and only a trend across drops distinguishes them.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Synthesis results from every drop
- PPA budgets from ARCH-06
- Constraints from SYN-01
- Power results from SYN-06
- Physical-aware results from SYN-04

PRODUCES

- Reporting framework with budget ingestion
- Per-drop timing, area and power reports
- Budget comparison per block
- Trend analysis across drops
- Escalation records

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-05

PRODUCES

Synthesis QoR report per drop against PPA targets. This activity is the deliverable, and "against targets" is what makes it usable.

SYN-02

FEEDS

N1 and N2 netlist drops with QoR delta reports. The delta report accompanying each drop comes from here.

RISKS AND WATCH-OUTS

- ◆ **QoR reported in absolute terms.** Numbers with no budget reference cannot be acted on, and everyone forms their own opinion of whether they are acceptable.
- ◆ **No escalation rule.** Divergence noticed and not escalated becomes a surprise at the final turn, when there is nothing left to do about it.
- ◆ **Reports not normalised across drops.** More RTL in a later drop makes the trend meaningless unless the comparison accounts for it.
- ◆ **Power reported at nominal only.** The number that matters is at the product's operating point, and reporting at nominal understates it consistently.
- ◆ **Reporting stopping between drops.** The interesting question is whether a block is improving, and that needs data between releases as well as at them.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous reporting	4.0	PPA budgets available from ARCH-06	SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback
Per-drop reports	2.5	First synthesis results available	
Budget comparison	2.0	Reporting framework agreed with the PPA lead	SYN-08 N2 netlist drop — timing and congestion converged, late RTL changes absorbed
Trend analysis	1.5		
Total	10.0	EXIT CRITERIA	PD-09 Multi-corner multi-mode timing closure
WHO IT TAKES	~2.0 FTE	- Every report references the budget, not only the absolute number - Divergence escalated on a rule, not on judgement - Trends normalised so drops are comparable	ARCH-06 PPA budget allocation per block
PPA reporting engineer Owns the framework and the reports			MEASURED BY
Synthesis engineers Data generation per drop			Blocks reported against budget
PPA lead Budget comparison and escalation decisions		DEPENDS ON	Escalations raised before the final turn
Block owners Act on their block's divergence		SYN-01 SDC constraint development and validation	Trend accuracy in predicting final-turn QoR
Program manager Consumes trends for schedule decisions		SYN-02 Technology mapping and optimisation — multi-Vt, multi-bit, useful skew	
		ARCH-06 PPA budget allocation per block	
		PDK-12 Compute farm, EDA license and storage capacity planning	

SYN-12

Netlist handoff and QoR delta review with physical design, per drop

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
22w cont. w2–w24 in stage, runs throughout	6.0 M/M planned	~1.2 FTE while it runs	6 2 in parallel	no has float	Synthesis lead PPA lead approves

PURPOSE

Hand each netlist to physical design as a **complete, accepted package** — netlist, constraints, UPF, DFT collateral, abstracts — and review the QoR delta together rather than throwing results over a wall.

The handoff is where two teams either share a picture of the design or maintain two. Six man-months across the stage buys a structured package and a joint review per drop, and it prevents the pattern where physical design spends a week discovering what synthesis already knew.

WORK FLOW

6 STEPS · 2 LANES

w2	w7	w12	w17	w22	w24
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Handoff package definition — contents and format	1w	main
2	Per-drop package assembly	1.5w	main
3	Acceptance criteria agreed with physical design	1.5w	parallel
4	QoR delta review with physical design per drop	1.5w	main
5	Issue tracking between drops	1.5w	parallel
6	Continuous handoff across the drop sequence	18w	main

Step 3 is what makes a handoff a transaction rather than a delivery. Acceptance criteria mean physical design can reject a package that is incomplete, and rejection at handoff is far cheaper than discovering the gap three days into a turn.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Netlists from every drop
- Constraints from SYN-01
- UPF and low-power implementation from SYN-07
- DFT collateral from DFT-06
- QoR reports from SYN-11

PRODUCES

- Handoff package definition and format
- Assembled packages per drop
- Acceptance criteria agreed with PD
- QoR delta review records
- Issue tracking between drops

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SYN-D8

PRODUCES

Physical design handoff package per drop. This activity is the deliverable, and acceptance rather than delivery is its standard.

SYN-D2

FEEDS

N1 and N2 netlist drops with QoR delta reports. The delta review is where the drop is actually transferred.

RISKS AND WATCH-OUTS

- ◆ **Packages delivered without acceptance.** Physical design discovers the gap mid-turn, and the turn absorbs the delay.
- ◆ **QoR reviewed asynchronously.** A report read separately produces two interpretations, and the disagreement surfaces at the next escalation.
- ◆ **Issues not tracked between drops.** Each handoff then starts fresh, and the same problems are raised at every drop.
- ◆ **Package contents changing between drops.** Physical design automates against the format, and an inconsistent package breaks their flow.
- ◆ **Handoff treated as a synthesis milestone.** It is a shared event; a handoff that PD has not accepted has not happened.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous handoff	2.5	● First netlist drop ready in SYN-03	PD-01 Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity
QoR delta review	1.5	● Acceptance criteria negotiable with physical design	
Package assembly	1.0	● Package format agreed	PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Acceptance criteria	1.0		
Total	6.0	EXIT CRITERIA	
WHO IT TAKES	~1.2 FTE	● Every package accepted rather than merely delivered ● QoR reviewed jointly per drop ● Issues tracked across drops rather than re-raised	PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
Synthesis lead Owns the handoff and the delta review			PD-13 Final turn on the FFN — full closure, no functional change admitted
Physical design lead Acceptance and feedback			
Configuration manager Package assembly and version control		DEPENDS ON	MEASURED BY
DFT engineer DFT collateral in the package		SYN-03 N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow	● Packages accepted at first presentation ● Issues carried between drops against re-raised
PPA lead Joint QoR interpretation		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback	
		SYN-11 Timing, area and power reporting against budget, per drop	● Days lost to incomplete handoffs

IMPLEMENT

12 Physical Design

PD

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w46 – w76	30 _w	16	449 _{M/M}	8 _{of 16}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
PD-01	Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity		7w 22 M/M	PD-D1 Flow setup release — MMMC environment, scripts and runtime baseline
PD-02	Floorplan, macro placement and partition definition		9w 32 M/M	PD-D2 Turn 1 and Turn 2 databases with QoR delta reports
PD-03	Power delivery network design and early IR analysis		8w 20 M/M	PD-D3 Final-turn routed database on the FFN, per block and top
PD-04	Bump and RDL planning with chip-package co-design feedback		8w 14 M/M	PD-D4 Floorplan and PDN specification
PD-05	Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established		8w 45 M/M	PD-D5 MCMM timing closure reports with violation burn-down across turns
PD-06	Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified		8w 45 M/M	PD-D6 Interim physical DRC / LVS clean
PD-07	Clock tree synthesis with skew and jitter budgeting, refined across turns		12w cont. 26 M/M	PD-D7 Bump map, RDL and package interface files
PD-08	Detailed routing and DRC convergence		16w cont. 40 M/M	PD-D8 ECO log and change control record
PD-09	Multi-corner multi-mode timing closure		22w cont. 60 M/M	PD-D9 Signoff-ready database handoff
PD-10	Signal and power integrity iteration — crosstalk, EM		10w 18 M/M	
PD-11	Chip power model (CPM/CPS) extraction and handoff to package co-verification		5w 6 M/M	
PD-12	Scan chain reordering and DFT-aware routing		6w 8 M/M	
PD-13	Final turn on the FFN — full closure, no functional change admitted		11w 55 M/M	
PD-14	Functional and timing ECO implementation		18w cont. 22 M/M	
PD-15	Chip finishing — seal ring, dummy and metal fill, alignment marks		5w 6 M/M	

PD-16	Hierarchical block closure and top-level assembly	14w	30 M/M
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PD-01

Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	22.0 M/M	~13.7 FTE	6	yes	Physical design flow lead
w0–w7 in stage	planned	while it runs	2 in parallel	no float	Synthesis liaison approves

PURPOSE

Build the flow — **scripts, MMMC environment, decks, runtimes, capacity** — and debug it on the No flow-flush netlist rather than on the netlist the schedule depends on.

This is why SYN-03 releases a netlist nobody expects to close. Every flow problem — a deck that does not load, a scenario that does not converge, a runtime that exceeds the night — is found here on a throwaway database, weeks before the same problem would have cost a turn.

WORK FLOW

6 STEPS · 2 LANES

w0	w2	w4	w6	w7
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Flow scope definition and script framework	1.5w	main
2	MMMC environment build across corners and modes	2w	main
3	Script, deck and library integration	1.5w	parallel
4	N0 flow-flush run, end to end	2w	main
5	Hierarchical and block-level flow definition	1.5w	parallel
6	Runtime baseline and capacity assessment	1.5w	main

Step 6 is what makes the schedule credible. A flow that runs correctly and takes forty hours per iteration sets a closure cadence nobody planned for, and knowing that number now is what lets `PKD-12` buy capacity before it is needed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- N0 netlist from SYN-03
- Golden environment and flow from PDK-13
- Constraints from SYN-01
- Corner and mode definition from PDK-11
- Compute and licence capacity from PDK-12

PRODUCES

- Physical design script framework
- MMMC environment across corners and modes
- Integrated decks and libraries
- N0 end-to-end run results
- Hierarchical flow definition
- Runtime baseline and capacity assessment

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D1

PRODUCES

Flow setup release — MMMC environment, scripts and runtime baseline. This activity is the deliverable, and the runtime baseline is the part the schedule depends on.

PD-D5

FEEDS

MCMM timing closure reports. The scenario set built here is what every later closure report is generated from.

RISKS AND WATCH-OUTS

- ◆ **Flow setup on the first real netlist.** Every flow problem then costs a turn, and the turns are the schedule.
- ◆ **MMMC scenarios incomplete.** A corner or mode missing from the setup is one the design is never closed at, and it is found at S0-02.
- ◆ **Runtime not measured.** A flow that works and takes two days per iteration produces a closure schedule nobody planned, discovered when closure starts.
- ◆ **Hierarchical flow deferred.** A design this size cannot be closed flat; deferring the block-level flow means building it under pressure.
- ◆ **Capacity assumed rather than booked.** The flow's peak concurrency is known here, and licences take weeks to procure.

EFFORT SPLIT	22 M/M	ENTRY CRITERIA	FEEDS INTO
MMMC environment build	6.0	N0 netlist available from SYN-03	PD-02 Floorplan, macro placement and partition definition
N0 flow-flush run	5.0	Golden environment released by PDK-13	PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Flow scope and scripts	4.0	Corner and mode definition final from PDK-11	SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback
Deck and library integration	3.5		PDK-12 Compute farm, EDA license and storage capacity planning
Hierarchical flow definition	2.0		
Runtime baseline	1.5		
Total	22.0		
WHO IT TAKES	~13.7 FTE	EXIT CRITERIA	MEASURED BY
Physical design flow lead Owns the flow, its scripts and its runtime		- Flow runs end to end on N0, not on a fragment - Runtime baseline measured and fed to capacity planning - Hierarchical and block-level flows both defined	- Flow steps running end to end on N0 - Runtime per iteration against the closure plan - Flow problems found on N0 against on N1
CAD methodology engineer Deck, library and environment integration			
Timing methodology engineer MMMC scenario construction			
CAD infrastructure Runtime, capacity and licence planning			
Synthesis liaison N0 content and flow feedback			
		SYN-03 N0 flow-flush netlist release — functionally incomplete, structurally representative, issued so PD can build the flow	
		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
		PDK-12 Compute farm, EDA license and storage capacity planning	
		PDK-13 Golden design environment release and flow regression	

12 PD · PHYSICAL DESIGN

PD-02

Floorplan, macro placement and partition definition

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w w5–w14 in stage	32.0 M/M planned	~15.4 FTE while it runs	6 2 in parallel	yes no float	Floorplan owner Synthesis liaison approves

PURPOSE

Decide **where everything physically goes** — partitions, macros, block boundaries, pins, utilisation — and freeze it, because everything downstream is built on it.

The floorplan is the most consequential and least reversible decision in the stage. Macro placement sets routing congestion, block boundaries set what can be closed independently, and pin placement sets the top-level routing burden. A floorplan changed at turn 2 invalidates the clock trees, the power grid and the routing built on it.

WORK FLOW

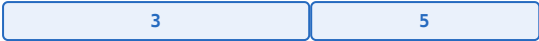
6 STEPS · 2 LANES

w5w7w9w11w13w14

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Partition definition against the design hierarchy	2w	main
2	Macro placement — memories, PHYs, analog	2.5w	main
3	Standard cell area planning and utilisation targets	2w	parallel
4	Block boundary and pin planning	2w	main
5	Keep-out, blockage and halo definition	1.5w	parallel
6	Floorplan review and freeze	2.5w	main

Step 2 is where a memory-dominated die is won or lost. Hundreds of macro instances have to be placed so their access paths are short, their power can be delivered and the logic between them can be routed — and moving one late moves everything around it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Floorplan intent and bump map from ARCH-08
- Macro abstracts from AMS-15 and PDK-04
- Partitioning from ARCH-02
- Netlist from SYN-05
- Area budgets from ARCH-06

PRODUCES

- Partition definition
- Macro placement
- Area and utilisation plan
- Block boundaries and pin placement
- Keep-out and blockage definition
- Frozen floorplan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D4

PRODUCES

Floorplan and PDN specification. The floorplan half; PD-03 supplies the power grid it is built against.

PD-D2

FEEDS

Turn 1 and Turn 2 databases. Every turn is placed and routed inside this floorplan; changing it invalidates the turns.

RISKS AND WATCH-OUTS

- ◆ **Macro placement not congestion-aware.** Macros placed for access and not for the logic between them produce routing congestion that cannot be fixed at PD-08.
- ◆ **Floorplan changed after turn 1.** Clock trees, power grid and routing are all built on it, and all of them are invalidated.
- ◆ **Utilisation set too high.** A dense floorplan closes area and cannot be routed or timing-closed, and the discovery comes after weeks of work.
- ◆ **Pin placement ignoring top-level routing.** Block pins facing the wrong way make top-level routing expensive for the whole rest of the program.
- ◆ **Partitions that do not match the design hierarchy.** Closing blocks that cut across logical boundaries makes timing budgets and equivalence checking harder for no gain.

EFFORT SPLIT	32 M/M	ENTRY CRITERIA	FEEDS INTO
Macro placement	9.0	- Floorplan intent and bump map from ARCH-08 - Macro abstracts available from AMS-15 - Flow running from PD-01	PD-03 Power delivery network design and early IR analysis
Partition definition	7.0		PD-04 Bump and RDL planning with chip-package co-design feedback
Block boundary and pin planning	6.0		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established
Area and utilisation planning	5.0	EXIT CRITERIA - Floorplan frozen before turn 1 starts - Macro placement validated against congestion, not only access - Pin placement reviewed for top-level routing cost	PD-16 Hierarchical block closure and top-level assembly
Review and freeze	3.0		SYN-04 Physical-aware synthesis with congestion feedback from the PD flow setup
Blockages and halos	2.0		DFT-09 Scan compression and chain routing feasibility with physical design
Total	32.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~15.4 FTE	PD-01 Flow setup and MMMC environment build on the NO flow-flush netlist — scripts, decks, runtimes, capacity	- Floorplan changes after freeze - Congestion at turn 1 against prediction - Utilisation achieved against target
Floorplan owner Owns partitions, macro placement and the freeze		ARCH-08 Chip-level floorplan intent, pin and bump budget	
Physical design engineers Area planning, pins and blockages		AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)	
Architecture liaison Confirms the floorplan matches the intent		PDK-04 Memory compiler evaluation and instance generation (SRAM, register file)	
Package liaison Bump map alignment with the floorplan		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback	
Synthesis liaison Congestion feedback into physical-aware synthesis			

12 PD · PHYSICAL DESIGN

PD-03

Power delivery network design and early IR analysis

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	20.0 M/M	~10.9 FTE	6	no	Power delivery engineer
w6-w14 in stage	planned	while it runs	2 in parallel	has float	Floorplan owner approves

PURPOSE

Build the **power delivery network** — grid topology, straps, switches, decap — and prove by early IR analysis that it can actually feed the die at its power envelope.

The PDN is built early because everything routes around it. Grid density trades directly against routing resource: too sparse and IR drop eats timing margin, too dense and there is nothing left to route with. Getting it wrong is discovered at S0-04, when the fix is a metal change across the whole die.

WORK FLOW

6 STEPS · 2 LANES

w6w8w10w12w14

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	PDN topology and grid definition per domain	1.5w	main
2	Grid implementation and strap insertion	2w	main
3	Power switch and always-on region planning	1.5w	parallel
4	Early static IR analysis	2w	main
5	Decap planning and placement	1.5w	parallel
6	PDN refinement and freeze	2.5w	main

Step 4 has to happen while the grid can still change. Static IR on a placed design with real power numbers tells you whether the grid works; running it after routing means any fix competes with a routed design for the same metal.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Power budget per domain from ARCH-06
- Power domain definition from ARCH-05 and SYN-07
- Floorplan from PD-02
- Bump and rail assignment from PD-04
- Power estimates from SYN-06

PRODUCES

- PDN topology and grid definition
- Implemented grid and straps
- Power switch and always-on plan
- Early static IR analysis results
- Decap plan
- Frozen PDN specification

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D4	PRODUCES	Floorplan and PDN specification. The power grid half of the deliverable, frozen alongside the floorplan it sits in.
PD-D5	FEEDS	MCMM timing closure reports. IR drop consumes timing margin, so the PDN's quality shows up in closure.

RISKS AND WATCH-OUTS

- ◆ **Grid density traded against routing without analysis.** Both constraints are real, and choosing by feel produces either IR problems or congestion.
- ◆ **IR analysed only statically.** Dynamic IR under real switching is worse, and SIPI-04 with the package model is worse again.
- ◆ **Decap placed where there is room.** Decoupling works where the current is drawn, not where the floorplan left space.
- ◆ **Power switches planned late.** They need area, always-on routing and their own control, and retrofitting them disturbs the floorplan.
- ◆ **PDN frozen before power numbers are real.** A grid designed against estimates and fed a design that draws more is a grid that fails at signoff.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Grid implementation	6.0	Floorplan available from PD-02	PD-08 Detailed routing and DRC convergence
Early IR analysis	5.0	Power budgets per domain from ARCH-06	PD-10 Signal and power integrity iteration — crosstalk, EM
Topology definition	3.5	Bump and rail assignment progressing in PD-04	PD-11 Chip power model (CPM/CPS) extraction and handoff to package co-verification
Decap planning	2.5		SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
Power switch planning	2.0		S0-04 EM and IR-drop signoff, static and dynamic
Refinement and freeze	1.0		
Total	20.0		
WHO IT TAKES	~10.9 FTE	EXIT CRITERIA	MEASURED BY
Power delivery engineer Owns the PDN and its analysis		- Static IR analysed on a placed design with real power numbers - Grid density justified against routing resource - PDN frozen with the floorplan, not after it	- Static IR against budget - Routing resource consumed by the grid - PDN changes after freeze
Physical design engineers Grid implementation and decap placement		DEPENDS ON	
Power architect Domain topology and switch strategy		PD-02 Floorplan, macro placement and partition definition	
SI/PI liaison Hands the grid to chip-package co-analysis		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
Floorplan owner Area for switches, straps and decap		ARCH-06 PPA budget allocation per block	
		SYN-07 Low-power synthesis — isolation and level-shifter insertion against UPF	

12 PD · PHYSICAL DESIGN

PD-04

Bump and RDL planning with chip-package co-design feedback

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w6-w14 in stage	14.0 M/M planned	~7.6 FTE while it runs	6 2 in parallel	no has float	Bump and RDL owner SI liaison approves

PURPOSE

Turn the architecture's bump budget into **an actual bump map and RDL**, iterating with package design until both sides can build what the other needs.

The bump map is the contract between the die and the package, and it is negotiated rather than declared. Signal escape on the package constrains bump position; power delivery constrains bump count; and the die's floorplan constrains where either can go. All three have to close at once.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Bump map ingestion from the architecture budget	1.5w	main
2	RDL and redistribution planning	2w	main
3	Power bump and rail assignment	1.5w	parallel
4	Package co-design iteration	2w	main
5	Signal escape planning with the substrate	1.5w	parallel
6	Bump map freeze and interface file release	2.5w	main

Step 6 is a hard date rather than a milestone. `PKGD-11` freezes package design against this map, and the substrate order that follows has a twenty-week lead time — a late bump map moves wafer-out by moving the substrate.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Bump budget and map intent from ARCH-08
- Package architecture from PKGD-01
- Floorplan and IO ring from PD-02
- Power delivery requirements from PD-03
- Substrate escape constraints from PKGD-04

PRODUCES

- Bump map with signal and power assignment
- RDL and redistribution plan
- Power bump and rail assignment
- Package co-design iteration record
- Frozen bump map and package interface files

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D7	PRODUCES	Bump map, RDL and package interface files. This activity is the deliverable, and it is consumed directly by package design.
PD-D4	FEEDS	Floorplan and PDN specification. Bump position constrains where power can be delivered, so the two documents have to agree.

RISKS AND WATCH-OUTS

- ◆ **Bump map frozen late.** Package design and the substrate order both wait on it, and the substrate lead time makes the delay a wafer-out delay.
- ◆ **Escape routing not checked on the package side.** A map that is optimal on the die and unroutable on the substrate has to be redone.
- ◆ **Power bump count reduced to fit signals.** The signals that fit are then fed by a PDN that cannot deliver, and SIPI-04 finds it.
- ◆ **RDL treated as free.** Redistribution has its own rules and resistance, and a plan that ignores them produces IR and EM problems in the RDL itself.
- ◆ **Co-design conducted by file exchange.** Two teams iterating asynchronously on the same interface converge slowly or not at all.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Package co-design iteration	4.0	● Bump budget available from ARCH-08 ● Package architecture selected in PKGD-01 ● Floorplan and IO ring available from PD-02	PKGD-02
RDL planning	3.5		PKGD-11
Bump map ingestion and assignment	2.5		
Power bump assignment	2.0		SIPI-02
Escape planning	2.0		SIPI-04
Total	14.0	EXIT CRITERIA	
WHO IT TAKES	~7.6 FTE	● Bump map frozen on the date package design needs it ● Escape routing confirmed feasible on the substrate ● Power bump count sufficient for the delivery requirement	AMS-03
Bump and RDL owner Owns the map and the interface files			MEASURED BY
Package designer Escape routing and substrate constraints			
Power delivery engineer Power bump count and rail assignment			
Floorplan owner Die-side placement constraints			
SI liaison Signal assignment for channel quality			
		DEPENDS ON	
		PD-02	● Bump map freeze against the package design need date ● Escape feasibility confirmed before freeze ● Power bumps against the delivery requirement
		PD-03	
		ARCH-08	
		PKGD-01	

PD-05

Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w8–w16 in stage	45.0 M/M planned	~24.4 FTE while it runs	8 3 in parallel	yes no float	Physical design turn lead Synthesis liaison approves

PURPOSE

Run the **first real turn on the N1 netlist** — place, build clocks, route — and establish the QoR baseline that says how far the design is from closing.

Turn 1 is the first honest measurement of the design's physical difficulty. Everything before it was estimate; this produces real congestion, real timing and real power on a real floorplan, and the gap it reveals is what the next two turns exist to close.

WORK FLOW

8 STEPS · 2 LANES

w8	w10	w12	w14	w16
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	N1 intake and turn setup	1w	main
2	Placement and optimisation	2w	main
3	Congestion analysis and mitigation	1.5w	parallel
4	Clock tree synthesis, first pass	2w	main
5	Timing analysis and first ECO round	1.5w	parallel
6	First full route	2w	main
7	Power analysis on the placed and routed design	1.5w	parallel
8	QoR baseline and feedback to synthesis	1w	main

Step 8 is what makes turn 1 worth its cost. The turn's purpose is not to close — it will not — but to produce a measured baseline and a specific list of what synthesis and RTL must change before turn 2.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- N1 netlist and handoff package from SYN-05 and SYN-12
- Frozen floorplan from PD-02
- PDN from PD-03
- Flow and MMMC from PD-01
- Constraints from SYN-01

PRODUCES

- Turn 1 placed and routed database
- Congestion analysis and mitigation results
- First-pass clock trees
- Timing and power analysis results
- QoR baseline

- Feedback list to synthesis and RTL

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D2	PRODUCES	Turn 1 and Turn 2 databases with QoR delta reports. The first turn, and the baseline the second is measured against.
PD-D5	FEEDS	MCMM timing closure reports. The violation burn-down starts here, with turn 1's numbers as its origin.

RISKS AND WATCH-OUTS

- ◆ **Turn 1 expected to close.** It will not, and treating the gap as a crisis rather than as data produces panic instead of a plan.
- ◆ **Feedback not specific.** "Congestion is high" changes nothing; "the compute cluster is over 90% utilisation in three regions and needs restructuring" changes the next drop.
- ◆ **Turn run on an incomplete netlist.** A baseline missing blocks understates the difficulty and misleads the closure plan.
- ◆ **Congestion mitigated locally.** Fixing congestion by rip-up and reroute hides a structural problem that only synthesis or floorplan can solve.
- ◆ **Clock tree built before the floorplan is stable.** A CTS on a floorplan that then changes is work thrown away.

EFFORT SPLIT	45 M/M	ENTRY CRITERIA	FEEDS INTO
Placement and optimisation	11.0	N1 handoff accepted from SYN-12	PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
First full route	10.0	Floorplan frozen in PD-02	PD-07 Clock tree synthesis with skew and jitter budgeting, refined across turns
Clock tree synthesis	8.0	PDN implemented in PD-03	PD-09 Multi-corner multi-mode timing closure
Timing analysis and ECO	7.0	EXIT CRITERIA	SYN-08 N2 netlist drop — timing and congestion converged, late RTL changes absorbed
Congestion analysis	5.0	QoR baseline measured on a substantially complete netlist	
QoR baseline and feedback	4.0	Feedback to synthesis specific enough to act on	
Total	45.0	Congestion characterised as structural or local	MEASURED BY - QoR gap at turn 1 against budget - Feedback items acted on in N2 - Netlist completeness at the baseline
WHO IT TAKES	~24.4 FTE	DEPENDS ON	
Physical design turn lead Owns the turn and its QoR baseline		PD-01 Flow setup and MMMC environment build on the N0 flow-flush netlist — scripts, decks, runtimes, capacity	
Placement and routing engineers Turn execution		PD-02 Floorplan, macro placement and partition definition	
Clock tree engineer First-pass CTS		PD-03 Power delivery network design and early IR analysis	
Timing engineer Analysis and first ECO round		SYN-05 N1 netlist drop — first quality turn against PD placement and congestion feedback	
Synthesis liaison Receives and acts on the feedback		SYN-12 Netlist handoff and QoR delta review with physical design, per drop	

12 PD · PHYSICAL DESIGN

PD-06

Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w14–w22 in stage	45.0 M/M planned	~24.4 FTE while it runs	8 3 in parallel	yes no float	Physical design turn lead Program manager approves

PURPOSE

Run the **second turn on N2 and converge** — congestion, timing and power — then quantify honestly how much closure risk remains for the final turn.

Turn 2 is where the design either starts converging or reveals that it will not. Its output is a number: how many violations remain, in which blocks, and whether the trend from turn 1 projects to closure. That number is what the program uses to decide whether the tapeout date holds.

WORK FLOW

8 STEPS · 2 LANES

w14	w16	w18	w20	w22
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	N2 intake and delta assessment against turn 1	1w	main
2	Placement with congestion mitigation	2w	main
3	Congestion hot-spot restructuring with synthesis	1.5w	parallel
4	Clock tree refinement	2w	main
5	Timing convergence and ECO rounds	1.5w	parallel
6	Route convergence	2w	main
7	Power and IR convergence	1.5w	parallel
8	Closure risk quantification for the final turn	1w	main

Step 3 is joint work with <code>SYN-04</code> rather than a physical design task. Congestion that survives placement optimisation is structural, and the only remedies are logic restructuring or a floorplan change — both of which belong to someone else.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- N2 netlist and handoff from SYN-08 and SYN-12
- Turn 1 database and QoR from PD-05
- Congestion feedback loop with SYN-04
- Clock trees from PD-07

PRODUCES

- Turn 2 placed and routed database
- Congestion mitigation and restructuring results
- Refined clock trees
- Timing and power convergence results

- Timing closure status from PD-09
- Closure risk quantification

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D2	PRODUCES	Turn 1 and Turn 2 databases with QoR delta reports. The second turn, and the last one before the netlist is frozen.
PD-D5	FEEDS	MCMM timing closure reports. The burn-down trend across turns is what predicts whether the final turn closes.

RISKS AND WATCH-OUTS

- ◆ **Closure risk understated at turn 2.** This is the last point at which a frequency or feature concession is cheap, and an optimistic number removes the option.
- ◆ **Congestion fixed locally rather than structurally.** Rip-up and reroute buys a turn and leaves the problem for the final one, when there is no time.
- ◆ **Turn 2 not compared against turn 1.** The absolute number matters less than the trend, and only the delta says whether the design is converging.
- ◆ **Power convergence deferred.** Power fixes change timing; leaving them to the final turn means re-closing timing after every one.
- ◆ **Late RTL changes absorbed into turn 2.** A functional change mid-turn invalidates the comparison the turn exists to produce.

EFFORT SPLIT	45 M/M	ENTRY CRITERIA	FEEDS INTO
Placement and congestion mitigation	11.0	- N2 handoff accepted from SYN-12 - Turn 1 QoR baseline available from PD-05 - Congestion feedback loop running with SYN-04	PD-13 Final turn on the FFN — full closure, no functional change admitted PD-09 Multi-corner multi-mode timing closure SYN-09 FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
Route convergence	10.0		
Clock tree refinement	8.0		
Timing convergence	8.0		
Power and IR convergence	4.0		
Risk quantification	4.0		
Total	45.0		
WHO IT TAKES	~24.4 FTE	EXIT CRITERIA - Convergence trend measured against turn 1, not only reported - Structural congestion addressed structurally - Closure risk stated with paths, blocks and options	MEASURED BY - Violation count trend from turn 1 - Congestion resolved structurally against locally - Closure risk at turn 2 against final-turn outcome
Physical design turn lead Owns the turn and the closure risk statement		DEPENDS ON PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established SYN-08 N2 netlist drop — timing and congestion converged, late RTL changes absorbed SYN-12 Netlist handoff and QoR delta review with physical design, per drop PD-07 Clock tree synthesis with skew and jitter budgeting, refined across turns	
Placement and routing engineers Convergence execution			
Synthesis liaison Structural restructuring in congested regions			
Timing engineer Timing convergence and ECO			
Program manager Consumes the risk quantification for the tapeout decision			

PD-07

Clock tree synthesis with skew and jitter budgeting, refined across turns

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w cont. w10–w22 in stage, runs throughout	26.0 M/M planned	~9.4 FTE while it runs	8 3 in parallel	no has float	Clock tree engineer DFT engineer approves

PURPOSE

Build and refine the **clock trees** across turns — skew, insertion delay, jitter budget, power — for a design with dozens of domains and crossings between them.

Clock tree synthesis is rebuilt every turn and improved each time. Skew directly buys or costs timing margin; insertion delay costs power and OCV; and cross-domain paths need a defined phase relationship that only a deliberately built tree provides. It is one of the few areas where a good result is worth several percent of frequency.

WORK FLOW

8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Clock architecture implementation plan	1.5w	main
2	Clock tree synthesis per domain	2.5w	main
3	Clock power optimisation	2w	parallel
4	Skew and insertion delay balancing	2.5w	main
5	Jitter and OCV budget management	2w	parallel
6	Cross-domain and useful skew handling	2.5w	main
7	Clock DRC and quality checking	2w	parallel
8	Continuous refinement across turns	3w	main

Step 6 is the hardest part and the one that pays. Useful skew moves margin from paths that have it to paths that need it, and cross-domain handling decides whether those paths can be closed at all — but both make the tree harder to build and harder to verify.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Clock architecture from ARCH-05
- OCC insertion from DFT-05
- Floorplan from PD-02
- Placement from PD-05 and PD-06
- PLL characteristics from AMS-02

PRODUCES

- Clock tree implementation plan
- Synthesised clock trees per domain
- Clock power optimisation results
- Skew and insertion delay results
- Jitter and OCV budget records

- Clock DRC and quality reports

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D5 FEEDS **MCMM timing closure reports.** Clock skew and insertion delay are inputs to every timing number in the report.

PD-D2 FEEDS **Turn 1 and Turn 2 databases.** Each turn contains its own clock tree, rebuilt and improved.

RISKS AND WATCH-OUTS

- ◆ **Clock power ignored.** The clock network is a large fraction of dynamic power on a design this size, and a tree optimised only for skew is expensive to run.
- ◆ **Useful skew applied without a plan.** It buys margin and complicates every subsequent ECO, because moving one path's timing moves its neighbours'.
- ◆ **Cross-domain phase relationship undefined.** Paths between domains cannot be closed reliably, and the failure appears as intermittent silicon behaviour.
- ◆ **Insertion delay allowed to grow.** Deep trees increase OCV derate and jitter accumulation, consuming the margin the skew balancing bought.
- ◆ **Tree quality not checked.** Clock DRC violations — transition, capacitance, pulse width — degrade the tree in ways timing analysis reports as margin loss without explaining.

EFFORT SPLIT	26 M/M	ENTRY CRITERIA	FEEDS INTO
Clock tree synthesis per domain	7.0	● Clock architecture available from ARCH-05	PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified
Skew and insertion delay balancing	6.0	● OCC logic inserted by DFT-05	PD-09 Multi-corner multi-mode timing closure
Refinement across turns	5.0	● Placement available from the current turn	PD-13 Final turn on the FFN — full closure, no functional change admitted
Clock power optimisation	3.0		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Jitter and OCV budget	2.5		
Clock DRC and quality	2.5		
Total	26.0		
WHO IT TAKES	~9.4 FTE		
Clock tree engineer Owns CTS, skew and tree quality		EXIT CRITERIA ● Clock power inside the ARCH-06 budget ● Cross-domain relationships defined and closed ● Clock DRC clean, not only timing-clean	
Clocking architect Domain relationships and jitter budget		DEPENDS ON	MEASURED BY ● Clock skew and insertion delay per domain ● Clock power against budget ● Clock DRC violations at each turn
Timing engineer Useful skew planning and OCV		PD-02 Floorplan, macro placement and partition definition	
Power engineer Clock power optimisation		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	
DFT engineer OCC integration and test-mode clocking		ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	
		DFT-05 On-chip clock controller design for at-speed testing	
		AMS-02 PLL / clock generator design and closure	

PD-08

Detailed routing and DRC convergence

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w cont. w12–w28 in stage, runs throughout	40.0 M/M planned	~10.9 FTE while it runs	8 3 in parallel	yes no float	Routing lead Floorplan owner approves

PURPOSE

Route the design and drive it to **DRC clean** — global, detailed, antenna, via optimisation — across every turn, on a die where routing resource is the scarcest commodity.

Routing is where the floorplan, the synthesis and the power grid are all judged. Sixteen weeks and forty man-months buys convergence on a design where congestion is structural rather than local, and where every metal layer is contested between signals, clocks and the power grid.

WORK FLOW

8 STEPS · 2 LANES

w12	w16	w20	w24	w28
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Routing strategy and layer planning	1.5w	main
2	Global routing and congestion assessment	3w	main
3	Congestion-driven rip-up and reroute	2.5w	parallel
4	Detailed routing	3.5w	main
5	Antenna and via optimisation	2.5w	parallel
6	DRC convergence	4w	main
7	Routing quality and manufacturability	2.5w	parallel
8	Continuous routing across turns	4w	main

Step 6 takes four weeks because DRC convergence on an advanced node is iterative and asymptotic. The last few hundred violations take as long as the first hundred thousand, and they are the ones that need manual attention.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Placement and clock trees from PD-05, PD-06 and PD-07
- Metal stack and routing rules from TECH-02 and PDK-02
- PDN from PD-03
- Scan routing requirements from PD-12
- DFM rules from PDK-02

PRODUCES

- Routing strategy and layer plan
- Globally and detail-routed database
- Congestion resolution records
- Antenna and via optimisation results
- DRC convergence results
- Routing quality and manufacturability findings

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D6	PRODUCES	Interim physical DRC / LVS clean. This activity is the deliverable at each turn, and clean here is what makes S0-03 tractable.
PD-D3	FEEDS	Final-turn routed database on the FFN. The final route is this activity's last execution.

RISKS AND WATCH-OUTS

- ◆ **Congestion treated as a routing problem.** Structural congestion cannot be routed around, and rip-up cycles spent on it consume time without converging.
- ◆ **Antenna violations left late.** Fixes need diodes or layer jumps, both of which need space that a fully routed design does not have.
- ◆ **DRC convergence assumed linear.** The last violations are the hardest, and planning as if the rate is constant produces a schedule that misses.
- ◆ **Manufacturability ignored until signoff.** DFM-clean routing is cheaper to produce than to retrofit, and S0-08 finds what was not.
- ◆ **Layer plan not respecting the PDN.** Signals and the power grid contend for the same upper layers, and a plan that ignores the grid produces conflicts everywhere.

EFFORT SPLIT	40 M/M	ENTRY CRITERIA	FEEDS INTO
Detailed routing	11.0	● Placement and clock trees available for the turn	PD-10 Signal and power integrity iteration — crosstalk, EM
DRC convergence	9.0	● Metal stack and routing rules available from PDK-02	PD-13 Final turn on the FFN — full closure, no functional change admitted
Global routing	7.0	● PDN implemented in PD-03	PD-15 Chip finishing — seal ring, dummy and metal fill, alignment marks
Rip-up and reroute	6.0		S0-03 Full-chip DRC, LVS, antenna and density verification
Antenna and via optimisation	4.0		S0-08 DFM and lithography checks, CMP and hotspot analysis
Routing quality	3.0		
Total	40.0		
WHO IT TAKES	~10.9 FTE		
Routing lead Owns routing strategy and DRC convergence		EXIT CRITERIA ● DRC clean at each turn, not only at the final one ● Congestion resolved structurally where it is structural ● Manufacturability addressed during routing, not after	MEASURED BY ● DRC violations remaining per turn ● Rip-up cycles spent on structural congestion ● Antenna violations at first detailed route
Routing engineers Global, detailed routing and violation fixing		DEPENDS ON PD-03 Power delivery network design and early IR analysis	
DFM engineer Manufacturability and litho-friendly routing		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	
Physical verification engineer DRC analysis and triage		PD-07 Clock tree synthesis with skew and jitter budgeting, refined across turns	
Floorplan owner Structural remedies for congestion		PDK-02 Design rule manual review; restricted and recommended rule disposition	
		PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control	

12 PD · PHYSICAL DESIGN

PD-09

Multi-corner multi-mode timing closure

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
22w cont. w8–w30 in stage, runs throughout	60.0 M/M planned	~11.8 FTE while it runs	8 3 in parallel	yes no float	Timing closure lead Program manager approves

PURPOSE

Close **timing across every corner and every mode** — setup, hold, cross-corner, cross-mode — and drive the violation burn-down that decides the tapeout date.

Twenty-two weeks and sixty man-months, the largest activity in the stage. Multi-corner multi-mode closure on a large die is a long grind of analysis, ECO and re-analysis, and its burn-down curve is the single best predictor the program has of whether it will tape out on time.

WORK FLOW

8 STEPS · 2 LANES

w8w13w18w23w28w30

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Timing setup and MMMC scenario definition	1.5w	main
2	Setup timing closure	3w	main
3	Violation burn-down tracking and reporting	3w	parallel
4	Hold timing closure	3w	main
5	Timing ECO generation and application	3w	parallel
6	Cross-corner and cross-mode convergence	3.5w	main
7	Signoff timing correlation with the SO flow	3w	parallel
8	Continuous closure across turns	11w	main

Step 7 prevents the worst surprise available. A design closed in the implementation tool and re-analysed in the signoff tool will show differences; discovering their size early lets the team close with the right margin rather than re-closing after `SO-02` reports.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Placed and routed databases from each turn
- Constraints from SYN-01
- Corner and derate definition from PDK-11
- Clock trees from PD-07
- Signoff correlation from SO-12

PRODUCES

- MMMC timing scenarios
- Setup and hold closure results
- Violation burn-down tracking
- Timing ECOs
- Cross-corner convergence results
- Signoff correlation findings

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D5	PRODUCES	MCMM timing closure reports with violation burn-down across turns. This activity is the deliverable, and the burn-down is the program's best schedule predictor.
PD-D3	FEEDS	Final-turn routed database. A database is only handed to signoff when this activity says it is closed.

RISKS AND WATCH-OUTS

- ◆ **Closing at some corners and deferring others.** The deferred corner is the one that fails, and it fails when there is no time to fix it.
- ◆ **Implementation and signoff timing diverging.** A design closed in one tool and judged in another can be weeks from closure without anyone knowing.
- ◆ **Hold fixed after setup.** Hold fixes add buffers that change setup, and doing them sequentially means iterating both.
- ◆ **Burn-down not tracked.** Without a trend nobody can distinguish a design that is converging slowly from one that is not converging at all.
- ◆ **ECO volume exceeding what the database can absorb.** Thousands of ECOs degrade the physical result, and at some point a re-run is cheaper than another ECO round.

EFFORT SPLIT	60 M/M	ENTRY CRITERIA	FEEDS INTO
Setup timing closure	15.0	● Routed database available from the current turn	PD-13 Final turn on the FFN — full closure, no functional change admitted
Hold timing closure	12.0	● Corner and derate definition final from PDK-11	PD-14 Functional and timing ECO implementation
Continuous closure across turns	12.0	● Signoff flow available for correlation from SO-01	SO-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Timing ECO generation	9.0		SO-12 Signoff corner correlation against foundry decks
Cross-corner convergence	7.0		DV-11 Gate-level simulation, functional and timing-annotated
Signoff correlation	5.0		
Total	60.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~11.8 FTE	● Closed at every corner and mode, not a subset	● Violations remaining per corner and mode
Timing closure lead		● Implementation and signoff timing correlated within a stated tolerance	● Implementation-to-signoff timing correlation
Owens closure, burn-down and the tapeout prediction		● Burn-down trend projecting closure before the tapeout date	● Burn-down rate against the tapeout date
STA engineers		DEPENDS ON	
Analysis, ECO generation and convergence		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	
Physical design engineers		PD-07 Clock tree synthesis with skew and jitter budgeting, refined across turns	
ECO application and re-route		SYN-01 SDC constraint development and validation	
Signoff methodology engineer		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
Correlation with the signoff flow			
Program manager			
Uses the burn-down for schedule decisions			

12 PD · PHYSICAL DESIGN

PD-10

Signal and power integrity iteration — crosstalk, EM

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	18.0 M/M	~7.8 FTE	6	no	SI/PI engineer
w16–w26 in stage	planned	while it runs	2 in parallel	has float	SIPI liaison approves

PURPOSE

Fix what the routing did to the electricals — **crosstalk, noise, electromigration, dynamic IR** — before signoff has to reject the design for them.

Signal and power integrity problems are created by routing and found by analysis. Crosstalk between adjacent nets, EM in high-current wires and dynamic IR under real switching are all consequences of physical decisions, and fixing them means changing routing that was closed for timing.

WORK FLOW

6 STEPS · 2 LANES

w16w18w20w22w24w26

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	SI setup and crosstalk analysis	1.5w	main
2	Crosstalk fixing — spacing, shielding, buffer sizing	2.5w	main
3	Noise and glitch analysis	2w	parallel
4	Electromigration analysis and fixing	2.5w	main
5	Dynamic IR iteration with the PDN	2w	parallel
6	SI and PI iteration and closure	3.5w	main

Step 6 is an iteration rather than a step, because every fix perturbs something else. Widening a wire for EM changes its coupling; adding shielding consumes routing resource; and both change timing, which sends the design back to `PD-09`.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Routed database from PD-08
- PDN from PD-03
- Switching activity from SYN-06 and DV-04
- Signoff SI/PI methodology from SO-05
- Package co-analysis findings from SIPI-04

PRODUCES

- Crosstalk analysis and fixes
- Noise and glitch analysis results
- EM analysis and routing fixes
- Dynamic IR results and PDN adjustments
- SI and PI closure records

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D6	FEEDS	Interim physical DRC / LVS clean. SI and EM fixes are routing changes, so they close together with DRC.
PD-D5	FEEDS	MCMM timing closure reports. Crosstalk-induced delay is part of timing, and SI fixes change timing in both directions.

RISKS AND WATCH-OUTS

- ◆ **SI analysed after timing closes.** Crosstalk delay is timing, and closing without it means closing against numbers that will move.
- ◆ **EM checked only on power.** High-toggle signal nets on advanced nodes carry EM exposure too, and they are usually the ones nobody checks.
- ◆ **Dynamic IR analysed with uniform activity.** Real switching is bursty and localised, and uniform assumptions understate the worst case badly.
- ◆ **Fixes applied without re-timing.** Shielding and spacing change delay, and a fix that breaks timing has moved the problem rather than solved it.
- ◆ **Package effects excluded.** Die-only dynamic IR is optimistic; SIPI-04 with package inductance is the number signoff will use.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
Crosstalk fixing	5.0	● Routed database available from PD-08	PD-13 Final turn on the FFN — full closure, no functional change admitted
EM analysis and fixing	4.5	● Switching activity available from SYN-06	S0-04 EM and IR-drop signoff, static and dynamic
SI and PI iteration	3.5	● Signoff SI/PI methodology agreed with SO-05	S0-05 Power and signal integrity signoff — crosstalk, noise
Noise and glitch analysis	2.5		SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
Dynamic IR iteration	2.5		
Total	18.0	EXIT CRITERIA	
WHO IT TAKES	~7.8 FTE	● Crosstalk included in timing closure, not analysed after it	MEASURED BY
SI/PI engineer		● EM checked on signal nets as well as power	● Crosstalk-induced delay against the timing budget
Owns crosstalk, noise, EM and IR analysis		● Dynamic IR analysed with realistic, localised activity	● EM violations at signoff
Routing engineers			● Dynamic IR against budget with realistic activity
Apply spacing, shielding and width fixes		DEPENDS ON	
Power delivery engineer		PD-03 Power delivery network design and early IR analysis	
PDN adjustments for dynamic IR		PD-08 Detailed routing and DRC convergence	
Timing engineer		PD-09 Multi-corner multi-mode timing closure	
Re-timing after SI fixes		SYN-06 Power optimisation — clock gating, operand isolation, Vt mix	
SIPI liaison			
Package-aware IR and channel context			

12 PD · PHYSICAL DESIGN

PD-11

Chip power model (CPM/CPS) extraction and handoff to package co-verification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w w14–w19 in stage	6.0 M/M planned	~5.2 FTE while it runs	5 1 in parallel	no has float	Power modelling engineer Configuration manager approves

PURPOSE

Extract the **chip power model** the package co-verification needs — per domain, per mode, with realistic switching — and hand it over as a versioned artifact.

SIPI-03 cannot analyse the die, package and board as one power delivery network without a model of the die's current draw. Producing that model is a small activity with an outsized dependency: without it the co-verification cannot start, and the co-verification gates mask release.

WORK FLOW5 STEPS · 2 LANES

w14	w15	w16	w17	w18	w19
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Power model requirement definition with SIPI	1w	main
2	CPM extraction per domain and operating mode	1.5w	main
3	Switching scenario selection for the extraction	1w	parallel
4	Model validation against internal IR analysis	1.5w	main
5	Handoff and version control	1w	main

Step 3 determines whether the model is useful. A CPM extracted under average activity understates the transient the package has to supply; extracting under the worst realistic burst is what makes `SIPI-04`'s answer meaningful.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Placed and routed database from PD-05 or PD-06
- PDN from PD-03
- Switching activity from DV-04 and SYN-06
- Model format requirements from SIPI-01
- Power domain definition from ARCH-05

PRODUCES

- Power model requirement definition
- Chip power model per domain and mode
- Switching scenario selection record
- Model validation against internal IR
- Versioned model handoff

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D4	FEEDS	Floorplan and PDN specification. The power model is the PDN's behaviour expressed for external consumers.
PD-D7	FEEDS	Bump map, RDL and package interface files. The model is part of what the package side needs alongside the physical interface.

RISKS AND WATCH-OUTS

- ◆ **Model extracted under average activity.** The package has to supply the transient, and an averaged model hides exactly the event that matters.
- ◆ **Delivered late.** Co-verification is on the critical path to mask release, and it cannot start without this model.
- ◆ **Model not validated.** A CPM that disagrees with the die's own IR analysis is wrong in one of the two places, and nobody knows which.
- ◆ **Single model for all modes.** Different DVFS points and workloads draw very differently, and one model cannot represent them.
- ◆ **No version control.** The package team analysing against a stale model produces conclusions about a die that no longer exists.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
CPM extraction	2.5	Placed and routed database available	SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode
Model validation	1.5	PDN implemented in PD-03	
Requirement definition	1.0	Model format agreed with SIPI-01	SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
Handoff and version control	1.0		
Total	6.0	EXIT CRITERIA	SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
WHO IT TAKES	~5.2 FTE	- Model extracted under realistic worst-case switching - Validated against the die's own IR analysis - Delivered versioned, on the date SIPI needs it	S0-06 Chip-package-system co-analysis signoff review against SIPI results
Power modelling engineer Owns extraction and validation		DEPENDS ON	MEASURED BY
Power delivery engineer Internal IR correlation		PD-03 Power delivery network design and early IR analysis	Model delivery against the SIPI need date
SIPI liaison Model format and scenario requirements		PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established	Validation agreement with internal IR
Verification liaison Realistic switching scenarios		SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode	Modes modelled against modes the product uses
Configuration manager Model versioning and handoff		DV-04 Chip-level and system-level scenario testing	

12 PD · PHYSICAL DESIGN

PD-12

Scan chain reordering and DFT-aware routing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w16–w22 in stage	8.0 M/M planned	~5.8 FTE while it runs	6 2 in parallel	no has float	DFT physical engineer Clock engineer approves

PURPOSE

Reorder and route the **scan chains physically** — shorter routes, less congestion — without destroying the chain-to-block mapping diagnosis depends on.

Scan chains as stitched by synthesis follow logical order and route terribly. Physical reordering fixes the routing and can scramble the relationship between chain position and physical location, which is exactly what failure diagnosis uses. Both constraints are real.

WORK FLOW

6 STEPS · 2 LANES

w16	w18	w20	w22
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Scan chain physical analysis and routing cost	1w	main
2	Chain reordering for routing efficiency	1.5w	main
3	Diagnosis mapping preservation	1w	parallel
4	DFT-aware routing and shielding of test paths	1.5w	main
5	Test clock routing and skew	1w	parallel
6	Chain verification after reorder	2w	main

Step 6 is not optional. Reordering rewrites the netlist's scan connectivity, and a chain broken by the reorder makes everything behind it untestable — a failure that would otherwise be discovered at ATE bring-up.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Scan-inserted netlist from DFT-06
- Physical DFT plan from DFT-09
- Placement from the current turn
- Routing resource from PD-08
- Test clock architecture from DFT-05

PRODUCES

- Chain routing cost analysis
- Reordered scan chains
- Diagnosis mapping record
- DFT-aware routing and shielding
- Test clock routing
- Post-reorder chain verification results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D6	FEEDS	Interim physical DRC / LVS clean. Scan routing is routing, and it closes with everything else.
PD-D3	FEEDS	Final-turn routed database. The chains in the final database are the ones ATPG patterns were generated against.

RISKS AND WATCH-OUTS

- ◆ **Reordering breaking diagnosis.** Failure analysis relies on mapping a failing bit to a physical location, and a scrambled chain removes that.
- ◆ **Chains not verified after reorder.** A break makes everything behind it untestable, and ATE bring-up is a costly place to find that.
- ◆ **Patterns generated before the final reorder.** Any chain change after DFT-07 invalidates the pattern set and costs weeks of regeneration.
- ◆ **Test clock skew ignored.** Scan shift reaches every flop, and skew on the shift clock produces shift failures that look like design bugs.
- ◆ **Chain imbalance after reorder.** Physical reordering can lengthen one chain and shorten another, and shift time follows the longest.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Chain reordering	2.5	● Scan-inserted netlist available from DFT-06	DFT-07ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure
DFT-aware routing	2.0	● Physical DFT plan agreed in DFT-09	PD-13Final turn on the FFN — full closure, no functional change admitted
Post-reorder verification	1.5	● Placement available for the turn	S0-03Full-chip DRC, LVS, antenna and density verification
Diagnosis mapping	1.0		
Test clock routing	1.0	EXIT CRITERIA	
Total	8.0	● Chains verified after every reorder	MEASURED BY
WHO IT TAKES	~5.8 FTE	● Diagnosis mapping preserved or explicitly traded	● Routing resource saved by reordering
DFT physical engineer		● Chain order frozen before ATPG generates patterns	● Chain integrity after reorder
Owns reordering and post-reorder verification			● Chain order changes after pattern generation
Routing engineers		DEPENDS ON	
Scan routing and shielding		DFT-06Scan insertion and DFT DRC during synthesis	
Diagnosis engineer		DFT-09Scan compression and chain routing feasibility with physical design	
Mapping preservation requirements		PD-08Detailed routing and DRC convergence	
ATPG engineer			
Pattern regeneration if chains change			
Clock engineer			
Test clock routing and skew			

PD-13

Final turn on the FFN — full closure, no functional change admitted

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
11w	55.0 M/M	~21.7 FTE	10	yes	Physical design lead
w19–w30 in stage	planned	while it runs	4 in parallel	no float	Signoff lead approves

PURPOSE

Run the **final turn on the FFN and close it completely** — timing, power, DRC, LVS — with no functional change admitted and no next turn to fall back on.

Everything the stage has done is preparation for this. Fifty-five man-months over eleven weeks, on a frozen netlist, closing every corner and every mode to signoff standard. The discipline that matters is refusing functional change: an ECO that alters behaviour restarts verification, and there is no schedule for that.

WORK FLOW

10 STEPS · 2 LANES

w19w22w25w28w30

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	FFN intake and delta assessment against N2	1w	main
2	Placement on the final netlist	2w	main
3	ECO-only change discipline and gatekeeping	1.5w	parallel
4	Final clock tree synthesis	2w	main
5	Power and IR final convergence	2w	parallel
6	Final routing	2.5w	main
7	DRC and LVS convergence	2w	parallel
8	Final timing closure across all corners and modes	2w	main
9	Closure reporting and remaining-violation disposition	1.5w	parallel
10	Database freeze and signoff handoff	1.5w	main

Step 3 runs across the whole turn and is a gate rather than a task. Every change request during the final turn has to be classified — timing ECO, functional change, or bug fix — and only the first is admitted without escalation.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- FFN and handoff package from SYN-09 and SYN-12
- Turn 2 database and risk from PD-06
- Clock trees from PD-07

PRODUCES

- Final placed, clocked and routed database
- ECO discipline records
- Final power and IR convergence results

- SI/PI fixes from PD-10
 - ECO infrastructure from PD-14
- DRC and LVS clean results
 - Final timing closure across corners and modes
 - Frozen signoff-ready database

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D3

PRODUCES

Final-turn routed database on the FFN, per block and top. This activity is the deliverable, and it is what signoff and tapeout consume.

PD-D5

FEEDS

MCMM timing closure reports. The final turn's closure report is the one Design Freeze is granted against.

RISKS AND WATCH-OUTS

- ◆

Functional change admitted during the final turn. It restarts verification and equivalence, and no schedule allows for that at this point.
- ◆

Closure gap larger than turn 2 predicted. If the prediction was optimistic, the discovery comes with no remaining turns and only concessions available.
- ◆

ECO volume degrading the database. Thousands of ECOs on a final database produce a physical result worse than a clean re-run, but a re-run costs weeks.
- ◆

Remaining violations dispositioned informally. Every violation that survives to signoff must be a named, accepted waiver rather than an oversight.
- ◆

Database frozen before LVS is clean. A handoff that fails LVS at S0-03 sends everything back with no time to spare.

EFFORT SPLIT	55 M/M	ENTRY CRITERIA	FEEDS INTO
Final timing closure	14.0	● FFN released by SYN-09 against frozen RTL	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Final routing	11.0	● Turn 2 closure risk understood from PD-06	S0-03 Full-chip DRC, LVS, antenna and density verification
Placement	9.0	● ECO infrastructure running from PD-14	S0-09 Final formal equivalence and LVS netlist consistency
Final clock tree synthesis	7.0		T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
Power and IR convergence	6.0		PD-16 Hierarchical block closure and top-level assembly
DRC and LVS convergence	5.0		
Freeze and handoff	3.0		
Total	55.0		
WHO IT TAKES	~21.7 FTE		
Physical design lead Owns the final turn and the freeze		EXIT CRITERIA ● Closed at every corner and mode to signoff standard ● No functional change admitted, or each one escalated and recorded ● Database frozen LVS-clean and accepted by signoff	MEASURED BY ● Functional changes admitted during the final turn ● Closure achieved against the turn-2 prediction ● Violations dispositioned as waivers against oversights
Placement, CTS and routing engineers Final turn execution		DEPENDS ON	
Timing closure lead Final closure across corners and modes		PD-06 Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified	
Program manager Gatekeeps functional change requests		PD-07 Clock tree synthesis with skew and jitter budgeting, refined across turns	
Signoff lead Accepts the database for signoff		PD-10 Signal and power integrity iteration — crosstalk, EM	
		SYN-09 FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point	
		SYN-12 Netlist handoff and QoR delta review with physical design, per drop	

12 PD · PHYSICAL DESIGN

PD-14

Functional and timing ECO implementation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
18w cont. w12–w30 in stage, runs throughout	22.0 M/M planned	~5.3 FTE while it runs	6 2 in parallel	no has float	ECO lead Configuration manager approves

PURPOSE

Run the **ECO machinery** — flow, spare cells, metal-only strategy, impact tracking — that every turn depends on and that the final turn depends on absolutely.

ECOs are how a closed design absorbs change without a full re-run. The infrastructure matters more than any individual ECO: a spare cell strategy decided at floorplan time determines whether a late fix is metal-only or needs a base layer change, and that distinction is worth several weeks and a mask set.

WORK FLOW

6 STEPS · 2 LANES

w12	w16	w20	w24	w28	w30
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	ECO flow and methodology setup	1.5w	main
2	Functional ECO implementation capability	2.5w	main
3	ECO impact assessment and tracking	2w	parallel
4	Timing ECO implementation	2.5w	main
5	Spare cell and metal-only ECO strategy	2w	parallel
6	Continuous ECO operation across turns	11.5w	main

Step 5 is decided early and matters late. Spare cells distributed through the floorplan cost area now and make a metal-only fix possible at tapeout; without them, a late functional fix needs base layers and a full mask set.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Change requests from RTL-10, DV-12 and timing closure
- Placed and routed databases from each turn
- Spare cell placement from PD-02
- ECO flow support from PDK-09
- Timing ECOs from PD-09

PRODUCES

- ECO flow and methodology
- Functional ECO implementation capability
- Impact assessment and tracking records
- Timing ECO implementation
- Spare cell and metal-only strategy
- ECO log across turns

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D8

PRODUCES

ECO log and change control record. This activity is the deliverable, and the log is what makes a late database traceable to the design that was verified.

PD-D3

FEEDS

Final-turn routed database. The final database is the FFN plus its ECO history, and the two are inseparable.

RISKS AND WATCH-OUTS

- ◆ **No spare cell strategy.** A late functional fix then requires base layers and a full mask set rather than a metal-only change.
- ◆ **ECOs applied without impact tracking.** A database with untracked changes cannot be reconciled against the netlist that was verified.
- ◆ **ECO volume unbounded.** Beyond some number, the physical result degrades enough that a clean re-run would be better — and nobody notices the crossing point.
- ◆ **Functional and timing ECOs not distinguished.** They have completely different verification consequences, and conflating them lets functional change in unnoticed.
- ◆ **Spare cells consumed early.** Using them for convenience during turn 1 leaves none for the fix that actually needs them at tapeout.

EFFORT SPLIT	22 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous ECO operation	8.0	ECO flow available from PDK-09	PD-13 Final turn on the FFN — full closure, no functional change admitted
Timing ECO implementation	5.0	Spare cell placement decided in PD-02	
Functional ECO implementation	4.0	Change control process running in RTL-10	T0-08 BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
Impact assessment and tracking	3.0		
Spare cell strategy	2.0		
Total	22.0	EXIT CRITERIA - Metal-only fix path available at tapeout - Every ECO tracked and reconcilable against the netlist - Functional and timing ECOs distinguished in the log	S0-09 Final formal equivalence and LVS netlist consistency
WHO IT TAKES	~5.3 FTE	DEPENDS ON PD-02 Floorplan, macro placement and partition definition PD-09 Multi-corner multi-mode timing closure PDK-09 Foundry reference flow bring-up and internal methodology deck RTL-10 Specification change control and ECO board	T0-03 Tapeout checklist completion and owner sign-off
ECO lead Owns the flow, the strategy and the log			MEASURED BY - Spare cells remaining at tapeout - ECOs tracked against ECOs applied - Metal-only fixes possible against fixes needed
Physical design engineers ECO implementation			
Timing engineers Timing ECO generation			
Verification liaison Functional ECO verification requirements			
Configuration manager ECO log and database traceability			

PD-15

Chip finishing — seal ring, dummy and metal fill, alignment marks

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w w24–w29 in stage	6.0 M/M planned	~5.2 FTE while it runs	6 2 in parallel	no has float	Chip finishing engineer Product engineering approves

PURPOSE

Finish the die as a **manufacturable object** — seal ring, metal fill, dummy, alignment marks, identification — and verify that none of it broke what was closed.

Chip finishing is mechanical and it is not free. Metal fill changes coupling capacitance and therefore timing; the seal ring consumes die edge; and dummy fill interacts with the analog matching that AMS-12 spent weeks building. Every one has to be applied and then re-verified.

WORK FLOW

6 STEPS · 2 LANES

w24	w25	w26	w27	w28	w29
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Seal ring and die edge implementation	1w	main
2	Metal fill and density compliance	1w	main
3	Fill impact on timing and extraction	1w	parallel
4	Alignment marks, logos and die identification	1.5w	main
5	Dummy and CMP compliance	1w	parallel
6	Finishing verification against the closed database	1.5w	main

Step 3 is why finishing is not a formality. Fill adds capacitance to every net near it, and a design closed before fill and taped out after it has been signed off against a database that no longer exists.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Routed database from PD-08 and PD-13
- Seal ring and frame rules from PDK-02
- Fill and density rules from the DRM
- Analog macro fill exclusions from AMS-12
- Die identification requirements from MP-09

PRODUCES

- Seal ring and die edge implementation
- Metal fill meeting density rules
- Fill timing impact analysis
- Alignment marks and die identification
- Dummy and CMP compliance
- Finishing verification results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D3

FEEDS

Final-turn routed database. The database handed to tapeout is the finished one, and the finishing is part of it.

PD-D6

FEEDS

Interim physical DRC / LVS clean. Density and fill rules are DRC rules, and finishing is where they are met.

RISKS AND WATCH-OUTS

- ◆ **Timing signed off before fill.** The taped-out database has fill and the signed-off one did not, which is a signoff against a different design.
- ◆ **Fill applied over analog matching structures.** It disturbs the parasitics AMS-12 tuned, and the damage is invisible until silicon characterisation.
- ◆ **Seal ring area not reserved.** Discovered late, it forces the die edge inward and moves the bump map.
- ◆ **Die identification omitted.** Traceability at yield analysis and field return depends on being able to identify the die, and it cannot be added later.
- ◆ **Finishing treated as a checkbox.** Each step changes the physical database, and each needs verification against what was closed.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Metal fill and density	2.0	● Routed database stable from PD-08	T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
Seal ring and die edge	1.5	● Fill and seal ring rules available from PDK-02	S0-03 Full-chip DRC, LVS, antenna and density verification
Finishing verification	1.0	● Analog fill exclusions defined by AMS-12	S0-08 DFM and lithography checks, CMP and hotspot analysis
Fill timing impact	0.8		MP-09 Production readiness review and change control (PCN) setup
Marks and identification	0.8		
Total	6.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.2 FTE	● Timing verified after fill, not before it ● Analog matching structures protected from fill ● Die identification present and readable	● Timing delta introduced by fill ● Density compliance across the die ● Analog structures affected by fill
Chip finishing engineer Owns seal ring, fill and marks		DEPENDS ON	
Physical verification engineer Density and DRC compliance		PD-08 Detailed routing and DRC convergence	
Timing engineer Fill impact on extraction and timing		PD-13 Final turn on the FFN — full closure, no functional change admitted	
Analog layout liaison Fill exclusions around matching structures		PDK-02 Design rule manual review; restricted and recommended rule disposition	
Product engineering Die identification and traceability requirements		AMS-12 Custom layout with DRC / LVS / antenna and fill closure	

12 PD · PHYSICAL DESIGN

PD-16

Hierarchical block closure and top-level assembly

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w	30.0 M/M	~9.3 FTE	8	yes	Hierarchical closure lead
w14–w28 in stage	planned	while it runs	3 in parallel	no float	Physical verification engineer approves

PURPOSE

Close the **blocks independently and assemble them into a top level** that meets timing across their boundaries — the hierarchical discipline a die this size requires.

A design of this size cannot be closed flat: runtime and capacity make it impossible. Hierarchical closure lets blocks converge in parallel, at the cost of interface budgets that have to be right. A block that closes internally and misses its boundary budget has not helped.

WORK FLOW

8 STEPS · 2 LANES

w14	w17	w20	w23	w26	w28
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Block closure criteria and handoff definition	1.5w	main
2	Block-level closure	3w	main
3	Inter-block timing budget and interface closure	2.5w	parallel
4	Block abstract and timing model generation	3w	main
5	Top-level routing and feedthrough management	2.5w	parallel
6	Top-level assembly	3w	main
7	Hierarchical timing correlation, block model against flat	2w	parallel
8	Top-level integration verification	3.5w	main

Step 7 is the check that keeps hierarchy honest. A block's timing model is an abstraction, and if it disagrees with a flat analysis of the same block then the top level has been closed against a fiction.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Partition definition from PD-02
- Block databases from each turn
- Interface budgets from ARCH-06
- Timing closure status from PD-09
- Final database from PD-13

PRODUCES

- Block closure criteria
- Closed block databases
- Inter-block timing budgets
- Block abstracts and timing models
- Top-level routing and feedthroughs
- Assembled and verified top level

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PD-D9

PRODUCES

Signoff-ready database handoff. This activity is the deliverable — the assembled top level signoff and tapeout actually consume.

PD-D3

FEEDS

Final-turn routed database on the FFN, per block and top. The per-block half of the deliverable comes from here.

RISKS AND WATCH-OUTS

- ◆ **Block models disagreeing with flat analysis.** The top level is then closed against an abstraction that does not represent the block.
- ◆ **Interface budgets set late.** Blocks close internally and fail at their boundaries, and the fix requires reopening blocks that were declared done.
- ◆ **Feedthroughs not planned.** Signals crossing a block have to be routed through it, and a block closed without feedthrough provision blocks the top level.
- ◆ **Blocks closed to different criteria.** Without an enforced standard, each block closes to what its owner considered sufficient, and the top level inherits the weakest.
- ◆ **Top-level assembly left to the end.** Assembly finds interface problems, and finding them after the final turn leaves no time to fix them.

EFFORT SPLIT	30 M/M	ENTRY CRITERIA	FEEDS INTO
Block-level closure	8.0	● Partitions defined in PD-02	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Top-level assembly	6.0	● Interface budgets available from ARCH-06	S0-03 Full-chip DRC, LVS, antenna and density verification
Integration verification	5.0	● Blocks converging in PD-09	S0-09 Final formal equivalence and LVS netlist consistency
Abstract and model generation	4.0		T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
Interface budgets	4.0		
Top-level routing	3.0		
Total	30.0		
WHO IT TAKES	~9.3 FTE	EXIT CRITERIA	MEASURED BY
Hierarchical closure lead Owns block criteria, budgets and assembly		● Block models correlated against flat analysis	● Block model to flat analysis correlation
Block physical owners Close their blocks to the criteria		● Every block closed to one shared criterion	● Interface violations at assembly
Top-level integration engineer Assembly, feedthroughs and top routing		● Top level assembled and verified before the final turn ends	● Blocks closed to the shared criterion
Timing engineer Hierarchical correlation against flat analysis			
Physical verification engineer Top-level integration verification			
		DEPENDS ON	
		PD-02 Floorplan, macro placement and partition definition	
		PD-09 Multi-corner multi-mode timing closure	
		PD-13 Final turn on the FFN — full closure, no functional change admitted	
		ARCH-06 PPA budget allocation per block	

IMPLEMENT

13 Signoff

S0

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w62 – w78	16 _w	12	159 _{M/M}	6 _{of 12}
ENGINEERING ACTIVITIES				KEY DELIVERABLES
S0-01	Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed		6w14 _{M/M}	S0-D1STA signoff reports across all corners and modes, with waiver list
S0-02	Multi-corner multi-mode static timing analysis and closure on the final turn		10w34 _{M/M}	S0-D2Clean DRC / LVS / antenna / density reports
S0-03	Full-chip DRC, LVS, antenna and density verification		9w26 _{M/M}	S0-D3EM/IR and SI/PI signoff reports
S0-04	EM and IR-drop signoff, static and dynamic		8w20 _{M/M}	S0-D4Reliability reports — ESD, latch-up, FIT
S0-05	Power and signal integrity signoff — crosstalk, noise		6w12 _{M/M}	S0-D5DFM and lithography hotspot report
S0-06	Chip-package-system co-analysis signoff review against SIPI results		4w5 _{M/M}	S0-D6Final formal equivalence report
S0-07	Reliability verification — ESD, latch-up, soft error / FIT		6w10 _{M/M}	S0-D7Signoff summary and Design Freeze package
S0-08	DFM and lithography checks, CMP and hotspot analysis		6w10 _{M/M}	
S0-09	Final formal equivalence and LVS netlist consistency		5w8 _{M/M}	
S0-10	Waiver review board and foundry waiver alignment		6w6 _{M/M}	
S0-11	Gate-level simulation with final SDF		5w8 _{M/M}	
S0-12	Signoff corner correlation against foundry decks		5w6 _{M/M}	

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S0-01

Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w0-w6 in stage	14.0 M/M planned	~10.1 FTE while it runs	5 2 in parallel	yes no float	Signoff lead Timing engineer approves

PURPOSE

Rehearse the entire signoff flow on the **turn 2 database** — decks, runtimes, capacity, violation triage — so the final turn is not the first time it runs at full chip.

Signoff is a large, slow, deck-heavy flow, and the first run always finds problems that have nothing to do with the design. Finding them on turn 2 costs a week of a database nobody is waiting for; finding them on the final turn costs a week of the tapeout schedule.

WORK FLOW

5 STEPS · 2 LANES

w0	w2	w4	w6
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MAIN SEQUENCE

1	2	5
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Signoff flow assembly and deck integration	1.5w	main
2	Dry run on the turn 2 database	2w	main
3	Runtime and capacity measurement	1w	parallel
4	Violation triage process and tooling	1w	parallel
5	Flow issue resolution and rehearsal findings	2.5w	main

Step 3 sizes the signoff campaign. Signoff runtime on a die this size is measured in days per corner, and knowing the number before the final turn is what lets the schedule be built rather than discovered.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Turn 2 database from PD-06
- Decks and runsets from PDK-10
- Corner definition from PDK-11
- Tool matrix from PDK-08
- Compute and licence capacity from PDK-12

PRODUCES

- Assembled signoff flow with integrated decks
- Dry run results on turn 2
- Runtime and capacity measurements
- Violation triage process and tooling
- Rehearsal findings and flow fixes

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D7	FEEDS	Signoff summary and Design Freeze package. The rehearsal is what makes the real campaign's schedule credible.
S0-D1	FEEDS	STA signoff reports. The flow that produces them is built and debugged here.

RISKS AND WATCH-OUTS

- ◆ **Dry run skipped for schedule.** The flow's first full-chip run then happens on the critical path, and every problem it finds costs tapeout days.
- ◆ **Runtime not measured.** A signoff campaign planned without knowing its runtime is planned against a guess.
- ◆ **Dry run on an unrepresentative database.** Turn 2 has to be close enough in size and structure that the runtimes and violation counts mean something.
- ◆ **Triage tooling built during the real campaign.** Thousands of violations arrive at once, and sorting them by hand is how a week disappears.
- ◆ **Findings not fixed before the final turn.** A rehearsal that identifies problems and does not resolve them has only produced a list.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Flow assembly and deck integration	4.5	● Turn 2 database available from PD-06	S0-02Multi-corner multi-mode static timing analysis and closure on the final turn
Dry run execution	3.5	● Decks validated and versioned by PDK-10	S0-03Full-chip DRC, LVS, antenna and density verification
Issue resolution	3.0	● Corner definition final from PDK-11	S0-04EM and IR-drop signoff, static and dynamic
Triage process and tooling	1.5		PD-09Multi-corner multi-mode timing closure
Runtime measurement	1.5	EXIT CRITERIA	MEASURED BY
Total	14.0	● Flow runs end to end at full chip before the final turn	● Flow issues found in rehearsal against in the real campaign
WHO IT TAKES	~10.1 FTE	● Runtime measured and the campaign scheduled against it	● Runtime per corner against the campaign plan
Signoff lead		● Rehearsal findings fixed, not only listed	● Days between rehearsal findings and their fixes
Signoff methodology engineers		DEPENDS ON	
Deck integration and flow assembly		PD-06Turn 2 on the N2 drop — congestion, timing and power convergence; closure risk quantified	
CAD infrastructure		PDK-08EDA tool version qualification for the node — synthesis, P&R, STA, PV, EM/IR	
Runtime and capacity measurement		PDK-10Rule deck, QRC tech file and DRC/LVS deck version control	
Physical design liaison		PDK-11Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	
Provides the turn 2 database and context		PDK-12Compute farm, EDA license and storage capacity planning	
Timing engineer			
Triage process and tooling			

13 S0 · SIGNOFF

S0-02

Multi-corner multi-mode static timing analysis and closure on the final turn

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	34.0 M/M	~14.8 FTE	7	yes	STA signoff lead
w5-w15 in stage	planned	while it runs	3 in parallel	no float	Signoff methodology engineer approves

PURPOSE

Prove the final database meets timing **at every corner and every mode, in the signoff tool**, and dispose of every remaining violation as a fix or a signed waiver.

This is the largest activity in the stage and the one Design Freeze most depends on. It differs from PD-09 in that it is judgement rather than optimisation: the signoff tool, the foundry-agreed derates and the full corner set, with no ECO loop to fall back on.

WORK FLOW

7 STEPS · 2 LANES

w5	w7	w9	w11	w13	w15
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	Signoff STA setup on the final database	1.5w	main
2	Setup timing analysis across all corners and modes	2.5w	main
3	Hold timing analysis across all corners and modes	2w	parallel
4	Violation triage and root-cause classification	2.5w	main
5	ECO iteration with physical design	2w	parallel
6	Cross-mode and test-mode timing signoff	2w	parallel
7	Timing signoff closure and waiver disposition	3.5w	main

Step 6 covers the modes that are easy to forget and expensive to miss. Scan shift, at-speed capture and any low-power mode each have their own timing, and a design signed off functionally and not in test mode produces parts that fail at ATE.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database from PD-13 and PD-16
- Constraints from SYN-01
- Corner and derate definition from PDK-11
- Correlation findings from SO-12
- Timing closure status from PD-09

PRODUCES

- Signoff STA setup
- Setup and hold analysis across all corners and modes
- Violation triage with root causes
- ECO requests to physical design
- Test-mode timing signoff
- Timing signoff reports and waiver list

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D1	PRODUCES	STA signoff reports across all corners and modes, with waiver list. This activity is the deliverable, and the waiver list is the part that carries risk into silicon.
S0-D7	FEEDS	Signoff summary and Design Freeze package. Timing signoff is the largest single input to the freeze decision.

RISKS AND WATCH-OUTS

- ◆ **Corners deferred to fit the schedule.** The deferred corner is the one that fails, and it fails in silicon where there is no fix.
- ◆ **Waivers granted without root cause.** A waived violation with no understood cause is a silicon risk being accepted blind.
- ◆ **Test modes not signed off.** Scan shift and at-speed timing failures produce parts that work functionally and fail at ATE.
- ◆ **Signoff and implementation disagreeing late.** If S0-12 has not correlated the two, the first signoff run can show a design weeks from closure.
- ◆ **ECO loop reopening after signoff starts.** Each ECO invalidates the analysis it was based on, and uncontrolled iteration never converges.

EFFORT SPLIT	34 M/M	ENTRY CRITERIA	FEEDS INTO
Setup timing analysis	10.0	● Final database frozen and handed over by PD-13	S0-10 Waiver review board and foundry waiver alignment
Hold timing analysis	7.0	● Corner and derate set agreed in PDK-11	S0-11 Gate-level simulation with final SDF
Violation triage and root cause	6.0	● Correlation established in SO-12	T0-02 Final full-chip verification re-run on the released database
Closure and waiver disposition	5.0		
Test-mode signoff	3.5		
ECO iteration	2.5		T0-03 Tapeout checklist completion and owner sign-off
Total	34.0		
WHO IT TAKES	~14.8 FTE	EXIT CRITERIA ● Every corner and mode analysed, none deferred ● Every waiver carries a root cause and a signature ● Test modes signed off alongside functional	MEASURED BY ● Corners and modes signed off against total ● Waivers with a root cause ● Violations remaining at Design Freeze
STA signoff lead Owns timing signoff and the waiver list		DEPENDS ON	
STA engineers Analysis, triage and root cause		S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed	
Physical design liaison ECO implementation		S0-12 Signoff corner correlation against foundry decks	
DFT engineer Test-mode timing requirements		PD-13 Final turn on the FFN — full closure, no functional change admitted	
Signoff methodology engineer Derates, corners and tool correlation		PD-16 Hierarchical block closure and top-level assembly	
		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)	

13 S0 · SIGNOFF

S0-03

Full-chip DRC, LVS, antenna and density verification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w w6–w15 in stage	26.0 M/M planned	~12.6 FTE while it runs	6 2 in parallel	yes no float	Physical verification lead Foundry liaison approves

PURPOSE

Run **full-chip physical verification** — DRC, LVS, antenna, density — to clean, on the database that will become masks.

Physical verification is binary and unforgiving: the foundry will not accept a database with violations it has not waived. On a die this size the runs take days, the violation counts start in the tens of thousands, and the last hundred are the ones that need a human.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12	w14	w15
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Full-chip DRC setup and hierarchical run strategy	1.5w	main
2	Full-chip DRC execution and violation triage	3w	main
3	Antenna verification and fixing	2w	parallel
4	LVS execution and mismatch debug	2.5w	main
5	Density and fill verification	1.5w	parallel
6	Physical verification closure and waiver preparation	2w	main

Step 4 is where the hardest problems hide. An LVS mismatch on a full chip can take days to localise, and the causes are usually elsewhere — a macro abstract that disagrees with its layout, or an ECO applied to one view and not the other.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database from PD-13 and PD-16
- Verification decks from PDK-10
- Macro layouts and abstracts from AMS-12 and AMS-15
- Chip finishing from PD-15
- Foundry rule set from PDK-02

PRODUCES

- DRC run strategy and results
- Antenna verification and fixes
- LVS results and mismatch resolution
- Density and fill verification
- Physical verification clean reports and waiver candidates

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D2

PRODUCES

Clean DRC / LVS / antenna / density reports. This activity is the deliverable, and it is a precondition for mask release rather than an opinion.

S0-D7

FEEDS

Signoff summary and Design Freeze package. A design with open physical violations cannot be frozen.

RISKS AND WATCH-OUTS

- ◆ **LVS mismatch traced late.** Full-chip mismatches take days to localise, and they are frequently caused by macro views rather than by the design.
- ◆ **Runs started too late.** Each full-chip run takes days, and the iteration count is what sets the schedule rather than any single run.
- ◆ **Antenna fixes needing space that is gone.** Diodes and layer jumps need room, and a fully routed design has none.
- ◆ **Waivers assumed acceptable.** Foundry waivers are granted case by case, and a design that assumes one will be granted may not tape out.
- ◆ **Verification on a database that then changes.** Any ECO after a clean run invalidates it, and re-running costs days.

EFFORT SPLIT	26 M/M	ENTRY CRITERIA	FEEDS INTO
DRC execution and triage	8.0	● Final database available from PD-13	S0-10 Waiver review board and foundry waiver alignment
LVS and mismatch debug	7.0	● Decks validated and versioned in PDK-10	T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
Antenna verification and fixing	4.5	● Chip finishing complete from PD-15	T0-02 Final full-chip verification re-run on the released database
Closure and waiver preparation	3.5		T0-09 BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack
Density and fill verification	3.0		
Total	26.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~12.6 FTE	● DRC, LVS, antenna and density all clean or formally waived ● LVS mismatches resolved, not waived ● No database change after the last clean run	● Violations remaining at freeze ● Full-chip run iterations required ● LVS mismatches resolved against waived
Physical verification lead Owns DRC, LVS and antenna closure		DEPENDS ON	
Physical verification engineers Run execution and violation triage		S0-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed	
Layout engineers Violation fixing		PD-13 Final turn on the FFN — full closure, no functional change admitted	
Macro owners Abstract and layout consistency for LVS		PD-15 Chip finishing — seal ring, dummy and metal fill, alignment marks	
Foundry liaison Waiver requests and rule interpretation		PD-16 Hierarchical block closure and top-level assembly	
		PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control	

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S0-04

EM and IR-drop signoff, static and dynamic

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w7–w15 in stage	20.0 M/M planned	~10.9 FTE while it runs	6 2 in parallel	yes no float	EM/IR signoff lead SIPI liaison approves

PURPOSE

Sign off **electromigration and IR drop** — static and dynamic, with real switching — because a design that meets timing and fails EM will not survive its warranty.

EM and IR are reliability and performance at once. IR drop consumes timing margin the STA assumed; EM limits how long the part will work. Both are analysed on the final database with real activity, and both can send routing back for changes that reopen timing.

WORK FLOW

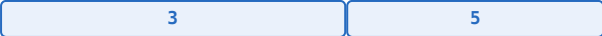
6 STEPS · 2 LANES

w7	w9	w11	w13	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	EM/IR setup and activity scenario definition	1.5w	main
2	Static IR drop analysis and closure	2w	main
3	Dynamic IR analysis under realistic switching	2w	parallel
4	Electromigration analysis on power and signal	2w	main
5	IR-aware timing impact assessment	1.5w	parallel
6	EM/IR closure and waiver disposition	2.5w	main

Step 5 closes a loop that is often left open. IR drop lowers the local supply and slows the logic, and unless that is back-annotated into timing the STA has been run at a voltage the die does not actually see.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database from PD-13
- PDN from PD-03
- Switching activity from DV-04 and SYN-06
- Chip power model and co-analysis from SIPI-04
- EM rules from PDK-02

PRODUCES

- EM/IR setup and activity scenarios
- Static IR analysis and closure
- Dynamic IR results under real switching
- EM analysis on power and signal nets
- IR-aware timing impact assessment
- EM/IR signoff reports

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-03	PRODUCES	EM/IR and SI/PI signoff reports. The EM and IR half of the deliverable; S0-05 supplies the SI/PI half.
S0-01	FEEDS	STA signoff reports. IR-aware timing is timing, and the two signoffs have to be reconciled rather than reported separately.

RISKS AND WATCH-OUTS

- ◆ **IR not back-annotated into timing.** The design is then signed off at a supply voltage it never sees, and the margin was never real.
- ◆ **Dynamic IR analysed with uniform activity.** Real switching is bursty and local, and uniform assumptions understate the worst case.
- ◆ **Package effects excluded.** Die-only IR is optimistic; the number that matters includes the package inductance SIPI-04 models.
- ◆ **Signal EM overlooked.** High-toggle nets on advanced nodes carry real EM exposure, and checking only power misses them.
- ◆ **EM fixes reopening timing.** Widening wires changes delay, and a fix applied after timing signoff invalidates it.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Static and dynamic IR analysis	7.0	● Final database available from PD-13	S0-02Multi-corner multi-mode static timing analysis and closure on the final turn
EM analysis	5.0	● Realistic switching activity available from DV-04	S0-10Waiver review board and foundry waiver alignment
Closure and waiver disposition	4.0	● Package co-analysis results available from SIPI-04	T0-02Final full-chip verification re-run on the released database
IR-aware timing assessment	2.5		MP-02Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Setup and scenarios	1.5		
Total	20.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~10.9 FTE	● IR back-annotated into timing signoff	● IR drop against budget with realistic activity
EM/IR signoff lead Owns EM and IR analysis and closure		● Dynamic IR analysed with realistic localised activity	● Timing margin consumed by IR
Power integrity engineers Static and dynamic IR analysis		● EM checked on signal nets as well as power	● EM violations at freeze
Reliability engineer EM limits and lifetime interpretation		DEPENDS ON	
Timing engineer IR-aware timing reconciliation		S0-01Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed	
SIPI liaison Package-aware IR context		PD-03Power delivery network design and early IR analysis	
		PD-10Signal and power integrity iteration — crosstalk, EM	
		PD-13Final turn on the FFN — full closure, no functional change admitted	
		SIPI-04Dynamic voltage-drop analysis with package inductance and real switching profiles	

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SO-05

Power and signal integrity signoff — crosstalk, noise

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	12.0 M/M	~8.7 FTE	6	no	SI signoff engineer
w9–w15 in stage	planned	while it runs	2 in parallel	has float	Signoff lead approves

PURPOSE

Sign off **signal integrity** — crosstalk delay, noise, glitch — on the final database, and confirm the numbers the timing signoff assumed.

Crosstalk is timing. Coupling between adjacent nets changes delay in both directions and can inject glitches that propagate as functional failures. The analysis is separate from STA and its results feed straight back into it, which makes the two inseparable at signoff.

WORK FLOW

6 STEPS · 2 LANES

w9	w11	w13	w15
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	SI signoff setup and coupling extraction	1w	main
2	Crosstalk delay analysis and timing impact	1.5w	main
3	Noise and glitch analysis	1.5w	parallel
4	Victim-aggressor analysis on critical nets	1.5w	main
5	Power integrity signoff reconciliation	1w	parallel
6	SI signoff closure and reporting	2w	main

Step 4 is where the analysis becomes actionable. A global crosstalk number tells nobody what to fix; a victim-aggressor list on the critical paths tells routing exactly which nets to space or shield.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database and extraction from PD-13
- Crosstalk fixes from PD-10
- Timing analysis from SO-02
- Noise rules and thresholds from PDK-02
- PI results from SO-04

PRODUCES

- SI signoff setup and coupling extraction
- Crosstalk delay analysis and timing impact
- Noise and glitch analysis results
- Victim-aggressor findings on critical nets
- SI signoff report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D3	PRODUCES	EM/IR and SI/PI signoff reports. The signal integrity half of the deliverable.
S0-D1	FEEDS	STA signoff reports. Crosstalk delay is part of the timing number, not a separate finding.

RISKS AND WATCH-OUTS

- ◆ **Crosstalk reported and not reconciled with timing.** Two separate reports means the design is signed off against timing that excludes coupling.
- ◆ **Glitch analysis omitted.** Noise on a static net can propagate as a functional failure that no timing analysis would show.
- ◆ **Analysis on a subset of nets.** Coupling is a property of adjacency, and the critical net set is not the same as the coupled net set.
- ◆ **Fixes applied after timing signoff.** Spacing and shielding change delay, and a late fix invalidates the timing it was meant to protect.
- ◆ **Thresholds set by tool default.** Noise thresholds are a library and technology property, and defaults are either optimistic or unnecessarily punitive.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Crosstalk delay analysis	3.5	● Final database and extraction available from PD-13	S0-02Multi-corner multi-mode static timing analysis and closure on the final turn
Victim-aggressor analysis	2.5	● Crosstalk fixes applied in PD-10	S0-10Waiver review board and foundry waiver alignment
Noise and glitch analysis	2.5	● Timing analysis running in SO-02	T0-02Final full-chip verification re-run on the released database
Closure and reporting	2.0		
Setup and extraction	1.5	EXIT CRITERIA	
Total	12.0	● Crosstalk delay included in the timing signoff numbers	MEASURED BY
WHO IT TAKES	~8.7 FTE	● Glitch analysis run, not only delay analysis	● Crosstalk delay against the timing budget
SI signoff engineer		● Noise thresholds set from library data, not defaults	● Glitch violations at freeze
Owns crosstalk and noise signoff			● Nets analysed against nets coupled
STA engineer		DEPENDS ON	
Timing reconciliation with crosstalk delay		S0-01Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed	
Routing engineers		S0-04EM and IR-drop signoff, static and dynamic	
Spacing and shielding fixes		PD-10Signal and power integrity iteration — crosstalk, EM	
Library engineer		PD-13Final turn on the FFN — full closure, no functional change admitted	
Noise thresholds and cell characterisation			
Signoff lead			
Closure and reporting			

13 S0 · SIGNOFF

S0-06

Chip-package-system co-analysis
signoff review against SIPI results

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	5.0 M/M	~5.4 FTE	5	no	Signoff lead
w11–w15 in stage	planned	while it runs	2 in parallel	has float	Timing engineer approves

PURPOSE

Review the **chip-package-system co-verification result** as a signoff item in its own right, rather than assuming the die-only analysis covers it.

SIPI analysed the die, package and board as one electrical system. This activity brings that result into the signoff record and reconciles it against the die-only numbers — because where they disagree, the system-level answer is the one silicon will produce.

WORK FLOW5 STEPS · 2 LANES

w11	w12	w13	w14	w15
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MAIN SEQUENCE

1	2	5
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	SIPI result intake and scope confirmation	0.8w	main
2	Die-only against system-level IR reconciliation	1.3w	main
3	Channel compliance review against the interface budgets	1w	parallel
4	Power-aware timing correlation review	1w	parallel
5	Co-verification signoff review and disposition	2w	main

Step 2 is the reconciliation that matters. Die-only IR analysis excludes package inductance, and the difference between the two numbers is margin the design either has or does not — a discrepancy that has to be resolved before mask release rather than explained afterwards.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Co-verification results from SIPI-11
- Die-only IR from SO-04
- Channel budgets from ARCH-04 and AMS-03
- Chip power model from PD-11
- Timing signoff from SO-02

PRODUCES

- SIPI result intake record
- Die-only against system-level reconciliation
- Channel compliance review findings
- Power-aware timing correlation review
- Co-verification signoff disposition

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D3	FEEDS	EM/IR and SI/PI signoff reports. The system-level results belong in the same report as the die-only ones, reconciled rather than filed separately.
S0-D7	FEEDS	Signoff summary and Design Freeze package. Co-verification signoff is a freeze precondition, since mask release follows it.

RISKS AND WATCH-OUTS

- ◆ **Co-verification treated as a package deliverable.** It constrains the die's signoff, and excluding it from the signoff record means freezing without it.
- ◆ **Discrepancies explained rather than resolved.** Where die-only and system-level disagree, one is wrong, and the difference is real margin.
- ◆ **Channel compliance reviewed against internal assumptions.** The interface specification's mask is what compliance means, and internal budgets can be more generous.
- ◆ **Review scheduled after Design Freeze.** The freeze is what gates mask release, and a co-verification finding afterwards has nowhere to go.
- ◆ **Power-aware timing not reviewed.** STA at nominal voltage and IR analysis showing droop are two views of the same design, and only the reconciliation is the truth.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Co-verification review and disposition	2.0	- Co-verification signed off in SIPI-11 - Die-only IR results available from SO-04	S0-10 Waiver review board and foundry waiver alignment
IR reconciliation	1.5	Timing signoff progressing in SO-02	T0-04 Open-issue risk assessment and waiver acceptance
Channel compliance review	0.8		T0-06 FEOL layer data preparation and MTO release to the mask shop
Timing correlation review	0.8		
Total	5.0		
WHO IT TAKES	~5.4 FTE	EXIT CRITERIA	MEASURED BY
Signoff lead Owns the review and its disposition		- Die-only and system-level results reconciled, not merely compared - Channel compliance reviewed against the specification mask - Review completed before Design Freeze, not after	- Discrepancies resolved against identified - Channel margin against the compliance mask - Review completion against the freeze date
SIPI lead Presents the co-verification results			
Power integrity engineer IR reconciliation between die and system			
Interface architect Channel compliance interpretation			
Timing engineer Power-aware timing correlation			
		S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn	
		S0-04 EM and IR-drop signoff, static and dynamic	
		SIPI-11 Co-verification signoff review and criteria disposition with design and package teams	
		PD-11 Chip power model (CPM/CPS) extraction and handoff to package co-verification	

13 S0 · SIGNOFF

S0-07

Reliability verification — ESD, latch-up, soft error / FIT

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	10.0 M/M	~7.2 FTE	5	no	Reliability signoff engineer
w8–w14 in stage	planned	while it runs	2 in parallel	has float	Layout engineers approves

PURPOSE

Verify the **reliability requirements** the product will be qualified against — ESD paths, latch-up structures, soft error rate — on the final database rather than in the qualification lab.

Qualification failures are the most expensive kind: they arrive after silicon, after packaging and after months of stress testing. Every one of these checks has a design-time equivalent, and running them here converts a possible respin into a layout fix.

WORK FLOW

5 STEPS · 2 LANES

w8	w10	w12	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Reliability requirement collection and rule setup	1w	main
2	ESD path verification through the full chip	1.5w	main
3	Latch-up structure and guard ring verification	1.5w	parallel
4	Soft error rate and FIT estimation	1.5w	parallel
5	Reliability closure and waiver disposition	3.5w	main

Step 2 traces the discharge path through the actual layout rather than confirming the strategy. A clamp that exists and a path that reaches it are different claims, and only the full-chip trace establishes the second.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database from PD-13
- ESD strategy and clamps from PDK-07
- Latch-up rules from PDK-02
- Soft error requirements from the qualification plan
- AMS reliability results from AMS-14

PRODUCES

- Reliability requirements and rule setup
- Full-chip ESD path verification results
- Latch-up and guard ring verification
- Soft error rate and FIT estimate
- Reliability signoff report and waivers

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D4	PRODUCES	Reliability reports — ESD, latch-up, FIT. This activity is the deliverable, and it is what MP-03 will be measured against.
S0-D7	FEEDS	Signoff summary and Design Freeze package. Reliability findings are part of the risk the freeze decision accepts.

RISKS AND WATCH-OUTS

- ◆ **ESD verified as a strategy rather than a path.** A clamp with no complete discharge path to it protects nothing, and only a full-chip trace finds the gap.
- ◆ **Latch-up checked only at the IO ring.** Internal structures near power domain boundaries can latch up too, and they are rarely checked.
- ◆ **Soft error rate not estimated.** FIT is a product requirement in many markets, and estimating it after qualification leaves no mitigation options.
- ◆ **Reliability waivers granted without lifetime analysis.** A waiver with no quantified impact cannot be defended when qualification fails.
- ◆ **Analysis on a database that then changes.** Late ECOs can break an ESD path, and nobody re-runs the check.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
ESD path verification	3.0	● Final database available from PD-13	S0-10 Waiver review board and foundry waiver alignment
Latch-up verification	2.0	● ESD strategy fixed by PDK-07	T0-02 Final full-chip verification re-run on the released database
Closure and waiver disposition	2.0	● Qualification reliability requirements known	MP-03 ESD (HBM, CDM) and latch-up qualification
Soft error and FIT estimation	1.5		MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Requirement setup	1.5	EXIT CRITERIA	
Total	10.0	● ESD paths traced through the full chip, not assumed from strategy	
WHO IT TAKES	~7.2 FTE	● Latch-up checked internally as well as at the ring	MEASURED BY
Reliability signoff engineer		● FIT estimated against the product requirement	● ESD paths verified against IO count
Owns ESD, latch-up and FIT verification			● Latch-up violations at freeze
ESD engineer		DEPENDS ON	● FIT estimate against requirement
Full-chip discharge path analysis		PD-13 Final turn on the FFN — full closure, no functional change admitted	
Physical verification engineer		PDK-02 Design rule manual review; restricted and recommended rule disposition	
Rule-based latch-up and guard checking		PDK-07 IO, ESD and latch-up library qualification	
Quality liaison		AMS-14 Reliability checks — EM/IR, aging, ESD, latch-up	
Qualification requirement alignment			
Layout engineers			
Fixes to paths and structures			

13 S0 · SIGNOFF

S0-08

DFM and lithography checks, CMP and hotspot analysis

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	10.0 M/M	~7.2 FTE	5	no	DFM engineer
w8-w14 in stage	planned	while it runs	2 in parallel	has float	Yield engineer approves

PURPOSE

Check the design against **manufacturability** — lithography hotspots, CMP density, DFM recommended rules — so the foundry can print it at yield.

A DRC-clean design can still be difficult to manufacture. Litho hotspots print differently from what was drawn, and CMP density variation produces thickness variation that moves timing. Both are checked with foundry-supplied models and both are cheaper to fix here than to yield-learn in production.

WORK FLOW

5 STEPS · 2 LANES

w8w10w12w14

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	DFM deck setup and model version confirmation	1w	main
2	Lithography hotspot detection and analysis	1.5w	main
3	CMP and density simulation	1.5w	parallel
4	Recommended rule compliance scoring	1.5w	parallel
5	Hotspot fixing and DFM closure	3.5w	main

Step 4 quantifies what `PDK-02` decided. Recommended-rule compliance is a score rather than a pass, and it correlates with yield — a design that scored 60% on the rules it chose to ignore is a design whose yield learning curve starts lower.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database from PD-13
- DFM decks and litho models from the foundry
- Rule dispositions from PDK-02
- Fill and density from PD-15
- Defect density model from TECH-07

PRODUCES

- DFM deck setup and model versions
- Litho hotspot analysis results
- CMP and density simulation results
- Recommended rule compliance score
- Hotspot fixes and DFM closure report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D5

PRODUCES

DFM and lithography hotspot report. This activity is the deliverable, and its compliance score is a yield predictor.

S0-D2

FEEDS

Clean DRC / LVS / antenna / density reports. DFM fixes are layout changes and close alongside physical verification.

RISKS AND WATCH-OUTS

- ◆ **DFM checked after routing is frozen.** Hotspot fixes are layout changes, and a frozen database has no room for them.
- ◆ **Recommended rules ignored without scoring.** The yield cost of ignoring them is real and unmeasured unless someone scores compliance.
- ◆ **Litho models outdated.** Foundry models improve through the node's life, and an old model finds the wrong hotspots.
- ◆ **CMP variation not linked to timing.** Thickness variation changes resistance and capacitance, and the timing impact is rarely fed back.
- ◆ **Fixes applied without re-verification.** A DFM fix is a layout change, and it can create DRC or timing problems of its own.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Hotspot fixing and closure	3.5	● Final database available from PD-13	S0-10 Waiver review board and foundry waiver alignment
Litho hotspot analysis	2.5	● DFM decks and current litho models available	T0-02 Final full-chip verification re-run on the released database
CMP and density simulation	2.0	● Fill applied in PD-15	FAB-06 Inline metrology and defect inspection monitoring
Rule compliance scoring	1.0		MP-05 Yield learning, failure pareto and defect analysis
Deck setup	1.0		
Total	10.0		
WHO IT TAKES	~7.2 FTE	EXIT CRITERIA	MEASURED BY
DFM engineer Owns hotspot analysis and DFM closure		● Hotspots fixed or accepted with a yield impact estimate ● Compliance scored rather than assumed ● Fixes re-verified against DRC and timing	● Hotspots remaining at freeze ● Recommended rule compliance score ● Yield impact of accepted hotspots
Physical verification engineer Deck setup and rule scoring			
Layout engineers Hotspot fixing			
Foundry liaison Model versions and hotspot interpretation			
Yield engineer Compliance score against the yield model			
		DEPENDS ON	
		PD-13 Final turn on the FFN — full closure, no functional change admitted	
		PD-15 Chip finishing — seal ring, dummy and metal fill, alignment marks	
		PDK-02 Design rule manual review; restricted and recommended rule disposition	
		TECH-07 Foundry roadmap and risk-production timing alignment	

13 S0 · SIGNOFF

S0-09

Final formal equivalence and LVS netlist consistency

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	8.0 M/M	~7.0 FTE	5	yes	Equivalence engineer
w10–w15 in stage	planned	while it runs	2 in parallel	no float	Signoff lead approves

PURPOSE

Prove for the last time that the **database being taped out is the design that was verified** — final formal equivalence and netlist-to-layout consistency.

Between the FFN and the tapeout database sit hundreds of ECOs, a scan reorder, fill and finishing. Each was individually correct; the question is whether the composition still implements the RTL that DV signed off. This is the last check that answers it.

WORK FLOW

5 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE

1	2	5
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Final equivalence setup against the FFN and the RTL	1w	main
2	RTL to final netlist equivalence	1.5w	main
3	ECO reconciliation against the change log	1w	parallel
4	Netlist to layout consistency confirmation	1w	parallel
5	Non-equivalence resolution and final report	2.5w	main

Step 3 pairs the formal result with the ECO log from `PD-14`. Equivalence proves the netlists match; reconciliation proves that every difference between them was an ECO somebody approved rather than a change nobody noticed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database and netlist from PD-13
- FFN from SYN-09
- ECO log from PD-14
- LVS results from SO-03
- Equivalence methodology from SYN-10

PRODUCES

- Final equivalence setup
- RTL to final netlist equivalence results
- ECO reconciliation record
- Netlist to layout consistency confirmation
- Final equivalence report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D6

PRODUCES

Final formal equivalence report. This activity is the deliverable, and it is the last evidence that verification's conclusions still apply.

S0-D7

GATES

Signoff summary and Design Freeze package. Freezing a database that has not been proved equivalent freezes an unverified design.

RISKS AND WATCH-OUTS

- ◆ **Equivalence run against the FFN rather than the RTL.** That proves the ECOs preserved the netlist and not that the netlist implements the verified design.
- ◆ **ECO log not reconciled.** A difference that no ECO explains is an unapproved change, and it is exactly what this check exists to find.
- ◆ **Non-equivalences waived under tapeout pressure.** Each waiver is a region where the silicon may not do what verification proved.
- ◆ **Run before the last ECO.** Any change after the equivalence run invalidates it, and re-running takes days nobody has.
- ◆ **Layout consistency assumed from LVS.** LVS compares layout to netlist; that the netlist is the right one is a separate claim.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Non-equivalence resolution and report	3.0	● Final database frozen by PD-13 ● ECO log complete from PD-14 ● LVS clean from SO-03	S0-10 Waiver review board and foundry waiver alignment
RTL to netlist equivalence	2.0		T0-02 Final full-chip verification re-run on the released database
ECO reconciliation	1.5		T0-03 Tapeout checklist completion and owner sign-off
Setup	1.0	EXIT CRITERIA	MEASURED BY
Layout consistency	0.5	● Equivalence proved against the verified RTL, not only the FFN ● Every netlist difference traced to an approved ECO ● Run performed after the last database change	● Non-equivalences at the final check ● Netlist differences traced to approved ECOs ● Database changes after the equivalence run
Total	8.0	DEPENDS ON	
WHO IT TAKES	~7.0 FTE	S0-03 Full-chip DRC, LVS, antenna and density verification	
Equivalence engineer Owns the final check and its report		PD-13 Final turn on the FFN — full closure, no functional change admitted	
Physical design lead ECO log and database provenance		PD-14 Functional and timing ECO implementation	
Verification lead Confirms the RTL reference is the verified one		SYN-09 FFN (final full netlist) release — RTL frozen, ECOs closed, no functional change beyond this point	
Physical verification engineer LVS and layout consistency		SYN-10 Formal equivalence checking per drop, RTL versus netlist	
Signoff lead Waiver decisions and closure			

13 S0 · SIGNOFF

S0-10

Waiver review board and foundry waiver alignment

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	6.0 M/M	~4.3 FTE	6	yes	Signoff lead
w10–w16 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Run the **waiver review board** — every violation that survives signoff, examined, owned and either fixed or accepted — and assemble the Design Freeze package.

Every large design tapes out with waivers. What distinguishes a controlled tapeout from a hopeful one is whether each waiver was understood, quantified and signed, or simply ran out of time. This board is where that distinction is made, and its package is what the Go / No-Go decision reads.

WORK FLOW

6 STEPS · 2 LANES

w10	w12	w14	w16
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Waiver intake and classification across all signoff domains	1w	main
2	Root cause and risk quantification per waiver	1.5w	main
3	Foundry waiver alignment and submission	1.5w	parallel
4	Board review and disposition	1.5w	main
5	Residual risk statement	1w	parallel
6	Signoff summary and Design Freeze package assembly	2w	main

Step 3 has a duration outside the program's control. Foundry waivers are reviewed on the foundry's schedule, and a design that assumes approval and does not receive it cannot release masks — which makes early submission worth more than a complete submission.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Waivers from every signoff activity
- Timing waivers from S0-02
- Physical verification waivers from S0-03
- Reliability and DFM waivers from S0-07 and S0-08
- Foundry waiver process from TECH-06

PRODUCES

- Classified waiver register
- Root cause and risk quantification per waiver
- Foundry waiver submissions and responses
- Board disposition records
- Residual risk statement
- Signoff summary and Design Freeze package

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D7

PRODUCES

Signoff summary and Design Freeze package. This activity is the deliverable, and it is what the tapeout decision is taken on.

S0-D1

FEEDS

STA signoff reports with waiver list. The timing waiver list is dispositioned here rather than inside the STA activity.

RISKS AND WATCH-OUTS

- ◆ **Waivers accumulated rather than reviewed.** A register nobody dispositioned is a set of accepted risks nobody accepted.
- ◆ **Foundry waivers submitted late.** Their review is on the foundry's schedule, and an unapproved waiver blocks mask release.
- ◆ **Risk quantified qualitatively.** "Low risk" is not a basis for a tapeout decision; a stated failure mode and probability is.
- ◆ **Board without authority.** A review that cannot refuse a waiver is a review that approves everything.
- ◆ **Residual risk not summarised.** The Go / No-Go decision needs one statement of what is being accepted, not a register of two hundred entries.

EFFORT SPLIT	6 M/M	EXIT CRITERIA	FEEDS INTO
Package assembly	1.5	● Every waiver has a root cause, a quantified risk and a signature ● Foundry waivers submitted and answered before freeze ● Residual risk summarised in one statement for the gate	T0-03 Tapeout checklist completion and owner sign-off
Root cause and quantification	1.5		T0-04 Open-issue risk assessment and waiver acceptance
Board review and disposition	1.5		T0-05 Go / No-Go decision meeting
Foundry alignment	1.0		MP-09 Production readiness review and change control (PCN) setup
Residual risk statement	0.5		
Total	6.0	DEPENDS ON	MEASURED BY
WHO IT TAKES	~4.3 FTE	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn	● Waivers dispositioned against raised ● Foundry waivers answered before freeze ● Waivers with a quantified risk
Signoff lead Chairs the board and owns the package		S0-03 Full-chip DRC, LVS, antenna and density verification	
Domain signoff owners Present and justify their waivers		S0-04 EM and IR-drop signoff, static and dynamic	
Foundry liaison Waiver submission and response		S0-05 Power and signal integrity signoff — crosstalk, noise	
Chief architect Product impact of accepted risk		S0-07 Reliability verification — ESD, latch-up, soft error / FIT	
Program manager Carries the residual risk into the Go / No-Go		S0-08 DFM and lithography checks, CMP and hotspot analysis	
		S0-09 Final formal equivalence and LVS netlist consistency	
ENTRY CRITERIA		● Signoff activities producing waiver candidates ● Foundry waiver process understood from TECH-06 ● Board membership named with authority to refuse	

13 S0 · SIGNOFF

S0-11

Gate-level simulation with final SDF

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	8.0 M/M	~7.0 FTE	5	no	Gate-level simulation engineer
w10–w15 in stage	planned	while it runs	2 in parallel	has float	Signoff lead approves

PURPOSE

Run **gate-level simulation with the final SDF** — the last dynamic check that the design behaves correctly with real timing on the database being taped out.

Static timing analysis proves paths meet constraints; it does not prove the design works. Simulating with final timing catches what STA structurally cannot — race conditions, reset behaviour under real delays, and any place where the constraints described a design different from the one built.

WORK FLOW

5 STEPS · 2 LANES

w10	w11	w12	w13	w14	w15
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MAIN SEQUENCE

1	2	5
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Final SDF generation and simulation setup	1w	main
2	Timing-annotated functional simulation	1.5w	main
3	Reset and initialisation sequence simulation	1w	parallel
4	Test mode simulation with final timing	1w	parallel
5	Failure debug and signoff reporting	2.5w	main

Step 3 is the check most likely to find something. Reset behaviour depends on relative delays across the die, and a sequence that works with estimated timing can fail with the real numbers — a failure that would otherwise appear at first power-on.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final database and SDF from PD-13 and PD-09
- Gate-level environment from DV-11
- Test patterns and modes from DFT-08
- Timing signoff results from SO-02
- Power intent from SYN-07

PRODUCES

- Final SDF and simulation setup
- Timing-annotated functional simulation results
- Reset and initialisation results
- Test mode simulation results
- Gate-level signoff report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

S0-D7	FEEDS	Signoff summary and Design Freeze package. Dynamic evidence on the final database belongs in the freeze package alongside the static analysis.
S0-D1	FEEDS	STA signoff reports. A gate-level failure with clean STA means the constraints described a different design, which is a timing finding.

RISKS AND WATCH-OUTS

- ◆ **Skipped because STA is clean.** They answer different questions, and the failures gate-level simulation finds are invisible to static analysis.
- ◆ **Run on an earlier database.** A simulation against pre-final timing says nothing about the database being taped out.
- ◆ **Reset simulation omitted.** Reset ordering under real delays is a common silicon failure and one of the cheapest to catch here.
- ◆ **X failures forced away.** Initialising signals to make the simulation run hides exactly the uninitialised state that will fail at power-on.
- ◆ **Runtime prohibiting meaningful scenarios.** Full-timing simulation is slow, and a scenario set trimmed to fit may exclude the case that matters.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	DEPENDS ON
Failure debug and reporting	3.0	● Final database and SDF available from PD-13 and PD-09 ● Gate-level environment available from DV-11 ● Test patterns available from DFT-08	PD-09Multi-corner multi-mode timing closure
Timing-annotated simulation	2.0		PD-13Final turn on the FFN — full closure, no functional change admitted
Reset and initialisation simulation	1.5		DV-11Gate-level simulation, functional and timing-annotated
Test mode simulation	1.0		DFT-08Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
SDF and setup	0.5		
Total	8.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~7.0 FTE	● Simulated against the final SDF, not an earlier one ● Reset and initialisation simulated with real delays ● X failures fixed rather than forced	S0-10Waiver review board and foundry waiver alignment
Gate-level simulation engineer			T0-02Final full-chip verification re-run on the released database
Owens the runs and the report			BU-02Power-on, power sequencing and basic health check
Timing engineer			
SDF generation and timing interpretation			
Verification engineers			MEASURED BY
Failure debug			● Scenarios simulated with final timing ● Failures found that STA could not show ● X issues fixed against forced
DFT engineer			
Test mode simulation			
Signoff lead			
Accepts the dynamic evidence into the package			

13 S0 · SIGNOFF

S0-12

Signoff corner correlation against foundry decks

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	6.0 M/M	~5.2 FTE	5	no	Signoff methodology engineer
w2–w7 in stage	planned	while it runs	2 in parallel	has float	Foundry liaison approves

PURPOSE

Correlate the **signoff flow against the foundry's own decks and the implementation tool**, so that a number produced here means the same thing everywhere it is read.

Three parties analyse this design: the implementation tool, the signoff tool and the foundry. If they disagree, the design can be closed in one and rejected by another. Establishing the correlation early is what lets physical design close with the right margin instead of re-closing after signoff reports.

WORK FLOW

5 STEPS · 2 LANES

w2	w3	w4	w5	w6	w7
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MAIN SEQUENCE

1	2	5
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Correlation methodology and reference case selection	1w	main
2	Implementation to signoff tool correlation	1.5w	main
3	Foundry deck and derate correlation	1w	parallel
4	Extraction and parasitic correlation	1w	parallel
5	Correlation findings and margin guidance	2.5w	main

Step 5 is the output physical design actually uses. Knowing that signoff reads 40 ps more pessimistic than implementation on long paths lets `PD-09` close with that margin built in, rather than closing to zero and discovering the gap at `SO-02`.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Implementation timing results from PD-09
- Signoff flow from SO-01
- Foundry decks and derates from PDK-11
- Extraction decks from PDK-10
- Library models from PDK-03

PRODUCES

- Correlation methodology and reference cases
- Implementation to signoff correlation results
- Foundry deck and derate correlation
- Extraction and parasitic correlation
- Margin guidance for physical design

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SO-01	FEEDS	STA signoff reports. Correlation is what makes the signoff numbers comparable to the ones physical design closed against.
SO-07	FEEDS	Signoff summary and Design Freeze package. A stated correlation is part of what makes the signoff result defensible to the foundry.

RISKS AND WATCH-OUTS

- ◆ **Correlation established after signoff starts.** The gap is then discovered on the final database, when closing it means re-opening closure.
- ◆ **Correlation on unrepresentative paths.** Tool differences concentrate on long, heavily coupled paths, and correlating on short clean ones shows agreement that does not generalise.
- ◆ **Extraction differences overlooked.** Parasitic differences between flows can exceed timing tool differences and are less often checked.
- ◆ **Margin guidance not adopted.** Correlation that physical design does not close against has produced information nobody used.
- ◆ **Foundry decks assumed identical to internal ones.** Derates and corner definitions can differ in detail, and the foundry's version is the one that decides acceptance.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	DEPENDS ON
Correlation findings and guidance	2.0	● Implementation timing available from PD-09 ● Signoff flow assembled in SO-01 ● Foundry decks and derates available from PDK-11	SO-01 Signoff flow dry run on the Turn 2 database — decks, runtimes, capacity and violation triage rehearsed
Implementation to signoff correlation	1.5		PD-09 Multi-corner multi-mode timing closure
Extraction correlation	1.0		PDK-03 Standard cell library selection and qualification (track height, multi-Vt, multi-bit flops)
Foundry deck correlation	1.0		PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control
Methodology and reference cases	0.5		PDK-11 Signoff corner and derate definition agreed with foundry (PVT, OCV/AOCV/POCV)
Total	6.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~5.2 FTE	● Correlation established on representative, difficult paths ● Margin guidance published and adopted by physical design ● Foundry deck differences understood, not assumed away	SO-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Signoff methodology engineer			PD-09 Multi-corner multi-mode timing closure
Owns correlation and margin guidance			T0-02 Final full-chip verification re-run on the released database
STA engineers			
Reference case analysis in both flows			
Extraction engineer			
Parasitic correlation			
Physical design liaison			
Adopts margin guidance into closure			
Foundry liaison			
Deck and derate alignment			
		MEASURED BY	
			● Correlation gap on representative paths ● Margin guidance adopted in closure ● Surprises at first signoff run

MANUFACTURE

14

Tapeout & Mask Release (MTO)

T0

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w78 – w86	8 _w	11	47 _{M/M}	8 of 11

ENGINEERING ACTIVITIES					KEY DELIVERABLES	
T0-01	Final GDSII / OASIS assembly on the FFN closure database, with layer map verification		2w	6 _{M/M}	T0-D1	Released GDSII / OASIS database with checksum record
T0-02	Final full-chip verification re-run on the released database		2w	8 _{M/M}	T0-D2	Tapeout checklist with sign-off matrix
T0-03	Tapeout checklist completion and owner sign-off		3w	5 _{M/M}	T0-D3	Open issue and risk acceptance record
T0-04	Open-issue risk assessment and waiver acceptance		2w	3 _{M/M}	T0-D4	Go / No-Go decision minutes
T0-05	Go / No-Go decision meeting		0.5w	1 _{M/M}	T0-D5	FEOL MTO release package and mask order confirmation
T0-06	FEOL layer data preparation and MTO release to the mask shop		1.5w	3 _{M/M}	T0-D6	BEOL ECO log covering the fix window
T0-07	FEOL mask order confirmation and mask shop scheduling		1w	1 _{M/M}	T0-D7	BEOL MTO release package and mask order confirmation
T0-08	BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut		4w	10 _{M/M}	T0-D8	Full mask set completion record
T0-09	BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack		2.5w	6 _{M/M}		
T0-10	BEOL layer data preparation and MTO release		1.5w	3 _{M/M}		
T0-11	BEOL mask order confirmation and full mask set completion tracking		1w	1 _{M/M}		

14 T0 · TAPEOUT & MASK RELEASE (MT0)

T0-01

Final GDSII / OASIS assembly on the FFN closure database, with layer map verification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	6.0 M/M	~13.0 FTE	4	yes	Tapeout engineer
w0-w2 in stage	planned	while it runs	1 in parallel	no float	Foundry liaison approves

PURPOSE

Assemble the **final GDSII or OASIS database** from the closed design — every block, every macro, every layer — and verify that the layer map the foundry will read matches the one the design was built with.

This is a mechanical activity with no tolerance for error. A layer misassigned in the stream-out produces masks that are wrong in a way no verification would have caught, because the design was correct and the translation was not.

WORK FLOW4 STEPS · 2 LANES

w0	w1	w2
----	----	----

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Closure database intake and provenance check	0.5w	main
2	GDSII / OASIS stream out from the closed database	0.8w	main
3	Macro and hard IP GDS merge	0.5w	parallel
4	Layer map verification against the foundry definition	0.8w	main

Step 4 is checked against the foundry's layer map rather than the internal one. The two are usually identical and occasionally are not, and the difference is only visible at the mask shop — by which point the masks exist.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Frozen signoff database from PD-13 and PD-16
- Hard macro GDS from AMS-15
- Foundry layer map and stream-out rules
- Chip finishing from PD-15
- Signoff clean status from SO-03

PRODUCES

- Streamed GDSII / OASIS database
- Merged macro and IP layers
- Layer map verification record
- Database checksum and provenance record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D1

PRODUCES

Released GDSII / OASIS database with checksum record. This activity is the deliverable, and the checksum is what makes every later reference unambiguous.

T0-D2

FEEDS

Tapeout checklist with sign-off matrix. Database assembly and layer verification are checklist entries in their own right.

RISKS AND WATCH-OUTS

- ◆ **Layer map mismatch.** A misassigned layer produces masks that are wrong in a way no design verification would have found.
- ◆ **Macro GDS from the wrong version.** An abstract that matched and a layout that did not is discovered at LVS, or worse, not at all.
- ◆ **Stream-out from an unfrozen database.** The design has to stop moving before it is streamed, or the streamed version is not the signed-off one.
- ◆ **No checksum.** Without one, the question "is this the database we signed off" has no answer.
- ◆ **Assembly compressed to hours.** Stream-out on a die this size takes real time, and rushing it is how a merge error is missed.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Stream-out	2.5	● Signoff database frozen by PD-13	T0-02 Final full-chip verification re-run on the released database
Layer map verification	1.5	● Physical verification clean from SO-03	T0-03 Tapeout checklist completion and owner sign-off
Macro merge	1.0	● Foundry layer map and format confirmed	T0-06 FEOL layer data preparation and MTO release to the mask shop
Intake and provenance	1.0		T0-08 BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
Total	6.0	EXIT CRITERIA ● Layer map verified against the foundry's definition, not the internal one ● Every macro GDS traced to its released version ● Database checksummed and recorded	MEASURED BY ● Layer map discrepancies found ● Macro versions traced to release ● Stream-out reruns required
WHO IT TAKES	~13.0 FTE	DEPENDS ON PD-13 Final turn on the FFN — full closure, no functional change admitted PD-15 Chip finishing — seal ring, dummy and metal fill, alignment marks PD-16 Hierarchical block closure and top-level assembly SO-03 Full-chip DRC, LVS, antenna and density verification AMS-15 Hard macro abstraction and view generation (LEF, .lib, CDL, GDS, behavioural model)	
Tapeout engineer Owns assembly, stream-out and provenance			
Physical design lead Confirms the database is the frozen one			
Macro owners GDS versions for hard IP			
CAD engineer Stream-out flow and layer map configuration			
Foundry liaison Layer map and format requirements			

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-02

Final full-chip verification re-run on the released database

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	8.0 M/M	~17.4 FTE	5	yes	Physical verification engineer
w0–w2 in stage	planned	while it runs	2 in parallel	no float	Foundry liaison approves

PURPOSE

Re-run **full-chip physical verification on the released database** — not on the one signoff checked, but on the exact bytes going to the mask shop.

Between the signoff run and the released database sit stream-out, macro merge and possibly a late fix. Each is a chance to introduce something. This re-run is short, expensive in compute and non-negotiable: it is the only check that the thing being released is the thing that was verified.

WORK FLOW

5 STEPS · 2 LANES

w0w1w2

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Verification setup on the released database	0.5w	main
2	Full-chip DRC re-run	0.8w	main
3	Density and fill re-check	0.5w	parallel
4	LVS and antenna re-run	0.8w	main
5	Checksum confirmation against the released database	0.5w	parallel

Step 5 sounds pedantic and prevents the worst class of error. Verifying one database and releasing another has happened on real programs, and the only defence is confirming the checksum of what was verified against the checksum of what was sent.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Released database from TO-01
- Verification decks from PDK-10
- Signoff results from SO-03
- Waiver list from SO-10
- Compute capacity for full-chip runs

PRODUCES

- Verification results on the released database
- Density and fill confirmation
- LVS and antenna re-run results
- Checksum confirmation record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TO-01	FEEDS	Released GDSII / OASIS database with checksum record. The release is only complete when the released bytes have been verified.
TO-02	FEEDS	Tapeout checklist with sign-off matrix. Verification on the released database is one of the checklist's non-waivable entries.

RISKS AND WATCH-OUTS

- ◆ **Skipped because signoff was clean.** Signoff checked a different database, and the difference between them is exactly what this run exists to catch.
- ◆ **Run on the signoff database rather than the released one.** That repeats the earlier check and proves nothing new.
- ◆ **Compute capacity unavailable.** A full-chip run takes days and needs to be scheduled, not requested.
- ◆ **New violations discovered with no time to fix.** Which is why S0-03 has to close cleanly rather than nearly.
- ◆ **Checksums not compared.** Verifying one file and releasing another is a documented failure mode with a trivial defence.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	DEPENDS ON
DRC re-run	3.0	- Database released by TO-01 - Decks versioned and matching the submission - Compute capacity scheduled	TO-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
LVS and antenna re-run	2.5		S0-03 Full-chip DRC, LVS, antenna and density verification
Setup	1.0		S0-09 Final formal equivalence and LVS netlist consistency
Density re-check	1.0		PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control
Checksum confirmation	0.5		
Total	8.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~17.4 FTE	- Verification run on the exact released database - Checksums confirmed identical between verified and released - No violations outstanding that were not already waived	TO-03 Tapeout checklist completion and owner sign-off
Physical verification engineer Owns the re-run and its results			TO-05 Go / No-Go decision meeting
Tapeout engineer Database provenance and checksums			TO-06 FEOL layer data preparation and MTO release to the mask shop
CAD infrastructure Compute capacity for full-chip runs			
Signoff lead Disposition of anything newly found			
Foundry liaison Deck versions matching the submission			MEASURED BY
			- New violations found on the released database - Checksum agreement - Run completion before the Go / No-Go

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-03

Tapeout checklist completion and owner sign-off

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	5.0 M/M	~7.2 FTE	4	yes	Program manager
w0-w3 in stage	planned	while it runs	1 in parallel	no float	Package lead approves

PURPOSE

Walk the **tapeout checklist** — every domain, every entry, evidence attached, owner signed — so the Go / No-Go decision is taken on facts rather than on confidence.

The checklist is the program's collected memory of what has to be true before masks are cut. Its value is entirely in the evidence: an entry ticked because the owner believes it is true is worth nothing, and the difference between a checklist and a ritual is whether anyone reads the attachments.

WORK FLOW

4 STEPS · 2 LANES

w0	w1	w2	w3
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Checklist assembly per domain	0.8w	main
2	Evidence collection per item	1w	main
3	Gap chasing and escalation	0.8w	parallel
4	Owner sign-off collection	1.3w	main

Step 3 runs in parallel because it is where the time goes. Chasing an owner who has not signed, or an entry whose evidence does not exist, is the real work — the checklist itself takes an afternoon.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Signoff results from SO-02 through SO-11
- DFT signoff from DFT-08
- DV closure from DV-12
- Verification results from TO-02
- Waiver dispositions from SO-10

PRODUCES

- Assembled tapeout checklist
- Evidence attached per item
- Gap and escalation log
- Owner sign-off matrix

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-02

PRODUCES

Tapeout checklist with sign-off matrix. This activity is the deliverable, and the sign-off matrix is what makes accountability explicit.

T0-D4

GATES

Go / No-Go decision minutes. The decision reads the checklist; entries without evidence are decisions being taken blind.

RISKS AND WATCH-OUTS

- ◆ **Entries ticked without evidence.** A checklist of assertions is a ritual, and it produces confidence rather than information.
- ◆ **Sign-off by proxy.** An entry signed by someone other than the accountable owner has moved the signature without moving the accountability.
- ◆ **Checklist assembled at tapeout.** It should be maintained across the program; assembling it in the last week means discovering gaps with no time to close them.
- ◆ **Domains missing from the checklist.** Test, package and reliability entries are easy to omit from a design-centred checklist and expensive to miss.
- ◆ **Gaps escalated too late.** An unsigned entry three days before the Go / No-Go is a decision the meeting cannot take.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	DEPENDS ON
Evidence collection	2.0	● Signoff activities producing evidence	T0-01Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
Sign-off collection	1.5	● Checklist template maintained across the program	T0-02Final full-chip verification re-run on the released database
Checklist assembly	1.0	● Domain owners identified	S0-10Waiver review board and foundry waiver alignment
Gap escalation	0.5		DFT-08Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)
Total	5.0	EXIT CRITERIA	DV-12Bug triage and disposition board
WHO IT TAKES	~7.2 FTE	● Every entry carries evidence, not an assertion	FEEDS INTO
Program manager		● Signatures from accountable owners, not proxies	T0-04Open-issue risk assessment and waiver acceptance
Owns the checklist and the sign-off matrix		● Gaps escalated with time to close them	T0-05Go / No-Go decision meeting
Domain owners			MEASURED BY
Sign their own entries with evidence			● Entries with attached evidence
Signoff lead			● Signatures from accountable owners
Design-side evidence			● Gaps open at the Go / No-Go
DFT and test leads			
Test-side entries			
Package lead			
Package and assembly entries			

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-04

Open-issue risk assessment and waiver acceptance

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	3.0 M/M	~6.5 FTE	4	no	Program manager
w1–w3 in stage	planned	while it runs	1 in parallel	has float	Chief architect approves

PURPOSE

Take every open issue the program is carrying — **bugs, waivers, unclosed items** — and turn it into an accepted risk with a name against it.

No large program tapes out with nothing open. The question is whether what remains was assessed and accepted, or simply ran out of time. This activity converts a list of loose ends into a set of decisions, each with an owner, a consequence and a mitigation.

WORK FLOW

4 STEPS · 2 LANES

w1	w2	w3
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Open item consolidation across design, verification and test	0.5w	main
2	Risk classification and consequence assessment	0.8w	main
3	Mitigation options per item	0.5w	parallel
4	Risk acceptance and owner assignment	0.8w	main

Step 3 is what gives the Go / No-Go meeting something to decide. An item with a mitigation — a test screen, a software workaround, a bin split — is a different decision from one without, and identifying which is which is this activity's contribution.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Open bug list and risk statement from DV-12
- Waiver register from SO-10
- Test coverage gaps from DFT-07
- Package and assembly open items from PKGD-11
- Program risk register

PRODUCES

- Consolidated open item list
- Risk classification with consequences
- Mitigation options per item
- Risk acceptance record with named owners

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D3

PRODUCES

Open issue and risk acceptance record. This activity is the deliverable, and named ownership is what distinguishes acceptance from omission.

T0-D4

FEEDS

Go / No-Go decision minutes. The decision is largely a decision about this list.

RISKS AND WATCH-OUTS

- ◆ **Items carried without assessment.** An open issue nobody classified is a risk being accepted without anyone accepting it.
- ◆ **Risk stated qualitatively.** "Low risk" gives the decision room nothing; a failure mode, a probability and an affected population gives it a decision.
- ◆ **Mitigations not identified.** Many silicon risks can be screened at test or worked around in software, and the option is worth knowing before the gate.
- ◆ **Test and package items excluded.** Risk consolidation that covers only design misses the categories that most often cause a late problem.
- ◆ **Acceptance without an owner.** An accepted risk with no name against it has nobody watching for it at bring-up.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Risk classification	1.3	Bug and waiver lists available from DV-12 and SO-10	T0-05 Go / No-Go decision meeting
Acceptance and ownership	0.8	Test and package open items collected	BU-10 Errata capture, workaround definition and documentation
Consolidation	0.5	Risk classification scheme agreed	MP-09 Production readiness review and change control (PCN) setup
Mitigation options	0.5		
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~6.5 FTE	- Every open item classified with a stated consequence - Mitigation options identified where they exist - Each accepted risk has a named owner	- Items with a quantified consequence - Accepted risks with a named owner - Mitigations identified against items carried
Program manager Owns consolidation and the acceptance record		DEPENDS ON	
Verification lead Bug risk and consequence		DV-12 Bug triage and disposition board	
Signoff lead Waiver risk		SO-10 Waiver review board and foundry waiver alignment	
Product engineering Screening and mitigation options at test		DFT-07 ATPG pattern generation — stuck-at, transition, cell-aware — and coverage closure	
Chief architect Product impact of each accepted risk			

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-05

Go / No-Go decision meeting

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
0.5w	1.0 M/M	~8.7 FTE	3	yes	Executive sponsor
w1.5-w2 in stage	planned	while it runs	0 in parallel	no float	Product engineering approves

PURPOSE

Hold the **Go / No-Go decision** — the moment the program commits several million dollars of masks and a quarter of schedule to the design as it stands.

Half a week of activity and one man-month, preceded by three weeks of preparation. That ratio is correct: a gate argued in the room rather than before it is a gate whose outcome depends on who attended. The meeting exists to take a decision, record its conditions and name who owns them.

WORK FLOW3 STEPS · 1 LANE

w1.5w2

MAIN SEQUENCE

#	STEP	TAT	LANE
1	Decision pack presentation — checklist, risks, readiness	0.2w	main
2	Decision, conditions and dissent capture	0.2w	main
3	Minutes, distribution and condition assignment	0.1w	main

Step 2 captures dissent deliberately. The objection nobody minuted is the one that resurfaces at first silicon with nothing to point at, and recording it costs a line.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Tapeout checklist from TO-03
- Risk acceptance record from TO-04
- Verification results from TO-02
- Foundry readiness from TECH-05
- Program schedule position from DEF-08

PRODUCES

- Decision pack
- Go / No-Go decision with conditions
- Dissent record
- Minutes and condition assignments

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D4	PRODUCES	Go / No-Go decision minutes. This activity is the deliverable, and the conditions attached are as important as the decision itself.
T0-D5	GATES	FEOL MTO release package. Mask data is not released before this decision, whatever the schedule pressure.

RISKS AND WATCH-OUTS

- ◆ **Decision taken because the date arrived.** A gate that cannot say no is not a gate, and the organisation learns that quickly.
- ◆ **Conditions accepted verbally.** A Go granted "subject to" something unrecorded is an unconditional Go.
- ◆ **Wrong people in the room.** The decision needs someone who can say no and someone who owns the consequence, and both have to attend.
- ◆ **Dissent unrecorded.** It is the cheapest insurance available and it is routinely omitted.
- ◆ **Material presented for the first time in the room.** A pack circulated on the day produces a decision based on whoever speaks most confidently.

EFFORT SPLIT	1 M/M	ENTRY CRITERIA	FEEDS INTO
Decision pack presentation	0.4	● Checklist complete with signatures from TO-03	TO-06 FEOL layer data preparation and MTO release to the mask shop
Decision and conditions	0.4	● Risk acceptance record available from TO-04	TO-08 BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
Minutes and distribution	0.2	● Decision pack circulated in advance	
Total	1.0		
WHO IT TAKES	~8.7 FTE	EXIT CRITERIA	MEASURED BY
Executive sponsor Chairs the decision and owns the outcome		● Decision recorded with conditions and their owners	
Program manager Presents the pack and records the minutes		● Dissent captured in the minutes	● Conditions attached and assigned
Chief architect Design readiness and product risk		● Conditions assigned before mask data is released	● Dissent recorded
Signoff lead Signoff status and residual risk			● Days between pack circulation and decision
Product engineering Manufacturability and screening options			
		DEPENDS ON	
		TO-02 Final full-chip verification re-run on the released database	
		TO-03 Tapeout checklist completion and owner sign-off	
		TO-04 Open-issue risk assessment and waiver acceptance	

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-06

FEOL layer data preparation and
MTO release to the mask shop

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
1.5w	3.0 M/M	~8.7 FTE	3	yes	Tapeout engineer
w2-w3.5 in stage	planned	while it runs	0 in parallel	no float	Procurement approves

PURPOSE

Prepare and release the **front-end layer data** to the mask shop — the first of the two mask tape-outs, and the one that starts the fab clock.

FEOL goes first because it is the longest lead item and the least likely to change. Releasing it now starts mask writing while the back-end layers are still absorbing late fixes, which is the entire reason the two releases are separated.

WORK FLOW3 STEPS · 1 LANE

w2w3w3.5

MAIN SEQUENCE

#	STEP	TAT	LANE
1	FEOL layer selection and data preparation	0.5w	main
2	Mask data preparation handoff — fracture, OPC scope	0.5w	main
3	Submission to the foundry and acceptance confirmation	0.5w	main

Step 1's layer selection has to be exact. A layer released with FEOL that later needs a fix cannot be fixed in the BEOL window, and a layer held back that the mask shop needed delays the whole set.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Released database from TO-01
- Go / No-Go decision from TO-05
- Foundry submission format and MDP requirements
- Layer split definition from TECH-02
- Mask order and slot from TECH-05

PRODUCES

- FEOL layer data set
- Mask data preparation handoff package
- Submission record and foundry acceptance

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D5PRODUCES**FEOL MTO release package and mask order confirmation.** This activity is the deliverable, and it is what starts the mask and fab clock.

T0-D1FEEDS**Released GDSII / OASIS database.** The FEOL release is a subset of it, and the two must be traceable to the same checksum.

RISKS AND WATCH-OUTS

- ◆ **Layer split wrong.** A layer released as FEOL that later needs a fix cannot use the BEOL window, and the fix costs a new mask.
- ◆ **Released before the Go / No-Go.** Schedule pressure makes this tempting and it removes the gate's authority entirely.
- ◆ **MDP requirements misunderstood.** Fracture and OPC expectations differ per foundry, and a non-compliant submission is rejected.
- ◆ **Submission not acknowledged.** A release nobody confirmed received is a release that may not have started the clock.
- ◆ **Traceability lost between database and submission.** The submitted layers have to be provably from the checksummed database.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Data preparation	1.5	● Go / No-Go decision taken in TO-05	T0-07 FEOL mask order confirmation and mask shop scheduling
MDP handoff	1.0	● Database verified in TO-02	FAB-01 FEOL mask set fabrication, inspection and qualification
Submission and acceptance	0.5	● Mask slot confirmed by TECH-05	T0-08 BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~8.7 FTE	● Layer split matches the FEOL/BEOL plan exactly ● Submission acknowledged by the foundry in writing ● Released layers traceable to the database checksum	● Submission accepted at first attempt ● Release date against the plan ● Layer split errors found after release
Tapeout engineer Owns layer preparation and submission		DEPENDS ON	
Foundry liaison Submission, acceptance and MDP interface		T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification	
Physical verification engineer Layer content confirmation		T0-05 Go / No-Go decision meeting	
Program manager Gate compliance before release		TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement	
Procurement Mask order alignment			

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-07

FEOL mask order confirmation and mask shop scheduling

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
1w	1.0 M/M	~4.3 FTE	3	no	Procurement
w3-w4 in stage	planned	while it runs	0 in parallel	has float	Finance partner approves

PURPOSE

Confirm the **FEOL mask order** and get a mask shop schedule the program can plan wafer start against.

One man-month of administration that fixes a date the whole downstream schedule hangs on. Mask writing takes weeks, and knowing when the FEOL set will be ready is what tells FAB-02 when wafers can start.

WORK FLOW

3 STEPS · 1 LANE

w3

w4

MAIN SEQUENCE

1	2	3
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#	STEP	TAT	LANE
1	Mask order confirmation and purchase authorisation	0.4w	main
2	Mask shop schedule confirmation	0.3w	main
3	Delivery date communication to the program	0.3w	main

Step 3 sounds trivial and is where schedules break. A mask completion date known only to the person who confirmed it is a date the rest of the program plans without.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- FEOL submission record from TO-06
- Mask cost sheet from TECH-04
- Capacity and slot commitment from TECH-05
- Purchase authorisation
- Mask shop schedule

PRODUCES

- Confirmed mask order
- Mask shop schedule with completion date
- Delivery date communication to the program

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-05	FEEDS	FEOL MTO release package and mask order confirmation. The order confirmation half of the release record.
T0-08	FEEDS	Full mask set completion record. FEOL completion is the first half of the set.

RISKS AND WATCH-OUTS

- ◆ **Order placed without confirmed pricing.** Mask cost is a large line item and an unconfirmed order produces an invoice nobody budgeted.
- ◆ **Mask shop schedule assumed.** Mask writing capacity is shared, and the completion date is the shop's to give rather than the program's to assume.
- ◆ **Completion date not communicated.** Wafer start planning depends on it, and a date held by one person is a date the program does not have.
- ◆ **No escalation path.** Mask shop delays happen, and knowing who to call is worth arranging before it is needed.
- ◆ **Order and submission mismatched.** The order has to name the exact layer set that was submitted.

EFFORT SPLIT	1 M/M	ENTRY CRITERIA	FEEDS INTO
Order confirmation	0.5	● FEOL data submitted and accepted in TO-06	FAB-01 FEOL mask set fabrication, inspection and qualification
Schedule confirmation	0.3	● Mask pricing confirmed in TECH-04	FAB-02 Wafer start on FEOL mask availability
Communication	0.2	● Purchase authority available	TO-11 BEOL mask order confirmation and full mask set completion tracking
Total	1.0		
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA	MEASURED BY
Procurement Order confirmation and authorisation		● Order confirmed against the exact submitted layer set	● Days from submission to order confirmation
Foundry liaison Mask shop schedule and escalation		● Mask completion date received and communicated	● Mask completion date against plan
Program manager Communicates the date into the schedule		● Escalation contact identified	● Cost against the mask budget
Tapeout engineer Order and submission consistency			
Finance partner Cost against the mask budget			
		DEPENDS ON	
		TO-06 FEOL layer data preparation and MTO release to the mask shop	
		TECH-04 Wafer price, mask and NRE quotation; MPW and volume pricing	
		TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement	

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-08

BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w w2–w6 in stage	10.0 M/M planned	~10.9 FTE while it runs	6 2 in parallel	yes no float	Physical design lead Signoff lead approves

PURPOSE

Use the **four weeks between the two mask releases** to make metal-layer fixes — the last window in which a design change is possible without a new front-end mask set.

This window is the reason the releases are split. FEOL masks are being cut and cannot change; BEOL is still open, and any fix implementable in metal can still be made. It is a genuine and finite opportunity, and it closes on a date.

WORK FLOW

6 STEPS · 2 LANES

w2	w3	w4	w5	w6
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Fix window scope and admission rules	0.5w	main
2	Metal-layer ECO implementation	1.5w	main
3	Fix prioritisation and risk assessment	1w	parallel
4	ECO verification per fix	1w	main
5	Equivalence and consistency check per fix	1w	parallel
6	Window close and BEOL database freeze	1w	main

Step 1 has to be strict. Every late fix carries risk, and a window with no admission rules fills with changes that individually look safe and collectively make the BEOL database something nobody has verified as a whole.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Released database from TO-01
- Late findings from bring-up preparation, verification and signoff
- Spare cell and metal-only strategy from PD-14
- Go / No-Go conditions from TO-05
- Equivalence methodology from SO-09

PRODUCES

- Fix window scope and admission rules
- Implemented metal-layer ECOs
- Fix prioritisation and risk record
- Per-fix verification results
- Frozen BEOL database

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D6	PRODUCES	BEOL ECO log covering the fix window. This activity is the deliverable, and the log is what makes the BEOL database traceable.
T0-D7	FEEDS	BEOL MTO release package. The database released to BEOL is the FEOL database plus this window's fixes.

RISKS AND WATCH-OUTS

- ◆ **Window used as general contingency.** Fixes admitted without rules accumulate into a database that differs from the verified one in ways nobody assessed as a whole.
- ◆ **Fix requiring base layers.** If PD-14's spare cell strategy was not put in place, the fix cannot be made in metal and the window is useless.
- ◆ **Verification per fix but not in combination.** Individually correct changes can interact, and the combined database needs a check of its own.
- ◆ **Window extended.** BEOL mask release has a date; extending the window pushes the whole mask set and with it wafer start.
- ◆ **Equivalence skipped for speed.** A metal ECO can change function, and the only detector is an equivalence run.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	DEPENDS ON
ECO implementation	4.0	● FEOL released in T0-06	T0-01 Final GDSII / OASIS assembly on the FFN closure database, with layer map verification
ECO verification	2.5	● Admission rules agreed before the window opens	T0-05 Go / No-Go decision meeting
Prioritisation and risk	1.5	● Spare cell capacity available from PD-14	T0-06 FEOL layer data preparation and MTO release to the mask shop
Equivalence checks	1.5		PD-14 Functional and timing ECO implementation
Window close	0.5		
Total	10.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~10.9 FTE	● Every fix implementable in metal, none requiring base layers	T0-09 BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack
Physical design lead Owns implementation and the window		● Combined database verified, not only individual fixes	T0-10 BEOL layer data preparation and MTO release
Program manager Admission rules and prioritisation		● Window closed on its date, not extended	BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule
Verification lead Per-fix and combined verification			MEASURED BY
Equivalence engineer Equivalence per fix and at window close			● Fixes admitted against fixes requested
Signoff lead Re-verification scope for changed layers			● Fixes implementable in metal
			● Window closed on the planned date

14 TO · TAPEOUT & MASK RELEASE (MT0)

T0-09

BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2.5w	6.0 M/M	~10.4 FTE	4	yes	Physical verification engineer
w3.5–w6 in stage	planned	while it runs	1 in parallel	no float	Signoff lead approves

PURPOSE

Re-verify the **metal stack after the fix window** — DRC, LVS, antenna, density on the changed layers — before the back-end data is released.

The BEOL database is not the one that was signed off. It carries the fix window's changes, and every one of them touched routing. This re-verification is what makes the BEOL release as trustworthy as the FEOL one, and it has to fit inside the window's tail.

WORK FLOW

4 STEPS · 2 LANES

w3.5

w4.5

w5.5

w6

MAIN SEQUENCE

1

2

4

IN PARALLEL

3

#	STEP	TAT	LANE
1	Re-verification scope on the changed layers	0.5w	main
2	DRC and antenna verification on the metal stack	1w	main
3	Timing impact confirmation for the applied fixes	0.8w	parallel
4	LVS and density re-check	1w	main

Step 1 decides between a targeted re-run and a full one. Targeted is faster and risks missing an interaction; full is safe and may not fit the window. The scope decision has to be made on what the fixes actually touched.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Frozen BEOL database from TO-08
- ECO log from TO-08
- Verification decks from PDK-10
- Timing analysis capability from SO-02
- Signoff baseline from SO-03

PRODUCES

- Re-verification scope decision
- DRC and antenna results on the metal stack
- Timing impact confirmation
- LVS and density re-check results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D7	FEEDS	BEOL MTO release package. The release is only defensible if the changed layers were re-verified.
T0-D6	FEEDS	BEOL ECO log. The log and the re-verification results are read together — what changed, and what was checked afterwards.

RISKS AND WATCH-OUTS

- ◆ **Scope too narrow.** A targeted re-run that misses an interaction between two fixes releases a violation into the mask set.
- ◆ **Timing not re-confirmed.** Metal ECOs change delay, and a fix that breaks timing has replaced one problem with another.
- ◆ **Re-verification not fitting the window.** Full-chip runs take days, and a window planned without them cannot accommodate them.
- ◆ **LVS skipped because the change was metal-only.** Metal changes connectivity, which is exactly what LVS checks.
- ◆ **Results arriving after the release.** Verifying after submission is verifying something already committed to masks.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	DEPENDS ON
DRC and antenna verification	2.5	● BEOL database frozen by TO-08	T0-08 BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut
LVS and density re-check	2.0	● ECO log complete	
Scope definition	0.8	● Verification decks and capacity available	PDK-10 Rule deck, QRC tech file and DRC/LVS deck version control
Timing confirmation	0.8		
Total	6.0	EXIT CRITERIA	S0-02 Multi-corner multi-mode static timing analysis and closure on the final turn
WHO IT TAKES	~10.4 FTE	● Scope justified against what the fixes touched	FEEDS INTO
Physical verification engineer Owns the re-verification		● Timing re-confirmed for every applied fix	T0-10 BEOL layer data preparation and MTO release
Physical design lead Scope decision based on what changed		● Results complete before BEOL data is released	
Timing engineer Timing impact of the applied fixes			MEASURED BY
Tapeout engineer Release gating on the results			● Violations found after the fix window
Signoff lead Disposition of anything newly found			● Timing changes introduced by fixes
			● Re-verification completion before release

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-10

BEOL layer data preparation and MTO release

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
1.5w	3.0 M/M	~8.7 FTE	3	yes	Tapeout engineer
w6–w7.5 in stage	planned	while it runs	0 in parallel	no float	Configuration manager approves

PURPOSE

Prepare and release the **back-end layer data** — the second mask tape-out, and the moment the design is fully committed to silicon.

BEOL release is the real tapeout. After it nothing about the design can change without a new mask set, and the milestone the program has been working toward for eighty weeks is this submission being accepted.

WORK FLOW

3 STEPS · 1 LANE

w6

w7

w7.5

MAIN SEQUENCE

#	STEP	TAT	LANE
1	BEOL layer data preparation from the re-verified database	0.5w	main
2	Mask data preparation handoff	0.5w	main
3	Submission to the foundry and acceptance confirmation	0.5w	main

The layer set here has to be exactly the complement of what `TO-06` released. A layer released twice or not at all produces a mask set that is either wrong or incomplete, and the mask shop will not necessarily notice.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Re-verified BEOL database from TO-09
- FEOL layer split from TO-06
- Foundry submission requirements
- Mask order framework from TO-07
- ECO log from TO-08

PRODUCES

- BEOL layer data set
- Mask data preparation handoff package
- Submission record and foundry acceptance

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TO-D7	PRODUCES	BEOL MTO release package and mask order confirmation. This activity is the deliverable, and the milestone the stage carries — Tapeout.
TO-D6	FEEDS	BEOL ECO log. The released database is the FEOL one plus the logged fixes, and the two travel together.

RISKS AND WATCH-OUTS

- ◆ **Layer set not complementary to FEOL.** A layer released twice or omitted produces an incomplete or contradictory mask set.
- ◆ **Released before re-verification completes.** The fix window's changes are then committed unchecked.
- ◆ **Submission rejected on format.** BEOL layers have their own OPC and fracture requirements, and a rejection costs days against a running FEOL clock.
- ◆ **No acceptance confirmation.** An unconfirmed release may not have started the second mask clock.
- ◆ **Traceability broken.** The BEOL database differs from the checksummed original, and the difference has to be exactly the ECO log.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Data preparation	1.5	● BEOL re-verification complete in TO-09	T0-11 BEOL mask order confirmation and full mask set completion tracking
MDP handoff	1.0	● FEOL layer split known from TO-06	
Submission and acceptance	0.5	● Foundry submission requirements confirmed	FAB-03 BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing
Total	3.0		
WHO IT TAKES	~8.7 FTE	EXIT CRITERIA	MEASURED BY
Tapeout engineer Owns preparation and submission		● Layer set exactly complementary to the FEOL release ● Submission acknowledged in writing ● Database difference from the original traced to the ECO log	● Submission accepted at first attempt ● Layer set completeness against the split ● Release date against plan
Foundry liaison Submission, acceptance and MDP interface			
Physical verification engineer Confirms re-verification complete			
Program manager Milestone confirmation and communication			
Configuration manager Database traceability and ECO log			
		DEPENDS ON	
		T0-08 BEOL design fix window — metal-layer ECOs and late fixes while FEOL masks are being cut	
		T0-09 BEOL layer re-verification — DRC, LVS, antenna and density on the fixed metal stack	
		T0-06 FEOL layer data preparation and MTO release to the mask shop	

14 TO · TAPEOUT & MASK RELEASE (MTO)

T0-11

BEOL mask order confirmation and full mask set completion tracking

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
1w w7-w8 in stage	1.0 M/M planned	~4.3 FTE while it runs	3 0 in parallel	no has float	Procurement Operations planner approves

PURPOSE

Confirm the **BEOL order and track the mask set to completion**, so that FAB-05 knows when back-end layers will be available and the line does not stall mid-lot.

The BEOL masks are needed several weeks into wafer processing, not at wafer start. Tracking them to completion is what confirms that the parallel schedule the FEOL/BEOL split assumed actually holds.

WORK FLOW

3 STEPS · 1 LANE

w7

w8

MAIN SEQUENCE

1	2	3
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#	STEP	TAT	LANE
1	BEOL mask order confirmation	0.4w	main
2	Full mask set completion tracking	0.3w	main
3	Mask availability communication to fabrication	0.3w	main

Step 2 tracks the whole set rather than the second half. A mask set is only usable when every layer is complete, and the last layer to finish is the one that decides.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- BEOL submission record from TO-10
- Mask order framework from TO-07
- Mask shop schedules
- Wafer processing plan from FAB-04
- Cost sheet from TECH-04

PRODUCES

- Confirmed BEOL mask order
- Full mask set completion tracking
- Mask availability communication to fabrication

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

T0-D8

PRODUCES

Full mask set completion record. This activity is the deliverable, and it is what confirms the split schedule held.

T0-D7

FEEDS

BEOL MTO release package and mask order confirmation. The order half of the BEOL release record.

RISKS AND WATCH-OUTS

- ◆ **BEOL masks late against the wafer schedule.** Wafers reach the metallisation steps and stop, which is worse than starting late.
- ◆ **Completion tracked per layer rather than per set.** The set is usable when the last layer is done, and per-layer optimism hides that.
- ◆ **Availability not communicated to fabrication.** The line plans against the date, and a date nobody passed on is a date the line does not have.
- ◆ **Order mismatched to submission.** The order has to name exactly the layers submitted, and BEOL sets are large.
- ◆ **No contingency for a mask defect.** Masks fail inspection sometimes, and a remake takes weeks.

EFFORT SPLIT	1 M/M	ENTRY CRITERIA	DEPENDS ON
Order confirmation	0.4	● BEOL data submitted and accepted in TO-10	TO-07 FEOL mask order confirmation and mask shop scheduling
Set completion tracking	0.4	● Mask order framework established in TO-07	TO-10 BEOL layer data preparation and MTO release
Communication	0.2	● Wafer processing plan available from FAB-04	FEEDS INTO
Total	1.0		
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA	
Procurement Order confirmation		● Order confirmed against the submitted layer set	FAB-03 BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing
Foundry liaison Mask shop tracking and escalation		● Set completion tracked to the last layer	FAB-05 Back-end-of-line processing
Program manager Communicates availability into the fab plan		● Availability communicated to fabrication against the line schedule	FAB-09 Hot-lot management and WIP tracking
Tapeout engineer Order and submission consistency			MEASURED BY
Operations planner Wafer schedule alignment			● Mask set completion against the wafer schedule
			● Layers remaining at each status check
			● Mask remakes required

MANUFACTURE

15 Fabrication

FAB

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w79 – w98	19 _w	10	30 _{M/M}	6 of 10

ENGINEERING ACTIVITIES				KEY DELIVERABLES		
FAB-01	FEOL mask set fabrication, inspection and qualification	7w	4	M/M	FAB-D1	FEOL and BEOL mask sets with qualification reports
FAB-02	Wafer start on FEOL mask availability	1w	1	M/M	FAB-D2	Processed engineering-lot wafers
FAB-03	BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing	6w	4	M/M	FAB-D3	Inline, PCM and WAT data package
FAB-04	Front-end wafer processing	8w	3	M/M	FAB-D4	Wafer acceptance disposition record
FAB-05	Back-end-of-line processing	5w	2	M/M	FAB-D5	Wafer-out forecast versus actual log
FAB-06	Inline metrology and defect inspection monitoring	14w cont.	5	M/M	FAB-D6	First Silicon availability notice
FAB-07	E-test / PCM data review	3w	3	M/M		
FAB-08	Wafer acceptance test and lot disposition	3w	3	M/M		
FAB-09	Hot-lot management and WIP tracking	18w cont.	4	M/M		
FAB-10	Wafer shipment and logistics to sort and assembly	2w	1	M/M		

15 FAB · FABRICATION

FAB-01

FEOL mask set fabrication, inspection and qualification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	4.0 M/M	~2.5 FTE	6	yes	Foundry liaison
w0–w7 in stage	planned	while it runs	2 in parallel	no float	Procurement approves

PURPOSE

Get the **front-end mask set written, inspected and qualified** — seven weeks of the mask shop's calendar that the program can watch but not accelerate.

EUV mask sets at this node are slow and expensive to produce, and a single mask failing inspection means a remake measured in weeks. The program's role is oversight rather than execution: confirm the data was accepted, review inspection results and escalate when a layer slips.

WORK FLOW

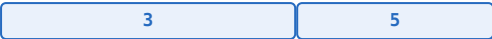
6 STEPS · 2 LANES

w0w2w4w6w7

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Mask data acceptance and mask data preparation	1w	main
2	FEOL mask writing	3w	main
3	Pellicle and handling preparation	1.5w	parallel
4	Mask inspection and defect repair	2w	main
5	Mask defect disposition and remake decisions	1w	parallel
6	Mask qualification and release to the line	1w	main

Step 5 is where a schedule is saved or lost. A mask defect can sometimes be repaired, sometimes waived as non-printing, and sometimes requires a remake — and the three outcomes differ by weeks.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- FEOL data submission from TO-06
- Mask order and schedule from TO-07
- Foundry mask inspection criteria
- Layer criticality assessment
- Mask cost and remake terms from TECH-04

PRODUCES

- Mask data preparation acceptance
- Written FEOL mask set
- Inspection results per layer
- Defect disposition decisions
- Qualified mask set released to the line

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D1	PRODUCES	FEOL and BEOL mask sets with qualification reports. The front-end half of the deliverable.
FAB-D5	FEEDS	Wafer-out forecast versus actual log. Mask completion is the first date the wafer-out forecast is built on.

RISKS AND WATCH-OUTS

- ◆ **Mask defect requiring a remake.** Weeks of schedule, and the program has no way to accelerate the replacement.
- ◆ **Inspection criteria not understood.** Whether a defect is printing depends on layer and position, and the disposition needs design input the foundry does not have.
- ◆ **Critical layers not prioritised.** Mask shops write in a sequence, and the layers on the critical path can sometimes be pulled forward if anyone asks.
- ◆ **Data preparation rejected late.** MDP problems surface days after submission and consume the schedule silently.
- ◆ **No visibility into progress.** A seven-week activity with no status reporting means the first news is either completion or a delay.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Mask writing oversight	1.5	● FEOL data submitted and accepted in TO-06	FAB-02 Wafer start on FEOL mask availability
Inspection review	1.0	● Mask order confirmed in TO-07	FAB-04 Front-end wafer processing
Qualification	0.8	● Inspection and disposition process agreed with the foundry	FAB-09 Hot-lot management and WIP tracking
MDP acceptance	0.8		
Total	4.0		
WHO IT TAKES	~2.5 FTE	EXIT CRITERIA	MEASURED BY
Foundry liaison Mask shop interface and escalation		● Every mask inspected and dispositioned, not only written	● Mask completion against plan
Tapeout engineer MDP acceptance and defect disposition input		● Set qualified and released to the line	● Defects requiring remake
Program manager Schedule tracking and communication		● Completion date communicated against the wafer start plan	● Days between defect detection and disposition
Physical verification engineer Printing assessment on reported defects			
Procurement Remake authorisation and cost			
		DEPENDS ON	
		T0-06 FEOL layer data preparation and MTO release to the mask shop	
		T0-07 FEOL mask order confirmation and mask shop scheduling	

15 FAB · FABRICATION

FAB-02

Wafer start on FEOL mask availability

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
1w	1.0 M/M	~4.3 FTE	3	yes	Operations planner
w7–w8 in stage	planned	while it runs	0 in parallel	no float	Program manager approves

PURPOSE

Start wafers **the moment the front-end masks are qualified** — lot definition, starting material, split strategy — because every day of delay here is a day at the far end of the program.

A single week of activity that gates the remaining eleven. The decisions it carries are small and consequential: how many wafers, what splits, which lots are hot, and whether any material is held back for a second attempt.

WORK FLOW

3 STEPS · 1 LANE

w7

w8

MAIN SEQUENCE

#	STEP	TAT	LANE
1	Wafer start authorisation and lot definition	0.4w	main
2	Starting material and lot split decision	0.3w	main
3	Start confirmation and schedule communication	0.3w	main

Step 2 decides the split strategy — whether all wafers run the same process conditions or some are deliberately skewed. Splits cost wafers and buy information about process sensitivity that no simulation can provide.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Qualified FEOL mask set from FAB-01
- Capacity and hot-lot terms from TECH-05
- Wafer quantity plan from the engineering lot requirement
- Split strategy from TECH-07 node risk
- Starting material specification

PRODUCES

- Wafer start authorisation
- Lot definition with split strategy
- Start confirmation and schedule communication

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D2

FEEDS

Processed engineering-lot wafers. The lot definition here determines what comes out at the end.

FAB-D5

FEEDS

Wafer-out forecast versus actual log. Wafer start is the point from which the forecast is measured.

RISKS AND WATCH-OUTS

- ◆ **Start delayed after masks are ready.** Every day is a day at the far end of a schedule with no recovery mechanism.
- ◆ **Wafer quantity too small.** Bring-up, characterisation, qualification and customer samples all draw from this lot, and running short means a second lot and a quarter.
- ◆ **No process splits.** A single condition gives one data point; splits reveal process sensitivity while there is still time to act on it.
- ◆ **Hot-lot status not invoked.** The priority negotiated in TECH-05 has to be requested at start, not assumed.
- ◆ **Starting material not specified.** Substrate type and specification affect device behaviour, and defaulting is how an unexpected variable enters.

EFFORT SPLIT	1 M/M	ENTRY CRITERIA	DEPENDS ON
Authorisation	0.4	● FEOL masks qualified by FAB-01	FAB-01 FEOL mask set fabrication, inspection and qualification
Lot split decision	0.4	● Capacity and hot-lot terms confirmed in TECH-05	TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement
Confirmation	0.2	● Wafer quantity agreed against downstream needs	
Total	1.0		
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA	FEEDS INTO
Operations planner Lot definition and material		● Wafers started within days of mask qualification	FAB-04 Front-end wafer processing
Foundry liaison Start authorisation and hot-lot invocation		● Quantity sufficient for bring-up, qual and samples	FAB-09 Hot-lot management and WIP tracking
Product engineering Wafer quantity against downstream needs		● Hot-lot priority invoked where it was negotiated	
Device engineer Split strategy			MEASURED BY
Program manager Start confirmation and communication			● Days between mask qualification and wafer start
			● Wafers started against the downstream requirement
			● Splits included against those planned

15 FAB · FABRICATION

FAB-03

BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	4.0 M/M	~2.9 FTE	5	no	Foundry liaison
w4-w10 in stage	planned	while it runs	1 in parallel	has float	Physical verification engineer approves

PURPOSE

Write and qualify the **back-end mask set while front-end processing runs** — the parallelism the FEOL/BEOL split was designed to create.

This activity is the payoff of the two-stage release. BEOL masks are written during the weeks the wafers spend in front-end processing, so the fix window cost the program nothing in schedule as long as these masks are ready when the lots reach metallisation.

WORK FLOW

5 STEPS · 2 LANES

w4w6w8w10

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	BEOL mask data acceptance and preparation	0.5w	main
2	BEOL mask writing	2.5w	main
3	Defect disposition on inspected layers	1.5w	parallel
4	Inspection and qualification	2w	main
5	Release to the line ahead of metallisation	1w	main

Step 5's date is the one that matters. The masks do not need to exist at wafer start; they need to exist before the lots reach the first metal layer, and that is several weeks of margin the split created deliberately.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- BEOL data submission from TO-10
- Mask order from TO-11
- Front-end processing schedule from FAB-04
- Foundry inspection criteria
- Metallisation start estimate

PRODUCES

- BEOL mask data acceptance
- Written BEOL mask set
- Inspection and defect disposition results
- Qualified BEOL masks released to the line

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D1	PRODUCES	FEOL and BEOL mask sets with qualification reports. The back-end half of the deliverable.
FAB-D5	FEEDS	Wafer-out forecast versus actual log. A BEOL mask delay stops lots mid-process, which the forecast has to reflect.

RISKS AND WATCH-OUTS

- ◆ **Masks late against metallisation.** Lots stop mid-process, which wastes fab capacity and is harder to recover from than a late start.
- ◆ **Parallelism assumed rather than tracked.** The split only pays off if the BEOL schedule is actively compared against the front-end progress.
- ◆ **Defect disposition slower on BEOL layers.** There are more of them, and a per-layer process that worked for FEOL can bottleneck here.
- ◆ **Metallisation start estimate stale.** The masks are needed against a moving date, and planning against the original estimate is planning against fiction.
- ◆ **No priority on the first metal layers.** They are needed first and can often be written first if anyone asks.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	DEPENDS ON
Mask writing oversight	1.5	● BEOL data submitted and accepted in TO-10	TO-10 BEOL layer data preparation and MTO release
Inspection and qualification	1.5	● Mask order confirmed in TO-11	TO-11 BEOL mask order confirmation and full mask set completion tracking
Data acceptance	0.5	● Front-end processing progressing in FAB-04	FAB-04 Front-end wafer processing
Release	0.5		
Total	4.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~2.9 FTE	● Masks qualified before lots reach metallisation	FAB-05 Back-end-of-line processing
Foundry liaison		● Progress tracked against the front-end schedule, not in isolation	FAB-09 Hot-lot management and WIP tracking
Mask shop tracking and priority		● First metal layers prioritised in the writing sequence	MEASURED BY
Tapeout engineer			● Mask availability against the metallisation date
Data acceptance and defect disposition			● Weeks of margin at metallisation
Operations planner			● Layers requiring remake
Metallisation date tracking			
Program manager			
Parallelism tracking and escalation			
Physical verification engineer			
Printing assessment on defects			

15 FAB · FABRICATION

FAB-04

Front-end wafer processing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	3.0 M/M	~1.6 FTE	6	yes	Foundry liaison
w7–w15 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Watch the wafers through **front-end processing** — transistor formation, contacts, local interconnect — and respond when the line reports something unexpected.

Eight weeks in which the program does almost nothing and cannot afford to do nothing. The lots are the foundry's to run; what belongs to the program is understanding the inline data, recognising an excursion early and deciding what to do about it.

WORK FLOW

6 STEPS · 2 LANES

w7	w9	w11	w13	w15
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Lot release and front-end entry confirmation	1w	main
2	Transistor formation processing	3w	main
3	Inline monitoring at front-end critical steps	2w	parallel
4	Contact and local interconnect processing	3w	main
5	Excursion response and disposition	1.5w	parallel
6	Front-end exit and handoff to back end	1w	main

Step 5 needs a standing decision framework. When a lot goes out of control limits the choices are scrap, rework, continue at risk or split — and making that decision in the moment, without a framework, tends to produce whichever answer is least disruptive rather than best.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Started lots from FAB-02
- Inline metrology data from FAB-06
- Foundry process control data
- Excursion response framework
- Hot-lot priority from TECH-05

PRODUCES

- Front-end processing progress
- Inline monitoring findings at critical steps
- Excursion responses and lot dispositions
- Front-end exit confirmation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D2	FEEDS	Processed engineering-lot wafers. Front-end processing is where the transistors are actually made.
FAB-D3	FEEDS	Inline, PCM and WAT data package. Front-end inline data is the earliest indication of how the lot is running.

RISKS AND WATCH-OUTS

- ◆ **Excursion response improvised.** Without a framework the decision defaults to whatever is least disruptive rather than what is best for the lot.
- ◆ **Inline data not reviewed by the program.** The foundry monitors its process; only the design team can say whether a deviation matters for this device.
- ◆ **Hot-lot priority lost mid-flow.** Priority granted at start can quietly lapse, and nobody notices until the cycle time slips.
- ◆ **Scrap decision taken too late.** A lot continuing through eight weeks of processing after a fatal excursion wastes capacity that a second lot needed.
- ◆ **No second lot in reserve.** If the first lot is lost, having no material started means restarting the whole cycle.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Monitoring and data review	1.3	● Wafers started in FAB-02	FAB-05 Back-end-of-line processing
Excursion response	0.8	● Inline monitoring plan running from FAB-06	FAB-03 BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing
Lot management	0.5	● Excursion response framework agreed	FAB-07 E-test / PCM data review
Handoff	0.5		FAB-09 Hot-lot management and WIP tracking
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~1.6 FTE	● Every excursion dispositioned against a framework, not improvised	● Cycle time against the hot-lot plan
Foundry liaison	Lot status, escalation and priority	● Inline data reviewed by the design team, not only the foundry	● Excursions dispositioned within the framework
Device engineer	Inline data interpretation for this device	● Hot-lot priority maintained through the flow	● Wafers surviving front-end processing
Yield engineer	Excursion assessment and disposition	DEPENDS ON	
Operations planner	Lot management and reserve material	FAB-02 Wafer start on FEOL mask availability	
Program manager	Schedule impact and communication	FAB-06 Inline metrology and defect inspection monitoring	

15 FAB · FABRICATION

FAB-05

Back-end-of-line processing

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w w14–w19 in stage	2.0 M/M planned	~1.7 FTE while it runs	5 1 in parallel	yes no float	Foundry liaison Package liaison approves

PURPOSE

Complete the wafers through **back-end-of-line processing** — metallisation, vias, top metal, passivation — and finish the lot.

The back end is where the design's routing becomes physical. It is also where the BEOL masks are consumed, which makes it the point at which the FEOL/BEOL split either paid off invisibly or produces a stall that everyone notices.

WORK FLOW

5 STEPS · 2 LANES

w14	w15	w16	w17	w18	w19
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Back-end entry and mask availability confirmation	0.5w	main
2	Metallisation and via formation	2w	main
3	Back-end inline monitoring	1.5w	parallel
4	Top metal and passivation	1.5w	main
5	Wafer completion and exit	1w	main

Step 1 is a confirmation with real consequences. If `FAB-03` has not delivered qualified masks by this point the lots wait in the line, and waiting mid-process is more expensive than waiting at the start.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Front-end wafers from FAB-04
- Qualified BEOL masks from FAB-03
- Inline monitoring from FAB-06
- Process control data
- Metallisation process specification

PRODUCES

- Metallised and via-formed wafers
- Back-end inline monitoring findings
- Completed wafers with passivation
- Lot exit confirmation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D2	PRODUCES	Processed engineering-lot wafers. This activity is where the wafers are finished — the deliverable the whole stage exists to produce.
FAB-D3	FEEDS	Inline, PCM and WAT data package. Back-end inline data completes the process record for the lot.

RISKS AND WATCH-OUTS

- ◆ **BEOL masks not ready at back-end entry.** Lots stall mid-process, consuming fab capacity and adding cycle time nobody planned.
- ◆ **Metallisation defects detected late.** Back-end defects affect yield directly, and inline detection is what allows a response.
- ◆ **Passivation and top metal treated as routine.** They affect bump attachment and package assembly, and problems here surface at ASSY-03.
- ◆ **Lot completion not confirmed against quantity.** Wafers lost during processing reduce the number available downstream, and the count has to be checked rather than assumed.
- ◆ **Process changes between split lots not tracked.** If splits were run, which wafer had which condition has to be traceable to the end.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	DEPENDS ON
Monitoring	0.8	● Front-end complete from FAB-04	FAB-03 BEOL mask set fabrication, inspection and qualification, in parallel with front-end processing
Entry checks	0.5	● BEOL masks qualified and released by FAB-03	
Completion	0.5	● Inline monitoring running from FAB-06	FAB-04 Front-end wafer processing
Handoff	0.3		FAB-06 Inline metrology and defect inspection monitoring
Total	2.0	EXIT CRITERIA	
WHO IT TAKES	~1.7 FTE	● Wafers completed without a mask-driven stall	FEEDS INTO
Foundry liaison Back-end lot status and escalation		● Split traceability maintained to lot exit	FAB-07 E-test / PCM data review
Device engineer Metallisation and inline data interpretation		● Wafer count confirmed against the downstream requirement	FAB-08 Wafer acceptance test and lot disposition
Yield engineer Defect response in the back end			FAB-10 Wafer shipment and logistics to sort and assembly
Operations planner Lot tracking and split traceability			MEASURED BY
Package liaison Passivation and bump readiness			● Days stalled waiting for masks
			● Wafers completed against started
			● Back-end defect density

15 FAB · FABRICATION

FAB-06

Inline metrology and defect inspection monitoring

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w cont. w5–w19 in stage, runs throughout	5.0 M/M planned	~1.6 FTE while it runs	6 2 in parallel	no has float	Yield engineer Product engineering approves

PURPOSE

Watch the **inline metrology and defect inspection data** across the whole run, so an excursion is caught while the lot can still be saved rather than at final test.

The foundry monitors its process against its own control limits; the program needs to know whether a deviation matters for this device. A defect density excursion on a layer the design uses heavily is a different event from one on a layer it barely touches, and only the design team can tell the difference.

WORK FLOW

6 STEPS · 2 LANES

w5	w8	w11	w14	w17	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Monitoring plan and critical step selection	1w	main
2	Inline metrology data review	2w	main
3	Excursion detection and escalation	2w	parallel
4	Defect inspection review and classification	2w	main
5	Defect Pareto and source attribution	2w	parallel
6	Continuous monitoring across the lot	9w	main

Step 1 chooses where to look. Monitoring every step equally wastes attention; selecting the layers and dimensions this design is most sensitive to — minimum-pitch metal, critical gate dimensions, via layers — is what makes the monitoring informative.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Foundry inline metrology data
- Defect inspection results
- Design sensitivity analysis from TECH-03 and SO-08
- Process control limits
- Yield model from TECH-07

PRODUCES

- Monitoring plan with critical step selection
- Inline metrology review findings
- Excursion detections and escalations
- Defect classification and Pareto
- Source attribution findings

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D3	PRODUCES	Inline, PCM and WAT data package. The inline half of the deliverable, and the earliest signal the program gets.
FAB-D4	FEEDS	Wafer acceptance disposition record. Inline findings inform the acceptance decision rather than being separate from it.

RISKS AND WATCH-OUTS

- ◆ **Monitoring every step equally.** Attention spread evenly finds nothing; the design's sensitivities decide where to look.
- ◆ **Data reviewed only by the foundry.** Process control answers whether the line is in control, not whether this device is affected.
- ◆ **Excursions detected after the affected step.** Once the lot has moved on, the response options narrow sharply.
- ◆ **Defects classified but not attributed.** Knowing the count without the source gives no basis for a corrective action.
- ◆ **Monitoring stopping at front-end exit.** Back-end defects affect yield as much and are often less well monitored.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous monitoring	2.0	● Lots in process from FAB-04	FAB-04 Front-end wafer processing
Defect classification	1.3	● Data access agreed with the foundry	FAB-05 Back-end-of-line processing
Metrology review	1.0	● Design sensitivity analysis available	FAB-07 E-test / PCM data review
Escalation	0.8		FAB-08 Wafer acceptance test and lot disposition
Total	5.0	EXIT CRITERIA	MP-05 Yield learning, failure pareto and defect analysis
WHO IT TAKES	~1.6 FTE	● Monitoring focused on the design's actual sensitivities ● Excursions escalated while the lot can still be acted on ● Defects attributed to a source, not only counted	MEASURED BY
Yield engineer Owns monitoring, classification and Pareto			● Excursions detected before the affected step completes
Device engineer Metrology interpretation for this device		DEPENDS ON	● Defects attributed to a source
Foundry liaison Data access and excursion escalation		FAB-02 Wafer start on FEOL mask availability	
DFM engineer Design sensitivity to specific defect modes		TECH-07 Foundry roadmap and risk-production timing alignment	
Product engineering Yield implication of observed defects		S0-08 DFM and lithography checks, CMP and hotspot analysis	

15 FAB · FABRICATION

FAB-07

E-test / PCM data review

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	3.0 M/M	~4.3 FTE	4	no	Device engineer
w15–w18 in stage	planned	while it runs	1 in parallel	has float	Product engineering approves

PURPOSE

Read the **e-test and PCM data** — device parameters measured on structures in the scribe line — and compare them against what the models predicted.

PCM data arrives before packaged parts and answers process questions weeks earlier than the lab can. Threshold voltages, resistances, capacitances and ring oscillator frequencies measured on real silicon are the first hard evidence of whether the design's assumptions about the process were right.

WORK FLOW4 STEPS · 2 LANES

w15	w16	w17	w18
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MAIN SEQUENCE

1	2	4
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IN PARALLEL

3

#	STEP	TAT	LANE
1	PCM structure and parameter list confirmation	0.5w	main
2	E-test data collection and review	1w	main
3	Across-wafer and lot-to-lot variability analysis	1w	parallel
4	Parameter comparison against target and model	1.5w	main

Step 4 is where the value is. A threshold voltage 30 mV from target changes timing across the whole design, and knowing that before parts arrive lets bring-up be planned around it rather than surprised by it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- E-test and PCM data from the foundry
- Model predictions from PDK-03 and TECH-03
- Test chip correlation from TC-07
- Process target specifications
- Split lot definitions from FAB-02

PRODUCES

- PCM parameter list and structures
- E-test data review
- Across-wafer and lot variability analysis
- Parameter comparison against model and target

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D3	PRODUCES	Inline, PCM and WAT data package. The PCM half of the deliverable, and the earliest device-level data available.
FAB-D4	FEEDS	Wafer acceptance disposition record. Parameter deviations are part of what acceptance is decided on.

RISKS AND WATCH-OUTS

- ◆ **PCM data not reviewed until parts arrive.** It answers process questions weeks earlier and costs nothing extra to read.
- ◆ **Comparison against target only.** The design was built against models; comparing against the model is what predicts silicon behaviour.
- ◆ **Variability ignored.** Across-wafer parameter spread predicts the bin distribution, and the mean alone does not.
- ◆ **Split lots analysed together.** If splits were run, aggregating them averages away exactly the information the splits were for.
- ◆ **No link to the test chip results.** TC-07 correlated models against silicon; this data either confirms that correlation or contradicts it.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Parameter comparison	1.3	● Wafers reaching e-test in FAB-05	FAB-08 Wafer acceptance test and lot disposition
Data review	1.0	● Model predictions available from PDK-03	BU-07 Shmoo across voltage, frequency and temperature
Variability analysis	0.8	● Split lot definitions known from FAB-02	MP-08 Process corner and split-lot validation
Total	3.0		MP-05 Yield learning, failure pareto and defect analysis
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA	MEASURED BY
Device engineer Owns PCM analysis and model comparison		● Parameters compared against models, not only against targets	● Parameters within model prediction
Yield engineer Variability and distribution analysis		● Split lots analysed separately	● Across-wafer variability against expectation
Foundry liaison Data access and parameter interpretation		● Findings communicated before packaged parts arrive	● Days before part delivery that findings were available
Library engineer Model correlation implications			
Product engineering Bin distribution prediction			
		DEPENDS ON	
		FAB-05 Back-end-of-line processing	
		FAB-06 Inline metrology and defect inspection monitoring	
		TC-07 Silicon characterisation and correlation to simulation	

15 FAB · FABRICATION

FAB-08

Wafer acceptance test and lot disposition

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	3.0 M/M	~4.3 FTE	4	yes	Product engineering
w16–w19 in stage	planned	while it runs	1 in parallel	no float	Validation liaison approves

PURPOSE

Run **wafer acceptance testing and decide whether the lot is accepted** — the formal gate between the fab and everything downstream.

WAT is the contractual boundary. Parameters within specification mean the foundry delivered what was agreed; parameters outside mean a conversation about disposition, rework or credit. Either way the decision has to be taken deliberately rather than by default.

WORK FLOW

4 STEPS · 2 LANES

w16	w17	w18	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Acceptance criteria confirmation with the foundry	0.5w	main
2	WAT execution and data review	1w	main
3	Out-of-specification escalation with the foundry	0.8w	parallel
4	Lot disposition and release decision	1.5w	main

Step 4 can accept a lot that is out of specification. On an engineering lot the priority is usually silicon in the lab rather than perfect silicon, and accepting a marginal lot with the deviation documented is often the right call — but it must be a call.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Completed wafers from FAB-05
- WAT specification and acceptance criteria
- PCM findings from FAB-07
- Inline findings from FAB-06
- Downstream schedule pressure from the program plan

PRODUCES

- Confirmed acceptance criteria
- WAT execution results
- Out-of-specification escalation record
- Lot disposition and release decision

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D4

PRODUCES

Wafer acceptance disposition record. This activity is the deliverable, and the record is what a later yield discussion refers back to.

FAB-D2

GATES

Processed engineering-lot wafers. Wafers are not released downstream until this decision is taken.

RISKS AND WATCH-OUTS

- ◆ **Acceptance by default.** A lot released without a decision has been accepted by nobody, and the deviation is discovered later with no record.
- ◆ **Criteria not agreed in advance.** Negotiating acceptance criteria after the data exists favours whoever is more motivated.
- ◆ **Marginal lot rejected on principle.** On an engineering lot, silicon in the lab is usually worth more than a lot within every limit.
- ◆ **Deviations not documented.** A parameter out of specification affects bring-up interpretation, and the bring-up team has to know.
- ◆ **Disposition delayed.** Every day the lot sits undispositioned is a day of bring-up not happening.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
WAT review	1.3	● Wafers completed in FAB-05	FAB-10 Wafer shipment and logistics to sort and assembly
Disposition	1.0	● Acceptance criteria agreed with the foundry	ASSY-01 Known-good-die sort and selection
Escalation	0.5	● PCM and inline findings available	BU-01 Sample receipt, incoming inspection and board mounting
Criteria confirmation	0.3		MP-05 Yield learning, failure pareto and defect analysis
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~4.3 FTE	● Disposition decided deliberately and recorded ● Deviations documented and communicated to bring-up ● Decision taken without holding the lot unnecessarily	● Days between WAT data and disposition ● Deviations documented against those found ● Wafers accepted against wafers completed
Product engineering Owns acceptance and disposition		DEPENDS ON	
Device engineer WAT data interpretation		FAB-05 Back-end-of-line processing	
Foundry liaison Escalation and commercial disposition		FAB-06 Inline metrology and defect inspection monitoring	
Program manager Schedule trade-off in the decision		FAB-07 E-test / PCM data review	
Validation liaison Impact of deviations on bring-up			

15 FAB · FABRICATION

FAB-09

Hot-lot management and WIP tracking

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
18w cont. w1–w19 in stage, runs throughout	4.0 M/M planned	~1.0 FTE while it runs	6 2 in parallel	no has float	Program manager Downstream stage leads approves

PURPOSE

Track the lots and **maintain an honest wafer-out forecast**, because ten downstream activities are planning against a date this activity owns.

Assembly, test, bring-up and qualification all schedule against wafer-out. The forecast will move — fab schedules do — and the program's exposure is not the movement but the surprise. Four man-months across eighteen weeks buys a date everyone can plan against and early warning when it changes.

WORK FLOW

6 STEPS · 2 LANES

w1	w5	w9	w13	w17	w19
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	WIP tracking setup and reporting cadence	1w	main
2	Hot-lot priority management	2w	main
3	Escalation on schedule deviation	2w	parallel
4	Wafer-out forecast maintenance	2w	main
5	Forecast communication to downstream stages	2w	parallel
6	Continuous tracking across the run	13w	main

Step 5 is the reason this activity exists. A forecast held by the fab liaison and not distributed leaves `ASSY`, `TEST` and `BU` planning against the last number they heard, which is usually the original one.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Foundry WIP reporting
- Lot status from FAB-04 and FAB-05
- Mask availability from FAB-01 and FAB-03
- Hot-lot terms from TECH-05
- Downstream schedule dependencies from DEF-08

PRODUCES

- WIP tracking and reporting cadence
- Hot-lot priority management record
- Schedule deviation escalations
- Wafer-out forecast, maintained
- Forecast communications to downstream stages

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D5	PRODUCES	Wafer-out forecast versus actual log. This activity is the deliverable, and its value is in being current rather than in being right the first time.
FAB-D6	FEEDS	First Silicon availability notice. The notice is the forecast's final form.

RISKS AND WATCH-OUTS

- ◆ **Forecast not updated.** Downstream stages plan against a stale date and discover the change when the parts do not arrive.
- ◆ **Hot-lot priority lapsing.** Priority is maintained by attention, and a lot that quietly loses it takes standard cycle time.
- ◆ **Deviations not escalated.** A slip identified and not raised becomes a slip everyone discovers at the same time, too late to replan.
- ◆ **Forecast presented without confidence.** A date with no stated uncertainty invites downstream teams to plan to the day.
- ◆ **Tracking limited to the fab.** Wafer-out is not the same as parts available; sort, logistics and assembly all sit between.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Continuous tracking	1.8	● Lots started in FAB-02	ASSY-01 Known-good-die sort and selection
Forecast maintenance	1.0	● WIP data access agreed with the foundry	TEST-03 Probe card design, fabrication and qualification
Hot-lot management	0.8	● Downstream dependencies mapped from DEF-08	BU-01 Sample receipt, incoming inspection and board mounting
Escalation	0.5		PTV-12 Feedback into package design and assembly process window freeze
Total	4.0		
WHO IT TAKES	~1.0 FTE	EXIT CRITERIA	MEASURED BY
Program manager Owns the forecast and its communication		● Forecast updated at every status change, not monthly	● Forecast accuracy against actual wafer-out
Foundry liaison WIP data, priority and escalation		● Hot-lot priority actively maintained	● Days between a status change and downstream notification
Operations planner Logistics and downstream dependency mapping		● Downstream stages replanning against the current date	● Cycle time against the hot-lot plan
Product engineering Sort and assembly readiness against the date			
Downstream stage leads Consume the forecast and replan against it			
		FAB-01 FEOL mask set fabrication, inspection and qualification	
		FAB-02 Wafer start on FEOL mask availability	
		TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement	
		DEF-08 Program schedule, resource and staffing plan drafting	

15 FAB · FABRICATION

FAB-10

Wafer shipment and logistics to sort and assembly

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	1.0 M/M	~2.2 FTE	3	yes	Operations planner
w17–w19 in stage	planned	while it runs	0 in parallel	no float	Program manager approves

PURPOSE

Move the wafers from the fab to **sort and assembly** — shipment, customs, documentation — and confirm First Silicon is actually available.

Wafers finished in a fab on another continent are not parts in a lab. Shipment, export documentation and customs sit between wafer-out and anyone touching silicon, and they are the part of the schedule most often left out of the plan entirely.

WORK FLOW

3 STEPS · 1 LANE

w17	w18	w19
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MAIN SEQUENCE

#	STEP	TAT	LANE
1	Shipment planning and documentation	0.5w	main
2	Customs and export control handling	0.8w	main
3	Delivery to sort and assembly	0.8w	main

Step 2 is where a week can vanish. Export control classification and customs paperwork for engineering samples crossing borders is a real process, and starting it when the wafers are ready rather than while they are in the line is what makes it a delay.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Accepted lots from FAB-08
- Export control classification from TECH-06
- Shipping and logistics arrangements from TECH-09
- Sort and assembly readiness from ASSY-01
- Delivery destination requirements

PRODUCES

- Shipment plan and documentation
- Customs and export clearance
- Delivery confirmation to sort and assembly
- First Silicon availability notice

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

FAB-D6	PRODUCES	First Silicon availability notice. This activity is the deliverable, and it carries the stage's milestone.
FAB-D2	FEEDS	Processed engineering-lot wafers. Wafers become useful when they arrive, not when they are finished.

RISKS AND WATCH-OUTS

- ◆ **Customs paperwork started at wafer-out.** It can be prepared during processing and takes a week if it is not.
- ◆ **Export classification unresolved.** Advanced semiconductors carry export controls, and an unclassified shipment does not move.
- ◆ **Destination not ready.** Wafers arriving at a sort facility with no probe card or no capacity wait there instead of in the fab.
- ◆ **Shipment split without tracking.** Lots split across shipments for risk are useful only if which wafers went where is traceable.
- ◆ **Milestone declared at wafer-out rather than at delivery.** First Silicon means silicon someone can touch, and declaring it early hides a week of transit.

EFFORT SPLIT	1 M/M	ENTRY CRITERIA	DEPENDS ON
Customs and documentation	0.5	● Lots accepted in FAB-08 ● Export classification available from TECH-06 ● Sort and assembly destination ready	FAB-08 Wafer acceptance test and lot disposition
Shipment planning	0.3		TECH-06 Foundry legal engagement — NDA, design agreement, IP licensing frame
Delivery tracking	0.3		TECH-09 OSAT and backend supply chain preliminary alignment
Total	1.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~2.2 FTE	● Documentation prepared before wafer-out, not after ● Delivery confirmed at the destination ● First Silicon declared on delivery, not on wafer-out	ASSY-01 Known-good-die sort and selection
Operations planner Owns shipment and logistics			TEST-05 Wafer sort test program development
Compliance officer Export control classification and clearance			BU-01 Sample receipt, incoming inspection and board mounting
Foundry liaison Release and shipping from the fab			MEASURED BY
Product engineering Destination readiness			● Days between wafer-out and delivery ● Customs and export delays ● Destination readiness at arrival
Program manager Milestone declaration and communication			

INTEGRATE

16

Package & Substrate Design

PKGD

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w24 – w76	52 _w	11	110 _{M/M}	6 _{of 11}

ENGINEERING ACTIVITIES				
PKGD-01	Package architecture selection — 2.5D interposer type, organic substrate, layer count	8w	10	M/M
PKGD-02	Bump map, pitch and power-ground planning with physical design	10w	12	M/M
PKGD-03	Interposer / RDL routing design	14w	18	M/M
PKGD-04	Substrate stack-up, escape routing and package DRC	14w	18	M/M
PKGD-05	Package routing for signal integrity — length matching, reference plane and via design	10w	12	M/M
PKGD-06	Package PDN and decap footprint design (electrical closure owned by SIPI)	9w	10	M/M
PKGD-07	Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	12w	12	M/M
PKGD-08	Test vehicle requirement definition and handoff to PTV	4w	3	M/M
PKGD-09	Substrate and interposer supplier selection and lead-time booking	10w	4	M/M
PKGD-10	OSAT selection and assembly process definition	9w	5	M/M
PKGD-11	Package design freeze, DRC and tooling release	6w	6	M/M

KEY DELIVERABLES	
PKGD-D1	Package architecture specification
PKGD-D2	Bump map and interposer / RDL database
PKGD-D3	Substrate design files (Gerber / ODB++) and stack-up
PKGD-D4	Package electrical design intent and model handoff to co-verification
PKGD-D5	Thermal and mechanical (warpage) simulation reports
PKGD-D6	Test vehicle requirement specification for PTV
PKGD-D7	Substrate and interposer PO with committed lead time
PKGD-D8	OSAT assembly process flow and agreement
PKGD-D9	Package Design Freeze package

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-01

Package architecture selection — 2.5D interposer type, organic substrate, layer count

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	10.0 M/M	~5.4 FTE	6	yes	Package architect
w0-w8 in stage	planned	while it runs	2 in parallel	no float	Chief architect approves

PURPOSE

Choose the **package architecture** — interposer type, substrate construction, layer count, body size — and commit to it, because everything from bump pitch to assembly capacity follows.

This decision has the same character as the monolithic-versus-chiplet call in ARCH-02: it looks like an engineering choice and it is a supply chain commitment. A 2.5D silicon interposer is a different supplier, a different capacity queue and a different cost structure from an organic bridge, and switching later is not practical.

WORK FLOW

6 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Package requirement consolidation — die size, power, interfaces, thermal	1.5w	main
2	Architecture option definition — interposer, bridge, organic	2w	main
3	Capability and capacity screen per option	1.5w	parallel
4	Substrate construction and layer count study	2w	main
5	Cost comparison per option with the DEF-04 model	1.5w	parallel
6	Architecture selection and specification	2.5w	main

Step 3 can eliminate an option outright. Advanced interposer capacity is allocated quarters ahead to a small number of customers, and an architecture the program cannot get capacity for is not an option however good it looks technically.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture assumption from DEF-01
- Partitioning and die count from ARCH-02
- Backend supply chain findings from TECH-09
- Bandwidth and HBM requirements from DEF-05
- Cost model from DEF-04

PRODUCES

- Package requirement consolidation
- Architecture options with capability assessment
- Capacity screen per option
- Substrate construction study
- Cost comparison
- Package architecture specification

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKG-D1	PRODUCES	Package architecture specification. This activity is the deliverable, and it is a supply chain commitment as much as a technical one.
PKG-D7	FEEDS	Substrate and interposer PO with committed lead time. The architecture decides which supplier queue the program is standing in.

RISKS AND WATCH-OUTS

- ◆ **Architecture chosen without capacity.** Advanced packaging capacity is scarce and allocated far ahead; a technically ideal option with no allocation is not an option.
- ◆ **Decision taken after the bump map.** Package architecture constrains bump pitch and field, and deciding afterwards means renegotiating the die.
- ◆ **Cost compared on the package alone.** Interposer, substrate, assembly yield and test all differ per architecture, and comparing one line misleads.
- ◆ **Thermal requirement not in the selection.** Package thermal resistance is architecture-dependent and sets the achievable power envelope.
- ◆ **Single option evaluated.** Without an alternative the program has no negotiating position and no fallback if capacity is refused.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO	
Architecture option definition	2.5	● Partitioning and die count known from ARCH-02 ● Backend supply chain findings available from TECH-09 ● Bandwidth and HBM requirements from DEF-05	PKGD-02 Bump map, pitch and power-ground planning with physical design	
Substrate construction study	2.0		PKGD-04 Substrate stack-up, escape routing and package DRC	
Selection and specification	2.0		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
Capacity screen	1.5		PKGD-09 Substrate and interposer supplier selection and lead-time booking	
Cost comparison	1.0		PD-04 Bump and RDL planning with chip-package co-design feedback	
Requirement consolidation	1.0	EXIT CRITERIA ● Architecture selected with capacity indicated, not assumed ● At least two options evaluated and costed end to end ● Specification issued before bump planning starts	PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress	
Total	10.0		DEPENDS ON	MEASURED BY
WHO IT TAKES ~5.4 FTE			DEF-01 Market and customer requirements consolidation	● Options evaluated against the one chosen ● Capacity indicated for the selected architecture ● Architecture changes after selection
Package architect Owns the architecture selection and specification			DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)	
Backend supply chain lead Capability and capacity screening			ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
Thermal engineer Thermal resistance per option		TECH-09 OSAT and backend supply chain preliminary alignment		
Product cost analyst Cost comparison across options		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)		
Chief architect Die-side implications of the choice				

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-02

Bump map, pitch and power-ground planning with physical design

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	6	yes	Package designer
w8-w18 in stage	planned	while it runs	2 in parallel	no float	Package architect approves

PURPOSE

Negotiate the **bump map with physical design** — pitch, count, power-to-signal ratio, position — until both the die and the package can build what the other needs.

The bump field is where two design teams share one constraint. The die wants bumps where its logic is; the package wants them where its escape routing can reach; and the power delivery requirement takes most of the field regardless. Convergence takes iteration, and the iteration has to finish before PD-04 freezes.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14	w16	w18
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Bump pitch and field definition against the architecture	2w	main
2	Power-to-signal ratio derivation from the power envelope	1.5w	main
3	Signal bump assignment per interface	2w	parallel
4	Escape feasibility check against the substrate	2w	main
5	Die-side placement iteration with physical design	2w	parallel
6	Bump map convergence and handoff	4.5w	main

Step 4 is the check that usually forces the iteration. A bump assignment that works on the die can require escape routing the substrate cannot provide at its layer count, and finding that early is what keeps the map from being redone after freeze.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture from PKGD-01
- Bump budget and intent from ARCH-08
- Power envelope and delivery requirement from PD-03
- Interface lane counts from ARCH-04
- Substrate layer count from PKGD-04

PRODUCES

- Bump pitch and field definition
- Power-to-signal ratio
- Signal bump assignment per interface
- Escape feasibility findings
- Converged bump map

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-02	PRODUCES	Bump map and interposer / RDL database. The bump map half of the deliverable; PKGD-03 supplies the interposer routing.
PKGD-04	FEEDS	Package electrical design intent. Bump assignment determines channel quality before any routing is drawn.

RISKS AND WATCH-OUTS

- ◆ **Bump map converged on the die side only.** A map that cannot escape on the substrate has to be redone, and by then the die may be placed around it.
- ◆ **Power bumps reduced to fit signals.** The interfaces then work and the die cannot be fed, which SIPI-04 finds much later.
- ◆ **Pitch chosen from the previous program.** Pitch depends on the interposer technology and the assembly process, both of which may have changed.
- ◆ **Signal assignment ignoring channel quality.** Which bump a high-speed signal uses affects its escape path and its crosstalk exposure.
- ◆ **Iteration continuing past the freeze date.** PD-04 and the substrate order both depend on this closing on time.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Bump pitch and field definition	3.0	● Package architecture selected in PKGD-01 ● Bump budget available from ARCH-08 ● Power delivery requirement known from PD-03	PKGD-03 Interposer / RDL routing design
Signal assignment	2.5		PKGD-05 Package routing for signal integrity — length matching, reference plane and via design
Escape feasibility	2.5		PD-04 Bump and RDL planning with chip-package co-design feedback
Convergence and handoff	2.0		SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
Power ratio derivation	1.0		
Die-side iteration	1.0	EXIT CRITERIA	
Total	12.0	● Escape feasibility confirmed on the substrate, not only on the die ● Power-to-signal ratio meets the delivery requirement ● Map converged before the PD-04 freeze date	MEASURED BY ● Iterations to convergence ● Escape feasibility at first check ● Power bumps against the delivery requirement
WHO IT TAKES	~5.2 FTE	DEPENDS ON	
Package designer Owns the package-side bump map		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	
Physical design liaison Die-side placement and constraints		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	
Power delivery engineer Power bump count and distribution		ARCH-08 Chip-level floorplan intent, pin and bump budget	
SI engineer Signal assignment for channel quality		PD-03 Power delivery network design and early IR analysis	
Package architect Arbitrates when the two sides cannot converge			

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-03

Interposer / RDL routing design

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w	18.0 M/M	~5.6 FTE	7	yes	Interposer designer
w16–w30 in stage	planned	while it runs	3 in parallel	no float	Package architect approves

PURPOSE

Route the **interposer and its redistribution layers** — the silicon layer that carries the die-to-HBM channels and everything else the substrate cannot.

On a 2.5D part the interposer carries the highest-density and highest-performance connections in the package. HBM channels alone run to thousands of nets at fine pitch, and their routing quality decides whether the memory interface trains at speed.

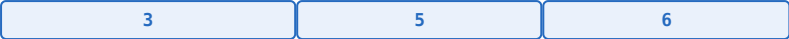
WORK FLOW7 STEPS · 2 LANES

w16w19w22w25w28w30

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Interposer layer stack and routing resource plan	2w	main
2	HBM channel routing and length management	4w	main
3	Die-to-die and remaining signal routing	3w	parallel
4	Power and ground distribution on the interposer	3w	main
5	RDL routing and via planning	2.5w	parallel
6	Interposer DRC and manufacturability	2.5w	parallel
7	Interposer database release	5w	main

Step 2 dominates because HBM channels are numerous, fine-pitch and length-sensitive at once. Getting them routed with matched lengths and clean references is most of the interposer's difficulty and most of its schedule.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Bump map from PKGD-02
- HBM channel requirements from ARCH-03 and ARCH-04
- Interposer technology rules from the supplier
- Power delivery requirement from PD-03
- SI requirements from SIPI-07

PRODUCES

- Interposer layer stack and routing plan
- Routed HBM channels with length management
- Die-to-die and signal routing
- Interposer power distribution
- RDL routing and via plan
- Released interposer database

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D2	PRODUCES	Bump map and interposer / RDL database. The interposer half of the deliverable — the routed silicon layer itself.
PKGD-D4	FEEDS	Package electrical design intent. Interposer routing is the highest-performance part of the package's electrical behaviour.

RISKS AND WATCH-OUTS

- ◆ **HBM channel lengths unmatched.** Skew across a wide parallel interface eats the training margin, and the failure appears at bring-up as a memory that will not train at speed.
- ◆ **Reference plane discontinuities.** A channel crossing a plane split has an impedance discontinuity, and on HBM there are thousands of channels to get right.
- ◆ **Interposer power distribution underprovisioned.** Current for two HBM stacks and the die passes through the interposer, and its resistance is part of the IR budget.
- ◆ **Manufacturability checked late.** Interposer rules differ from die rules, and a design that violates them is a supplier rejection rather than a warning.
- ◆ **Routing started before the bump map converges.** Every bump change is an interposer rework, and the rework is expensive at this density.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
HBM channel routing	5.0	Bump map converged in PKGD-02	PKGD-05 Package routing for signal integrity — length matching, reference plane and via design
Power distribution	3.5	Interposer technology rules available from the supplier	PKGD-11 Package design freeze, DRC and tooling release
Signal and die-to-die routing	3.0	Channel requirements known from SIPI-07	SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
RDL and via planning	2.5		SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
Database release	2.0		PTV-05 TV die fabrication and interposer / substrate vehicle build
DRC and manufacturability	2.0		
Total	18.0	EXIT CRITERIA - HBM channel lengths matched within the training budget - Reference planes continuous under high-speed channels - Interposer DRC clean against supplier rules	MEASURED BY - Channel length skew against the training budget - Reference plane violations - Interposer DRC violations at supplier review
WHO IT TAKES	~5.6 FTE	DEPENDS ON PKGD-02 Bump map, pitch and power-ground planning with physical design ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels) ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCIe, HBM3/3E) PD-03 Power delivery network design and early IR analysis	
Interposer designer Owns interposer routing and its release			
SI engineer Channel length, reference and impedance requirements			
Power delivery engineer Interposer power distribution			
Supplier liaison Interposer technology rules and manufacturability			
Package architect Layer stack and resource allocation			

16 PKGD · PACKAGE & SUBSTRATE DESIGN

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w w18–w32 in stage	18.0 M/M planned	~5.6 FTE while it runs	7 3 in parallel	yes no float	Substrate designer Power integrity engineer approves

PURPOSE

Design the **substrate** — stack-up, escape routing, power planes, package DRC — the organic carrier that connects the interposer to the board.

The substrate is the part with the longest lead time and the least flexibility once ordered. Its layer count is fixed by escape routing needs and its cost scales with it, so the design converges on the smallest stack-up that can carry the escape — a trade made once and lived with.

WORK FLOW7 STEPS · 2 LANES

w18	w21	w24	w27	w30	w32
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	Stack-up definition and layer count decision	2w	main
2	Escape routing from the interposer footprint	4w	main
3	Power and ground plane design	3w	parallel
4	Board-side ball assignment and breakout	3w	main
5	Via structure and drill planning	2.5w	parallel
6	Package DRC against supplier rules	2.5w	parallel
7	Substrate database release	5w	main

Step 1 is the cost decision. Every substrate layer adds cost across the product's lifetime, and the layer count is set by whether the escape routing fits — which is not known until step 2 is attempted. The two steps iterate.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture from PKGD-01
- Interposer footprint from PKGD-03
- Board requirements and ball map from EVB-02
- Supplier design rules from PKGD-09
- Power delivery requirement from SIPI-05

PRODUCES

- Substrate stack-up and layer count
- Escape routing from the interposer
- Power and ground plane design
- Board-side ball assignment
- Via and drill plan
- Released substrate database

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D3	PRODUCES	Substrate design files (Gerber / ODB++) and stack-up. This activity is the deliverable, and it is what the supplier manufactures from.
PKGD-D7	FEEDS	Substrate and interposer PO with committed lead time. The order cannot be placed against a design that is not finished.

RISKS AND WATCH-OUTS

- ◆ **Layer count set before escape is attempted.** A stack-up chosen for cost that cannot carry the escape has to be increased, and the cost model was built on the smaller number.
- ◆ **Supplier rules obtained late.** Substrate rules vary by supplier and technology, and designing to generic rules produces a design one supplier can build.
- ◆ **Power planes fragmented by escape routing.** Escape vias perforate the planes, and the resulting current path is what SIPI-03 will analyse.
- ◆ **Ball map not agreed with the board.** The substrate's board-side interface is the board's problem too, and changing it late affects both.
- ◆ **Design released before DRC closes.** A supplier rejection costs weeks against a twenty-week lead time.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
Escape routing	5.0	● Package architecture selected in PKGD-01 ● Interposer footprint available from PKGD-03 ● Supplier design rules obtained via PKGD-09	PKGD-05 Package routing for signal integrity — length matching, reference plane and via design
Ball assignment and breakout	3.5		PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)
Power and ground planes	3.0		PKGD-11 Package design freeze, DRC and tooling release
Stack-up definition	2.5		SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
Via and drill planning	2.0		PTV-05 TV die fabrication and interposer / substrate vehicle build
DRC and release	2.0		
Total	18.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.6 FTE	● Layer count justified by an attempted escape, not assumed ● Package DRC clean against the actual supplier's rules ● Ball map agreed with the board design	● Layer count against the cost model assumption
Substrate designer Owns stack-up, routing and release			● DRC violations at supplier review
Package architect Layer count and cost trade			● Escape routing completed at the chosen layer count
Supplier liaison Design rules and manufacturability			
Board designer Ball map and board-side interface		DEPENDS ON	
Power integrity engineer Plane design and current paths		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	
		PKGD-03 Interposer / RDL routing design	
		PKGD-09 Substrate and interposer supplier selection and lead-time booking	

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-05

Package routing for signal integrity — length matching, reference plane and via design

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	12.0 M/M	~5.2 FTE	6	no	SI engineer
w26–w36 in stage	planned	while it runs	2 in parallel	has float	Supplier liaison approves

PURPOSE

Route the package for **signal integrity** — length matching, reference continuity, via design, crosstalk separation — so the channels can carry what the interfaces need.

Package routing is where high-speed channels are made or ruined. A via with a long stub, a trace crossing a plane split, or a differential pair routed without its partner all cost margin that the PHY then has to make up. SIPI-07 will measure the result; this activity is what determines it.

WORK FLOW6 STEPS · 2 LANES

w26	w28	w30	w32	w34	w36
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Channel budget allocation to the package	1.5w	main
2	Length matching and skew control per interface	2.5w	main
3	Reference plane continuity and return path design	2w	parallel
4	Via structure design and stub management	2w	main
5	Crosstalk separation and shielding	2w	parallel
6	Routing review against the SI requirements	4w	main

Step 4 is the one most often underestimated. Via stubs resonate at frequencies that matter for the fastest interfaces, and back-drilling or blind via structures are decisions with cost implications that belong to the substrate design rather than to routing.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Interposer and substrate routing from PKGD-03 and PKGD-04
- Channel budgets from AMS-03 and ARCH-04
- SI requirements and targets from SIPI-07
- Interface compliance requirements
- Supplier via and drill capability

PRODUCES

- Channel budget allocation to the package
- Length-matched routing per interface
- Reference plane continuity design
- Via structures with managed stubs
- Crosstalk separation plan
- SI routing review record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D4	PRODUCES	Package electrical design intent and model handoff to co-verification. The signal half of the deliverable — the design intent SIPI then verifies.
PKGD-D3	FEEDS	Substrate design files and stack-up. SI-driven routing is part of what the substrate files contain.

RISKS AND WATCH-OUTS

- ◆ **Via stubs unmanaged.** They resonate in band for the fastest interfaces, and the remedy — back-drilling or blind vias — is a substrate cost decision made too late.
- ◆ **Length matching without reference continuity.** Matched traces over a split plane have matched delay and mismatched impedance.
- ◆ **Channel budget not allocated.** Die, package and board each consume part of the loss budget, and without allocation each assumes the others left room.
- ◆ **Crosstalk managed by spacing alone.** On a dense substrate there is no spare space, and the answer is usually reference and via design instead.
- ◆ **Review after the substrate is released.** SI findings then have nowhere to go except into the PHY's margin.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Length matching and skew	3.0	● Interposer and substrate routing progressing ● Channel budgets available from AMS-03 ● SI targets defined by SIPI-07	PKGD-11 Package design freeze, DRC and tooling release
Via structure design	2.5		SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
Reference and return path	2.0		SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)
Crosstalk separation	2.0		S0-06 Chip-package-system co-analysis signoff review against SIPI results
Budget allocation	1.5		
Review	1.0	EXIT CRITERIA	
Total	12.0	● Channel budget explicitly allocated between die, package and board ● Via stubs managed rather than accepted ● Routing reviewed against SI requirements before substrate release	MEASURED BY
WHO IT TAKES	~5.2 FTE		● Channel loss against the package's allocated budget ● Via stub length on critical interfaces ● Length skew per interface
SI engineer Owns channel budgets and routing requirements			
Package designer Implements length, reference and via design		DEPENDS ON	
Interposer designer Interposer-side channel routing		PKGD-03 Interposer / RDL routing design	
PHY liaison Channel budget from the die side		PKGD-04 Substrate stack-up, escape routing and package DRC	
Supplier liaison Via and drill capability		ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	
		AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY)	

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-06

Package PDN and decap footprint design (electrical closure owned by SIPI)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	10.0 M/M	~4.8 FTE	6	no	Package power integrity engineer
w28-w37 in stage	planned	while it runs	2 in parallel	has float	Supplier liaison approves

PURPOSE

Design the **package power delivery** — plane structure, via arrays, decoupling footprints — knowing that its electrical closure belongs to SIPI rather than to this activity.

The package sits between the die's instantaneous current demand and the board's slower supply. Its planes, vias and decoupling determine the impedance in the mid-frequency band where neither the die's on-chip decap nor the board's bulk capacitors help. Designing it is here; proving it works is co-verification.

WORK FLOW6 STEPS · 2 LANES

w28w30w32w34w36w37

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Package PDN topology and plane allocation	2w	main
2	Power via array design and current capability	2w	main
3	Decoupling capacitor footprint placement	1.5w	parallel
4	Plane perforation and current path management	2w	main
5	Interposer to substrate power transition	1.5w	parallel
6	PDN design intent handoff to co-verification	3w	main

Step 3 places footprints rather than choosing values. Which capacitors go where is decided by `SIPI-05` against the impedance target; the package's job is to provide mounting positions close enough to the current draw for those choices to be available.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Substrate stack-up from PKGD-04
- Power delivery requirement from PD-03
- Bump and ball power assignment from PKGD-02
- Impedance targets from SIPI-03
- Supplier capability for via arrays and mounting

PRODUCES

- Package PDN topology and plane allocation
- Power via array design
- Decoupling footprint placement
- Plane perforation and current path analysis
- Interposer to substrate power transition design
- PDN design intent handoff

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D4	PRODUCES	Package electrical design intent and model handoff to co-verification. The power half of the deliverable, handed to SIPI for closure.
PKGD-D3	FEEDS	Substrate design files and stack-up. Plane allocation and via arrays are substrate content.

RISKS AND WATCH-OUTS

- ◆ **Decoupling footprints too far from the current draw.** Mounting position determines the loop inductance, and a capacitor mounted far away is a capacitor that does not help.
- ◆ **Plane perforation ignored.** Escape vias perforate power planes, and the remaining current path can be far worse than the plane suggests.
- ◆ **Via array sized for DC only.** Transient current is what the package has to supply, and DC sizing understates the requirement.
- ◆ **Closure assumed to be someone else's problem.** If the design intent is not analysable, SIPI cannot close it and the iteration comes back here.
- ◆ **Interposer to substrate transition overlooked.** The power path crosses two interfaces, and the transition is where its resistance concentrates.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
PDN topology and planes	2.5	● Substrate stack-up defined in PKGD-04	PKGD-11 Package design freeze, DRC and tooling release
Power via arrays	2.0	● Power delivery requirement known from PD-03	SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
Plane perforation management	2.0	● Impedance targets available from SIPI-03	SIPI-05 Decoupling capacitor budget and placement optimisation across die, package and board
Decoupling footprints	1.5		S0-06 Chip-package-system co-analysis signoff review against SIPI results
Power transition design	1.0		
Handoff	1.0		
Total	10.0		
WHO IT TAKES	~4.8 FTE	EXIT CRITERIA	MEASURED BY
Package power integrity engineer Owns package PDN design		● Decoupling footprints placed for loop inductance, not convenience	● Package PDN impedance against target
Substrate designer Plane and via implementation		● Plane perforation analysed rather than assumed away	● Decoupling loop inductance
SIPI liaison Impedance targets and closure feedback		● Design intent handed over in a form SIPI can analyse	● Current path resistance through perforated planes
Die power delivery engineer Die-side current profile		DEPENDS ON	
Supplier liaison Via array and mounting capability		PKGD-02 Bump map, pitch and power-ground planning with physical design	
		PKGD-04 Substrate stack-up, escape routing and package DRC	
		PD-03 Power delivery network design and early IR analysis	
		SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target	

PKGD-07

Thermal and mechanical simulation
— warpage, co-planarity, TIM, lid

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	12.0 M/M	~4.3 FTE	7	no	Thermal engineer
w20–w32 in stage	planned	while it runs	3 in parallel	has float	Die power liaison approves

PURPOSE

Simulate the **thermal and mechanical behaviour** of the package — heat path, warpage, co-planarity, TIM and lid — before any of it is built.

A package at this power dissipates hundreds of watts through a stack of materials with different expansion coefficients. The thermal path decides the achievable power envelope; the mechanical behaviour decides whether the assembly survives reflow. Both are simulated here and measured on real vehicles in PTV.

WORK FLOW

7 STEPS · 2 LANES

w20	w23	w26	w29	w32
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MAIN SEQUENCE

1	2	5	7
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IN PARALLEL

3	4	6
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#	STEP	TAT	LANE
1	Thermal model construction — die, interposer, substrate, lid	2w	main
2	Steady-state thermal simulation and Rjc estimation	2.5w	main
3	Hotspot and transient thermal analysis	2w	parallel
4	TIM selection and interface resistance study	2w	parallel
5	Warpage simulation across the reflow profile	3w	main
6	Mechanical stress and co-planarity analysis	2w	parallel
7	Thermal and mechanical design recommendations	4.5w	main

Step 5 predicts what `PTV-07` will measure. Warpage across the reflow profile is the mechanism behind most 2.5D assembly failures, and simulating it before the vehicles are built is what makes the vehicle results interpretable rather than merely surprising.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture from PKGD-01
- Power map and hotspot data from PD-03 and SYN-06
- Material properties from the supplier
- Assembly reflow profile from PKGD-10
- Thermal requirements from DEF-03

PRODUCES

- Thermal model of the package stack
- Steady-state thermal results and Rjc estimate
- Hotspot and transient analysis
- TIM selection and interface study
- Warpage simulation across reflow

- Mechanical stress and co-planarity results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D5	PRODUCES	Thermal and mechanical (warpage) simulation reports. This activity is the deliverable, and PTV is what validates it.
PKGD-D6	FEEDS	Test vehicle requirement specification for PTV. The simulations decide which mechanical and thermal risks the vehicles must measure.

RISKS AND WATCH-OUTS

- ◆ **Thermal simulated at uniform power.** An accelerator's power is concentrated in the compute clusters, and uniform assumptions understate the hotspot badly.
- ◆ **Warpage simulated at room temperature only.** The failure happens at reflow peak, and the room-temperature number says nothing about it.
- ◆ **TIM performance taken from the datasheet.** Interface resistance depends on bond line thickness and pressure, both of which are assembly properties.
- ◆ **Material properties assumed.** Substrate and underfill properties vary by supplier and lot, and simulation is only as good as the inputs.
- ◆ **Results not fed into the vehicle plan.** Simulation identifies which risks to measure, and a vehicle plan built independently measures the wrong things.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Warpage simulation	3.0	Package architecture selected in PKGD-01	PKGD-08 Test vehicle requirement definition and handoff to PTV
Steady-state thermal	2.5	Power map available from PD-03	PKGD-10 OSAT selection and assembly process definition
TIM and interface study	2.0	Material properties obtained from suppliers	PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement
Hotspot and transient analysis	1.5		
Mechanical stress analysis	1.5		
Recommendations	1.5		
Total	12.0		
WHO IT TAKES	~4.3 FTE	EXIT CRITERIA - Thermal simulated against the real power map, not uniform power - Warpage simulated across the full reflow profile - Findings handed to PTV as vehicle requirements	PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)
Thermal engineer Owns thermal modelling and TIM selection			SIPI-10 Electro-thermal co-analysis coupled with the package thermal model
Mechanical engineer Warpage, stress and co-planarity		DEPENDS ON	MEASURED BY - Predicted R_{jc} against the PTV measurement - Predicted warpage against measured - Hotspot temperature against the thermal requirement
Package architect Design recommendations from the findings		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	
Supplier liaison Material properties and reflow profile		PD-03 Power delivery network design and early IR analysis	
Die power liaison Power map and hotspot distribution		DEF-03 PPA target definition — frequency, power envelope, die area budget	

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-08

Test vehicle requirement definition and handoff to PTV

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	4	no	Package architect
w10–w14 in stage	planned	while it runs	1 in parallel	has float	OSAT liaison approves

PURPOSE

Tell **PTV what the vehicles have to measure** — which risks, which structures, which conditions — early enough that the vehicles can be designed, built and stressed before wafer-out.

The test vehicle programme has a fifty-two week window and a hard gate at product wafer-out. Its requirements have to arrive while there is time to act on them, which is why this activity sits at week ten of a fifty-two week stage rather than after the package design is finished.

WORK FLOW

4 STEPS · 2 LANES

w10	w11	w12	w13	w14
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Package risk identification for vehicle coverage	1w	main
2	Vehicle type requirements — mechanical, thermal, electrical	1.5w	parallel
3	Measurement condition and structure specification	1.5w	main
4	Requirement handoff and PTV plan review	1.5w	main

Step 1 has to be done on an incomplete package design, which is uncomfortable and necessary. The architecture is known, the materials are known and the assembly process is broadly known — enough to say what will be risky, which is all the vehicle plan needs.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture from PKGD-01
- Thermal and warpage predictions from PKGD-07
- Assembly process direction from PKGD-10
- CPI and ULK risk from TECH-03
- Previous-program package failures

PRODUCES

- Package risk list for vehicle coverage
- Vehicle type requirements
- Measurement conditions and structure specification
- Requirement handoff to PTV

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D6

PRODUCES

Test vehicle requirement specification for PTV. This activity is the deliverable, and its timing matters as much as its content.

PKGD-D5

FEEDS

Thermal and mechanical simulation reports. The vehicles measure what the simulations predicted, so the two are specified together.

RISKS AND WATCH-OUTS

- ◆ **Requirements issued after the package design completes.** The vehicle programme then cannot finish before wafer-out, and the gate is missed.
- ◆ **Risks identified from the design rather than from experience.** The failures that matter are the ones previous programs hit, and they are rarely visible in a design review.
- ◆ **Electrical vehicle omitted.** Daisy-chain continuity and assembly yield are as important as thermal and mechanical, and they need their own vehicle.
- ◆ **Conditions unspecified.** A vehicle measured at the wrong temperature or reflow profile answers a question nobody asked.
- ◆ **No feedback loop defined.** The vehicle results have to change something, and the mechanism for that has to be agreed up front.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Risk identification	1.0	● Package architecture selected in PKGD-01 ● First thermal and warpage predictions available ● Assembly process direction known from PKGD-10	PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress
Measurement specification	0.8		PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon
Vehicle type requirements	0.8		PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement
Handoff and review	0.5		PTV-04 Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures
Total	3.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.3 FTE	● Requirements issued with time for the vehicle programme to finish before wafer-out ● Risks drawn from experience as well as from the design ● Feedback mechanism into package design agreed	● Weeks between requirement issue and the wafer-out gate ● Risks covered by vehicles against risks identified ● Vehicle findings that changed the package design
Package architect Owns the risk list and the requirements		DEPENDS ON	
Thermal and mechanical engineers Vehicle requirements from the simulations		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	
Reliability engineer Failure modes worth measuring		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
PTV lead Receives the requirements and confirms feasibility		TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	
OSAT liaison Assembly process realism in the vehicle plan			

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-09

Substrate and interposer supplier selection and lead-time booking

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	4.0 M/M	~1.7 FTE	6	yes	Backend supply chain lead
w22-w32 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Select the **substrate and interposer suppliers and book their lead time** — a twenty-week commitment that sets the date the package can be built.

Substrate lead time is the longest single external dependency in the back half of the program, and it is ordered against a design that is not finished. Selecting the supplier early, understanding their rules and booking capacity is what makes the order placeable the moment the design freezes.

WORK FLOW6 STEPS · 2 LANES

w22	w24	w26	w28	w30	w32
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MAIN SEQUENCE

1	2	5	6
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	Supplier long-list and capability screen	1.5w	main
2	Design rule and technology fit assessment	2w	main
3	Capacity and lead time confirmation	2w	parallel
4	Commercial negotiation and pricing	2w	parallel
5	Supplier selection and qualification plan	2.5w	main
6	Purchase order and lead time booking	4w	main

Step 6 takes four weeks because the order cannot be placed until the design is frozen, and the freeze is in `PKGD-11`. Everything before it exists so that the order can be placed the same week the design is released rather than a month later.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture from PKGD-01
- Substrate design requirements from PKGD-04
- Backend supply chain landscape from TECH-09
- Volume forecast and ramp from DEF-08
- Cost targets from DEF-04

PRODUCES

- Supplier capability screen
- Design rule and technology fit assessment
- Capacity and lead time confirmation
- Commercial terms
- Supplier selection and qualification plan
- Purchase order with committed lead time

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D7	PRODUCES	Substrate and interposer PO with committed lead time. This activity is the deliverable, and the committed date is what wafer-out has to be planned against.
PKGD-D3	FEEDS	Substrate design files and stack-up. Supplier rules determine what the substrate design can contain.

RISKS AND WATCH-OUTS

- ◆ **Order placed after wafer-out is committed.** A twenty-week substrate against a fixed wafer-out date means wafers waiting for a package.
- ◆ **Supplier rules obtained after design starts.** A substrate designed to generic rules may be unbuildable by the chosen supplier.
- ◆ **Capacity assumed.** Advanced substrate capacity is constrained, and a supplier who can build it may not be able to build it when needed.
- ◆ **Single supplier with no alternative.** A supply disruption then has no mitigation, and substrates are not quickly re-sourced.
- ◆ **Lead time quoted without a commitment.** An indicative lead time is not a date, and the difference appears when the schedule depends on it.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Supplier selection and qualification	1.0	● Package architecture selected in PKGD-01	PKGD-04 Substrate stack-up, escape routing and package DRC
Design rule assessment	0.8	● Substrate requirements known from PKGD-04	PKGD-11 Package design freeze, DRC and tooling release
Capacity and lead time	0.8	● Volume and ramp forecast available	PTV-05 TV die fabrication and interposer / substrate vehicle build
Purchase order	0.8		ASSY-04 Interposer and substrate assembly, underfill, molding
Commercial negotiation	0.5	EXIT CRITERIA	
Capability screen	0.3	● Lead time committed in writing, not indicative	
Total	4.0	● Supplier rules available before substrate design completes	MEASURED BY
WHO IT TAKES	~1.7 FTE	● Order placeable the week the design freezes	● Committed lead time against the wafer-out date
Backend supply chain lead		DEPENDS ON	● Weeks between design freeze and order placement
Owns supplier selection and the order		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	● Suppliers qualified against those screened
Substrate designer		TECH-09 OSAT and backend supply chain preliminary alignment	
Technology fit and rule assessment		DEF-08 Program schedule, resource and staffing plan drafting	
Procurement			
Commercial terms and purchase order			
Quality liaison			
Supplier qualification plan			
Program manager			
Lead time against the wafer-out date			

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-10

OSAT selection and assembly process definition

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	5.0 M/M	~2.4 FTE	6	no	Backend supply chain lead
w24–w33 in stage	planned	while it runs	2 in parallel	has float	PTV lead approves

PURPOSE

Select the **OSAT and define the assembly process** — bonding, underfill, molding, lid attach — because the process determines what the package design has to accommodate.

Assembly is not a service the package is handed to; it is a set of constraints the package is designed within. Bond force, reflow profile, underfill flow and lid attach all impose requirements, and choosing the OSAT late means designing a package that has to be adapted to whoever ends up building it.

WORK FLOW

6 STEPS · 2 LANES

w24	w26	w28	w30	w32	w33
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MAIN SEQUENCE

1	2	5	6
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IN PARALLEL

3	4
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#	STEP	TAT	LANE
1	OSAT capability screen for the package architecture	1.5w	main
2	Assembly process flow definition	2w	main
3	Bonding, underfill and molding process parameters	2w	parallel
4	Known-good-die handoff and responsibility model	1.5w	parallel
5	OSAT selection and capacity commitment	2w	main
6	Assembly agreement and process freeze plan	3.5w	main

Step 4 settles a question that becomes expensive later. When an assembled unit fails, whether the die or the assembly is responsible determines who bears the cost, and agreeing it while nobody has lost money is far easier than afterwards.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package architecture from PKGD-01
- Backend supply chain findings from TECH-09
- Thermal and mechanical requirements from PKGD-07
- Volume and ramp forecast from DEF-08
- Known-good-die strategy from TEST-01

PRODUCES

- OSAT capability screen
- Assembly process flow definition
- Process parameter specification
- Known-good-die responsibility model
- OSAT selection and capacity commitment
- Assembly agreement

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D8	PRODUCES	OSAT assembly process flow and agreement. This activity is the deliverable, and the agreement includes the responsibility model.
PKGD-D5	FEEDS	Thermal and mechanical simulation reports. The reflow profile the simulations use comes from the assembly process defined here.

RISKS AND WATCH-OUTS

- ◆ **OSAT selected after the package is designed.** The design then has to be adapted to the assembler's process rather than developed with it.
- ◆ **Advanced packaging capacity not committed.** The same constraint as the interposer, and it applies to assembly capacity independently.
- ◆ **Known-good-die responsibility undefined.** The first failed lot is the wrong moment to discover nobody agreed who owns the die.
- ◆ **Process parameters left to the OSAT.** Bond force and reflow profile affect warpage and CPI, and the package design has to know them.
- ◆ **Process frozen without vehicle data.** PTV-12 freezes the process window on evidence; freezing earlier means freezing on assumption.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Assembly process definition	1.5	● Package architecture selected in PKGD-01	PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
OSAT selection	1.0	● Backend landscape known from TECH-09	PKGD-11 Package design freeze, DRC and tooling release
Process parameters	1.0	● Volume and ramp forecast available	PTV-06 OSAT assembly of vehicles with process window DOE
Agreement and freeze plan	0.8		ASSY-01 Known-good-die sort and selection
Known-good-die model	0.8		ASSY-03 Die attach, micro-bump and thermo-compression bonding
Total	5.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~2.4 FTE	● OSAT selected with committed capacity	● Assembly capacity committed against the ramp requirement
Backend supply chain lead		● Process parameters known to the package design, not assumed	● Process parameters specified against those assumed
Owns OSAT selection and the agreement		● Known-good-die responsibility agreed in writing	● Responsibility model agreed before first build
Package architect		DEPENDS ON	
Process requirements from the design		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	
Quality liaison		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
Known-good-die responsibility and failure ownership		TECH-09 OSAT and backend supply chain preliminary alignment	
Procurement			
Capacity commitment and commercial terms			
PTV lead			
Process window validation plan			

16 PKGD · PACKAGE & SUBSTRATE DESIGN

PKGD-11

Package design freeze, DRC and tooling release

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w46–w52 in stage	6.0 M/M planned	~4.3 FTE while it runs	6 2 in parallel	yes no float	Package architect Procurement approves

PURPOSE

Freeze the **package design**, close its DRC, release the tooling — and start the twenty-week substrate lead time that everything downstream waits on.

The freeze is a schedule event more than a design one. The substrate order cannot be placed against a moving design, and the lead time that follows is what sets when packages can be built. Every week of delay here is a week at the far end, after wafer-out, where it is most visible and least recoverable.

WORK FLOW

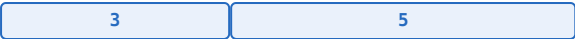
6 STEPS · 2 LANES

w46w48w50w52

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Design completeness review across interposer and substrate	1w	main
2	Final package DRC and supplier rule compliance	1.5w	main
3	Electrical intent confirmation against SIPI findings	1w	parallel
4	Tooling and manufacturing data release	1.5w	main
5	Supplier design review and acceptance	1.5w	parallel
6	Design freeze declaration and order release	2w	main

Step 5 runs in parallel and can send the design back. Suppliers review for manufacturability against their own process, and their acceptance is what makes the order placeable — a design frozen internally and rejected by the supplier has not frozen anything.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Interposer database from PKGD-03
- Substrate database from PKGD-04
- SI and PDN design from PKGD-05 and PKGD-06
- Co-verification findings from SIPI-11
- Supplier PO framework from PKGD-09

PRODUCES

- Design completeness review record
- Final package DRC results
- Electrical intent confirmation
- Released tooling and manufacturing data
- Supplier acceptance record
- Package Design Freeze declaration

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PKGD-D9

PRODUCES

Package Design Freeze package. This activity is the deliverable, and it carries the stage's milestone.

PKGD-D7

GATES

Substrate and interposer PO with committed lead time. The order is released on this freeze; nothing before it starts the lead time.

RISKS AND WATCH-OUTS

- ◆ **Freeze slipping.** Twenty weeks of substrate lead time follow it, and the delay lands after wafer-out where nothing can absorb it.
- ◆ **Frozen before co-verification completes.** A SIPI finding after freeze is a package change against an order already placed.
- ◆ **Supplier acceptance assumed.** A design frozen internally and rejected on manufacturability has not frozen anything except the schedule.
- ◆ **Tooling data incomplete.** Manufacturing data is more than the design files, and a missing item stalls the order after it is placed.
- ◆ **Late bump map change from the die side.** PD-04 has to be frozen first, and a die-side change after this point is extremely expensive.

EFFORT SPLIT	6 M/M	EXIT CRITERIA	FEEDS INTO
Tooling and data release	1.5	● Supplier has accepted the design, not only received it ● Order released the same week as the freeze ● Co-verification findings incorporated before freeze, not after	ASSY-04 Interposer and substrate assembly, underfill, molding
Final DRC	1.5		PTV-12 Feedback into package design and assembly process window freeze
Freeze and order release	1.0		MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Completeness review	1.0		
Supplier review	0.5		
Electrical confirmation	0.5		
Total	6.0	DEPENDS ON	MEASURED BY
WHO IT TAKES Package architect Owns the freeze and the release Package and substrate designers Completeness and DRC closure SIPI liaison Electrical intent confirmation Supplier liaison Design review and acceptance Procurement Order release on freeze	~4.3 FTE		● Freeze date against the substrate lead time requirement ● Supplier acceptance at first review ● Package changes requested after freeze
	PKGD-03 Interposer / RDL routing design		
	PKGD-04 Substrate stack-up, escape routing and package DRC		
	PKGD-05 Package routing for signal integrity — length matching, reference plane and via design		
	PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)		
	PKGD-09 Substrate and interposer supplier selection and lead-time booking		
ENTRY CRITERIA	PKGD-10 OSAT selection and assembly process definition		PD-04 Bump and RDL planning with chip-package co-design feedback
	PD-04		
	SIPI-11 Co-verification signoff review and criteria disposition with design and package teams		
● Interposer and substrate designs complete ● Co-verification signed off in SIPI-11 ● Bump map frozen in PD-04			

INTEGRATE

17

Package Test Vehicle (CPI / TTV)

PTV

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w36 – w88	52 _w	12	96 _{M/M}	6 of 12		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
PTV-01	Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress		6w	5 M/M	PTV-D1	Test vehicle plan and risk coverage matrix
PTV-02	Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon		9w	8 M/M	PTV-D2	MTV, TTV and daisy-chain vehicle designs
PTV-03	Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement		9w	8 M/M	PTV-D3	Built vehicle lots with assembly travelers
PTV-04	Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures		8w	6 M/M	PTV-D4	Warpage and co-planarity data across the reflow profile
PTV-05	TV die fabrication and interposer / substrate vehicle build		16w	6 M/M	PTV-D5	CPI stress assessment report — ULK, bump and BEOL integrity
PTV-06	OSAT assembly of vehicles with process window DOE		11w	12 M/M	PTV-D6	Thermal characterisation report with model correlation (Rjc, TIM, hotspot map)
PTV-07	Warpage and co-planarity measurement across the reflow profile (shadow moiré)		7w	8 M/M	PTV-D7	Board-level reliability data on vehicles
PTV-08	Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity		11w	14 M/M	PTV-D8	Frozen assembly process window definition
PTV-09	Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model		9w	10 M/M	PTV-D9	Package validation complete record — gate for product wafer-out
PTV-10	Board-level reliability on vehicles — temperature cycle, drop, bend		14w	8 M/M		
PTV-11	Daisy-chain continuity test and assembly yield learning		9w	6 M/M		
PTV-12	Feedback into package design and assembly process window freeze		7w	5 M/M		

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-01

Test vehicle strategy and risk item definition
— CPI, warpage, thermal, ULK/BEOL stress

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	5.0 M/M	~3.6 FTE	6	yes	PTV lead
w0–w6 in stage	planned	while it runs	2 in parallel	no float	OSAT liaison approves

PURPOSE

Decide **which package risks the vehicles will actually retire** — chip-package interaction, warpage, thermal, assembly yield — and what evidence would count as retiring them.

The vehicle programme has one shot and a hard gate at product wafer-out. Its value depends entirely on measuring the risks that would otherwise be discovered on real dies, and a vehicle plan built from what is easy to measure rather than from what is risky produces reassurance instead of information.

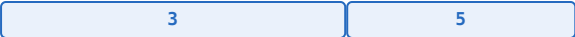
WORK FLOW6 STEPS · 2 LANES

w0w2w4w6

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Risk intake from package design and previous programs	1w	main
2	Risk prioritisation against product impact	1.5w	main
3	Evidence definition — what would retire each risk	1.5w	parallel
4	Vehicle type allocation per risk	1.5w	main
5	Schedule feasibility against the wafer-out gate	1w	parallel
6	Vehicle plan and risk coverage matrix	2w	main

Step 5 checks the whole programme against one date. Fifty-two weeks of vehicle work has to finish before product wafers come out at week 98, and if the arithmetic does not close then the plan has to shrink rather than the gate move.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vehicle requirements from PKGD-08
- Thermal and warpage predictions from PKGD-07
- CPI and ULK risk from TECH-03
- Assembly process direction from PKGD-10
- Wafer-out forecast from FAB-09

PRODUCES

- Package risk list with product impact
- Evidence definition per risk
- Vehicle type allocation
- Schedule feasibility finding
- Test vehicle plan and risk coverage matrix

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D1

PRODUCES

Test vehicle plan and risk coverage matrix. This activity is the deliverable, and the matrix is what every later result is reported against.

PTV-D9

FEEDS

Package validation complete record. Validation is complete when the matrix's risks are retired, which is why the matrix has to be right.

RISKS AND WATCH-OUTS

- ◆ **Plan built from what is measurable.** Easy measurements produce data; the risk list produces information, and the two are not the same set.
- ◆ **Schedule not checked against the gate.** A fifty-two week programme that finishes after wafer-out has validated nothing in time to matter.
- ◆ **Evidence undefined.** Without agreeing in advance what would retire a risk, the results become an argument rather than a conclusion.
- ◆ **Previous-program failures not consulted.** The package failure modes that recur are known, and they are the ones worth measuring.
- ◆ **Vehicle count driven by budget rather than by risk.** Fewer vehicles means fewer risks retired, and the trade should be explicit.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Risk prioritisation	1.5	● Vehicle requirements received from PKGD-08 ● Thermal and warpage predictions available ● Wafer-out forecast available from FAB-09	PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon
Vehicle allocation	1.0		PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement
Plan and coverage matrix	1.0		PTV-04 Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures
Evidence definition	0.8		PTV-12 Feedback into package design and assembly process window freeze
Risk intake	0.8		
Total	5.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~3.6 FTE	● Every carried risk has defined retiring evidence ● Programme schedule closes before the wafer-out gate ● Plan reflects prioritised risk rather than measurement convenience	● Risks with defined retiring evidence ● Schedule margin against the wafer-out gate ● Vehicle types allocated against risks identified
PTV lead Owns the plan and the coverage matrix			
Package architect Risk intake from the design			
Reliability engineer Failure modes worth measuring			
Program manager Schedule feasibility against the gate			
OSAT liaison Vehicle build feasibility		PKGD-07 Thermal and mechanical simulation — warpage, coplanarity, TIM, lid	
		PKGD-08 Test vehicle requirement definition and handoff to PTV	
		TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-02

Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w w5–w14 in stage	8.0 M/M planned	~3.9 FTE while it runs	6 2 in parallel	no has float	Mechanical engineer OSAT liaison approves

PURPOSE

Design the **mechanical test vehicle** — dummy die stack, representative bump array, substrate coupon — that carries the package's mechanical behaviour without any of its electronics.

The MTV exists to be stressed. Its die are blank, its interposer is representative and its purpose is to reproduce the mechanical stack faithfully enough that warpage, co-planarity and CPI measured on it predict what the product will do.

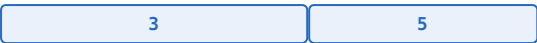
WORK FLOW6 STEPS · 2 LANES

w5w7w9w11w13w14

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Mechanical representativeness requirements	1.5w	main
2	Dummy die and die stack design	2w	main
3	Bump array design representative of the product	2w	parallel
4	Interposer and substrate coupon design	2.5w	main
5	Measurement feature integration — strain, reference marks	1.5w	parallel
6	MTV design release	3w	main

Step 1 defines what "representative" means for mechanics, which is narrower than it sounds. Die thickness, bump metallurgy, underfill and substrate construction matter; the circuitry does not. Getting that distinction right is what makes the MTV cheap.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vehicle plan from PTV-01
- Package architecture from PKGD-01
- Mechanical predictions from PKGD-07
- Die thickness and bump specification from PD-04
- Material specifications from suppliers

PRODUCES

- Mechanical representativeness requirements
- Dummy die and stack design
- Representative bump array
- Interposer and substrate coupon design
- Measurement feature integration
- Released MTV design

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D2

PRODUCES

MTV, TTV and daisy-chain vehicle designs. The mechanical vehicle of the three.

PTV-D4

FEEDS

Warpage and co-planarity data across the reflow profile. What the MTV measures is bounded by how it was designed.

RISKS AND WATCH-OUTS

- ◆ **Vehicle not mechanically representative.** Different die thickness or underfill produces warpage that does not predict the product's.
- ◆ **Measurement features omitted.** A vehicle with no reference marks or strain features can be measured externally and not internally.
- ◆ **Bump array simplified.** Bump density and distribution drive local stress, and a uniform array does not reproduce the product's concentrations.
- ◆ **Design released before the product's stack is settled.** A vehicle built on an assumption that changes measures the wrong package.
- ◆ **Coupon size unrepresentative.** Warpage scales with body size, and a smaller coupon understates it.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Coupon and substrate design	2.0	● Vehicle plan available from PTV-01	PTV-05TV die fabrication and interposer / substrate vehicle build
Dummy die and stack	1.8	● Package architecture and materials known	PTV-07Warpage and co-planarity measurement across the reflow profile (shadow moiré)
Bump array design	1.5	● Die thickness and bump specification available from PD-04	PTV-08Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity
Design release	1.5		
Measurement features	1.3		
Total	8.0	EXIT CRITERIA	
WHO IT TAKES	~3.9 FTE	● Mechanically representative in the dimensions that drive warpage	MEASURED BY
Mechanical engineer		● Measurement features integrated, not added externally	● Representativeness against the product stack
Owens representativeness and the MTV design		● Released against a settled product stack	● Measurement features integrated against those required
Package designer			● Design changes after release
Coupon and substrate design		DEPENDS ON	
Materials engineer		PTV-01Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress	
Die, underfill and substrate material match		PKGD-01Package architecture selection — 2.5D interposer type, organic substrate, layer count	
Metrology engineer		PKGD-07Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
Measurement feature requirements			
OSAT liaison			
Build feasibility			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-03

Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w w5–w14 in stage	8.0 M/M planned	~3.9 FTE while it runs	6 2 in parallel	no has float	Thermal engineer OSAT liaison approves

PURPOSE

Design the **thermal test vehicle** — embedded heaters and temperature sensors laid out to match the product's power map — so the package's thermal path can be measured rather than simulated.

A TTV is a die that only makes heat, in the pattern the real die will make it. Uniform heaters measure an average the product will never experience; heaters matched to the compute cluster distribution measure the hotspot that decides the achievable power envelope.

WORK FLOW

6 STEPS · 2 LANES

w5	w7	w9	w11	w13	w14
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Power map translation into heater zones	1.5w	main
2	Heater array design and drive requirements	2w	main
3	RTD and temperature sensor placement	2w	parallel
4	TTV die and interposer integration	2.5w	main
5	Calibration approach for the sensors	1.5w	parallel
6	TTV design release	3w	main

Step 3 decides what the vehicle can see. Sensors placed only at the die centre measure a number the datasheet can quote; sensors distributed across the zones measure the gradient that determines whether the hotspot is survivable.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vehicle plan from PTV-01
- Power map and hotspot distribution from PD-03 and SYN-06
- Thermal predictions from PKGD-07
- Package architecture from PKGD-01
- Sensor and heater process capability

PRODUCES

- Heater zone map from the power distribution
- Heater array design and drive requirements
- Sensor placement plan
- TTV die and interposer integration
- Sensor calibration approach
- Released TTV design

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D2	PRODUCES	MTV, TTV and daisy-chain vehicle designs. The thermal vehicle of the three.
PTV-D6	FEEDS	Thermal characterisation report with model correlation. What the TTV can measure bounds what the correlation can claim.

RISKS AND WATCH-OUTS

- ◆ **Uniform heaters.** They measure an average power distribution the product will never have, and the hotspot goes unmeasured.
- ◆ **Sensors too few or badly placed.** Thermal gradients are what matter, and a single centre sensor cannot see one.
- ◆ **Sensor calibration not planned.** An uncalibrated RTD gives a relative number, and thermal resistance needs an absolute one.
- ◆ **Heater drive capability underspecified.** Reproducing hundreds of watts through a test vehicle needs real power delivery, and the setup has to supply it.
- ◆ **Power map taken from an early estimate.** The distribution changes through implementation, and a TTV built on the architecture-era map measures the wrong hotspot.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
TTV die integration	2.0	● Vehicle plan available from PTV-01	PTV-05 TV die fabrication and interposer / substrate vehicle build
Heater array design	1.8	● Power map available from PD-03	
Sensor placement	1.5	● Thermal predictions available from PKGD-07	PTV-09 Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model
Design release	1.5		
Power map translation	1.3		
Total	8.0		
WHO IT TAKES	~3.9 FTE	EXIT CRITERIA	MEASURED BY
Thermal engineer Owns heater zoning and sensor placement		● Heater zones match the product's real power distribution	● Heater zone match against the power map ● Sensors placed against gradients of interest ● Power delivered against the product's envelope
TTV designer Heater array and die integration		● Sensors placed to see gradients, not only a peak	
Metrology engineer Sensor calibration approach		● Calibration approach defined before build	
Die power liaison Power map and hotspot distribution			
OSAT liaison Build feasibility			
		DEPENDS ON	
		PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress	
		PKGD-01 Package architecture selection — 2.5D interposer type, organic substrate, layer count	
		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-04

Daisy-chain electrical vehicle design —
interconnect continuity and bump yield structures

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w6–w14 in stage	6.0 M/M planned	~3.3 FTE while it runs	6 2 in parallel	no has float	Package test engineer Yield engineer approves

PURPOSE

Design the **daisy-chain electrical vehicle** — continuity structures through every interconnect level — so assembly yield and interconnect integrity can be measured electrically.

Daisy chains turn assembly quality into a resistance measurement. Every bump, every via and every ball becomes part of a chain that either conducts or does not, which makes an otherwise invisible assembly process observable and quantifiable.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Interconnect level inventory — bump, TSV, via, ball	1w	main
2	Daisy chain topology and segmentation design	2w	main
3	Kelvin structures for resistance measurement	1.5w	parallel
4	Chain-to-location mapping for defect localisation	1.5w	main
5	Probe and measurement access design	1.5w	parallel
6	Daisy-chain vehicle design release	3.5w	main

Step 2's segmentation decides the diagnostic resolution. One long chain says whether the assembly worked; many short segments say where it failed, and the difference is what turns a yield number into a corrective action.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vehicle plan from PTV-01
- Bump map and interconnect structure from PKGD-02 and PKGD-03
- Assembly process from PKGD-10
- Substrate ball map from PKGD-04
- Measurement capability at the OSAT

PRODUCES

- Interconnect level inventory
- Daisy chain topology with segmentation
- Kelvin resistance structures
- Chain-to-location mapping
- Probe and measurement access design
- Released daisy-chain vehicle design

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D2	PRODUCES	MTV, TTV and daisy-chain vehicle designs. The electrical vehicle of the three.
PTV-D3	FEEDS	Built vehicle lots with assembly travelers. The daisy-chain vehicle is what makes the build's quality measurable.

RISKS AND WATCH-OUTS

- ◆ **Chains too long.** A single break makes the whole chain open, and the yield number carries no information about where or why.
- ◆ **No Kelvin structures.** Two-wire resistance includes the probe path, and marginal interconnects are exactly where the difference matters.
- ◆ **Chain-to-location mapping omitted.** A failing chain that cannot be located gives no basis for a process change.
- ◆ **Not all interconnect levels covered.** Bump, TSV, via and ball each fail differently, and a vehicle covering only bumps misses the rest.
- ◆ **Measurement access not planned.** A vehicle that can only be measured with equipment the OSAT does not have is a vehicle that is not measured.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	DEPENDS ON
Chain topology and segmentation	1.5	● Vehicle plan available from PTV-01	PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress
Design release	1.3	● Interconnect structure known from PKGD-02 and PKGD-03	
Chain-to-location mapping	1.3	● OSAT measurement capability confirmed	PKGD-02 Bump map, pitch and power-ground planning with physical design
Kelvin structures	1.0		PKGD-03 Interposer / RDL routing design
Measurement access	1.0		PKGD-10 OSAT selection and assembly process definition
Total	6.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~3.3 FTE	● Chains segmented finely enough to localise a failure	PTV-05 TV die fabrication and interposer / substrate vehicle build
Package test engineer Owns chain design and measurement		● Every interconnect level covered by at least one structure	PTV-11 Daisy-chain continuity test and assembly yield learning
Package designer Chain routing through the interconnect levels		● Measurement possible with equipment the OSAT actually has	
Failure analysis engineer Localisation requirements			MEASURED BY
OSAT liaison Measurement capability and access			● Localisation resolution of the chain design
Yield engineer Segmentation for yield attribution			● Interconnect levels covered
			● Measurability at the OSAT

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-05

TV die fabrication and interposer / substrate vehicle build

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	6.0 M/M	~1.6 FTE	6	yes	PTV lead
w14–w30 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Build the **vehicle hardware** — TV die, interposers, substrates — through the same suppliers and processes the product will use.

Sixteen weeks and six man-months: mostly waiting, on the same supplier queues the product will stand in. Using the real suppliers and the real processes is the point — a vehicle built through a different route measures a package the product will never have.

WORK FLOW

6 STEPS · 2 LANES

w14	w18	w22	w26	w30
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Vehicle build planning and quantity definition	1.5w	main
2	TV die fabrication — dummy, TTV and daisy-chain	6w	main
3	Interposer vehicle build	5w	parallel
4	Substrate vehicle build	5.5w	main
5	Incoming inspection and material qualification	2.5w	parallel
6	Material readiness for assembly	3w	main

Step 4 sits on the same substrate queue as the product, which is why it needs planning rather than ordering. Vehicle substrates competing with product substrates for the same capacity is a scheduling problem the programme creates for itself if nobody sequences them.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Vehicle designs from PTV-02, PTV-03 and PTV-04
- Supplier capacity from PKGD-09
- Substrate and interposer processes from PKGD-03 and PKGD-04
- Build quantity from the DOE plan in PTV-06
- Material specifications

PRODUCES

- Vehicle build plan and quantities
- Fabricated TV die
- Interposer vehicles
- Substrate vehicles
- Incoming inspection records
- Materials ready for assembly

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-03	PRODUCES	Built vehicle lots with assembly travelers. The material half of the deliverable; PTV-06 assembles them.
PTV-09	FEEDS	Package validation complete record. Nothing can be validated before the vehicles exist.

RISKS AND WATCH-OUTS

- ◆ **Vehicles built through a different process.** A package assembled from materials the product will not use measures a package that will not exist.
- ◆ **Vehicle builds competing with product builds.** Same suppliers, same queue; without sequencing one delays the other.
- ◆ **Quantity too small for the DOE.** The process window study needs enough units to vary conditions, and a build sized for a single condition cannot support it.
- ◆ **Incoming inspection skipped.** A vehicle built from out-of-specification material produces results that describe the material rather than the package.
- ◆ **Build schedule not tracked against the gate.** Sixteen weeks of build inside a fifty-two week programme leaves limited room for a slip.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	DEPENDS ON
Substrate and interposer builds	2.0	● Vehicle designs released from PTV-02, PTV-03 and PTV-04	PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon
TV die fabrication	1.5	● Supplier capacity booked through PKGD-09	PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement
Incoming inspection	1.0	● DOE quantity requirement known from PTV-06	PTV-04 Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures
Build planning	0.8		PKGD-09 Substrate and interposer supplier selection and lead-time booking
Material readiness	0.8		
Total	6.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~1.6 FTE	● Vehicles built through the product's own suppliers and processes	PTV-06 OSAT assembly of vehicles with process window DOE
PTV lead		● Quantity sufficient for the assembly DOE	PTV-12 Feedback into package design and assembly process window freeze
Owns the build plan and schedule		● Materials inspected before assembly, not after	MEASURED BY
Supply chain lead			● Build completion against the gate schedule
Supplier sequencing against product builds			● Units built against the DOE requirement
Package designer			● Incoming inspection findings
Vehicle design support during build			
Quality engineer			
Incoming inspection and material qualification			
Program manager			
Schedule tracking against the wafer-out gate			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-06

OSAT assembly of vehicles with process window DOE

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
11w	12.0 M/M	~4.7 FTE	7	yes	Assembly process engineer
w28–w39 in stage	planned	while it runs	3 in parallel	no float	PTV lead approves

PURPOSE

Assemble the vehicles at the OSAT across a **designed experiment** — bond force, reflow profile, underfill, cure — so the assembly process window is measured rather than assumed.

The process window is what ASSY will run to. Establishing it needs deliberate variation: units assembled at the edges of each parameter, not just at nominal, so that the window's boundaries are known rather than the centre being confirmed.

WORK FLOW

7 STEPS · 2 LANES

w28	w31	w34	w37	w39
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MAIN SEQUENCE

1	2	4	7
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IN PARALLEL

3	5	6
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#	STEP	TAT	LANE
1	DOE design across the assembly parameters	1.5w	main
2	Bonding and die attach across conditions	2.5w	main
3	Underfill and cure across conditions	2w	parallel
4	Molding, lid and TIM attach	2w	main
5	Inline inspection during assembly — X-ray, CSAM	2w	parallel
6	Assembly traveler and condition traceability	2w	parallel
7	Assembled lot release to characterisation	5w	main

Step 6 is what makes the DOE usable. Every unit has to be traceable to the exact conditions it was assembled under, and an assembly traveler that records the lot but not the condition turns a designed experiment back into a single build.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Built vehicle materials from PTV-05
- Assembly process definition from PKGD-10
- Process parameter ranges from the OSAT
- Thermal and mechanical predictions from PKGD-07
- Inspection capability at the OSAT

PRODUCES

- Assembly DOE design
- Assembled units across conditions
- Inline inspection results
- Assembly travelers with condition traceability
- Assembled lots released to characterisation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D3	PRODUCES	Built vehicle lots with assembly travelers. The assembly half of the deliverable, with the condition traceability that makes it useful.
PTV-D8	FEEDS	Frozen assembly process window definition. The window is derived from this DOE, so its design determines what can be known.

RISKS AND WATCH-OUTS

- ◆ **Assembly at nominal only.** That confirms the centre and says nothing about the boundaries, which is where production will occasionally operate.
- ◆ **Condition traceability lost.** Units that cannot be attributed to their assembly conditions turn the DOE into an expensive single build.
- ◆ **DOE too large for the unit count.** Varying every parameter needs units the build did not produce, and an under-powered DOE gives ambiguous results.
- ◆ **Inline inspection omitted.** Assembly defects found only at final characterisation cannot be attributed to a process step.
- ◆ **OSAT running its standard process.** The point is deliberate variation, and an assembler defaulting to its recipe produces no window information.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Bonding and die attach	3.0	● Vehicle materials ready from PTV-05	PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)
Lot release and handling	2.5	● Assembly process defined in PKGD-10	PTV-08 Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity
Underfill and cure	2.0	● OSAT parameter ranges agreed	PTV-09 Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model
Inline inspection	2.0	EXIT CRITERIA	PTV-10 Board-level reliability on vehicles — temperature cycle, drop, bend
Molding and lid attach	1.5	● Units assembled across the parameter ranges, not only at nominal	PTV-11 Daisy-chain continuity test and assembly yield learning
DOE design and travelers	1.0	● Every unit traceable to its assembly conditions	PTV-12 Feedback into package design and assembly process window freeze
Total	12.0	● Inline inspection performed during assembly, not after	MEASURED BY
WHO IT TAKES	~4.7 FTE	DEPENDS ON	● Conditions covered against the DOE design
Assembly process engineer Owns the DOE and its execution		PTV-05 TV die fabrication and interposer / substrate vehicle build	● Units traceable to conditions
OSAT process engineer Parameter ranges and assembly execution		PKGD-10 OSAT selection and assembly process definition	● Assembly defects caught inline
Quality engineer Traveler discipline and traceability			
Metrology engineer Inline inspection during assembly			
PTV lead DOE scope against unit count			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-07

Warpage and co-planarity measurement across the reflow profile (shadow moiré)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w38–w45 in stage	8.0 M/M planned	~5.0 FTE while it runs	6 2 in parallel	no has float	Mechanical engineer Quality engineer approves

PURPOSE

Measure **warpage and co-planarity across the reflow profile** — not at room temperature, but at every point in the thermal cycle where the package deforms.

Warpage is the mechanism behind most 2.5D assembly failures. A package flat at room temperature can bow by more than a hundred microns at reflow peak, opening joints that then re-close as it cools — leaving a unit that looks fine and has intermittent connections.

WORK FLOW

6 STEPS · 2 LANES

w38	w40	w42	w44	w45
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Measurement setup — shadow moiré or equivalent	1w	main
2	Room-temperature baseline across units	1.5w	main
3	Warpage measurement across the reflow profile	2w	main
4	Co-planarity and ball flatness measurement	1.5w	parallel
5	Condition-to-warpage correlation from the DOE	1.5w	parallel
6	Comparison against the PKGD-07 simulation	2.5w	main

Step 6 is what makes the measurement reusable. A warpage number is one package; a validated simulation model predicts the next one, and correlating the two is what turns a measurement into a capability.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assembled units with condition traceability from PTV-06
- Warpage simulation from PKGD-07
- Reflow profile from PKGD-10
- Measurement equipment and capability
- Co-planarity specification from the assembly requirements

PRODUCES

- Measurement setup and method
- Room-temperature warpage baseline
- Warpage across the reflow profile
- Co-planarity and ball flatness results
- Condition correlation from the DOE
- Simulation comparison

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D4	PRODUCES	Warpage and co-planarity data across the reflow profile. This activity is the deliverable, and the reflow-profile part is what distinguishes it from a room-temperature check.
PTV-D8	FEEDS	Frozen assembly process window definition. Warpage against condition is one of the window's defining constraints.

RISKS AND WATCH-OUTS

- ◆ **Measured only at room temperature.** The failure occurs at reflow peak, and the room-temperature number is not predictive of it.
- ◆ **Simulation comparison skipped.** Without correlation the model stays unvalidated, and the next package has to be measured from scratch.
- ◆ **Co-planarity ignored.** Ball flatness determines whether the package mounts to the board reliably, and it is a separate property from warpage.
- ◆ **Too few units to see condition effects.** Warpage varies unit to unit, and distinguishing a condition effect from noise needs population.
- ◆ **Results not linked to assembly conditions.** Knowing the warpage without knowing which condition produced it gives no basis for a window.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Warpage across reflow	2.0	● Assembled units available from PTV-06	PTV-12 Feedback into package design and assembly process window freeze
Simulation comparison	1.8	● Simulation predictions available from PKGD-07	PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
Room-temperature baseline	1.5	● Measurement equipment available	ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology
Co-planarity measurement	1.5		
Condition correlation	1.3	EXIT CRITERIA	
Total	8.0	● Warpage measured across the full reflow profile	MEASURED BY
WHO IT TAKES	~5.0 FTE	● Simulation correlated against measurement	● Peak warpage against the limit
Mechanical engineer		● Results attributed to assembly conditions	● Simulation error against measurement
Owns warpage measurement and correlation		DEPENDS ON	● Condition effects distinguishable from noise
Metrology engineer		PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon	
Measurement setup and execution		PTV-06 OSAT assembly of vehicles with process window DOE	
Simulation engineer		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
Model comparison and update			
Assembly process engineer			
Condition attribution			
Quality engineer			
Co-planarity against specification			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-08

Chip-package interaction stress evaluation —
low-k / ULK crack, bump and BEOL integrity

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
11w	14.0 M/M	~5.5 FTE	6	yes	Reliability engineer
w37–w48 in stage	planned	while it runs	2 in parallel	no float	Die liaison approves

PURPOSE

Evaluate **chip-package interaction** — the stress the package puts into the die's low-k dielectric stack, and whether the bumps and back-end metal survive it.

CPI is the failure mode that makes 2.5D packaging risky at advanced nodes. Ultra-low-k dielectrics are mechanically weak, the package expands differently from the silicon, and the resulting stress concentrates exactly where the bumps meet the top metal. It cracks or it does not, and simulation alone is not enough to know.

WORK FLOW6 STEPS · 2 LANES

w37	w40	w43	w46	w48
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	CPI stress evaluation plan and structure selection	1.5w	main
2	Thermal cycling stress application	3w	main
3	Bump and BEOL integrity inspection	2.5w	parallel
4	Low-k and ULK crack detection	2.5w	main
5	Cross-section and failure analysis on affected units	2w	parallel
6	CPI assessment and design implications	4w	main

Step 5 is destructive and necessary. Electrical and acoustic methods say something is wrong; only cross-sectioning shows what failed and where, and the design implication depends entirely on which layer cracked.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assembled MTV units from PTV-06
- Die stack and ULK information from TECH-03
- Warpage results from PTV-07
- Thermal cycling capability
- Underfill and material properties

PRODUCES

- CPI evaluation plan and structure selection
- Thermal cycling stress results
- Bump and BEOL integrity findings
- ULK crack detection results
- Cross-section and failure analysis
- CPI assessment with design implications

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D5	PRODUCES	CPI stress assessment report — ULK, bump and BEOL integrity. This activity is the deliverable, and it is the most consequential result the vehicle programme produces.
PTV-D9	FEEDS	Package validation complete record. CPI is usually the risk the gate most depends on.

RISKS AND WATCH-OUTS

- ◆ **CPI evaluated by simulation alone.** Material behaviour at these dimensions is not predictable to the accuracy the decision needs.
- ◆ **Insufficient thermal cycles.** CPI failures accumulate; a short cycle count shows nothing and reassures wrongly.
- ◆ **Cross-sectioning skipped.** Detection without localisation gives no design implication, and the implication is the output that matters.
- ◆ **Only nominal units stressed.** The DOE's edge conditions are where CPI is most likely, and testing only the centre misses it.
- ◆ **Findings arriving after package design freezes.** A CPI problem needs an underfill, bump or design change, none of which are available after PKGD-11.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
CPI assessment and implications	3.5	● Assembled units available from PTV-06	PTV-12 Feedback into package design and assembly process window freeze
Thermal cycling	3.0	● Die stack information available from TECH-03	PKGD-11 Package design freeze, DRC and tooling release
Crack detection	2.5	● Thermal cycling and FA capability available	MP-04 Package qualification — MSL, drop, bend, board-level reliability
Cross-section and FA	2.5		
Bump and BEOL inspection	2.5		
Total	14.0	EXIT CRITERIA	
WHO IT TAKES	~5.5 FTE	● Stress applied to a cycle count that would reveal accumulation	MEASURED BY
Reliability engineer Owns CPI evaluation and assessment		● Failures cross-sectioned and localised, not only detected	● Thermal cycles survived against the qualification requirement
Failure analysis engineer Cross-section and crack detection		● Findings delivered before the package design freeze	● Failures localised against detected
Materials engineer Underfill and stress mitigation options		DEPENDS ON	● Weeks between findings and package design freeze
Package architect Design implications from the findings		PTV-02 Mechanical test vehicle (MTV) design — bump array, dummy die stack, substrate coupon	
Die liaison BEOL and ULK stack implications		PTV-06 OSAT assembly of vehicles with process window DOE	
		PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)	
		TECH-03 Density / performance / leakage benchmarking against product targets (DTCO study)	

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-09

Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	10.0 M/M	~4.8 FTE	6	no	Thermal engineer
w38–w47 in stage	planned	while it runs	2 in parallel	has float	Product engineering approves

PURPOSE

Characterise the package's **thermal performance** — junction-to-case and junction-to-ambient resistance, TIM behaviour, hotspot mapping — and correlate it against the simulation.

The thermal path decides the achievable power envelope, and the product is sold on that envelope. Measuring it on a TTV that reproduces the real power distribution is what turns a simulated Rjc into a number the datasheet can carry.

WORK FLOW

6 STEPS · 2 LANES

w38	w40	w42	w44	w46	w47
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Thermal measurement setup and sensor calibration	1.5w	main
2	Steady-state Rjc and Rja measurement	2w	main
3	Hotspot mapping across the heater zones	2w	parallel
4	TIM performance and bond line study	2w	main
5	Transient thermal response measurement	1.5w	parallel
6	Correlation against the PKGD-07 thermal model	3.5w	main

Step 4 measures the interface nobody can simulate reliably. TIM performance depends on bond line thickness, pressure and pump-out over time, and the datasheet value is a starting point rather than an answer.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assembled TTV units from PTV-06
- Thermal simulation from PKGD-07
- Sensor calibration approach from PTV-03
- Thermal measurement equipment
- TIM material options

PRODUCES

- Thermal measurement setup and calibration
- Rjc and Rja measurements
- Hotspot maps across zones
- TIM performance and bond line results
- Transient thermal response
- Model correlation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D6	PRODUCES	Thermal characterisation report with model correlation. This activity is the deliverable, and the correlation is what makes the model reusable.
PTV-D8	FEEDS	Frozen assembly process window definition. TIM bond line is an assembly parameter, and its thermal effect is measured here.

RISKS AND WATCH-OUTS

- ◆ **TIM performance taken from the datasheet.** Bond line thickness and pressure are assembly parameters, and the achieved performance differs from the specification.
- ◆ **Hotspot not measured.** The average junction temperature is not what limits the part; the hottest zone is.
- ◆ **Simulation not correlated.** The model then remains unvalidated, and every future thermal question needs another vehicle.
- ◆ **Transient response ignored.** An accelerator's power is bursty, and the transient behaviour determines whether the package can absorb a workload spike.
- ◆ **Measured at a single ambient condition.** Rja depends on airflow and ambient, and one condition does not characterise the range the product will see.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Model correlation	2.5	● Assembled TTV units available from PTV-06	PTV-12 Feedback into package design and assembly process window freeze
Rjc and Rja measurement	2.0	● Thermal simulation available from PKGD-07	PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid
TIM study	2.0	● Sensors calibrated per the PTV-03 approach	SIPI-10 Electro-thermal co-analysis coupled with the package thermal model
Hotspot mapping	1.8		MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Transient measurement	1.8		
Total	10.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~4.8 FTE	● TIM performance measured as assembled, not as specified	● Measured Rjc against simulation
Thermal engineer		● Hotspot mapped rather than averaged	● Hotspot temperature against the envelope requirement
Owns characterisation and model correlation		● Thermal model correlated and updated	● TIM performance as assembled against specification
Metrology engineer		DEPENDS ON	
Measurement setup and calibration		PTV-03 Thermal test vehicle (TTV) design — embedded heater array and RTD sensor placement	
Materials engineer		PTV-06 OSAT assembly of vehicles with process window DOE	
TIM options and bond line study		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
Simulation engineer			
Model update from measurement			
Product engineering			
Envelope implications for the datasheet			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-10

Board-level reliability on vehicles
— temperature cycle, drop, bend

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
14w	8.0 M/M	~2.5 FTE	6	yes	Reliability engineer
w36–w50 in stage	planned	while it runs	2 in parallel	no float	Package architect approves

PURPOSE

Stress the vehicles through **board-level reliability** — temperature cycling, drop, bend — because the package has to survive being mounted in a system, not only being assembled.

Board-level reliability is where the package meets the customer's manufacturing process. Solder joints crack under thermal cycling, boards flex during handling, and a package that passes component qualification and fails board-level testing is a package the customer cannot use.

WORK FLOW

6 STEPS · 2 LANES

w36	w39	w42	w45	w48	w50
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Board-level test plan and coupon board design	2w	main
2	Coupon board fabrication and vehicle mounting	2.5w	main
3	Temperature cycling to failure	4w	main
4	Drop and shock testing	2.5w	parallel
5	Bend and flex testing	2w	parallel
6	Failure mode analysis and characteristic life	5.5w	main

Step 3 runs to failure rather than to a pass criterion. Characteristic life — the cycle count at which a defined fraction fails — is what a customer needs to assess the package in their system, and a pass at a fixed cycle count does not provide it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assembled units from PTV-06
- Board-level reliability standards (JEDEC)
- Customer board and mounting requirements
- Coupon board design capability
- Reliability chamber capacity

PRODUCES

- Board-level test plan and coupon boards
- Mounted vehicle assemblies
- Temperature cycling results to failure
- Drop and shock results
- Bend and flex results
- Failure mode analysis and characteristic life

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D7

PRODUCES

Board-level reliability data on vehicles. This activity is the deliverable, and characteristic life is what customers will ask for.

PTV-D9

FEEDS

Package validation complete record. Board-level survival is part of what "validated" has to mean.

RISKS AND WATCH-OUTS

- ◆ **Testing to a pass criterion rather than to failure.** A pass tells the customer nothing about margin; characteristic life does.
- ◆ **Coupon board unrepresentative.** Board thickness and pad design affect solder joint stress, and a test board unlike the customer's predicts poorly.
- ◆ **Started too late to finish.** Temperature cycling to failure takes months, and it is the longest test in the vehicle programme.
- ◆ **Failure modes not analysed.** Knowing units failed at a cycle count without knowing how gives no route to improvement.
- ◆ **Only one mounting condition.** Different customers mount differently, and the package's sensitivity to that is worth knowing.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Failure analysis and characteristic life	2.5	● Assembled units available from PTV-06 ● Board-level standards and customer requirements known ● Reliability chamber capacity booked	PTV-12 Feedback into package design and assembly process window freeze
Temperature cycling	2.0		MP-04 Package qualification — MSL, drop, bend, board-level reliability
Coupon board and mounting	1.5		MP-11 Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety
Drop and shock	1.0		
Bend and flex	1.0		
Total	8.0	EXIT CRITERIA ● Testing run to failure, with characteristic life established ● Coupon boards representative of customer mounting ● Failure modes identified, not only counted	MEASURED BY ● Characteristic life against the requirement ● Failure modes identified against failures observed ● Test completion against the wafer-out gate
WHO IT TAKES	~2.5 FTE	DEPENDS ON	
Reliability engineer Owns board-level testing and analysis		PTV-06 OSAT assembly of vehicles with process window DOE	
Board engineer Coupon board design and mounting			
Failure analysis engineer Failure mode identification			
Quality liaison Standards compliance and customer requirements			
Package architect Design implications from failures			

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-11

Daisy-chain continuity test and assembly yield learning

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	6.0 M/M	~2.9 FTE	6	no	Package test engineer
w39–w48 in stage	planned	while it runs	2 in parallel	has float	Quality engineer approves

PURPOSE

Measure **assembly yield electrically** through the daisy chains, localise what failed, and turn that into process learning before the product is assembled.

Assembly yield on a 2.5D package with hundreds of thousands of interconnects is not guaranteed. The daisy chains make it measurable, and the segmentation makes it attributable — which is what lets the OSAT fix a process problem rather than accept a yield number.

WORK FLOW

6 STEPS · 2 LANES

w39	w41	w43	w45	w47	w48
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Continuity test setup and measurement plan	1w	main
2	Chain continuity measurement across units	2w	main
3	Kelvin resistance measurement on interconnects	1.5w	parallel
4	Failure localisation from chain segmentation	2w	main
5	Condition-to-yield correlation from the DOE	1.5w	parallel
6	Assembly yield learning and process feedback	4w	main

Step 5 is why the DOE was run. Yield varying with bond force or reflow profile identifies the parameter that matters, and that is the parameter the process window in <code>PTV-12</code> will be built around.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assembled daisy-chain units from PTV-06
- Chain-to-location mapping from PTV-04
- Assembly conditions from the travelers
- Measurement capability at the OSAT
- Yield expectations from the cost model

PRODUCES

- Continuity test setup and plan
- Chain continuity results across units
- Kelvin resistance measurements
- Failure localisation findings
- Condition-to-yield correlation
- Assembly yield learning and process feedback

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D3	FEEDS	Built vehicle lots with assembly travelers. The travelers' condition data is what makes this analysis possible.
PTV-D8	FEEDS	Frozen assembly process window definition. Yield against condition is the window's most direct evidence.

RISKS AND WATCH-OUTS

- ◆ **Yield measured without localisation.** A number with no location gives the OSAT nothing to act on.
- ◆ **Marginal interconnects passing continuity.** A high-resistance joint conducts and will fail in the field; Kelvin measurement is what distinguishes it.
- ◆ **Condition correlation not performed.** The DOE's whole purpose is to link yield to parameters, and skipping the analysis wastes the build.
- ◆ **Feedback arriving after the process freezes.** Learning that cannot change the process window is learning with no consequence.
- ◆ **Sample size too small for yield estimation.** Assembly yield is a rate, and estimating it needs enough units for the rate to be meaningful.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Yield learning and feedback	1.5	● Assembled daisy-chain units available from PTV-06 ● Chain-to-location mapping available from PTV-04 ● Assembly condition travelers complete	PTV-12 Feedback into package design and assembly process window freeze
Continuity measurement	1.3		ASSY-07 Assembly yield analysis and process tuning
Failure localisation	1.3		MP-05 Yield learning, failure pareto and defect analysis
Kelvin measurement	1.0		
Condition correlation	1.0		
Total	6.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~2.9 FTE	● Failures localised to interconnect level and position ● Marginal joints identified by resistance, not only by continuity ● Feedback delivered before the process window freezes	● Assembly yield against the cost model assumption ● Failures localised against detected ● Yield variation explained by DOE conditions
Package test engineer Owns continuity measurement and localisation			
Yield engineer Yield estimation and condition correlation			
OSAT process engineer Process feedback and corrective action			
Failure analysis engineer Physical confirmation of localised failures			
Quality engineer Marginal interconnect criteria		DEPENDS ON	
		PTV-04 Daisy-chain electrical vehicle design — interconnect continuity and bump yield structures	
		PTV-06 OSAT assembly of vehicles with process window DOE	

17 PTV · PACKAGE TEST VEHICLE (CPI / TTV)

PTV-12

Feedback into package design and assembly process window freeze

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	5.0 M/M	~3.1 FTE	6	yes	PTV lead
w45–w52 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Freeze the **assembly process window**, feed the findings back into package design, and declare package validation complete — the gate product wafer-out has to clear.

This is the activity the whole stage exists for. It converts a large body of measurement into two things: a process window ASSY will run to, and a statement that the package has been validated. Product wafers must not come out before it is issued, because the alternative is learning about the package on real dies.

WORK FLOW

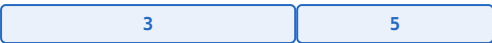
6 STEPS · 2 LANES

w45w47w49w51w52

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Result consolidation across all vehicle characterisation	1w	main
2	Risk retirement assessment against the coverage matrix	1.5w	main
3	Package design feedback and change requests	1.5w	parallel
4	Assembly process window definition and freeze	2w	main
5	Residual risk statement for unretired items	1w	parallel
6	Package validation complete declaration	2.5w	main

Step 2 is the honest accounting. Some risks in the matrix will not have been retired — a vehicle failed, a measurement was inconclusive — and the declaration has to say which, rather than declaring completion because the date arrived.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Warpage results from PTV-07
- CPI assessment from PTV-08
- Thermal characterisation from PTV-09
- Board-level reliability from PTV-10
- Assembly yield learning from PTV-11

PRODUCES

- Consolidated vehicle results
- Risk retirement assessment against the matrix
- Package design feedback and change requests
- Frozen assembly process window
- Residual risk statement
- Package validation complete declaration

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

PTV-D8	PRODUCES	Frozen assembly process window definition. The window ASSY will build the product to, derived from evidence rather than from the OSAT's standard recipe.
PTV-D9	PRODUCES	Package validation complete record — gate for product wafer-out. This activity carries the stage's milestone and the gate.

RISKS AND WATCH-OUTS

- ◆ **Completion declared because the date arrived.** The gate exists to be capable of holding wafer-out, and a declaration that cannot say no is not a gate.
- ◆ **Unretired risks not stated.** A validation record that implies full coverage when some risks remain open misleads exactly the decision it informs.
- ◆ **Feedback arriving after PKGD-11.** Package design changes are not available after the freeze, so vehicle findings have to land before it.
- ◆ **Process window frozen too wide.** A window that accommodates every observed condition is not a window; the boundaries have to exclude what produced failures.
- ◆ **Window not transferred to the OSAT as a specification.** A window that exists in a report and not in the assembly instructions will not be followed.

EFFORT SPLIT	5 M/M	EXIT CRITERIA	FEEDS INTO
Validation declaration	1.3	● Every matrix risk marked retired or explicitly not retired	ASSY-01 Known-good-die sort and selection
Process window definition	1.0	● Process window frozen with boundaries that exclude observed failures	ASSY-07 Assembly yield analysis and process tuning
Risk retirement assessment	1.0	● Window transferred to the OSAT as a specification, not a report	FAB-10 Wafer shipment and logistics to sort and assembly
Design feedback	1.0		
Residual risk statement	0.8		
Total	5.0		MP-04 Package qualification — MSL, drop, bend, board-level reliability
WHO IT TAKES	~3.1 FTE	DEPENDS ON	MEASURED BY
PTV lead Owns consolidation, the window and the declaration		PTV-01 Test vehicle strategy and risk item definition — CPI, warpage, thermal, ULK/BEOL stress	● Risks retired against the coverage matrix
Package architect Design feedback and change requests		PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)	● Weeks of margin against the wafer-out gate
Assembly process engineer Process window definition		PTV-08 Chip-package interaction stress evaluation — low-k / ULK crack, bump and BEOL integrity	● Process window adopted by the OSAT
Reliability engineer Risk retirement assessment		PTV-09 Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model	
Program manager Gate decision against wafer-out		PTV-10 Board-level reliability on vehicles — temperature cycle, drop, bend	
ENTRY CRITERIA		PTV-11 Daisy-chain continuity test and assembly yield learning	
● Characterisation complete from PTV-07 through PTV-11 ● Coverage matrix available from PTV-01 ● Wafer-out date known from FAB-09			

INTEGRATE

18

Chip-Package-System Co-Verification

SIPI

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w50 – w76	26 _w	11	85 _{M/M}	7 of 11

ENGINEERING ACTIVITIES					KEY DELIVERABLES	
SIPI-01	Chip power model (CPM/CPS) extraction per power domain and operating mode	7w	8	M/M	SIPI-D1	Chip power model (CPM/CPS) release per domain and mode
SIPI-02	Package and board model extraction — 3D EM, S-parameter and RLC network	8w	10	M/M	SIPI-D2	Extracted package and board electrical models
SIPI-03	Die-package-board PDN co-simulation — DC and AC impedance versus target	9w	12	M/M	SIPI-D3	PDN impedance and dynamic IR co-simulation report
SIPI-04	Dynamic voltage-drop analysis with package inductance and real switching profiles	8w	10	M/M	SIPI-D4	Decap budget and placement specification across die, package and board
SIPI-05	Decoupling capacitor budget and placement optimisation across die, package and board	7w	7	M/M	SIPI-D5	Channel compliance report per interface, with margins
SIPI-06	SSN / SSO and simultaneous-switching noise analysis at the IO ring	5w	5	M/M	SIPI-D6	Eye, jitter and BER budget closure record
SIPI-07	High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path	9w	11	M/M	SIPI-D7	Power-aware STA correlation report
SIPI-08	Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)	8w	9	M/M	SIPI-D8	Chip-package-system co-verification signoff — tapeout gate
SIPI-09	Power-aware timing correlation — voltage drop back-annotated into STA	5w	5	M/M		
SIPI-10	Electro-thermal co-analysis coupled with the package thermal model	5w	5	M/M		
SIPI-11	Co-verification signoff review and criteria disposition with design and package teams	4w	3	M/M		

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SIPI-01

Chip power model (CPM/CPS) extraction per power domain and operating mode

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	8.0 M/M	~5.0 FTE	6	yes	Power modelling engineer
w0–w7 in stage	planned	while it runs	2 in parallel	no float	Configuration manager approves

PURPOSE

Get a **chip power model** that represents the die's current demand honestly — per domain, per mode, under the switching the product actually produces.

Everything in this stage depends on this model. A CPM extracted under averaged activity understates the transient the package must supply, and every downstream conclusion inherits that error without any indication that it did.

WORK FLOW

6 STEPS · 2 LANES

w0	w2	w4	w6	w7
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Model requirement definition with physical design	1w	main
2	Switching scenario selection from real workloads	1.5w	main
3	Domain and mode decomposition	1.5w	parallel
4	CPM extraction and assembly	2w	main
5	Model validation against die-level IR analysis	1.5w	parallel
6	Model release and version control	2.5w	main

Step 2 decides everything the stage can conclude. The worst realistic transient — a workload burst arriving after an idle period — is what the package has to absorb, and a model built on steady-state activity cannot represent it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Placed and routed database from PD-05 and PD-06
- Chip power model extraction from PD-11
- Switching activity from DV-04 and SYN-06
- Power domain definition from ARCH-05
- Model format requirements from the analysis tools

PRODUCES

- Model requirements and format definition
- Switching scenario selection
- Domain and mode decomposition
- Extracted chip power model
- Validation against die-level IR
- Versioned model release

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D1

PRODUCES

Chip power model (CPM/CPS) release per domain and mode. This activity is the deliverable, and everything else in the stage consumes it.

SIPI-D3

FEEDS

PDN impedance and dynamic IR co-simulation report. The model is the die-side input to every co-simulation.

RISKS AND WATCH-OUTS

- ◆ **Model built on averaged activity.** The transient is what the package must supply, and averaging removes exactly the event of interest.
- ◆ **Single model for all modes.** DVFS points and workloads draw very differently, and one model represents none of them well.
- ◆ **Not validated against die-level IR.** A CPM that disagrees with the die's own analysis is wrong somewhere, and nobody knows where.
- ◆ **Delivered late.** The stage cannot start without it, and the stage gates mask release.
- ◆ **Version drift.** Analysis run against a stale model produces conclusions about a die that has since changed.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
CPM extraction	2.5	Placed database available from PD-05	SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
Model validation	1.5	Switching activity available from DV-04	
Scenario selection	1.5	Model format requirements agreed	SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
Requirement definition	1.5		
Release and version control	1.0		SIPI-09 Power-aware timing correlation — voltage drop back-annotated into STA
Total	8.0	EXIT CRITERIA - Model built on the worst realistic transient, not an average - Separate models per domain and mode - Validated against die-level IR before release	PD-11 Chip power model (CPM/CPS) extraction and handoff to package co-verification
WHO IT TAKES	~5.0 FTE	DEPENDS ON PD-05 Turn 1 on the N1 drop — placement, CTS and first full route; QoR baseline established PD-11 Chip power model (CPM/CPS) extraction and handoff to package co-verification DV-04 Chip-level and system-level scenario testing ARCH-05 Power domain, clock domain and DVFS architecture; UPF intent draft	MEASURED BY - Modes modelled against modes the product uses - Validation agreement against die-level IR - Model delivery against the stage start
Power modelling engineer Owns extraction and validation			
Physical design liaison Database and die-level IR correlation			
Verification liaison Realistic switching scenarios			
SI/PI lead Model requirements for co-analysis			
Configuration manager Model versioning			

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SIPI-02

Package and board model extraction —
3D EM, S-parameter and RLC network

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	10.0 M/M	~5.4 FTE	6	yes	SI/PI extraction engineer
w2-w10 in stage	planned	while it runs	2 in parallel	no float	CAD infrastructure approves

PURPOSE

Extract **electrical models of the package and the board** — 3D field solution, S-parameters, RLC networks — so the system can be analysed as one circuit rather than three assumptions.

Each of the three parties has been designing against a model of the other two. Extraction replaces those assumptions with a solved answer: what the interposer, substrate and board actually do electrically, at the frequencies the design cares about.

WORK FLOW

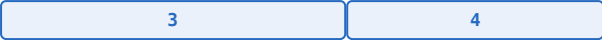
6 STEPS · 2 LANES

w2w4w6w8w10

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Extraction scope and frequency range definition	1.5w	main
2	Interposer model extraction	2w	main
3	Substrate model extraction	2w	parallel
4	Board and socket model extraction	1.5w	parallel
5	PDN network model assembly	2w	main
6	Model validation and correlation	2.5w	main

Step 1's frequency range decides the extraction cost and its usefulness. PDN analysis needs DC to the switching harmonics; channel analysis needs far higher. Extracting once for both is efficient and extracting for the wrong range wastes the effort entirely.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Interposer database from PKGD-03
- Substrate database from PKGD-04
- Board design from EVB-03
- Bump and ball assignment from PKGD-02 and PD-04
- 3D field solver capability

PRODUCES

- Extraction scope and frequency definition
- Interposer electrical model
- Substrate electrical model
- Board and socket models
- Assembled PDN network model
- Model validation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D2

PRODUCES

Extracted package and board electrical models. This activity is the deliverable, and it is what turns three design intents into one analysable circuit.

SIPI-D5

FEEDS

Channel compliance report per interface. Channel simulation runs on these models rather than on idealised transmission lines.

RISKS AND WATCH-OUTS

- ◆ **Frequency range too narrow.** A model extracted for PDN analysis cannot support channel simulation, and re-extraction is expensive.
- ◆ **Extraction on an unfinished design.** Package routing changes invalidate the model, and re-extraction is measured in weeks.
- ◆ **Board model omitted.** The board is part of the PDN and part of the channel, and excluding it makes the analysis die-and-package only.
- ◆ **Models not validated.** A 3D extraction has convergence and meshing choices, and an unvalidated result can be confidently wrong.
- ◆ **Socket and connector effects ignored.** The bring-up platform includes them, and channel margin measured at bring-up will reflect them.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Interposer extraction	2.5	● Interposer and substrate designs stable from PKGD-03 and PKGD-04 ● Board design available from EVB-03 ● Field solver capacity available	SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target
Substrate extraction	2.0		SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
PDN network assembly	2.0		SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring
Model validation	1.5	EXIT CRITERIA ● Frequency range covers both PDN and channel needs ● Board and socket included, not only die and package ● Models validated rather than assumed correct	SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path
Board and socket models	1.0		MEASURED BY ● Frequency coverage against analysis needs ● Model validation error ● Re-extractions required after design change
Scope definition	1.0		
Total	10.0		
WHO IT TAKES	~5.4 FTE	DEPENDS ON	
SI/PI extraction engineer Owns extraction and model validation		PKGD-02 Bump map, pitch and power-ground planning with physical design	
Package designer Interposer and substrate geometry		PKGD-03 Interposer / RDL routing design	
Board designer Board, socket and connector models		PKGD-04 Substrate stack-up, escape routing and package DRC	
SI lead Frequency range and scope definition		PD-04 Bump and RDL planning with chip-package co-design feedback	
CAD infrastructure Field solver capacity		EVB-03 PCB layout with high-speed channel SI/PI simulation	

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SIPI-03

Die-package-board PDN co-simulation
— DC and AC impedance versus target

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	12.0 M/M	~5.8 FTE	6	yes	Power integrity engineer
w8–w17 in stage	planned	while it runs	2 in parallel	no float	SI/PI lead approves

PURPOSE

Simulate the **die, package and board as one power delivery network** and check its impedance against target across the frequency range that matters.

The three levels each cover a different band: on-die decap handles the highest frequencies, package decoupling the middle, board bulk capacitance the lowest. Analysed separately each looks adequate; analysed together the gaps between them appear, and those gaps are where the supply actually sags.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Impedance target derivation per domain	1.5w	main
2	Combined die-package-board PDN assembly	2w	main
3	DC resistance and static drop analysis	1.5w	parallel
4	AC impedance analysis across frequency	2.5w	main
5	Anti-resonance identification and analysis	1.5w	parallel
6	PDN findings and improvement recommendations	3w	main

Step 5 finds what separate analyses cannot. Anti-resonances form between the package's inductance and the die's capacitance, at frequencies neither level would identify alone, and they are where impedance peaks far above target.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Chip power model from SIPI-01
- Package and board models from SIPI-02
- Die PDN from PD-03
- Package PDN design from PKGD-06
- Board power design from EVB-05

PRODUCES

- Impedance targets per domain
- Combined PDN model
- DC resistance and static drop results
- AC impedance across frequency
- Anti-resonance analysis
- PDN findings and recommendations

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D3	PRODUCES	PDN impedance and dynamic IR co-simulation report. The impedance half of the deliverable; SIPI-04 supplies the dynamic IR half.
SIPI-D4	FEEDS	Decap budget and placement specification. The impedance profile is what the decap budget is designed to flatten.

RISKS AND WATCH-OUTS

- ◆ **Levels analysed separately.** Each looks adequate in isolation and the transitions between them are where the impedance peaks.
- ◆ **Anti-resonances missed.** They occur at frequencies no single level would examine and can exceed target by a large factor.
- ◆ **Impedance target derived from a rule of thumb.** The target follows from the current step and the allowable ripple, and a generic value is either wasteful or inadequate.
- ◆ **DC analysis only.** Static drop is the easy half; the frequency-dependent behaviour is where the problems are.
- ◆ **Findings without recommendations.** An impedance plot that exceeds target tells the team there is a problem, not which level to fix.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
AC impedance analysis	3.0	● Chip power model released by SIPI-01	SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
Findings and recommendations	2.5	● Package and board models extracted in SIPI-02	SIPI-05 Decoupling capacitor budget and placement optimisation across die, package and board
PDN assembly	2.0	● Die PDN available from PD-03	PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)
Anti-resonance analysis	2.0		S0-06 Chip-package-system co-analysis signoff review against SIPI results
DC analysis	1.5	EXIT CRITERIA	
Target derivation	1.0	● Impedance analysed across the full frequency range as one network	
Total	12.0	● Anti-resonances identified and addressed	
WHO IT TAKES	~5.8 FTE	● Findings carry recommendations naming which level to change	MEASURED BY
Power integrity engineer		DEPENDS ON	● Impedance against target across frequency
Owns PDN co-simulation and findings		SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode	● Anti-resonance peaks identified
Die power delivery engineer		SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network	● Recommendations adopted by the responsible level
Die-side PDN and on-chip decap		PD-03 Power delivery network design and early IR analysis	
Package power engineer		PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)	
Package PDN and decoupling			
Board designer			
Board PDN and bulk capacitance			
SI/PI lead			
Target derivation and recommendation priority			

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SIPI-04

Dynamic voltage-drop analysis with package inductance and real switching profiles

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	10.0 M/M	~5.4 FTE	6	yes	Power integrity engineer
w12–w20 in stage	planned	while it runs	2 in parallel	no float	Package power engineer approves

PURPOSE

Analyse **dynamic voltage droop with the package's inductance included**, under the switching profiles the product actually produces — the number the die-only analysis is optimistic about.

Die-only dynamic IR excludes the package's series inductance, which is the dominant impedance at the frequencies a workload burst excites. The difference between the two numbers is real margin, and it is the reason this stage exists rather than trusting S0-04.

WORK FLOW

6 STEPS · 2 LANES

w12w14w16w18w20

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Transient scenario selection from the workload profiles	1.5w	main
2	Combined transient simulation with package inductance	2.5w	main
3	Droop magnitude and duration analysis per domain	2w	main
4	Comparison against die-only IR results	1.5w	parallel
5	Worst-case scenario exploration	1.5w	parallel
6	Dynamic IR findings and margin assessment	2w	main

Step 4 is the comparison the program most needs. Where die-only analysis says 4.1% and the co-analysis says 5.0% against a 5% budget, the design has no margin — and the two numbers have to be reconciled before `SO-04` signs off on the optimistic one.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Chip power model from SIPI-01
- Combined PDN model from SIPI-03
- Die-only IR results from PD-03 and SO-04
- Workload transient profiles from DV-04
- Decap configuration from SIPI-05

PRODUCES

- Transient scenario selection
- Combined transient simulation results
- Droop magnitude and duration per domain
- Die-only comparison
- Worst-case scenario findings
- Dynamic IR margin assessment

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D3	PRODUCES	PDN impedance and dynamic IR co-simulation report. The dynamic IR half of the deliverable, and the number signoff should be using.
SIPI-D7	FEEDS	Power-aware STA correlation report. Droop is what gets back-annotated into timing in SIPI-09.

RISKS AND WATCH-OUTS

- ◆ **Package inductance excluded.** It dominates at the frequencies a burst excites, and excluding it produces a comfortable and wrong answer.
- ◆ **Transient scenario too gentle.** The worst case is a burst after idle, and a scenario built from steady-state activity never reaches it.
- ◆ **Droop duration ignored.** A deep short droop and a shallow long one have different timing consequences, and only the magnitude is usually reported.
- ◆ **Die-only comparison not made.** Without it nobody knows whether S0-04's number is optimistic, and the two signoffs disagree silently.
- ◆ **Findings arriving after timing signoff.** Droop consumes timing margin, and a late finding invalidates closure.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Transient simulation	3.0	● Chip power model and PDN model available	SIPI-05 Decoupling capacitor budget and placement optimisation across die, package and board
Droop analysis	2.0	● Die-only IR results available from PD-03	SIPI-09 Power-aware timing correlation — voltage drop back-annotated into STA
Findings and margin assessment	2.0	● Workload transients available from DV-04	S0-04 EM and IR-drop signoff, static and dynamic
Die-only comparison	1.5		S0-06 Chip-package-system co-analysis signoff review against SIPI results
Scenario selection and exploration	1.5		
Total	10.0	EXIT CRITERIA	
WHO IT TAKES	~5.4 FTE	● Package inductance included in every result	MEASURED BY
Power integrity engineer Owns dynamic analysis and margin assessment		● Die-only and co-analysis numbers reconciled	● Droop against budget with package inductance
Die power delivery engineer Die-only comparison and reconciliation		● Findings delivered before timing signoff closes	● Gap between die-only and co-analysis results
Verification liaison Transient scenarios from real workloads		DEPENDS ON	● Timing margin consumed by droop
Timing engineer Timing consequence of the droop		SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode	
Package power engineer Package-side mitigation options		SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network	
		SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target	
		PD-10 Signal and power integrity iteration — crosstalk, EM	

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SIPI-05

Decoupling capacitor budget and placement optimisation across die, package and board

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w14–w21 in stage	7.0 M/M planned	~4.3 FTE while it runs	6 2 in parallel	no has float	Power integrity engineer Program manager approves

PURPOSE

Allocate the **decoupling budget across die, package and board** — how much capacitance, of what type, mounted where — so the three levels cover the frequency range between them rather than overlapping.

Decoupling is a system-level allocation that is usually made three times independently. On-die capacitance is expensive in area, package capacitors are limited by mounting positions, and board bulk is cheap and slow. Optimising them together is what covers the band; optimising them separately leaves the middle uncovered.

WORK FLOW

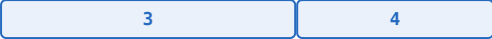
6 STEPS · 2 LANES

w14	w16	w18	w20	w21
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Frequency band allocation across the three levels	1.5w	main
2	On-die decap requirement and area cost	1.5w	main
3	Package capacitor selection and mounting position	1.5w	parallel
4	Board bulk capacitance requirement	1w	parallel
5	Combined impedance verification with the allocation	1.5w	main
6	Decap specification and handoff to each level	2.5w	main

Step 2 carries a real cost. On-die decoupling consumes die area that could have been logic, and the trade between area and package capacitors is a system decision rather than a physical design one.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- PDN impedance results from SIPI-03
- Dynamic droop results from SIPI-04
- On-die decap capability and area cost from PD-03
- Package mounting positions from PKGD-06
- Board capacitance options from EVB-05

PRODUCES

- Frequency band allocation
- On-die decap requirement with area cost
- Package capacitor selection and placement
- Board bulk capacitance requirement
- Combined impedance verification
- Decap specification per level

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D4	PRODUCES	Decap budget and placement specification across die, package and board. This activity is the deliverable, and it is the only place the three levels are allocated together.
SIPI-D3	FEEDS	PDN impedance and dynamic IR co-simulation report. The allocation is verified against the impedance target it was designed to meet.

RISKS AND WATCH-OUTS

- ◆ **Each level decoupled independently.** The bands overlap in places and leave gaps in others, and the gaps are where the impedance peaks.
- ◆ **Package capacitors placed where there is room.** Mounting position determines loop inductance, and a capacitor far from the current draw contributes little.
- ◆ **On-die decap area not costed.** It is die area, and treating it as free produces a design that traded logic for decoupling without anyone deciding to.
- ◆ **Allocation not verified.** A budget that has not been re-simulated against the impedance target is a proposal rather than a solution.
- ◆ **Specification not handed to each level.** A system allocation that stays in a report is not implemented by anyone.

EFFORT SPLIT	7 M/M	ENTRY CRITERIA	FEEDS INTO
Decap specification and handoff	1.5	● Impedance results available from SIPI-03 ● Droop results available from SIPI-04 ● Mounting and area constraints known from each level	PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI)
Band allocation	1.3		PD-03 Power delivery network design and early IR analysis
On-die requirement	1.3		EVB-05 Power delivery, VRM and telemetry design and bring-up
Package capacitor selection	1.3		SIPI-11 Co-verification signoff review and criteria disposition with design and package teams
Combined verification	1.0		
Board requirement	0.8		
Total	7.0	EXIT CRITERIA	
WHO IT TAKES	~4.3 FTE	● Bands allocated so the three levels cover the range without gaps ● Package capacitors placed for loop inductance ● Allocation verified against the impedance target before handoff	MEASURED BY
Power integrity engineer Owns the allocation and verification		DEPENDS ON SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles PKGD-06 Package PDN and decap footprint design (electrical closure owned by SIPI) PD-03 Power delivery network design and early IR analysis	● Impedance gaps between levels ● On-die decap area against the budget ● Specification adopted by each level
Die power delivery engineer On-die decap and area cost			
Package power engineer Capacitor selection and mounting			
Board designer Bulk capacitance and board placement			
Program manager Arbitrates the area-versus-cost trade			

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SIPI-06

SSN / SSO and simultaneous-switching noise analysis at the IO ring

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	5.0 M/M	~4.3 FTE	6	no	SI/PI engineer
w15–w20 in stage	planned	while it runs	2 in parallel	has float	Interface architect approves

PURPOSE

Analyse **simultaneous switching noise at the IO ring** — the supply disturbance many outputs switching together inject into their own reference.

SSN is a local problem with system consequences. When a wide interface switches together, the current through the shared return path moves the reference the receivers are measuring against, and the resulting noise appears as jitter and reduced eye margin on the very interface that caused it.

WORK FLOW

6 STEPS · 2 LANES

w15	w16	w17	w18	w19	w20
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	SSN analysis scope — which interfaces switch together	1w	main
2	IO ring power and ground network modelling	1.5w	main
3	Simultaneous switching simulation per interface	1.5w	parallel
4	Ground bounce and reference shift analysis	1.5w	main
5	Mitigation assessment — bump assignment, decoupling	1w	parallel
6	SSN findings and interface impact	1w	main

Step 5 usually points back at the bump map. The most effective SSN mitigation is more or better-placed power and ground bumps for that interface, which is a `PKGD-02` decision — and one that has to be made before the map freezes.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package and board models from SIPI-02
- IO library characteristics from PDK-07
- Bump and ball assignment from PKGD-02
- Interface switching patterns from ARCH-04
- On-die IO power distribution from PD-03

PRODUCES

- SSN analysis scope
- IO ring power and ground models
- Simultaneous switching simulation results
- Ground bounce and reference shift analysis
- Mitigation assessment
- SSN findings per interface

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D5	FEEDS	Channel compliance report per interface. SSN degrades the eye of the interface that causes it, so it belongs in the same compliance assessment.
SIPI-D3	FEEDS	PDN impedance and dynamic IR co-simulation report. SSN is a local PDN problem and is reported alongside the global one.

RISKS AND WATCH-OUTS

- ◆ **SSN analysed per pin rather than per interface.** The problem is simultaneity, and a single-pin analysis cannot see it.
- ◆ **Reference shift not included in the eye budget.** Ground bounce moves the receiver's threshold, and excluding it overstates the margin.
- ◆ **Mitigation identified after bump map freeze.** The most effective remedy is bump assignment, and it is unavailable after PKGD-02 closes.
- ◆ **Worst-case switching pattern not simulated.** All outputs switching the same direction simultaneously is the worst case and is rarely the default pattern.
- ◆ **IO ring modelled as ideal.** The ring's own power distribution is what carries the return current, and idealising it removes the effect being analysed.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
IO ring modelling	1.3	● Package models available from SIPI-02	SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)
Switching simulation	1.0	● IO characteristics available from PDK-07	SIPI-11 Co-verification signoff review and criteria disposition with design and package teams
Ground bounce analysis	1.0	● Bump assignment available from PKGD-02	S0-05 Power and signal integrity signoff — crosstalk, noise
Mitigation assessment	1.0		
Findings	0.8		
Total	5.0	EXIT CRITERIA	
WHO IT TAKES	~4.3 FTE	● Worst-case simultaneous switching simulated per interface	MEASURED BY
SI/PI engineer		● Reference shift included in the interface eye budget	● Ground bounce against the reference budget
IO library engineer		● Mitigations identified before the bump map freezes	● Eye margin consumed by SSN
Package designer			● Mitigations available before bump map freeze
Die power engineer		DEPENDS ON	
Interface architect		SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network	
		PKGD-02 Bump map, pitch and power-ground planning with physical design	
		PDK-07 IO, ESD and latch-up library qualification	
		PD-03 Power delivery network design and early IR analysis	

SIPI-07

High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	11.0 M/M	~5.3 FTE	6	yes	SI engineer
w8–w17 in stage	planned	while it runs	2 in parallel	no float	Interface architect approves

PURPOSE

Simulate the **high-speed channels end to end with extracted package models** — insertion loss, crosstalk, return path, reflections — rather than against the idealised channel the PHY was designed to.

Every PHY closes its margin against an assumed channel. This activity produces the real one: interposer, substrate, board, socket, connectors and all their discontinuities. Where the real channel is worse than assumed, the margin the PHY reported never existed.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE

1	2	3	6
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IN PARALLEL

4	5
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#	STEP	TAT	LANE
1	Channel topology definition per interface	1.5w	main
2	End-to-end channel assembly from extracted models	2w	main
3	Insertion loss and return loss analysis	2w	main
4	Crosstalk analysis — near and far end	2w	parallel
5	Return path and discontinuity analysis	1.5w	parallel
6	Channel characterisation against the PHY assumption	3.5w	main

Step 6 is the reconciliation that matters. The PHY in <code>AMS-03</code> closed against a channel budget; this activity says what the channel actually is, and any gap between them is margin the interface does not have.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package and board models from SIPI-02
- Channel budgets from AMS-03 and ARCH-04
- Package SI routing from PKGD-05
- Interface specifications and compliance masks
- PHY equalisation capability from AMS-03

PRODUCES

- Channel topology per interface
- End-to-end channel models
- Insertion and return loss results
- Crosstalk analysis results
- Return path and discontinuity findings
- Channel characterisation against PHY assumptions

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D5	PRODUCES	Channel compliance report per interface, with margins. This activity is the deliverable, and the margin is stated against the real channel.
SIPI-D2	FEEDS	Extracted package and board electrical models. Channel simulation is the primary consumer of the extraction.

RISKS AND WATCH-OUTS

- ◆ **Channel simulated on nominal models only.** Manufacturing variation moves impedance and loss, and the compliance margin has to survive it.
- ◆ **PHY assumption not reconciled.** The PHY closed against a budget; if the real channel is worse, the interface has no margin and nobody has noticed.
- ◆ **Crosstalk analysed only at the near end.** Far-end crosstalk on a long dense channel can dominate, and it behaves differently.
- ◆ **Socket and connector excluded.** The bring-up platform includes them, and margin measured at bring-up will reflect them.
- ◆ **Findings after the package design freeze.** Channel improvements are routing changes, and they are unavailable after PKGD-11.

EFFORT SPLIT	11 M/M	ENTRY CRITERIA	FEEDS INTO
Channel characterisation	3.0	Package and board models extracted in SIPI-02	SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)
Insertion and return loss	2.0	Channel budgets available from AMS-03	PKGD-05 Package routing for signal integrity — length matching, reference plane and via design
Channel assembly	2.0	Package SI routing complete in PKGD-05	AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY)
Crosstalk analysis	2.0		
Return path analysis	1.0		
Topology definition	1.0		
Total	11.0	EXIT CRITERIA - Channels simulated end to end including socket and connectors - Real channel reconciled against the PHY's assumed budget - Findings delivered before the package design freeze	S0-06 Chip-package-system co-analysis signoff review against SIPI results
WHO IT TAKES	~5.3 FTE	DEPENDS ON SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network PKGD-05 Package routing for signal integrity — length matching, reference plane and via design ARCH-04 Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E) AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCle, HBM PHY)	MEASURED BY - Channel loss against the PHY's assumed budget - Crosstalk against the interface allowance - Discontinuities identified and addressed
SI engineer Owns channel simulation and characterisation			
PHY designer Equalisation capability and channel assumptions			
Package designer Routing changes from the findings			
Board designer Board-side channel contribution			
Interface architect Compliance requirements per interface			

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SIPI-08

Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	9.0 M/M	~4.9 FTE	6	yes	SI engineer
w15–w23 in stage	planned	while it runs	2 in parallel	no float	SI/PI lead approves

PURPOSE

Close the **eye, jitter and BER budget per interface** — combining channel loss, crosstalk, SSN, PHY equalisation and jitter into one number that either meets the compliance mask or does not.

This is where all the electrical work resolves into a pass or a fail. Every contribution — the channel from SIPI-07, the noise from SIPI-06, the PLL jitter from AMS-02 and the PHY’s own equalisation — lands in a single budget, and the interface either has margin or it is shipping on hope.

WORK FLOW

6 STEPS · 2 LANES

w15	w17	w19	w21	w23
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Budget structure and contribution allocation per interface	1.5w	main
2	Jitter budget assembly — random, deterministic, correlated	2w	main
3	Statistical eye analysis with equalisation	2w	parallel
4	BER estimation and margin against the mask	2w	main
5	Sensitivity to channel and process variation	1.5w	parallel
6	Budget closure and interface sign-off	2.5w	main

Step 2 is where interfaces are usually lost. Jitter contributions from the PLL, the PHY, the channel and the SSN do not add linearly, and a budget that sums them naively either over-designs or, more often, over-promises.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Channel characterisation from SIPI-07
- SSN findings from SIPI-06
- PLL jitter from AMS-02
- PHY equalisation and BER capability from AMS-03
- Interface compliance masks

PRODUCES

- Budget structure and contribution allocation
- Assembled jitter budget per interface
- Statistical eye analysis results
- BER estimation and mask margin
- Variation sensitivity analysis
- Interface budget closure records

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D6	PRODUCES	Eye, jitter and BER budget closure record. This activity is the deliverable — the statement that each interface will work.
SIPI-D5	FEEDS	Channel compliance report per interface. Compliance is a budget conclusion rather than a channel property alone.

RISKS AND WATCH-OUTS

- ◆ **Jitter contributions summed naively.** Random and deterministic components combine differently, and getting it wrong produces a budget that is confidently wrong.
- ◆ **Equalisation assumed at its maximum.** A PHY closing only with full equalisation has no adaptation range left for a worse-than-nominal channel.
- ◆ **Variation sensitivity ignored.** The budget has to hold across manufacturing spread, not only at nominal.
- ◆ **SSN excluded from the budget.** It is a real contribution and it degrades the interface that generates it.
- ◆ **Interface failing the mask with no route to fix.** Late findings leave only channel changes, PHY changes or a rate reduction, and all three are expensive.

EFFORT SPLIT	9 M/M	ENTRY CRITERIA	FEEDS INTO
Budget closure and sign-off	2.0	Channel characterisation available from SIPI-07	SIPI-11 Co-verification signoff review and criteria disposition with design and package teams
Jitter budget assembly	2.0	SSN findings available from SIPI-06	S0-06 Chip-package-system co-analysis signoff review against SIPI results
BER estimation	1.8	PHY and PLL contributions available from AMS-02 and AMS-03	BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Statistical eye analysis	1.8		
Variation sensitivity	1.0		
Budget structure	0.5		
Total	9.0	EXIT CRITERIA - Every interface closed against its compliance mask with stated margin - Jitter combined correctly rather than summed - Budget holds across variation, not only at nominal	MP-11 Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety
WHO IT TAKES	~4.9 FTE		MEASURED BY - Eye margin against the compliance mask - Equalisation range remaining at closure - Budget margin under variation
SI engineer Owns the budget and its closure		DEPENDS ON	
PHY designer Equalisation, jitter and BER capability		SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring	
Interface architect Compliance mask and rate decisions		SIPI-07 High-speed channel simulation with extracted package models — insertion loss, crosstalk, return path	
Clocking architect PLL jitter contribution		AMS-02 PLL / clock generator design and closure	
SI/PI lead Sign-off on interface closure		AMS-03 SerDes and PHY design or vendor hardening (PCIe/CXL, UCIe, HBM PHY)	

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SIPI-09

Power-aware timing correlation — voltage drop back-annotated into STA

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	5.0 M/M	~4.3 FTE	6	no	Power integrity engineer
w18–w23 in stage	planned	while it runs	2 in parallel	has float	SI/PI lead approves

PURPOSE

Back-annotate the **voltage droop into static timing analysis** — because a design signed off at nominal voltage was signed off at a voltage the die does not see.

Timing and power are analysed by different teams with different tools, and the connection between them is usually a margin someone assumed. This activity replaces the assumption: the droop map from SIPI-04 applied to the timing analysis, so that closure reflects the supply the logic actually has.

WORK FLOW

6 STEPS · 2 LANES

w18	w19	w20	w21	w22	w23
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Droop map generation per region and mode	1w	main
2	Voltage-aware timing analysis setup	1.5w	main
3	Timing re-analysis with back-annotated droop	1.5w	parallel
4	Margin impact assessment per timing path group	1.5w	main
5	Reconciliation with the SO-02 signoff margin	1.5w	parallel
6	Power-aware timing findings and recommendations	1w	main

Step 5 is the reconciliation. If timing signoff already carries a flat voltage margin and this analysis shows the droop is worse in some regions and better in others, the flat margin was both too generous and insufficient — and the recommendation is regional rather than global.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Dynamic droop results from SIPI-04
- Timing analysis from PD-09 and SO-02
- Voltage-aware timing tool capability
- Region and instance mapping from PD-02
- Operating point definitions from ARCH-05

PRODUCES

- Droop map per region and mode
- Voltage-aware timing setup
- Re-analysed timing with droop
- Margin impact per path group
- Reconciliation with signoff margin
- Power-aware timing recommendations

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D7	PRODUCES	Power-aware STA correlation report. This activity is the deliverable, and it connects two signoffs that otherwise run independently.
SIPI-D3	FEEDS	PDN impedance and dynamic IR co-simulation report. The timing consequence is what makes the IR number actionable.

RISKS AND WATCH-OUTS

- ◆ **Flat voltage margin used instead.** Droop is regional, and a flat margin is simultaneously wasteful where droop is low and inadequate where it is high.
- ◆ **Analysis run after timing closure.** Findings then require reopening closure, which is expensive at this point in the schedule.
- ◆ **Droop map too coarse.** Region granularity determines whether the analysis can distinguish a hot spot from an average.
- ◆ **Reconciliation skipped.** Two signoffs with different voltage assumptions produce a design that satisfies neither cleanly.
- ◆ **Recommendations global rather than regional.** The remedy is usually local — more decoupling or a PDN change in one region — and a global response over-corrects.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Timing re-analysis	1.3	● Droop results available from SIPI-04	SIPI-11 Co-verification signoff review and criteria disposition with design and package teams
Margin impact assessment	1.0	● Timing analysis available from PD-09	
Droop map generation	1.0	● Voltage-aware timing capability available	SO-02 Multi-corner multi-mode static timing analysis and closure on the final turn
Reconciliation	1.0		
Findings	0.8	EXIT CRITERIA	SO-06 Chip-package-system co-analysis signoff review against SIPI results
Total	5.0	● Droop applied regionally rather than as a flat margin	
WHO IT TAKES	~4.3 FTE	● Reconciled with the SO-02 signoff assumption	MEASURED BY
Power integrity engineer		● Recommendations targeted at the regions that need them	● Timing margin change from droop back-annotation
Droop map generation			● Regional variation in droop impact
STA engineer		DEPENDS ON	● Agreement with the signoff voltage assumption
Voltage-aware timing analysis		SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles	
Signoff methodology engineer		PD-09 Multi-corner multi-mode timing closure	
Reconciliation with signoff margin		SO-02 Multi-corner multi-mode static timing analysis and closure on the final turn	
Physical design liaison			
Regional remedies			
SI/PI lead			
Findings and recommendation priority			

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-10

Electro-thermal co-analysis coupled with the package thermal model

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	5.0 M/M	~4.3 FTE	6	no	Thermal engineer
w18–w23 in stage	planned	while it runs	2 in parallel	has float	SI/PI lead approves

PURPOSE

Couple the **electrical and thermal analyses** — because leakage rises with temperature, temperature rises with power, and the two feed each other until they settle somewhere the isolated analyses did not predict.

Electro-thermal coupling matters at this power density. A hotspot raises local leakage, which raises local power, which raises the hotspot. Analysing power at a fixed temperature and temperature at a fixed power finds an equilibrium that is optimistic in exactly the region that matters.

WORK FLOW6 STEPS · 2 LANES

w18	w19	w20	w21	w22	w23
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Coupling methodology and iteration scheme	1w	main
2	Thermal model integration with the package model	1.5w	main
3	Leakage-temperature feedback modelling	1w	parallel
4	Coupled electro-thermal iteration to convergence	1.5w	main
5	Hotspot and gradient assessment	1w	parallel
6	Findings against the isolated analyses	1w	main

Step 4 iterates to convergence, and whether it converges is itself information. A design where the leakage-temperature loop does not settle is a design with a thermal runaway risk, which is a different and more serious finding than a high temperature.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package thermal model from PKGD-07
- Thermal characterisation from PTV-09
- Chip power model from SIPI-01
- Leakage characterisation from PDK-03
- Power map from PD-03

PRODUCES

- Coupling methodology and iteration scheme
- Integrated electro-thermal model
- Leakage-temperature feedback model
- Coupled analysis results
- Hotspot and gradient assessment
- Comparison against isolated analyses

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

- SIPI-D3

FEEDS

PDN impedance and dynamic IR co-simulation report. Temperature affects both resistance and leakage, and the coupled result is the honest one.
- SIPI-D7

FEEDS

Power-aware STA correlation report. Temperature affects timing as well as voltage, and both belong in the same correlation.

RISKS AND WATCH-OUTS

- ◆

Analyses run at fixed temperature and fixed power. The equilibrium the coupled analysis finds is worse, and it is the one that occurs.
- ◆

Leakage-temperature model too coarse. The feedback is exponential, and a linear approximation understates the hotspot.
- ◆

Non-convergence dismissed as a numerical issue. It can indicate a real thermal runaway risk and deserves investigation rather than a damping factor.
- ◆

Package thermal model unvalidated. If PTV-09 has not correlated it, the coupled analysis inherits an unknown error.
- ◆

Gradient ignored in favour of peak temperature. Gradients drive mechanical stress and timing variation across the die, and the peak alone does not capture them.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Coupled iteration	1.3	● Package thermal model available from PKGD-07	SIPI-11 Co-verification signoff review and criteria disposition with design and package teams
Thermal model integration	1.0	● Thermal characterisation available from PTV-09	S0-06 Chip-package-system co-analysis signoff review against SIPI results
Leakage feedback modelling	1.0	● Chip power model available from SIPI-01	MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Hotspot assessment	1.0		
Findings and comparison	0.8		
Total	5.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~4.3 FTE	● Coupled analysis converged, or non-convergence investigated ● Compared against the isolated analyses to quantify the difference ● Gradients assessed alongside peak temperature	● Temperature difference between coupled and isolated analysis ● Convergence of the coupled iteration ● Gradient magnitude across the die
Thermal engineer Thermal model and coupling		DEPENDS ON	
Power integrity engineer Electrical side of the coupled analysis		SIPI-01 Chip power model (CPM/CPS) extraction per power domain and operating mode	
Device engineer Leakage-temperature characterisation		PKGD-07 Thermal and mechanical simulation — warpage, co-planarity, TIM, lid	
Package thermal liaison Package model and validation status		PTV-09 Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model	
SI/PI lead Findings against the isolated analyses		PD-03 Power delivery network design and early IR analysis	

18 SIPI · CHIP-PACKAGE-SYSTEM CO-VERIFICATION

SIPI-11

Co-verification signoff review and criteria disposition with design and package teams

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	3.0 M/M	~3.3 FTE	6	yes	SI/PI lead
w22–w26 in stage	planned	while it runs	2 in parallel	no float	Signoff lead approves

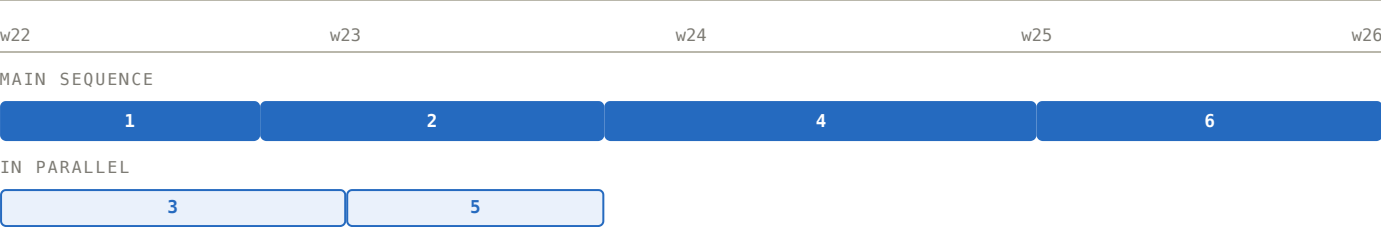
PURPOSE

Hold the **co-verification signoff** — every result reviewed against named exit criteria, dispositioned, and signed — because mask release depends on it.

This is the gate the stage exists to reach. Without a signoff with criteria, co-verification is a collection of analyses that each conclude something and together conclude nothing. The disposition here is what 50–06 reviews and what makes the MTO gate at week 78 meaningful.

WORK FLOW

6 STEPS · 2 LANES



#	STEP	TAT	LANE
1	Exit criteria confirmation across all analyses	0.8w	main
2	Result consolidation and margin summary	1w	main
3	Open item disposition with die, package and board owners	1w	parallel
4	Signoff review with design and package teams	1.3w	main
5	Residual risk statement	0.8w	parallel
6	Co-verification signoff declaration	1w	main

Step 3 is a three-party disposition. Every open item belongs to the die, the package or the board, and closing it requires the owner in the room — which is why co-verification signoff is a joint review rather than a report circulated for comment.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Results from SIPI-03 through SIPI-10
- Exit criteria defined at stage start
- Package design freeze schedule from PKGD-11
- Signoff schedule from SO-06
- Program risk register

PRODUCES

- Confirmed exit criteria
- Consolidated results and margin summary
- Open item dispositions with owners
- Signoff review record
- Residual risk statement
- Co-verification signoff declaration

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

SIPI-D8	PRODUCES	Chip-package-system co-verification signoff — tapeout gate. This activity is the deliverable, and it carries the stage's milestone.
SIPI-D6	GATES	Eye, jitter and BER budget closure record. An interface that has not closed its budget is an open item at this gate.

RISKS AND WATCH-OUTS

- ◆ **Signoff without exit criteria.** The analyses each conclude something and together conclude nothing that can be signed.
- ◆ **Open items dispositioned without their owners.** An item closed by the analysis team rather than by the die, package or board owner has not actually been closed.
- ◆ **Signoff after the package design freeze.** Findings then have nowhere to go, and the freeze was taken without them.
- ◆ **Residual risk not stated.** Some margin will be thin; saying which and how thin is what lets the tapeout decision be informed.
- ◆ **Declared to meet the date.** The gate exists to be able to hold mask release, and one that cannot is not a gate.

EFFORT SPLIT	3 M/M	EXIT CRITERIA	FEEDS INTO
Signoff review	1.0	● Every analysis assessed against a named exit criterion	SO-06 Chip-package-system co-analysis signoff review against SIPI results
Result consolidation	0.8	● Open items dispositioned by their owners, not by the analysis team	PKGD-11 Package design freeze, DRC and tooling release
Open item disposition	0.5	● Signoff complete before the package design freeze and the MTO gate	TO-04 Open-issue risk assessment and waiver acceptance
Declaration	0.5		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Residual risk statement	0.3		
Total	3.0	DEPENDS ON	
WHO IT TAKES	~3.3 FTE	SIPI-03 Die-package-board PDN co-simulation — DC and AC impedance versus target	MEASURED BY
SI/PI lead Chairs the signoff and owns the declaration		SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles	● Analyses assessed against exit criteria
Die power and timing owners Die-side dispositions		SIPI-05 Decoupling capacitor budget and placement optimisation across die, package and board	● Open items closed by their owners
Package architect Package-side dispositions		SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring	● Signoff date against the package freeze and MTO gate
Board designer Board-side dispositions		SIPI-08 Eye diagram, jitter and BER budget closure per interface (PCIe/CXL, HBM, die-to-die)	
Signoff lead Carries the result into SO-06 and the freeze		SIPI-09 Power-aware timing correlation — voltage drop back-annotated into STA	
ENTRY CRITERIA		SIPI-10 Electro-thermal co-analysis coupled with the package thermal model	
● Analyses complete from SIPI-03 through SIPI-10			
● Exit criteria defined and agreed			
● Die, package and board owners available to disposition			

INTEGRATE

19 Assembly & Package Build

ASSY

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH		
w98 – w106	8 _w	9	36 _{M/M}	5 _{of 9}		
ENGINEERING ACTIVITIES				KEY DELIVERABLES		
ASSY-01	Known-good-die sort and selection		3w	5 _{M/M}	ASSY-D1	Assembled units — bring-up, qualification and sample lots
ASSY-02	HBM stack procurement and incoming inspection		3w	2 _{M/M}		
ASSY-03	Die attach, micro-bump and thermo-compression bonding		3w	6 _{M/M}	ASSY-D3	Assembly yield report and failure pareto
ASSY-04	Interposer and substrate assembly, underfill, molding		3w	5 _{M/M}	ASSY-D4	Package-level inspection and test data
ASSY-05	Lid / TIM attach and ball attach		2w	3 _{M/M}	ASSY-D5	Unit allocation record across bring-up, qual and customers
ASSY-06	Assembly inline inspection — X-ray, CSAM, warpage metrology		2w	3 _{M/M}		
ASSY-07	Assembly yield analysis and process tuning		5w	6 _{M/M}		
ASSY-08	Package-level open / short and continuity test		2w	3 _{M/M}		
ASSY-09	Unit build for bring-up, qualification and customer samples		3w	3 _{M/M}		

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-01

Known-good-die sort and selection

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	5.0 M/M	~7.2 FTE	5	yes	Product engineering
w0-w3 in stage	planned	while it runs	1 in parallel	no float	Quality engineer approves

PURPOSE

Sort the wafers and pick the **known-good die** that will go into packages — because on a 2.5D part a bad die does not fail alone, it takes an interposer, two HBM stacks and an assembly with it.

Known-good-die screening is what makes multi-die packaging economically viable. The coverage achieved at sort determines how many assembled units fail for reasons that were detectable on the wafer, and each of those is a package's worth of material lost.

WORK FLOW

5 STEPS · 2 LANES

w0	w1	w2	w3
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Wafer receipt and sort setup on the probe station	0.5w	main
2	Wafer sort execution against the KGD criteria	1w	main
3	Marginal die disposition	0.8w	parallel
4	Known-good-die selection and wafer mapping	1w	main
5	Die release to assembly with traceability	0.5w	main

Step 3 is where the economics live. A marginal die is cheap to scrap and expensive to assemble; the criteria for rejecting one have to be set against the assembled unit's cost rather than against the die's.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Accepted wafers from FAB-08 and FAB-10
- Wafer sort program from TEST-05
- Probe card from TEST-03
- KGD criteria from PKGD-10
- Assembly quantity requirement from ASSY-09

PRODUCES

- Sort execution results
- Marginal die disposition record
- Known-good-die map per wafer
- Die released to assembly with traceability

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D1	FEEDS	Assembled units. Everything assembled starts from a die this activity selected.
ASSY-D3	FEEDS	Assembly yield report and failure pareto. Assembly failures traceable to die quality belong against the sort coverage that let them through.

RISKS AND WATCH-OUTS

- ◆ **KGD coverage below what the package economics need.** Every escaped die costs an assembled unit, and the coverage target follows from that cost rather than from a test convention.
- ◆ **Marginal criteria set against die cost.** On a 2.5D part the assembled unit is worth many times the die, and the reject threshold should reflect that.
- ◆ **Traceability lost between wafer and package.** An assembly failure that cannot be traced back to a die and its wafer position cannot be attributed.
- ◆ **Sort program not ready.** The probe card and program come from TEST, and a delay there stops assembly rather than delaying test.
- ◆ **Sorting only for function.** Parametric marginality — leakage, Vmin — predicts field failure and is worth screening even when the die functions.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Sort execution	2.0	● Wafers delivered from FAB-10	ASSY-03 Die attach, micro-bump and thermo-compression bonding
KGD selection and mapping	1.3	● Sort program and probe card ready from TEST-03 and TEST-05	ASSY-07 Assembly yield analysis and process tuning
Sort setup	0.8	● KGD criteria agreed with the OSAT	ASSY-09 Unit build for bring-up, qualification and customer samples
Marginal disposition	0.5		
Release and traceability	0.5		
Total	5.0		
WHO IT TAKES	~7.2 FTE	EXIT CRITERIA	MEASURED BY
Product engineering		● Coverage sufficient for the assembled unit's economics	● KGD coverage achieved
Owns sort criteria and die disposition		● Marginal criteria set against package cost, not die cost	● Assembly failures attributable to escaped die
Test engineering		● Every released die traceable to wafer and position	● Die traceability to wafer position
Sort program and probe card execution			
Yield engineer		DEPENDS ON	
Wafer map analysis and marginal criteria		FAB-08 Wafer acceptance test and lot disposition	
Assembly liaison		FAB-10 Wafer shipment and logistics to sort and assembly	
Die release, quantity and traceability		TEST-03 Probe card design, fabrication and qualification	
Quality engineer		TEST-05 Wafer sort test program development	
KGD criteria against assembled unit cost		PKGD-10 OSAT selection and assembly process definition	

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-02

HBM stack procurement and incoming inspection

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	2.0 M/M	~2.9 FTE	5	no	Procurement
w0–w3 in stage	planned	while it runs	1 in parallel	has float	Supplier liaison approves

PURPOSE

Procure and inspect the **HBM stacks** — bought components that go into the package alongside the program's own die, and that fail in ways the program does not control.

HBM is a purchased part with its own supply constraints and its own quality distribution. Incoming inspection matters because an HBM stack that fails after assembly costs the whole unit, and the supplier's outgoing quality is not the program's to verify after the fact.

WORK FLOW

5 STEPS · 2 LANES

w0	w1	w2	w3
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	HBM procurement confirmation and delivery	0.5w	main
2	Incoming inspection against specification	1w	main
3	Supplier test data review	0.8w	parallel
4	Stack qualification and release to assembly	1w	main
5	Inventory and lot traceability	0.5w	main

Step 3 costs almost nothing and catches a class of problem inspection cannot. Supplier test data shows the distribution the delivered lot came from, and a lot at the edge of specification is worth knowing about before it is bonded in.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- HBM purchase orders from MP-10 and procurement
- Supplier test data and certificates
- HBM specification and requirements from ARCH-03
- Incoming inspection capability
- Assembly quantity requirement

PRODUCES

- Delivered HBM stacks
- Incoming inspection results
- Supplier test data review
- Qualified stacks released to assembly
- Inventory and lot traceability record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D1

FEEDS

Assembled units. HBM is part of every assembled unit, and its quality is part of the unit's.

ASSY-D4

FEEDS

Package-level inspection and test data. Incoming HBM data belongs with the assembly's own inspection record.

RISKS AND WATCH-OUTS

- ◆ **HBM supply constrained.** It is a scarce component with long lead times, and a shortage stops assembly regardless of how many die are ready.
- ◆ **Incoming inspection skipped.** A marginal stack bonded into a unit costs the whole unit, and the fault is only visible afterwards.
- ◆ **Supplier data not reviewed.** The distribution the lot came from predicts its behaviour, and the data is provided at no cost.
- ◆ **Lot traceability not maintained.** An HBM-related failure that cannot be traced to a supplier lot cannot be escalated usefully.
- ◆ **Specification mismatch discovered at assembly.** Stack height, bump pitch and thermal characteristics all have to match the package design.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	FEEDS INTO
Incoming inspection	0.8	● HBM ordered with committed delivery	ASSY-03 Die attach, micro-bump and thermo-compression bonding
Stack qualification	0.5	● Specification and inspection criteria defined	ASSY-07 Assembly yield analysis and process tuning
Procurement coordination	0.4	● Inspection capability available	MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Traceability	0.3		
Total	2.0	EXIT CRITERIA	
WHO IT TAKES	~2.9 FTE	● Every lot inspected and its supplier data reviewed ● Stacks confirmed against the package specification ● Lot traceability maintained into assembly	MEASURED BY ● Stacks rejected at incoming inspection ● Supply delivered against the assembly schedule ● HBM-attributed failures after assembly
Procurement HBM supply and delivery		DEPENDS ON	
Quality engineer Incoming inspection and supplier data		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	
Package engineer Specification compliance		TECH-09 OSAT and backend supply chain preliminary alignment	
Operations planner Inventory and traceability		PKGD-10 OSAT selection and assembly process definition	
Supplier liaison Escalation on quality or supply			

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-03

Die attach, micro-bump and thermo-compression bonding

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	6.0 M/M	~8.7 FTE	5	yes	OSAT process engineer
w2–w5 in stage	planned	while it runs	1 in parallel	no float	Yield engineer approves

PURPOSE

Bond the die and HBM stacks to the interposer — **micro-bump preparation and thermo-compression bonding** — inside the process window the vehicles established.

This is the step the whole PTV programme existed to de-risk. Bond force, temperature and time were characterised on vehicles; here they are applied to real die, and the process window is either followed or the vehicle work was wasted.

WORK FLOW

5 STEPS · 2 LANES

w2	w3	w4	w5
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Process setup to the frozen window	0.5w	main
2	Micro-bump preparation and flux application	1w	main
3	Bond parameter monitoring during the run	0.8w	parallel
4	Thermo-compression bonding of die and stacks	1w	main
5	Post-bond inspection	0.5w	main

Step 3 records what actually happened. The frozen window says what the parameters should be; monitoring says what they were, and a unit that failed can only be attributed if its actual bond conditions were captured.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Known-good die from ASSY-01
- Qualified HBM stacks from ASSY-02
- Interposers from PKGD-09 supply
- Frozen process window from PTV-12
- Assembly process flow from PKGD-10

PRODUCES

- Process setup to the frozen window
- Prepared micro-bumps
- Bond parameter monitoring data
- Bonded die and stack assemblies
- Post-bond inspection results
- Assembly travelers with actual conditions

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D2

PRODUCES

Assembly travelers and process data. This activity is where the traveler starts and where the actual bond conditions are recorded.

ASSY-D1

FEEDS

Assembled units. Bonding is the step that makes a set of components into a package.

RISKS AND WATCH-OUTS

- ◆ **Running outside the frozen window.** The PTV programme established boundaries; operating outside them discards the evidence it produced.
- ◆ **Bond parameters not recorded per unit.** A failed unit cannot be attributed to its conditions, and the yield analysis loses its most useful variable.
- ◆ **Bump preparation inconsistent.** Flux and surface condition affect joint formation, and variability here shows up as scattered yield.
- ◆ **Post-bond inspection sampled too thinly.** Bonding defects are localised, and a sample that misses them lets them through to underfill where they are unfixable.
- ◆ **Process drift across the run.** Long runs drift, and without monitoring the drift is only visible in the yield trend afterwards.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Thermo-compression bonding	2.5	● Known-good die released from ASSY-01	ASSY-04 Interposer and substrate assembly, underfill, molding
Bump preparation	1.5	● HBM stacks qualified in ASSY-02	ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology
Process setup	1.0	● Process window frozen by PTV-12	ASSY-07 Assembly yield analysis and process tuning
Post-bond inspection	0.5		
Parameter monitoring	0.5		
Total	6.0		
WHO IT TAKES	~8.7 FTE	EXIT CRITERIA	MEASURED BY
OSAT process engineer Executes bonding to the window		● Every unit bonded inside the frozen window	● Units bonded within the window
Assembly process engineer Window compliance and monitoring		● Actual bond parameters recorded per unit	● Bond parameter capture rate
Quality engineer Post-bond inspection and traveler discipline		● Post-bond inspection coverage sufficient to catch localised defects	● Post-bond defects found against escaped
Package engineer Process deviation decisions			
Yield engineer Parameter-to-yield attribution			
		DEPENDS ON	
		ASSY-01 Known-good-die sort and selection	
		ASSY-02 HBM stack procurement and incoming inspection	
		PTV-12 Feedback into package design and assembly process window freeze	
		PKGD-10 OSAT selection and assembly process definition	

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-04

Interposer and substrate assembly, underfill, molding

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	5.0 M/M	~7.2 FTE	5	yes	OSAT process engineer
w3–w6 in stage	planned	while it runs	1 in parallel	no float	Reliability liaison approves

PURPOSE

Attach the interposer to the substrate and **underfill and mold** the assembly — the steps that make it mechanically robust and that lock in whatever went before.

Underfill is irreversible. Once it has flowed and cured, a bonding defect underneath is permanent and invisible to everything except acoustic imaging. The step's quality — void-free flow, complete fillet, correct cure — determines whether the mechanical stress the package will see is distributed or concentrated.

WORK FLOW5 STEPS · 2 LANES

w3w4w5w6

MAIN SEQUENCE

1

2

4

5

IN PARALLEL

3

#	STEP	TAT	LANE
1	Interposer to substrate attach	0.5w	main
2	Underfill dispense and flow	1w	main
3	Void and fillet inspection	0.8w	parallel
4	Underfill cure and molding	1w	main
5	Post-mold inspection	0.5w	main

Step 3 is the last chance to catch a bonding problem. Acoustic imaging before cure can still identify a void or an incomplete fill; after cure the unit is what it is, and the defect surfaces during thermal cycling months later.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Bonded assemblies from ASSY-03
- Substrates from PKGD-09 supply
- Underfill and mold compound specifications
- Frozen process window from PTV-12
- Warpage data from PTV-07

PRODUCES

- Interposer-to-substrate attached assemblies
- Underfilled assemblies
- Void and fillet inspection results
- Cured and molded units
- Post-mold inspection results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D1	FEEDS	Assembled units. After molding the unit is mechanically complete.
ASSY-D4	FEEDS	Package-level inspection and test data. Void and fillet inspection results are part of the unit's quality record.

RISKS AND WATCH-OUTS

- ◆ **Underfill voids.** They concentrate stress and are the mechanism behind delayed CPI failures, and they are invisible after cure without acoustic imaging.
- ◆ **Incomplete fillet.** The fillet distributes stress at the die edge, and an incomplete one puts that stress into the die corner instead.
- ◆ **Cure profile deviation.** Under-cured underfill has different mechanical properties, and the difference appears as reliability rather than as a visible defect.
- ◆ **Warpage during cure exceeding the window.** The vehicle programme measured it; running outside that window reintroduces the risk that was retired.
- ◆ **Inspection only after molding.** Once molded the assembly cannot be examined usefully, and the inspection has to happen before.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	DEPENDS ON
Underfill dispense and cure	2.0	● Bonded assemblies from ASSY-03	ASSY-03 Die attach, micro-bump and thermo-compression bonding
Interposer attach	1.3	● Substrates delivered against the PKGD-09 order	PKGD-09 Substrate and interposer supplier selection and lead-time booking
Molding	1.0	● Process window and materials specified	PTV-12 Feedback into package design and assembly process window freeze
Inspection	0.8		
Total	5.0	EXIT CRITERIA	
WHO IT TAKES	~7.2 FTE	● Void and fillet inspected before cure, not after	FEEDS INTO
OSAT process engineer		● Cure profile within the frozen window	ASSY-05 Lid / TIM attach and ball attach
Underfill, cure and molding execution		● Warpage during the process inside the validated range	ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology
Assembly process engineer			ASSY-07 Assembly yield analysis and process tuning
Window compliance and profile control			
Quality engineer			MEASURED BY
Void and fillet inspection			● Void rate at inspection
Materials engineer			● Fillet completeness
Underfill and mold compound behaviour			● Cure profile deviation from the window
Reliability liaison			
Defect implications for qualification			

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ASSY-05

Lid / TIM attach and ball attach

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	3.0 M/M	~6.5 FTE	4	yes	OSAT process engineer
w5–w7 in stage	planned	while it runs	1 in parallel	no float	Product engineering approves

PURPOSE

Finish the package — **TIM, lid and ball attach** — and produce units that can be mounted, cooled and tested.

These are the last steps and they determine two things the product is judged on: the thermal interface that decides the achievable power envelope, and the ball array that decides whether the package mounts reliably. Both were characterised on vehicles and both are executed here.

WORK FLOW

4 STEPS · 2 LANES

w5	w6	w7
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MAIN SEQUENCE

1	2	4
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IN PARALLEL

3

#	STEP	TAT	LANE
1	TIM application and lid attach	0.5w	main
2	Ball attach and reflow	0.8w	main
3	Lid flatness and co-planarity check	0.5w	parallel
4	Final visual and dimensional inspection	0.8w	main

Step 1 sets the thermal performance the product will actually have. Bond line thickness under the lid is a process parameter, and the difference between the characterised value and the achieved one is watts of headroom.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Molded assemblies from ASSY-04
- TIM material and lid components
- Ball attach process from the frozen window
- TIM characterisation from PTV-09
- Dimensional specification from PKGD-11

PRODUCES

- Lidded units with applied TIM
- Ball-attached packages
- Lid flatness and co-planarity results
- Final inspection results
- Completed assembled units

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D1	PRODUCES	Assembled units — bring-up, qualification and sample lots. This activity is where a unit becomes a finished package.
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ASSY-D4

FEEDS

Package-level inspection and test data. Dimensional and flatness data belongs with the unit's record.

RISKS AND WATCH-OUTS

- ◆ **TIM bond line thicker than characterised.** Thermal resistance rises, and the power envelope the product was sold on shrinks.
- ◆ **Lid flatness out of specification.** The cooling solution then contacts unevenly, which produces hotspots the thermal design did not predict.
- ◆ **Ball co-planarity outside the mounting window.** The package does not mount reliably, and the customer discovers it during their assembly.
- ◆ **Dimensional drift from the specification.** The package has to fit sockets, heatsinks and board footprints that were designed to the drawing.
- ◆ **Final inspection sampled rather than complete.** On an engineering build every unit is precious, and a defect shipped to bring-up costs lab days.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Lid and TIM attach	1.3	● Molded assemblies from ASSY-04	ASSY-06Assembly inline inspection — X-ray, CSAM, warpage metrology
Ball attach	1.0	● TIM and lid components available	
Inspection	0.8	● Dimensional specification from PKGD-11	ASSY-08Package-level open / short and continuity test
Total	3.0		ASSY-09Unit build for bring-up, qualification and customer samples
WHO IT TAKES	~6.5 FTE	EXIT CRITERIA	MEASURED BY
OSAT process engineer		● TIM bond line within the characterised range	● TIM bond line against the characterised value
Lid, TIM and ball attach execution		● Lid flatness and ball co-planarity inside specification	● Flatness and co-planarity yield
Thermal engineer		● Every unit inspected, not a sample	● Dimensional conformance
TIM bond line and thermal implications			
Quality engineer			
Flatness, co-planarity and dimensional inspection		DEPENDS ON	
		ASSY-04Interposer and substrate assembly, underfill, molding	
		PTV-09Thermal characterisation — Rjc / Rja, TIM performance, hotspot mapping against the thermal model	
		PKGD-11Package design freeze, DRC and tooling release	

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-06

Assembly inline inspection — X-ray, CSAM, warpage metrology

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	3.0 M/M	~6.5 FTE	4	no	Quality engineer
w3–w5 in stage	planned	while it runs	1 in parallel	has float	Package engineer approves

PURPOSE

Inspect the assemblies **without taking them apart** — X-ray for joints, acoustic imaging for delamination and voids, metrology for warpage — because most assembly defects are invisible from outside.

A 2.5D package hides everything that matters. The joints are under the die, the underfill is under the interposer, and the only way to see any of it non-destructively is through imaging. What this activity catches is what does not reach bring-up as an unexplained failure.

WORK FLOW4 STEPS · 2 LANES

w3w4w5

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Inspection plan and sampling strategy	0.5w	main
2	X-ray inspection of joints and voids	0.8w	main
3	Defect classification and disposition	0.5w	parallel
4	Acoustic imaging and warpage metrology	0.8w	main

Step 1's sampling strategy is a real decision on a small build. Full inspection is affordable at engineering-lot quantities and not at production volume, and starting with full coverage is what tells you what sampling would have missed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assemblies at each process stage from ASSY-03 to ASSY-05
- Inspection criteria from PTV vehicle results
- X-ray and CSAM capability at the OSAT
- Warpage limits from PTV-07
- Defect classification scheme

PRODUCES

- Inspection plan and sampling strategy
- X-ray inspection results
- Defect classification and disposition
- Acoustic imaging and warpage results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D4	PRODUCES	Package-level inspection and test data. This activity is the deliverable — the non-destructive record of what is inside each unit.
ASSY-D3	FEEDS	Assembly yield report and failure pareto. Inspection findings are the physical evidence behind the yield numbers.

RISKS AND WATCH-OUTS

- ◆ **Sampling too thin on an engineering build.** Full inspection is affordable at these quantities, and a defect reaching bring-up costs far more than the inspection would have.
- ◆ **Defects classified inconsistently.** Without a scheme the same feature is a defect on Monday and acceptable on Tuesday, and the yield data becomes noise.
- ◆ **Inspection only at the end.** Defects are cheapest to catch at the step that created them, and end-of-line inspection cannot attribute them.
- ◆ **Warpage limits not applied.** The vehicle programme set them; a unit outside them is a reliability risk regardless of whether it functions.
- ◆ **Findings not fed back to process.** Inspection that produces a pass/fail and no process signal wastes most of its value.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	DEPENDS ON
X-ray and acoustic imaging	1.3	● Assemblies available at each process stage	ASSY-03 Die attach, micro-bump and thermo-compression bonding
Warpage metrology	0.8	● Inspection criteria derived from PTV results	ASSY-04 Interposer and substrate assembly, underfill, molding
Defect classification	0.5	● Imaging capability available at the OSAT	ASSY-05 Lid / TIM attach and ball attach
Inspection plan	0.5		PTV-07 Warpage and co-planarity measurement across the reflow profile (shadow moiré)
Total	3.0		
WHO IT TAKES	~6.5 FTE	EXIT CRITERIA	FEEDS INTO
Quality engineer Owns inspection and classification		● Inspection at each process step, not only at the end	ASSY-07 Assembly yield analysis and process tuning
OSAT metrology X-ray, CSAM and warpage measurement		● Defects classified against a consistent scheme	ASSY-08 Package-level open / short and continuity test
Assembly process engineer Process feedback from findings		● Findings fed back to the process that created them	MP-04 Package qualification — MSL, drop, bend, board-level reliability
Failure analysis engineer Ambiguous defect interpretation			
Package engineer Acceptance criteria against vehicle data			MEASURED BY
			● Coverage of inspection against build quantity
			● Defects caught at the creating step
			● Classification consistency

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-07

Assembly yield analysis and process tuning

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	6.0 M/M	~5.2 FTE	6	no	Yield engineer
w3–w8 in stage	planned	while it runs	2 in parallel	has float	Product cost analyst approves

PURPOSE

Analyse the **assembly yield**, build the failure pareto, find the root causes with the OSAT and tune the process inside the window the vehicles defined.

First-build assembly yield on a 2.5D part is rarely what the cost model assumed. This activity turns the gap into a set of attributed causes and process adjustments — and compares the result against what the vehicle programme predicted, which is how the vehicle work gets validated in turn.

WORK FLOW

6 STEPS · 2 LANES

w3	w4	w5	w6	w7	w8
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Yield data collection across process steps	0.8w	main
2	Failure pareto by mode and process step	1.3w	main
3	Comparison against the PTV yield prediction	1w	parallel
4	Root cause analysis with the OSAT	1.5w	main
5	Corrective action tracking	1w	parallel
6	Process tuning inside the frozen window	1.5w	main

Step 6 stays inside the window deliberately. Tuning beyond the validated boundaries would abandon the evidence the vehicle programme produced, and a yield improvement bought that way carries an unquantified reliability risk.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Yield data from every assembly step
- Inspection results from ASSY-06
- Continuity results from ASSY-08
- PTV yield prediction from PTV-11
- Frozen process window from PTV-12

PRODUCES

- Yield data by process step
- Failure pareto by mode
- Comparison against vehicle prediction
- Root cause findings
- Corrective actions and their tracking
- Process tuning record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D3

PRODUCES

Assembly yield report and failure pareto. This activity is the deliverable, and the pareto is what the OSAT acts on.

ASSY-D2

FEEDS

Assembly travelers and process data. Yield attribution depends on the traveler data, and the analysis feeds back into what travelers should capture.

RISKS AND WATCH-OUTS

- ◆ **Yield reported without attribution.** A number tells the program it has a problem; a pareto by process step tells the OSAT what to change.
- ◆ **Tuning outside the validated window.** It may improve yield and it discards the reliability evidence the vehicles produced.
- ◆ **Vehicle prediction not compared.** The comparison is what validates or invalidates the whole PTV programme, and it is cheap to make.
- ◆ **Root cause pursued without the OSAT.** Process knowledge sits with the assembler, and analysis without them produces hypotheses rather than causes.
- ◆ **Corrective actions untracked.** An agreed action that nobody follows up is an action that did not happen.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	DEPENDS ON
Root cause analysis	2.0	● Assembly running with data collection in place ● Inspection and continuity results available ● Vehicle yield prediction available from PTV-11	ASSY-03 Die attach, micro-bump and thermo-compression bonding
Failure pareto	1.5		ASSY-04 Interposer and substrate assembly, underfill, molding
Process tuning	1.3		ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology
Data collection	0.8		ASSY-08 Package-level open / short and continuity test
Vehicle comparison	0.5		PTV-11 Daisy-chain continuity test and assembly yield learning
Total	6.0	EXIT CRITERIA	PTV-12 Feedback into package design and assembly process window freeze
WHO IT TAKES	~5.2 FTE	● Failures attributed to process steps, not only counted ● Tuning kept inside the validated window ● Yield compared against the vehicle prediction	FEEDS INTO
Yield engineer Owns yield analysis and the pareto			MP-05 Yield learning, failure pareto and defect analysis
OSAT process engineer Root cause and process adjustment			MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Quality engineer Failure classification and corrective action tracking			PTV-12 Feedback into package design and assembly process window freeze
Package engineer Window compliance for any tuning			MEASURED BY
Product cost analyst Yield against the cost model assumption			● Assembly yield against the cost model ● Failures attributed to a process step ● Corrective actions completed

19 ASSY · ASSEMBLY & PACKAGE BUILD

ASSY-08

Package-level open / short and continuity test

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
2w	3.0 M/M	~6.5 FTE	4	no	Package test engineer
w6–w8 in stage	planned	while it runs	1 in parallel	has float	Product engineering approves

PURPOSE

Test the assembled packages **electrically for opens, shorts and marginal joints** before they go anywhere — the cheapest possible screen on the most expensive units the program has.

A package-level continuity test costs minutes and catches assembly failures that would otherwise consume lab days at bring-up. On units this expensive, screening before shipping is not optional economics.

WORK FLOW

4 STEPS · 2 LANES

w6w7w8

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Test setup and continuity criteria	0.5w	main
2	Open and short test execution	0.8w	main
3	Marginal resistance screening	0.5w	parallel
4	Failure disposition and unit release	0.8w	main

Step 3 separates working from reliably working. A joint that conducts with elevated resistance passes a continuity test and fails in the field, and screening for it here is far cheaper than a field return.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Completed units from ASSY-05
- Continuity criteria from PTV-11
- Test hardware and fixtures
- Package pin map from PKGD-04
- Failure disposition process

PRODUCES

- Test setup and criteria
- Open and short results per unit
- Marginal resistance screening results
- Failure disposition and released units

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D4

PRODUCES

Package-level inspection and test data. The electrical half of the deliverable, alongside the imaging from ASSY-06.

ASSY-01

GATES

Assembled units. A unit that fails continuity is not released to bring-up, however scarce units are.

RISKS AND WATCH-OUTS

- ◆ **Screening skipped because units are scarce.** A failed unit consumed at bring-up costs more lab time than the whole screen would have.
- ◆ **Continuity only, no resistance.** Marginal joints conduct and fail later, and only a resistance measurement distinguishes them.
- ◆ **Criteria not derived from vehicle data.** PTV-11 measured what good and marginal look like, and the criteria should come from there.
- ◆ **Failures not fed to yield analysis.** An electrical failure is yield data and belongs in the pareto.
- ◆ **Test fixture damaging units.** On packages this expensive the fixture itself has to be qualified.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Test execution	1.3	Completed units from ASSY-05	ASSY-07 Assembly yield analysis and process tuning
Failure disposition	0.8	Continuity criteria derived from PTV-11	ASSY-09 Unit build for bring-up, qualification and customer samples
Test setup	0.5	Test fixture available and qualified	BU-01 Sample receipt, incoming inspection and board mounting
Marginal screening	0.5	EXIT CRITERIA	
Total	3.0	- Every unit screened before release - Marginal joints identified by resistance, not only continuity - Failures fed into the yield pareto	MEASURED BY
WHO IT TAKES	~6.5 FTE	DEPENDS ON	- Units screened against units built - Marginal units identified - Fixture-induced damage
Package test engineer Owns the screen and its criteria		ASSY-05 Lid / TIM attach and ball attach	
Quality engineer Disposition and release decisions		PTV-11 Daisy-chain continuity test and assembly yield learning	
Yield engineer Failure data into the pareto		PKGD-04 Substrate stack-up, escape routing and package DRC	
OSAT test Execution and fixturing			
Product engineering Release decisions on marginal units			

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ASSY-09

Unit build for bring-up, qualification and customer samples

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	3.0 M/M	~4.3 FTE	5	yes	Program manager
w5–w8 in stage	planned	while it runs	1 in parallel	no float	Operations planner approves

PURPOSE

Allocate the finished units — **bring-up, qualification, customer samples** — and build to that allocation, because the first build never has enough for everyone.

Three groups need units simultaneously and all of them are on the critical path of something. Bring-up needs them to start debugging, qualification needs them to start a months-long stress campaign, and customers were promised samples. The allocation is a program decision, not an assembly one.

WORK FLOW

5 STEPS · 2 LANES

w5	w6	w7	w8
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Allocation requirement collection from each consumer	0.5w	main
2	Unit allocation and reservation	1w	main
3	Contention resolution and escalation	0.5w	parallel
4	Build to allocation across lots	1w	main
5	Release and traceability per allocation	0.5w	main

Step 3 is a program decision escalated deliberately. Qualification starting late delays mass production; bring-up starting late delays everything; and customer samples are often contractual. Resolving that trade below the program level puts it in the wrong hands.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Screened units from ASSY-08
- Bring-up requirement from BU-01
- Qualification vehicle requirement from MP-01
- Customer sample commitments from DEF-01
- Build quantity from the assembly run

PRODUCES

- Allocation requirements per consumer
- Unit allocation and reservation record
- Contention resolution decisions
- Units built to allocation
- Release and traceability per allocation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

ASSY-D5	PRODUCES	Unit allocation record across bring-up, qual and customers. This activity is the deliverable, and it settles a contention that would otherwise be settled by whoever asks loudest.
ASSY-D1	PRODUCES	Assembled units — bring-up, qualification and sample lots. The units as allocated lots rather than as an undifferentiated build.

RISKS AND WATCH-OUTS

- ◆ **Allocation decided by whoever asks first.** The three consumers have different downstream consequences, and the trade belongs to the program.
- ◆ **Qualification starved.** Reliability stress runs for months; starting it late moves mass production directly.
- ◆ **Customer samples cut.** They are frequently contractual, and missing them has commercial consequences the program may not see.
- ◆ **Build quantity below the sum of needs.** Which is normal, and it means the shortfall has to be decided rather than discovered.
- ◆ **Traceability lost per allocation.** A qualification failure has to be traceable to its lot and its assembly conditions.

EFFORT SPLIT	3 M/M	ENTRY CRITERIA	FEEDS INTO
Allocation and reservation	1.0	● Units screened and released from ASSY-08	BU-01 Sample receipt, incoming inspection and board mounting
Build to allocation	0.8	● Requirements collected from all three consumers	MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Requirement collection	0.5	● Build quantity known	MP-04 Package qualification — MSL, drop, bend, board-level reliability
Contention resolution	0.5		BU-12 Customer sample readiness and release package
Release and traceability	0.3		
Total	3.0	EXIT CRITERIA	
WHO IT TAKES	~4.3 FTE	● Allocation decided at program level, not by request order	MEASURED BY
Program manager		● Qualification quantity sufficient to start its campaign	● Units allocated against requested per consumer
Owns the allocation and its contention		● Every allocated unit traceable to its lot and conditions	● Qualification start date against the plan
Validation lead			● Traceability maintained per allocation
Bring-up requirement		DEPENDS ON	
Quality lead		ASSY-01 Known-good-die sort and selection	
Qualification vehicle requirement		ASSY-05 Lid / TIM attach and ball attach	
Product marketing		ASSY-08 Package-level open / short and continuity test	
Customer sample commitments			
Operations planner			
Build execution and traceability			

VALIDATE & RAMP

20

Validation Hardware / EVB

EVB

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w52 – w90	38 _w	10	92 _{M/M}	5 of 10
ENGINEERING ACTIVITIES				KEY DELIVERABLES
EVB-01	Validation platform requirements and topology definition		5w 5 M/M	EVB-D1 Validation platform specification
EVB-02	EVB schematic design — power tree, clocking, host interface, connectors		10w 14 M/M	EVB-D2 EVB schematics, BOM and layout database
EVB-03	PCB layout with high-speed channel SI/PI simulation		12w 18 M/M	EVB-D3 Fabricated and assembled boards, rev A/B with quantity plan
EVB-04	PCB fabrication and assembly		8w 4 M/M	EVB-D4 Board bring-up report and known issues
EVB-05	Power delivery, VRM and telemetry design and bring-up		7w 8 M/M	EVB-D5 Debug and trace access documentation
EVB-06	Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers		7w 8 M/M	EVB-D6 Lab setup and instrument reservation plan
EVB-07	Board bring-up with socketed or dummy parts		7w 10 M/M	
EVB-08	Thermal solution and cooling for the lab platform		7w 7 M/M	
EVB-09	Lab instrumentation reservation and test rack build		7w 8 M/M	
EVB-10	Minimum host-side enablement for power-on (BSP handoff boundary)		7w 10 M/M	

20 EVB · VALIDATION HARDWARE / EVB

EVB-01

Validation platform requirements and topology definition

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	5.0 M/M	~4.3 FTE	6	yes	Validation lead
w0–w5 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Decide **what the silicon will be validated on** — how many platforms, in what configuration, at what quantity — before anyone draws a schematic.

The board is the only way to touch the silicon, and its capability bounds everything bring-up and characterisation can do. A platform that cannot reach a rail, cannot socket a part, or cannot be temperature-soaked closes off measurements that the qualification plan assumes are available.

WORK FLOW

6 STEPS · 2 LANES

w0	w1	w2	w3	w4	w5
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Validation requirement collection from bring-up, characterisation and qual	1w	main
2	Platform topology options — socketed, soldered, mezzanine	1w	main
3	Reuse assessment against existing platforms	1w	parallel
4	Topology selection with its trade-offs recorded	1.5w	main
5	Quantity and revision plan	1w	parallel
6	Platform specification	1.5w	main

Step 3 is where a schedule can be bought. An existing platform that carries 70% of the requirement, adapted, beats a new design by a quarter — and the assessment costs a week.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Bring-up requirements from BU-02 through BU-08 planning
- Characterisation plan from TEST-07
- Package pin map from PKGD-04
- Power envelope from ARCH-08
- Interface list from ARCH-03

PRODUCES

- Consolidated validation requirements
- Platform topology options with trade-offs
- Reuse assessment
- Topology selection record
- Quantity and revision plan
- Validation platform specification

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D1	PRODUCES	Validation platform specification. This activity is the deliverable, and everything the board team builds derives from it.
EVB-D3	INFORMS	Fabricated and assembled boards. The quantity and revision plan decides how many boards exist and when.

RISKS AND WATCH-OUTS

- ◆ **Requirements collected from bring-up only.** Characterisation and qualification also live on this platform, and their needs — soak capability, instrumented rails — differ.
- ◆ **Socket versus solder decided late.** It changes the board, the mechanical design and the unit consumption, and reversing it costs a board revision.
- ◆ **Quantity underestimated.** Boards are the constraint on parallel debug, and a second fabrication run costs two months.
- ◆ **Reuse not assessed.** An adaptable existing platform is the cheapest schedule available and it is often never looked for.
- ◆ **Specification written after design started.** The board then documents what was built rather than bounding what should be.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Requirement collection	1.5	● Package pin map stable enough to plan against	EVB-02 EVB schematic design — power tree, clocking, host interface, connectors
Topology selection	1.3	● Validation and characterisation intent understood	EVB-05 Power delivery, VRM and telemetry design and bring-up
Specification	1.0	● Power envelope from ARCH-08 available	EVB-06 Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers
Reuse assessment	0.8		EVB-08 Thermal solution and cooling for the lab platform
Quantity plan	0.5		EVB-09 Lab instrumentation reservation and test rack build
Total	5.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~4.3 FTE	● Requirements collected from all three consumers, not just bring-up	● Requirements traceable into the specification
Validation lead Owns the requirement and the specification		● Socket-versus-solder decided and its consequences accepted	● Reuse opportunities identified
Hardware architect Topology options and trade-offs		● Quantity plan sized against parallel debug needs	● Board quantity against parallel debug demand
Characterisation engineer Instrumentation and soak requirements		DEPENDS ON	
Package engineer Pin map and mechanical constraints		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	
Program manager Quantity, revision and budget		ARCH-08 Chip-level floorplan intent, pin and bump budget	
		PKGD-04 Substrate stack-up, escape routing and package DRC	
		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps	

20 EVB · VALIDATION HARDWARE / EVB

EVB-02

EVB schematic design — power tree, clocking, host interface, connectors

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	14.0 M/M	~6.1 FTE	8	yes	Hardware engineer
w4–w14 in stage	planned	while it runs	3 in parallel	no float	Hardware lead approves

PURPOSE

Design the **EVB schematic** — power tree, host interface, socket, clocking, debug headers — and its bill of materials.

The schematic is where the platform's capability is fixed. A rail without a sense point cannot be measured, a debug header omitted cannot be added, and a component with a 40-week lead time turns a board schedule into a procurement schedule.

WORK FLOW

8 STEPS · 2 LANES

w4	w6	w8	w10	w12	w14
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Block diagram and interface partition	1w	main
2	Power tree and VRM schematic	2.5w	main
3	Clocking and reference schematic	2w	parallel
4	Host interface and connector schematic	2.5w	main
5	Debug and instrumentation header design	2w	parallel
6	Socket and DUT interface schematic	2w	main
7	BOM construction and component sourcing check	1.5w	parallel
8	Schematic review and release to layout	2w	main

Step 7 runs alongside deliberately. Component lead times are discovered during sourcing, and a part found to be 40 weeks out while the schematic is still editable can be substituted; found after layout it cannot.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Platform specification from EVB-01
- Package pin map from PKGD-04
- Power rail requirements from PD-08
- Interface specifications from ARCH-03
- Debug requirements from EVB-06

PRODUCES

- Block diagram and interface partition
- Power tree and VRM schematic
- Clocking and reference schematic
- Host interface schematic
- Debug and instrumentation headers
- Socket and DUT interface

- Bill of materials with sourcing status
- Reviewed and released schematic

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D2	PRODUCES	EVB schematics, BOM and layout database. The schematic half of the deliverable.
EVB-D1	INFORMS	Validation platform specification. Schematic decisions feed back into what the specification can promise.

RISKS AND WATCH-OUTS

- ◆ **Rails without sense points.** A rail that cannot be measured cannot be characterised, and the omission is only discovered during bring-up.
- ◆ **Long-lead components found after layout.** The substitution then costs a board revision instead of a schematic edit.
- ◆ **Debug headers designed for the debugger nobody uses.** The tooling has to be agreed with validation rather than assumed.
- ◆ **Pin map changing after schematic release.** The package pin map has to be frozen for the schematic to be worth reviewing.
- ◆ **Power tree sized without the real power envelope.** A VRM sized to an early estimate cannot supply the silicon's actual transients.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	DEPENDS ON
Power tree design	3.5	● Platform specification approved from EVB-01 ● Package pin map frozen from PKGD-04 ● Power rail requirements available from PD-08	EVB-01 Validation platform requirements and topology definition
Host interface	3.0		PKGD-04 Substrate stack-up, escape routing and package DRC
Socket interface	2.5		PD-08 Detailed routing and DRC convergence
Clocking	2.0		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)
Review and release	1.5		
BOM and sourcing	1.5	EXIT CRITERIA	FEEDS INTO
Total	14.0	● Every rail instrumented with a sense point ● BOM sourced with lead times confirmed against the schedule ● Schematic reviewed by validation, not only by hardware	EVB-03 PCB layout with high-speed channel SI/PI simulation
WHO IT TAKES	~6.1 FTE		EVB-04 PCB fabrication and assembly
Hardware engineer Owns the schematic			EVB-05 Power delivery, VRM and telemetry design and bring-up
Power engineer Power tree and VRM design			
Validation engineer Debug and instrumentation requirements			
Component engineer BOM sourcing and lead times			MEASURED BY
Hardware lead Schematic review and release			● Rails with sense points against total rails ● Long-lead components identified before layout ● Review findings against post-fabrication rework

20 EVB · VALIDATION HARDWARE / EVB

EVB-03

PCB layout with high-speed channel SI/PI simulation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	18.0 M/M	~6.5 FTE	8	yes	PCB layout engineer
w12–w24 in stage	planned	while it runs	3 in parallel	no float	Fabrication liaison approves

PURPOSE

Lay out the PCB with **signal and power integrity simulated on the critical channels**, because a validation board carrying PCIe Gen6 or CXL is itself a high-speed design.

The board is in series with every interface measurement bring-up will make. A channel with a marginal board contribution produces link failures that look like silicon problems, and separating the two costs weeks of debug time that the simulation would have avoided.

WORK FLOW

8 STEPS · 2 LANES

w12	w15	w18	w21	w24
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Stack-up and impedance planning	1.5w	main
2	Placement and floorplan	2.5w	main
3	SI simulation on the critical channels	2w	parallel
4	High-speed channel routing	3w	main
5	PI simulation on the board PDN	2w	parallel
6	Power plane and PDN layout	2.5w	main
7	Thermal and mechanical layout constraints	2w	parallel
8	DRC, review and release to fabrication	2.5w	main

Steps 3 and 5 run against the routing rather than after it. A channel simulated after the board is routed produces a report; simulated during, it produces a routing change.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Released schematic and BOM from EVB-02
- Package model from SIPI-02
- Channel budget from SIPI-04
- Board PDN target from SIPI-06
- Mechanical constraints from EVB-08

PRODUCES

- Stack-up and impedance plan
- Placement and floorplan
- SI simulation results on critical channels
- Routed high-speed channels
- PI simulation on the board PDN
- Power plane layout

- Thermal and mechanical constraint compliance
- DRC-clean layout database released to fabrication

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D2	PRODUCES	EVB schematics, BOM and layout database. The layout half of the deliverable — what fabrication is built from.
EVB-D3	FEEDS	Fabricated and assembled boards. Fabrication takes the released layout database directly.

RISKS AND WATCH-OUTS

- ◆ **Board channel contribution not budgeted.** The SIPI channel budget covers package and board together, and a board that consumes more than its share leaves the silicon nothing.
- ◆ **SI simulated after routing.** The result is a report rather than a design change, and fixing it costs a revision.
- ◆ **Board PDN inadequate for the transient.** The silicon's di/dt is on the package model; a board PDN sized without it droops.
- ◆ **Stack-up chosen for cost before impedance.** Controlled impedance on the high-speed layers is not negotiable, and a cheaper stack-up that misses it is not cheaper.
- ◆ **Mechanical constraints applied last.** The cooling solution and socket need space, and finding that out after routing costs a revision.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	DEPENDS ON
High-speed channel routing	5.0	Schematic released from EVB-02	EVB-02 EVB schematic design — power tree, clocking, host interface, connectors
SI simulation	3.5	Package model available from SIPI-02	
Power plane layout	3.0	Channel and PDN budgets from SIPI-04 and SIPI-06	SIPI-02 Package and board model extraction — 3D EM, S-parameter and RLC network
Placement	2.5		
PI simulation	2.0		SIPI-04 Dynamic voltage-drop analysis with package inductance and real switching profiles
DRC and release	2.0		
Total	18.0	EXIT CRITERIA - Board channel contribution inside its share of the SIPI budget - SI and PI simulated during routing, not after - Mechanical and thermal constraints met without a revision	SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring
WHO IT TAKES	~6.5 FTE		EVB-08 Thermal solution and cooling for the lab platform
PCB layout engineer Owns the layout			FEEDS INTO
SI/PI engineer Channel and PDN simulation			EVB-04 PCB fabrication and assembly
Hardware engineer Schematic intent and constraint definition			EVB-07 Board bring-up with socketed or dummy parts
Mechanical engineer Thermal and mechanical constraints			MEASURED BY
Fabrication liaison Stack-up feasibility and DFM		- Board channel loss against its budget share - Simulation findings resolved in layout - DRC and DFM clean at release	

20 EVB · VALIDATION HARDWARE / EVB

EVB-04

PCB fabrication and assembly

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	4.0 M/M	~2.2 FTE	6	yes	Hardware engineer
w23–w31 in stage	planned	while it runs	2 in parallel	no float	Vendor liaison approves

PURPOSE

Get the boards **fabricated, assembled and inspected** — an eight-week vendor cycle that is mostly waiting and entirely on the critical path to bring-up.

This is procurement work with an engineering tail. The board has to arrive before silicon does, its quantity has to match the parallel-debug plan, and every board that arrives faulty is a debug station that does not exist.

WORK FLOW

6 STEPS · 2 LANES

w23	w25	w27	w29	w31
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	Fabrication vendor selection and data release	1w	main
2	PCB fabrication	3w	main
3	Component procurement against lead times	1.5w	parallel
4	Board assembly	2.5w	main
5	First article inspection	1w	parallel
6	Incoming inspection and acceptance	1.5w	main

Step 5 catches assembly problems on one board before they are replicated across the whole run. It costs a day and it is routinely skipped in the interest of speed, which is how twenty identically-wrong boards get built.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Released layout database from EVB-03
- BOM with sourcing from EVB-02
- Quantity plan from EVB-01
- Fabrication and assembly vendor capacity
- Inspection criteria

PRODUCES

- Vendor selection and released fabrication data
- Fabricated bare boards
- Procured components
- Assembled boards
- First article inspection result
- Incoming inspection and acceptance record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D3	PRODUCES	Fabricated and assembled boards, rev A/B with quantity plan. This activity is the deliverable.
EVB-D2	INFORMS	EVB schematics, BOM and layout database. Fabrication findings — DFM issues, substitutions — feed back into the database.

RISKS AND WATCH-OUTS

- ◆ **Long-lead components blocking assembly.** The board is fabricated and waiting on one part, which is the most common way this activity slips.
- ◆ **First article skipped.** An assembly error is then replicated across the entire run and discovered at bring-up.
- ◆ **Quantity cut for cost.** Boards are the constraint on parallel debug, and the saving is repaid in serialised lab time.
- ◆ **Boards arriving after silicon.** Silicon waiting on a board is the worst schedule outcome the program can produce.
- ◆ **Controlled impedance not verified.** The stack-up was designed for it; the fabricated board has to be coupon-tested to confirm it.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	DEPENDS ON
Assembly oversight	1.3	● Layout database released from EVB-03	EVB-03 PCB layout with high-speed channel SI/PI simulation
Fabrication oversight	1.0	● Components ordered against confirmed lead times	EVB-02 EVB schematic design — power tree, clocking, host interface, connectors
Procurement	1.0	● Vendor capacity booked	EVB-01 Validation platform requirements and topology definition
Inspection	0.8		
Total	4.0	EXIT CRITERIA	
WHO IT TAKES	~2.2 FTE	● Boards delivered before silicon arrives	FEEDS INTO
Hardware engineer Fabrication data release and acceptance		● First article inspected before the run completes	EVB-07 Board bring-up with socketed or dummy parts
Component engineer Procurement and lead-time management		● Controlled impedance verified by coupon	BU-01 Sample receipt, incoming inspection and board mounting
Quality engineer First article and incoming inspection			MEASURED BY
Program manager Quantity and schedule against silicon arrival			● Board delivery against silicon arrival
Vendor liaison Fabrication and assembly execution			● Boards accepted against boards built
			● First article findings caught before the run

20 EVB · VALIDATION HARDWARE / EVB

EVB-05

Power delivery, VRM and telemetry design and bring-up

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	8.0 M/M	~5.0 FTE	6	no	Power engineer
w8–w15 in stage	planned	while it runs	2 in parallel	has float	Test engineer approves

PURPOSE

Design and bring up the **power delivery, VRM and telemetry** on the platform — the rails the silicon runs on and the instrumentation that measures them.

Characterisation is mostly a power measurement. Shmoo needs rails that can be swept, power validation needs rails that can be measured accurately, and both need protection that trips before an expensive part is destroyed rather than after.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14	w15
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Power tree requirement from the die and package	1w	main
2	VRM selection and design	1.5w	main
3	Transient response design against the SIPI budget	1.5w	parallel
4	Telemetry and monitoring design	1.5w	main
5	Protection and sequencing design	1.5w	parallel
6	Power bring-up and validation on the board	3w	main

Step 5 is what protects the units. A sequencing error or an overcurrent event on a first-silicon part costs a package worth many thousands, and the protection has to be validated before a real part is ever inserted.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Power rail requirements from PD-08
- Board PDN design from EVB-03
- Transient budget from SIPI-06
- Shmoo range from BU-07 planning
- Package power delivery from PKGD-05

PRODUCES

- Power tree requirement
- VRM design
- Transient response design
- Telemetry and monitoring capability
- Protection and sequencing design
- Power bring-up and validation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D2	FEEDS	EVB schematics, BOM and layout database. The power subsystem design is part of the schematic package.
EVB-D4	FEEDS	Board bring-up report and known issues. Power bring-up results are the first part of the board's own bring-up record.

RISKS AND WATCH-OUTS

- ◆ **Rails not sweepable across the shmoo range.** Characterisation then cannot reach the corners the qualification plan requires.
- ◆ **Protection untested before a real part goes in.** A sequencing fault destroys a first-silicon unit, and there are very few of them.
- ◆ **Telemetry accuracy insufficient.** Power measurements at the resolution characterisation needs demand real instrumentation, not a rough monitor.
- ◆ **Transient response inadequate.** The die's di/dt is known from SIPI; a VRM that cannot follow it produces droop that looks like a silicon failure.
- ◆ **Sequencing not matching the silicon's requirement.** The power-up order comes from the design, and getting it wrong is a common first-board fault.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
VRM design	2.5	● Power rail requirements from PD-08	EVB-07 Board bring-up with socketed or dummy parts
Power bring-up	2.0	● Transient budget from SIPI-06	BU-02 Power-on, power sequencing and basic health check
Telemetry design	1.5	● Shmoo range defined by validation	BU-07 Shmoo across voltage, frequency and temperature
Transient response	1.3	EXIT CRITERIA	MEASURED BY
Protection and sequencing	0.8	● Rails sweepable across the full shmoo range	● Rail sweep range against the shmoo requirement
Total	8.0	● Protection validated before any real part is inserted	● Protection trip validated before first insertion
WHO IT TAKES	~5.0 FTE	● Telemetry accurate enough for characterisation	● Telemetry accuracy against characterisation need
Power engineer Owns the power subsystem		DEPENDS ON	
Hardware engineer Board integration and bring-up		EVB-01 Validation platform requirements and topology definition	
SI/PI engineer Transient and PDN requirements		PD-08 Detailed routing and DRC convergence	
Validation engineer Shmoo and telemetry requirements		SIPI-06 SSN / SSO and simultaneous-switching noise analysis at the IO ring	
Test engineer Measurement accuracy needs		PKGD-05 Package routing for signal integrity — length matching, reference plane and via design	

20 EVB · VALIDATION HARDWARE / EVB

EVB-06

Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w10–w17 in stage	8.0 M/M planned	~5.0 FTE while it runs	6 2 in parallel	no has float	Validation engineer Mechanical engineer approves

PURPOSE

Build the **debug access** — JTAG and trace pods, interposers, probe points — because silicon debug is bounded by what can be observed.

When first silicon does something unexpected, the difference between a day and a month is whether the state that explains it can be read out. The DFT infrastructure provides the internal access; this activity provides the physical path to it.

WORK FLOW

6 STEPS · 2 LANES

w10	w12	w14	w16	w17
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Debug requirements from validation and DFT	1w	main
2	JTAG and trace pod interface design	1.5w	main
3	Logic analyser and scope access design	1.5w	parallel
4	Interposer and probe access design	1.5w	main
5	Software debug tool integration	1w	parallel
6	Debug infrastructure validation	3w	main

Step 6 validates the debug path before it is needed. A trace pod that does not work is discovered during the crisis it was meant to resolve, which is the worst possible moment.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Debug and trace architecture from DFT-08
- Silicon debug requirements from BU-09 planning
- Platform specification from EVB-01
- Package pin map from PKGD-04
- Debug tooling from the DFT flow

PRODUCES

- Debug requirements
- JTAG and trace pod interface
- Logic analyser and scope access
- Interposer and probe access design
- Software debug tool integration
- Debug infrastructure validation results
- Debug and trace access documentation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D5

PRODUCES

Debug and trace access documentation. This activity is the deliverable, and it is what the bring-up team reads at 2am.

EVB-D2

FEEDS

EVB schematics, BOM and layout database. Debug headers and access points are schematic and layout content.

RISKS AND WATCH-OUTS

- ◆ **Debug access designed without DFT.** The internal observability comes from DFT-08, and the board access has to match what it exposes.
- ◆ **Trace bandwidth insufficient.** Trace is only useful if it can keep up with the events being traced, and the bandwidth is a board design parameter.
- ◆ **Probe points unreachable once the cooling solution is mounted.** The thermal solution and the probe access compete for the same physical space.
- ◆ **Debug path unvalidated before silicon.** It is then debugged during the emergency it exists to resolve.
- ◆ **Software tooling not integrated.** A hardware debug path with no software to drive it is not a debug path.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Pod interface design	2.5	● DFT debug architecture available from DFT-08 ● Silicon debug requirements understood ● Platform specification from EVB-01	EVB-02 EVB schematic design — power tree, clocking, host interface, connectors
Infrastructure validation	2.0		EVB-07 Board bring-up with socketed or dummy parts
Probe access design	1.5		BU-09 Silicon debug of anomalies and failure analysis requests
Analyser access	1.3		
Tool integration	0.8		
Total	8.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.0 FTE	● Debug access matching what DFT actually exposes ● Trace bandwidth sufficient for the events being traced ● Debug path validated before silicon, not during a crisis	● Debug access coverage against DFT observability ● Trace bandwidth against event rate ● Debug path validated before silicon arrival
Validation engineer Owns debug requirements			
DFT engineer Internal observability and access protocol			
Hardware engineer Board debug interface design			
Software engineer Debug tooling integration			
Mechanical engineer Probe access against the thermal solution		DEPENDS ON	
		EVB-01 Validation platform requirements and topology definition	
		DFT-08 Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL)	
		PKGD-04 Substrate stack-up, escape routing and package DRC	

20 EVB · VALIDATION HARDWARE / EVB

EVB-07

Board bring-up with socketed or dummy parts

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w30–w37 in stage	10.0 M/M planned	~6.2 FTE while it runs	6 2 in parallel	yes no float	Hardware engineer Mechanical engineer approves

PURPOSE

Bring the **board itself up** — with dummy parts or a socketed loopback — so that when silicon arrives, every failure is the silicon's.

This is the single highest-leverage activity in the stage. A board debugged before first silicon means bring-up starts on a known-good platform; a board debugged alongside first silicon means every anomaly has two possible causes and each takes days to separate.

WORK FLOW

6 STEPS · 2 LANES

w30	w32	w34	w36	w37
-----	-----	-----	-----	-----

MAIN SEQUENCE

1	2	4	6
---	---	---	---

IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Bring-up plan and dummy part fitting	1w	main
2	Power tree bring-up and rail verification	1.5w	main
3	Telemetry and protection validation	1.5w	parallel
4	Clocking and reference validation	1.5w	main
5	Thermal solution fit check	1.5w	parallel
6	Interface loopback and signal validation	3w	main

The whole activity is scheduled to complete before silicon arrives. Every hour spent here is an hour not spent, at ten times the cost, debugging a board while first silicon waits on the bench.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Assembled boards from EVB-04
- Power subsystem from EVB-05
- Debug infrastructure from EVB-06
- Thermal solution from EVB-08
- Host enablement from EVB-10

PRODUCES

- Bring-up plan and fitted dummy parts
- Power tree bring-up results
- Telemetry and protection validation
- Clocking and reference validation
- Thermal fit check
- Interface loopback and signal validation results
- Board bring-up report and known issues

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D4	PRODUCES	Board bring-up report and known issues. This activity is the deliverable, and the known-issues list is what stops board faults being mistaken for silicon faults.
EVB-D3	GATES	Fabricated and assembled boards. A board is not released to silicon bring-up until it has passed this.

RISKS AND WATCH-OUTS

- ◆ **Board bring-up deferred until silicon arrives.** Every anomaly then has two candidate causes, and separating them costs days each.
- ◆ **Known issues not documented.** The bring-up team then rediscovers each board quirk independently, once per engineer.
- ◆ **Dummy parts not representative.** A loopback that does not load the rails or exercise the channels validates less than it appears to.
- ◆ **Only one board brought up.** Board-to-board variation is real, and a fleet where only one is characterised is a fleet with unknown members.
- ◆ **Thermal fit checked after silicon insertion.** A cooling solution that does not fit is discovered with an irreplaceable part underneath it.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	DEPENDS ON
Interface validation	3.0	● Assembled boards accepted from EVB-04 ● Power and debug subsystems designed ● Dummy parts or loopback fixtures available	EVB-04 PCB fabrication and assembly
Power bring-up	2.5		EVB-05 Power delivery, VRM and telemetry design and bring-up
Clocking validation	2.0		EVB-06 Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers
Telemetry validation	1.5		EVB-08 Thermal solution and cooling for the lab platform
Thermal fit	1.0		EVB-10 Minimum host-side enablement for power-on (BSP handoff boundary)
Total	10.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~6.2 FTE	● Board bring-up complete before silicon arrives ● Known issues documented for the bring-up team ● Every board in the fleet brought up, not just one	BU-01 Sample receipt, incoming inspection and board mounting
Hardware engineer Owns board bring-up			BU-02 Power-on, power sequencing and basic health check
Power engineer Rail and protection verification			
Validation engineer Known-issue capture for the bring-up team			
SI engineer Loopback and signal validation			
Mechanical engineer Thermal fit			MEASURED BY
			● Board bring-up completion against silicon arrival ● Boards brought up against boards built ● Silicon-bring-up issues later traced to the board

20 EVB · VALIDATION HARDWARE / EVB

EVB-08

Thermal solution and cooling for the lab platform

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	7.0 M/M	~4.3 FTE	6	no	Thermal engineer
w16–w23 in stage	planned	while it runs	2 in parallel	has float	Hardware engineer approves

PURPOSE

Design and validate the **cooling for the lab platform** — which is not the product's thermal solution, but has to remove the same power.

A part that thermally throttles during characterisation produces data about the cooling rather than about the silicon. The lab solution can be oversized and ugly in ways the product's cannot, and it should be, because measurement validity depends on it.

WORK FLOW

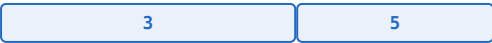
6 STEPS · 2 LANES

w16	w18	w20	w22	w23
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Thermal requirement from the package envelope	1w	main
2	Cooling solution selection	1.5w	main
3	Airflow and enclosure design	1.5w	parallel
4	Mounting and interface design	1.5w	main
5	Temperature control and soak capability	1w	parallel
6	Thermal validation on the platform	3w	main

Step 5 is what makes qualification-adjacent measurement possible. Characterisation needs the part held at a temperature, not merely kept cool, and a solution that only cools cannot produce a hot-corner shmoo.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package thermal envelope from PKGD-08
- Power envelope from ARCH-08
- Platform specification from EVB-01
- Characterisation temperature range from TEST-07
- Mechanical constraints from EVB-03

PRODUCES

- Thermal requirement
- Cooling solution selection
- Airflow and enclosure design
- Mounting and interface design
- Temperature control and soak capability
- Thermal validation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D1	FEEDS	Validation platform specification. Thermal capability is part of what the platform can promise.
EVB-D4	FEEDS	Board bring-up report and known issues. Thermal validation results belong in the platform's own record.

RISKS AND WATCH-OUTS

- ◆ **Cooling sized to the product's solution rather than to the lab's need.** The lab platform can be oversized, and measurement validity is worth more than realism.
- ◆ **No temperature control, only cooling.** Characterisation needs the part held at a set point, including a hot one.
- ◆ **Throttling during characterisation.** The data then describes the cooling rather than the silicon, and it is often not noticed.
- ◆ **Mounting competing with probe access.** The debug points and the heatsink want the same space, and the conflict has to be resolved in layout.
- ◆ **Thermal validated only at nominal power.** The interesting measurements are at maximum power, which is where the solution is stressed.

EFFORT SPLIT	7 M/M	ENTRY CRITERIA	DEPENDS ON
Cooling selection	2.0	● Package thermal envelope from PKGD-08 ● Characterisation temperature range from TEST-07 ● Platform specification from EVB-01	EVB-01 Validation platform requirements and topology definition
Thermal validation	2.0		PKGD-08 Test vehicle requirement definition and handoff to PTV
Mounting design	1.5		ARCH-08 Chip-level floorplan intent, pin and bump budget
Airflow design	1.0		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps
Temperature control	0.5		
Total	7.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~4.3 FTE	● No throttling at maximum power during characterisation ● Temperature held at set point, including the hot corner ● Mounting compatible with debug probe access	EVB-03 PCB layout with high-speed channel SI/PI simulation
Thermal engineer Owns the lab cooling solution			EVB-07 Board bring-up with socketed or dummy parts
Mechanical engineer Mounting, enclosure and airflow			BU-07 Shmoo across voltage, frequency and temperature
Validation engineer Temperature range and soak requirement			
Package engineer Package thermal envelope			
Hardware engineer Board integration and probe access			MEASURED BY
			● Throttling events during characterisation ● Temperature set-point range achieved ● Probe access retained with cooling mounted

20 EVB · VALIDATION HARDWARE / EVB

EVB-09

Lab instrumentation reservation and test rack build

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	8.0 M/M	~5.0 FTE	6	no	Lab manager
w24–w31 in stage	planned	while it runs	2 in parallel	has float	Facilities approves

PURPOSE

Reserve the instruments and **build the lab racks** — because characterisation needs equipment that is expensive, shared and booked months ahead.

A high-speed oscilloscope or a BERT is a capital item with a queue in front of it. Discovering during bring-up that the scope with the right bandwidth is committed elsewhere for six weeks does not delay the measurement by six weeks — it delays everything downstream of it.

WORK FLOW

6 STEPS · 2 LANES

w24	w26	w28	w30	w31
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Instrument requirement from the characterisation plan	1w	main
2	Instrument reservation and procurement	1.5w	main
3	Automation and control infrastructure	1.5w	parallel
4	Test rack build and cabling	1.5w	main
5	Lab space and power provisioning	1w	parallel
6	Rack integration and calibration	3w	main

Step 3 decides whether characterisation is manual or automatic. A shmoo run by hand takes days per corner; the same shmoo automated runs overnight, and the automation is written here rather than improvised during bring-up.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Characterisation plan from TEST-07
- Bring-up plan from BU-02 through BU-08
- Platform specification from EVB-01
- Instrument pool and capital budget
- Lab space and infrastructure

PRODUCES

- Instrument requirement list
- Instrument reservations and procurement
- Automation and control infrastructure
- Built test racks with cabling
- Lab space and power provisioning
- Rack integration and calibration record
- Lab setup and instrument reservation plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D6

PRODUCES

Lab setup and instrument reservation plan. This activity is the deliverable, and it is what stops bring-up queueing for a scope.

EVB-D4

INFORMS

Board bring-up report and known issues. The lab setup is the context every bring-up measurement is made in.

RISKS AND WATCH-OUTS

- ◆ **Instruments booked too late.** High-bandwidth equipment is shared and queued, and a six-week wait delays everything downstream.
- ◆ **Automation deferred to bring-up.** It is then written under time pressure by people who should be debugging silicon.
- ◆ **Racks not calibrated.** Measurements from an uncalibrated setup are unusable for characterisation and often not questioned.
- ◆ **Lab power or space insufficient.** A rack that cannot be powered is a rack that does not exist, and facilities lead times are long.
- ◆ **Instrument bandwidth below the signal rate.** A scope that cannot resolve the edge produces a plausible and wrong measurement.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Rack build and cabling	2.0	● Characterisation plan available from TEST-07 ● Bring-up measurement needs understood ● Capital budget approved	BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Integration and calibration	2.0		BU-07 Shmoo across voltage, frequency and temperature
Instrument reservation	1.8		BU-08 Performance validation against the architecture model
Automation infrastructure	1.5		
Space provisioning	0.8	EXIT CRITERIA ● Instruments reserved against the bring-up window, not requested during it ● Automation working before bring-up starts ● Racks calibrated and their calibration current	MEASURED BY ● Instrument availability against the bring-up window ● Measurements automated against manual ● Calibration currency at bring-up start
Total	8.0		
WHO IT TAKES	~5.0 FTE	DEPENDS ON	
Lab manager Owns instrument reservation and lab space		EVB-01 Validation platform requirements and topology definition	
Characterisation engineer Instrument requirements and calibration		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps	
Automation engineer Control and data-capture infrastructure			
Validation lead Prioritisation across competing bookings			
Facilities Power and space provisioning			

20 EVB · VALIDATION HARDWARE / EVB

EVB-10

Minimum host-side enablement for power-on (BSP handoff boundary)

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w w26–w33 in stage	10.0 M/M planned	~6.2 FTE while it runs	6 2 in parallel	no has float	Validation software engineer Bring-up lead approves

PURPOSE

Provide the **minimum host-side software** needed to power on and talk to the part — boot, register access, enumeration, logging — without taking on the product's software stack.

SW/FW is outside this program's scope, but silicon cannot be brought up without something on the other end of the link. The boundary matters: this activity delivers exactly enough to reach first register read and no more, and where that line sits should be agreed rather than discovered.

WORK FLOW

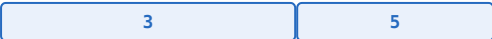
6 STEPS · 2 LANES

w26	w28	w30	w32	w33
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Handoff boundary definition with the firmware team	1w	main
2	Minimal boot and initialisation software	1.5w	main
3	Host driver stub and enumeration	1.5w	parallel
4	Register access and diagnostic tooling	1.5w	main
5	Logging and data capture	1w	parallel
6	Host enablement validation on the platform	3w	main

Step 1 is the activity's most valuable hour. Without an explicit boundary, hardware validation writes software the firmware team is also writing, and each assumes the other is doing it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Register map from RTL-06
- Boot sequence from ARCH-06
- Debug tooling from EVB-06
- Firmware team's scope and schedule
- Platform specification from EVB-01

PRODUCES

- Handoff boundary definition
- Minimal boot and initialisation software
- Host driver stub and enumeration
- Register access and diagnostic tooling
- Logging and data capture
- Host enablement validation results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

EVB-D4	FEEDS	Board bring-up report and known issues. Host enablement is validated as part of the platform and reported with it.
EVB-D5	FEEDS	Debug and trace access documentation. The diagnostic tooling is how debug access is actually used.

RISKS AND WATCH-OUTS

- ◆ **Boundary with firmware left implicit.** Either the work is duplicated or nobody does it, and both are discovered at power-on.
- ◆ **Register map drifting from the RTL.** A diagnostic tool built against a stale map reads the wrong addresses and reports plausible nonsense.
- ◆ **Scope creeping into the product stack.** Minimal enablement becomes a driver project, and the validation team stops validating.
- ◆ **Logging insufficient for post-mortem.** A failure that cannot be reconstructed from the log has to be reproduced, which on first silicon may not be possible.
- ◆ **Enablement unvalidated before silicon.** Software bugs then present as silicon bugs during the most expensive week of the program.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Boot software	3.0	● Register map available from RTL-06	EVB-07 Board bring-up with socketed or dummy parts
Enablement validation	2.5	● Boot sequence defined in ARCH-06	BU-04 Boot, firmware load and functional smoke test
Diagnostic tooling	2.0	● Firmware team's scope known	BU-02 Power-on, power sequencing and basic health check
Driver stub	1.5		
Logging	1.0		
Total	10.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~6.2 FTE	● Boundary with firmware explicit and agreed	● Boundary disputes with firmware during bring-up
Validation software engineer Owns host enablement		● Register access validated against the current map	● Register map currency against RTL
Firmware liaison Boundary definition and handoff		● Enablement validated on the platform before silicon	● Bring-up issues later traced to host software
Design engineer Register map and boot sequence		DEPENDS ON	
Validation engineer Diagnostic tooling requirements		EVB-01 Validation platform requirements and topology definition	
Bring-up lead Scope discipline against the product stack		EVB-06 Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers	
		RTL-06 Lint, CDC and RDC closure	
		ARCH-06 PPA budget allocation per block	

VALIDATE & RAMP

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Test Development

TEST

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w54 – w96	42 _w	11	174 _{M/M}	4 _{of 11}

ENGINEERING ACTIVITIES				KEY DELIVERABLES		
TEST-01	Test plan and coverage strategy — wafer sort, final test, system-level test	8w	12	M/M	TEST-D1	Test plan and test coverage matrix
TEST-02	ATE platform selection and tester time booking	6w	4	M/M	TEST-D2	Qualified probe card and load board
TEST-03	Probe card design, fabrication and qualification	20w	16	M/M	TEST-D3	Wafer sort and final test programs, release-tagged
TEST-04	Load board / DUT board design, fabrication and bring-up	17w	16	M/M	TEST-D4	Characterisation test suite
TEST-05	Wafer sort test program development	16w	30	M/M	TEST-D5	ATE-ready pattern set with debug log
TEST-06	Final / package test program development	16w	28	M/M	TEST-D6	Test time and test cost model
TEST-07	Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps	12w	20	M/M	TEST-D7	Test data infrastructure and yield database
TEST-08	DFT pattern porting to ATE format and pattern debug	10w	16	M/M		
TEST-09	Test time and cost optimisation — parallelism, site count	9w	12	M/M		
TEST-10	Correlation between ATE, bench and system results	8w	10	M/M		
TEST-11	Test data infrastructure — STDF collection and yield database	8w	10	M/M		

TEST-01

Test plan and coverage strategy —
wafer sort, final test, system-level test

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w w0–w8 in stage	12.0 M/M planned	~6.5 FTE while it runs	8 3 in parallel	yes no float	Test architect Cost analyst approves

PURPOSE

Decide **what will be tested, where, and to what defect level** — the plan every other activity in the stage implements.

Test is where quality is bought and where cost per unit is set, and the two trade directly. The plan is the document where that trade is made explicitly: a DPPM target implies coverage, coverage implies content, content implies test time, and test time is money on every unit ever shipped.

WORK FLOW8 STEPS · 2 LANES

w0	w2	w4	w6	w8
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Coverage requirement from the specification and DFT	1w	main
2	Test flow architecture — sort, final, characterisation	2w	main
3	Tester resource estimation	1.5w	parallel
4	Coverage matrix construction	2w	main
5	DPPM target derivation from the coverage	1.5w	parallel
6	Test insertion and binning strategy	1.5w	main
7	Burn-in and screening decision	1w	parallel
8	Plan review and release	1.5w	main

Step 5 is the one that closes the loop. Coverage without a DPPM target is an engineering preference; a DPPM target derived from the coverage is a commitment the customer can be given.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Product specification and quality target from DEF-04
- DFT coverage from DFT-03 and DFT-05
- Package pin map from PKGD-04
- Quality and reliability requirement from MP-01
- Cost target from DEF-06

PRODUCES

- Coverage requirement
- Test flow architecture across insertions
- Tester resource estimate
- Test coverage matrix
- DPPM target derivation

- Test insertion and binning strategy
- Burn-in and screening decision
- Reviewed test plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D1	PRODUCES	Test plan and test coverage matrix. This activity is the deliverable, and every other TEST activity implements part of it.
TEST-D6	INFORMS	Test time and test cost model. The flow architecture and insertion strategy set the cost model's structure.

RISKS AND WATCH-OUTS

- ◆ **Coverage chosen without a DPPM target.** There is then no way to say whether the test is sufficient, only whether it is thorough.
- ◆ **Test time not budgeted at plan time.** Content accumulates and the cost per unit is discovered at production readiness when it is expensive to change.
- ◆ **Burn-in decided late.** It is a capital and cycle-time commitment, and reversing it after qualification planning is expensive.
- ◆ **Sort and final coverage overlapping without intent.** Testing the same thing twice costs money on every unit and often happens by accretion.
- ◆ **DFT coverage assumed rather than measured.** The plan then promises a DPPM the actual pattern set cannot deliver.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Coverage matrix	3.0	DFT coverage results available from DFT-03 and DFT-05	TEST-02 ATE platform selection and tester time booking
Test flow architecture	2.5	Quality target defined	TEST-05 Wafer sort test program development
Binning strategy	2.0	Package pin map from PKGD-04	TEST-06 Final / package test program development
DPPM derivation	1.8		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps
Review and release	1.5		TEST-09 Test time and cost optimisation — parallelism, site count
Resource estimation	1.3		
Total	12.0		
WHO IT TAKES	~6.5 FTE		
Test architect Owns the plan and the coverage matrix		EXIT CRITERIA - DPPM target derived from actual coverage, not asserted - Test time budgeted before content is written - Sort and final coverage split deliberately	MEASURED BY - Coverage against the DPPM target - Test time estimate against the cost target - Overlap between sort and final content
DFT engineer Structural coverage and pattern capability		DEPENDS ON	
Quality engineer DPPM target and screening strategy		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)	
Product engineering Binning and grading strategy		DEF-06 Competitive benchmarking and product positioning	
Cost analyst Test cost against the product cost target		DFT-03 MBIST / BIRA / BISR architecture for embedded memories	
		DFT-05 On-chip clock controller design for at-speed testing	
		PKGD-04 Substrate stack-up, escape routing and package DRC	

TEST-02

ATE platform selection and tester time booking

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	4.0 M/M	~2.9 FTE	6	no	Test engineering manager
w4–w10 in stage	planned	while it runs	2 in parallel	has float	ATE vendor liaison approves

PURPOSE

Choose the **ATE platform** and book the tester time — a capital and capacity decision that has to be made long before there is anything to test.

Tester capacity is a shared, scheduled resource. Program development time, characterisation time and production capacity are all booked against the same pool, and a program that has not committed its slots competes with programs that have.

WORK FLOW

6 STEPS · 2 LANES

w4

w6

w8

w10

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Tester requirement from the test plan	1w	main
2	ATE platform evaluation against the requirement	1.5w	main
3	Instrument option and pin-count analysis	1w	parallel
4	Platform selection and configuration	1.5w	main
5	Cost per tested unit model	1w	parallel
6	Tester time booking and capacity commitment	2w	main

Step 6 is why this runs early. Development, characterisation and production all draw on the same testers, and slots that are not booked are taken.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test plan and flow architecture from TEST-01
- Pin count and interface rates from PKGD-04
- Available ATE platforms and their capability
- Test capacity and capital budget
- Production volume forecast from DEF-05

PRODUCES

- Tester requirement specification
- ATE platform evaluation
- Instrument option and pin-count analysis
- Platform selection and configuration
- Cost per tested unit model
- Tester time bookings across development, characterisation and production

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D6	FEEDS	Test time and test cost model. The platform's cost per hour is the model's rate.
TEST-D3	INFORMS	Wafer sort and final test programs. Programs are written for a specific platform and its instrument set.

RISKS AND WATCH-OUTS

- ◆ **Tester time not booked ahead.** Capacity is shared, and a program without commitments waits behind programs with them.
- ◆ **Instrument options short of the interface rate.** High-speed content cannot be run, and the shortfall is discovered when the program is written.
- ◆ **Pin count insufficient for parallel sites.** Multi-site conversion later becomes impossible, and test cost stays where it started.
- ◆ **Platform chosen on capital cost alone.** Cost per tested unit depends on throughput as much as on the tester's price.
- ◆ **Development capacity not separated from production.** Program debug then competes with the ramp it is meant to enable.

EFFORT SPLIT	4 M/M	ENTRY CRITERIA	FEEDS INTO
Platform evaluation	1.3	● Test plan available from TEST-01	TEST-05 Wafer sort test program development
Tester time booking	1.0	● Pin count and interface rates known	TEST-06 Final / package test program development
Selection and configuration	0.8	● Capital budget and capacity forecast available	TEST-08 DFT pattern porting to ATE format and pattern debug
Pin-count analysis	0.5		MP-07 Test time reduction and multi-site conversion
Cost model	0.5	EXIT CRITERIA	
Total	4.0	● Development, characterisation and production slots all booked ● Instrument set covering the highest interface rate ● Pin count supporting the planned multi-site conversion	MEASURED BY ● Tester slots committed against the plan ● Instrument capability against interface rates ● Cost per tested unit against target
WHO IT TAKES	~2.9 FTE	DEPENDS ON	
Test engineering manager Owns platform selection and booking		TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test	
Test architect Requirement derivation from the plan		PKG-D-04 Substrate stack-up, escape routing and package DRC	
Operations planner Capacity commitment and scheduling		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)	
Cost analyst Cost per tested unit			
ATE vendor liaison Configuration and instrument options			

TEST-03

Probe card design, fabrication and qualification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
20w	16.0 M/M	~3.5 FTE	8	yes	Probe engineer
w10–w30 in stage	planned	while it runs	3 in parallel	no float	Operations planner approves

PURPOSE

Design, fabricate and qualify the **probe card** — the physical interface between the tester and the wafer, and the longest lead time in the stage.

Nothing can be sorted without it, and sort gates assembly. Twenty weeks of design and fabrication sit in front of the first wafer, which means the card has to be committed against a bump map that is frozen far earlier than most people expect.

WORK FLOW

8 STEPS · 2 LANES

w10	w14	w18	w22	w26	w30
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Probe card requirement and bump map extraction	2w	main
2	Probe card design	3w	main
3	Contact resistance and planarity analysis	3w	parallel
4	Probe card fabrication	8w	main
5	Touchdown life and cleaning strategy	3w	parallel
6	Qualification on the tester	4w	main
7	Spare card and repair plan	2.5w	parallel
8	Release to wafer sort	3w	main

Step 4's eight weeks are pure vendor lead time and cannot be compressed. Step 1's bump map therefore has to be frozen roughly five months before sort, which is well before most people believe the design is stable.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Bump map from PD-15 and PKGD-03
- Test plan and sort content from TEST-01
- ATE platform and interface from TEST-02
- Wafer sort touchdown plan
- Probe card vendor capability

PRODUCES

- Probe card requirement and bump map
- Probe card design
- Contact resistance and planarity analysis
- Fabricated probe card
- Touchdown life and cleaning strategy
- Tester qualification results
- Spare card and repair plan

- Probe card released to sort

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D2	PRODUCES	Qualified probe card and load board. The probe card half of the deliverable.
TEST-D3	GATES	Wafer sort and final test programs. A sort program cannot be debugged on silicon without a qualified card.

RISKS AND WATCH-OUTS

- ◆ **Bump map changing after card release.** Eight weeks of fabrication are lost, and sort — and therefore assembly — moves by that much.
- ◆ **Contact resistance out of specification.** Marginal contact produces failures that look like silicon defects and corrupts the whole sort yield.
- ◆ **Planarity insufficient across the touchdown area.** Some probes contact and some do not, and the pattern is subtle enough to be missed.
- ◆ **No spare card.** A damaged card with a ten-week replacement stops sort entirely, and cards are damaged routinely.
- ◆ **Touchdown life below the wafer volume.** Card degradation then appears as a yield trend that gets chased as a process problem.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	DEPENDS ON	
Probe card design	4.0	● Bump map frozen from PD-15 and PKGD-03 ● ATE platform selected in TEST-02 ● Vendor capacity booked against the lead time	TEST-01	Test plan and coverage strategy — wafer sort, final test, system-level test
Qualification	3.5		TEST-02	ATE platform selection and tester time booking
Requirement and bump map	2.5		PD-15	Chip finishing — seal ring, dummy and metal fill, alignment marks
Contact analysis	2.5		PKGD-03	Interposer / RDL routing design
Touchdown strategy	2.0			
Spare plan	1.5	EXIT CRITERIA	FEEDS INTO	
Total	16.0	● Card qualified on the tester before wafers arrive ● Contact resistance and planarity inside specification ● Spare card available against the sort volume	TEST-05	Wafer sort test program development
WHO IT TAKES	~3.5 FTE		ASSY-01	Known-good-die sort and selection
Probe engineer			FAB-08	Wafer acceptance test and lot disposition
Owens the card and its qualification				
Test engineer			MEASURED BY	● Card readiness against wafer arrival ● Contact resistance and planarity at qualification ● Touchdown life against planned wafer volume
Requirement and tester interface				
Physical design				
Bump map freeze and change control				
Probe card vendor				
Design and fabrication				
Operations planner				
Spare strategy and lead time				

TEST-04

Load board / DUT board design, fabrication and bring-up

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
17w	16.0 M/M	~4.1 FTE	8	no	Test hardware engineer
w12–w29 in stage	planned	while it runs	3 in parallel	has float	Socket vendor approves

PURPOSE

Design, build and bring up the **load board and socket** — the package-level equivalent of the probe card, and what every final-test and characterisation measurement passes through.

The load board carries the tester's signals to the package at full rate and delivers its power at full current. It is a high-speed board design in its own right, and its own contribution to a measurement has to be small enough that what is measured is the silicon.

WORK FLOW

8 STEPS · 2 LANES

w12w16w20w24w28w29

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Load board requirement and socket selection	2w	main
2	Load board design with signal integrity	3.5w	main
3	Socket contact and insertion life analysis	2.5w	parallel
4	Fabrication and assembly	5w	main
5	Power delivery design on the load board	2.5w	parallel
6	Bring-up on the tester	3.5w	main
7	Correlation fixture design	2w	parallel
8	Release to final test	3w	main

Step 5 is routinely underestimated. A high-power part draws current through the socket, and a load board whose power delivery droops produces failures at speed that are indistinguishable from silicon marginality.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Package pin map and ball map from PKGD-04
- Test plan and final content from TEST-01
- ATE platform interface from TEST-02
- Power envelope from ARCH-08
- Socket vendor capability

PRODUCES

- Load board requirement and socket selection
- Load board design
- Socket contact and life analysis
- Fabricated and assembled load board
- Load board power delivery design
- Tester bring-up results
- Correlation fixture

- Load board released to final test

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D2	PRODUCES	Qualified probe card and load board. The load board half of the deliverable.
TEST-D3	GATES	Wafer sort and final test programs. The final test program is debugged on this hardware.

RISKS AND WATCH-OUTS

- ◆ **Load board power delivery droop.** It produces at-speed failures that look exactly like silicon marginality and consume weeks of debug.
- ◆ **Socket insertion life below the test volume.** Contact degrades, yield trends downward, and the cause is looked for in the process.
- ◆ **Board channel loss consuming the measurement margin.** The interface test then fails on the board rather than on the part.
- ◆ **Socket selected before the package is dimensioned.** A socket that does not fit the finished package is discovered very late.
- ◆ **No correlation fixture.** ATE and bench results then cannot be reconciled, and both are distrusted.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	FEEDS INTO
Load board design	4.0	● Package pin and ball map from PKGD-04	TEST-06 Final / package test program development
Tester bring-up	3.0	● ATE platform selected in TEST-02	TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps
Fabrication and assembly	2.5	● Power envelope known from ARCH-08	TEST-10 Correlation between ATE, bench and system results
Power delivery design	2.5		
Socket analysis	2.0	EXIT CRITERIA	
Release	2.0	● Power delivery adequate at maximum test current	MEASURED BY
Total	16.0	● Socket insertion life above the planned test volume	● Power droop at maximum test current
WHO IT TAKES	~4.1 FTE	● Board channel contribution small against the measurement margin	● Socket life against test volume
Test hardware engineer			● Board channel loss against measurement margin
Owns the load board			
SI/PI engineer		DEPENDS ON	
Channel and power delivery on the board		TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test	
Test engineer		TEST-02 ATE platform selection and tester time booking	
Tester interface and bring-up		PKGD-04 Substrate stack-up, escape routing and package DRC	
Package engineer		ARCH-08 Chip-level floorplan intent, pin and bump budget	
Socket and package dimensional fit			
Socket vendor			
Contact design and insertion life			

TEST-05

Wafer sort test program development

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	30.0 M/M	~8.1 FTE	8	yes	Test engineer
w16–w32 in stage	planned	while it runs	3 in parallel	no float	Assembly liaison approves

PURPOSE

Write the **wafer sort test program** — the screen that decides which die are worth assembling into a package.

On a 2.5D part, sort is the most economically consequential test in the flow. Everything it lets through carries an interposer, two HBM stacks and an assembly with it, so its coverage is measured against package cost rather than against die cost.

WORK FLOW

8 STEPS · 2 LANES

w16	w20	w24	w28	w32
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Sort flow and test list definition	2w	main
2	DC and continuity test development	3.5w	main
3	Parametric and process monitor test development	2.5w	parallel
4	Scan and structural test integration	3.5w	main
5	Binning and wafer map output	2.5w	parallel
6	Functional and memory BIST content	3.5w	main
7	Test limits and guard-band setting	2.5w	parallel
8	Program integration and debug on the tester	3.5w	main

Step 7 is where the program's economics are set. Limits too tight throw away good die; too loose and packages are built on bad ones. The guard bands should come from measurement uncertainty and package cost, not from convention.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test plan and coverage matrix from TEST-01
- ATE patterns from TEST-08
- Probe card from TEST-03
- Memory BIST from DFT-06
- Scan architecture from DFT-03

PRODUCES

- Sort flow and test list
- DC and continuity tests
- Parametric and monitor tests
- Integrated scan and structural content
- Binning and wafer map output
- Functional and BIST content
- Test limits and guard bands

- Debugged sort program

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D3	PRODUCES	Wafer sort and final test programs, release-tagged. The sort half of the deliverable.
TEST-D1	FEEDS	Test plan and test coverage matrix. What the program actually achieves updates what the plan claimed.

RISKS AND WATCH-OUTS

- ◆ **Guard bands set by convention.** On a part where an escape costs an assembled package, the limits should be derived from that cost.
- ◆ **Coverage below what the package economics need.** Assembly yield then absorbs the difference, at many times the price.
- ◆ **Program debugged only in simulation.** Tester behaviour differs from the model, and the difference is found on the first wafer.
- ◆ **Wafer map output not matching the assembly requirement.** KGD selection depends on the map format, and a mismatch stops assembly.
- ◆ **Test time growing without check.** Sort time multiplies across every wafer ever run, and content accretes silently.

EFFORT SPLIT	30 M/M	ENTRY CRITERIA	FEEDS INTO
Scan and structural integration	6.5	● Test plan from TEST-01 ● ATE patterns available from TEST-08 ● Probe card qualified in TEST-03	ASSY-01 Known-good-die sort and selection
Functional and BIST	6.0		FAB-08 Wafer acceptance test and lot disposition
DC and continuity	5.5		MP-06 Production test program release and guard-band validation
Integration and debug	5.0		TEST-09 Test time and cost optimisation — parallelism, site count
Parametric development	4.0		
Binning and limits	3.0		
Total	30.0		
WHO IT TAKES	~8.1 FTE	EXIT CRITERIA	MEASURED BY
Test engineer Owns the sort program		● Guard bands derived from package cost and measurement uncertainty ● Coverage sufficient for the assembly economics ● Wafer map format agreed with assembly	● Sort coverage against the DPPM target ● Sort test time against budget ● Assembly failures attributable to sort escapes
DFT engineer Scan and BIST content integration		DEPENDS ON TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test	
Product engineering Limits, guard bands and binning		TEST-03 Probe card design, fabrication and qualification	
Yield engineer Parametric monitors and map output		TEST-08 DFT pattern porting to ATE format and pattern debug	
Assembly liaison Wafer map format and KGD criteria		DFT-03 MBIST / BIRA / BISR architecture for embedded memories	
		DFT-06 Scan insertion and DFT DRC during synthesis	

TEST-06

Final / package test program development

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
16w	28.0 M/M	~7.6 FTE	8	no	Test engineer
w20–w36 in stage	planned	while it runs	3 in parallel	has float	Yield engineer approves

PURPOSE

Write the **final test program** that runs on the assembled package — at speed, at temperature, across every interface the product ships with.

Final test is the last screen before a customer sees the part. It exercises what sort could not: the assembled interfaces, the HBM stacks, the full power envelope, and the at-speed behaviour of a part that is now mechanically complete.

WORK FLOW

8 STEPS · 2 LANES

w20	w24	w28	w32	w36
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Final test flow definition	2w	main
2	Package-level DC and continuity content	3.5w	main
3	Power and thermal test content	2.5w	parallel
4	At-speed functional content	3.5w	main
5	Binning and grading content	2.5w	parallel
6	Interface and HBM test content	3.5w	main
7	Final test limits and guard bands	2.5w	parallel
8	Program integration and debug	3.5w	main

Step 6 is where the 2.5D part differs most from a monolithic one. The HBM interfaces exist only after assembly, they cannot be tested at sort, and they are the most likely thing to be wrong about an assembled unit.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test plan from TEST-01
- Load board from TEST-04
- ATE patterns from TEST-08
- HBM interface specification from ARCH-03
- Binning strategy from TEST-01

PRODUCES

- Final test flow
- Package-level DC and continuity content
- Power and thermal test content
- At-speed functional content
- Binning and grading content
- Interface and HBM test content
- Final test limits and guard bands

- Debugged final test program

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D3 **PRODUCES** **Wafer sort and final test programs, release-tagged.** The final-test half of the deliverable.

TEST-D6 **FEEDS** **Test time and test cost model.** Final test time is the larger half of the per-unit test cost.

RISKS AND WATCH-OUTS

- ◆ **HBM interface content insufficient.** The interfaces only exist after assembly, and a marginal one shipped is a field failure.
- ◆ **At-speed content not run at temperature.** Timing marginality appears at the hot corner, and testing only at room temperature misses it.
- ◆ **Binning content added late.** Grading is how the product line is built, and retrofitting it changes the flow and its cost.
- ◆ **Test time not tracked against the cost model.** Final test dominates per-unit test cost and grows quietly.
- ◆ **Program debugged without real assembled units.** The first units are precious, and the program has to be ready when they exist.

EFFORT SPLIT	28 M/M	ENTRY CRITERIA	FEEDS INTO
At-speed functional	6.0	● Test plan from TEST-01	MP-06 Production test program release and guard-band validation
Interface and HBM content	6.0	● Load board released from TEST-04	
DC and continuity	5.0	● ATE patterns available from TEST-08	TEST-09 Test time and cost optimisation — parallelism, site count
Integration and debug	4.5		TEST-10 Correlation between ATE, bench and system results
Power and thermal	3.5		
Binning and limits	3.0		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Total	28.0		
WHO IT TAKES	~7.6 FTE	EXIT CRITERIA	MEASURED BY
Test engineer Owns the final test program		● HBM interfaces exercised at their operating rate ● At-speed content run across the temperature range ● Test time inside the cost model's allowance	● Final test coverage against the DPPM target ● Final test time against the cost model ● Escapes found at system level
Product engineering Binning, grading and limits		DEPENDS ON	
Interface engineer HBM and high-speed content		TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test	
Thermal engineer Temperature-dependent content		TEST-04 Load board / DUT board design, fabrication and bring-up	
Yield engineer Failure classification from final test		TEST-08 DFT pattern porting to ATE format and pattern debug	
		ARCH-03 Dataflow and memory hierarchy definition (on-die SRAM capacity, HBM channels)	

TEST-07

Characterisation test content —
shmoo, Vmin/Fmax, PVT sweeps

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	20.0 M/M	~7.2 FTE	8	no	Characterisation engineer
w24–w36 in stage	planned	while it runs	3 in parallel	has float	Quality engineer approves

PURPOSE

Build the **characterisation content** — shmoos, corners, parametric sweeps, datasheet parameter measurement — that turns silicon into a specified product.

Production test asks whether a part passes. Characterisation asks what the part actually does, across voltage, frequency and temperature, over a statistically meaningful sample. Every number in the datasheet comes from here, and so do the guard bands production test uses.

WORK FLOW

8 STEPS · 2 LANES

w24	w27	w30	w33	w36
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Characterisation requirement from qualification and marketing	1.5w	main
2	Shmoo and corner content	2.5w	main
3	Temperature and voltage sweep automation	2w	parallel
4	Parametric characterisation content	2.5w	main
5	Statistical sample plan	2w	parallel
6	Datasheet parameter test content	2.5w	main
7	Data analysis and reporting pipeline	1.5w	parallel
8	Content integration and correlation setup	3w	main

Step 5 decides whether the results mean anything. A datasheet number from three units is an anecdote; the sample size follows from the distribution being characterised and the confidence the specification claims.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Datasheet parameter list from DEF-04
- Qualification requirements from MP-01
- Test plan from TEST-01
- Load board and correlation fixture from TEST-04
- Lab instrumentation from EVB-09

PRODUCES

- Characterisation requirement
- Shmoo and corner content
- Sweep automation
- Parametric characterisation content
- Statistical sample plan

- Datasheet parameter content
- Data analysis and reporting pipeline
- Integrated characterisation suite

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D4	PRODUCES	Characterisation test suite. This activity is the deliverable, and every datasheet number traces to it.
TEST-D6	INFORMS	Test time and test cost model. Characterisation guard bands set how much production test time is spent on margin.

RISKS AND WATCH-OUTS

- ◆ **Sample size too small for the claim.** A datasheet parameter from a handful of units is a number without confidence behind it.
- ◆ **Corners not spanning the qualification range.** Qualification then stresses conditions characterisation never measured.
- ◆ **Sweeps not automated.** Manual shmoo consumes engineer-weeks per corner, and the corners get quietly reduced.
- ◆ **Datasheet parameters not traceable to a measurement.** A customer question about a number then has no answer.
- ◆ **Analysis pipeline built after the data.** Data is collected in a format nobody can analyse, and the campaign is re-run.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Shmoo and corner content	4.5	● Datasheet parameter list from DEF-04	BU-07 Shmoo across voltage, frequency and temperature
Datasheet parameters	4.0	● Qualification conditions from MP-01	BU-08 Performance validation against the architecture model
Parametric content	3.5	● Correlation fixture from TEST-04	MP-08 Process corner and split-lot validation
Integration and correlation	3.0	EXIT CRITERIA	MP-12 Documentation release — datasheet, user guide, errata, application notes
Sweep automation	2.5	● Sample size supporting the datasheet's confidence claim	EVB-01 Validation platform requirements and topology definition
Sample plan and reporting	2.5	● Corners spanning the qualification range	
Total	20.0	● Every datasheet number traceable to a measurement	
WHO IT TAKES	~7.2 FTE	DEPENDS ON	MEASURED BY
Characterisation engineer Owns the suite		TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test	● Sample size against the confidence claim
Product engineering Datasheet parameters and guard bands		TEST-04 Load board / DUT board design, fabrication and bring-up	● Corner coverage against qualification range
Statistician or yield engineer Sample plan and confidence		DEF-04 Die size, yield and product cost model (wafer, mask, package, test, ASP)	● Datasheet parameters with traceable measurements
Automation engineer Sweep automation and data pipeline		MP-01 Qualification plan definition against JEDEC / AEC standards	
Quality engineer Alignment with qualification conditions		EVB-09 Lab instrumentation reservation and test rack build	

TEST-08

DFT pattern porting to ATE format and pattern debug

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	16.0 M/M	~7.0 FTE	8	yes	Test engineer
w22–w32 in stage	planned	while it runs	3 in parallel	no float	Product engineering approves

PURPOSE

Port the **DFT patterns to the tester's format** and debug them there — the step between a pattern that simulates and a pattern that runs.

ATPG produces patterns against an idealised model. The tester has finite memory, real timing resolution, protocol constraints and a specific pin electronics behaviour. Every one of those turns a pattern that passes in simulation into one that fails on hardware, and finding out which is the work.

WORK FLOW

8 STEPS · 2 LANES

w22	w24	w26	w28	w30	w32
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	ATPG pattern handoff from DFT	1.5w	main
2	Pattern translation to the tester format	2w	main
3	Pattern volume and memory-depth reduction	1.5w	parallel
4	Timing and protocol mapping	2.5w	main
5	Compression and streaming setup	1.5w	parallel
6	Pattern simulation against the tester model	2w	main
7	Pattern regression and version control	1.5w	parallel
8	Pattern debug on silicon or emulation	2w	main

Step 3 is a hard constraint, not an optimisation. Tester pattern memory is finite, and a pattern set that exceeds it simply cannot be loaded — which is discovered at translation, not at ATPG.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- ATPG patterns from DFT-03
- Compression architecture from DFT-04
- ATE platform and memory depth from TEST-02
- Timing constraints from SO-03
- Tester timing model

PRODUCES

- Pattern handoff record
- Translated tester-format patterns
- Volume and memory reduction
- Timing and protocol mapping
- Compression and streaming setup
- Tester-model simulation results

- Pattern regression under version control
- Debugged ATE-ready pattern set

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D5

PRODUCES

ATE-ready pattern set with debug log. This activity is the deliverable.

TEST-D3

FEEDS

Wafer sort and final test programs. Both programs run these patterns as their structural content.

RISKS AND WATCH-OUTS

- ◆ **Pattern volume exceeding tester memory.** The set cannot be loaded at all, and reduction after ATPG loses coverage.
- ◆ **Timing not mappable to the tester's resolution.** At-speed patterns then run at the wrong speed, silently.
- ◆ **Patterns not simulated against the tester model.** Failures are then debugged on silicon at tester rates rather than in simulation.
- ◆ **Pattern set not versioned.** A yield excursion cannot be correlated to a pattern change nobody recorded.
- ◆ **Debug deferred until first silicon.** Pattern problems and silicon problems then arrive together and cannot be separated.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	FEEDS INTO
Timing and protocol mapping	4.0	● ATPG patterns released from DFT-03	TEST-05 Wafer sort test program development
Pattern debug	3.5	● ATE platform and memory depth known from TEST-02	TEST-06 Final / package test program development
Translation	3.0	● Tester timing model available	MP-06 Production test program release and guard-band validation
Tester-model simulation	2.5		
Volume reduction	1.5	EXIT CRITERIA	
Regression and version control	1.5	● Pattern set fitting tester memory without coverage loss	MEASURED BY
Total	16.0	● Timing mapped within the tester's resolution	● Coverage retained after volume reduction
WHO IT TAKES	~7.0 FTE	● Patterns simulated against the tester model before silicon	● Patterns passing tester-model simulation
Test engineer		DEPENDS ON	● Pattern failures found on silicon that simulation missed
Owns pattern porting and debug		DFT-03 MBIST / BIRA / BISR architecture for embedded memories	
DFT engineer		DFT-04 Boundary scan and IJTAG (1149.1 / 1687), TAP and debug access architecture	
ATPG handoff and coverage preservation		TEST-02 ATE platform selection and tester time booking	
ATE applications engineer		S0-03 Full-chip DRC, LVS, antenna and density verification	
Tester format, timing and memory			
CAD engineer			
Translation flow and regression			
Product engineering			
Coverage after reduction			

TEST-09

Test time and cost optimisation — parallelism, site count

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	12.0 M/M	~5.8 FTE	8	no	Test engineer
w8–w17 in stage	planned	while it runs	3 in parallel	has float	Operations planner approves

PURPOSE

Reduce **test time and test cost per unit** — an activity that starts early, because content is far cheaper to not write than to remove.

Test time is paid on every unit for the product's whole life. A second saved at final test on a million units is real money, and the savings available at plan time are much larger than the ones available after the program is written and qualified.

WORK FLOW

8 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Test time baseline measurement	1.5w	main
2	Test time contributor analysis	2w	main
3	Redundant coverage identification	1.5w	parallel
4	Content reduction and reordering	2w	main
5	Adaptive and skip-on-fail strategy	1.5w	parallel
6	Parallelism and multi-site assessment	1.8w	main
7	Tester resource cost per unit	1.3w	parallel
8	Cost model update and release	1.8w	main

Step 4's reordering is often the largest single win and costs nothing. Running the tests that fail most first means failing parts leave the tester early, and the average test time drops without any coverage change.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test plan and flow from TEST-01
- Sort content from TEST-05
- Final content from TEST-06
- Tester cost per hour from TEST-02
- Expected yield and failure distribution

PRODUCES

- Test time baseline
- Contributor analysis by test block
- Redundant coverage findings
- Content reduction and reordering
- Adaptive and skip-on-fail strategy
- Parallelism and multi-site assessment
- Cost per unit model

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D6

PRODUCES

Test time and test cost model. This activity is the deliverable.

TEST-D1

INFORMS

Test plan and test coverage matrix. Reduction decisions change what the plan promises, and the matrix has to follow.

RISKS AND WATCH-OUTS

- ◆ **Optimisation started after qualification.** Any content change then requires re-qualification, which is why this starts early.
- ◆ **Coverage traded away silently.** Time reduction that loses coverage has to be an explicit decision against the DPPM target.
- ◆ **Reordering not done.** It is free and it is the largest single lever, and it is routinely overlooked.
- ◆ **Multi-site assumed without pin count.** The tester configuration from TEST-02 has to support it, and retrofitting it does not work.
- ◆ **Cost model without a yield assumption.** Cost per good unit depends on yield, and a model without it understates the cost.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	DEPENDS ON
Contributor analysis	3.0	● Test plan and flow available from TEST-01 ● Content estimates or early programs available ● Tester cost per hour known	TEST-01 Test plan and coverage strategy — wafer sort, final test, system-level test
Content reduction	2.5		TEST-02 ATE platform selection and tester time booking
Parallelism assessment	2.0		TEST-05 Wafer sort test program development
Cost model	1.8		TEST-06 Final / package test program development
Redundancy identification	1.5		
Adaptive strategy	1.3	EXIT CRITERIA	FEEDS INTO
Total	12.0	● Optimisation done before qualification freezes the content ● Coverage trades made explicitly against the DPPM target ● Cost model including a yield assumption	MP-07 Test time reduction and multi-site conversion
WHO IT TAKES	~5.8 FTE		MP-06 Production test program release and guard-band validation
Test engineer			DEF-06 Competitive benchmarking and product positioning
Owns time and cost optimisation			
Product engineering			
Coverage trade decisions			MEASURED BY
Cost analyst			● Test time against the plan's budget ● Cost per good unit against target ● Coverage retained after reduction
Cost per unit model			
Yield engineer			
Failure distribution for reordering			
Operations planner			
Multi-site and capacity implications			

TEST-10

Correlation between ATE, bench and system results

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	10.0 M/M	~5.4 FTE	8	no	Product engineering
w32–w40 in stage	planned	while it runs	3 in parallel	has float	Quality engineer approves

PURPOSE

Correlate **ATE, bench and system results** so that a part passing on the tester behaves the same way in a customer's system.

Three measurement environments produce three numbers for the same parameter, and they never agree exactly. What matters is whether the disagreement is understood: an unexplained gap between tester and system is either an escape path or an over-screen, and both are expensive.

WORK FLOW

8 STEPS · 2 LANES

w32	w34	w36	w38	w40
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MAIN SEQUENCE

1	2	4	6	8
---	---	---	---	---

IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Correlation plan and unit selection	1w	main
2	ATE-to-bench correlation	2w	main
3	Measurement uncertainty budget	1.5w	parallel
4	ATE-to-system correlation	2w	main
5	Golden unit designation and retention	1.5w	parallel
6	Discrepancy analysis and root cause	1.5w	main
7	Correlation regression setup	1w	parallel
8	Guard-band adjustment and release	1.5w	main

Step 5 is what makes correlation repeatable. A retained set of golden units re-measured after any change is how tester drift, load board degradation and program regressions are caught before they become yield excursions.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Final test results from TEST-06
- Bench measurements from BU-08
- System-level results from BU-05
- Measurement uncertainty from TEST-07
- Load board and correlation fixture from TEST-04

PRODUCES

- Correlation plan and unit selection
- ATE-to-bench correlation results
- Measurement uncertainty budget
- ATE-to-system correlation results
- Designated golden units
- Discrepancy analysis and root causes

- Correlation regression
- Adjusted guard bands

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D4	FEEDS	Characterisation test suite. Correlation validates that characterisation numbers hold outside the tester.
TEST-D3	FEEDS	Wafer sort and final test programs. Guard-band adjustments from correlation go back into the programs.

RISKS AND WATCH-OUTS

- ◆ **Discrepancy accepted without root cause.** The gap is either an escape or an over-screen, and 'they just differ' resolves neither.
- ◆ **Correlation on too few units.** Two units that agree prove very little about the population.
- ◆ **No golden units retained.** Drift in the tester, the load board or the program then has nothing to be measured against.
- ◆ **System correlation skipped.** ATE-to-bench is the easy half; ATE-to-system is the one the customer experiences.
- ◆ **Guard bands not updated.** The correlation produced the information and nothing acted on it.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
ATE-to-bench correlation	2.5	● Final test program running from TEST-06	MP-06 Production test program release and guard-band validation
ATE-to-system correlation	2.3	● Bench and system results available from bring-up	MP-08 Process corner and split-lot validation
Discrepancy analysis	2.0	● Correlation fixture available from TEST-04	MP-12 Documentation release — datasheet, user guide, errata, application notes
Uncertainty budget	1.5		
Guard-band adjustment	1.3		
Golden unit retention	0.5		
Total	10.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~5.4 FTE	● Every material discrepancy root-caused, not accepted ● Golden units retained and re-measurable ● Guard bands updated from what correlation found	● Discrepancies root-caused against found ● Correlation sample size ● Guard-band changes traced to correlation findings
Product engineering Owns correlation and guard bands		DEPENDS ON	
Test engineer ATE-side measurement and regression		TEST-04 Load board / DUT board design, fabrication and bring-up	
Characterisation engineer Bench measurement and uncertainty		TEST-06 Final / package test program development	
Validation engineer System-level measurement		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps	
Quality engineer Escape and over-screen implications		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die	
		BU-08 Performance validation against the architecture model	

TEST-11

Test data infrastructure — STDF collection and yield database

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	10.0 M/M	~5.4 FTE	8	no	Test data engineer
w26–w34 in stage	planned	while it runs	3 in parallel	has float	IT infrastructure approves

PURPOSE

Build the **test data infrastructure** — STDF collection, parsing, a yield database and the analysis on top of it — so that test results become information rather than files.

Every yield decision the program makes for the rest of its life is made from this data. Without a pipeline, the data exists as per-lot files nobody can query, and questions that should take an hour take a week or go unasked.

WORK FLOW

8 STEPS · 2 LANES

w26	w28	w30	w32	w34
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MAIN SEQUENCE

1	2	4	6	8
---	---	---	---	---

IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Data requirement from yield, quality and characterisation	1w	main
2	STDF collection and parsing pipeline	2w	main
3	Traceability keys — lot, wafer, unit, assembly	1.5w	parallel
4	Yield database and schema	2w	main
5	Retention and archival policy	1.5w	parallel
6	Analysis and dashboard layer	1.5w	main
7	Alerting on yield excursions	1w	parallel
8	Deployment and access provisioning	1.5w	main

Step 3 is the piece that is hardest to add later. A unit's test result is only useful if it can be joined to its wafer position, its assembly lot and its bring-up history, and those keys have to be carried from the beginning.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Sort and final test output formats from TEST-05 and TEST-06
- Yield analysis requirements from MP-05
- Characterisation data needs from TEST-07
- Traceability scheme from ASSY-01 and ASSY-09
- Data infrastructure and storage

PRODUCES

- Data requirements
- STDF collection and parsing pipeline
- Traceability key scheme
- Yield database and schema
- Retention and archival policy
- Analysis and dashboard layer

- Excursion alerting
- Deployed and accessible infrastructure

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

TEST-D7	PRODUCES	Test data infrastructure and yield database. This activity is the deliverable.
TEST-D1	INFORMS	Test plan and test coverage matrix. What the data shows about actual failure modes updates the plan's assumptions.

RISKS AND WATCH-OUTS

- ◆ **Traceability keys added retroactively.** Historical data then cannot be joined, and the most valuable early lots become unanalysable.
- ◆ **Pipeline built after the first lots run.** The first silicon's data is the most informative the program will ever have, and it is the data most often lost.
- ◆ **Retention policy absent.** Field returns arrive years later, and the test record for that unit has to still exist.
- ◆ **Dashboards without an alerting layer.** A yield excursion is noticed at the weekly review rather than the same day.
- ◆ **Schema not covering characterisation.** Characterisation data has a different shape, and a sort-only schema forces it into a separate silo.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	DEPENDS ON
Collection pipeline	2.5	● Test output formats known from TEST-05 and TEST-06	TEST-05 Wafer sort test program development
Yield database	2.3	● Yield analysis requirements from MP-05	TEST-06 Final / package test program development
Dashboard layer	2.0	● Traceability scheme agreed with assembly	TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps
Traceability keys	1.5		
Deployment	1.0		
Retention policy	0.8		ASSY-09 Unit build for bring-up, qualification and customer samples
Total	10.0	EXIT CRITERIA ● Traceability keys carried from the first lot ● Pipeline running before first silicon data arrives ● Retention policy covering the field-return horizon	FEEDS INTO
WHO IT TAKES	~5.4 FTE		MP-05 Yield learning, failure pareto and defect analysis
Test data engineer Owns the pipeline and database			MP-09 Production readiness review and change control (PCN) setup
Yield engineer Analysis requirements and excursion definitions			FAB-09 Hot-lot management and WIP tracking
Quality engineer Retention policy and traceability			
Test engineer Output formats and collection			MEASURED BY ● First-silicon data captured against produced ● Traceability join completeness ● Time from excursion to alert
IT infrastructure Storage, access and deployment			

VALIDATE & RAMP

22

Silicon Bring-up

BU

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w102 – w120	18 _w	12	153 _{M/M}	6 of 12
ENGINEERING ACTIVITIES				KEY DELIVERABLES
BU-01	Sample receipt, incoming inspection and board mounting		1.5w 2 M/M	BU-D1 Bring-up report with per-milestone health status
BU-02	Power-on, power sequencing and basic health check		3w 8 M/M	BU-D2 Characterisation data set — V/F/T shmoo and power measurements
BU-03	Reset, clocking and PLL lock validation		3w 8 M/M	BU-D3 Interface compliance results with training margins
BU-04	Boot, firmware load and functional smoke test		4w 12 M/M	BU-D4 Errata list with workarounds
BU-05	Interface bring-up — PCIe/CXL link training, HBM training, die-to-die		8w 26 M/M	BU-D5 Failure analysis reports
BU-06	Memory subsystem and bandwidth validation		6w 16 M/M	BU-D6 Respin versus metal-fix decision record
BU-07	Shmoo across voltage, frequency and temperature		7w 16 M/M	BU-D7 Customer sample release package
BU-08	Performance validation against the architecture model		6w 16 M/M	
BU-09	Silicon debug of anomalies and failure analysis requests		10w 28 M/M	
BU-10	Errata capture, workaround definition and documentation		7w 10 M/M	
BU-11	Respin versus metal-fix decision analysis — ECO scope, cost, schedule		4w 5 M/M	
BU-12	Customer sample readiness and release package		5w 6 M/M	

22 BU · SILICON BRING-UP

BU-01

Sample receipt, incoming inspection and board mounting

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
1.5w	2.0 M/M	~5.8 FTE	4	yes	Bring-up lead
w0-w1.5 in stage	planned	while it runs	1 in parallel	no float	Program manager approves

PURPOSE

Receive the first units, **check them and mount them** — a short activity whose only job is to make sure the most valuable objects the program has ever produced are not damaged before they are powered.

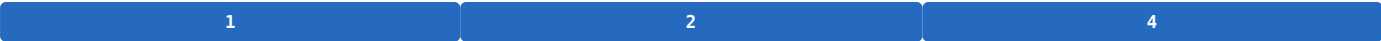
There are very few first-silicon units and each represents the whole program's schedule. A unit destroyed by a mounting error or an unnoticed shipping defect is not replaceable inside any useful timeframe.

WORK FLOW

4 STEPS · 2 LANES

w0	w1	w1.5
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Sample receipt and incoming inspection	0.5w	main
2	Board mounting and mechanical fit	0.5w	main
3	Unit tracking register and allocation record	0.5w	parallel
4	Pre-power checklist against the board's known issues	0.5w	main

Step 4 is the reason board bring-up happened first. The checklist is only meaningful because `EVB-07` already established which board behaviours are normal and which are faults.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Allocated units from ASSY-09
- Screened unit record from ASSY-08
- Brought-up boards from EVB-07
- Board known-issue list from EVB-07
- ESD and handling procedures

PRODUCES

- Incoming inspection record
- Units mounted on boards
- Unit tracking and allocation register
- Pre-power checklist completion

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D1

FEEDS

Bring-up report with per-milestone health status. Receipt and mounting are the report's first entries.

BU-05

INFORMS

Failure analysis reports. A unit's history starts here, and an FA needs to know how it was handled.

RISKS AND WATCH-OUTS

- ◆ **Handling damage on an irreplaceable unit.** ESD or mechanical damage at mounting costs a unit the program cannot replace.
- ◆ **Mounting onto an unvalidated board.** A board fault then destroys the part, and EVB-07 exists to prevent exactly this.
- ◆ **No unit tracking.** Every subsequent result has to be attributable to a specific unit and its history.
- ◆ **Incoming inspection skipped in the rush.** Shipping damage is real and is much cheaper to find before power than after.
- ◆ **All units mounted at once.** A systematic board or handling fault then takes the whole fleet rather than one unit.

EFFORT SPLIT	2 M/M	ENTRY CRITERIA	DEPENDS ON
Board mounting	0.8	- Units allocated and screened from ASSY-08 and ASSY-09 - Boards brought up and accepted from EVB-07 - ESD-controlled bench available	ASSY-08 Package-level open / short and continuity test
Incoming inspection	0.5		ASSY-09 Unit build for bring-up, qualification and customer samples
Pre-power checklist	0.5		EVB-07 Board bring-up with socketed or dummy parts
Unit tracking	0.3	EXIT CRITERIA - Every unit inspected before mounting - Units mounted only on validated boards - Unit register started and maintained	
Total	2.0		
WHO IT TAKES	~5.8 FTE		FEEDS INTO
Bring-up lead Owns the sequence and the unit register			BU-02 Power-on, power sequencing and basic health check
Hardware engineer Mounting and mechanical fit			MEASURED BY - Units damaged during receipt and mounting - Units traceable in the register - Boards validated before mounting
Quality engineer Incoming inspection			
Lab technician ESD-controlled handling			
Program manager Unit allocation and risk spreading			

22 BU · SILICON BRING-UP

BU-02

Power-on, power sequencing and basic health check

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	8.0 M/M	~11.6 FTE	5	yes	Bring-up lead
w1–w4 in stage	planned	while it runs	1 in parallel	no float	Program manager approves

PURPOSE

Apply power for the first time — **sequenced, current-limited and instrumented** — and establish that the part is alive.

First power-on is the moment the whole design becomes physical. It is also the moment a shorted rail or a sequencing error can destroy a unit in milliseconds, which is why it is done with limits, with measurement, and with the ability to stop.

WORK FLOW

5 STEPS · 2 LANES

w1	w2	w3	w4
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Power sequencing verification against the specification	0.5w	main
2	First power application with current limits	1w	main
3	Anomaly triage on the first application	0.8w	parallel
4	Rail voltage and current profiling	1w	main
5	Health check and go/no-go for the next step	0.5w	main

Step 2's current limits are what turn a fatal event into a diagnostic one. A rail that draws far more than expected trips the limit and the part survives to be investigated; without the limit it is destroyed and the information goes with it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Mounted units from BU-01
- Power subsystem and protection from EVB-05
- Power sequencing specification from ARCH-08
- Expected rail currents from PD-08
- Telemetry from EVB-05

PRODUCES

- Power sequencing verification
- First power application record
- Anomaly triage findings
- Rail voltage and current profiles
- Health check and go/no-go decision

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D1	FEEDS	Bring-up report with per-milestone health status. Power-on is the first milestone and its result sets the program's tone.
BU-D2	FEEDS	Characterisation data set. The first rail current measurements are the earliest real power data the program has.

RISKS AND WATCH-OUTS

- ◆ **Power applied without current limits.** A short then destroys the unit instead of revealing itself.
- ◆ **Sequencing not verified against the specification.** A wrong power-up order damages the part and the fault is the platform's, not the silicon's.
- ◆ **Rail currents not profiled.** The first real power data is available at exactly this moment and is easy to not capture.
- ◆ **Multiple units powered before the first is understood.** A systematic fault then takes several units instead of one.
- ◆ **Anomalies pushed past to reach the next milestone.** Something unexplained at power-on rarely stays harmless.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	DEPENDS ON
First power application	2.5	● Units mounted from BU-01	BU-01 Sample receipt, incoming inspection and board mounting
Rail profiling	2.0	● Power subsystem validated in EVB-05	EVB-05 Power delivery, VRM and telemetry design and bring-up
Sequencing verification	1.5	● Expected rail currents from PD-08	ARCH-08 Chip-level floorplan intent, pin and bump budget
Anomaly triage	1.3	EXIT CRITERIA	PD-08 Detailed routing and DRC convergence
Health check	0.8	● Current limits in place before first application	
Total	8.0	● Rail currents profiled against expectation	FEEDS INTO
WHO IT TAKES	~11.6 FTE	● No unexplained anomaly carried into the next milestone	BU-03 Reset, clocking and PLL lock validation
Bring-up lead			BU-09 Silicon debug of anomalies and failure analysis requests
Power engineer			MP-05 Yield learning, failure pareto and defect analysis
Design engineer			MEASURED BY
Validation engineer			● Units surviving first power-on
Program manager			● Rail currents against prediction
			● Anomalies resolved before proceeding

22 BU · SILICON BRING-UP

BU-03

Reset, clocking and PLL lock validation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
3w	8.0 M/M	~11.6 FTE	5	no	Bring-up engineer
w2–w5 in stage	planned	while it runs	1 in parallel	has float	Validation engineer approves

PURPOSE

Validate **reset, clocking and PLL lock** — the infrastructure everything else depends on, and the layer where a fault masquerades as a fault somewhere else.

A PLL that locks unreliably or a reset domain released in the wrong order produces symptoms all over the chip: boot failures, interface training failures, sporadic functional errors. Clearing this layer first means the failures found later are real.

WORK FLOW

5 STEPS · 2 LANES

w2	w3	w4	w5
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MAIN SEQUENCE

1	2	4	5
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IN PARALLEL

3

#	STEP	TAT	LANE
1	Reference clock and input validation	0.5w	main
2	PLL lock across the frequency range	1w	main
3	Reset sequence and domain release verification	0.8w	parallel
4	Clock tree and domain verification	1w	main
5	Jitter and stability measurement	0.5w	main

Step 5's jitter measurement is easy to skip because the part appears to work without it. It matters because interface training margin depends on clock quality, and a jitter problem found here explains a class of failures that would otherwise be chased in the SerDes.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Powered units from BU-02
- PLL specifications from AMS-04
- Clock architecture from ARCH-05
- Reset architecture from RTL-04
- Clock instrumentation from EVB-09

PRODUCES

- Reference clock validation
- PLL lock results across the range
- Reset sequence and domain release verification
- Clock tree and domain verification
- Jitter and stability measurements

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D1

FEEDS

Bring-up report with per-milestone health status. Clocking and reset form the second milestone.

BU-02

FEEDS

Characterisation data set. PLL lock range and jitter are datasheet-relevant measurements.

RISKS AND WATCH-OUTS

- ◆ **PLL lock validated only at nominal.** The lock range matters because shmoo and characterisation will operate outside nominal.
- ◆ **Reset domain release order not verified.** The resulting failures appear anywhere on the chip and are attributed to the wrong block.
- ◆ **Jitter unmeasured.** Interface training margin depends on it, and a jitter problem is then debugged inside the SerDes.
- ◆ **Clock tree assumed correct because the part boots.** A misconfigured divider is compatible with booting and incompatible with performance.
- ◆ **Clock quality measured through the board without accounting for it.** The board contributes jitter and its contribution has to be separated.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
PLL lock validation	2.5	- Units powered and healthy from BU-02 - PLL specification from AMS-04 - Clock instrumentation available from EVB-09	BU-04 Boot, firmware load and functional smoke test
Clock tree verification	2.0		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Reset sequence verification	1.5		BU-09 Silicon debug of anomalies and failure analysis requests
Jitter measurement	1.3		
Reference validation	0.8		
Total	8.0	EXIT CRITERIA	MEASURED BY
WHO IT TAKES	~11.6 FTE	- PLL lock verified across the full operating range - Reset domain release order confirmed against the architecture - Jitter measured with the board's contribution separated	- PLL lock range achieved against specification - Jitter against the interface budget - Reset-related failures found later
Bring-up engineer			
Owens clocking and reset validation			
AMS engineer			
PLL behaviour and lock characteristics			
Design engineer		DEPENDS ON	
Clock tree and reset architecture		BU-02	Power-on, power sequencing and basic health check
Characterisation engineer		AMS-04	LDO, bandgap and power-management cell design
Jitter measurement		ARCH-05	Power domain, clock domain and DVFS architecture; UPF intent draft
Validation engineer		RTL-04	Chip-level integration and top assembly
Instrumentation and data capture			

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BU-04

Boot, firmware load and functional smoke test

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	12.0 M/M	~13.0 FTE	6	yes	Bring-up lead
w4–w8 in stage	planned	while it runs	2 in parallel	no float	Program manager approves

PURPOSE

Boot the part — ROM, firmware load, register access, first functional response — and confirm that the design does what it was written to do.

First boot is the milestone that turns a powered die into a working processor. It is also the point at which the design, the firmware, the board and the host software are all exercised together for the first time, which is why it is the milestone most likely to expose an integration assumption nobody wrote down.

WORK FLOW

6 STEPS · 2 LANES

w4	w5	w6	w7	w8
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Boot ROM and initialisation sequence execution	0.8w	main
2	Firmware load and handoff	1.3w	main
3	Boot failure triage	1w	parallel
4	Register access and device identity read	1.3w	main
5	First bring-up report and status broadcast	1w	parallel
6	Functional smoke test	0.8w	main

Step 4's first successful register read is the moment the program knows the silicon is fundamentally sound. Everything before it is infrastructure and everything after it is degrees of correctness.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Clocking-validated units from BU-03
- Host enablement from EVB-10
- Boot ROM and firmware from the firmware team
- Register map from RTL-06
- Debug access from EVB-06

PRODUCES

- Boot ROM execution record
- Firmware load and handoff results
- Boot failure triage findings
- Register access and identity read
- First bring-up report
- Functional smoke test results

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D1

PRODUCES

Bring-up report with per-milestone health status. First boot is the report's central milestone and where it starts being published widely.

BU-D4

FEEDS

Errata list with workarounds. Boot-time deviations are the earliest errata the program records.

RISKS AND WATCH-OUTS

- ◆ **Boot failure with no state visible.** Without debug access the failure is a symptom with no information, and debug access comes from EVB-06.
- ◆ **Firmware and hardware assumptions differing.** The handoff is where two teams' expectations meet, and nobody wrote them down.
- ◆ **Register map drift.** The tool reads addresses the silicon does not implement and reports plausible garbage.
- ◆ **Smoke test too shallow.** A part that boots and does nothing else has passed a milestone without demonstrating much.
- ◆ **Status not broadcast.** First boot is the news the whole program is waiting for, and silence is read as failure.

EFFORT SPLIT	12 M/M	ENTRY CRITERIA	FEEDS INTO
Firmware load and handoff	3.5	● Clocking and reset validated in BU-03	BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
Register access	3.0	● Host enablement validated in EVB-10	
Boot sequence execution	2.5	● Firmware and boot ROM available	BU-06 Memory subsystem and bandwidth validation
Failure triage	1.8		BU-09 Silicon debug of anomalies and failure analysis requests
Smoke test	1.3		BU-10 Errata capture, workaround definition and documentation
Total	12.0		
WHO IT TAKES	~13.0 FTE		MEASURED BY
Bring-up lead Owns the boot milestone		● Successful register read and device identity confirmed	● Time from power-on to first register read
Firmware engineer Boot ROM, firmware load and handoff		● Boot-time deviations captured as errata candidates	● Boot failures with usable state captured
Design engineer Register map and failure interpretation		● Status broadcast to the program	● Smoke test coverage at first boot
Validation software engineer Host tooling and register access			
Program manager Status broadcast and expectation management			
		DEPENDS ON	
		BU-03 Reset, clocking and PLL lock validation	
		EVB-06 Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers	
		EVB-10 Minimum host-side enablement for power-on (BSP handoff boundary)	
		RTL-06 Lint, CDC and RDC closure	

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BU-05

Interface bring-up — PCIe/CXL link training, HBM training, die-to-die

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
8w	26.0 M/M	~14.1 FTE	8	yes	Interface engineer
w6–w14 in stage	planned	while it runs	3 in parallel	no float	Bring-up lead approves

PURPOSE

Bring up the **interfaces** — PCIe/CXL link training, HBM training, die-to-die — which on this part is the largest single body of bring-up work and the most likely to produce a respin decision.

Interfaces are where design, package, board and protocol all meet. Training failures are notoriously hard to attribute because every layer can cause them, and the margin measurements taken here are what tell the program whether the part is comfortable or on the edge.

WORK FLOW

8 STEPS · 2 LANES

w6	w8	w10	w12	w14
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
---	---	---

#	STEP	TAT	LANE
1	Interface bring-up order and plan	1w	main
2	PCIe/CXL link training	2w	main
3	Link margin and eye measurement	2w	parallel
4	HBM interface training	2.5w	main
5	Training failure debug	2w	parallel
6	Die-to-die interface bring-up	1.5w	main
7	Interoperability against host platforms	1.5w	parallel
8	Interface stability and compliance run	1w	main

Step 3 is not optional even when training succeeds. A link that trains with no margin trains today and fails at temperature or on a different host, and only the eye measurement distinguishes the two.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Booted units from BU-04
- SIPI channel budget and predictions from SIPI-04 and SIPI-05
- Debug and trace access from EVB-06
- Lab instrumentation from EVB-09
- Interface specifications from ARCH-03

PRODUCES

- Interface bring-up plan
- PCIe/CXL link training results
- Link margin and eye measurements
- HBM training results
- Training failure debug findings
- Die-to-die interface results

- Interoperability results
- Interface stability and compliance data

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D3	PRODUCES	Interface compliance results with training margins. This activity is the deliverable, and the margins are what say whether the part is comfortable.
BU-D1	FEEDS	Bring-up report with per-milestone health status. Interface training is the milestone that decides whether the product works as designed.

RISKS AND WATCH-OUTS

- ◆ **Training success accepted without margin measurement.** A zero-margin link works on the bench and fails in the field.
- ◆ **Failures not attributable across layers.** Design, package, board and protocol can all cause the same symptom, and the SIPI predictions are what separate them.
- ◆ **HBM training treated as a vendor problem.** The interface is the program's, and its training parameters have to be understood locally.
- ◆ **Interoperability tested against one host.** Customers use many, and a link that only works with the lab's host is not compliant.
- ◆ **Compliance run deferred.** Formal compliance is a gate for certification and takes lab time that has to be booked.

EFFORT SPLIT	26 M/M	ENTRY CRITERIA	FEEDS INTO
HBM training	7.0	- Units booted from BU-04 - SIPI predictions available from SIPI-04 and SIPI-05 - High-speed instrumentation available from EVB-09	BU-06 Memory subsystem and bandwidth validation
PCIe/CXL training	6.0		BU-08 Performance validation against the architecture model
Margin and eye measurement	4.5		BU-10 Errata capture, workaround definition and documentation
Training failure debug	4.0		TEST-10 Correlation between ATE, bench and system results
Die-to-die bring-up	2.5	EXIT CRITERIA - Every interface's margin measured, not only its training result - Failures attributed to a layer against the SIPI prediction - Interoperability confirmed on more than the lab's host	MP-11 Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety
Compliance and interoperability	2.0		
Total	26.0		
WHO IT TAKES	~14.1 FTE		MEASURED BY - Training margin against the SIPI prediction - Interfaces trained against interfaces designed - Interoperability across host platforms
Interface engineer Owns link and memory training		DEPENDS ON BU-04Boot, firmware load and functional smoke test SIPI-04Dynamic voltage-drop analysis with package inductance and real switching profiles SIPI-05Decoupling capacitor budget and placement optimisation across die, package and board EVB-06Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers EVB-09Lab instrumentation reservation and test rack build	
SI/PI engineer Margin attribution against the channel budget			
Design engineer SerDes and controller behaviour			
Validation engineer Compliance and interoperability			
Bring-up lead Prioritisation and escalation			

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BU-06

Memory subsystem and bandwidth validation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w8–w14 in stage	16.0 M/M planned	~11.6 FTE while it runs	6 2 in parallel	no has float	Memory engineer Thermal engineer approves

PURPOSE

Validate the **memory subsystem** — controller, address map, bandwidth, latency, ECC — because on an AI accelerator the memory system is most of the performance.

HBM training establishes that the links work; this establishes that the memory system delivers. Bandwidth and latency against the architecture model is the measurement the product's performance claims rest on, and it is where architectural assumptions meet a real memory controller.

WORK FLOW

6 STEPS · 2 LANES

w8	w10	w12	w14
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Memory controller initialisation and configuration	1w	main
2	Address map and pattern validation	1.5w	main
3	ECC and error handling validation	1.5w	parallel
4	Bandwidth measurement against the model	1.5w	main
5	Refresh and thermal behaviour	1.5w	parallel
6	Latency and efficiency characterisation	2w	main

Step 3 has to be done deliberately by injecting errors. ECC that has never corrected anything is ECC that has never been shown to work, and a broken correction path is silent until it matters.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Trained HBM interfaces from BU-05
- Memory subsystem model from ARCH-04
- Address map from RTL-05
- Memory test content from TEST-06
- Thermal capability from EVB-08

PRODUCES

- Controller initialisation and configuration
- Address map and pattern validation
- ECC and error handling validation
- Bandwidth measurements against the model
- Refresh and thermal behaviour data
- Latency and efficiency characterisation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D2	FEEDS	Characterisation data set. Memory bandwidth, latency and efficiency are core characterisation results.
BU-D3	FEEDS	Interface compliance results with training margins. Sustained memory operation is what proves the trained interfaces hold.

RISKS AND WATCH-OUTS

- ◆ **ECC never exercised.** A correction path that has never corrected an injected error has not been validated.
- ◆ **Bandwidth measured without the model to compare against.** A number without a target says nothing about whether the design met its intent.
- ◆ **Refresh behaviour at temperature not checked.** HBM refresh scales with temperature and the effect on bandwidth is real.
- ◆ **Address map validated only at the boundaries.** Aliasing errors sit in the middle of the map and pass boundary tests.
- ◆ **Efficiency ignored in favour of peak.** Peak bandwidth is a datasheet number; achieved efficiency is what workloads see.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	FEEDS INTO
Bandwidth measurement	4.5	● HBM interfaces trained in BU-05	BU-08Performance validation against the architecture model
Latency characterisation	3.5	● Architecture memory model available from ARCH-04	BU-10Errata capture, workaround definition and documentation
Address map validation	3.0	● Thermal control available from EVB-08	MP-12Documentation release — datasheet, user guide, errata, application notes
ECC validation	2.5		
Controller initialisation	1.5	EXIT CRITERIA	
Refresh behaviour	1.0	● ECC validated by injected errors, not by absence of failures	MEASURED BY
Total	16.0	● Bandwidth and latency compared against the architecture model	● Achieved bandwidth against the model
WHO IT TAKES	~11.6 FTE	● Efficiency characterised, not only peak	● Latency against the architectural target
Memory engineer		DEPENDS ON	● ECC correction verified by injection
Owns memory subsystem validation		BU-05Interface bring-up — PCIe/CXL link training, HBM training, die-to-die	
Architect		ARCH-04Interface and protocol selection (PCIe gen, CXL, UCle, HBM3/3E)	
Model comparison and efficiency expectation		RTL-05Clock, reset and power intent (UPF) implementation	
Design engineer		EVB-08Thermal solution and cooling for the lab platform	
Controller behaviour and address map			
Validation engineer			
ECC injection and error handling			
Thermal engineer			
Refresh and temperature interaction			

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BU-07

Shmoo across voltage, frequency and temperature

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	16.0 M/M	~9.9 FTE	6	no	Characterisation engineer
w9–w16 in stage	planned	while it runs	2 in parallel	has float	Design engineer approves

PURPOSE

Shmoo the part across voltage, frequency and temperature — the measurement that establishes how much margin the design actually has.

Every operating condition the product will be specified at comes out of this. The shmoo shows where the part stops working, and the distance between that boundary and the nominal operating point is the margin the datasheet, the guard bands and the qualification plan all depend on.

WORK FLOW

6 STEPS · 2 LANES

w9	w11	w13	w15	w16
----	-----	-----	-----	-----

MAIN SEQUENCE

1	2	4	6
---	---	---	---

IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Shmoo plan and automation setup	1w	main
2	Voltage-frequency shmoo at room temperature	1.5w	main
3	Power measurement across the sweep	1.5w	parallel
4	Temperature-extended shmoo	1.5w	main
5	Unit-to-unit variation across the fleet	1.5w	parallel
6	Shmoo analysis and margin extraction	3w	main

Step 5 is what separates a characteristic from an anecdote. One unit's shmoo is that unit; the fleet's spread is what the population looks like, and the specification has to be set against the population.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Booted and interface-trained units from BU-04 and BU-05
- Sweep automation from EVB-09
- Thermal control from EVB-08
- Rail sweep capability from EVB-05
- Characterisation content from TEST-07

PRODUCES

- Shmoo plan and automation
- Voltage-frequency shmoo at room
- Power measurements across the sweep
- Temperature-extended shmoo
- Unit-to-unit variation data
- Margin extraction and analysis

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D2

PRODUCES

Characterisation data set — V/F/T shmoo and power measurements. This activity is the deliverable.

BU-D1

FEEDS

Bring-up report with per-milestone health status. Margin is the strongest single indicator of whether the silicon is healthy.

RISKS AND WATCH-OUTS

- ◆ **Shmoo on a single unit.** The result describes that unit and is used as if it described the population.
- ◆ **Room temperature only.** The hot and cold corners are where the margin is smallest and where qualification will operate.
- ◆ **Manual sweeps.** The corner count then gets reduced to fit the available engineer time, silently.
- ◆ **Thermal throttling inside the sweep.** The shmoo then maps the cooling solution rather than the silicon.
- ◆ **Margin measured but not fed into guard bands.** The data exists and production test still uses convention.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	FEEDS INTO		
Temperature-extended shmoo	4.0	- Units functional from BU-04 and BU-05 - Sweep automation working from EVB-09 - Thermal control available from EVB-08	BU-08 Performance validation against the architecture model		
Voltage-frequency shmoo	3.5		BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule		
Margin extraction	3.0		MP-06 Production test program release and guard-band validation		
Power measurement	2.5		MP-08 Process corner and split-lot validation		
Unit-to-unit variation	2.0		TEST-10 Correlation between ATE, bench and system results		
Automation setup	1.0	<th>EXIT CRITERIA</th> <td rowspan="5"><th>MEASURED BY</th></td>	EXIT CRITERIA	<th>MEASURED BY</th>	MEASURED BY
Total	16.0				
WHO IT TAKES	~9.9 FTE				
Characterisation engineer Owns the shmoo campaign					
Validation engineer Automation and data capture					
Product engineering Margin interpretation and guard bands		- Margin at the worst corner - Units in the shmoo sample - Corners measured against planned			
Thermal engineer Temperature control and throttling checks					
Design engineer Failure mode interpretation at the boundaries					
		DEPENDS ON			
		BU-04 Boot, firmware load and functional smoke test			
		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die			
		EVB-05 Power delivery, VRM and telemetry design and bring-up			
		EVB-08 Thermal solution and cooling for the lab platform			
		EVB-09 Lab instrumentation reservation and test rack build			
		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps			

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BU-08

Performance validation against the architecture model

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	16.0 M/M	~11.6 FTE	6	no	Performance engineer
w11–w17 in stage	planned	while it runs	2 in parallel	has float	Product marketing approves

PURPOSE

Measure **real performance against the architecture model** — the check on whether the product does what it was sold as doing.

The whole program was justified by a performance projection. This is where that projection meets a physical part running real workloads, and any gap has to be attributed: to the model, to the design, to the memory system, or to the software driving it.

WORK FLOW

6 STEPS · 2 LANES

w11w13w15w17

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Benchmark and workload setup	1w	main
2	Performance measurement across workloads	1.5w	main
3	Power efficiency measurement	1.5w	parallel
4	Comparison against the architecture model	1.5w	main
5	Bottleneck profiling	1.5w	parallel
6	Gap analysis and attribution	2w	main

Step 6 is where the program learns something durable. A gap attributed to the model improves the next program's projections; a gap attributed to the design becomes an errata or a respin input; an unattributed gap teaches nothing.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Functional units from BU-04 through BU-06
- Architecture performance model from ARCH-02
- Workload definitions from DEF-03
- Power measurement from BU-07
- Lab instrumentation from EVB-09

PRODUCES

- Benchmark and workload setup
- Performance measurements across workloads
- Power efficiency measurements
- Comparison against the architecture model
- Bottleneck profiling results
- Gap analysis and attribution

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D2	FEEDS	Characterisation data set. Performance and efficiency measurements are part of the product's characterisation.
BU-D1	FEEDS	Bring-up report with per-milestone health status. Performance against target is the milestone the business cares about most.

RISKS AND WATCH-OUTS

- ◆ **Performance measured on unrepresentative workloads.** A benchmark that is not what customers run produces a number that does not predict their experience.
- ◆ **Gap reported without attribution.** Nobody can act on it, and the next program repeats the modelling error.
- ◆ **Software limiting the measurement.** An immature stack understates the silicon, and the difference has to be established rather than assumed.
- ◆ **Efficiency ignored.** Performance per watt is the metric this product class is bought on, and raw throughput alone misleads.
- ◆ **Measured at a single operating point.** Real deployment spans conditions, and performance at nominal is not the whole picture.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	FEEDS INTO
Performance measurement	4.0	● Memory subsystem validated in BU-06	BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule
Model comparison	3.5	● Architecture model available from ARCH-02	MP-12 Documentation release — datasheet, user guide, errata, application notes
Gap analysis	3.0	● Workloads defined in DEF-03	TEST-10 Correlation between ATE, bench and system results
Bottleneck profiling	2.5		
Efficiency measurement	2.0	EXIT CRITERIA	
Benchmark setup	1.0	● Every material gap attributed to a cause	MEASURED BY
Total	16.0	● Efficiency measured alongside throughput	● Performance against the architecture projection
WHO IT TAKES	~11.6 FTE	● Software's contribution to the measurement established	● Efficiency against the target
Performance engineer		DEPENDS ON	● Gaps attributed against gaps found
Owns measurement and attribution		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die	
Architect		BU-06 Memory subsystem and bandwidth validation	
Model comparison and gap interpretation		BU-07 Shmoo across voltage, frequency and temperature	
Validation software engineer		ARCH-02 Compute / memory / interconnect partitioning; monolithic vs chiplet decision	
Workload execution and software effects		DEF-03 PPA target definition — frequency, power envelope, die area budget	
Design engineer			
Bottleneck root cause			
Product marketing			
Workload representativeness			

22 BU · SILICON BRING-UP

BU-09

Silicon debug of anomalies and failure analysis requests

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	28.0 M/M	~12.2 FTE	8	yes	Silicon debug engineer
w4–w14 in stage	planned	while it runs	3 in parallel	no float	Failure analysis engineer approves

PURPOSE

Debug what is wrong — reproduce it, isolate it, extract the internal state, find the root cause, and get failure analysis where physical evidence is needed.

This is the activity the whole DFT and debug infrastructure investment exists to serve. It runs alongside every other bring-up activity because anomalies arrive from all of them, and its throughput largely determines how long bring-up takes.

WORK FLOW

8 STEPS · 2 LANES

w4	w6	w8	w10	w12	w14
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Anomaly intake and triage process	1w	main
2	Reproduction and isolation	2w	main
3	Simulation and emulation cross-check	2w	parallel
4	Internal state extraction via DFT and trace	2.5w	main
5	Design-team escalation and analysis	2w	parallel
6	Hypothesis test and root cause	2.5w	main
7	Debug knowledge capture	1.5w	parallel
8	Failure analysis request and disposition	2w	main

Step 3 is the highest-leverage step in the activity. An anomaly reproduced in simulation can be examined with full visibility and unlimited retries, which is an environment no amount of silicon debug can match.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Anomalies from every bring-up activity
- DFT and trace access from DFT-08 and EVB-06
- Simulation and emulation environments from DV-08 and DV-11
- Design database and RTL from RTL-09
- FA laboratory capability

PRODUCES

- Anomaly intake and triage records
- Reproduction and isolation results
- Simulation and emulation cross-check findings
- Extracted internal state
- Design-team analysis
- Root cause determinations

- Debug knowledge base
- Failure analysis reports and dispositions

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D5 **PRODUCES** **Failure analysis reports.** This activity is the deliverable, and physical FA is its most conclusive form.

BU-D4 **FEEDS** **Errata list with workarounds.** Every root cause found here becomes an errata entry or a fix.

RISKS AND WATCH-OUTS

- ◆ **Symptoms treated instead of causes.** A workaround without a root cause hides a mechanism that resurfaces in the field.
- ◆ **No path back to simulation.** Debug then happens only on silicon, with limited visibility and long turnaround per hypothesis.
- ◆ **Design team engaged late.** The people who wrote the logic can often recognise a symptom in minutes, and validation debugs alone for days first.
- ◆ **Units consumed by destructive FA too readily.** Physical analysis destroys the unit, and there are very few.
- ◆ **Findings not captured.** The same anomaly is then debugged twice by different engineers.

EFFORT SPLIT		28 M/M	EXIT CRITERIA		FEEDS INTO	
Root cause determination		7.0	● Root cause established rather than symptom suppressed ● Anomalies reproduced in simulation where possible ● Findings captured so they are debugged once		BU-10 Errata capture, workaround definition and documentation	
Internal state extraction		6.0			BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule	
Reproduction and isolation		5.0			MP-05 Yield learning, failure pareto and defect analysis	
Simulation cross-check		4.0				
FA and disposition		3.5	DEPENDS ON BU-02 Power-on, power sequencing and basic health check BU-03 Reset, clocking and PLL lock validation BU-04 Boot, firmware load and functional smoke test BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die DFT-08 Pattern validation by gate-level simulation and ATE format conversion (STIL/WGL) EVB-06 Debug infrastructure — JTAG and trace pods, interposers, instrumentation headers DV-08 Emulation and FPGA prototype bring-up — capacity, partitioning, speed DV-11 Gate-level simulation, functional and timing-annotated			
Knowledge capture		2.5				
Total		28.0				
WHO IT TAKES		~12.2 FTE				
Silicon debug engineer Owns anomaly resolution						
Design engineer Logic analysis and hypothesis generation						
DFT engineer State extraction and trace						
DV engineer Simulation and emulation reproduction						
Failure analysis engineer Physical analysis and disposition						
ENTRY CRITERIA ● Anomalies arriving from bring-up activities ● DFT and trace access working from EVB-06 ● Simulation and emulation environments available						
			Anomalies arrive from every bring-up activity; the dependencies listed are the ones that supply the debug capability rather than the problems.			

22 BU · SILICON BRING-UP

BU-10

Errata capture, workaround definition and documentation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
7w	10.0 M/M	~6.2 FTE	6	no	Product engineering
w10-w17 in stage	planned	while it runs	2 in parallel	has float	Technical writer approves

PURPOSE

Turn what is wrong into a **documented errata list with workarounds** — the honest account of the silicon's behaviour that customers and the next revision both need.

Every part ships with errata. What distinguishes a well-run program is that its errata are known, characterised, worked around and disclosed, rather than discovered by a customer. This activity is where debug findings become a product document.

WORK FLOW

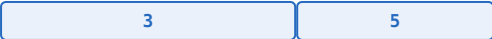
6 STEPS · 2 LANES

w10	w12	w14	w16	w17
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Errata intake and classification	1w	main
2	Impact and severity assessment	1.5w	main
3	Customer-visibility determination	1.5w	parallel
4	Workaround definition and validation	1.5w	main
5	Fix-versus-document decision input	1w	parallel
6	Errata documentation and release	3w	main

Step 4's validation is the part that gets skipped. A workaround that has been described but never run is a hypothesis, and shipping it as a mitigation is how a known issue becomes a field failure anyway.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Root causes from BU-09
- Failure analysis results from BU-09
- Design impact assessment from RTL-09
- Customer use models from DEF-03
- Documentation standards from MP-12

PRODUCES

- Errata intake and classification
- Impact and severity assessment
- Customer-visibility determination
- Validated workarounds
- Fix-versus-document input
- Released errata documentation

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D4	PRODUCES	Errata list with workarounds. This activity is the deliverable, and it is what ships alongside the product.
BU-D6	FEEDS	Respin versus metal-fix decision record. Which errata are tolerable with a workaround is the decision's main input.

RISKS AND WATCH-OUTS

- ◆ **Workarounds documented but never validated.** The mitigation then fails in the field and the errata gave false comfort.
- ◆ **Severity assessed by engineering alone.** Customer impact depends on use models, and the assessment needs that perspective.
- ◆ **Errata suppressed for commercial reasons.** Undisclosed known issues discovered by a customer damage far more than the disclosure would have.
- ◆ **Errata list not versioned against silicon revision.** Which errata apply to which stepping becomes ambiguous the moment a second stepping exists.
- ◆ **Documentation deferred to the end.** The context is freshest at root cause, and reconstructing it months later loses detail.

EFFORT SPLIT	10 M/M	ENTRY CRITERIA	FEEDS INTO
Workaround definition and validation	3.0	● Root causes available from BU-09 ● Customer use models from DEF-03 ● Documentation standard from MP-12	BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule
Errata documentation	2.5		BU-12 Customer sample readiness and release package
Impact assessment	2.0		MP-12 Documentation release — datasheet, user guide, errata, application notes
Visibility determination	1.5	EXIT CRITERIA	
Fix decision input	1.0	● Every workaround validated on silicon, not only described ● Severity assessed with customer use models in view ● Errata versioned against silicon revision	MEASURED BY
Total	10.0		● Workarounds validated against documented ● Errata found by customers after release ● Time from root cause to documented errata
WHO IT TAKES	~6.2 FTE	DEPENDS ON	
Product engineering Owns the errata list		BU-04 Boot, firmware load and functional smoke test	
Design engineer Mechanism and workaround definition		BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die	
Validation engineer Workaround validation on silicon		BU-06 Memory subsystem and bandwidth validation	
Product marketing Customer impact and disclosure		BU-09 Silicon debug of anomalies and failure analysis requests	
Technical writer Errata documentation			

22 BU · SILICON BRING-UP

BU-11

Respin versus metal-fix decision analysis — ECO scope, cost, schedule

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
4w	5.0 M/M	~5.4 FTE	6	yes	Program manager
w13–w17 in stage	planned	while it runs	2 in parallel	no float	Executive sponsor approves

PURPOSE

Decide whether the silicon needs a **respin, a metal fix, or nothing** — the single most consequential decision the program makes after tapeout.

A full-mask respin costs a mask set and roughly a quarter of schedule. A metal fix costs far less but can only reach defects in the routing layers. Shipping as-is with workarounds costs nothing now and possibly a great deal later. The decision has to weigh all three against evidence rather than optimism.

WORK FLOW

6 STEPS · 2 LANES

w13	w14	w15	w16	w17
-----	-----	-----	-----	-----

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Defect list and fix scoping	0.8w	main
2	Metal-fix feasibility assessment	1w	main
3	Full-mask respin scoping	0.8w	parallel
4	Cost, schedule and risk comparison	1.3w	main
5	Customer and market impact input	0.8w	parallel
6	Decision and its record	1w	main

Step 2 is bounded by physics rather than by will. A metal fix can only change routing layers, and whether the required change fits inside them is a question with a factual answer that has to be established before the decision is framed.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Errata list and severity from BU-10
- Root causes from BU-09
- Margin data from BU-07
- Performance gaps from BU-08
- Mask and fab cost from TECH-08

PRODUCES

- Defect list and fix scope
- Metal-fix feasibility assessment
- Full-mask respin scope
- Cost, schedule and risk comparison
- Customer and market impact assessment
- Recorded decision with its rationale

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D6

PRODUCES

Respin versus metal-fix decision record. This activity is the deliverable, and the record is what the program is held to.

BU-D4

FEEDS

Errata list with workarounds. Errata not fixed by the decision become permanent, documented behaviour.

RISKS AND WATCH-OUTS

- ◆ **Decision made on schedule pressure alone.** Shipping a part with a real defect to protect a date routinely costs more than the delay would have.
- ◆ **Metal-fix feasibility assumed.** It is a physical question about which layers the fix touches, and it is often answered optimistically.
- ◆ **Margin data not weighed.** A part with adequate function and no margin fails in the field, and the shmoo already said so.
- ◆ **Decision unrecorded.** Six months later nobody can reconstruct what was known and what was accepted.
- ◆ **Customer input excluded.** Whether an errata is tolerable depends on what customers do with the part.

EFFORT SPLIT	5 M/M	ENTRY CRITERIA	FEEDS INTO
Cost, schedule and risk comparison	1.5	● Errata list and severities available from BU-10 ● Margin and performance data from BU-07 and BU-08 ● Mask and fab costs available	BU-12 Customer sample readiness and release package
Metal-fix feasibility	1.3		MP-09 Production readiness review and change control (PCN) setup
Respin scoping	1.0		MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Decision record	0.8		
Market impact input	0.5		
Total	5.0	EXIT CRITERIA	
WHO IT TAKES ~5.4 FTE Program manager Owns the decision and its record Design lead Fix scope and metal-fix feasibility Product engineering Errata severity and margin evidence Product marketing Customer and market impact Executive sponsor Cost and schedule authority	● Metal-fix feasibility established physically, not assumed ● Margin evidence weighed alongside functional defects ● Decision recorded with what was known at the time		MEASURED BY
			● Decision made against the planned date ● Feasibility claims that held ● Defects found later that the decision did not consider
		DEPENDS ON	
		BU-07 Shmoo across voltage, frequency and temperature	
		BU-08 Performance validation against the architecture model	
		BU-09 Silicon debug of anomalies and failure analysis requests	
		BU-10 Errata capture, workaround definition and documentation	
		TECH-08 Second-source and node-migration strategy assessment	

22 BU · SILICON BRING-UP

BU-12

Customer sample readiness and release package

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
5w	6.0 M/M	~5.2 FTE	6	no	Product marketing
w13–w18 in stage	planned	while it runs	2 in parallel	has float	Program manager approves

PURPOSE

Get **samples to customers** with everything they need to use them — screened units, documentation, errata, and a channel to report back.

Customer samples start the evaluation cycle that decides design wins, and they start it months before mass production. What ships with the unit determines whether the customer's first week is productive or spent rediscovering issues the program already knows about.

WORK FLOW

6 STEPS · 2 LANES

w13	w14	w15	w16	w17	w18
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
---	---

#	STEP	TAT	LANE
1	Sample readiness criteria	0.8w	main
2	Unit selection and screening for samples	1.3w	main
3	Customer documentation and errata pack	1w	parallel
4	Release package assembly	1.5w	main
5	Feedback channel establishment	1w	parallel
6	Sample shipment and support setup	1.5w	main

Step 5 is the one that pays back most. A customer who finds something and has an obvious route to report it gives the program early field data; one who does not gives it a design loss with no explanation.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Allocated sample units from ASSY-09
- Errata list from BU-10
- Decision record from BU-11
- Preliminary datasheet from MP-12
- Customer commitments from DEF-01

PRODUCES

- Sample readiness criteria
- Screened and selected sample units
- Customer documentation and errata pack
- Assembled release package
- Feedback channel
- Shipped samples with support in place

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

BU-D7

PRODUCES

Customer sample release package. This activity is the deliverable — units plus everything needed to use them.

BU-D4

FEEDS

Errata list with workarounds. The errata pack that ships with samples is the errata list's first external use.

RISKS AND WATCH-OUTS

- ◆ **Samples shipped without errata.** The customer spends their first week rediscovering known issues and loses confidence in the part.
- ◆ **Units not screened before shipping.** A marginal sample creates a customer-visible failure that the program cannot investigate remotely.
- ◆ **Documentation lagging the silicon.** A datasheet describing behaviour the sample does not have generates support load and doubt.
- ◆ **No feedback channel.** Customer findings then arrive as rumour or not at all, and the program loses its earliest field data.
- ◆ **Samples shipped before the respin decision.** If a respin follows, customers are evaluating a stepping that will not ship.

EFFORT SPLIT	6 M/M	ENTRY CRITERIA	FEEDS INTO
Release package assembly	1.8	● Sample units allocated from ASSY-09	MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Unit selection and screening	1.3	● Errata list released from BU-10	
Documentation pack	1.3	● Respin decision recorded in BU-11	MP-12 Documentation release — datasheet, user guide, errata, application notes
Shipment and support	1.0		
Feedback channel	0.8		
Total	6.0		
WHO IT TAKES	~5.2 FTE	EXIT CRITERIA	MEASURED BY
Product marketing Owns customer sample release		● Errata pack shipping with every sample	● Samples shipped against commitment
Product engineering Unit screening and readiness criteria		● Units screened before shipment	● Customer issues that were already documented
Applications engineer Customer support and feedback channel		● Feedback channel established and staffed	● Feedback received through the channel
Technical writer Documentation and errata pack		DEPENDS ON	
Program manager Timing against the respin decision		ASSY-09 Unit build for bring-up, qualification and customer samples	
		BU-10 Errata capture, workaround definition and documentation	
		BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule	

VALIDATE & RAMP

23 Qualification & Production

MP

WINDOW	DURATION	ACTIVITIES	EFFORT	ON CRITICAL PATH
w106 – w132	26 _w	12	208 _{M/M}	6 of 12
ENGINEERING ACTIVITIES				KEY DELIVERABLES
MP-01	Qualification plan definition against JEDEC / AEC standards		6w 8 M/M	MP-D1 Qualification plan and JEDEC-compliant qualification report
MP-02	Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB		18w 30 M/M	MP-D2 Reliability and package qualification data packages
MP-03	ESD (HBM, CDM) and latch-up qualification		6w 8 M/M	MP-D3 Production test program release
MP-04	Package qualification — MSL, drop, bend, board-level reliability		12w 18 M/M	MP-D4 Yield model versus cost target report
MP-05	Yield learning, failure pareto and defect analysis		20w cont. 40 M/M	MP-D5 Production readiness review sign-off
MP-06	Production test program release and guard-band validation		9w 20 M/M	MP-D6 Ramp plan and supply commitment
MP-07	Test time reduction and multi-site conversion		9w 16 M/M	MP-D7 Compliance certificates
MP-08	Process corner and split-lot validation		9w 14 M/M	MP-D8 Datasheet and product documentation set
MP-09	Production readiness review and change control (PCN) setup		6w 8 M/M	MP-D9 Mass Production release record
MP-10	Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM		12w 12 M/M	
MP-11	Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety		12w 18 M/M	
MP-12	Documentation release — datasheet, user guide, errata, application notes		10w 16 M/M	

23 MP · QUALIFICATION & PRODUCTION

MP-01

Qualification plan definition against JEDEC / AEC standards

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w0–w6 in stage	8.0 M/M planned	~5.8 FTE while it runs	6 2 in parallel	yes no float	Reliability engineer Operations planner approves

PURPOSE

Define **what qualification means for this product** — which standards apply, which stresses run, on how many units, and against what acceptance criteria.

Qualification is the evidence that the part survives its intended life. The plan is where that evidence is designed: too little and a customer's own qualification finds what yours did not, too much and months of chamber time are spent on stresses nobody required.

WORK FLOW

6 STEPS · 2 LANES

w0w2w4w6

MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Standard selection — JEDEC, AEC and customer applicability	1w	main
2	Stress matrix and sample size definition	1.5w	main
3	Qualification vehicle and lot definition	1w	parallel
4	Lab and chamber capacity booking	1.5w	main
5	Customer qualification requirement collection	1w	parallel
6	Plan review and release	2w	main

Step 4 is why this activity runs first. HTOL occupies a chamber for a thousand hours, and chambers are shared; a booking made late pushes the whole qualification campaign and therefore mass production.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Product requirements and quality target from DEF-04
- Application and use conditions from DEF-03
- Customer qualification requirements
- Package construction from PKGD-11
- Unit availability from ASSY-09

PRODUCES

- Applicable standard selection
- Stress matrix with sample sizes
- Qualification vehicle and lot definition
- Chamber and lab bookings
- Customer requirement record
- Released qualification plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D1	PRODUCES	Qualification plan and JEDEC-compliant qualification report. The plan half of the deliverable; the report follows from executing it.
MP-D5	INFORMS	Production readiness review sign-off. The plan defines what evidence the readiness review will be asked to accept.

RISKS AND WATCH-OUTS

- ◆ **Chamber capacity booked late.** Stress duration is fixed and capacity is shared, so a late booking moves mass production directly.
- ◆ **Sample sizes below the confidence the claim needs.** A reliability claim from too few units does not support the specification.
- ◆ **Customer requirements collected after the plan.** A customer-specific stress discovered later restarts a months-long campaign.
- ◆ **Three lots not planned.** Standards generally require multiple production lots, and a single-lot qualification is not accepted.
- ◆ **Use conditions assumed rather than specified.** The acceleration factors depend on them, and wrong conditions make the whole campaign non-representative.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
Stress matrix and sample sizes	2.5	● Product use conditions defined in DEF-03 ● Package construction known from PKGD-11 ● Unit allocation agreed in ASSY-09	MP-02Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB
Plan review and release	1.8		MP-03ESD (HBM, CDM) and latch-up qualification
Standard selection	1.5		MP-04Package qualification — MSL, drop, bend, board-level reliability
Lab and chamber booking	1.0		MP-09Production readiness review and change control (PCN) setup
Vehicle definition	0.8		TEST-07Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps
Customer requirements	0.5	EXIT CRITERIA	
Total	8.0	● Chamber capacity booked against the full stress duration ● Sample sizes supporting the reliability claim ● Customer-specific requirements included before release	MEASURED BY ● Chamber capacity secured against the plan ● Sample size against the confidence claim ● Customer requirements captured before execution
WHO IT TAKES	~5.8 FTE	DEPENDS ON	
Reliability engineer Owns the qualification plan		DEF-03PPA target definition — frequency, power envelope, die area budget	
Quality manager Standard applicability and acceptance criteria		DEF-04Die size, yield and product cost model (wafer, mask, package, test, ASP)	
Package engineer Package-specific stress requirements		PKGD-11Package design freeze, DRC and tooling release	
Product marketing Customer qualification requirements		ASSY-09Unit build for bring-up, qualification and customer samples	
Operations planner Unit availability and chamber capacity			

23 MP · QUALIFICATION & PRODUCTION

MP-02

Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
18w	30.0 M/M	~7.2 FTE	8	yes	Reliability engineer
w5–w23 in stage	planned	while it runs	3 in parallel	no float	Lab technician approves

PURPOSE

Run the **reliability stresses** — HTOL, high-temperature storage, temperature cycling, uHAST, THB — that demonstrate the silicon survives its specified life.

This is eighteen weeks of mostly waiting, and it is the longest fixed-duration item between first silicon and mass production. HTOL alone is a thousand hours; it cannot be shortened, only started earlier or run in parallel.

WORK FLOW

8 STEPS · 2 LANES

w5	w9	w13	w17	w21	w23
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Sample preparation and pre-stress characterisation	1.5w	main
2	Stress chamber loading and campaign start	2w	main
3	uHAST and THB execution	3w	parallel
4	HTOL execution	6w	main
5	Interim readouts at defined intervals	3w	parallel
6	Temperature cycle and HTS execution	4.5w	main
7	Failure analysis on stress failures	2.5w	parallel
8	Final readout, analysis and report	4w	main

Step 5's interim readouts are what turn a pass/fail campaign into a learning one. A parameter drifting at 500 hours predicts a failure at 1000 that would otherwise arrive as a surprise at the end of a four-month run.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Qualification plan from MP-01
- Qualification units from ASSY-09
- Pre-stress characterisation from TEST-07
- Chamber capacity from MP-01
- Failure analysis capability

PRODUCES

- Pre-stress characterisation baseline
- Loaded stress campaign
- uHAST and THB results
- HTOL results
- Interim readout data
- Temperature cycle and HTS results

- Stress failure analyses
- Reliability data package

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D2

PRODUCES

Reliability and package qualification data packages. The silicon-reliability half of the deliverable.

MP-D1

FEEDS

Qualification plan and JEDEC-compliant qualification report. These are the results the report is written from.

RISKS AND WATCH-OUTS

- ◆ **Campaign started late.** The duration is fixed, so every week of delay is a week of mass production delay, one for one.
- ◆ **No pre-stress characterisation.** Drift cannot be measured without a baseline, and a failure cannot be distinguished from a part that was always marginal.
- ◆ **Interim readouts skipped.** Failures then arrive only at the end, with no time left to respond before the ramp.
- ◆ **Stress failures not analysed.** A failure without a mechanism cannot be judged as systematic or as a one-off, and the distinction decides the ramp.
- ◆ **Units insufficient for the full matrix.** A stress dropped for lack of units is a gap in the qualification the customer will find.

EFFORT SPLIT	30 M/M	ENTRY CRITERIA	FEEDS INTO
HTOL execution	8.0	● Qualification plan released from MP-01	MP-09 Production readiness review and change control (PCN) setup
Temperature cycle and HTS	6.0	● Units available from ASSY-09	
Readout and analysis	5.0	● Chambers booked and available	MP-12 Documentation release — datasheet, user guide, errata, application notes
uHAST and THB	4.5	EXIT CRITERIA	MP-05 Yield learning, failure pareto and defect analysis
Interim readouts	3.5		
Failure analysis	3.0	● Pre-stress baseline recorded for every unit	MEASURED BY
Total	30.0	● Interim readouts taken at the planned intervals	
WHO IT TAKES	~7.2 FTE	● Every stress failure analysed to a mechanism	
Reliability engineer		DEPENDS ON	● Campaign start against plan
Owens the stress campaign			
Test engineer		MP-01	
Pre-stress and readout measurement			
Failure analysis engineer		ASSY-09	
Stress failure mechanisms			● Failures per stress against the acceptance criterion
Quality engineer		TEST-07	
Acceptance against the standard			
Lab technician			● Failures analysed to a mechanism
Chamber operation and readout scheduling			

23 MP · QUALIFICATION & PRODUCTION

MP-03

ESD (HBM, CDM) and latch-up qualification

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w w6–w12 in stage	8.0 M/M planned	~5.8 FTE while it runs	6 2 in parallel	no has float	ESD engineer Failure analysis engineer approves

PURPOSE

Qualify the part's **ESD and latch-up robustness** — HBM and CDM discharge, and latch-up immunity — the handling-survival properties every downstream user depends on.

ESD failures happen in assembly lines, in labs and in the field, and they are almost never traced back to the part that could not take them. The classification levels are what tell a customer's manufacturing engineer how to handle the device.

WORK FLOW

6 STEPS · 2 LANES

w6	w8	w10	w12
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MAIN SEQUENCE

1	2	4	6
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IN PARALLEL

3	5
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#	STEP	TAT	LANE
1	ESD and latch-up test plan and pin classification	1w	main
2	HBM ESD characterisation	1.5w	main
3	Pin classification and protection review	1w	parallel
4	CDM ESD characterisation	1.5w	main
5	Failure analysis on ESD failures	1w	parallel
6	Latch-up test and report	2w	main

Step 4 matters more than step 2 on a package this size. CDM stress scales with package capacitance, and a large 2.5D package charges to a far higher energy than a small one — the CDM level is usually the binding constraint.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- ESD protection design from AMS-09 and PD-11
- Package construction from PKGD-11
- Qualification plan from MP-01
- Qualification units from ASSY-09
- ESD test laboratory capability

PRODUCES

- ESD and latch-up test plan
- HBM ESD classification
- Pin classification and protection review
- CDM ESD classification
- ESD failure analyses
- Latch-up test results and report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D2	FEEDS	Reliability and package qualification data packages. ESD and latch-up classification are part of the qualification package.
MP-D8	FEEDS	Datasheet and product documentation set. ESD classification is a datasheet parameter every handler reads.

RISKS AND WATCH-OUTS

- ◆ **CDM underestimated for the package size.** A large package stores far more charge, and the CDM level is usually the binding one.
- ◆ **ESD failures not analysed.** Which pin and which protection element failed is what makes the result actionable for a fix.
- ◆ **Latch-up tested only at room temperature.** Latch-up susceptibility increases with temperature, and the hot corner is the real test.
- ◆ **Classification below what customer assembly requires.** A part that cannot survive a customer's line is a part they cannot buy.
- ◆ **Pin classification incomplete.** Untested pin combinations are unqualified pins, and there is no assumption to fall back on.

EFFORT SPLIT	8 M/M	ENTRY CRITERIA	FEEDS INTO
HBM ESD characterisation	2.0	● Qualification plan from MP-01	MP-09 Production readiness review and change control (PCN) setup
CDM ESD characterisation	2.0	● Units available from ASSY-09	
Latch-up test	1.8	● ESD test laboratory booked	MP-12 Documentation release — datasheet, user guide, errata, application notes
Pin classification	1.0		
Test plan	0.8	EXIT CRITERIA	
Failure analysis	0.5	● CDM evaluated against the package's actual capacitance	MEASURED BY
Total	8.0	● Latch-up tested at the hot corner	● ESD classification achieved against target
WHO IT TAKES	~5.8 FTE	● Classification meeting customer assembly requirements	● Latch-up immunity at maximum temperature
ESD engineer Owns ESD and latch-up qualification		DEPENDS ON	● Pin combinations tested against total
Circuit designer Protection design and failure interpretation		MP-01 Qualification plan definition against JEDEC / AEC standards	
Package engineer Package contribution to CDM		ASSY-09 Unit build for bring-up, qualification and customer samples	
Quality engineer Classification against customer requirements		AMS-09 Redundancy and repair scheme integration with the BISR architecture	
Failure analysis engineer ESD failure localisation		PD-11 Chip power model (CPM/CPS) extraction and handoff to package co-verification	
		PKGD-11 Package design freeze, DRC and tooling release	

23 MP · QUALIFICATION & PRODUCTION

MP-04

Package qualification — MSL, drop, bend, board-level reliability

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	18.0 M/M	~6.5 FTE	8	no	Package reliability engineer
w8–w20 in stage	planned	while it runs	3 in parallel	has float	Failure analysis engineer approves

PURPOSE

Qualify the **package** — moisture sensitivity, board-level reliability, drop and bend — which on a 2.5D structure is where the reliability risk actually concentrates.

The silicon's reliability is a well-understood function of process and use conditions. The package's is not: a large interposer on an organic substrate with two HBM stacks has thermal expansion mismatches on every interface, and this is where they are proven or found out.

WORK FLOW8 STEPS · 2 LANES

w8	w11	w14	w17	w20
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	MSL preconditioning and moisture classification	1.5w	main
2	Board-level reliability — temperature cycling on board	2.5w	main
3	Solder joint reliability modelling correlation	2w	parallel
4	Drop and bend testing	3w	main
5	Warpage and co-planarity tracked through stress	2w	parallel
6	Package-level readout and acoustic imaging	2.5w	main
7	Failure analysis on package failures	1.5w	parallel
8	Analysis and qualification report	2.5w	main

Step 3 is what turns this campaign into a predictive model rather than a single data point. Correlating measured joint life against the model means the next package variant can be assessed without repeating twelve weeks of testing.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Qualification plan from MP-01
- Package units from ASSY-09
- PTV reliability results from PTV-10
- Package construction from PKGD-11
- Board-level test capability

PRODUCES

- MSL classification
- Board-level reliability results
- Solder joint reliability model correlation
- Drop and bend results
- Warpage and co-planarity through stress

- Package readout and acoustic imaging
- Package failure analyses
- Package qualification report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D2	PRODUCES	Reliability and package qualification data packages. The package half of the deliverable.
MP-D1	FEEDS	Qualification plan and JEDEC-compliant qualification report. Package qualification is a required section of the report.

RISKS AND WATCH-OUTS

- ◆ **MSL classification worse than the assembly process needs.** A high moisture sensitivity forces bake-and-handle procedures on every customer line.
- ◆ **Board-level reliability short of the application requirement.** The joints, not the die, then set the product's life.
- ◆ **PTV results not compared.** The vehicles predicted this behaviour, and the comparison validates or invalidates the whole vehicle programme.
- ◆ **Warpage measured only before stress.** The interesting change is what stress does to it, and that requires measurement through the campaign.
- ◆ **Failures attributed to the package without analysis.** A CPI failure and a solder joint failure need different fixes and look similar from outside.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	FEEDS INTO
Board-level reliability	4.5	● Qualification plan from MP-01	MP-09 Production readiness review and change control (PCN) setup
Drop and bend testing	3.5	● Package units available from ASSY-09	
MSL preconditioning	3.0	● PTV reliability data available from PTV-10	MP-12 Documentation release — datasheet, user guide, errata, application notes
Readout and acoustic imaging	2.5		
Joint reliability correlation	2.5	EXIT CRITERIA	PTV-12 Feedback into package design and assembly process window freeze
Warpage tracking	2.0	● MSL classification compatible with customer assembly ● Board-level life meeting the application requirement ● Results compared against the PTV prediction	MEASURED BY
Total	18.0	DEPENDS ON	● MSL level achieved ● Board-level cycles to failure against requirement ● Correlation between measured and modelled joint life
WHO IT TAKES	~6.5 FTE		
Package reliability engineer Owns package qualification			
Package engineer Construction and failure mechanism			
Materials engineer Interface and joint behaviour		MP-01 Qualification plan definition against JEDEC / AEC standards	
Quality engineer Standard compliance and acceptance		ASSY-09 Unit build for bring-up, qualification and customer samples	
Failure analysis engineer Package failure localisation		PTV-10 Board-level reliability on vehicles — temperature cycle, drop, bend	
		PKGD-11 Package design freeze, DRC and tooling release	
		ASSY-06 Assembly inline inspection — X-ray, CSAM, warpage metrology	

23 MP · QUALIFICATION & PRODUCTION

MP-05

Yield learning, failure pareto and defect analysis

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
20w cont. w2–w22 in stage, runs throughout	40.0 M/M planned	~8.7 FTE while it runs	8 3 in parallel	yes no float	Yield engineer Cost analyst approves

PURPOSE

Run **yield learning** continuously through the ramp — pareto, defect analysis, corrective action, and a yield model checked against the cost target.

Yield is the difference between a product that makes money and one that does not, and first-ramp yield is never where the cost model assumed. This activity runs for the whole stage because yield learning is a rate, not an event: what matters is how fast the pareto's top item is removed and replaced by the next.

WORK FLOW

8 STEPS · 2 LANES

w2	w6	w10	w14	w18	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Yield data pipeline and baseline establishment	2w	main
2	Failure pareto by bin and failure mode	3.5w	main
3	Wafer-level spatial signature analysis	3w	parallel
4	Defect analysis and root cause	4.5w	main
5	Assembly and test yield split	3w	parallel
6	Corrective action and yield improvement	5w	main
7	Excursion detection and response	3w	parallel
8	Yield model against the cost target	5w	main

Step 5 is the split that decides who acts. Die yield, assembly yield and test yield have different owners and different fixes, and a combined number tells none of them what to do.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Test data infrastructure from TEST-11
- Fab yield data from FAB-09
- Assembly yield from ASSY-07
- Bring-up failure analysis from BU-09
- Cost model from DEF-06

PRODUCES

- Yield data pipeline and baseline
- Failure pareto by bin and mode
- Spatial signature analysis
- Defect root causes
- Assembly and test yield split
- Corrective actions and improvement record

- Excursion detection and responses
- Yield model against cost target

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D4	PRODUCES	Yield model versus cost target report. This activity is the deliverable, and the model is what the business plan is checked against.
MP-D5	FEEDS	Production readiness review sign-off. Yield at the readiness review is one of its principal acceptance criteria.

RISKS AND WATCH-OUTS

- ◆ **Yield reported as one number.** Die, assembly and test yield have different owners, and a combined figure directs nobody.
- ◆ **Spatial signatures not analysed.** Edge, centre and radial patterns each point at a different process cause and are visible only on the wafer map.
- ◆ **Corrective actions not verified.** An action taken without confirming the pareto moved is an action that may have done nothing.
- ◆ **Excursions detected at the weekly review.** By then a week of material is affected, and detection has to be automatic.
- ◆ **Yield model not checked against the cost target.** The gap between achieved and assumed yield is the product's margin, and it can go unnoticed for months.

EFFORT SPLIT	40 M/M	ENTRY CRITERIA	FEEDS INTO
Defect analysis and root cause	10.0	● Test data infrastructure running from TEST-11	MP-09 Production readiness review and change control (PCN) setup
Corrective action and improvement	9.0	● Fab and assembly yield data flowing	MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Failure pareto	7.0	● Cost model available from DEF-06	MP-06 Production test program release and guard-band validation
Yield model	6.0		
Spatial signature analysis	4.5	EXIT CRITERIA	
Excursion response	3.5	● Yield split by die, assembly and test ● Corrective actions verified against the pareto ● Yield model reconciled with the cost target	MEASURED BY
Total	40.0	DEPENDS ON	● Yield against the cost model assumption ● Pareto top-item removal rate ● Time from excursion to detection
WHO IT TAKES	~8.7 FTE		
Yield engineer Owns yield learning and the pareto			
Process engineer Fab-side defect root cause		TEST-11 Test data infrastructure — STDF collection and yield database	
Assembly engineer Assembly yield contribution		FAB-09 Hot-lot management and WIP tracking	
Test engineer Test yield and bin analysis		ASSY-07 Assembly yield analysis and process tuning	
Cost analyst Yield model against the business case		BU-09 Silicon debug of anomalies and failure analysis requests	
		DEF-06 Competitive benchmarking and product positioning	

23 MP · QUALIFICATION & PRODUCTION

MP-06

Production test program release and guard-band validation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	20.0 M/M	~9.7 FTE	8	yes	Product engineering
w10–w19 in stage	planned	while it runs	3 in parallel	no float	Operations approves

PURPOSE

Release the **production test program** — frozen, guard-band-validated, qualified and version-controlled — for every unit the product will ever ship.

After this release, a change to test content requires re-qualification. That makes the release a genuine freeze, and it makes the guard-band validation that precedes it the last chance to be sure the limits neither throw away good parts nor let bad ones through.

WORK FLOW

8 STEPS · 2 LANES

w10	w12	w14	w16	w18	w19
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Release candidate program freeze	1w	main
2	Guard-band validation against characterisation	2w	main
3	Test escape and over-kill analysis	1.5w	parallel
4	Production flow validation on volume lots	2w	main
5	Limit adjustment and re-validation	1.5w	parallel
6	Program qualification and sign-off	2w	main
7	Production tester porting and correlation	1.3w	parallel
8	Release and version control	2w	main

Step 3 quantifies both directions of error. Escapes cost the customer's trust and over-kill costs yield, and a limit set without measuring both is optimising one at the other's expense without knowing it.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Sort and final programs from TEST-05 and TEST-06
- Characterisation data from TEST-07 and BU-07
- Correlation results from TEST-10
- Yield data from MP-05
- Production tester fleet from TEST-02

PRODUCES

- Frozen release candidate
- Guard-band validation results
- Escape and over-kill analysis
- Production flow validation on volume lots
- Limit adjustments
- Program qualification and sign-off
- Tester porting and correlation

- Released, version-controlled production programs

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D3	PRODUCES	Production test program release. This activity is the deliverable, and after it the content is frozen.
MP-D5	FEEDS	Production readiness review sign-off. A released and qualified test program is a precondition for the review.

RISKS AND WATCH-OUTS

- ◆ **Guard bands not validated against characterisation.** Limits then rest on convention rather than on what the silicon actually does.
- ◆ **Over-kill unmeasured.** Good parts are discarded on every lot and the loss is attributed to yield rather than to the limits.
- ◆ **Flow validated on one lot.** Lot-to-lot variation is the thing the flow has to tolerate, and one lot cannot show it.
- ◆ **Porting across the tester fleet without correlation.** Tester-to-tester differences then present as yield differences by machine.
- ◆ **Released without version control.** A yield excursion cannot be correlated with a program change nobody can identify.

EFFORT SPLIT	20 M/M	ENTRY CRITERIA	FEEDS INTO
Guard-band validation	5.0	● Programs debugged from TEST-05 and TEST-06 ● Characterisation and correlation data available ● Volume lots available for flow validation	MP-07 Test time reduction and multi-site conversion
Production flow validation	4.0		MP-09 Production readiness review and change control (PCN) setup
Program qualification	3.5		MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Escape and over-kill analysis	3.0	EXIT CRITERIA ● Guard bands derived from characterisation, not convention ● Both escape and over-kill quantified ● Flow validated across multiple lots and testers	
Release and version control	2.5		
Tester porting	2.0		
Total	20.0		MEASURED BY ● Escape rate against the DPPM target ● Over-kill rate at the released limits ● Tester-to-tester yield spread after porting
WHO IT TAKES	~9.7 FTE	DEPENDS ON	
Product engineering Owns the release and its guard bands		TEST-05 Wafer sort test program development	
Test engineer Program freeze, porting and correlation		TEST-06 Final / package test program development	
Quality engineer Qualification and sign-off		TEST-09 Test time and cost optimisation — parallelism, site count	
Yield engineer Escape and over-kill quantification		TEST-10 Correlation between ATE, bench and system results	
Operations Fleet deployment and version control		BU-07 Shmoo across voltage, frequency and temperature	
		MP-05 Yield learning, failure pareto and defect analysis	

23 MP · QUALIFICATION & PRODUCTION

MP-07

Test time reduction and multi-site conversion

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	16.0 M/M	~7.7 FTE	8	no	Test engineer
w14–w23 in stage	planned	while it runs	3 in parallel	has float	Yield engineer approves

PURPOSE

Convert production test to **multi-site** and cut test time — the change that makes the cost per unit sustainable at volume.

Single-site testing is fine for a hundred units and ruinous for a million. Multi-site conversion multiplies throughput per tester and divides the capital needed for the ramp, and it is a real engineering exercise because parallel sites interfere with each other in ways single-site testing never reveals.

WORK FLOW

8 STEPS · 2 LANES

w14	w16	w18	w20	w22	w23
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Multi-site feasibility on the production hardware	1w	main
2	Multi-site load board and probe card conversion	2w	main
3	Test time reduction on the released content	1.5w	parallel
4	Parallel program adaptation	2w	main
5	Site-to-site variation analysis	1.5w	parallel
6	Multi-site correlation and validation	2w	main
7	Throughput and capacity model update	1.3w	parallel
8	Release of the converted flow	2w	main

Step 5 is what makes the conversion trustworthy. Sites differ — in contact, in power delivery, in noise coupling — and a systematic site bias shows up as a yield difference that gets chased as a process problem for weeks.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Released production programs from MP-06
- Tester pin count and configuration from TEST-02
- Test time analysis from TEST-09
- Load board and probe card designs from TEST-03 and TEST-04
- Volume forecast from MP-10

PRODUCES

- Multi-site feasibility assessment
- Converted multi-site hardware
- Test time reduction on released content
- Adapted parallel programs
- Site-to-site variation analysis
- Multi-site correlation and validation

- Updated throughput and capacity model
- Released converted flow

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D3	FEEDS	Production test program release. The multi-site flow is a controlled revision of the released program.
MP-D6	FEEDS	Ramp plan and supply commitment. Throughput per tester is what the capacity commitment is built from.

RISKS AND WATCH-OUTS

- ◆ **Site-to-site bias undetected.** It appears as a yield difference and is investigated as a process problem for weeks.
- ◆ **Pin count insufficient for the site count.** The tester configuration from TEST-02 bounds this, and it cannot be changed now.
- ◆ **Parallel power delivery inadequate.** Four parts drawing full current through one load board droop in ways one part does not.
- ◆ **Conversion done after the ramp starts.** Capital is then committed at single-site throughput and cannot be recovered.
- ◆ **Time reduction changing coverage.** Any content change after MP-06 requires re-qualification, which is expensive at this point.

EFFORT SPLIT	16 M/M	ENTRY CRITERIA	DEPENDS ON
Multi-site hardware conversion	4.0	● Production programs released from MP-06	MP-06 Production test program release and guard-band validation
Parallel program adaptation	3.5	● Tester configuration supporting the site count	TEST-02 ATE platform selection and tester time booking
Correlation and validation	3.0	● Test time analysis available from TEST-09	TEST-03 Probe card design, fabrication and qualification
Test time reduction	2.5		TEST-04 Load board / DUT board design, fabrication and bring-up
Site variation analysis	1.8		TEST-09 Test time and cost optimisation — parallelism, site count
Capacity model	1.3		
Total	16.0	EXIT CRITERIA	FEEDS INTO
WHO IT TAKES	~7.7 FTE	● Site-to-site bias measured and within tolerance	MP-09 Production readiness review and change control (PCN) setup
Test engineer		● Parallel power delivery adequate at full current	MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM
Owns multi-site conversion		● Conversion complete before capital is committed to the ramp	
Test hardware engineer			
Load board and probe card conversion			
Product engineering			
Site correlation and coverage preservation			
Operations planner			MEASURED BY
Capacity and throughput model			● Throughput multiple achieved against site count
Yield engineer			● Site-to-site yield spread
Site bias detection			● Cost per tested unit after conversion

23 MP · QUALIFICATION & PRODUCTION

MP-08

Process corner and split-lot validation

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
9w	14.0 M/M	~6.8 FTE	8	no	Product engineering
w8-w17 in stage	planned	while it runs	3 in parallel	has float	Yield engineer approves

PURPOSE

Validate the design across **process corners** using split lots — material deliberately fabricated at the edges of the process window.

Production silicon spans the process distribution, and the units bring-up characterised came from one lot near its centre. Split lots produce the fast and slow corners on purpose, which is the only way to know whether the design's margin survives the population it will actually be built from.

WORK FLOW

8 STEPS · 2 LANES

w8	w10	w12	w14	w16	w17
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Split-lot design across process corners	1w	main
2	Split lot fabrication and tracking	2w	main
3	Parametric spread and Cpk analysis	1.5w	parallel
4	Corner characterisation on split material	2w	main
5	Design margin at the corners	1.5w	parallel
6	Corner-to-model correlation	2w	main
7	Process window recommendation	1.3w	parallel
8	Corner sign-off and report	2w	main

Step 6 closes a loop opened right back at signoff. The design was timed against corner models; measuring real corner silicon says whether those models were pessimistic, accurate, or — the dangerous case — optimistic.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Split lot material from FAB-06
- Corner models from PDK-05
- Characterisation content from TEST-07
- Shmoo baseline from BU-07
- Process window from TECH-05

PRODUCES

- Split-lot design
- Fabricated and tracked split lots
- Parametric spread and Cpk analysis
- Corner characterisation results
- Design margin at the corners
- Corner-to-model correlation

- Process window recommendation
- Corner sign-off report

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D4	FEEDS	Yield model versus cost target report. Corner yield is what the model has to predict across the real distribution.
MP-D1	FEEDS	Qualification plan and JEDEC-compliant qualification report. Corner validation is part of demonstrating the design is production-worthy.

RISKS AND WATCH-OUTS

- ◆ **Corner models optimistic against real silicon.** The design was signed off against those models, and a gap here means the margin does not exist.
- ◆ **Split lots not actually at the corners.** Material that drifted back toward centre proves nothing about the edges.
- ◆ **Cpk not computed.** Capability against the specification is what predicts production yield, and a pass/fail result does not.
- ◆ **Corner characterisation skipped for schedule.** The population's edges are then only seen in production, as an unexplained yield tail.
- ◆ **Process window recommendation not fed back to the fab.** The data exists and the process keeps running at its original centre point.

EFFORT SPLIT	14 M/M	ENTRY CRITERIA	FEEDS INTO
Corner characterisation	3.5	● Split lot material available from FAB-06	MP-06 Production test program release and guard-band validation
Split lot execution	3.0	● Corner models available from PDK-05	
Corner-to-model correlation	2.5	● Characterisation content ready from TEST-07	MP-09 Production readiness review and change control (PCN) setup
Cpk analysis	2.0		
Design margin assessment	1.8	EXIT CRITERIA	MP-05 Yield learning, failure pareto and defect analysis
Window recommendation	1.3		
Total	14.0	● Split material verified to be at the intended corners	MEASURED BY
WHO IT TAKES	~6.8 FTE	● Cpk computed against the specification	● Corner spread achieved against intended
Product engineering Owns corner validation		● Model-to-silicon correlation established at both corners	● Cpk against the specification
Process engineer Split lot design and execution		DEPENDS ON	● Model-to-silicon gap at the corners
Characterisation engineer Corner measurement		FAB-06 Inline metrology and defect inspection monitoring	
Design engineer Model correlation and margin assessment		PDK-05 Compiler instance PPA characterisation against block budgets — density, Vmin, access time, leakage	
Yield engineer Cpk and distribution analysis		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps	
		BU-07 Shmoo across voltage, frequency and temperature	
		TECH-05 Fab capacity and tapeout slot reservation; hot-lot policy agreement	

23 MP · QUALIFICATION & PRODUCTION

MP-09

Production readiness review and change control (PCN) setup

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
6w	8.0 M/M	~5.8 FTE	6	yes	Program manager
w18–w24 in stage	planned	while it runs	2 in parallel	no float	Executive sponsor approves

PURPOSE

Hold the **production readiness review** — the cross-functional gate where every function states whether it is ready — and stand up change control for everything that follows.

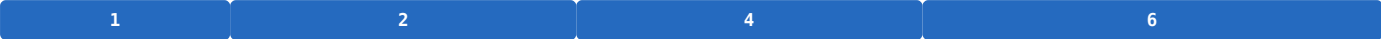
This is the program's last formal decision point. After it, the product is in production and every change goes through customer notification. The review's value is that it forces each function to say ready or not ready on the record, which is a different act from being asked in a meeting whether there are any concerns.

WORK FLOW

6 STEPS · 2 LANES

w18	w20	w22	w24
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Readiness criteria and evidence collection	1w	main
2	Cross-functional readiness review	1.5w	main
3	Change control and document baseline	1w	parallel
4	Gap closure and conditional item tracking	1.5w	main
5	Escalation of unresolved items	1w	parallel
6	Sign-off and PCN process establishment	2w	main

Step 4's conditional items are where a review either works or becomes theatre. A conditional pass with an owner, a date and a consequence is a real mechanism; a conditional pass without them is an unconditional pass with a note attached.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Qualification results from MP-02, MP-03 and MP-04
- Test program release from MP-06
- Yield model from MP-05
- Supply commitment from MP-10
- Compliance certificates from MP-11

PRODUCES

- Readiness criteria and collected evidence
- Cross-functional review record
- Change control and document baseline
- Gap closure and conditional item tracking
- Escalation record
- Signed-off readiness and established PCN process
- Mass production release record

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D5	PRODUCES	Production readiness review sign-off. This activity is the deliverable — the record of who declared ready and on what evidence.
MP-D9	PRODUCES	Mass Production release record. The formal statement that the product is in production, with its conditions.

RISKS AND WATCH-OUTS

- ◆ **Review held without evidence.** Function-by-function assurances are not the same as qualification data, and a review that accepts them is not a gate.
- ◆ **Conditional items without owners or dates.** The condition is then never closed and the pass was unconditional in practice.
- ◆ **Change control not established.** Post-production changes then reach customers without notification, which is a contractual failure.
- ◆ **Dissent not recorded.** A function that said not-ready and was overruled is the program's most valuable record if something goes wrong.
- ◆ **Document baseline not frozen.** Which datasheet, which errata, which test program was released becomes ambiguous immediately.

EFFORT SPLIT	8 M/M	DEPENDS ON	MEASURED BY
Cross-functional review	2.3	MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB	● Readiness criteria met against total ● Conditional items closed by their date ● Post-release issues that the review had evidence for
Evidence collection	1.8		
Gap closure and tracking	1.5	MP-03 ESD (HBM, CDM) and latch-up qualification	
Sign-off and PCN setup	1.3	MP-04 Package qualification — MSL, drop, bend, board-level reliability	
Change control baseline	0.8	MP-05 Yield learning, failure pareto and defect analysis	
Escalation	0.5	MP-06 Production test program release and guard-band validation	
Total	8.0	MP-07 Test time reduction and multi-site conversion	
WHO IT TAKES ~5.8 FTE		MP-08 Process corner and split-lot validation	
Program manager Owns the review and its record		MP-10 Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM	
Quality manager Readiness criteria and evidence standards		MP-11 Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety	
Function leads Ready or not-ready declarations		MP-12 Documentation release — datasheet, user guide, errata, application notes	
Document control Baseline freeze and change control		BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule	
Executive sponsor Sign-off authority and escalation		The review is the convergence point for the whole stage — it consumes the output of nearly every other MP activity.	
ENTRY CRITERIA		FEEDS INTO	
● Qualification results available from MP-02 through MP-04 ● Test program released from MP-06 ● Supply and compliance status available		<i>Terminal — nothing downstream inside the template.</i>	
EXIT CRITERIA			
● Every conditional item with an owner, a date and a consequence ● Change control and PCN process established before release ● Dissenting positions recorded, not resolved off the record			

23 MP · QUALIFICATION & PRODUCTION

MP-10

Capacity, supply chain and ramp commitment — wafer starts, substrate, HBM

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w	12.0 M/M	~4.3 FTE	8	yes	Operations manager
w10–w22 in stage	planned	while it runs	3 in parallel	no float	Program manager approves

PURPOSE

Commit the **supply chain and the ramp** — wafers, substrates, HBM, assembly and test capacity — against a demand forecast, with the lead times and second sources that make the commitment real.

A ramp is a chain of commitments made months before the volume they serve. Substrates have long lead times, HBM is allocated across the industry, and assembly capacity is booked. Any one of them short stops the ramp, and none of them can be fixed quickly.

WORK FLOW

8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Demand forecast and volume commitment	1.5w	main
2	Wafer, substrate and HBM supply commitment	2.5w	main
3	Lead time and buffer analysis	2w	parallel
4	Assembly and test capacity commitment	2.5w	main
5	Cost per unit at volume	2w	parallel
6	Ramp plan and build schedule	2.5w	main
7	Logistics and distribution setup	1.5w	parallel
8	Supply chain risk and second-source plan	3w	main

Step 8 runs last but covers everything before it. Each commitment made in steps 2 and 4 is a single point of failure, and the plan has to say what happens if any one of them fails — before it does.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Demand forecast from DEF-05 and product marketing
- Yield model from MP-05
- Test throughput from MP-07
- HBM supply position from ASSY-02

PRODUCES

- Demand forecast and volume commitment
- Wafer, substrate and HBM supply commitments
- Lead time and buffer analysis
- Assembly and test capacity commitments

- Substrate and interposer supply from PKGD-09
- Cost per unit at volume
 - Ramp plan and build schedule
 - Logistics and distribution setup
 - Supply chain risk and second-source plan

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D6	PRODUCES	Ramp plan and supply commitment. This activity is the deliverable.
MP-D4	FEEDS	Yield model versus cost target report. Cost per unit at volume is where the yield model meets the actual supply cost.

RISKS AND WATCH-OUTS

- ◆ **HBM allocation insufficient.** It is allocated across the industry and a shortfall stops the ramp regardless of everything else.
- ◆ **Substrate lead time underestimated.** Advanced substrates are long-lead and capacity-constrained, and the buffer has to reflect that.
- ◆ **Ramp planned against an optimistic yield.** Volume commitments then require more wafer starts than were booked.
- ◆ **No second source anywhere in the chain.** Every single-source node is a single point of failure for the whole product.
- ◆ **Capacity committed before the readiness review.** A conditional readiness outcome then meets an unconditional supply commitment.

EFFORT SPLIT	12 M/M	EXIT CRITERIA	FEEDS INTO
Supply commitment	3.0	● Every long-lead item committed with buffer	MP-09 Production readiness review and change control (PCN) setup
Capacity commitment	2.5	● Ramp built on the achieved yield, not the assumed one	
Ramp plan	2.0	● Single-source nodes identified with a contingency	
Risk and second-source plan	1.8		
Lead time analysis	1.5		
Cost at volume	1.3		
Total	12.0		
WHO IT TAKES	~4.3 FTE	DEPENDS ON	MEASURED BY
Operations manager Owns the ramp and supply commitments		MP-05 Yield learning, failure pareto and defect analysis	● Long-lead items committed against required
Procurement Supplier commitments and second sources		MP-06 Production test program release and guard-band validation	● Ramp achieved against plan
Product marketing Demand forecast		MP-07 Test time reduction and multi-site conversion	● Single-source nodes with a contingency
Cost analyst Cost per unit at volume		ASSY-02 HBM stack procurement and incoming inspection	
Program manager Ramp against readiness		PKGD-09 Substrate and interposer supplier selection and lead-time booking	
ENTRY CRITERIA		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)	
● Demand forecast available ● Yield model available from MP-05 ● Test throughput known from MP-07		BU-11 Respin versus metal-fix decision analysis — ECO scope, cost, schedule	
		BU-12 Customer sample readiness and release package	

23 MP · QUALIFICATION & PRODUCTION

MP-11

Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
12w w10–w22 in stage	18.0 M/M planned	~6.5 FTE while it runs	8 3 in parallel	no has float	Compliance engineer Product marketing approves

PURPOSE

Obtain the **compliance and certification** the product cannot ship without — interface compliance, EMC and safety, and materials declarations.

These are gates rather than engineering activities: without a PCIe or CXL compliance listing, without EMC certification, without RoHS and REACH declarations, the product cannot be sold in most markets regardless of how well it works.

WORK FLOW

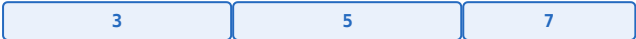
8 STEPS · 2 LANES

w10	w13	w16	w19	w22
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MAIN SEQUENCE



IN PARALLEL



#	STEP	TAT	LANE
1	Applicable standards and certification scope	1.5w	main
2	PCIe/CXL compliance testing at a plugfest	2.5w	main
3	Compliance failure remediation	2w	parallel
4	EMC and safety testing	2.5w	main
5	Documentation and declaration preparation	2w	parallel
6	Materials compliance — RoHS, REACH, conflict minerals	2.5w	main
7	Export classification	1.5w	parallel
8	Certificate issue and registration	3w	main

Step 2 depends on an event calendar the program does not control. Plugfests happen on fixed dates a few times a year, and missing one costs a quarter rather than a week.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Interface compliance results from BU-05
- Materials declarations from the supply chain
- Package construction from PKGD-11
- Applicable regulatory requirements
- Certification body and plugfest schedules

PRODUCES

- Applicable standards and scope
- PCIe/CXL compliance test results
- Remediation record
- EMC and safety test results
- Compliance documentation and declarations

- Materials compliance declarations
- Export classification
- Issued certificates and registrations

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D7

PRODUCES

Compliance certificates. This activity is the deliverable, and without it the product cannot ship.

MP-D8

FEEDS

Datasheet and product documentation set. Compliance statements and classifications are part of the product documentation.

RISKS AND WATCH-OUTS

- ◆ **Plugfest date missed.** They occur a few times a year on fixed dates, and missing one delays the listing by a quarter.
- ◆ **Materials declarations not collected from suppliers.** Every component in the package needs one, and collecting them takes months.
- ◆ **Compliance failure discovered at the plugfest.** Pre-compliance testing exists precisely so that formal testing is a confirmation.
- ◆ **Export classification left late.** A high-performance accelerator may be controlled, and the classification determines which markets exist.
- ◆ **Certification scope assessed for one region.** Requirements differ by market, and a missing certification closes a region entirely.

EFFORT SPLIT	18 M/M	ENTRY CRITERIA	DEPENDS ON
PCIe/CXL compliance	4.5	● Interface results available from BU-05	BU-05 Interface bring-up — PCIe/CXL link training, HBM training, die-to-die
EMC and safety testing	4.0	● Package construction and materials known from PKGD-11	PKGD-11 Package design freeze, DRC and tooling release
Materials compliance	3.5	● Target markets defined	ASSY-02 HBM stack procurement and incoming inspection
Certificate issue and registration	2.5		DEF-05 Memory and interface bandwidth requirements (HBM stacks, PCIe/CXL lanes, D2D)
Failure remediation	2.0	EXIT CRITERIA	
Documentation and export	1.5	● Compliance listing obtained, not just tested	
Total	18.0	● Materials declarations collected for every component	FEEDS INTO
WHO IT TAKES	~6.5 FTE	● Export classification completed before shipment	MP-09 Production readiness review and change control (PCN) setup
Compliance engineer Owns certification and declarations			MP-12 Documentation release — datasheet, user guide, errata, application notes
Interface engineer Compliance testing and remediation			
Regulatory affairs Standards applicability and export classification			MEASURED BY
Supply chain Supplier materials declarations			● Certifications obtained against required
Product marketing Target markets and their requirements			● Plugfest attendance against schedule
			● Markets open at launch

23 MP · QUALIFICATION & PRODUCTION

MP-12

Documentation release — datasheet, user guide, errata, application notes

TAT	EFFORT	IMPLIED TEAM	STEPS	CRITICAL PATH	OWNER
10w	16.0 M/M	~7.0 FTE	8	no	Technical writer
w14–w24 in stage	planned	while it runs	3 in parallel	has float	Product marketing approves

PURPOSE

Release the **product documentation** — datasheet, user guide, errata, application notes — the artefacts a customer actually receives alongside the silicon.

The documentation is the product as far as most customers experience it. A datasheet parameter that cannot be traced to a measurement, a programming model that omits a required sequence, or an errata list that lags the silicon each cost far more support effort than the writing would have.

WORK FLOW

8 STEPS · 2 LANES

w14	w16	w18	w20	w22	w24
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MAIN SEQUENCE

1	2	4	6	8
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IN PARALLEL

3	5	7
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#	STEP	TAT	LANE
1	Documentation scope and ownership	1.5w	main
2	Datasheet from characterisation data	2w	main
3	Parameter traceability to measurement	1.5w	parallel
4	User guide and programming model	2w	main
5	Localisation and format production	1.5w	parallel
6	Errata and application notes	2w	main
7	Revision and distribution control	1.5w	parallel
8	Technical review, approval and release	2.5w	main

Step 3 is what makes the datasheet defensible. Every number should point at a measurement, its conditions and its sample size, so that a customer's question about a parameter has an answer rather than a search.

INPUTS AND WORKING ARTIFACTS

CONSUMES

- Characterisation data from TEST-07 and BU-07
- Errata list from BU-10
- Performance results from BU-08
- Compliance classifications from MP-11
- Programming model from RTL-06 and ARCH-06

PRODUCES

- Documentation scope and ownership
- Datasheet
- Parameter traceability record
- User guide and programming model documentation
- Localised and formatted outputs

- Errata and application notes
- Revision and distribution control
- Reviewed and released documentation set

Working artifacts are how the activity carries its own work. They are not the stage's key deliverables — how the two relate is below.

RELATIONSHIP TO THE STAGE'S KEY DELIVERABLES

MP-D8	PRODUCES	Datasheet and product documentation set. This activity is the deliverable.
MP-D9	FEEDS	Mass Production release record. A released documentation set is part of what the release record attests to.

RISKS AND WATCH-OUTS

- ◆ **Datasheet parameters without traceable measurements.** A customer question then has no answer, and the number cannot be defended.
- ◆ **Errata lagging the silicon.** The most useful document ships out of date, which is worse than shipping it late.
- ◆ **Programming model incomplete.** Required initialisation sequences omitted from the guide become support cases, one per customer.
- ◆ **Documentation written from specifications rather than measurements.** It then describes the intended product rather than the one shipping.
- ◆ **No revision control on released documents.** Which datasheet revision a customer holds becomes unknowable.

EFFORT SPLIT	16 M/M	EXIT CRITERIA	FEEDS INTO
Datasheet	4.5	● Every datasheet parameter traceable to a measurement ● Errata current with the shipping silicon ● Documentation under revision and distribution control	MP-09 Production readiness review and change control (PCN) setup
User guide	3.5		BU-12 Customer sample readiness and release package
Errata and application notes	2.8		
Review and release	2.5		
Parameter traceability	1.8	DEPENDS ON	MEASURED BY ● Parameters with traceable measurements ● Errata currency against silicon revision ● Support cases attributable to documentation gaps
Revision control	1.0		
Total	16.0		
WHO IT TAKES	~7.0 FTE		
Technical writer Owns the documentation set		TEST-07 Characterisation test content — shmoo, Vmin/Fmax, PVT sweeps	
Product engineering Datasheet parameters and traceability		TEST-10 Correlation between ATE, bench and system results	
Applications engineer User guide and application notes		BU-07 Shmoo across voltage, frequency and temperature	
Design engineer Programming model accuracy		BU-08 Performance validation against the architecture model	
Product marketing Approval and release		BU-10 Errata capture, workaround definition and documentation	
ENTRY CRITERIA		MP-11 Compliance and certification — PCIe/CXL compliance, RoHS/REACH, safety	
		MP-02 Reliability stress — HTOL, HTS, temperature cycle, uHAST, THB	
● Characterisation data available from TEST-07 and BU-07 ● Errata list released from BU-10 ● Compliance classifications from MP-11			